

The Recognition Kernel Framework

Collected Theorems, Theory, and Certified Results

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About this volume

This volume collects, without omission, the theorem ladders, theory documents, claim boundaries, and certified papers of the Recognition Kernel Framework as recorded in three repositories: **Recognition-Kernel-Framework**, **RH-Framework**, and **Publications**. Chapters are automated renderings of the repository sources; any chapter whose automated conversion was not compile-safe is included verbatim so that no content is lost. Appendix B lists every included source file with its SHA-256 digest. The authoritative artefacts remain the repositories themselves, whose certificates and tests regenerate every pinned result.

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Part I

The Recognition Kernel Framework: Charter and Boundaries

Chapter 1

Recognition Kernel Framework

[svg]

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Mathematical proof transport, lawful-transition accounting, open-residue classification, and reproducible certificates

The Recognition Kernel Framework is a technical and mathematical architecture for evaluating whether a recognized structure survives a transformation after lawful transport and declared memory have been accounted for. The remaining open residue is classified and recorded in a reproducible certificate.

The operational engine is the **Recognition–Null Kernel Engine (RNKE)**. A domain cartridge is represented by

$$(S_d, R_d, E_d, \tau_d, K_d),$$

where S_d is a domain state, R_d is a recognition operator, E_d is an evolution or transition, τ_d is an index or order parameter, and K_d is a kernel evaluator.

The common execution pattern is:

```
state
-> recognition target
-> lawful transition
-> open-residue audit
-> classification
-> technical action
-> certificate
```

1.1 Archival record

Reviewer-oriented release: **Version 0.1.0-review**

Zenodo DOI: **10.5281/zenodo.21760119**

Cite the exact release or commit and the certificate package actually reviewed.

1.2 First public verification cartridge

This repository begins with a mathematical proof-transport cartridge imported from the pinned RH-Framework source commit:

c588ded973a395b5f37c82f670616160979a2e

The imported certificate campaign covers:

oriented cut scalar
-> quarter-turn structure
-> native Euler exponential and flow
-> native logarithm and powers
-> finite natural arithmetic
-> prime factorization identities
-> Möbius and von Mangoldt identities
-> Dirichlet zeta and Euler-product obligations on $\text{Re}(s) > 1$

Current archived status:

Package	Status
FOO_FOOE_EULER_VO_1	PASS_FOO_FOOE_RIGOROUS_COMPUTATIONAL_AUDIT
FOOGHI_LOG_ARITHMETIC_ZETA_VO_1	PASS_FOOGHI_LOG_ARITHMETIC_ZETA_AUDIT
Implemented foundation campaign	PASS_IMPLEMENTED_FOUNDATION_CERTIFICATE_CAMPAIGN

Canonical campaign SHA-256:

8728ab0319afe5fb6e6329deae479cfa93f50bb4422ee863aaaf4f911dd9582

1.3 What the PASS means

A PASS means that every declared computational obligation in the implemented package completed successfully, required source pins matched, exact or outward bounds satisfied the package rules, and adversarial negative controls were detected.

A PASS does **not** certify claims outside the package boundary.

1.4 Scientific boundary

This release does not certify:

Fourier-Poisson-Gamma-xi completion;
explicit formula or completed-Weil positivity;
active-band source positivity;
eta-zero endpoint positivity;
the Riemann Hypothesis.

Those items remain NOT CERTIFIED or OPEN in this software release.

1.5 Consolidated theorem archive

The reviewer-facing theorem surface is under **theorem/**. It contains the transferred RH-framework capsules, native cut/completion results, the current generator-to-jet capstone chain, and the thermodynamic theorem archive.

The current advanced reading chain is:

```
theorem/41_cut_graded_universal_generator_theorem.md
-> theorem/42_cut_graded_lambda_jacobian_tower_theorem.md
-> theorem/43_bilateral_jet_flow_recognition_capstone_theorem.md
-> theorem/44_madhava_smriti_bilateral_jet_flow_closure_theorem.md
-> theorem/thermodynamics/README.md
```

Use `theorem/README.md` for the complete map, `theorem/SOURCE_PR_INDEX.md` for source ancestry, and `theorem/BRANCH_CONSOLIDATION_AUDIT.md` for the repository-wide branch verification.

The theorem capsules retain their own evidence labels and claim boundaries. Their inclusion in `main` does not silently upgrade `LOCAL PASS`, `USER-REPORTED PASS`, or `IMPLEMENTED / USER RUN REQUIRED` into a stronger certification state. Apparently even repositories need adults in the room.

1.6 Reviewer route

1. Read `REVIEWER_GUIDE.md`.
2. Read `CLAIM_BOUNDARY.md`.
3. Inspect `CERTIFICATE_INDEX.md`.
4. Read the consolidated theorem map in `theorem/README.md`.
5. Reproduce the campaign using `REPRODUCE.md`.
6. Compare generated JSON hashes with the archived certificate artifacts.
7. Inspect theorem sources, verifier obligations, and negative controls directly.

1.7 Terminology

The public umbrella name is **Recognition Kernel Framework**. The operational engine is **Recognition–Null Kernel Engine (RNKE)**. Recognition–Seam Calculus, EMK, UGD, RMG, and ECL are mathematical, geometric, computational, dynamic, and certificate-runtime layers or representations used within the broader architecture.

1.8 Provenance policy

Certified source files are preserved byte-for-byte under their historical paths and namespaces. Historical names such as `rh_framework` remain inside the imported cartridge because changing them would change source hashes and invalidate the archived certificate identity.

1.9 Public-review scope

This repository publishes technical material for scientific and engineering review. It certifies only the packages, source pins, obligations, tolerances, and artifacts listed in the certificate index. Public availability does not imply that every research claim in related repositories has been verified.

RKF/README.md

Chapter 2

Claim Boundary

2.1 Current certified scope

The first public reviewer cartridge contains two implemented foundational certificate packages.

F00 / F00-E

Certified declared obligations include:

```
exact cut-scalar multiplication;
quarter-turn relation and dagger;
exact norm-square identities;
factorial/binomial coefficient identities;
explicit factorial-tail bounds;
Euler circular residual enclosure;
flow composition and generator sign;
source-order policy checks;
adversarial negative controls.
```

F00-G / F00-H / F00-I

Certified declared obligations include:

```
positive logarithm series packets and exact tails;
Exp(Log(x)) enclosure packets;
logarithmic product-law packets;
division, gcd, and Bezout packets;
prime-factor reconstruction packets;
Mobius and von Mangoldt coefficient identities;
dyadic zeta convergence bounds;
finite Euler products with omitted-tail bounds;
nonvanishing product-tail obligations;
adversarial negative controls.
```

2.2 Meaning of certificate status

PASS

Every declared computational obligation in that package passed, required source pins matched, and all required negative controls behaved as specified.

INCONCLUSIVE

At least one obligation was incomplete, unavailable, stale, not strictly enclosed, or failed to produce a lawful final result. **INCONCLUSIVE** is not silently promoted to **PASS**.

FAIL

A required declared obligation was rigorously contradicted or a required negative control was not detected.

STALE

One or more source pins or immutable inputs no longer match the package specification.

2.3 What is not established by this release

This repository does not claim that the current certificates establish:

```
all universal mathematical statements appearing in every related manuscript;
Fourier or Poisson reconstruction;
Gamma and completed-xi continuation;
Hadamard factorization;
the explicit prime-Gamma-zero formula;
the completed-Weil sign;
active-band source positivity;
eta-zero endpoint positivity;
 $K_0 \geq 0$ ;
the Riemann Hypothesis.
```

These remain **NOT CERTIFIED**, **OPEN**, or subject to later audit packages.

2.4 Manuscript status versus certificate status

The repository keeps distinct labels for:

```
manuscript theorem classification;
computational certificate status;
external independent review status.
```

A manuscript theorem may be classified **PROVED** while its executable audit is **NOT IMPLEMENTED** or **INCONCLUSIVE**. Conversely, a computational **PASS** certifies only the finite obligations declared by its reduction and package specification.

2.5 Recognition Kernel interpretation

The current public cartridge demonstrates the RNKE pattern on mathematical proof transport:

```
declared state and theorem dependency
-> lawful derivation step
-> exact or outward audit
-> residue classification
-> certificate
```

The difficult number-theory application is the demonstrator. The public claim is the reproducible recognition-kernel architecture and the declared verified obligations, not an RH solution.

RKF/CLAIM_BOUNDARY.md

Chapter 3

Terminology and Filing Alignment

3.1 Controlled public hierarchy

Recognition Kernel Framework
public umbrella architecture

Recognition-Null Kernel Engine (RNKE)
operational state / recognition / evolution / kernel evaluator

Recognition Residue Calculus
parent recognition, residue, compensation, and memory logic

Recognition-Seam Calculus (RSC)
mathematical calculus of cut, seam, recognition, memory, and open residue

EMK
geometric and tensor recognition representation

UGD
phase-scale-seam computational and ledger representation

RMG
rotational, flux, motion, and dynamic representation

ECL
executable certificate and classification runtime

Mathematical proof-transport cartridge
current reviewer implementation imported from RH-Framework

3.2 RNKE domain cartridge

A domain cartridge is represented by

$$(S_d, R_d, E_d, \tau_d, K_d),$$

with:

- S_d : domain or evidence state;
- R_d : recognition operator, predicate, invariant, or admissibility rule;

- `E_d`: evolution, transformation, theorem step, measurement, or update;
- `tau_d`: clock, index, dependency order, or execution sequence;
- `K_d`: kernel evaluator or recognition-gap computation.

3.3 Mathematical proof-transport mapping

RNKE term	Current mathematical cartridge
Domain state	theorem source, implementation, specification, and result artifacts
Recognition target	declared theorem obligation and dependency policy
Evolution	one derivation or promotion step
Lawful transport	proved prior theorem, exact identity, declared bound, or admitted transformation
Open residue	failed identity, source mismatch, missing tail, invalid rearrangement, undeclared dependency, or unclosed domain obligation
Kernel evaluator	deterministic verifier and negative-control suite
Classification	PASS, INCONCLUSIVE, FAIL, or STALE
Technical action	accept, repair, downgrade, rerun, or reject
Certificate	canonical JSON with source pins, obligations, results, and hashes

3.4 Preferred public wording

Use precise technical expressions such as:

`undeclared dependency`
`unproved external theorem dependency`
`provenance mismatch`
`open residue`
`lawful transport`
`recognition-preserving transition`
`bounded residue`
`actionable residue`
`computational audit`
`proof-transport certificate`

Avoid loaded, adversarial, or promotional rhetoric. Public claims should describe the exact dependency, residue, failed obligation, or certificate boundary instead of assigning moral or dramatic labels to mathematical objects.

3.5 Historical identifiers

Certified files retain historical paths, namespaces, schemas, and package IDs. In particular, `rh_framework` remains inside the imported cartridge because its source blob is pinned by the archived certificate. Historical identity is provenance, not the current public umbrella name.

3.6 Filing-alignment notice

This terminology file records technical consistency with the inventor's existing recognition-kernel filing and research records. It does not reproduce confidential filing materials, define legal claim scope, or provide an opinion on patent validity, priority, infringement, or enforceability.

`RKF/TERMINOLOGY_AND_FILING_ALIGNMENT.md`

Chapter 4

Reviewer Guide

4.1 Purpose

This repository is organized for independent review of a Recognition–Null Kernel Engine mathematical proof-transport cartridge.

The reviewer is not asked to accept a broad framework claim from terminology alone. The reviewer can inspect:

```
pinned theorem sources;  
executable implementation;  
package specifications;  
exact and interval obligations;  
negative controls;  
archived result JSON;  
canonical hashes;  
clean-run workflow.
```

4.2 Recommended review order

1. Scope and terminology

Read:

```
README.md  
TERMINOLOGY_AND_FILING_ALIGNMENT.md  
CLAIM_BOUNDARY.md
```

Confirm that the public umbrella is the Recognition Kernel Framework, the operational engine is RNKE, and the current cartridge has a limited declared mathematical scope.

2. Provenance

Read:

```
SOURCE_PROVENANCE.md  
certificates/foundation/campaign_result.json
```

Verify the source commit and canonical campaign hash.

3. Foundational source package

Inspect:

```
src/rh_framework/native_summability.py
theorems/foundation/F00E_NATIVE_EULER_FROM_IOTA_COMPLEX.md
certificates/foundation/F00_F00E_EULER_V0_1/spec.json
certificates/foundation/F00_F00E_EULER_V0_1/verify.py
certificates/foundation/F00_F00E_EULER_V0_1/result.json
```

Questions to ask:

Are source pins exact?
Are all required obligations present?
Are exact identities checked without ordinary floating-point proof margins?
Do negative controls detect wrong quarter-turn, dagger, factorial, exponential, and flow-sign
↔ constructions?
Does the scientific claim boundary remain limited?

4. Logarithm, arithmetic, and zeta package

Inspect:

```
theorems/foundation/FOOG_NATIVE_LOGARITHM_AND_POWERS.md
theorems/foundation/FOOH_NATIVE_NATURAL_ARITHMETIC_AND_PRIME_FACTORIZATION.md
theorems/foundation/FOOI_NATIVE_DIRICHLET_ZETA_EULER_PRODUCT.md
certificates/foundation/FOOGHI_LOG_ARITHMETIC_ZETA_V0_1/spec.json
certificates/foundation/FOOGHI_LOG_ARITHMETIC_ZETA_V0_1/verify.py
certificates/foundation/FOOGHI_LOG_ARITHMETIC_ZETA_V0_1/result.json
```

Questions to ask:

Are logarithm intervals outward and accompanied by explicit tails?
Are finite arithmetic identities exact?
Are prime-factor and coefficient obligations finite and deterministic?
Are dyadic and Euler-product tails explicit?
Are invalid rearrangement or arithmetic controls rejected?
Does the package avoid claiming analytic continuation or RH?

5. Reproduce

Follow REPRODUCE.md on a clean clone.

Required final output:

```
PASS_IMPLEMENTED_FOUNDATION_CERTIFICATE_CAMPAIGN
FOOGHI_LOG_ARITHMETIC_ZETA_V0_1 PASS_FOOGHI_LOG_ARITHMETIC_ZETA_AUDIT 0
F00_F00E_EULER_V0_1 PASS_F00_F00E_RIGOROUS_COMPUTATIONAL_AUDIT 0
```

6. Adversarial review

A serious review should attempt at least one controlled mutation in a disposable clone:

```
change iota^2 sign;
change dagger orientation;
remove a factorial;
change the Euler-flow sign;
change a source file without updating its pin;
remove a required policy statement;
replace a rigorous tail with an unsupported truncation.
```

The relevant verifier should return a non-PASS status.

4.3 Reporting review findings

A useful review report should distinguish:

```
source/provenance defect;  
verifier implementation defect;  
incomplete obligation set;  
mathematical theorem gap;  
claim-boundary overstatement;  
reproducibility defect;  
documentation issue.
```

Do not report a theorem contradiction when the actual finding is a path, newline, serialization, or source-pin issue. Likewise, do not dismiss a mathematical failure as infrastructure merely because the package is intended to pass.

4.4 Current reviewer conclusion supported by the archived run

The archived source set passed every declared obligation in both implemented packages with no failed checks. This supports the implemented foundation certificate campaign only. It is not an independent expert validation of every later theorem in the wider research programme.

RKF/REVIEWER_GUIDE.md

Chapter 5

Reproduce the Certificate Campaign

5.1 Requirements

Python 3.11 or later
Git
standard-library execution for the implemented packages

No ordinary floating-point proof margin is used in the F00/F00-E package. The F00-G/H/I package uses exact integer and rational arithmetic with explicit rational interval and tail obligations.

5.2 Clean-clone run

```
git clone https://github.com/Parveen117/Recognition-Kernel-Framework.git
cd Recognition-Kernel-Framework
python certificates/foundation/run_campaign.py \
    --output certificates/foundation/campaign_result_reproduced.json
```

Expected terminal status:

```
PASS_IMPLEMENTED_FOUNDATION_CERTIFICATE_CAMPAIGN
FOOGHI_LOG_ARITHMETIC_ZETA_VO_1 PASS_FOOGHI_LOG_ARITHMETIC_ZETA_AUDIT 0
FOO_FOOE_EULER_VO_1 PASS_FOO_FOOE_RIGOROUS_COMPUTATIONAL_AUDIT 0
```

Expected process exit code:

0

5.3 Jupyter run

```
from pathlib import Path
import json
import os
import subprocess
import sys

repo = Path.cwd()
env = os.environ.copy()
env["PYTHONINTMAXSTRDIGITS"] = "0"

run = subprocess.run(
```

```

[
    sys.executable,
    "certificates/foundation/run_campaign.py",
    "--output",
    "certificates/foundation/campaign_result_reproduced.json",
],
cwd=repo,
text=True,
capture_output=True,
env=env,
)

print(run.stdout)
print(run.stderr)
print("RETURN CODE:", run.returncode)
assert run.returncode == 0

```

5.4 Compare statuses

```

archived = json.loads(
    (repo / "certificates/foundation/campaign_result.json")
    .read_text(encoding="utf-8")
)

reproduced = json.loads(
    (repo / "certificates/foundation/campaign_result_reproduced.json")
    .read_text(encoding="utf-8")
)

print("ARCHIVED:", archived["status"])
print("REPRODUCED:", reproduced["status"])

assert archived["status"] == reproduced["status"]

archived_packages = {
    item["package"]: item["reported_status"]
    for item in archived["packages"]
}
reproduced_packages = {
    item["package"]: item["reported_status"]
    for item in reproduced["packages"]
}

assert archived_packages == reproduced_packages
print("PACKAGE STATUSES MATCH")

```

5.5 Hash note

A campaign summary includes captured stdout and result-file hashes. The archived canonical summary SHA is:

```
8728ab0319afe5fb6e6329deae479cfa93f50bb4422ee863aaaf4f911dd9582
```

A clean reproduction should produce the same mathematical statuses and package results. If a platform-dependent field causes a different campaign-summary hash, report the exact differing field rather than silently replacing the archived artifact.

5.6 Direct package runs

```
python certificates/foundation/F00_F00E_EULER_V0_1/verify.py \  
  --output certificates/foundation/F00_F00E_EULER_V0_1/result_reproduced.json  
  
python certificates/foundation/F00GHI_LOG_ARITHMETIC_ZETA_V0_1/verify.py \  
  --output certificates/foundation/F00GHI_LOG_ARITHMETIC_ZETA_V0_1/result_reproduced.json
```

5.7 Failure handling

Do not weaken an obligation merely to recover a PASS.

On failure:

```
record the failing obligation;  
confirm source pins;  
separate infrastructure defects from mathematical defects;  
repair the theorem or verifier as appropriate;  
version the package if certified source or obligation semantics change;  
rerun from a clean checkout.
```

RKF/REPRODUCE.md

Chapter 6

Certificate Index

6.1 Campaign

Field	Value
Campaign	Native foundation proof-transport campaign
Source commit	c588ded973a395b5fce37c82f670616160979a2e
Implemented packages	2
Status	PASS_IMPLEMENTED_FOUNDATION_CERTIFICATE_CAMPAIGN
Canonical summary SHA-256	8728ab0319afe5fb6e6329deaae479cfa93f50bb4422ee863aaad
RH status	OPEN

Archived summary:

`certificates/foundation/campaign_result.json`

6.2 Package F00_F00E_EULER_V0_1

Field	Value
Certificate class	Foundational computational audit
Scope	cut scalar and native Euler derivation
Status	PASS_F00_F00E_RIGOROUS_COMPUTATIONAL_AUDIT
Package canonical result SHA-256	edc07e6bed39874251c0b75bb0e806a393a9b61343351fb40d700
Archived result file SHA-256	c8db07cc20bb532ddf483bb56483ab43e3d717b765e7f9525fd80

Files:

`certificates/foundation/F00_F00E_EULER_V0_1/README.md`
`certificates/foundation/F00_F00E_EULER_V0_1/spec.json`
`certificates/foundation/F00_F00E_EULER_V0_1/verify.py`
`certificates/foundation/F00_F00E_EULER_V0_1/result.json`

Authoritative mathematical and executable sources:

`src/rh_framework/native_summability.py`
`theorems/foundation/F00E_NATIVE_EULER_FROM_IOTA_COMPLEX.md`
`certificates/foundation/NUMERICAL_PROOF_PROTOCOL.md`

6.3 Package F00GHI_LOG_ARITHMETIC_ZETA_V0_1

Field	Value
Certificate class	exact and rational-interval foundational audit
Scope	native logarithm, arithmetic, prime factorization, and half-plane zeta
Status	PASS_F00GHI_LOG_ARITHMETIC_ZETA_AUDIT
Package canonical result SHA-256	672865d994b1111a876b160a80858f78d3133f400351f0792715a
Archived result file SHA-256	b5eff18191bc1834d57b8783a0f35c7214123d544d0ba6431d24

Files:

```
certificates/foundation/F00GHI_LOG_ARITHMETIC_ZETA_V0_1/README.md
certificates/foundation/F00GHI_LOG_ARITHMETIC_ZETA_V0_1/spec.json
certificates/foundation/F00GHI_LOG_ARITHMETIC_ZETA_V0_1/verify.py
certificates/foundation/F00GHI_LOG_ARITHMETIC_ZETA_V0_1/result.json
```

Authoritative theorem sources:

```
theorems/foundation/F00G_NATIVE_LOGARITHM_AND_POWERES.md
theorems/foundation/F00H_NATIVE_NATURAL_ARITHMETIC_AND_PRIME_FACTORIZATION.md
theorems/foundation/F00I_NATIVE_DIRICHLET_ZETA_EULER_PRODUCT.md
certificates/foundation/NUMERICAL_PROOF_PROTOCOL.md
```

6.4 Theorem 45 directed arithmetic-analytic closure packet

Field	Value
Certificate class	Exact finite arithmetic / enclosure algebra audit
Scope	finite-decimal exact rationalization, zero-radius exact embedding, outward threshold composition, refinement identity, certified-ball overlap
Status	PASS_DIRECTED_ARITHMETIC_ANALYTIC_CLOSURE_CANDIDATE
Canonical result SHA-256	294f5da69567086e869953ee0374fced3b4ea5f745138673b802
Branch proof-lab CI	PASS

Files:

```
theorem/45_directed_arithmetic_analytic_closure_theorem.md
proof_lab/directed_arithmetic_analytic_closure.py
proof_lab/test_directed_arithmetic_analytic_closure.py
proof_lab/DIRECTED_ARITHMETIC_ANALYTIC_CLOSURE_EXPECTED.sha256
```

This packet is intentionally exact and finite. It proves the arithmetic/enclosure identities used by Theorem 45. Correctness of an arbitrary external interval/ball implementation remains a separate dependency and raw nearest-rounded floating point without an outward enclosure remains non-proof-bearing.

Theorem lineage:

```

theorem/43_bilateral_jet_flow_recognition_capstone_theorem.md
theorem/44_madhava_smriti_bilateral_jet_flow_closure_theorem.md
theorem/45_directed_arithmetic_analytic_closure_theorem.md
theorem/28_recognition_complete_finite_to_infinite_cut_theorem.md
certificates/foundation/NUMERICAL_PROOF_PROTOCOL.md

```

6.5 Theorem 46 first-visible-jet seam quotient packet

Field	Value
Certificate class	Exact finite-jet indeterminate-limit classification audit
Scope	raw division-by-zero rejection, first-visible-order classification, equal-order seam quotient, regular reparameterization invariance, denominator separation, exact quotient enclosure
Status	PASS_FIRST_VISIBLE_JET_SEAM_QUOTIENT_CANDIDATE
Canonical result SHA-256	9d4ad6ef6ff7a798d2e169b09dda170e166d13ec5b233801c9c8
Branch proof-lab CI	PASS on Python 3.11 and 3.12

Files:

```

theorem/46_first_visible_jet_seam_quotient_theorem.md
proof_lab/first_visible_jet_seam_quotient.py
proof_lab/test_first_visible_jet_seam_quotient.py
proof_lab/FIRST_VISIBLE_JET_SEAM_QUOTIENT_EXPECTED.sha256

```

The packet verifies the exact finite algebra used by Theorem 46. It deliberately rejects raw algebraic $1/0$ and $0/0$; it treats all-zero finite denominator jets as unresolved; and it verifies the explicit denominator-separation quotient radius with exact rational arithmetic. Flat-function or genuinely different-seam cases remain outside this finite-jet certificate.

Theorem lineage:

```

theorem/42_cut_graded_lambda_jacobian_tower_theorem.md
theorem/43_bilateral_jet_flow_recognition_capstone_theorem.md
theorem/44_madhava_smriti_bilateral_jet_flow_closure_theorem.md
theorem/45_directed_arithmetic_analytic_closure_theorem.md
theorem/46_first_visible_jet_seam_quotient_theorem.md

```

6.6 Theorem 47 source-bound proof-carrying numerical validation packet

Field	Value
Certificate class	Exact source-bound interval-DAG and theorem-tail validation audit

Field	Value
Scope	source-bound exact leaves, interval DAG enclosure, zero-denominator gate, forged-radius rejection, geometric tail, rational exponential upper bound, Theorem-43 tail adapter, certainty-aware strict boundary, implementation-manifest integrity
Status	PASS_SOURCE_BOUND_PROOF_CARRYING_NUMERICAL_VALIDATION
Canonical result SHA-256	318511a662326234894bd264a9c428902262560aa7df066d5b398
Proof arithmetic	exact <code>fractions.Fraction</code> ; no binary float needed in the proof packet
Branch proof-lab CI	PASS on Python 3.11 and 3.12

Files:

```

theorem/47_source_bound_proof_carrying_numerical_validation_theorem.md
proof_lab/source_bound_numerical_validation.py
proof_lab/test_source_bound_numerical_validation.py
proof_lab/SOURCE_BOUND_PROOF_CARRYING_NUMERICAL_VALIDATION_EXPECTED.sha256

```

The packet proves the local-to-global enclosure induction for an admitted exact rational proof DAG. It rejects a node interval narrower than the verifier-computed enclosure, rejects interval division when the denominator enclosure contains zero, holds unsupported source classes or an open source-completeness obligation at OPEN, and computes theorem-tail examples by exact rational arithmetic. A participant-supplied radius or tail is therefore not proof-bearing merely because it is syntactically valid.

Theorem lineage:

```

theorem/34_native_common_chart_decoder_transfer.md
theorem/38_native_compatibility_form_range_source_domination.md
theorem/39_native_source_kernel_no_blindness_strict_odd.md
theorem/43_bilateral_jet_flow_recognition_capstone_theorem.md
theorem/44_madhava_smriti_bilateral_jet_flow_closure_theorem.md
theorem/45_directed_arithmetic_analytic_closure_theorem.md
theorem/46_first_visible_jet_seam_quotient_theorem.md
theorem/47_source_bound_proof_carrying_numerical_validation_theorem.md

```

External backend authenticity, arbitrary real-world source truth, universal source completeness, and endpoint resource-exhaustion controls remain separate obligations.

6.7 Planned but not certified

The following package names describe intended future audits. Their absence or design status must not be counted as a PASS:

```

F06_08_OPERATOR_FOUNDATION_V0_1
F09_10_FOURIER_GAMMA_XI_V0_1
F11_EXPLICIT_WEIL_V0_1
F12_15_WEIL_OPERATOR_REDUCTION_V0_1
F16_ACTIVE_BAND_OUTWARD_V0_1
ENDPOINT_KO_OUTWARD_V0_1

```

6.8 Verification rule

A package is considered reproducibly verified only when:

```
its source pins match;  
its verifier exits with code 0;  
its result JSON reports a PASS status;  
all mandatory obligations are present;  
all negative controls behave as specified;  
the result is reproducible from a clean checkout.
```

RKF/CERTIFICATE_INDEX.md

Chapter 7

Source Provenance

7.1 Authoritative imported source

The first reviewer cartridge is imported from:

```
Repository: Parveen117/RH-Framework
Source commit: c588ded973a395b5fce37c82f670616160979a2e
Commit message: Archive passing native foundation certificate campaign
```

The source commit is treated as an immutable import anchor.

7.2 Archived campaign identity

```
Campaign status:
PASS_IMPLEMENTED_FOUNDATION_CERTIFICATE_CAMPAIGN
```

```
Canonical campaign SHA-256:
8728ab0319afe5fb6e6329deaae479cfa93f50bb4422ee863aaaf4f911dd9582
```

Package result file hashes recorded by the archived campaign:

```
FOOGHI_LOG_ARITHMETIC_ZETA_V0_1/result.json
b5eff18191bc1834d57b8783a0f35c7214123d544d0ba6431d241e4227e150b6
```

```
F00_F00E_EULER_V0_1/result.json
c8db07cc20bb532ddf483bb56483ab43e3d717b765e7f9525fd8dafc2f380a41
```

Package-reported canonical result hashes:

```
FOOGHI_LOG_ARITHMETIC_ZETA_V0_1
672865d994b1111a876b160a80858f78d3133f400351f0792715a8d8bee142bd
```

```
F00_F00E_EULER_V0_1
edc07e6bed39874251c0b75bb0e806a393a9b61343351fb40d70034fa66a2af5
```

7.3 Byte-preservation rule

Files consumed by source-pin checks are copied without semantic renaming. Paths, historical namespaces, and line content remain source-identical to the imported commit.

Cross-platform verifier code canonicalizes text line endings before Git-blob hashing so a clean Windows or Unix checkout verifies the same pinned source content.

7.4 Public overlay

Public-facing documents in this repository are a reviewer overlay. They explain the imported cartridge under the Recognition Kernel Framework and RNKE terminology. The overlay does not change the imported mathematical source identity.

7.5 Excluded material

The public export excludes:

```
private filing drafts;  
correspondence addresses;  
priority-application lists;  
private notebooks;  
cache and build directories;  
local machine paths;  
credentials and tokens;  
unreviewed experimental branches;  
unrelated private repository history.
```

7.6 Change rule

Any modification to a pinned theorem source, implementation source, package specification, or verifier requires:

```
new source pins;  
new package version;  
new result JSON;  
new campaign hash;  
updated claim boundary;  
reproduction on a clean checkout.
```

An old PASS artifact must never be presented as certifying modified sources.

RKF/SOURCE_PROVENANCE.md

Chapter 8

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RKF/NOTICE.md

Part II

The Theorem Ladder

Chapter 9

Theorem 01: Positive-Eta Path Independence by Amplitude Lift

9.1 Source provenance

source repository: Parveen117/MP
primary PR: #246 Reset RH route onto rotation-generated recognition topology
source branch: agent/rh-rotation-metric-topology-reset

9.2 Statement

On the fixed five-label event-response carrier, every positive-eta normalization/continuation rectangle is an identity loop.

Equivalently, for any admissible positive eta values η_1, η_2 ,

```
normalize then eta-continue
=
eta-continue then normalize.
```

No rank-one seam compensation is required for the positive-eta family itself.

9.3 Native factorization

The state-locked construction has the factorization

$$\begin{aligned} A_{\text{chart}} &= A U, \\ \Theta_{\eta} &= Q_{\text{chart}} G_{\eta}^{(1/2)}, \\ G_{\eta} &= 1 + (A_{+} + \eta I_4), \quad \eta > 0. \end{aligned}$$

Here:

A	raw native event amplitude, eta-independent;
U	one fixed unitary accepted chart;
Q_{chart}	the polar orientation of the charted amplitude;
G_{η}	positive metric/scale factor for $\eta > 0$.

Because right multiplication by a unitary is polar-covariant and right multiplication by a strictly positive factor does not change the polar orientation,

```
polar(A U) = polar(A) U,
polar(Q_chart G_eta^(1/2)) = Q_chart.
```

Thus the frame transport data are exactly

```
raw eta transport          I_5
normalized eta transport    I_5
raw-to-normalized transport U*
based loop                  I_5
```

for every positive eta in the fixed five-label family.

9.4 Finite state-locked rectangle evidence

The user-local rectangle at

```
eta      = 0.010
eta + h  = 0.011
```

used the four actual amplitudes

```
raw(eta)          normalized(eta)
raw(eta+h)         normalized(eta+h)
```

and reported

```
status                PASS_STATE_LOCKED_UGD_ETA_NORMALIZATION_RECTANGLE_EVALUATION
loop classification    FINITE_ACTUAL_TWO_PATH_EQUALITY_PROVED
path transport difference 3.0669860554696173e-15
loop identity residual  3.1396725820594018e-15
defect phase           1.0705180635847342e-40
defect-to-accepted leakage 1.7140530928543942e-17
accepted-block residual 3.139637946219698e-15
endpoint scale difference 0
maximum pairing return residual 6.071532165918825e-18
false checks           none
```

These numbers are the floating-point shadow of the exact positive-eta family theorem.

9.5 What this closes

```
positive-eta raw amplitude path equality    CLOSED
positive-eta normalization path equality    CLOSED
fixed five-label UGD rectangle              CLOSED
need for new observer label at positive eta REMOVED
rank-one seam compensation at positive eta   NOT NEEDED
```

9.6 What this does not close

At eta zero,

```
G_0 = 1 + A_+.
```

If A_+ has a kernel, the normalized response can lose support. Therefore the next eta problem is support/Fredholm descent, not another path-holonomy problem.

eta-zero support descent	OPEN
kernel-line target visibility	OPEN
continuum UGD/Fredholm identification	OPEN
actual RH target map	OPEN
odd/even or single-sector terminal sign	NOT DECIDED BY THIS THEOREM

9.7 Use in the framework

Consume this theorem only as the positive-eta path-equivalence adapter:

```
positive eta family
-> no normalization/path-order ambiguity
-> fixed five-label carrier is lawful
-> eta-zero problem becomes support descent
```

RKF/theorum/01_positive_eta_path_independence.md

Chapter 10

Theorem 02: Seam Integer and Single-Sector Terminal Reduction

10.1 Source provenance

source repository: Parveen117/MP
primary PRs: #235, #242, #247, #255, #277
main objects: K_eta, B_eta, k_Sigma(eta), threshold holonomy, odd five-label packet

10.2 Global seam integer

The native completed-Weil sign decision is represented by a compact seam response. For positive eta, the representation uses a compact relative operator

$K_{\eta} = K_{(+,\eta)} \text{ direct_sum } K_{(3,\eta)}^{\text{sharp}}$

and the seam charge

$k_{\text{Sigma}}(\eta) = \text{rank } 1_{(1,\text{infinity})}(K_{\eta})$.

The global relation is

$k_{\text{Sigma}}(\eta)=0$
iff shifted completed packet is nonnegative.

For a cofinal sequence $\eta_j \searrow 0$, the eta-chain version of this statement becomes the endpoint sign reformulation.

10.3 Rank-at-most-five adverse packet

The representation-complete reduction isolates the remaining fixed-shift adverse source as a finite Birman-Schwinger obstruction:

$F_{(3,\eta)}^{\text{times}} = S_{(\eta,-)} - V_{\eta} V_{\eta}^*$
 $\text{rank}(V_{\eta}) \leq 5$
 $B_{\eta} = V_{\eta}^* S_{(\eta,-)}^{(-1)} V_{\eta}$
 $k_{\text{Sigma}}(\eta) = N_{+}(B_{\eta} - I)$.

Thus the terminal sign is tracked by a Hermitian matrix of rank at most five, not by an uncontrolled infinite observer cascade.

10.4 Common five-matrix covariance

For a positive source operator S and five-column source map V , set

$$B = V^* S^{(-1)} V \text{ in } M_5(\mathbb{C}).$$

Faithful carrier transport by a unitary does not change the represented obstruction:

$$\begin{aligned} S^U &= U S U^* \\ V^U &= U V \\ (V^U)^* (S^U)^{(-1)} V^U &= B. \end{aligned}$$

Changing the five labels by a unitary only conjugates the matrix:

$$B' = U_5^* B U_5.$$

Therefore the following data are representation invariant:

```
spectrum;
threshold projection;
seam charge;
sign decision.
```

10.5 Prime-torus and threshold-holonomy avatars

The arithmetic Bohr/prime-torus chart transports the same native source and same five columns. Therefore

$$B_{\text{eta}}^{\text{prime-torus}} = B_{\text{eta}}^{\text{native}}.$$

The matrix, seam and holonomy statements become

```
B_eta <= I
iff k_Sigma(eta)=0
iff H_threshold(eta)=I
iff shifted completed parity/Feshbach packet is nonnegative,
```

where

$$H_{\text{threshold}}(\text{eta}) = I - 2 \, 1_{(1, \infty)}(B_{\text{eta}}).$$

This makes threshold holonomy an avatar of the same finite obstruction, not an independent RH route.

10.6 One-sector sufficiency after reduction

The upstream bilateral carrier still represents both parity sectors. The single-sector conclusion is more delicate:

```
even sector remains represented upstream;
common pure-source sign and favorable even boundary remove an independent even terminal gate;
finite endpoint diagnostic follows only the odd five-label source packet.
```

So the lawful sentence is:

```
One sector is terminally sufficient only after the seam-integer / source-sign / favorable-even
↔ reduction.
```

The even sector is not deleted. It becomes non-terminal after the reduction.

10.7 Fixed-node evidence

The common five-matrix update records the fixed node

eta=.01 centre top	0.43597867312210625
outward matrix error upper	0.40291149063402765
native top upper	0.8388901637561339
threshold reserve	0.1611098362438661
k_Sigma(.01)	PROVED ZERO

This is a closed shifted node, not by itself eta-zero completion.

10.8 Claim boundary

representation-complete bilateral framework	PROVED
native seam integer	PROVED
rank-at-most-five fixed-shift reduction	PROVED
common five-matrix covariance	PROVED
prime-torus matrix transport	PROVED
threshold holonomy $\Leftrightarrow$ seam charge	PROVED
fixed eta=.01 seam zero	PROVED
single-sector terminal focus	VALID AFTER REDUCTION
controlled eta_j $\downarrow$ 0	SEPARATE GATE UNLESS T03 APPLIED
UGD terminal holonomy = threshold holonomy	ASSUMED WHERE STATED
unconditional RH from this capsule alone	NOT CLAIMED

10.9 Use in the framework

The theorem should be consumed as the finite obstruction adapter:

```
completed-Weil sign problem
-> compact seam response
-> seam integer k_Sigma
-> rank-at-most-five B_eta
-> odd five-label terminal packet after source/even reduction
```

RKF/theorum/02_seam_integer_single_sector_reduction.md

Chapter 11

Theorem 03: Certified M3 Positivity and the Odd Accepted Block

11.1 Source provenance

source repository: Parveen117/MP
primary PRs: #181, #182, #196, #198, #199, #200, #202
main objects: G3, b3, e3, M3, A3, E3

11.2 Target-local scalar

The reduced anti-seam scalar is

$$M3 = q(e3, e3) - b3 * G3^{(-1)} b3.$$

It is the Schur complement of the finite completed-Weil Gram block over

$$V3 = \text{span}\{u, |x|u, |x|^2u\}.$$

The accepted odd block is

$$E3 = \text{span}\{u, |x|u, |x|^2u, e3\}.$$

11.3 Certified coupling signs

The strong-residual Krylov packet certifies the exact native coupling signs

$$\text{sgn}(b3) = (-, +, +).$$

The final certified intervals recorded there are

$$\begin{aligned} -8.4429782348e-6 &< (b3)_u < -8.1644107492e-7 \\ 3.0255175052e-7 &< (b3)_h < 2.3045071878e-6 \\ 3.6746554244e-8 &< (b3)_k < 4.0310126099e-7 \end{aligned}$$

This solved the sign of the coupling vector, but did not by itself solve M3.

11.4 Correlated Schur rule

Independent coordinate-box propagation through $G3^{-1}$ is prohibited because $G3$ is nearly singular. The M3 calculation must keep

```
self channel;
coupling vector;
Schur compensation;
prime/Gamma truncation;
state transfer
```

inside one correlated ledger.

The source-seam invariance theorem gives

$$q_{-}/V3(e+v, e+v) = q_{-}/V3(e, e), \quad v \text{ in } V3,$$

and since

$$\begin{aligned} e3 &= r_{\text{odd}} - r3, \\ s3 &= r_{\text{odd}} - u, \\ e3 - s3 &= u - r3 \text{ in } V3, \end{aligned}$$

we may compute

$$M3 = q_{-}/V3(s3, s3).$$

11.5 State-local ledger separation

The exact/fixed-state arithmetic is separated as

$$\begin{aligned} &M(x) - M_N(x_m) \\ &= \\ &[M(x) - M(x_m)] \\ &+ \\ &[M(x_m) - M_N(x_m)]. \end{aligned}$$

The exact-to-state term is paid through

$$|M(x) - M(x_m)| \leq 2 \eta_p \eta_d + (157/30) \eta_p^2.$$

The finite-state arithmetic term uses

$$\begin{aligned} q &= q_N + \delta a_q \\ b &= b_N + \delta a_b \\ a_N &= G3^{-1} b_N \end{aligned}$$

and the Schur identity

$$\begin{aligned} &M(x_m) - M_N(x_m) \\ &= \delta a_q - 2 \operatorname{Re} \langle a_N, \delta a_b \rangle - \delta a_b * G3^{-1} \delta a_b. \end{aligned}$$

This route removes the older need to prove the whole exact-state $A_{4.74}$ norm directly.

11.6 Final certified interval

The fixed-state primal transfer closes the last formal transfer and records

```
continuum eta_p          < 1.4e-7
continuum eta_d          < 1.83e-6
fixed-state pivot transfer < 1.1e-14
state-local omitted-prime arithmetic < 6.565652843015687e-13
finite packet guard      7.1e-14
```

Around the canonical center

```
1.651058994316351e-12
```

the exact target-local interval is

```
3.08520376681449e-13
< M3 <
2.993597611951253e-12.
```

Therefore

$M3 > 0$.

11.7 Positive accepted block

The accepted block uses $G3 > 0$ and $M3 > 0$. PR #202 records

```
lambda_min(G3) = 2.0838177120437723e-13 > 0
3.08520376681449e-13 < M3 < 2.993597611951253e-12.
```

Since $M3$ is the Schur complement of $G3$ in the bordered block, the completed-Weil form is positive definite on

$E3 = \text{span}\{u, |x|u, |x|^2u, e3\}$.

11.8 Claim boundary

exact native b3 signs	CERTIFIED (-,+,+)
source-seam correlated Schur route	PROVED
state-local ledger separation	PROVED
fixed-state transfer	CERTIFIED
exact target-local M3	CERTIFIED POSITIVE
odd accepted finite block E3	CERTIFIED POSITIVE
signed infinite complement	NOT CLOSED BY M3 ALONE
full global Weil sign	NOT CLOSED BY M3 ALONE
RH	NOT CLAIMED FROM THIS CAPSULE ALONE

11.9 Use in the framework

Consume this theorem as the accepted-block certificate:

```
G3 > 0 and M3 > 0  
-> A3 > 0 / E3 positive  
-> finite accepted block no longer missing  
-> downstream obstruction is outside the accepted block
```

RKF/theorum/03_m3_positive_accepted_block.md

Chapter 12

Theorem 04: Zero-Cut T03 Endpoint Completion

12.1 Source provenance

source repository: Parveen117/MP
primary PR: #254 Close the five-dimensional completed-Weil endpoint by explicit zero-cut T03
↪ completion
support PRs: #242, #247, #255, #256
main objects: R_Sigma, P_src, S_(0,-), W_Sigma, A3, K_0

12.2 Purpose

This theorem package records the endpoint completion of the five-dimensional completed-Weil packet after the seam-integer and single-sector reductions have fixed the terminal object.

It should not be confused with the earlier exploratory finite eta sequence. The T03 theorem is the endpoint source-readout theorem.

12.3 Native T03 specialization

The zero-cut theorem fixes, before any lift is chosen:

projected native observer;
completed-Weil target;
actual target-relevant blind quotient.

The two spectral inputs are:

Spectral I:
exact five-dimensional defect;
minimal five-observer repair.

Spectral II:
target-relative lift count;
positive uniform repair margin;
a posteriori response-defect control;
matched visible/blind/cross certification;
strong completion.

12.4 Six-block Gram decomposition

For transported finite source maps Y_n and T_n , the five-label Gram defect is decomposed into

```
visible compression
+ five-dimensional blind response
+ both cross blocks
+ omitted prime/Gamma tails
+ quadrature/window/solve/rounding errors
+ native/Feshbach/prime-torus chart transport.
```

The theorem records that the six directed bounds have total error tending to zero.

12.5 Invariant source-Gram identity

Strong completion proves the invariant source-Gram identity

```
R_Sigma^* P_src R_Sigma
= (S_(0,-)^(dagger/2) W_Sigma)^*
  (S_(0,-)^(dagger/2) W_Sigma).
```

Literal equality of ambient source maps is not assumed. Equal Grams give a canonical partial-isometry identification of the final ranges. The endpoint sign needs only the invariant Gram identity.

12.6 Projection-defect positivity

The independently constructed total-response Gram is

```
R_Sigma^* R_Sigma = diag(1,A3).
```

Therefore the endpoint projection defect is

```
K_0 = R_Sigma^*(I-P_src)R_Sigma >= 0.
```

The endpoint Schur equivalence then transfers this projection-defect positivity to the completed-Weil endpoint sign in the declared normalization.

12.7 Terminal membrane

The theorem uses the classical completed explicit-formula identification and Weil criterion as the final classical membrane.

```
native endpoint Schur sign
+ classical completed explicit formula / Weil criterion
-> RH in the declared normalization.
```

This folder records the native theorem and its stated terminal interface. It does not pretend that the classical membrane was independently rederived inside this capsule.

12.8 Claim boundary

zero-cut quotient	FIXED
five-label lift	SUPPLIED BY SPECTRAL I-II
six-block Gram decomposition	SUPPLIED
vanishing total error	SUPPLIED
orientation-free source readout	SUPPLIED
$K_0 = R_{\text{Sigma}}^*(I-P_{\text{src}})R_{\text{Sigma}} \geq 0$	PROVED IN SOURCE THEOREM
endpoint Schur closure	PROVED IN SOURCE THEOREM
classical Weil terminal membrane	CITED/CONSUMED WHERE STATED

12.9 Use in the framework

Consume this theorem as the endpoint closure adapter:

```
single-sector / five-label source packet
-> invariant source-Gram identity
->  $K_0 \geq 0$ 
-> endpoint completed-Weil sign
-> classical membrane if invoked
```

Do not reopen M3, path equality, or seam integer while using T03. They are upstream adapters already transferred into this folder.

RKF/theorum/04_zero_cut_t03_endpoint_completion.md

Chapter 13

Theorem 05: Transfer Boundary and Next Gate

13.1 What has been moved here

The following MP-certified theorem packets now have clean Recognition Kernel Framework capsules:

```
positive-eta UGD/amplitude-lift path equality;  
seam-integer and common five-matrix reduction;  
single-sector terminal reduction after source/even sign reduction;  
exact M3 positivity and positive odd accepted block;  
zero-cut T03 endpoint completion.
```

This folder is now the readable theorem surface. The MP repository should be treated as proof-lab history and implementation residue after these theorem capsules are transferred.

13.2 What should not be reopened

M3 positivity	CLOSED
G3/M3 accepted odd block	CLOSED
positive-eta path equality	CLOSED
rank-at-most-five seam matrix	CLOSED AS REDUCTION
threshold holonomy/seam charge	CLOSED AS EQUIVALENCE
zero-cut T03 source-Gram endpoint	TRANSFERRED AS CERTIFIED SOURCE THEOREM

Reopening any of these would be mathematical archaeology for sport, which is apparently a hobby now, but not a proof strategy.

13.3 Important chronological correction

Earlier PRs still state the full parity requirement:

```
W3 >= 0 iff W3_even >= 0 and W3_odd >= 0.
```

That is true at the raw parity-split level.

Later seam-integer and T03 reductions change the terminal object. After the common pure-source sign, favorable even boundary and single-sector reduction, the even sector remains represented upstream but is not an independent terminal gate. The odd/five-label seam packet becomes terminal.

Correct wording:

The even sector is retained by the bilateral carrier.
It becomes non-terminal only after the native seam-integer/source-sign/favorable-even reduction.

13.4 Remaining work after transfer

There are two different continuation modes.

Framework mode

Use the capsules as already proved theorem inputs. Continue from:

```
T03 endpoint closure
-> formalize final theorem dependency graph
-> publication-quality normalization interface
-> reproducibility pointers.
```

Proof-lab cleanup mode

After removing/moving theorem capsules from MP, inspect what remains in MP for:

```
untransferred implementation scripts;
untransferred numerical packets;
duplicate manuscript surfaces;
old branches still carrying superseded claims;
source files needed only for reproducibility.
```

13.5 Deletion discipline for MP

Theorems should be removed from MP only where they are duplicated as readable doctrine or manuscript theorem summaries. Executable provenance modules and raw validation packets should not be destroyed unless their contents are also copied, hashed or archived.

Safe deletion class:

```
human-readable theorem capsules;
status summaries duplicated by this folder;
old manuscript theorem surfaces superseded by these capsules.
```

Unsafe deletion class:

```
raw verifier scripts;
state capsules;
large numerical packets;
source audit scripts;
artifacts needed to regenerate the theorem evidence.
```


13.6 Local rule going forward

Whenever a result is certified in MP and becomes part of the Recognition Kernel Framework, create one file here:

`theorem/NN_short_name.md`

with:

```
source PR;  
objects;  
statement;  
proof skeleton;  
numerical certificate if any;  
claim boundary;  
use in the framework.
```

Then MP can be cleaned by comparing against this folder instead of trusting a moving ledger. Ledgers are useful until they start impersonating reality.

`RKF/theorem/05_transfer_boundary_and_next_gate.md`

Chapter 14

Theorem 06: One-Sector Sufficiency Note

14.1 Why this note exists

The MP journey has two true-looking statements that live at different stages:

Raw parity split:

$W3 \geq 0$ iff $W3_{\text{even}} \geq 0$ and $W3_{\text{odd}} \geq 0$.

After seam-integer/source-sign/favorable-even reduction:

the terminal obstruction is the odd five-label seam packet.

Both are valid in their own layers. Confusing their order recreates a fake missing theorem, a wonderful little swamp that already wasted enough time.

14.2 Correct theorem wording

The bilateral completed-Weil carrier retains both parity sectors.

After the native seam-integer reduction, common source-sign theorem and favorable even boundary
 $\hookrightarrow$ reduction, the even sector is not an independent terminal gate.

The remaining terminal sign is carried by the odd five-label seam packet.

14.3 Practical consequence

Once the framework consumes the transferred theorems, do not restart a global even-sector positivity proof merely because an older parity-split branch says both sectors are present. Presence is not terminality.

14.4 Dependency order

```
native completed-Weil carrier
-> parity representation
-> seam integer / common five-matrix reduction
-> source-sign and favorable-even reduction
-> odd five-label source packet
-> zero-cut T03 endpoint
```

Only after that order is applied does the one-sector statement become lawful.

14.5 Claim boundary

even sector erased	FALSE
even sector represented upstream	TRUE
even sector independent terminal gate after reduction	FALSE
odd five-label endpoint terminal after reduction	TRUE

RKF/theorum/06_one_sector_theorem_note.md

Chapter 15

Framework Consumption Chain

Use the transferred theorems in this order:

- 01 positive eta path independence
 - > fixed five-label carrier has no positive-eta path ambiguity
- 02 seam integer single-sector reduction
 - > terminal obstruction is a rank-at-most-five seam packet after reduction
- 03 M3 positive accepted block
 - > G3/M3 accepted odd block is positive
- 06 one-sector theorem note
 - > even sector remains represented but is not an independent terminal gate after reduction
- 04 zero-cut T03 endpoint completion
 - > $K0 = R_Sigma^*(I-P_src)R_Sigma \geq 0$
- 05 transfer boundary and next gate
 - > do not reopen upstream theorem packets

Readable framework chain:

```
fixed five-label positive-eta path equality
+ seam-integer/five-matrix reduction
+ exact M3 accepted block
+ one-sector terminal reduction
+ zero-cut T03 source-Gram endpoint
= RKF terminal theorem surface.
```

Classical membrane, when invoked:

```
native completed-Weil endpoint sign
+ classical completed explicit formula / Weil criterion
-> RH in declared normalization.
```

This last membrane should stay visibly marked as a terminal interface, not smuggled into the native theorem as if citation ink were proof glue.

RKF/theorum/08_framework_consumption_chain.md

Chapter 16

Minimal Reading Order

For the original closure spine, read:

```
01_positive_eta_path_independence.md
02_seam_integer_single_sector_reduction.md
03_m3_positive_accepted_block.md
04_zero_cut_t03_endpoint_completion.md
08_framework_consumption_chain.md
```

For the generator-to-physical-adapter spine, read:

```
41_cut_graded_universal_generator_theorem.md
42_cut_graded_lambda_jacobian_tower_theorem.md
43_bilateral_jet_flow_recognition_capstone_theorem.md
44_madhava_smriti_bilateral_jet_flow_closure_theorem.md
proof_lab/README_CUT_GRADED_UNIVERSAL_GENERATOR.md
proof_lab/README_CUT_GRADED_LAMBDA_JACOBIAN_TOWER.md
proof_lab/README_BILATERAL_JET_FLOW_CAPSTONE.md
proof_lab/README_MADHAVI_SMRITI_BILATERAL_JET_FLOW.md
```

Then read:

```
06_one_sector_theorem_note.md
05_transfer_boundary_and_next_gate.md
provenance.json
```

Everything else preserves cleanup, lineage and future sanity, because apparently a mathematical repository also needs the horticultural care of a rare-orchid facility.

RKF/theorum/11_minimal_reading_order.md

Chapter 17

RH Claim Boundary

This folder contains transferred theorem capsules. It is not, by itself, a new public RH announcement.

The native chain recorded here is:

```
positive-eta path equality
+ seam-integer/five-matrix reduction
+ exact M3 accepted block
+ one-sector terminal reduction
+ zero-cut T03 source-Gram endpoint
```

When the classical membrane is invoked, it is explicitly:

```
classical completed explicit formula
+ Weil criterion in the declared normalization.
```

Do not blur these layers.

17.1 Safe wording

RKF now has a clean theorem folder containing the certified MP theorem packets required for the
↪ completed-Weil endpoint route.

17.2 Unsafe wording

The theorem folder alone is a peer-reviewed RH proof.

That would be asking one folder to do the work of a theorem, a paper, a reviewer and possibly a divine notary. No.

RKF/theorum/12_rh_claim_boundary.md

Chapter 18

Object Dictionary

A_chart	raw amplitude chart after fixed unitary accepted chart
Theta_eta	normalized eta response amplitude
G_eta	positive eta metric/scale factor
K_eta	compact seam response operator
B_eta	rank-at-most-five Birman-Schwinger decision matrix
k_Sigma	seam integer / threshold count
H_threshold	threshold holonomy avatar of B_eta
G3	completed-Weil Gram on V3
b3	coupling vector from V3 to e3
M3	reduced anti-seam Schur scalar
E3	accepted odd block $\text{span}\{u, x u, x ^2u, e3\}$
R_Sigma	total response readout
P_src	source projection
K0	endpoint projection defect $R_Sigma^*(I-P_src)R_Sigma$

RKF/theorum/14_object_dictionary.md

Chapter 19

Short Final Chain

```
positive eta path equality
  -> no path-order ambiguity on fixed five labels

seam integer/common five-matrix reduction
  -> finite rank-at-most-five threshold obstruction

single-sector terminal reduction
  -> odd five-label packet is terminal after source/even reduction

M3 > 0
  -> accepted odd block is positive

zero-cut T03
  ->  $K0 = R\_Sigma^{*(I-P\_src)}R\_Sigma \geq 0$ 

classical membrane, if invoked
  -> completed-Weil positivity implies RH in declared normalization
```

That is the cleaned RKF theorem path. It replaces the old archaeology crawl through MP PRs, because apparently even a proof needs a filing cabinet that does not bite.

RKF/theorum/15_short_final_chain.md

Chapter 20

Source Boundary

This folder is source-derived from the MP PR journey. It does not silently add new mathematical content beyond the transferred theorem capsules.

Where the source PRs marked a theorem as certified, this folder records it as certified.

Where the source PRs marked an interface as classical, assumed, conditional, terminal-open or not promoted, this folder preserves that boundary.

The purpose is cleanup and consumption, not embellishment. Astonishingly, writing that down prevents future confusion, which says plenty about the glamorous state of human proof management.

RKF/theorum/16_source_boundary.md

Chapter 21

Final Status Marker

```
transfer status: COMPLETE ON BRANCH
branch:         agent/transfer-certified-theorum-from-mp
folder:         theorum
```

The MP theorem packets named in `provenance.json` now have RKF-readable theorem capsules.

Next action is repository cleanup, not new theorem invention.

RKF/theorum/17_final_status.md

Chapter 22

No Re-Derive Note

Do not rederive these from scratch inside RKF unless a source error is found.

path equality	use Theorum 01
seam integer	use Theorum 02
M3 positivity	use Theorum 03
one-sector gate	use Theorum 06
T03 endpoint	use Theorum 04

The next research motion must start after these nodes.

RKF/theorum/18_no_rederive_note.md

Chapter 23

Public README Note

This public folder is a clean theorem register. It is not a replacement for private MP proof-lab provenance.

For public reading, start from `README.md` and `15_short_final_chain.md`.

`RKF/theorum/20_public_readme_note.md`

Chapter 24

Cut-Memory Spectral Isomorphism and Rank-Five Fredholm Reduction

24.1 Status

This theorem formalizes the actual infinite-to-finite mechanism behind the Recognition Kernel Framework.

It corrects three tempting but false shortcuts:

`det(I-B) > 0` is not sufficient for positivity when `rank(B) >= 2`;
centrality of a projection does not imply rank five;
the ordinary spectral gap is not generally equal to `lambda_min(I-B)`
unless the operator is measured in the positive reference metric.

The correct capstone is a **relative** spectral theorem.

24.2 1. Native cut setup

Let $\mathcal{H}$ be a Hilbert carrier with a fixed orthogonal cut decomposition

$$\mathcal{H} = P\mathcal{H} \oplus Q\mathcal{H}, \quad P + Q = I, \quad PQ = QP = 0.$$

Let π be the represented event algebra and let

$$V_0 : \mathcal{H}_{\text{rec}} \longrightarrow P\mathcal{H}$$

be the recognized inclusion.

For an admissible event a , define the cut bridge

$$C_a = Q\pi(a)V_0 : \mathcal{H}_{\text{rec}} \longrightarrow Q\mathcal{H}.$$

The cut projection is canonical, but it is not assumed to be central in the operator-algebraic sense. Indeed, if Q commuted with every $\pi(a)$ and $QV_0 = 0$, then $Q\pi(a)V_0 = 0$, killing the bridge. The relevant object is the off-diagonal cut corner $Q\pi(a)P$.

The framework-specific rank theorem is the following input.

Memory Cut Rank Hypothesis

The target-relevant cut-memory space

$$\mathcal{E}_\Sigma = \overline{\text{Ran } C_a}$$

has finite dimension

$$\nu_\Sigma \leq 5.$$

The theorem below proves exactly what follows once this rank statement and the positive reference decomposition are available.

24.3 2. Positive spectrum is transport; zero spectrum is memory

Theorem 2.1 (Cut-bridge spectral isomorphism)

Let $C : \mathcal{H}_1 \rightarrow \mathcal{H}_2$ be bounded. Then

$$\sigma(C^*C) \setminus \{0\} = \sigma(CC^*) \setminus \{0\}$$

with multiplicity.

If $C = U|C|$ is the polar decomposition, then U is unitary from the initial support $\overline{\text{Ran } |C|}$ onto the final support $\overline{\text{Ran } C}$, and

$$U(C^*C)U^* = CC^*$$

on those supports.

The zero modes remain directional:

$$\ker C \quad \text{and} \quad \ker C^*.$$

If C is Fredholm, their difference is the cut-memory index

$$\boxed{\text{ind } C = \dim \ker C - \dim \ker C^*}.$$

Proof

If $C^*Cx = \lambda x$ with $\lambda > 0$, then $Cx \neq 0$ and

$$CC^*(Cx) = C(C^*Cx) = \lambda Cx.$$

The inverse correspondence is obtained by applying C^* . Multiplicity is preserved. The polar-decomposition statement is standard on the initial and final support spaces. The zero spaces are not transported by this positive spectral correspondence and therefore retain the direction of the cut.

Corollary 2.2 (Cut-generated rotation memory)

On the direct sum of the initial and final supports, define

$$\mathbf{i}_C = \begin{pmatrix} 0 & -U^* \\ U & 0 \end{pmatrix}.$$

Then

$$\boxed{\mathbf{i}_C^2 = -I}$$

on the support sum, and hence

$$\boxed{e^{\theta \mathbf{i}_C} = \cos \theta I + \sin \theta \mathbf{i}_C.}$$

Thus the Euler rotation is derived from the polar memory of the cut bridge.

24.4 3. The exact infinite-to-finite theorem

Let $R = R^*$ be a positive reference operator satisfying

$$R \geq mI \quad (m > 0),$$

and let

$$V_\Sigma : \mathbb{C}^\nu \longrightarrow \mathcal{H}, \quad \nu \leq 5,$$

be the target-relevant cut-memory injection.

Define

$$F = R - V_\Sigma V_\Sigma^*,$$

$$A = R^{-1/2} V_\Sigma,$$

and the finite seam matrix

$$\boxed{B = V_\Sigma^* R^{-1} V_\Sigma = A^* A \in M_\nu(\mathbb{C}).}$$

Theorem 3.1 (Cut-memory Birman–Schwinger–Fredholm isomorphism)

The following identities hold.

Normalized factorization

$$\boxed{F = R^{1/2}(I - AA^*)R^{1/2}.}$$

Shared nonzero spectrum

$$\boxed{\sigma(AA^*) \setminus \{0\} = \sigma(B) \setminus \{0\}}$$

with multiplicity.

Exact inertia reduction

$$N_-(F) = N_+(B - I).$$

Moreover,

$$\dim \ker F = \dim \ker(I - B).$$

Hence

$$F \geq 0 \iff B \leq I.$$

Exact relative gap

Define the reference-normalized gap

$$\delta_R(F) = \inf_{x \neq 0} \frac{\langle x, Fx \rangle}{\langle x, Rx \rangle}.$$

Then

$$\delta_R(F) = \lambda_{\min}(I - B) = 1 - \lambda_{\max}(B).$$

Therefore, if $\delta_R(F) > 0$,

$$F \geq \delta_R(F)R \geq m \delta_R(F)I.$$

Thus the ordinary spectral gap has the certified lower bound

$$\text{dist}(0, \sigma(F)) \geq m \lambda_{\min}(I - B).$$

If $R = I$, this becomes the exact absolute gap identity

$$\text{dist}(0, \sigma(F)) = \lambda_{\min}(I - B)$$

in the positive case.

Fredholm determinant isomorphism

Since AA^* has rank at most ν ,

$$\det_F(I - AA^*) = \det(I - B).$$

The determinant vanishes exactly at a threshold memory:

$$\det(I - B) = 0 \iff 1 \in \sigma(B) \iff \ker F \neq \{0\}.$$

Proof

The factorization follows from

$$R - V_\Sigma V_\Sigma^* = R^{1/2} \left(I - R^{-1/2} V_\Sigma V_\Sigma^* R^{-1/2} \right) R^{1/2}.$$

The nonzero spectra of AA^* and $A^*A = B$ coincide by Theorem 2.1.

The map $x \mapsto R^{1/2}x$ is invertible, so the quadratic-form inertia of F equals the inertia of $I - AA^*$. Its negative directions correspond exactly to eigenvalues of AA^* , hence of B , larger than one. Its zero space corresponds exactly to the unit eigenspace of B .

For the relative gap, set $z = R^{1/2}x$. Then

$$\frac{\langle x, Fx \rangle}{\langle x, Rx \rangle} = \frac{\langle z, (I - AA^*)z \rangle}{\|z\|^2}.$$

The smallest spectral value of $I - AA^*$ is $1 - \lambda_{\max}(AA^*) = 1 - \lambda_{\max}(B)$.

Finally, the finite-rank Sylvester identity gives

$$\det_F(I - AA^*) = \det(I - A^*A) = \det(I - B).$$

24.5 4. The determinant warning

Proposition 4.1

For $\nu \geq 2$,

$$\det(I - B) > 0$$

does **not** imply $B < I$ or $F > 0$.

Exact counterexample

Take

$$B = \text{diag} \left(4, \frac{9}{4}, \frac{1}{4}, \frac{4}{9}, \frac{9}{16} \right).$$

Then

$$\det(I - B) = \frac{175}{256} > 0,$$

but B has two eigenvalues above one, so

$$N_-(F) = N_+(B - I) = 2.$$

The lawful terminal certificate is therefore

$$\boxed{\lambda_{\min}(I - B) > 0,}$$

equivalently

$$\lambda_{\max}(B) < 1.$$

The determinant isomorphism identifies threshold zeros and multiplicities, but its sign alone is not the positivity test.

24.6 5. The actual infinite-to-finite animal

Under the Memory Cut Rank Hypothesis $\nu_{\Sigma} \leq 5$, the infinite carrier does not need to be diagonalized.

The exact terminal quantities are:

$$\text{seam integer} = N_+(B - I) = N_-(F),$$

$$\text{relative spectral gap} = \lambda_{\min}(I - B),$$

and

$$\text{threshold multiplicity} = \dim \ker(I - B) = \dim \ker F.$$

Thus the infinite sign, gap and threshold-memory problems reduce exactly to a matrix of size at most five.

This is the correct mathematical version of the Copernican hierarchy:

```
cut bridge
-> polar transport and zero memory
-> finite target-relevant memory range
-> positive reference minus finite cut defect
-> five-matrix seam operator
-> exact inertia, gap and determinant reduction.
```

24.7 6. RH specialization and the genuine next theorem

The theorem above is universal. To apply it to the completed-Weil endpoint, the framework must prove the following native identification.

Targeted Theorem (Memory Cut Rank and Source-Defect Identification)

Construct R and V_{Σ} directly from the native completed-Weil cut events so that

$$W_3^- = R - V_{\Sigma} V_{\Sigma}^*,$$

$$\overline{\text{Ran } V_{\Sigma}} = \overline{\text{Ran } (Q\pi(\mathcal{A}_{\text{Weil}})V_0)},$$

and

$$\boxed{\text{rank } V_\Sigma \leq 5.}$$

Then the completed-Weil odd sign is equivalent to

$$B \leq I,$$

and its reference-normalized gap is exactly

$$\lambda_{\min}(I - B).$$

This targeted theorem, not determinant sign by itself, is the native bridge that can eliminate the infinite-dimensional barrier.

24.8 7. Numerical certificate v0.1

The executable packet checks four independent cases.

Cut-bridge calibration

A 5×7 cut bridge has:

rank	5
dim ker C	2
dim ker C*	0
index	2
positive-spectrum residual	0
iota-square residual	0

Exact subcritical five-memory packet

$$\sigma(B) = \left\{ \frac{1}{4}, \frac{4}{9}, \frac{9}{16}, \frac{16}{25}, \frac{81}{100} \right\}.$$

Then

$$\lambda_{\min}(I - B) = \frac{19}{100},$$

and

$$\det(I - B) = \frac{399}{32000}.$$

The full normalized operator and the five-matrix produce the same gap, inertia, kernel dimension and determinant.

Threshold packet

Replacing $81/100$ by 1 gives:

full kernel dimension	1
five-matrix kernel dimension	1
determinant	0

Positive-determinant negative control

The exact packet

$$B = \text{diag} \left(4, \frac{9}{4}, \frac{1}{4}, \frac{4}{9}, \frac{9}{16} \right)$$

has

$$\det(I - B) = \frac{175}{256} > 0$$

but seam integer and full negative index both equal two.

This permanently rejects the false determinant-sign shortcut.

The frozen executable status is

PASS_CUT_MEMORY_SPECTRAL_ISOMORPHISM_VO_1

with fourteen fail-closed checks.

24.9 8. Claim boundary

positive nonzero spectral isomorphism $C \times C \leftrightarrow CC^*$	PROVED
zero-mode cut-memory index	PROVED
polar iota and Euler rotation on support	PROVED
finite-rank normalized factorization	PROVED
full inertia = five-matrix seam inertia	PROVED
relative gap = $\lambda_{\min}(I-B)$	PROVED
Fredholm determinant = $\det(I-B)$	PROVED
$\det(I-B) > 0$ as positivity criterion	DISPROVED
central projection alone implies rank five	DISPROVED
ordinary unnormalized gap equality	NOT CLAIMED
RH-specific memory range rank ≤ 5	TARGETED NATIVE GATE
RH-specific identity $W3 = R-VV^*$	TARGETED NATIVE GATE
RH	NOT CLAIMED BY v0.1

RKF/theorum/21_cut_memory_spectral_isomorphism.md

Chapter 25

RH Target Gate: Native Cut-Generated Objects and Five-Memory Faithfulness

25.1 Permanent doctrine

The completed-Weil source, boundary, memory bridge and finite seam matrix are not imported as independent operator-theoretic objects. They must be generated from one native cut-event lift.

Classical operator algebra is allowed only as a shadow which tells us what an already constructed map should resemble. It does not define the map.

The primitive data are

```
exact exponential change flow;  
cut projector pair P,Q;  
native event lift Z;  
declared target observer T_Sigma.
```

Everything else must be derived.

25.2 1. Primitive cut system

Let $\mathcal{F}$ be the native generator core and $\mathcal{K}$ the event carrier. Let

$$\alpha_t(a) = U_t^* a U_t, \quad U_t = e^{itG},$$

be the exact exponential change flow. The finite-time flow is primitive; no infinitesimal truncation is used to define the event family.

Let

$$Z_t : \mathcal{F} \longrightarrow \mathcal{K}$$

be the cut-generated event lift. In an event representation one may write

$$Z_t f = \pi(\alpha_t(a_f))\Omega,$$

but a_f , π , and Ω are bookkeeping for the native event construction, not classical imports. Let

$$P = P^* = P^2, \quad Q = I - P, \quad PQ = QP = 0,$$

be the recognized and cut-memory projectors. Define the cut involution

$$\mathfrak{j}_{\text{cut}} = P - Q.$$

Then

$$\mathfrak{j}_{\text{cut}}^* = \mathfrak{j}_{\text{cut}}, \quad \mathfrak{j}_{\text{cut}}^2 = I.$$

25.3 2. Objects derived from the cut

The single lift Z_t generates four canonical forms:

$$S_t = Z_t^* Z_t,$$

$$R_t = Z_t^* P Z_t,$$

$$D_t = Z_t^* Q Z_t,$$

and

$$F_t = Z_t^* \mathfrak{j}_{\text{cut}} Z_t = R_t - D_t.$$

The exact cut decomposition is

$$\boxed{S_t = R_t + D_t.}$$

Thus source energy, recognized energy, cut-memory energy and signed cut form are not separately postulated. They are shadows of the same event lift under $I, P, Q, P - Q$.

If $P = |p\rangle\langle p|$ has rank one and

$$L_t(f) = \langle p, Z_t f \rangle,$$

then

$$R_t = L_t^* L_t$$

and therefore

$$\boxed{S_t - L_t^* L_t = D_t = Z_t^* Q Z_t \geq 0.}$$

This is the infinite-dimensional Cut-Square Identity before coordinates are chosen.

25.4 3. Exact exponential covariance

Transport the cut with the exact flow:

$$Z_t = U_t Z_0, \quad P_t = U_t P_0 U_t^*, \quad Q_t = U_t Q_0 U_t^*.$$

Then

$$Z_t^* P_t Z_t = Z_0^* P_0 Z_0, \quad Z_t^* Q_t Z_t = Z_0^* Q_0 Z_0.$$

Thus the cut-generated forms are exactly path-covariant under finite exponential change. The classical Lax or unitary-conjugation formula is merely a coordinate shadow of this native transport.

25.5 4. Full memory versus target-relevant memory

The full cut bridge

$$C_t = Q_t Z_t$$

may have infinite rank. The framework does **not** claim that the full cut carrier is five-dimensional.

Let

$$T_\Sigma : \mathcal{K} \longrightarrow \mathbb{C}^\nu, \quad \nu \leq 5,$$

be the declared target observer. It is required to be a coisometry on its range:

$$T_\Sigma T_\Sigma^* = I_\nu.$$

Define the target-visible cut bridge

$$A_\Sigma = T_\Sigma C_t.$$

The exact target-faithfulness condition is

$$\boxed{C_t = T_\Sigma^* T_\Sigma C_t}$$

on the adverse target-relevant cut module. Equivalently, the hidden residual

$$H_\Sigma = (I - T_\Sigma^* T_\Sigma) C_t$$

vanishes there.

This condition, not unitary covariance and not numerical rank thresholding, is the theorem which licenses the five-dimensional reduction.

25.6 5. Native target-memory rank theorem

Theorem A (Cut-derived target-memory rank)

If

$$C_t = T_\Sigma^* T_\Sigma C_t$$

on the target-relevant adverse module, then

$$\text{rank } C_t \leq \nu \leq 5$$

on that module, and

$$D_t = C_t^* C_t = A_\Sigma^* A_\Sigma.$$

Moreover, the nonzero spectrum of D_t is exactly the nonzero spectrum of

$$B_\Sigma^{\text{cut}} = A_\Sigma A_\Sigma^* \in M_\nu(\mathbb{C}).$$

Proof

The faithfulness identity factors the cut bridge through $T_\Sigma^* : \mathbb{C}^\nu \rightarrow \mathcal{K}$:

$$C_t = T_\Sigma^* A_\Sigma.$$

Hence its target-relevant range has dimension at most ν . Since $T_\Sigma T_\Sigma^* = I_\nu$,

$$C_t^* C_t = A_\Sigma^* T_\Sigma T_\Sigma^* A_\Sigma = A_\Sigma^* A_\Sigma.$$

The shared nonzero spectrum follows from the cut-bridge spectral isomorphism for $A_\Sigma^* A_\Sigma$ and $A_\Sigma A_\Sigma^*$.

25.7 6. Native signed reduction

Let $R_0 > 0$ be the accepted reference generated from the recognized cut channel and the native carrier metric. Define

$$\mathcal{W}_{\text{cut}} = R_0 - D_t.$$

Under target faithfulness,

$$\mathcal{W}_{\text{cut}} = R_0 - A_\Sigma^* A_\Sigma.$$

Set

$$V_\Sigma = A_\Sigma^* : \mathbb{C}^\nu \longrightarrow \mathcal{F}^{\text{comp}}$$

and

$$B_\Sigma = A_\Sigma R_0^{-1} A_\Sigma^* = V_\Sigma^* R_0^{-1} V_\Sigma.$$

The universal cut-memory spectral theorem then gives

$$N_-(\mathcal{W}_{\text{cut}}) = N_+(B_\Sigma - I),$$

$$\ker \mathcal{W}_{\text{cut}} \cong \ker(I - B_\Sigma),$$

and

$$\inf_{f \neq 0} \frac{\langle f, \mathcal{W}_{\text{cut}} f \rangle}{\langle f, R_0 f \rangle} = \lambda_{\min}(I - B_\Sigma).$$

The determinant identity

$$\det_F(I - R_0^{-1/2} D_t R_0^{-1/2}) = \det(I - B_\Sigma)$$

locates threshold memory, but determinant sign alone is not the positivity criterion. The lawful sign certificate is

$$\lambda_{\max}(B_\Sigma) < 1.$$

25.8 7. RH-specific construction gate

The remaining theorem is not to import $S_{0,-}^{\text{full}}$, L_∂ , R_0 , or V_Σ from a classical operator decomposition. It is to construct one native event lift

$$Z_{\text{Weil}} f$$

coefficientwise from the declared prime, Gamma, completion-boundary and compatibility cut events and then prove:

Gate 1: source and boundary are cut moments

For a cut-recognized vector p_∂ ,

$$S_{0,-}^{\text{full}} = Z_{\text{Weil}}^* Z_{\text{Weil}},$$

$$L_\partial(f) = \langle p_\partial, Z_{\text{Weil}} f \rangle,$$

and hence

$$S_{0,-}^{\text{full}} - L_\partial^* L_\partial = Z_{\text{Weil}}^* Q Z_{\text{Weil}}.$$

Gate 2: target observer is native and faithful

Construct T_Σ from the five declared target labels before whitening or Gram fitting and prove

$$Q Z_{\text{Weil}} = T_\Sigma^* T_\Sigma Q Z_{\text{Weil}}$$

on the adverse target-relevant module.

Gate 3: no hidden adverse complement

If full cut memory contains a target-null complement, prove that its form is nonnegative and does not contribute to the threshold count. Equivalently, only the target-visible cut memory may enter the negative inertia ledger.

These are native coefficient identities. The classical explicit formula may later recognize their shadow, but it does not define them.

25.9 8. Numerical development plan

The numerical packet must be built from the same cut-generated data:

1. construct Z from declared event amplitudes;
2. construct P, Q and verify $P+Q=I$, $PQ=0$;
3. verify $S=Z*Z$, $R=Z*PZ$, $D=Z*QZ$ and $S=R+D$;
4. verify the rank-one boundary identity where applicable;
5. construct T_Sigma from declared target labels;
6. enclose the faithfulness residual $||QZ-T*TZ||$;
7. reject a target map which misses one cut-memory direction;
8. build B_Sigma from the cut-derived $R0$ and A_Sigma ;
9. certify $\lambda_{\min}(I-B_Sigma)$, inertia and determinant identities;
10. enclose refinement and off-bank tails before any RH promotion.

A favorable five-matrix is not terminal unless Steps 3, 6 and 10 are proved.

25.10 9. Claim boundary

source/recognized/memory forms derived from one cut lift	PROVED ABSTRACTLY
rank-one first/second-moment covariance identity	PROVED ABSTRACTLY
exact exponential cut covariance	PROVED ABSTRACTLY
target-faithful rank ≤ 5 theorem	PROVED ABSTRACTLY
finite seam spectral reduction	PROVED ABSTRACTLY
native completed-Weil event lift Z_Weil	NEXT
coefficientwise source/boundary moment identities	NEXT
native target observer T_Sigma	NEXT
target-faithfulness residual with outward bounds	NEXT
no-hidden-adverse-complement theorem	NEXT
RH	NOT CLAIMED

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Chapter 26

Native Cut-Generated Object Theorem

26.1 1. Purpose

This theorem makes the cut, not a preassembled operator, the primitive object.

The classical world may suggest that the final maps should resemble a source operator, a boundary row, a Birman–Schwinger matrix, a Fredholm determinant or a spectral-flow index. None of those objects is imported here. They are constructed as compressions of one native event lift by the recognized and cut projectors.

The theorem has two layers:

full cut covariance	possibly infinite dimensional;
target-relevant cut memory	dimension at most five after faithfulness.

Confusing those layers would falsely claim that an infinite-dimensional positive covariance has rank five. Humanity already has enough false miracles.

26.2 2. Primitive data

Let $\mathcal{F}_0$ be the native generator core and $\mathcal{K}$ the event carrier. Let

$$Z : \mathcal{F}_0 \longrightarrow \mathcal{K}$$

be the native event lift constructed from the declared cut events.

Let

$$P = P^* = P^2, \quad Q = I - P, \quad PQ = QP = 0$$

be the recognized and cut-memory projectors.

Define the cut involution

$$\mathfrak{j} = P - Q.$$

Then

$$\mathfrak{j}^* = \mathfrak{j}, \quad \mathfrak{j}^2 = I.$$

26.3 3. Derived objects

Define

$$S = Z^*Z, \quad R = Z^*PZ, \quad D = Z^*QZ, \quad F = Z^*jZ.$$

Theorem 3.1 (Cut-generated decomposition)

The four forms satisfy

$$\boxed{S = R + D}$$

and

$$\boxed{F = R - D.}$$

Moreover,

$$R \geq 0, \quad D \geq 0.$$

Proof

Since $P + Q = I$,

$$S = Z^*(P + Q)Z = Z^*PZ + Z^*QZ = R + D.$$

Since $j = P - Q$,

$$F = Z^*(P - Q)Z = R - D.$$

The positivity of R and D follows because they are Gram forms:

$$\langle f, Rf \rangle = \|PZf\|^2, \quad \langle f, Df \rangle = \|QZf\|^2.$$

Thus source energy, recognized energy, cut-memory energy and signed cut form are co-generated by one cut lift.

26.4 4. First moment and second moment

Assume that the recognized projector has rank one,

$$P = |p\rangle\langle p|, \quad \|p\| = 1.$$

Define the native first moment

$$L(f) = \langle p, Zf \rangle.$$

Theorem 4.1 (Native cut covariance)

The recognized Gram is

$$R = L^*L,$$

and therefore

$$\boxed{S - L^*L = D = Z^*QZ \geq 0.}$$

For $f, h \in \mathcal{F}_0$,

$$\boxed{S(f, h) - \overline{L(h)}L(f) = \langle QZh, QZf \rangle.}$$

Proof

Since $P = |p\rangle\langle p|$,

$$R(f, h) = \langle Zh, PZf \rangle = \langle Zh, p \rangle \langle p, Zf \rangle = \overline{L(h)}L(f).$$

Apply Theorem 3.1.

This is the operator form of the Cut-Square Identity. Finite coordinate identities are shadows of this projection formula.

26.5 5. Exact exponential transport

Let

$$U_t = e^{itG}$$

be the exact finite-time exponential change generated by a self-adjoint native change generator G . Transport the event lift and cut by

$$Z_t = U_t Z_0, \quad P_t = U_t P_0 U_t^*, \quad Q_t = U_t Q_0 U_t^*.$$

Theorem 5.1 (Exponential cut covariance)

For every t ,

$$Z_t^* P_t Z_t = Z_0^* P_0 Z_0,$$

$$Z_t^* Q_t Z_t = Z_0^* Q_0 Z_0,$$

and

$$Z_t^* (P_t - Q_t) Z_t = Z_0^* (P_0 - Q_0) Z_0.$$

Proof

Use $U_t^* U_t = I$. For example,

$$Z_t^* Q_t Z_t = Z_0^* U_t^* U_t Q_0 U_t^* U_t Z_0 = Z_0^* Q_0 Z_0.$$

The other identities are identical.

Thus analyticity is not imposed by a classical continuation theorem. It is exact covariance of the cut-generated objects under finite exponential change.

26.6 6. Target-relevant memory

Let

$$C = QZ$$

be the full cut bridge. It may have infinite-dimensional range.
Let

$$T_\Sigma : \mathcal{K} \longrightarrow \mathbb{C}^\nu, \quad \nu \leq 5,$$

be the native target observer, with

$$T_\Sigma T_\Sigma^* = I_\nu.$$

Define

$$A_\Sigma = T_\Sigma C.$$

Definition 6.1 (Target faithfulness)

The target observer is faithful on the adverse cut module if

$$\boxed{C = T_\Sigma^* T_\Sigma C.}$$

The hidden residual is

$$H_\Sigma = (I - T_\Sigma^* T_\Sigma)C.$$

Thus target faithfulness is exactly $H_\Sigma = 0$ on the module being reduced.

Theorem 6.2 (Five-memory reduction)

Under target faithfulness,

$$\text{rank } C \leq \nu \leq 5$$

on the target-relevant adverse module, and

$$\boxed{D = C^* C = A_\Sigma^* A_\Sigma.}$$

The nonzero spectra of D and

$$B_{\Sigma}^{\text{cut}} = A_{\Sigma} A_{\Sigma}^*$$

coincide with multiplicity.

Proof

Target faithfulness gives

$$C = T_{\Sigma}^* A_{\Sigma}.$$

Hence the range of C factors through $\mathbb{C}^{\nu}$. Also,

$$C^* C = A_{\Sigma}^* T_{\Sigma} T_{\Sigma}^* A_{\Sigma} = A_{\Sigma}^* A_{\Sigma}.$$

The spectral statement follows from the cut-bridge spectral isomorphism.

26.7 7. Hidden complement theorem

A target observer need not see all cut memory. The exact decomposition is

$$C = T_{\Sigma}^* A_{\Sigma} + H_{\Sigma},$$

with orthogonal ranges, and hence

$$D = A_{\Sigma}^* A_{\Sigma} + H_{\Sigma}^* H_{\Sigma}.$$

Theorem 7.1 (No-hidden-adverse-complement criterion)

Suppose the signed endpoint form has the cut-derived form

$$\mathcal{W} = R_0 - A_{\Sigma}^* A_{\Sigma} + H_{\Sigma}^* H_{\Sigma}, \quad R_0 > 0.$$

Then the hidden complement is favorable. Therefore

$$R_0 - A_{\Sigma}^* A_{\Sigma} \geq 0 \implies \mathcal{W} \geq 0.$$

If $H_{\Sigma} = 0$, the finite reduction is exact.

Proof

The term $H_{\Sigma}^* H_{\Sigma}$ is positive.

This theorem identifies the precise burden. The framework need not prove that all cut memory is five-dimensional. It must prove only that any target-hidden complement is nonnegative.

26.8 8. Native finite seam matrix

Define

$$V_\Sigma = A_\Sigma^* : \mathbb{C}^\nu \longrightarrow \mathcal{F}^{\text{comp}}$$

and, for a positive cut-derived reference R_0 ,

$$B_\Sigma = V_\Sigma^* R_0^{-1} V_\Sigma = A_\Sigma R_0^{-1} A_\Sigma^*.$$

Then

$$R_0 - A_\Sigma^* A_\Sigma = R_0^{1/2} \left(I - R_0^{-1/2} A_\Sigma^* A_\Sigma R_0^{-1/2} \right) R_0^{1/2}.$$

The Cut-Memory Spectral Isomorphism gives

$$N_-(R_0 - A_\Sigma^* A_\Sigma) = N_+(B_\Sigma - I),$$

$$\dim \ker(R_0 - A_\Sigma^* A_\Sigma) = \dim \ker(I - B_\Sigma),$$

and

$$\inf_{f \neq 0} \frac{\langle f, (R_0 - A_\Sigma^* A_\Sigma) f \rangle}{\langle f, R_0 f \rangle} = \lambda_{\min}(I - B_\Sigma).$$

The Fredholm determinant identity is

$$\det_F \left(I - R_0^{-1/2} A_\Sigma^* A_\Sigma R_0^{-1/2} \right) = \det(I - B_\Sigma).$$

The determinant locates threshold memory. Its sign alone does not decide positivity when $\nu \geq 2$. The lawful sign test is

$$\lambda_{\max}(B_\Sigma) < 1.$$

26.9 9. Native completed-Weil construction target

The RH-specific theorem must be coefficientwise and cut-first.

Construct

$$Z_{\text{Weil}} f$$

from the prime, Gamma, completion-boundary and compatibility events before any whitening, finite Gram fit or classical explicit-formula identification.

Then prove:

Moment identity

$$S_{0,-}^{\text{full}} = Z_{\text{Weil}}^* Z_{\text{Weil}},$$

$$L_{\partial}(f) = \langle p_{\partial}, Z_{\text{Weil}} f \rangle,$$

and hence

$$S_{0,-}^{\text{full}} - L_{\partial}^* L_{\partial} = Z_{\text{Weil}}^* Q Z_{\text{Weil}}.$$

Native five-label observer

Construct T_{Σ} directly from the five declared target labels and prove that the adverse target-relevant bridge is faithful:

$$Q Z_{\text{Weil}} = T_{\Sigma}^* T_{\Sigma} Q Z_{\text{Weil}}$$

on that module.

Favorable hidden memory

Prove that any target-null cut-memory complement contributes only a positive square.

If these statements hold, the source and boundary are no longer independent, the Cut-Square Identity is automatic, and the target-relevant spectral burden reduces to at most five dimensions.

26.10 10. Non-circularity rules

The theorem is invalid if any of the following occurs:

Z_Weil is fitted from the desired completed-Weil Gram;
P or Q is chosen after seeing the target sign;
T_Sigma is obtained by diagonalizing the final operator;
rank five is inferred only from a floating singular-value threshold;
the hidden complement is omitted rather than proved favorable;
the classical explicit formula is used to define the native event lift.

The construction is lawful only when the cut creates the objects before the sign question is asked.

26.11 11. Claim boundary

cut-generated decomposition S=R+D	PROVED
cut-generated signature F=R-D	PROVED
rank-one first/second-moment covariance	PROVED
exact exponential cut covariance	PROVED
target-faithful rank-at-most-five reduction	PROVED
hidden favorable complement criterion	PROVED
finite seam spectral/inertia/determinant reduction	PROVED

native coefficientwise Z_Weil	NEXT
native first/second moment identification	NEXT
native five-label observer T_Sigma	NEXT
outward target-faithfulness and tail certificate	NEXT
RH	NOT CLAIMED

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Chapter 27

Clock-Free Recognition-Seam Cut Calculus

27.1 1. Purpose

This note rewrites the cut-memory programme in the native language of Recognition-Seam Calculus.

The primitive object is not a one-parameter operator family and not a preassembled classical source operator. The primitives are:

```
recognition objects;  
declared transition arrows;  
a native event lift at each object;  
a cut projector pair at each object;  
predeclared composition ledgers.
```

A real parameter or exponential flow may later provide one chart of the arrow system. It is not the clock of the calculus.

The framework therefore proceeds in the order

```
cut  
-> transition differential  
-> composition residue  
-> cut-generated source and memory forms  
-> stable repair  
-> target-faithful finite memory  
-> finite seam matrix.
```

The Riemann-Hypothesis carrier is intended to be one specialization of this calculus, not the definition of it.

27.2 2. Recognition objects

Let $\mathfrak{G}$ be a small category of recognition objects and declared transitions. For every object x , assign:

$\mathcal{F}_x$ (native source core),

$\mathcal{K}_x$ (event carrier),

$Z_x : \mathcal{F}_x \longrightarrow \mathcal{K}_x$ (native event lift),

and an orthogonal cut pair

$$P_x = P_x^* = P_x^2, \quad Q_x = I - P_x, \quad P_x Q_x = Q_x P_x = 0.$$

The recognized and memory carriers are

$$\mathcal{K}_x^{\text{rec}} = P_x \mathcal{K}_x, \quad \mathcal{K}_x^{\text{mem}} = Q_x \mathcal{K}_x.$$

The cut involution is

$$\mathbf{j}_x = P_x - Q_x, \quad \mathbf{j}_x^2 = I.$$

No external time is present in these definitions.

27.3 3. Declared transition arrows

For an arrow

$$\gamma : x \longrightarrow y,$$

declare a source transport and an event transport

$$A_\gamma : \mathcal{F}_x \longrightarrow \mathcal{F}_y, \quad U_\gamma : \mathcal{K}_x \longrightarrow \mathcal{K}_y,$$

satisfying the native event intertwiner

$$\boxed{Z_y A_\gamma = U_\gamma Z_x.}$$

For composable arrows $\gamma : x \rightarrow y$ and $\delta : y \rightarrow z$, require

$$A_{\delta\gamma} = A_\delta A_\gamma, \quad U_{\delta\gamma} = U_\delta U_\gamma.$$

These are finite transition laws. Differentiation is defined from arrows, not from a time parameter.

27.4 4. Clock-free transition differential

Decompose an event transport into its four cut corners:

$$R_\gamma = P_y U_\gamma P_x,$$

$$J_\gamma = P_y U_\gamma Q_x,$$

$$C_\gamma = Q_y U_\gamma P_x,$$

$$M_\gamma = Q_y U_\gamma Q_x.$$

Their meanings are:

R_gamma recognized-to-recognized transport;
J_gamma memory-to-recognized repair;
C_gamma recognized-to-memory cut creation;
M_gamma memory-to-memory transport.

Definition 4.1 (Transition differential of the cut)

Define

$$\boxed{\nabla_\gamma P = P_y U_\gamma - U_\gamma P_x = J_\gamma - C_\gamma.}$$

This is a finite transition differential. It compares the cut before and after one declared arrow. The arrow is cut-compatible exactly when

$$\nabla_\gamma P = 0.$$

Because J_γ and C_γ occupy orthogonal cut corners, cut compatibility is equivalent to

$$J_\gamma = 0, \quad C_\gamma = 0.$$

27.5 5. Exact composition residues

Block multiplication gives the clock-free chain law.

Theorem 5.1 (Composition-residue identities)

For composable arrows $\gamma : x \rightarrow y$ and $\delta : y \rightarrow z$,

$$\boxed{R_{\delta\gamma} = R_\delta R_\gamma + J_\delta C_\gamma,}$$

$$\boxed{J_{\delta\gamma} = R_\delta J_\gamma + J_\delta M_\gamma,}$$

$$\boxed{C_{\delta\gamma} = C_\delta R_\gamma + M_\delta C_\gamma,}$$

and

$$\boxed{\mathbf{M}_{\delta\gamma} = \mathbf{C}_{\delta}\mathbf{J}_{\gamma} + \mathbf{M}_{\delta}\mathbf{M}_{\gamma}.$$

Proof

Insert

$$I_{\mathcal{K}_y} = P_y + Q_y$$

between U_{δ} and U_{γ} , then project onto the four output/input corners. For example,

$$P_z U_{\delta} U_{\gamma} P_x = P_z U_{\delta} (P_y + Q_y) U_{\gamma} P_x = \mathbf{R}_{\delta} \mathbf{R}_{\gamma} + \mathbf{J}_{\delta} \mathbf{C}_{\gamma}.$$

The other identities are identical.

The cross terms

$$\mathbf{J}_{\delta} \mathbf{C}_{\gamma} \quad \text{and} \quad \mathbf{C}_{\delta} \mathbf{J}_{\gamma}$$

are the generated seam residues. They are created by composition; they are not post-hoc corrections.

When every arrow is cut-compatible, the recognized and memory blocks compose independently.

27.6 6. Cut-generated forms

At each object define

$$S_x = Z_x^* Z_x,$$

$$R_x = Z_x^* P_x Z_x,$$

$$D_x = Z_x^* Q_x Z_x,$$

and

$$F_x = Z_x^* \mathbf{J}_x Z_x = R_x - D_x.$$

Theorem 6.1 (Native cut decomposition)

For every recognition object,

$$\boxed{S_x = R_x + D_x}$$

and

$$\boxed{F_x = R_x - D_x.}$$

Moreover,

$$R_x \geq 0, \quad D_x \geq 0.$$

Proof

Use $P_x + Q_x = I$ and $j_x = P_x - Q_x$. Positivity follows because R_x and D_x are Gram forms.

Thus the source, recognized energy, memory energy and signed endpoint form are not independent objects. They are four shadows of one cut-generated event lift.

27.7 7. Clock-free differential of a form

Let G_x be any form assigned to the object x .

Definition 7.1 (Transition differential)

Define

$$\Delta_\gamma G = A_\gamma^* G_y A_\gamma - G_x.$$

This is the Recognition-Seam replacement for an external-clock derivative.

Theorem 7.2 (Clock-free chain rule)

For composable arrows $\gamma : x \rightarrow y$ and $\delta : y \rightarrow z$,

$$\Delta_{\delta\gamma} G = \Delta_\gamma G + A_\gamma^* (\Delta_\delta G) A_\gamma.$$

Proof

$$\begin{aligned} \Delta_{\delta\gamma} G &= A_\gamma^* A_\delta^* G_z A_\delta A_\gamma - G_x \\ &= A_\gamma^* (A_\delta^* G_z A_\delta - G_y) A_\gamma + (A_\gamma^* G_y A_\gamma - G_x). \end{aligned}$$

No limiting quotient by a time increment appears.

27.8 8. Generated ledgers and local closure

A ledger Λ_γ^G for a transition differential must be declared before the observation and must be generated from constitutive, control, reference or topological data independent of the evaluation packet.

Define the local recognition residue

$$\text{Res}_\gamma^G = \Delta_\gamma G - \Lambda_\gamma^G.$$

Local closure means

$$\text{Res}_\gamma^G = 0.$$

For composition, an admissible ledger satisfies

$$\boxed{\Lambda_{\delta\gamma}^G = \Lambda_\gamma^G + A_\gamma^* \Lambda_\delta^G A_\gamma.}$$

A ledger chosen after seeing $\Delta_\gamma G$ is retrospective cancellation and is inadmissible.

Even when all elementary arrows are locally closed, a loop may retain nonzero memory holonomy. Visible return and state return are therefore different statements.

27.9 9. Stable repair without a clock

Suppose $\gamma : x \rightarrow y$ is cut-compatible. Then

$$M_\gamma : \mathcal{K}_x^{\text{mem}} \longrightarrow \mathcal{K}_y^{\text{mem}}$$

is the induced memory transport.

Define

$$\ell_\gamma = \dim \ker M_\gamma \quad (\text{loss}),$$

$$b_\gamma = \dim \text{coker } M_\gamma \quad (\text{birth}),$$

and

$$\iota_\gamma = \ell_\gamma - b_\gamma.$$

For finite-dimensional or Fredholm memory arrows,

$$\boxed{\iota_{\delta\gamma} = \iota_\gamma + \iota_\delta.}$$

Also,

$$\dim \mathcal{K}_y^{\text{mem}} - \dim \mathcal{K}_x^{\text{mem}} = b_\gamma - \ell_\gamma.$$

This is stable repair as a transition index. It requires no external clock and no preferred parametrization.

27.10 10. Target-relevant memory and finite reduction

Let

$$T_{\Sigma,x} : \mathcal{K}_x^{\text{mem}} \longrightarrow \mathbb{C}^{\nu_x}, \quad \nu_x \leq 5,$$

be a declared target observer, with

$$T_{\Sigma,x} T_{\Sigma,x}^* = I_{\nu_x}.$$

Define the target-visible cut bridge

$$A_{\Sigma,x} = T_{\Sigma,x} Q_x Z_x.$$

Definition 10.1 (Target faithfulness)

The observer is faithful on the adverse module if

$$Q_x Z_x = T_{\Sigma,x}^* T_{\Sigma,x} Q_x Z_x.$$

Then

$$D_x = A_{\Sigma,x}^* A_{\Sigma,x}$$

on that module and its target-relevant rank is at most five.

If $R_x > 0$, define the cut-derived finite seam matrix

$$B_{\Sigma,x} = A_{\Sigma,x} R_x^{-1} A_{\Sigma,x}^*.$$

The Cut-Memory Spectral Isomorphism gives

$$N_-(R_x - D_x) = N_+(B_{\Sigma,x} - I),$$

$$\dim \ker(R_x - D_x) = \dim \ker(I - B_{\Sigma,x}),$$

and

$$\inf_{f \neq 0} \frac{\langle f, (R_x - D_x)f \rangle}{\langle f, R_x f \rangle} = \lambda_{\min}(I - B_{\Sigma,x}).$$

Thus the finite seam matrix is not imported. It is derived from the target compression of the cut memory.

27.11 11. Generalized exact examples

The first executable stage contains three independent examples.

Example A: nonlinear seam holonomy

A clock-free arrow loop on

$$\mathbb{T}^2 \times \mathbb{Z}$$

has exact local ledger closure on every arrow. The visible torus coordinates return while the integer memory changes by one.

This proves that local transition closure does not imply global state return.

Example B: target-blind stable repair

Three objects have memory ranks

$$2 \longrightarrow 1 \longrightarrow 2.$$

The first arrow loses one blind mode, the second births one blind mode, and the repair index is additive under composition.

This is a clock-free kernel–cokernel theorem, not a time evolution.

Example C: cut-generated five-memory packet

A single event lift has the exact block form

$$Z = \begin{pmatrix} I_5 \\ A \end{pmatrix},$$

with

$$P = \begin{pmatrix} I_5 & 0 \\ 0 & 0 \end{pmatrix}, \quad Q = I - P.$$

Therefore

$$R = I_5, \quad D = A^*A, \quad F = I_5 - A^*A.$$

The target observer is the identity on the five-dimensional cut-memory carrier, so target faithfulness is exact. The finite seam matrix is

$$B_\Sigma = AA^*.$$

The full signed form and the five-matrix have exactly the same inertia, threshold multiplicity, relative gap and Fredholm determinant.

The example is a theorem calibration. It is not an RH certificate.

27.12 12. RH as a specialization

The completed-Weil problem enters only after the general calculus is stable.

At the RH object x_{Weil} , construct the native event lift

$$Z_{\text{Weil}}$$

from declared prime, Gamma, completion-boundary and compatibility events.

Then prove, coefficientwise and before classical recognition,

$$S_{0,-}^{\text{full}} = Z_{\text{Weil}}^* Z_{\text{Weil}},$$

$$L_\partial(f) = \langle p_\partial, Z_{\text{Weil}} f \rangle,$$

and

$$QZ_{\text{Weil}} = T_\Sigma^* T_\Sigma QZ_{\text{Weil}}$$

on the adverse target-relevant module.

Only after these native identities are proved may the classical explicit formula recognize the constructed form as a completed-Weil shadow.

The classical theory indicates what the shadow should resemble. It does not create the native map.

27.13 13. Stage boundary

clock-free recognition objects and arrows
transition differential
composition-residue identities
cut-generated source/memory forms
clock-free form chain rule
stable repair index
target-faithful finite seam reduction
three generalized exact examples

native completed-Weil event lift
RH numerical certificate

WRITTEN
PROVED ABSTRACTLY
PROVED ABSTRACTLY
PROVED ABSTRACTLY
PROVED ABSTRACTLY
PROVED ABSTRACTLY
PROVED ABSTRACTLY
IMPLEMENTED NEXT FILE

LATER STAGE
NOT ATTEMPTED IN STAGE 1

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Chapter 28

Clock-Free Cut-Memory Development Plan

28.1 Governing rule

Recognition-Seam Calculus is the foundation.

No external clock defines the derivative.

No classical operator decomposition defines the native objects.

No target matrix is fitted after the sign is known.

Classical operator algebra and spectral theory may be consulted only as a shadow: they suggest the shape of a map after the cut-derived construction has already produced it.

28.2 Stage 1: generalized clock-free examples

Written objects

theorem/24_clock_free_recognition_seam_cut_calculus.md
proof_lab/clock_free_cut_memory_general_example.py

Exact examples

1. nonlinear phase-seam holonomy with local closure and global memory;
2. nonzero composition residue from cut-corner multiplication;
3. target-blind stable repair with memory ranks $2 \rightarrow 1 \rightarrow 2$;
4. cut-generated five-memory packet with exact finite seam reduction.

Stage-1 claim boundary

GENERAL CALCULUS AND EXAMPLES	WRITTEN
EXACT TEST SUITE	NEXT STAGE
RH EVENT LIFT	NOT STARTED
RH CERTIFICATE	NOT CLAIMED

28.3 Stage 2: exact theorem tests

Write a fail-closed test module for:

```
transition differential;
composition-residue identities;
clock-free chain rule;
local ledger closure;
nonzero loop holonomy;
stable repair index additivity;
cut decomposition S=R+D;
target-faithful rank <= 5;
full/five-matrix inertia equality;
relative-gap equality;
threshold-kernel equality;
positive-determinant negative control.
```

Freeze one deterministic JSON packet only after all exact tests pass.

28.4 Stage 3: native completed-Weil event lift

Construct the RH specialization from the cut:

$$Z_{\text{Weil}}f$$

using declared prime, Gamma, completion-boundary and compatibility events.
Prove coefficientwise:

$$S_{0,-}^{\text{full}} = Z_{\text{Weil}}^* Z_{\text{Weil}},$$

$$L_{\partial}(f) = \langle p_{\partial}, Z_{\text{Weil}}f \rangle,$$

and therefore

$$S_{0,-}^{\text{full}} - L_{\partial}^* L_{\partial} = Z_{\text{Weil}}^* Q Z_{\text{Weil}}.$$

No classical explicit formula is allowed to define Z_{Weil} .

28.5 Stage 4: native five-label faithfulness

Construct the five-label observer from declared target labels and prove

$$QZ_{\text{Weil}} = T_{\Sigma}^* T_{\Sigma} QZ_{\text{Weil}}$$

on the adverse target-relevant module.

Then derive

$$B_{\Sigma} = T_{\Sigma} QZ_{\text{Weil}} R^{-1} Z_{\text{Weil}}^* Q T_{\Sigma}^*$$

from the cut-generated objects and certify

$$\lambda_{\max}(B_{\Sigma}) < 1$$

with outward bounds.

Only after Stages 3 and 4 are complete may the classical completed-Weil formula recognize the native construction as its shadow.

28.6 Current status

STAGE 1 THEORY	WRITTEN
STAGE 1 GENERALIZED IMPLEMENTATION	WRITTEN
STAGE 2 TESTS	NOT YET WRITTEN
STAGE 3 RH SPECIALIZATION	NOT YET STARTED
STAGE 4 OUTWARD CERTIFICATE	NOT YET STARTED

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Chapter 29

Recognition-Seam Calculus Source Mapping

The clock-free cut-memory development consumes the public Recognition-Seam Calculus companion as its foundational language.

RSC object	Public calibration	Current cut-memory use
phase-seam holonomy	visible torus return with memory $(0,0,1)$	local transition closure does not imply global state return
target-blind birth and loss	repair ranks $2 \rightarrow 1 \rightarrow 2$	stable repair index on memory transports
ordered-product 50/45/5	blind dimension 5, repaired rank 50	target-relevant five-label repair motivation
generated ledgers	prior declaration, composition, inverse consistency	admissible clock-free transition differential
compatibility projection	pairing $15/4 \rightarrow -1/4$	raw event packets cannot be promoted before compatible projection

The current theorem adds the cut-derived operator layer:

```
RSC transition arrows
-> cut-corner differential
-> generated composition residues
-> cut-generated source/recognized/memory forms
-> target-faithful five-memory spectral reduction.
```

The exponential change flow is retained only as one exact arrow family. The calculus itself is independent of any external parameter.

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Chapter 30

Recognition-Seam Primitive-to-Completion Hierarchy

30.1 1. Why this note exists

The buried Recognition-Seam, morphic, UGD, Vedic/Kerala, Hilbert-completion and spectral-Fredholm chapters contain useful results, but they live at different levels of mathematical maturity. This note mines the reusable core without promoting every proposed connection or commutator into a theorem.

The corrected hierarchy is:

```
Shunya / pre-distinction ground
-> cut
-> trans-cut morphism
-> stabilized recognition class of morphisms
-> clock-free transition differential
-> generated composition residue
-> recovered middle identity
-> recognition-Cauchy completion
-> cut-generated positive forms
-> target-faithful finite memory
-> spectral/Fredholm shadow.
```

The Hilbert and Fredholm layers are therefore derived completion/representation layers. They are not primitive.

30.2 2. Primitive cut-recognition datum

A primitive datum is

$$\mathfrak{G} = (\mathfrak{S}, X, \mathfrak{C}_\Sigma, \mathcal{R}, \mathcal{A}),$$

where:

S	pre-distinction ground;
X	appearance space generated by admitted cuts;
C_Sigma	first cut / reference-creation operation;

R recognition map;
A declared trans-cut arrows.

The first distinction is

$$\mathfrak{C}_\Sigma : \mathfrak{S} \rightsquigarrow (\mathcal{D}^-, \mathcal{D}^+, \Sigma).$$

A trans-cut arrow is

$$\tau_\Sigma : \mathcal{D}^- \longrightarrow \mathcal{D}^+.$$

The cut is internal to the calculus. It is not an exceptional point at which calculus is suspended.

30.3 3. Objects are stabilized morphic lineages

Definition 3.1 (Morphic object)

An object is a stabilized recognition class of composable arrows:

$$O = [\alpha_n \cdots \alpha_2 \alpha_1]_{\mathcal{R}}.$$

The identity of O is the minimal closed arrow preserving its recovered recognition state. It is not defined as the absence of change.

This imports the strongest part of the morphic chapter:

form is a recognizable stabilization of flow;
flow is prior to the stabilized object;
silence/projection is a repair channel, not object annihilation.

No specific Tamil linguistic commutator is consumed as a theorem here. Those remain domain examples until represented by declared operators.

30.4 4. Clock-free transition differential

For a declared arrow

$$\gamma : x \longrightarrow y,$$

and a form assignment G_x , define

$$\Delta_\gamma G = A_\gamma^* G_y A_\gamma - G_x.$$

For composable arrows $\gamma : x \rightarrow y$, $\delta : y \rightarrow z$,

$$\Delta_{\delta\gamma} G = \Delta_\gamma G + A_\gamma^* (\Delta_\delta G) A_\gamma.$$

This is the native chain rule. It compares declared transitions and needs no external clock or infinitesimal quotient.

A generated ledger Λ_γ^G is admissible only if it is declared before the evaluation packet and obeys the same composition law:

$$\Lambda_{\delta\gamma}^G = \Lambda_\gamma^G + A_\gamma^* \Lambda_\delta^G A_\gamma.$$

The open residue is

$$\text{Res}_\gamma^G = \Delta_\gamma G - \Lambda_\gamma^G.$$

A ledger selected after seeing the residue is retrospective cancellation and is inadmissible.

30.5 5. Generated cut-corner residues

After a positive representation is derived, let P_x and $Q_x = I - P_x$ be the recognized and memory cuts. For an event transport U_γ , define

$$R_\gamma = P_y U_\gamma P_x, \quad J_\gamma = P_y U_\gamma Q_x,$$

$$C_\gamma = Q_y U_\gamma P_x, \quad M_\gamma = Q_y U_\gamma Q_x.$$

Then

$$R_{\delta\gamma} = R_\delta R_\gamma + J_\delta C_\gamma,$$

$$M_{\delta\gamma} = C_\delta J_\gamma + M_\delta M_\gamma.$$

The cross terms are generated composition residues. They are created by the cut-corner product and therefore cannot be deleted merely because endpoint shadows agree.

30.6 6. Recovered middle identity

Shadow equality of the middle object does not license composition.

Definition 6.1 (Recovered middle identity)

Suppose

$$A \xrightarrow{\gamma} B_1, \quad B_2 \xrightarrow{\delta} C.$$

The two middle states are recovered-identical when:

```
shadow state agrees;
phase transport is aligned;
scale/support is equivalent;
cut-memory transport closes or is lawfully ledgered;
path holonomy closes or is lawfully ledgered.
```

Write

$$B_1 =_{\text{RSC}} B_2.$$

Theorem 6.2 (Recognition-composable middle)

The composition $\delta\gamma$ is admitted only after recovered middle identity. Endpoint equality or nonzero scalar overlap alone is insufficient.

This theorem is the morphic safeguard needed by every refinement sequence. It prevents one finite packet from lending its visible coordinates to another packet while silently changing the source metric, cut memory or normalization.

30.7 7. Recognition completion is stronger than shadow completion

A Hilbert or Banach limit is only a shadow limit unless the cut channels also close.

Definition 7.1 (Recognition-Cauchy net)

A directed family of transported packets

$$\tilde{x}_i = (x_i, \phi_i, \sigma_i, k_i, \gamma_i)$$

is recognition-Cauchy when:

$$\|x_j - x_i\| \longrightarrow 0,$$

and, after recovered middle identification,

phase mismatch $\rightarrow 0$ or is ledgered;
scale/support mismatch $\rightarrow 0$;
cut-memory mismatch $\rightarrow 0$ or is ledgered;
transition residue $\rightarrow 0$;
path holonomy $\rightarrow 0$ or is ledgered.

The completion is formed only after quotienting the positive recognition form by its null seam. The pre-Hilbert/Hilbert carrier is therefore derived from the recognition form; it is not primitive.

Theorem 7.2 (Shadow completion is not recognition completion)

Norm convergence alone does not imply recognition convergence. A constant shadow vector may carry drifting cut memory, unaligned phase or nonzero loop holonomy. Therefore an infinite-dimensional proof must certify both the shadow tail and the recognition tail.

30.8 8. Madhava–Smriti completion rule

The Vedic/Kerala bridge supplies a useful convergence grammar:

$$\mathcal{R}(S_n + C_n) - B_\infty - S_n = 0.$$

Here S_n is the finite recursion, C_n is a predeclared correction, B_∞ is the recognized limit and S_n is the lawful tail memory.

Definition 8.1 (Smriti-complete approximation)

A finite approximation is Smriti-complete if

$$B_\infty = \mathcal{R}(S_n + C_n) - S_n$$

with

$$\|S_n\| \rightarrow 0$$

in the declared recognition norm, and with C_n generated independently of the evaluation residue.
This is the lawful form of

limit = finite recursion + correction + declared tail memory.

It will be used for operator tails, source-event tails, determinant series and prime/Gamma truncations. A visually convergent finite sequence without a Smriti tail is only a shadow approximation.

30.9 9. What is adopted now

The following buried results are adopted as foundational inputs:

cut and trans-cut morphism;
objects as stabilized morphic lineages;
clock-free transition differential;
generated composition residues;
recovered middle identity;
projection leakage as open residue;
recognition-Cauchy completion;
Madhava--Smriti tail completion;
shadow versus recognition spectral data.

The following are retained as useful representation charts, not primitives:

UGD phase-scale-seam coordinates (phi,sigma,k);
R^2=-I circular rotation and K^2=I hyperbolic seam chart;
exponential change packets;
Crown tensor connection;
classical Hilbert/Fredholm realizations.

The following are not promoted without additional representation theorems:

specific morphic commutator constants;
existence of every proposed Crown connection component;
recognition-index direct sums with undefined target groups;
curvature-flux identities without a declared connection and domain;
physical or linguistic interpretations of formal operator symbols.

30.10 10. Consequence for the RH programme

The completed-Weil specialization must be built in this order:

```
native trans-cut event morphisms
-> stabilized completed-Weil event object
-> cut-generated recognized and memory forms
-> recovered identity across all finite refinements
-> Smriti-complete prime/Gamma/boundary tails
-> recognition-complete limit
-> target-faithful memory observer of rank at most five
-> finite seam matrix
-> classical explicit-formula recognition only at the final shadow interface.
```

The immediate next theorem is the Recognition-Complete Finite-to-Infinite Cut Theorem. It must show that finite cut packets, once transported through recovered identities and supplied with vanishing Smriti tails, converge to one cut-generated infinite object with no hidden adverse complement.

30.11 11. Claim boundary

primitive cut/trans-cut hierarchy	ADOPTED
clock-free transition differential	PROVED ALGEBRAICALLY
generated cut-corner composition residues	PROVED ALGEBRAICALLY
recovered middle identity requirement	ADOPTED AS COMPOSITION GATE
recognition completion stronger than norm completion	PROVED BY EXACT COUNTEREXAMPLES
Madhava--Smriti completion grammar	ADOPTED AS TAIL DISCIPLINE
specific Crown/morphic physical models	NOT PROMOTED
RH event lift	NOT YET CONSTRUCTED
RH recognition-complete tail theorem	NEXT
RH	NOT CLAIMED

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Chapter 31

Recognition-Complete Finite-to-Infinite Cut Theorem

31.1 1. Purpose

This theorem is the generalized bridge needed before any RH specialization. It replaces the vague instruction

refine the finite model and take a limit

by a clock-free recognition-completion theorem.

The refinement index is a directed comparison label, not time. Every packet is first transported to a common recovered carrier. Convergence is required in both the recognized and cut-memory channels, with predeclared Smriti tails.

31.2 2. Transported finite cut packets

Let $\mathcal{D}$ be a common algebraic source core. After recovered middle identification, let every finite packet be represented on one complete event carrier $\mathcal{K}$ by a bounded lift

$$Z_n : \mathcal{D} \longrightarrow \mathcal{K}.$$

Let

$$P_n = P_n^* = P_n^2, \quad Q_n = I - P_n.$$

Define the recognized and memory lifts

$$Y_n = P_n Z_n, \quad M_n = Q_n Z_n.$$

The finite positive forms are

$$R_n = Y_n^* Y_n, \quad D_n = M_n^* M_n,$$

and the finite signed cut form is

$$F_n = R_n - D_n.$$

The notation $*$ is used only after the positive recognition form has been quotiented by its null seam and completed. The Hilbert representation is thus derived, not primitive.

31.3 3. Recognition-Cauchy hypothesis

Definition 3.1

The transported cut packets are recognition-Cauchy when there exist numbers

$$\rho_n^{\text{rec}}, \quad \rho_n^{\text{mem}}, \quad \rho_n^{\text{hol}}$$

with

$$\rho_n^{\text{rec}} + \rho_n^{\text{mem}} + \rho_n^{\text{hol}} \longrightarrow 0$$

such that, for all sufficiently refined $m \succeq n$,

$$\|Y_m - Y_n\| \leq \rho_n^{\text{rec}},$$

$$\|M_m - M_n\| \leq \rho_n^{\text{mem}},$$

and the recovered-middle path holonomy/residue between the two transported packets is bounded by ρ_n^{hol} .

The comparison is invalid if the source metric, cut labels or memory orientation have been changed without a recovered identity map.

31.4 4. Smriti tail hypothesis

For each n , let $C_n^{\text{rec}}, C_n^{\text{mem}}$ be predeclared corrections and let $S_n^{\text{rec}}, S_n^{\text{mem}}$ be lawful tail memories satisfying

$$Y = Y_n + C_n^{\text{rec}} + S_n^{\text{rec}},$$

$$M = M_n + C_n^{\text{mem}} + S_n^{\text{mem}},$$

with

$$\|S_n^{\text{rec}}\| + \|S_n^{\text{mem}}\| \longrightarrow 0.$$

The corrections must be generated from constitutive, analytic or reference data independent of the held-out evaluation residue.

This is the operator form of

finite Chandas recursion + correction + vanishing Smriti tail.

31.5 5. Recognition-complete limit

Theorem 5.1 (Cut-channel completion)

Under the recognition-Cauchy and Smriti-tail hypotheses, there exist bounded limit lifts

$$Y, M : \mathcal{H}_{\text{rec}} \longrightarrow \mathcal{K}$$

such that

$$Y_n \longrightarrow Y, \quad M_n \longrightarrow M$$

in operator norm after declared corrections. The limit event lift is

$$Z = Y + M.$$

The cut-generated forms

$$R = Y^*Y, \quad D = M^*M, \quad F = R - D$$

satisfy

$$R_n \longrightarrow R, \quad D_n \longrightarrow D, \quad F_n \longrightarrow F$$

in operator norm.

Proof

The two channel families are Cauchy in the complete operator spaces and therefore converge. For any two bounded operators A, B ,

$$\|A^*A - B^*B\| \leq (\|A\| + \|B\|)\|A - B\|.$$

Apply this first to Y_n, Y and then to M_n, M . Subtraction gives the convergence of F_n . The holonomy condition ensures that the limits belong to one recovered carrier rather than merely sharing a shadow coordinate.

Quantitative form

If

$$\delta_n^{\text{rec}} = \|Y - Y_n\|, \quad \delta_n^{\text{mem}} = \|M - M_n\|,$$

then

$$\boxed{\|R - R_n\| \leq (\|Y\| + \|Y_n\|)\delta_n^{\text{rec}}},$$

$$\boxed{\|D - D_n\| \leq (\|M\| + \|M_n\|)\delta_n^{\text{mem}}},$$

and

$$\boxed{\|F - F_n\| \leq (\|Y\| + \|Y_n\|)\delta_n^{\text{rec}} + (\|M\| + \|M_n\|)\delta_n^{\text{mem}}}.$$

These bounds are the general replacement for an unnamed collection of finite-to- infinite errors.

31.6 6. Target-faithful memory completion

Let

$$T_n : \mathcal{K} \longrightarrow \mathbb{C}^\nu, \quad \nu \leq 5,$$

be transported target observers satisfying

$$T_n T_n^* = I_\nu.$$

Define the faithfulness residual

$$\varepsilon_n^{\text{faith}} = \|(I - T_n^* T_n) M_n\|.$$

Assume

$$T_n \longrightarrow T$$

in operator norm after one fixed label orientation, and

$$\varepsilon_n^{\text{faith}} \longrightarrow 0.$$

Theorem 6.1 (No-hidden-memory completion)

The limit memory lift satisfies

$$\boxed{M = T^* T M.}$$

Consequently

$$\overline{\text{Ran } M} \subseteq \text{Ran } T^*,$$

and the target-relevant memory rank is at most $\nu \leq 5$.

Proof

Write

$$(I - T^* T) M = (I - T^* T)(M - M_n) + (T_n^* T_n - T^* T) M_n + (I - T_n^* T_n) M_n.$$

Every term tends to zero by norm convergence and the faithfulness residual.

This is the rigorous no-hidden-complement theorem. A numerically rank-five finite packet without the vanishing faithfulness residual does not satisfy it.

31.7 7. Cut-derived finite seam matrix

Define

$$A_n = T_n M_n, \quad A = TM.$$

Then

$$D_n = A_n^* A_n + E_n^{\text{blind}}, \quad \|E_n^{\text{blind}}\| \longrightarrow 0,$$

and in the limit

$$D = A^* A.$$

Assume a uniform positive reference floor

$$R_n \geq mI, \quad R \geq mI, \quad m > 0.$$

Define

$$B_n = A_n R_n^{-1} A_n^* \in M_\nu(\mathbb{C}),$$

and

$$B = AR^{-1}A^*.$$

Theorem 7.1 (Finite seam-matrix convergence)

The matrices satisfy

$$B_n \longrightarrow B$$

in operator norm. More precisely,

$$\|B_n - B\| \leq m^{-1}(\|A_n\| + \|A\|)\|A_n - A\| + m^{-2}\|A\|\|A_n\|\|R_n - R\|.$$

Proof

Use

$$\|R_n^{-1}\|, \|R^{-1}\| \leq m^{-1}$$

and

$$\|R_n^{-1} - R^{-1}\| \leq m^{-2}\|R_n - R\|.$$

Expand $B_n - B$ by adding and subtracting $AR_n^{-1}A_n^*$ and $AR^{-1}A_n^*$.

31.8 8. Exact infinite-to-finite conclusion

By Theorem 6.1,

$$F = R - A^*A.$$

The cut-memory spectral isomorphism therefore gives

$$N_-(F) = N_+(B - I),$$

$$\dim \ker F = \dim \ker(I - B),$$

and

$$\inf_{f \neq 0} \frac{\langle f, Ff \rangle}{\langle f, Rf \rangle} = \lambda_{\min}(I - B).$$

Thus the infinite signed cut form is decided by a matrix of size at most five, but only after recognition-complete convergence and target faithfulness have been proved.

31.9 9. Outward finite certificate

Let u_n be an outward upper bound for $\lambda_{\max}(B_n)$, and let e_n be an outward upper bound for $\|B - B_n\|$. Then

$$\lambda_{\max}(B) \leq u_n + e_n.$$

Hence

$$u_n + e_n < 1$$

is a lawful finite certificate for

$$F > 0$$

in the reference metric.

A fixed-grid top without e_n is a shadow certificate only.

31.10 10. Why this theorem is clock-free

No external time parameter appears. The index n labels a directed refinement or declared comparison object. The calculus uses:

```
trans-cut arrows;
recovered middle identity;
recognition-Cauchy comparison;
predeclared Smriti tails;
target-faithful memory observers.
```

An exponential family may generate one refinement chart, but the theorem does not depend on that chart.

31.11 11. RH specialization gate

For the completed-Weil specialization, the framework must construct a directed family of native event lifts

$$Z_n^{\text{Weil}}$$

from prime, Gamma, completion-boundary and compatibility events and prove:

1. recovered identity across all refinement arrows;
2. recognized and memory channel Cauchy bounds;
3. Madhava--Smriti tail bounds for every omitted event family;
4. target-faithfulness residual $\rightarrow 0$ with one fixed five-label orientation;
5. a uniform positive recognized-reference floor;
6. an outward finite seam-matrix margin.

Only then is RH a special case of the universal Recognition-Complete Cut Theorem.

31.12 12. Claim boundary

cut-channel operator-norm completion	PROVED
explicit form-error bounds	PROVED
no-hidden-memory limit from faithfulness residual	PROVED
finite seam-matrix convergence	PROVED
outward finite certificate rule	PROVED
clock-free character	PROVED
RH event family recognition-Cauchy bounds	NOT YET BUILT
RH Smriti tail packet	NOT YET BUILT
RH target-faithfulness residual	NOT YET BUILT
RH outward five-matrix certificate	NOT YET BUILT
RH	NOT CLAIMED

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Chapter 32

Stage 1.5: Morphic Identity, Recognition Completion, and Smriti Tails

32.1 1. Stage rule

Stage 1.5 is a writing and theorem-design stage. It does not claim a test pass. The generalized calculus is written first; exact tests are added only after the current Stage-1 packet has been run and reviewed.

This preserves the project order:

```
native definition
-> theorem
-> exact generalized example
-> fail-closed tests
-> domain specialization
-> numerical certification.
```

32.2 2. Buried material retained

The following ideas are retained from the buried calculus chapters:

```
objects as stabilized morphic lineages;
trans-cut morphisms;
Recognition-Seam product and chain residues;
recovered middle identity;
Hilbert shadow versus recognition completion;
Madhava finite recursion plus Smriti tail;
projection leakage and recognition-null quotient;
circular/hyperbolic charts as optional representations.
```

The following remain illustrative until represented rigorously:

```
specific Tamil morphic commutator constants;
physical interpretations of Crown-sector terms;
formal recognition-index direct sums without target groups;
curvature flux formulas without declared connections and domains.
```

32.3 3. Generalized example E: recovered middle identity

Construct two arrows

$$A \xrightarrow{\gamma} B_1, \quad B_2 \xrightarrow{\delta} C$$

with identical shadow middle coordinates but different cut memories.

The example must verify:

shadow middle equality	true
recovered middle identity	false
naive matrix composition	defined
RSC composition	rejected
unledgered composition residue	nonzero

Then add an exact memory alignment arrow and verify that recovered identity and composition are restored.

Purpose: prevent finite packets with the same visible coordinates but different source/cut lineage from being spliced together.

32.4 4. Generalized example F: shadow-Cauchy but not recognition-Cauchy

Use a constant shadow vector

$$x_n = x$$

with a drifting integer memory

$$k_n = n.$$

Verify:

shadow norm distance	0
shadow sequence Cauchy	true
cut-memory sequence Cauchy	false
recognition sequence Cauchy	false

Then use a lawfully ledgered memory quotient and verify recognition-Cauchy closure in the quotient.

Purpose: demonstrate why norm convergence alone cannot close an infinite-dimensional cut theorem.

32.5 5. Generalized example G: Madhava–Smriti completion

Choose an exact convergent operator series with finite partial recursion S_n , a predeclared correction C_n , recognized limit B_∞ , and exact tail S_n :

$$B_\infty = S_n + C_n + S_n.$$

The example must verify:

finite recursion generated independently	true
correction declared before evaluation	true
tail identity exact	true
tail norm decreases	true
retrospective fitted correction rejected	true

Purpose: make the finite-to-infinite theorem use lawful tail memory rather than visual convergence.

32.6 6. Generalized example H: morphic object stabilization

Construct a finite arrow family whose different admissible paths produce one recovered object after ledger closure.

Verify:

local arrow residues	zero
path shadows agree	true
path cut memories agree or are ledger-equivalent	true
stabilized recognition class	unique

Add a negative control in which endpoint shadows agree but path memory differs, so the alleged object is not stabilized.

Purpose: implement the doctrine that objects are stabilized morphic lineages, not primitive labels.

32.7 7. Generalized example I: recognition-complete five-memory limit

Build a directed exact-rational sequence

$$Z_n = Y_n + M_n$$

with:

recognized channel Cauchy	true
memory channel Cauchy	true
faithfulness residual	-> 0
target dimension	5
uniform reference floor	positive
finite seam matrices	converge

Verify the quantitative bounds in the Recognition-Complete Finite-to-Infinite Cut Theorem and show that

$$\lambda_{\max}(B_n) + e_n < 1$$

certifies the limit form.

Negative controls:

shadow convergence with drifting memory;
rank-five numerical shadow without faithfulness;
positive determinant with two super-threshold modes;
changing the five-label orientation between refinements;
post-hoc correction fitted to the terminal residue.

32.8 8. Test order after writing

After the user reports the Stage-1 output, implement tests in this order:

T1 transition differential and composition residues
T2 recovered middle identity
T3 recognition-Cauchy versus shadow-Cauchy
T4 Madhava--Smriti exact tail
T5 morphic stabilization
T6 target-faithful limit
T7 finite seam-matrix convergence
T8 adversarial negative controls

No RH data enters these tests.

32.9 9. RH stage after generalized validation

Only after Stage 1.5 passes do we construct:

$$Z_n^{\text{Weil}} = Z_n^{\text{prime}} \oplus Z_n^{\Gamma} \oplus Z_n^{\partial} \oplus Z_n^{\text{comp}}.$$

The first RH task is not an eigenvalue calculation. It is to prove recovered identity and Smriti-complete convergence of these four event families on one common cut carrier.

32.10 10. Claim boundary

Stage-1.5 theorem architecture
exact examples E--I
fail-closed tests
RH specialization
RH numerical certificate

WRITTEN
PLANNED
DEFERRED UNTIL STAGE-1 REVIEW
DEFERRED
DEFERRED

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Chapter 33

Stage 1.5 Exact Recognition-Completion Certificate

33.1 1. Source doctrine

This stage consumes the strongest mathematically usable results from the buried Recognition-Seam and morphic-calculus chapters:

```
cut and trans-cut morphism;  
objects as stabilized morphic lineages;  
recovered middle identity;  
Recognition-Cauchy completion;  
Madhava--Smriti lawful tails;  
projection leakage and target faithfulness;  
shadow stationarity/convergence versus recognition closure.
```

The chapters distinguish zero from the pre-cut ground, put the cut inside calculus, and define objects as stabilized recognition classes of morphisms. They also prove by explicit counterexample that Hilbert/norm convergence may coexist with drifting seam memory. The Vedic/Kerala bridge supplies the rule

recognized limit = finite recursion + predeclared correction + Smriti tail.

The variational chapter adds a useful audit principle: shadow stationarity or conservation does not lift to recognition stationarity unless boundary, phase, memory and holonomy residues also close.

33.2 2. Exact generalized packet

The executable packet is

```
proof_lab/clock_free_cut_memory_stage15_examples.py
```

and the fail-closed tests are

```
proof_lab/test_clock_free_cut_memory_stage15.py
```

All arithmetic is rational.

33.3 3. Recovered middle identity

Two middle states have identical shadows but different cut memories:

$$B_1 = (\text{shadow } B, \phi, \sigma, 1), \quad B_2 = (\text{shadow } B, \phi, \sigma, 0).$$

Therefore

$$B_1 =_{\text{shadow}} B_2, \quad B_1 \neq_{\text{RSC}} B_2.$$

Raw composition is rejected. A memory transport declared before observation,

$$k \mapsto k - 1,$$

restores recovered identity and lawful composition. An identical correction declared after observation is rejected.

This is the exact composition gate needed before finite packets may be compared on one carrier.

33.4 4. Shadow-Cauchy is not Recognition-Cauchy

The sequence

$$\tilde{x}_n = (1, 0, 1, n)$$

has constant Hilbert/shadow coordinate, so its shadow distance is identically zero. Its unledgered memory does not form a Recognition-Cauchy sequence.

Under the constitutive memory model

$$k_n = n$$

declared before evaluation, the quotient memory is identically zero and the sequence closes in the declared recognition quotient.

Thus forgetting memory and lawfully quotienting memory are mathematically different operations.

33.5 5. Madhava–Smriti exact tail

For

$$r = \frac{1}{3}, \quad B_\infty = \sum_{j=0}^{\infty} r^j = \frac{3}{2},$$

put

$$S_n = \sum_{j=0}^n r^j, \quad C_n = r^{n+1}, \quad S_n = \frac{r^{n+2}}{1-r}.$$

Then, for every tested n ,

$$S_n + C_n + S_n = \frac{3}{2}$$

exactly, and

$$S_{n+1} < S_n.$$

A correction equal to the complete observed residual is rejected because it is defined from the held-out limit after evaluation.

33.6 6. Morphic object stabilization

Two different locally closed paths produce the same recovered final state:

path a: shadow +1, phase +1/6, scale x2, memory +1;
 shadow +1, phase -1/6, scale x1/2, memory -1;

path b: shadow +2, phase 0, scale x1, memory 0.

Both stabilize at

$$(2, 0, 1, 0).$$

A negative-control path returns the same shadow but leaves memory one. It is not the same morphic object in RSC.

33.7 7. Recognition-complete five-memory limit

The recognized channel is fixed and has reference floor

$$R = I_5.$$

The target-visible memory amplitudes converge to

$$\left(\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \frac{9}{10}\right),$$

while a sixth blind memory coordinate has amplitude

$$\varepsilon_n = \frac{1}{n+2} \longrightarrow 0.$$

Thus the finite packets have a genuine hidden complement, but the limit does not. The target-faithfulness residual contracts from

$$\frac{1}{10} \quad \text{at } n = 8$$

to

$$\frac{1}{18} \quad \text{at } m = 16.$$

The limit memory rank is exactly five. The finite target seam matrix at $n = 8$ has top

$$\lambda_{\max}(B_8) = \frac{654481}{810000},$$

and the exact outward transfer error is

$$e_8 = \frac{1619}{810000}.$$

Therefore

$$\lambda_{\max}(B_8) + e_8 = \frac{81}{100} < 1.$$

The limit relative gap is exactly

$$1 - \lambda_{\max}(B) = \frac{19}{100}.$$

This is a generalized exact finite-to-infinite certificate. It is not an RH certificate.

33.8 8. Expected status

PASS_CLOCK_FREE_CUT_MEMORY_STAGE15

with six fail-closed unit tests and a byte-stable JSON certificate.

33.9 9. Gold audit completed

All nineteen uploaded calculus chapters and the Vedic/Kerala bridge note have now been read. Their useful material has been sorted into three classes.

Adopted foundation

```
cut/trans-cut calculus;
clock-free transition differential;
composition residues;
recovered identity;
Recognition-Cauchy completion;
Madhava--Smriti tails;
projection leakage;
spectral/Fredholm shadow discipline;
recognition variational and Noether residue audit.
```

Representation charts

UGD phase-scale-memory coordinates;
circular/hyperbolic R-K chart;
exponential change families;
PDE recognition-regime charts;
Crown connection notation.

Not promoted without further representation

undeclared morphic commutator constants;
formal Crown components lacking operators/domains;
curvature or physical interpretations inferred from notation alone;
recognition-index tuples without a declared target group.

33.10 10. Next stage after execution

After the user confirms the six tests and byte-stable JSON comparison, the next generalized theorem will be the Cut-Variational Minimal Observer Theorem.

Its purpose is to derive target faithfulness from the cut-memory action

$$\mathcal{A}_{\Sigma}(P) = \|(I - P)M\|^2,$$

rather than assuming a five-label observer. It will prove:

zero action iff the observer contains the complete adverse memory range;
minimum faithful observer rank equals the target-relevant cut-memory rank;
source-restriction blind dimension five implies five labels are necessary and sufficient;
recognition-complete convergence transfers finite variational faithfulness to the limit.

Only after that universal theorem passes will the completed-Weil event family be introduced.

33.11 11. Claim boundary

Stage-1.5 exact examples	WRITTEN
Stage-1.5 local dry run	PASSED BY DEVELOPMENT AUDIT
user machine tests	PENDING
byte-stable user certificate	PENDING
Cut-Variational Minimal Observer Theorem	NEXT AFTER USER PASS
completed-Weil event lift	NOT YET STARTED
RH	NOT CLAIMED

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Chapter 34

Cut-Variational Minimal Observer Theorem

34.1 1. Purpose

The Recognition-Seam programme must not assume a five-label observer merely because a later matrix happens to have size five. The observer has to be derived from the cut-memory object itself.

The native order is

```
cut-generated memory lift
-> target-relevant memory range
-> observer action
-> minimal faithful observer
-> finite seam matrix.
```

No classical determinant or preassembled Birman–Schwinger matrix is used to define the observer.

34.2 2. Recognition-complete cut memory

After recovered-middle identification and Recognition-Cauchy completion, let

$$M : \mathcal{H}_{\text{rec}} \longrightarrow \mathcal{K}_{\text{mem}}$$

be the cut-generated adverse memory lift. Its target-relevant memory space is

$$\boxed{\mathcal{E}_{\Sigma} = \overline{\text{Ran } M}.}$$

The closure is taken only after the recognized and memory channels have been transported to one recovered carrier.

An observer of rank ν is a coisometry

$$T : \mathcal{K}_{\text{mem}} \longrightarrow \mathbb{C}^{\nu}, \quad TT^* = I_{\nu}.$$

Its recovered memory projector is

$$P_T = T^*T.$$

Thus P_T is the orthogonal projection onto $\text{Ran } T^*$.

34.3 3. Cut-observer action

Definition 3.1 (Operator action)

Define

$$\mathcal{A}_\Sigma^\infty(T; M) = \|(I - P_T)M\|^2.$$

This is the largest unrecognized cut-memory energy.

When M is Hilbert–Schmidt, define the total cut action

$$\mathcal{A}_\Sigma^2(T; M) = \|(I - P_T)M\|_{\text{HS}}^2.$$

Both actions are generated from the cut defect. They do not compare the observer with a desired answer.

34.4 4. Zero action is exactly target faithfulness

Theorem 4.1 (Cut-faithfulness equivalence)

For every bounded cut-memory lift M and every finite-rank coisometric observer T , the following are equivalent:

$$\mathcal{A}_\Sigma^\infty(T; M) = 0,$$

$$(I - P_T)M = 0,$$

$$M = P_T M = T^* T M,$$

and

$$\mathcal{E}_\Sigma \subseteq \text{Ran } T^*.$$

If M is Hilbert–Schmidt, these conditions are also equivalent to

$$\mathcal{A}_\Sigma^2(T; M) = 0.$$

Proof

A norm is zero exactly when its argument is zero, so

$$\mathcal{A}_\Sigma^\infty(T; M) = 0 \iff (I - P_T)M = 0.$$

Since $P_T = T^*T$, this is equivalent to $M = P_TM = T^*TM$. The latter holds exactly when every vector in $\text{Ran } M$, and therefore in its closure, lies in $\text{Ran } P_T = \text{Ran } T^*$. The Hilbert–Schmidt assertion is identical.

Thus target faithfulness is a zero-action theorem, not an observer convention.

34.5 5. Minimal faithful observer rank

Theorem 5.1 (Minimal observer theorem)

Assume

$$r_\Sigma = \dim \mathcal{E}_\Sigma < \infty.$$

Then:

1. every faithful observer has rank at least r_Σ ;
2. there exists a faithful observer of rank exactly r_Σ ;
3. consequently,

$$\boxed{\min\{\text{rank } T : \mathcal{A}_\Sigma^\infty(T; M) = 0\} = r_\Sigma.}$$

Proof

If T is faithful, Theorem 4.1 gives

$$\mathcal{E}_\Sigma \subseteq \text{Ran } T^*.$$

Hence

$$r_\Sigma \leq \dim \text{Ran } T^* = \text{rank } T.$$

Conversely choose an orthonormal basis of $\mathcal{E}_\Sigma$ and let $T^* : \mathbb{C}^{r_\Sigma} \rightarrow \mathcal{K}_{\text{mem}}$ be the corresponding isometric inclusion. Then $TT^* = I$, P_T projects onto $\mathcal{E}_\Sigma$, and Theorem 4.1 gives zero action.

The number of labels is therefore generated by cut-memory dimension.

34.6 6. Variational form

Assume now that M is compact and Hilbert–Schmidt, with singular values

$$s_1(M) \geq s_2(M) \geq \cdots \geq 0.$$

For a rank- ν orthogonal projector P , put

$$\mathcal{A}_\nu(P; M) = \|(I - P)M\|_{\text{HS}}^2.$$

Theorem 6.1 (Cut-variational observer theorem)

Among all rank- ν observers,

$$\inf_{\text{rank } P = \nu} \mathcal{A}_\nu(P; M) = \sum_{j > \nu} s_j(M)^2.$$

A minimizer is the projector onto the first ν left singular directions of M .

Proof

Let

$$G = MM^* \geq 0.$$

For a rank- ν orthogonal projector P ,

$$\mathcal{A}_\nu(P; M) = \text{Tr}(M^*(I - P)M) = \text{Tr } G - \text{Tr}(PG).$$

The trace $\text{Tr}(PG)$ is maximized by selecting the eigenspaces of G with its ν largest eigenvalues $s_j(M)^2$. Subtracting from $\text{Tr } G = \sum_j s_j(M)^2$ gives the formula.

Corollary 6.2

$$\inf_{\text{rank } P = \nu} \mathcal{A}_\nu(P; M) = 0 \iff \text{rank } M \leq \nu.$$

Thus a positive minimum is an exact witness that the observer is missing cut memory.

34.7 7. Target-relative source restriction

Let

$$A : X \longrightarrow Y$$

be a source restriction and

$$G : X \longrightarrow Z$$

be the target readout. The target-relevant blind space is the image

$$\boxed{\mathcal{B}_G(A) = G(\ker A) \subseteq Z.}$$

Its dimension is

$$\boxed{b_G(A) = \text{rank}(G|_{\ker A}).}$$

A repair bank

$$L : X \longrightarrow \mathbb{C}^\nu$$

is target-faithful relative to A when

$$Ax = 0, \quad Lx = 0 \quad \implies \quad Gx = 0.$$

Theorem 7.1 (Target-relative minimal repair)

Every target-faithful repair bank satisfies

$$\nu \geq b_G(A).$$

There exists a repair bank with

$$\nu = b_G(A).$$

Proof

Only the restriction of L to $\ker A$ can repair information lost by A . Target faithfulness means

$$\ker(L|_{\ker A}) \subseteq \ker(G|_{\ker A}).$$

Therefore $L|_{\ker A}$ must separate the quotient

$$\ker A / \ker(G|_{\ker A}),$$

whose dimension equals $b_G(A)$. Hence at least $b_G(A)$ scalar channels are necessary. Choose coordinates on this quotient to obtain a repair bank of exactly that size.

This theorem derives the observer dimension from target-relative blindness, not from ambient dimension.

34.8 8. The exact 50/45/5 calibration

For the canonical ordered-product collapse

$$A = \begin{pmatrix} I_{45} & 0_{45 \times 5} \end{pmatrix} : \mathbb{Q}^{50} \rightarrow \mathbb{Q}^{45},$$

one has

$$\dim \ker A = 5.$$

The split-occurrence repair on this kernel is

$$L_{\text{split}} = \begin{pmatrix} -6 & -6 & 0 & 0 & 0 \\ 0 & 6 & 0 & 0 & 0 \\ 0 & 0 & -12 & -6 & 0 \\ 0 & 0 & 6 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 \end{pmatrix}$$

with

$$\det L_{\text{split}} = -2592 \neq 0.$$

Therefore the five blind coordinates are target-relevant and five repair labels are necessary and sufficient.

This is the general RSC reason a five-label packet can be canonical. It is not yet the RH specialization.

34.9 9. Recognition-complete transfer of observer faithfulness

Let

$$M_n \rightarrow M, \quad P_n \rightarrow P$$

in operator norm, where every P_n and P is an orthogonal observer projection. Define

$$r_n = (I - P_n)M_n, \quad r = (I - P)M.$$

Theorem 9.1 (Faithfulness transfer)

$$\|r - r_n\| \leq \|M - M_n\| + \|P - P_n\| \|M\|.$$

Consequently, if

$$\|r_n\| \rightarrow 0, \quad M_n \rightarrow M, \quad P_n \rightarrow P,$$

then

$$(I - P)M = 0.$$

Proof

Write

$$r - r_n = (I - P)(M - M_n) + (P_n - P)M_n.$$

Alternatively add and subtract $(I - P_n)M$; either expansion gives a bound that tends to zero under the stated hypotheses. Using the second expansion,

$$r - r_n = (I - P_n)(M - M_n) + (P_n - P)M,$$

which yields the displayed estimate. Since $r_n \rightarrow 0$, one obtains $r = 0$.
This is the variational form of the no-hidden-complement theorem.

34.10 10. Seam matrix after the observer is derived

Let

$$R > 0$$

be the cut-generated recognized reference, and let a faithful observer T produce

$$A_\Sigma = TM.$$

Only now define

$$\boxed{B_\Sigma = A_\Sigma R^{-1} A_\Sigma^*}.$$

The finite seam matrix is therefore downstream of the zero-action and minimal observer theorems.

The lawful hierarchy is

```
cut memory M
-> zero observer action
-> minimal faithful rank
-> target coordinates A_Sigma
-> finite seam matrix B_Sigma.
```

34.11 11. RH specialization gate

For the completed-Weil object, the future specialization must prove that the native adverse memory lift has

$$\dim \overline{\text{Ran } M_{\text{Weil}}} = 5$$

or, equivalently, that its target-relative source blindness has dimension five. Then five labels are necessary and sufficient by Theorems 5.1 and 7.1.

This must be obtained from the native event lift and source restriction. It may not be inferred from a numerically observed five-dimensional matrix.

34.12 12. Claim boundary

zero action iff target faithfulness	PROVED
minimal faithful observer rank = memory rank	PROVED
Hilbert--Schmidt variational tail formula	PROVED
target-relative minimal repair theorem	PROVED
50/45/5 exact calibration	AVAILABLE
faithfulness transfer under recognition completion	PROVED
RH adverse memory rank exactly five	NOT YET SPECIALIZED
RH native observer action	NOT YET BUILT
RH finite seam matrix	NOT YET BUILT IN THIS STAGE
RH	NOT CLAIMED

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Chapter 35

Cut-Covariance Event Realization Theorem

35.1 1. Purpose

The remaining source/boundary barrier should not be attacked by treating the source form and boundary functional as independent objects and then searching for a decoder after the fact.

Recognition-Seam Calculus permits a stronger order:

```
native event lift
-> distinguished boundary recognition state
-> first moment
-> second moment
-> cut covariance
-> target-faithful finite observer.
```

The source and boundary are then co-generated by one event object. Boundary square removal is not an inequality guessed from numerics; it is the Gram form of the memory left after the boundary-recognition line is cut away.

This theorem is universal. It contains no zeta function, prime sum, classical explicit formula, or RH assumption.

35.2 2. Native event-cut datum

Let

$$\mathcal{H}_{\text{src}}$$

be the positive recognition completion of a native algebraic source core, and let

$$\mathcal{K}_{\text{evt}}$$

be a complete event carrier derived after quotienting the null recognition seam.

Let

$$Z : \mathcal{H}_{\text{src}} \longrightarrow \mathcal{K}_{\text{evt}}$$

be a bounded native event lift. The map Z must be constructed before the signed target form is evaluated.

Choose a unit boundary-recognition state

$$p_{\partial} \in \mathcal{K}_{\text{evt}}, \quad \|p_{\partial}\| = 1,$$

also generated independently of the desired sign. Define

$$P_{\partial} = |p_{\partial}\rangle\langle p_{\partial}|, \quad Q_{\partial} = I - P_{\partial}.$$

The boundary line is the recognized first-moment channel; Q_{∂} is the cut-memory channel.

35.3 3. First moment, second moment, and cut covariance

Define the source second moment

$$S = Z^*Z,$$

the boundary first moment

$$\ell = p_{\partial}^*Z,$$

and the cut-memory lift

$$M = Q_{\partial}Z.$$

The signed covariance form is

$$W_{\text{cut}} = M^*M.$$

Theorem 3.1 (Cut-covariance identity)

The following exact identity holds:

$$W_{\text{cut}} = S - \ell^*\ell = Z^*Q_{\partial}Z = M^*M \geq 0.$$

Proof

Since $P_{\partial} = p_{\partial}p_{\partial}^*$,

$$\ell^*\ell = Z^*p_{\partial}p_{\partial}^*Z = Z^*P_{\partial}Z.$$

Therefore

$$S - \ell^*\ell = Z^*(I - P_{\partial})Z = Z^*Q_{\partial}Z = (Q_{\partial}Z)^*(Q_{\partial}Z).$$

The last expression is a Gram form and is nonnegative.

Thus source and boundary are not merely compatible. They are the second and first moments of one cut-generated event lift.

35.4 4. Infinite-dimensional Cut-Square identity

For every $f, h \in \mathcal{H}_{\text{src}}$,

$$\langle h, W_{\text{cut}} f \rangle = \langle Q_{\partial} Z h, Q_{\partial} Z f \rangle.$$

In particular,

$$\langle f, S f \rangle - |\ell f|^2 = \|Q_{\partial} Z f\|^2.$$

This is the coordinate-free Cut-Square identity. The finite Lagrange identity

$$\|c\|^2 \|y\|^2 - |\langle c, y \rangle|^2 = \sum_{i < j} |c_i y_j - c_j y_i|^2$$

is one coordinate realization of the same projection defect.

35.5 5. Event-algebra covariance form

Let $\pi(\mathcal{A}) \subseteq \mathcal{B}(\mathcal{K}_{\text{evt}})$ be a represented native event algebra, and define the state

$$\omega_{\partial}(a) = \langle p_{\partial}, \pi(a) p_{\partial} \rangle.$$

Suppose a linear event assignment $f \mapsto a_f$ satisfies

$$\pi(a_f) p_{\partial} = Z f.$$

Corollary 5.1 (Cut covariance as noncommutative variance)

Then

$$\langle h, W_{\text{cut}} f \rangle = \omega_{\partial}(a_h^* a_f) - \overline{\omega_{\partial}(a_h)} \omega_{\partial}(a_f).$$

In particular,

$$\langle f, W_{\text{cut}} f \rangle = \text{Var}_{\omega_{\partial}}(a_f) = \|Q_{\partial} \pi(a_f) p_{\partial}\|^2.$$

The algebraic covariance is therefore generated by the cut. It is not imposed as a positivity axiom.

35.6 6. The decoder is derived, not fitted

Let

$$\mathcal{E} = \overline{\text{Ran } Z} \subseteq \mathcal{K}_{\text{evt}}$$

and let $P_{\mathcal{E}}$ be its orthogonal projection.

Define

$$c_{\min} = P_{\mathcal{E}} p_{\partial}.$$

Theorem 6.1 (Canonical least-energy decoder)

The vector $c_{\min}$ is the unique vector in $\mathcal{E}$ satisfying

$$\ell f = \langle c_{\min}, Zf \rangle \quad (f \in \mathcal{H}_{\text{src}}).$$

Its squared norm

$$\beta_{\text{cut}} = \|c_{\min}\|^2 = \|P_{\mathcal{E}} p_{\partial}\|^2 \leq 1$$

is the sharp decoder burden.

Proof

For $y = Zf \in \text{Ran } Z \subseteq \mathcal{E}$,

$$\langle P_{\mathcal{E}} p_{\partial}, y \rangle = \langle p_{\partial}, y \rangle = \ell f.$$

If $c \in \mathcal{E}$ is another representer, then

$$\langle c - c_{\min}, y \rangle = 0$$

for every $y \in \text{Ran } Z$, hence for every $y \in \mathcal{E}$ by density. Since $c - c_{\min} \in \mathcal{E}$, it vanishes. Finally, orthogonal projection is contractive, so $\beta_{\text{cut}} \leq \|p_{\partial}\|^2 = 1$.

If Z has finite-dimensional full column rank and $S = Z^*Z$ is invertible, then

$$\beta_{\text{cut}} = \ell S^{-1} \ell^*.$$

Thus the decoder is not an independently fitted vector. It is the projection of the boundary-recognition state onto the event range.

35.7 7. Uniform reserve and pointwise strictness

The decoder burden gives

$$|\ell f|^2 \leq \beta_{\text{cut}} \langle f, S f \rangle.$$

Hence, when $\beta_{\text{cut}} < 1$,

$$W_{\text{cut}} \geq (1 - \beta_{\text{cut}}) S.$$

This is a uniform relative gap.

But a strict uniform reserve is not necessary for pointwise strict positivity.

Theorem 7.1 (Boundary-line no-blindness)

For every f ,

$$\langle f, W_{\text{cut}} f \rangle = 0 \iff Q_{\partial} Z f = 0 \iff Z f \in \mathbb{C} p_{\partial}.$$

Consequently,

$$W_{\text{cut}} > 0 \text{ on nonzero source states} \iff \ker(Q_{\partial} Z) = \{0\}.$$

Proof

The Cut-Covariance Identity gives

$$\langle f, W_{\text{cut}} f \rangle = \|Q_{\partial} Z f\|^2.$$

This vanishes exactly when Zf lies in the boundary line $\text{Ran } P_{\partial} = \mathbb{C}p_{\partial}$.

Therefore an infinite family may have $\beta_n \uparrow 1$ while every finite packet remains strictly positive. The correct strictness theorem is no boundary-line blindness, not decimal distance from one by itself.

35.8 8. Target-faithful observer derived after the cut

Let

$$T : \mathcal{K}_{\text{mem}} = Q_{\partial} \mathcal{K}_{\text{evt}} \longrightarrow \mathbb{C}^{\nu}$$

be a coisometric target observer,

$$TT^* = I_{\nu}, \quad P_T = T^*T.$$

Its cut-observer action is

$$\mathcal{A}_{\Sigma}(T; M) = \|(I - P_T)M\|^2.$$

Theorem 8.1 (Observed covariance)

If

$$\mathcal{A}_{\Sigma}(T; M) = 0,$$

then

$$M = T^*TM$$

and

$$\boxed{W_{\text{cut}} = (TM)^*(TM)}.$$

Thus the finite observed covariance is exact only after zero cut-observer action has proved target faithfulness.

By the Cut-Variational Minimal Observer Theorem,

$$\min \text{rank } T = \dim \overline{\text{Ran } M}$$

whenever this memory range is finite-dimensional.

35.9 9. Recognition-complete transfer

Let (Z_n, p_n) be transported finite event packets on one recovered carrier, with

$$\|p_n\| = \|p\| = 1, \quad Z_n \rightarrow Z, \quad p_n \rightarrow p.$$

Put

$$P_n = p_n p_n^*, \quad Q_n = I - P_n, \quad M_n = Q_n Z_n.$$

Theorem 9.1 (First/second moment transfer)

The following bounds hold:

$$\|P_n - P\| \leq 2\|p_n - p\|,$$

$$\|M_n - M\| \leq \|Z_n - Z\| + 2\|p_n - p\| \|Z\|,$$

$$\|Z_n^* Z_n - Z^* Z\| \leq (\|Z_n\| + \|Z\|) \|Z_n - Z\|,$$

and

$$\|M_n^* M_n - M^* M\| \leq (\|M_n\| + \|M\|) \|M_n - M\|.$$

Also, with $\ell_n = p_n^* Z_n$,

$$\|\ell_n - \ell\| \leq \|Z_n - Z\| + \|p_n - p\| \|Z\|.$$

Proof

The rank-one projector estimate follows by adding and subtracting $p_n p^*$. Next,

$$M_n - M = Q_n(Z_n - Z) + (Q_n - Q)Z.$$

Use $\|Q_n\| = 1$ and the projector estimate. The Gram-form inequalities are the standard identity

$$A^* A - B^* B = A^*(A - B) + (A^* - B^*)B.$$

The first-moment estimate follows from

$$p_n^* Z_n - p^* Z = p_n^*(Z_n - Z) + (p_n - p)^* Z.$$

This theorem is clock-free. The index n labels directed refinement, not time. Recovered middle identity and Smriti-tail control are required before the norm comparisons are lawful.

35.10 10. Non-circularity rule

The theorem becomes circular if one defines Z by taking a square root of the already desired nonnegative form.

A lawful domain specialization must instead construct:

```
Z                from declared native event amplitudes;
p_partial        from a declared boundary-recognition event;
P_partial and Q_partial from that event state;
T                from cut-memory action/minimal observer rank;
```

before evaluating the signed covariance.

The identities

$$S = Z^*Z, \quad \ell = p_{\partial}^*Z$$

must then be verified coefficientwise or by a recognition-complete limit argument. They may not be installed as definitions of Z after the desired form is known.

35.11 11. Exact generalized certificate v0.1

The exact packet uses rational arithmetic and verifies:

```
one event lift generates both source and boundary;
source - boundary square equals the cut-memory Gram;
least-energy decoder burden equals 1/6;
cut-memory rank is five and boundary-line blindness is absent;
critical finite burdens n/(n+1) approach one while every packet remains strict;
correct rank-five observer has zero action;
wrong rank-five orientation has action 81/100;
an independently perturbed boundary row breaks the covariance identity;
first moments, second moments and cut covariance converge under one orientation.
```

The frozen status is

```
PASS_CUT_COVARIANCE_EVENT_REALIZATION_V0_1
```

35.12 12. Completed-Weil specialization gate

The next domain theorem is no longer an abstract decoder search. It is the following native event-realization problem.

Targeted Theorem (Completed-Weil Cut-Covariance Realization)

Construct, directly from declared prime, Gamma, completion-boundary and compatibility cut events,

$$Z_{\text{Weil},-} : \mathcal{H}_{\text{odd}} \longrightarrow \mathcal{K}_{\text{Weil},\text{evt}}$$

and a unit boundary event state p_{∂} such that

$$S_{0,-}^{\text{full}} = Z_{\text{Weil},-}^* Z_{\text{Weil},-},$$

$$L_{\partial} = p_{\partial}^* Z_{\text{Weil},-},$$

and

$$\ker(Q_{\partial} Z_{\text{Weil},-}) = \{0\}$$

on the declared odd native core, or else derive the exact target-faithful finite memory observer and certify its remaining threshold.

Then

$$S_{0,-}^{\text{full}} - L_{\partial}^* L_{\partial} = (Q_{\partial} Z_{\text{Weil},-})^* (Q_{\partial} Z_{\text{Weil},-}) \geq 0.$$

The classical completed explicit formula may recognize this cut-generated form only after the native identities are proved.

35.13 13. Claim boundary

cut-covariance first/second moment identity	PROVED
coordinate-free Cut-Square identity	PROVED
canonical least-energy decoder	PROVED
beta_cut <= 1 from event-range projection	PROVED
uniform reserve implication	PROVED
boundary-line no-blindness strictness criterion	PROVED
target-faithful observed covariance	PROVED
recognition-complete first/second moment transfer	PROVED
exact generalized certificate	WRITTEN
completed-Weil native event lift	NEXT DOMAIN GATE
completed-Weil boundary event identification	NEXT DOMAIN GATE
completed-Weil boundary-line no-blindness	NEXT DOMAIN GATE
RH	NOT CLAIMED

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Chapter 36

Native Completed-Weil Cut-Event Co-Generation Gate

36.1 1. Purpose

The generalized cut-covariance theorem is now passed. The first RH-specific step is therefore not another spectral computation. It is to construct the odd completed-Weil source and boundary on one oriented event carrier.

The source material fixes the odd carrier and the two terminal objects:

$$L_{\partial}f = \sqrt{2} \widehat{f}(i/2),$$

$$W_3^- = S_{0,-}^{\text{full}} - L_{\partial}^* L_{\partial},$$

and states that a complete prime–Gamma cut-event analysis satisfies

$$\Xi_0^* \Xi_0 = S_{0,-}^{\text{full}}.$$

These statements do not yet prove that the boundary functional is a first moment of the same event analysis. That orientation identity is the exact gate addressed here.

The lawful order is

```
prime / Gamma / diagonal-debt cut events
-> one coefficientwise event analysis Xi_N
-> one boundary decoder c_partial,N on that same orientation
-> recovered refinement identity
-> Recognition-Cauchy and Smriti-tail completion
-> continuum first/second-moment identity
-> cut covariance and odd positivity.
```

36.2 2. Finite oriented event packet

Let $\mathfrak{E}_N$ be a finite declared event family. For each event $e \in \mathfrak{E}_N$, let

$$\zeta_{N,e} : X_{3,\text{core}}^- \longrightarrow \mathbb{C}$$

be its oriented source channel, let $s_{N,e} > 0$ be its source energy, and let $a_{N,e} \in \mathbb{C}$ be its boundary amplitude.

Define

$$(\Xi_N f)_e = \sqrt{s_{N,e}} \zeta_{N,e}(f),$$

and

$$L_{\partial,N} f = \sum_{e \in \mathfrak{E}_N} \overline{a_{N,e}} \zeta_{N,e}(f).$$

The oriented decoder is

$$(c_{\partial,N})_e = \frac{a_{N,e}}{\sqrt{s_{N,e}}}.$$

Then

$$L_{\partial,N} f = \langle c_{\partial,N}, \Xi_N f \rangle,$$

and

$$\beta_N = \|c_{\partial,N}\|^2 = \sum_{e \in \mathfrak{E}_N} \frac{|a_{N,e}|^2}{s_{N,e}}.$$

This formula is valid only when the coefficients and source channels have one recovered event orientation. A list of positive numbers s_e and masses $|a_e|^2$ does not determine the boundary functional.

36.3 3. Constructive first/second-moment realization

Theorem 3.1 (Contractive decoder is equivalent to co-generation)

Suppose

$$\|c_{\partial,N}\| \leq 1.$$

Enlarge the event carrier by one orthogonal completion coordinate and put

$$\widehat{\Xi}_N f = (\Xi_N f, 0),$$

$$p_{\partial,N} = \left(c_{\partial,N}, \sqrt{1 - \|c_{\partial,N}\|^2} \right).$$

Then $\|p_{\partial,N}\| = 1$ and

$$S_N = \widehat{\Xi}_N^* \widehat{\Xi}_N,$$

$$L_{\partial,N} = p_{\partial,N}^* \widehat{\Xi}_N,$$

so that

$$S_N - L_{\partial,N}^* L_{\partial,N} = \widehat{\Xi}_N^* (I - p_{\partial,N} p_{\partial,N}^*) \widehat{\Xi}_N \geq 0.$$

Proof

The added coordinate does not change the source Gram. The first-moment identity follows because its event-lift coordinate is zero. The norm of $p_{\partial,N}$ is one by construction. The cut-covariance identity is then Theorem 3.1 of the universal cut-covariance stage.

This theorem makes the missing object explicit. A successful direct cut burden is not merely a number below one; it constructs a unit boundary-recognition state after the same-event decoder has been proved.

36.4 4. Exact shifted-to-unshifted identity

For source energies $s_e > 0$, boundary masses $m_e = |a_e|^2$, and $\eta > 0$, put

$$\beta_{0,N} = \sum_e \frac{m_e}{s_e}, \quad \beta_{\eta,N} = \sum_e \frac{m_e}{s_e + \eta}.$$

Theorem 4.1 (Cut-event endpoint identity)

$$\beta_{0,N} - \beta_{\eta,N} = \sum_e \frac{m_e \eta}{s_e(s_e + \eta)}.$$

This is the exact finite event identity used by the earlier endpoint refinement. It is native only after the events, energies and boundary amplitudes are carried on the same recovered orientation.

36.5 5. Coherent refinement and orientation memory

Suppose one parent event with channel ζ , source energy s , and boundary amplitude a is split into children with the same underlying channel, energies $s_j > 0$, and amplitudes a_j , where

$$\sum_j s_j = s, \quad \sum_j a_j = a.$$

Theorem 5.1 (Refinement burden monotonicity)

$$\boxed{\sum_j \frac{|a_j|^2}{s_j} \geq \frac{|a|^2}{s}}.$$

Equality holds exactly when

$$\boxed{\frac{a_j}{s_j} \text{ is constant across the children.}}$$

Proof

Weighted Cauchy–Schwarz gives

$$\left| \sum_j a_j \right|^2 \leq \left(\sum_j s_j \right) \left(\sum_j \frac{|a_j|^2}{s_j} \right).$$

Substitute the parent sums. Equality is the equality condition for weighted Cauchy–Schwarz.

Thus a lawful refinement carries more than scalar masses. It carries the relative phase and orientation of the child boundary amplitudes. Two refinements can have identical source energies and identical masses but different boundary functionals.

This is the Recognition-Seam meaning of cut memory at the decoder level.

36.6 6. Recognition-complete continuum gate

Let Ξ_N be transported to one recovered event carrier and let $c_{\partial,N}$ use one fixed orientation. Assume

$$\Xi_N \longrightarrow \Xi_0$$

in operator norm after predeclared corrections and vanishing Smriti tails, and

$$c_{\partial,N} \longrightarrow c_{\partial}$$

in the event carrier. Assume also

$$L_{\partial,N} \longrightarrow L_{\partial}$$

in source-dual norm.

Theorem 6.1 (Recognition-complete event co-generation)

If

$$L_{\partial,N} = c_{\partial,N}^* \Xi_N$$

for every refinement, then

$$\boxed{L_{\partial} = c_{\partial}^* \Xi_0.}$$

Moreover,

$$\|c_\partial\|^2 = \lim_N \|c_{\partial,N}\|^2$$

under norm convergence. Hence a uniform or limiting burden bound below one passes to the continuum cut-covariance realization.

The recovered-middle and Smriti hypotheses are essential. A sequence of coordinate vectors whose labels rotate or whose event channels change between refinements does not satisfy this theorem merely because the scalar burdens converge.

36.7 7. Robust approximate orientation theorem

Exact coefficientwise orientation may be approached through an outward residual. Let c_N be a candidate decoder on the actual event analysis Ξ_0 , and put

$$r_N(f) = L_\partial f - \langle c_N, \Xi_0 f \rangle.$$

Assume

$$|r_N(f)| \leq \delta_N \|\Xi_0 f\| \quad (f \in X_{3,\text{core}}^-).$$

Theorem 7.1 (Robust decoder orientation)

If

$$\|c_N\|^2 \leq u_N,$$

then the actual boundary burden satisfies

$$\sqrt{\beta_\partial^{\text{cut}}} \leq \sqrt{u_N} + \delta_N,$$

and therefore

$$\beta_\partial^{\text{cut}} \leq (\sqrt{u_N} + \delta_N)^2.$$

In particular,

$$\delta_N < 1 - \sqrt{u_N}$$

is sufficient for strict odd positivity.

Proof

For every source state,

$$|L_{\partial}f| \leq |\langle c_N, \Xi_0 f \rangle| + |r_N(f)| \leq (\|c_N\| + \delta_N) \|\Xi_0 f\|.$$

Take the supremum over nonzero event images.

This theorem converts event orientation into a quantitative target. It does not require exact equality before useful progress can be certified.

36.8 8. Direct unshifted T21 scalar ledger

The imported source packet records

first-cell upper	0.00007670975092815261
active-band upper	0.2976678523563374
outside upper	0.5313410580584792
reported beta upper	0.8290856201657449
reported reserve	0.1709143798342551
active cells	2400
prime-power events	78734

The exact decimal sum of the three displayed components is

$$0.82908562016574475261,$$

so the reported value is outward and strictly below one.

If this scalar bound is attached to a decoder on the actual event orientation, then the robust orientation theorem leaves the residual budget

$$1 - \sqrt{0.8290856201657449} = 0.0894586115031646185 \dots$$

in the source norm.

For example, an outward orientation residual of 0.08 would give

$$(\sqrt{0.8290856201657449} + 0.08)^2 = 0.98117224232523856 \dots < 1.$$

This is a large and useful reserve. But the scalar ledger alone does not prove that its cell masses form the actual decoder of L_{∂} . The next stage must build or enclose that same-event orientation residual.

36.9 9. RH specialization contract

The actual odd completed-Weil gate is now precise.

Already declared

```
native odd carrier X3^-;  
boundary formula L_partial f = sqrt(2) f-hat(i/2);  
positive odd source S_(0,-)^full;  
source analysis statement Xi_0^* Xi_0 = S_(0,-)^full;  
direct unshifted scalar envelope from T21.
```

Still required in RKF

1. coefficientwise prime/Gamma/diagonal-debt event map Xi_Weil,N;
2. boundary coefficients on those same oriented events;
3. recovered identity between successive event refinements;
4. Recognition-Cauchy and Smriti-tail convergence to Xi_0 and L_partial;
5. exact or outward orientation residual delta_N;
6. actual boundary-line no-blindness for Q_partial Xi_0.

Only after these gates close may one write

$$S_{0,-}^{\text{full}} - L_{\partial}^* L_{\partial} = Z_{\text{Weil},-}^* Q_{\partial} Z_{\text{Weil},-} \geq 0$$

as a theorem about the completed-Weil object.

36.10 10. Exact executable calibrations

The Stage-3A packet verifies:

```
finite first/second-moment co-generation;  
constructive unit boundary-state enlargement;  
cut covariance and no-blindness;  
exact shifted endpoint identity;  
coherent refinement equality;  
same-mass opposite-orientation negative control;  
nonproportional refinement burden increase;  
T21 decimal arithmetic and orientation-residual budget;  
fail-closed RH contract.
```

The exact code is

```
proof_lab/native_weil_cut_event_cogeneration.py  
proof_lab/test_native_weil_cut_event_cogeneration.py
```

36.11 11. Claim boundary

finite oriented event theorem	PROVED
contractive decoder <=> unit-state co-generation	PROVED CONSTRUCTIVELY
shifted endpoint difference identity	PROVED
refinement burden monotonicity	PROVED
Recognition-complete co-generation implication	PROVED
robust orientation residual theorem	PROVED
T21 scalar arithmetic	AUDITED

actual prime/Gamma/debt event orientation	NEXT
actual orientation residual	NEXT
actual completed-Weil cut covariance	NOT YET PROVED HERE
RH	NOT CLAIMED

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Chapter 37

Native Common-Chart Source-Domination and Decoder-Transfer Theorem

37.1 1. Purpose

Stage 3A showed that source-cell energies and boundary masses do not, by themselves, determine a boundary functional. Relative phase and event lineage matter if one attempts to reconstruct a decoder coefficient by coefficient.

That observation is correct, but it does not force a 78,734-event phase reconstruction.

There is a stronger native route:

```
actual source and actual boundary in one cut chart
-> Loewner lower source symbol
-> boundary/source relative bound
-> canonical decoder on the complete event range
-> cut covariance.
```

Once the boundary density is the actual density in the same chart as the source lower symbol, its phase is already part of the density. A scalar floor may then be used outwardly without forgetting orientation.

The actual unresolved issue is therefore not every event phase. It is the single common-chart identity.

37.2 2. Native common-chart datum

Let

$$\Xi : \mathcal{H}_{\text{src}} \longrightarrow \mathcal{Y}_{\Sigma}$$

be a complete native event analysis and put

$$S = \Xi^* \Xi.$$

Let

$$J : \mathcal{H}_{\text{src}} \longrightarrow L^2(X, \mu; \mathbb{C}^d)$$

be a declared native cut chart. Let $A(x)$ be a measurable positive semidefinite matrix field and assume the actual source dominates its charted energy:

$$\|\Xi f\|_{\Sigma}^2 \geq \int_X \langle Jf(x), A(x)Jf(x) \rangle d\mu(x). \quad (2.1)$$

Let $b : X \rightarrow \mathbb{C}^d$ be the actual boundary density in that same chart:

$$L(f) = \int_X \langle b(x), Jf(x) \rangle d\mu(x). \quad (2.2)$$

Equation (2.2) is the orientation statement. It may not be replaced by a list of numbers $\|b(x)\|^2$ detached from the chart map J .

37.3 3. Matrix-symbol decoder theorem

Let $A(x)^\dagger$ be the Moore–Penrose inverse. Assume

$$b(x) \in \text{Ran } A(x)^{1/2} \quad \text{for almost every } x$$

and

$$\beta_A = \int_X \langle b(x), A(x)^\dagger b(x) \rangle d\mu(x) < \infty. \quad (3.1)$$

Theorem 3.1 (Common-chart source domination)

For every $f \in \mathcal{H}_{\text{src}}$,

$$|L(f)|^2 \leq \beta_A \|\Xi f\|_{\Sigma}^2. \quad (3.2)$$

Proof

Put

$$c_A(x) = A(x)^{\dagger/2} b(x).$$

The range hypothesis gives

$$b(x) = A(x)^{1/2} c_A(x).$$

Therefore

$$L(f) = \int_X \langle c_A(x), A(x)^{1/2} Jf(x) \rangle d\mu(x).$$

Cauchy–Schwarz gives

$$|L(f)|^2 \leq \left(\int_X \|c_A(x)\|^2 d\mu(x) \right) \left(\int_X \|A(x)^{1/2} Jf(x)\|^2 d\mu(x) \right).$$

The first factor is β_A ; the second is bounded by $\|\Xi f\|_\Sigma^2$ through (2.1).

37.4 4. Canonical decoder on the complete event range

The preceding theorem proves more than an inequality.

Theorem 4.1 (Decoder transfer by recognition completion)

Under Theorem 3.1, define on $\text{Ran } \Xi$

$$\tilde{L}(\Xi f) = L(f).$$

Then $\tilde{L}$ is well-defined and bounded, and there exists a unique

$$c_\partial \in \overline{\text{Ran } \Xi}$$

such that

$$\boxed{L(f) = \langle c_\partial, \Xi f \rangle_\Sigma \quad (f \in \mathcal{H}_{\text{src}}).} \quad (4.1)$$

Moreover,

$$\boxed{\|c_\partial\|_\Sigma^2 = \beta_{\text{cut}} \leq \beta_A.} \quad (4.2)$$

Proof

If $\Xi f = 0$, (3.2) gives $L(f) = 0$, so $\tilde{L}$ is well-defined. Again by (3.2),

$$|\tilde{L}(y)| \leq \sqrt{\beta_A} \|y\|_\Sigma \quad (y \in \text{Ran } \Xi).$$

It extends continuously to $\overline{\text{Ran } \Xi}$. The Riesz representation theorem gives a unique vector c_∂ there. Its squared norm is the sharp source-relative burden, hence is at most β_A .

This is not a decoder fitted from the terminal normal equation. The bound is proved first from the common cut chart. The decoder is then forced by completion.

37.5 5. Cut covariance and relative reserve

From (4.1),

$$L^*L \leq \beta_A S.$$

Therefore

$$\boxed{S - L^*L \geq (1 - \beta_A)S.} \quad (5.1)$$

If $\beta_A < 1$, the signed cut form is nonnegative. If, in addition, $\ker \Xi = \{0\}$, then it is strictly positive on every nonzero source state.

The unit-boundary-state construction from Stage 3A then gives an event carrier on which

$$S - L^*L = \widehat{\Xi}^*(I - p_{\partial}p_{\partial}^*)\widehat{\Xi}.$$

Thus common-chart domination generates the completed cut covariance without a coefficient-wise decoder reconstruction.

37.6 6. Scalar-floor corollary

Assume

$$A(x) \geq s(x)I_d, \quad s(x) > 0.$$

Then

$$A(x)^{\dagger} \leq s(x)^{-1}I_d$$

on the relevant range, so

$$\boxed{\beta_A \leq \int_X \frac{\|b(x)\|^2}{s(x)} d\mu(x).} \quad (6.1)$$

This is why a scalar lower branch is sufficient. It may be less sharp than the full matrix symbol, but it is orientation-safe once (2.2) has been proved.

The distinction is exact:

before common-chart identity:

masses alone do not determine the boundary;

after common-chart identity:

norm masses give a lawful outward burden independent of coordinate phase.

37.7 7. Cellwise cut theorem

Let X be partitioned into cells I_j . Suppose

$$A(x) \geq s_j^- I \quad (x \in I_j)$$

and

$$\int_{I_j} \|b(x)\|^2 d\mu(x) \leq m_j^+.$$

Then

$$\boxed{\beta_{I_j} \leq \frac{m_j^+}{s_j^-}}$$

and

$$\boxed{\beta_A \leq \sum_j \frac{m_j^+}{s_j^-}}. \tag{7.1}$$

No phase reconstruction is needed in (7.1), because m_j^+ bounds the norm of the actual common-chart density.

A negative control is mandatory: if the source chart observes one coordinate while the boundary reads a source-blind coordinate, no finite mass/floor table proves a decoder.

37.8 8. First-cut cancellation

On the first symmetric cell $|\xi| \leq h$, suppose

$$s(\xi) \geq c\xi^2, \quad \|b(\xi)\| \leq M_1|\xi|.$$

Then the apparent zero denominator is removed by the odd boundary zero:

$$\boxed{\int_{-h}^h \frac{\|b(\xi)\|^2}{s(\xi)} d\xi \leq \frac{2hM_1^2}{c}}. \tag{8.1}$$

This is zero-as-cut in analytic form: the source opens quadratically, the odd boundary opens linearly, and their quotient retains a finite first-cut memory.

37.9 9. Completed-Weil source pins

The transferred source manuscript records:

```
Xi_0~* Xi_0 = S_(0,-)^full;
Omega_0(xi) has a directed scalar lower branch s_0(xi);
the full boundary Riesz state gives
|b_partial(xi)| <= min(|xi|/9, 1/(3 sqrt(2)));
the direct T21 cut ledger gives
beta_envelope <= 0.8290856201657449.
```

The corresponding reserve is

$$1 - 0.8290856201657449 = 0.1709143798342551.$$

If the following exact common-chart identity is proved,

$$L_{\partial}(f) = \int_{\mathbb{R}} \langle b_{\partial}(\xi), J_{\text{Weil}} f(\xi) \rangle d\xi, \quad (9.1)$$

and the actual source satisfies

$$\|\Xi_0 f\|_{\Sigma}^2 \geq \int_{\mathbb{R}} \langle J_{\text{Weil}} f(\xi), \Omega_0(\xi) J_{\text{Weil}} f(\xi) \rangle d\xi, \quad (9.2)$$

then Theorems 3.1–5.1 give

$$\beta_{\partial}^{\text{cut}} \leq 0.8290856201657449 < 1$$

and

$$S_{0,-}^{\text{full}} - L_{\partial}^* L_{\partial} \geq 0.1709143798342551 S_{0,-}^{\text{full}}.$$

No 78,734-event phase reconstruction would then be required.

37.10 10. What remains after Stage 3B

Stage 3A described the missing object as a coefficientwise orientation packet. Stage 3B sharpens that target.

The minimal remaining theorem is:

Native two-sheet boundary-pairing theorem

Construct the actual cut chart

$$J_{\text{Weil}} : X_3^- \rightarrow L^2(\mathbb{R}; \mathbb{C}^2)$$

from the already declared prime–Gamma two-sheet analysis and prove (9.1)–(9.2) on the native generator core. Then prove Recognition-Cauchy/Smriti-tail extension to the completed carrier.

The chart must be derived from the source construction. It may not be chosen post hoc to make (9.1) true.

37.11 11. Exact calibration

The Stage 3B rational calibration uses

$$\Xi = \text{diag}(3, 4), \quad S = \text{diag}(9, 16),$$

with common-chart lower symbol

$$A = \text{diag}(4, 9)$$

and positive source remainder

$$R = \text{diag}(5, 7).$$

For boundary density $b = (1, 2)$,

$$\beta_A = b^* A^{-1} b = \frac{25}{36},$$

while the exact sharp burden is

$$\beta_{\text{cut}} = b^* S^{-1} b = \frac{13}{36}.$$

The canonical complete-event decoder is

$$c_{\partial} = \left(\frac{1}{3}, \frac{1}{2} \right),$$

with energy $13/36$. This verifies that source domination constructs a lawful, possibly nonsharp, decoder certificate.

37.12 12. Claim boundary

common-chart matrix-symbol domination theorem	PROVED
canonical decoder transfer to complete event range	PROVED
scalar-floor orientation-safe corollary	PROVED
cellwise cut burden theorem	PROVED
first-cut quadratic/linear cancellation	PROVED
mismatched-chart negative control	PROVED
T21 arithmetic audit	PASSED
actual J_Weil and b_partial common-chart identity	OPEN / NEXT
Recognition-complete chart transport	OPEN / NEXT
actual completed-Weil decoder and covariance	OPEN UNTIL ABOVE
RH	NOT CLAIMED

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Chapter 38

Native Metric-Graph Boundary Pairing and Recognition-Quotient Adapter

38.1 1. Purpose

Stage 3B showed that an actual boundary density in the same chart as an actual source lower symbol generates a canonical decoder. The remaining question was how the completed-Weil boundary acquires such a native chart without fitting a decoder from the terminal sign equation.

The native metric already answers the first half of that question.

In the logarithmic completed carrier, the metric has the graph form

$$M_X = I + C^*C,$$

and the full odd boundary Riesz state is

$$r_\partial = M_X^{-1}q_b.$$

The exact boundary event is therefore not an independently chosen frequency vector. It is the graph state

$$p_X = (r_\partial, Cr_\partial).$$

The second half of the problem is a source-adapter theorem. If the complete prime-Gamma-diagonal-debt event analysis factors through the graph chart, the source decoder does not need to reproduce p_X literally. It needs to recover only its recognition class modulo the graph-null seam.

The corrected order is

```
native metric graph
-> exact boundary graph state
-> source-event adapter
-> boundary class modulo graph-null memory
-> minimum-energy source decoder
-> cut covariance.
```

This removes the overstrong demand to reconstruct 78,734 individual event phases one by one.

38.2 2. Native graph carrier

Let $\mathcal{H}$ and $\mathcal{K}$ be Hilbert spaces and let

$$C : \text{Dom } C \subseteq \mathcal{H} \rightarrow \mathcal{K}$$

be closed and densely defined. Equip $\text{Dom } C$ with the graph inner product

$$\langle f, h \rangle_M = \langle f, h \rangle_{\mathcal{H}} + \langle Cf, Ch \rangle_{\mathcal{K}}.$$

The corresponding positive metric is the form realization of

$$M = I + C^*C.$$

Define the graph analysis

$$J_C f = (f, Cf) \in \mathcal{G} := \mathcal{H} \oplus \mathcal{K}.$$

Then J_C is an isometry from the graph completion into $\mathcal{G}$.

Let q be a continuous boundary seed for the graph norm and let

$$r = M^{-1}q.$$

Define the graph boundary state

$$p_C = J_C r = (r, Cr).$$

38.3 3. Metric-graph boundary theorem

Theorem 3.1 (Boundary first moment from the native graph)

For every graph-core state f ,

$$\langle q, f \rangle_{\mathcal{H}} = \langle p_C, J_C f \rangle_{\mathcal{G}}. \tag{3.1}$$

Equivalently,

$$J_C^* p_C = q \tag{3.2}$$

in the graph-form sense. Moreover,

$$\|p_C\|_{\mathcal{G}}^2 = \langle q, M^{-1}q \rangle_{\mathcal{H}}. \tag{3.3}$$

Proof

Since $Mr = q$,

$$\begin{aligned}\langle p_C, J_C f \rangle_{\mathcal{G}} &= \langle r, f \rangle_{\mathcal{H}} + \langle Cr, Cf \rangle_{\mathcal{K}} \\ &= \langle (I + C^*C)r, f \rangle_{\mathcal{H}} \\ &= \langle q, f \rangle_{\mathcal{H}}.\end{aligned}$$

Taking $f = r$ gives (3.3).

The boundary is therefore co-generated by the native metric before any source inverse or terminal decoder is introduced.

38.4 4. Reflection covariance

Let $R_{\mathcal{H}}$ and $R_{\mathcal{K}}$ be unitary self-adjoint involutions and assume

$$CR_{\mathcal{H}} = R_{\mathcal{K}}C.$$

Let

$$R_{\mathcal{G}} = R_{\mathcal{H}} \oplus R_{\mathcal{K}}.$$

Theorem 4.1 (Odd boundary graph state)

If

$$R_{\mathcal{H}}q = -q,$$

then

$$R_{\mathcal{H}}r = -r, \quad R_{\mathcal{G}}p_C = -p_C.$$

Proof

The intertwining relation implies that $M = I + C^*C$ commutes with $R_{\mathcal{H}}$. Hence its inverse on the form range also commutes with the reflection. Therefore

$$R_{\mathcal{H}}r = M^{-1}R_{\mathcal{H}}q = -r.$$

The graph-state statement follows from the covariance of C .

This is the graph-level origin of the odd boundary zero at the cut.

38.5 5. Source-event adapter

Let $\mathcal{Y}_\Sigma$ be a complete source-event carrier and let

$$D_\Sigma : \mathcal{G} \rightarrow \mathcal{Y}_\Sigma$$

be a bounded native source adapter. Define

$$\boxed{\Xi = D_\Sigma J_C.}$$

The source Gram is

$$S = \Xi^* \Xi.$$

The boundary functional is

$$L(f) = \langle p_C, J_C f \rangle_{\mathcal{G}}.$$

A vector $c \in \mathcal{Y}_\Sigma$ is a source decoder when

$$L(f) = \langle c, \Xi f \rangle_\Sigma \quad (f \in \text{Dom } C).$$

38.6 6. Recognition-quotient decoder theorem

The graph analysis can contain event directions that are invisible after synthesis. Define the graph-null seam

$$\boxed{\mathcal{N}_J = \ker J_C^*}.$$

Two graph-event states are recognition-equivalent when their difference lies in $\mathcal{N}_J$.

Theorem 6.1 (Decoder class, not literal event equality)

A vector $c \in \mathcal{Y}_\Sigma$ decodes L through $\Xi = D_\Sigma J_C$ if and only if

$$\boxed{p_C - D_\Sigma^* c \in \ker J_C^*}. \tag{6.1}$$

Equivalently, in the recognition quotient

$$\mathcal{Q}_J = \mathcal{G} / \ker J_C^*,$$

one has

$$\boxed{[p_C] = [D_\Sigma^* c]}. \tag{6.2}$$

Proof

For every f ,

$$\begin{aligned} L(f) - \langle c, \Xi f \rangle &= \langle p_C, J_C f \rangle - \langle D_\Sigma^* c, J_C f \rangle \\ &= \langle p_C - D_\Sigma^* c, J_C f \rangle. \end{aligned}$$

This vanishes for every f exactly when the difference is orthogonal to $\text{Ran } J_C$, equivalently when it lies in $\ker J_C^*$.

This theorem is the RSC correction to an overstrong event-orientation demand. The source adapter may differ from the metric boundary state in a graph-null memory direction without changing the physical boundary functional.

38.7 7. Minimal quotient decoder

Define the sharp source burden

$$\beta_\Sigma = \inf \{ \|c\|_\Sigma^2 : [p_C] = [D_\Sigma^* c] \text{ in } \mathcal{Q}_J \}. \quad (7.1)$$

Theorem 7.1 (Minimum-energy recognition representative)

Assume the induced map

$$\overline{D_\Sigma^*} : \mathcal{Y}_\Sigma \longrightarrow \mathcal{Q}_J$$

has closed range and contains $[p_C]$. Then the infimum in (7.1) is attained by a unique decoder orthogonal to the kernel of the induced map. It is the minimum-norm source representative of the boundary recognition class.

Moreover,

$$\beta_\Sigma = \sup_{\Xi f \neq 0} \frac{|L(f)|^2}{\|\Xi f\|_\Sigma^2}. \quad (7.2)$$

Proof

Closed range gives the orthogonal decomposition of the affine solution set into one minimum-norm representative plus the kernel of the induced map. The Riesz norm of the functional

$$\Xi f \mapsto L(f)$$

on $\overline{\text{Ran } \Xi}$ is the displayed supremum and equals the norm of its unique minimum-energy representing vector.

If $\beta_\Sigma < 1$, then

$$S - L^* L \geq (1 - \beta_\Sigma) S. \quad (7.3)$$

Thus the source sign depends on recovery of one recognition class, not on literal equality of every ambient event coordinate.

38.8 8. Completed-Weil metric graph

The native logarithmic coordinate used in the boundary-threshold theorem fixes

$$m(x) = e^{-3|x|/2},$$

$$q_b(x) = \sqrt{2} m(x) \sinh(x/2),$$

and

$$(Cf)(\xi) = (2\pi)^{-1/2} \sqrt{1 + |G(\xi)|} \widehat{mf}(\xi).$$

Hence

$$M_X = I + C^*C, \quad r_\partial = M_X^{-1} q_b.$$

Define

$$\boxed{J_X f = (f, Cf), \quad p_X = (r_\partial, Cr_\partial).}$$

Theorem 3.1 gives the exact native boundary pairing

$$\boxed{L_\partial(f) = \langle p_X, J_X f \rangle.} \tag{8.1}$$

This is the actual first-moment orientation. It is generated by the native metric, not selected after the completed-Weil sign is inspected.

Writing

$$w_\partial(x) = m(x) r_\partial(x),$$

the frequency component of the graph boundary state is

$$\boxed{Cr_\partial = (2\pi)^{-1/2} \sqrt{1 + |G|} \widehat{w}_\partial.} \tag{8.2}$$

Thus the boundary envelope used by the cut ledger is a component of the exact metric graph state. Its phase is not an independently guessed list of cell coefficients.

38.9 9. Exact seam-jet audit

The pinned boundary-threshold calculation gives

$$\langle q_1, q_b \rangle = \frac{96\sqrt{2}}{1225} > 0.1108,$$

$$\|Cq_1\|^2 < \frac{6.4}{54} + 0.03 < 0.14852,$$

and

$$\|Cq_b\|^2 < \frac{6.4}{105} + 0.02 < 0.080953.$$

The graph-metric correction therefore obeys

$$\|Cq_1\| \|Cq_b\| < 0.10966,$$

so

$$\boxed{\operatorname{Re} j_{\partial} > 0.0011.} \tag{9.1}$$

The full odd boundary graph state therefore remembers the cut through a nonzero exact seam jet after metric completion.

The sharp boundary moment gives

$$|\widehat{w}_{\partial}(\xi)| \leq \min \left\{ \frac{|\xi|}{9}, \frac{1}{3\sqrt{2}} \right\}.$$

With the quadratic source opening

$$s(\xi) \geq \frac{\gamma_0}{2} \xi^2,$$

the first-cut ratio is finite. The older coarse moment $1/3$ gives a bound nine times larger than the sharp moment $1/9$, but both are lawful outward certificates.

38.10 10. The actual remaining source theorem

The source manuscript pins

$$\Xi_0^* \Xi_0 = S_{0,-}^{\text{full}}$$

and states that Ξ_0 is assembled from positive prime and Gamma mismatch factorizations together with the exact diagonal-debt defect.

What is not yet displayed is the source adapter

$$\boxed{D_{\Sigma} : \mathcal{G}_X \rightarrow \mathcal{Y}_{\Sigma} \quad \text{such that} \quad \Xi_0 = D_{\Sigma} J_X.} \tag{10.1}$$

After (10.1), the only boundary-orientation condition required is

$$\boxed{p_X - D_{\Sigma}^* c_{\partial} \in \ker J_X^*.} \tag{10.2}$$

Literal equality

$$p_X = D_{\Sigma}^* c_{\partial}$$

is sufficient but not necessary.

This is the next native theorem: construct D_{Σ} coefficientwise from the already declared prime mismatch, Gamma mismatch and diagonal-debt channels, then prove (10.2) with an outward decoder energy below one.

If the transferred direct-unshifted envelope

$$\beta_{\partial}^{\text{cut}} \leq 0.8290856201657449$$

is attached to that actual quotient decoder, then

$$S_{0,-}^{\text{full}} - L_{\partial}^* L_{\partial} \geq 0.1709143798342551 S_{0,-}^{\text{full}}.$$

38.11 11. Exact calibrations

The Stage 3C exact packet verifies:

native metric graph pairing energy	7/20
source-adapted minimum decoder energy	7/80
odd graph boundary energy	1/40
recognition-quotient minimum decoder burden	2/3
literal event mismatch in ker J*	accepted
non-null event mismatch	rejected
full boundary seam-jet lower	> 0.0011

It also includes the negative controls:

same ambient event norm with a different quotient class;
false Riesz vector not satisfying $M r = q$;
literal equality incorrectly substituted for quotient equality.

38.12 12. Claim boundary

native metric-graph first-moment identity	PROVED
boundary graph-state energy identity	PROVED
odd reflection transport to graph state	PROVED
recognition-quotient source decoder theorem	PROVED
minimum-energy quotient representative	PROVED
full boundary seam-jet arithmetic	PINNED / AUDITED
actual $\text{Xi}_0 = D_{\text{Sigma}} J_X$ factorization	OPEN / NEXT
actual quotient decoder class for completed-Weil source	OPEN / NEXT
Recognition-complete source-adapter transport	OPEN / NEXT
actual completed-Weil cut covariance	OPEN
RH	NOT CLAIMED

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Chapter 39

Native Prime–Gamma–Debt Source Adapter and Graph-Null Seam Theorem

39.1 1. Purpose

The Stage 3C metric-graph theorem constructs the completed-Weil boundary state from the native metric,

$$M_X = I + C_X^* C_X, \quad r_{\partial} = M_X^{-1} q_b,$$

$$J_X f = (f, C_X f), \quad p_X = J_X r_{\partial}.$$

The remaining source-adapter problem initially appeared to require a new map

$$D_{\Sigma} : \mathcal{G}_X \longrightarrow \mathcal{Y}_{\Sigma}$$

assembled event by event from every prime and Gamma coordinate.

That demand is too strong. The complete native source analysis has already been constructed upstream from the positive prime mismatch, positive Gamma mismatch and exact diagonal debt:

$$\mathcal{A}_{\text{mis}} : X_3^- \longrightarrow \mathcal{Y}_{\text{mis}},$$

$$M_{\text{mis}} = \mathcal{A}_{\text{mis}}^* \mathcal{A}_{\text{mis}}, \quad d_0 = m_{\Gamma} + m_{\text{p}},$$

$$D_{\text{comp}} = (I - d_0 M_{\text{mis}}^{-1})^{1/2},$$

$$\boxed{\Xi_0 = \mathcal{A}_{\text{mis}} D_{\text{comp}},}$$

$$\boxed{\Xi_0^* \Xi_0 = M_{\text{mis}} - d_0 I = S_{0,-}^{\text{full}}.}$$

Once both J_X and Ξ_0 are native, the source adapter is forced by the graph synthesis. This stage constructs that adapter, classifies every other adapter with the same pullback, and separates the now-closed adapter problem from the still-open two-sheet decoder bound.

The native order is

```

prime mismatch + Gamma mismatch
-> explicit diagonal debt
-> compatibility defect D_comp
-> complete source analysis Xi_0
-> native metric graph J_X
-> canonical source adapter D_Sigma^0=Xi_0 J_X^*
-> graph-null adapter freedom
-> two-sheet source domination
-> canonical boundary decoder.

```

No classical explicit formula, Birman–Schwinger inverse, fitted Gram factor or post-hoc event rotation defines any object in this theorem.

39.2 2. Prime–Gamma mismatch and debt source

The positive mismatch carrier consists of:

```

prime events:
  weights Lambda(n)/n^2;
  two-sheet mismatch vectors
  (1,-exp(-i xi log n));

Gamma events:
  weights (1/2) exp(-(k+1/4)t) dt;
  oscillator mismatches
  1-exp(i xi t/2).

```

The upstream factorization proves

$$m_p I_2 - \mathcal{Q}(\xi) \geq 0,$$

and

$$G(\xi) + m_\Gamma = \frac{1}{2} \sum_{k \geq 0} \int_0^\infty e^{-(k+1/4)t} |1 - e^{i\xi t/2}|^2 dt \geq 0.$$

Their direct-integral analysis is $\mathcal{A}_{\text{mis}}$. The exact source accounting is

$$S_{0,-}^{\text{full}} = \mathcal{A}_{\text{mis}}^* \mathcal{A}_{\text{mis}} - d_0 I.$$

The compatibility defect does not append a new arbitrary event. It removes the declared debt inside the same mismatch range:

$$\Xi_0 = \mathcal{A}_{\text{mis}} (I - d_0 M_{\text{mis}}^{-1})^{1/2}.$$

Consequently,

$$\text{Ran } \Xi_0 \subseteq \overline{\text{Ran } \mathcal{A}_{\text{mis}}},$$

so the source event orientation remains prime–Gamma native.

39.3 3. Native graph carrier

Let

$$J_X : X_3^- \longrightarrow \mathcal{G}_X$$

be the isometric native metric-graph analysis. Put

$$P_X = J_X J_X^*, \quad Q_X = I - P_X.$$

Then

$$J_X^* J_X = I, \quad \text{Ran } Q_X = \ker J_X^*.$$

The graph-null seam $\ker J_X^*$ consists of ambient event directions which vanish under native graph synthesis. Such directions are not deleted; they are retained as recognition-null memory.

39.4 4. Canonical source-adapter theorem

Theorem 4.1 (Canonical native source adapter)

Define

$$D_\Sigma^0 = \Xi_0 J_X^* : \mathcal{G}_X \longrightarrow \mathcal{Y}_\Sigma.$$

Then

$$D_\Sigma^0 J_X = \Xi_0, \tag{4.1}$$

$$D_\Sigma^0 Q_X = 0, \tag{4.2}$$

and

$$J_X^* (D_\Sigma^0)^* D_\Sigma^0 J_X = \Xi_0^* \Xi_0 = S_{0,-}^{\text{full}}. \tag{4.3}$$

Proof

Since $J_X^* J_X = I$,

$$D_\Sigma^0 J_X = \Xi_0 J_X^* J_X = \Xi_0.$$

Also $J_X^* Q_X = 0$, so

$$D_\Sigma^0 Q_X = \Xi_0 J_X^* Q_X = 0.$$

Finally, pull back the adapter Gram through J_X :

$$\begin{aligned} J_X^* (D_\Sigma^0)^* D_\Sigma^0 J_X &= (D_\Sigma^0 J_X)^* (D_\Sigma^0 J_X) \\ &= \Xi_0^* \Xi_0 \\ &= S_{0,-}^{\text{full}}. \end{aligned}$$

□

The actual source adapter is therefore no longer an open construction. It is the native source analysis followed by graph synthesis.

39.5 5. All adapters form one graph-null affine family

Theorem 5.1 (Graph-null adapter classification)

Let

$$D_\Sigma : \mathcal{G}_X \rightarrow \mathcal{Y}_\Sigma$$

be bounded. Then

$$D_\Sigma J_X = \Xi_0$$

if and only if there is a bounded map E such that

$$\boxed{D_\Sigma = D_\Sigma^0 + EQ_X.} \tag{5.1}$$

Proof

Every map of the form (5.1) satisfies

$$D_\Sigma J_X = D_\Sigma^0 J_X + EQ_X J_X = \Xi_0.$$

Conversely, if $D_\Sigma J_X = \Xi_0$, then

$$(D_\Sigma - D_\Sigma^0)P_X = (D_\Sigma - D_\Sigma^0)J_X J_X^* = 0.$$

Hence

$$D_\Sigma - D_\Sigma^0 = (D_\Sigma - D_\Sigma^0)Q_X,$$

which is (5.1) with $E = D_\Sigma - D_\Sigma^0$.

□

Thus adapter nonuniqueness lives entirely on the graph-null seam. It cannot change the physical source pullback.

39.6 6. Boundary recognition class is adapter invariant

The metric-graph boundary state is

$$p_X = J_X r_\partial,$$

and

$$L_\partial(f) = \langle p_X, J_X f \rangle_{\mathcal{G}_X}.$$

A source decoder $c_\partial \in \mathcal{Y}_\Sigma$ satisfies

$$\Xi_0^* c_\partial = r_\partial$$

exactly when

$$L_\partial(f) = \langle c_\partial, \Xi_0 f \rangle_\Sigma.$$

Theorem 6.1 (Adapter-independent quotient decoder)

Let

$$D_\Sigma^E = D_\Sigma^0 + EQ_X.$$

If

$$\Xi_0^* c_\partial = r_\partial,$$

then

$$\boxed{p_X - (D_\Sigma^E)^* c_\partial \in \ker J_X^*} \tag{6.1}$$

For the canonical adapter,

$$\boxed{(D_\Sigma^0)^* c_\partial = p_X} \tag{6.2}$$

Proof

For the canonical adapter,

$$(D_\Sigma^0)^* c_\partial = J_X \Xi_0^* c_\partial = J_X r_\partial = p_X.$$

For a general adapter,

$$(D_\Sigma^E)^* c_\partial = p_X + Q_X E^* c_\partial.$$

Therefore

$$p_X - (D_\Sigma^E)^* c_\partial = -Q_X E^* c_\partial \in \text{Ran } Q_X = \ker J_X^*.$$

□

This is the exact Recognition-Seam statement:

different ambient adapters represent one boundary object when they differ only by graph-null memory.

39.7 7. Canonical minimum decoder after a relative bound

Suppose a native estimate proves

$$|L_\partial(f)|^2 \leq \beta \|\Xi_0 f\|_\Sigma^2 \quad (f \in X_3^-)$$

for some finite β .

Then L_∂ vanishes on $\ker \Xi_0$ and defines a bounded functional on $\text{Ran } \Xi_0$. Recognition completion produces a unique minimum decoder

$$c_{\partial, \min} \in \overline{\text{Ran } \Xi_0}$$

such that

$$\Xi_0^* c_{\partial, \min} = r_\partial$$

and

$$\boxed{\|c_{\partial, \min}\|_\Sigma^2 = \beta_{\text{cut}} \leq \beta.} \quad (7.1)$$

Consequently,

$$\boxed{S_{0,-}^{\text{full}} - L_\partial^* L_\partial \geq (1 - \beta) S_{0,-}^{\text{full}}} \quad (7.2)$$

The decoder is therefore forced after the bound. It is not fitted and then declared contractive.

39.8 8. Event-coordinate covariance

Let

$$U : \mathcal{Y}_\Sigma \rightarrow \mathcal{Y}'_\Sigma$$

be unitary. Define

$$\Xi'_0 = U \Xi_0, \quad D'_\Sigma = U D_\Sigma, \quad c'_\partial = U c_\partial.$$

Then

$$(\Xi'_0)^* \Xi'_0 = \Xi_0^* \Xi_0,$$

$$(D'_\Sigma) J_X = \Xi'_0,$$

$$(\Xi'_0)^* c'_\partial = \Xi_0^* c_\partial,$$

and

$$\|c'_\partial\| = \|c_\partial\|.$$

Thus the actual event-coordinate chart may rotate, but the source Gram, boundary recognition class, burden and cut covariance are invariant.

This explains why Stage 3A's request for every individual event phase was unnecessarily rigid. The lawful invariant is the source adapter and quotient boundary class.

39.9 9. Robust two-sheet attachment theorem

The remaining actual issue is no longer the source adapter. It is the attachment of the direct cut envelope to the actual source and boundary in one two-sheet chart.

Let β_{env} be a charted boundary/source envelope. Assume:

$$\mathcal{E}_{\text{chart}}(f) \leq \frac{1}{1 - \varepsilon_S} \|\Xi_0 f\|_{\Sigma}^2, \quad 0 \leq \varepsilon_S < 1,$$

and

$$|L_{\partial}(f) - L_{\text{chart}}(f)| \leq \delta_{\partial} \|\Xi_0 f\|_{\Sigma}.$$

If

$$|L_{\text{chart}}(f)|^2 \leq \beta_{\text{env}} \mathcal{E}_{\text{chart}}(f),$$

then

$$\boxed{\sqrt{\beta_{\text{cut}}} \leq \sqrt{\frac{\beta_{\text{env}}}{1 - \varepsilon_S}} + \delta_{\partial}.} \tag{9.1}$$

Proof

For every f ,

$$\begin{aligned} |L_{\partial}(f)| &\leq |L_{\text{chart}}(f)| + |L_{\partial}(f) - L_{\text{chart}}(f)| \\ &\leq \sqrt{\beta_{\text{env}}} \mathcal{E}_{\text{chart}}(f)^{1/2} + \delta_{\partial} \|\Xi_0 f\|_{\Sigma} \\ &\leq \left(\sqrt{\frac{\beta_{\text{env}}}{1 - \varepsilon_S}} + \delta_{\partial} \right) \|\Xi_0 f\|_{\Sigma}. \end{aligned}$$

Take the sharp supremum.

□

For the imported direct-unshifted envelope

$$\beta_{\text{env}} = 0.8290856201657449,$$

the exact relative reserve is

$$1 - \beta_{\text{env}} = 0.1709143798342551.$$

If the source attachment is exact, the allowed boundary residual is

$$\boxed{\delta_{\partial} < 1 - \sqrt{\beta_{\text{env}}} = 0.0894586115031646185 \dots}$$

If the source chart loses at most five percent, the residual budget remains

$$\delta_{\partial} < 0.0658042467246887 \dots$$

and at ten percent source loss it remains

$$\delta_{\partial} < 0.0402051028325867 \dots$$

Thus the next theorem need not be a symbolic miracle. It may be an exact attachment or a directed residual enclosure with a substantial budget.

39.10 10. Exact Stage 3D calibration

The exact rational packet uses six mismatch rows:

```
prime singular amplitudes  5, 9, 12
Gamma singular amplitudes 12, 12, 16
```

so

$$M_{\text{mis}} = \text{diag}(169, 225, 400).$$

With debt

$$d_0 = 144$$

and compatibility defect

$$D_{\text{comp}} = \text{diag}\left(\frac{5}{13}, \frac{3}{5}, \frac{4}{5}\right),$$

the source Gram is

$$\Xi^* \Xi = \text{diag}(25, 81, 256) = M_{\text{mis}} - 144I.$$

A rational graph isometry J is used to construct

$$D_{\Sigma}^0 = \Xi J^*.$$

The packet verifies:

```
D_Sigma^0 J = Xi;
D_Sigma^0 Q_graph = 0;
all alternate adapters differ by E Q_graph;
alternate boundary states differ only by ker(J*);
event-coordinate rotations preserve the source and decoder;
minimum decoder burden = 361/900;
relative reserve = 539/900;
cut covariance is positive definite.
```

These are exact Fraction calculations.

39.11 11. Actual completed-Weil status

The source material supports:

prime mismatch factorization	PROVED
Gamma oscillator mismatch factorization	PROVED
diagonal debt d_0	PROVED
compatibility defect D_comp	PROVED
Xi_0=A_mis D_comp	PROVED
Xi_0^*Xi_0=S_(0,-)^full	PROVED
native metric graph J_X	PASSED / HASH STABLE
canonical adapter D_Sigma^0=Xi_0 J_X^*	PROVED HERE
Xi_0=D_Sigma^0 J_X	PROVED HERE

Therefore:

ACTUAL SOURCE ADAPTER = CONSTRUCTED.

Still open:

actual two-sheet Loewner source domination;
 actual boundary density in that same chart;
 promotion of the T21 scalar envelope to beta_cut;
 actual minimum completed-Weil decoder;
 actual cut covariance;
 RH.

39.12 12. Next theorem: Stage 3E

The next development is the

Native Two-Sheet Source-Domination Attachment Theorem.

It must construct a two-sheet chart

$$J_{2\text{sh}} : X_3^- \rightarrow L^2(\mathbb{R}; \mathbb{C}^2)$$

from the already declared prime/Gamma mismatch analysis and prove:

$$\|\Xi_0 f\|_{\Sigma}^2 \geq \int_{\mathbb{R}} \langle J_{2\text{sh}} f(\xi), A_0(\xi) J_{2\text{sh}} f(\xi) \rangle d\xi, \quad (12.1)$$

together with

$$L_{\partial}(f) = \int_{\mathbb{R}} \langle b_{\partial}(\xi), J_{2\text{sh}} f(\xi) \rangle d\xi. \quad (12.2)$$

The directed cell floors and actual boundary envelope must then be shown to bound (12.1)–(12.2), exactly or with the residuals in Theorem 9.1.

If this stage attaches the existing envelope

$$\beta_{\text{env}} \leq 0.8290856201657449,$$

then Recognition completion forces

$$\|c_{\partial, \min}\|^2 \leq 0.8290856201657449 < 1$$

and

$$S_{0,-}^{\text{full}} - L_{\partial}^* L_{\partial} \geq 0.1709143798342551 S_{0,-}^{\text{full}}.$$

39.13 13. Claim boundary

PRIME--GAMMA MISMATCH/DEBT SOURCE	SOURCE-DERIVED / PROVED
CANONICAL METRIC-GRAPH SOURCE ADAPTER	PROVED
CLASSIFICATION OF ALL SOURCE ADAPTERS	PROVED
GRAPH-NUL RECOGNITION-QUOTIENT INVARIANCE	PROVED
EVENT-COORDINATE COVARIANCE	PROVED
ROBUST TWO-SHEET ATTACHMENT BUDGET	PROVED
EXACT RATIONAL STAGE 3D PACKET	WRITTEN
ACTUAL TWO-SHEET SOURCE DOMINATION	OPEN / NEXT
ACTUAL BOUNDARY COMMON-CHART IDENTITY	OPEN / NEXT
T21 ENVELOPE AS ACTUAL DECODER BOUND	OPEN
ACTUAL COMPLETED-WEIL CUT COVARIANCE	OPEN
RIEMANN HYPOTHESIS	NOT CLAIMED

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Chapter 40

Native Two-Sheet Hardy Boundary and Compatibility-Transfer Theorem

40.1 1. Purpose

The native prime-Gamma-debt source adapter is now constructed:

$$\Xi_0 = \mathcal{A}_{\text{mis}} D_{\text{comp}}, \quad \Xi_0^* \Xi_0 = S_{0,-}^{\text{full}}.$$

The native metric graph also generates the physical odd boundary state. The remaining problem is not the existence of a source adapter and not the existence of the boundary functional. It is to place them in one actual source-dominating two-sheet chart.

This stage proves the boundary half of that attachment exactly. The Hardy strip already carried by the native exponential spatial energy has an explicit Cauchy boundary kernel. Therefore the completion-boundary evaluation is a first moment of the declared two-sheet Hardy boundary values, not an imported row.

The stage then identifies the only remaining displacement. The prime-Gamma mismatch uses the raw two-sheet chart, while the physical source uses that chart after the compatibility defect. The boundary density must therefore be transported through the same defect on its positive support.

The corrected order is

```
native Hardy strip
-> exact two-sheet Cauchy boundary density
-> raw prime--Gamma two-sheet mismatch chart
-> compatibility defect on that chart
-> support-range test for the boundary density
-> transformed boundary density
-> source domination / direct T21 envelope
-> completed-Weil cut covariance.
```

40.2 2. Native two-sheet Hardy chart

Let

$$S_a = \{z \in \mathbb{C} : |\text{Im } z| < a\}, \quad a = \frac{3}{2}.$$

For a native odd state, the exponential spatial channel gives a Hardy function $\Phi_f \in H^2(S_a)$. Put

$$\Phi_f^-(\xi) = \Phi_f(\xi - ia), \quad \Phi_f^+(\xi) = \Phi_f(\xi + ia),$$

and define the two-sheet chart

$$J_{2\text{sh}}f = (\Phi_f^-, \Phi_f^+) \in L^2\left(\mathbb{R}, \frac{d\xi}{2\pi}; \mathbb{C}^2\right).$$

The source construction upstream uses these same two Hardy boundary channels for the prime two-sheet mismatch and Gamma oscillator mismatch.

40.3 3. Strip Cauchy boundary density

Let $|y| < a$. Cauchy's formula on an expanding rectangle in the strip gives

$$\begin{aligned} \Phi(iy) = & \int_{\mathbb{R}} \frac{-i \Phi^-(\xi)}{\xi - i(a+y)} \frac{d\xi}{2\pi} \\ & + \int_{\mathbb{R}} \frac{i \Phi^+(\xi)}{\xi + i(a-y)} \frac{d\xi}{2\pi}. \end{aligned} \tag{3.1}$$

The signs swap if the opposite Fourier convention is used. The two sheet masses and their sum do not change.

Theorem 3.1 (Exact strip boundary first moment)

There is a unique two-sheet Cauchy density $k_{a,y}$ in the Hardy boundary space such that

$$\Phi(iy) = \langle k_{a,y}, J_{2\text{sh}}\Phi \rangle.$$

Its sheet masses are

$$\|k_{a,y}^-\|^2 = \frac{1}{2(a+y)}, \quad \|k_{a,y}^+\|^2 = \frac{1}{2(a-y)}, \tag{3.2}$$

and

$$\|k_{a,y}\|^2 = \frac{a}{a^2 - y^2}. \tag{3.3}$$

Proof

Equation (3.1) is a bounded functional on the two boundary lines. The Riesz representer is obtained by reading off the two Cauchy kernels. Since

$$\int_{\mathbb{R}} \frac{d\xi}{2\pi(\xi^2 + c^2)} = \frac{1}{2c},$$

one obtains (3.2); addition gives (3.3).
For

$$a = \frac{3}{2}, \quad |y| = \frac{1}{2},$$

one sheet has mass $1/4$, the other has mass $1/2$, and

$$\|k_{3/2, 1/2}\|^2 = \frac{3}{4}.$$

Reflection swaps the two sheet masses and preserves their sum.
The actual odd completion boundary is

$$L_{\partial}f = \sqrt{2} \hat{f}(i/2).$$

Therefore

$$L_{\partial}f = \langle b_{\partial, 2\text{sh}}, J_{2\text{sh}}f \rangle, \quad b_{\partial, 2\text{sh}} = \sqrt{2} k_{3/2, \pm 1/2}, \quad (3.4)$$

followed by the declared odd/reflection projection. The sign of the evaluation height depends only on the pinned Fourier convention. Equation (3.4) is unchanged as a physical odd functional.

Thus the actual boundary is now attached to the declared two-sheet Hardy chart.

40.4 4. Compatibility displacement of the source chart

Let $J = J_{2\text{sh}}$. The raw positive mismatch analysis factors through this chart. Write it schematically as

$$\mathcal{A}_{\text{mis}} = \mathcal{M}_{\text{mis}}J.$$

The completed pure source is not the raw mismatch. It is obtained after the exact diagonal debt is paid:

$$\Xi_0 = \mathcal{A}_{\text{mis}}D_{\text{comp}}.$$

On the two-sheet chart range define

$$C_{\text{comp}} = JD_{\text{comp}}J^*. \quad (4.1)$$

Then

$$JD_{\text{comp}} = C_{\text{comp}}J \quad (4.2)$$

and

$$\Xi_0 = \mathcal{M}_{\text{mis}}C_{\text{comp}}J.$$

The boundary density in (3.4) pairs with Jf , while the source event analysis sees $C_{\text{comp}}Jf$. This is the precise remaining attachment problem.

40.5 5. Compatibility-transfer theorem

Theorem 5.1 (Boundary transfer through the source defect)

Let $C \geq 0$ be a contraction on a chart space and let

$$L(f) = \langle b, Jf \rangle.$$

The following are equivalent:

1. $b \in \text{Ran } C^*$ on the positive support;
2. there exists a minimum-norm density

$$\tilde{b} = C^{\dagger,*}b$$

such that

$$L(f) = \langle \tilde{b}, CJf \rangle;$$

3. b vanishes on $\ker C$.

When these conditions hold, every raw source domination

$$\|\Xi f\|^2 \geq \langle CJf, ACJf \rangle$$

gives

$$\boxed{|L(f)|^2 \leq \langle \tilde{b}, A^{\dagger}\tilde{b} \rangle \|\Xi f\|^2.} \tag{5.1}$$

Proof

The equivalence is the support form of the Moore–Penrose range theorem. Under it,

$$b = C^*\tilde{b},$$

so

$$L(f) = \langle \tilde{b}, CJf \rangle.$$

Apply Cauchy–Schwarz in the positive A -metric.

This theorem explains why an untransformed boundary mass table cannot be attached blindly to the post-debt source. The density consumed by the source ledger must be $\tilde{b}$, not merely b .

40.6 6. Zero remembers the compatibility cut

If C has a kernel, a component of b in that kernel cannot be decoded by the physical source. It is not a small numerical error. It is directional cut memory.

Corollary 6.1 (Zero-cut support gate)

$$\boxed{P_{\ker C} b = 0} \tag{6.1}$$

is necessary and sufficient for a finite compatibility-transferred boundary burden. A nonzero kernel component forces the source-relative burden to be infinite.

This is the operator form of the doctrine

positive spectrum transports;

zero spectrum remembers which side of the compatibility cut was lost.

40.7 7. Robust source-loss and boundary-residual theorem

Let $\beta_{\text{env}} < 1$ be a chart envelope for a candidate transformed boundary density. Suppose the actual source dominates the chart with relative loss $0 \leq \varepsilon_S < 1$, and the remaining boundary pairing residual has source-relative norm at most δ_∂ . Then

$$\boxed{\sqrt{\beta_{\text{actual}}} \leq \sqrt{\frac{\beta_{\text{env}}}{1 - \varepsilon_S}} + \delta_\partial.} \tag{7.1}$$

Therefore the actual decoder is contractive whenever the right side is below one.

For the direct T21 envelope

$$\beta_{\text{env}} = 0.8290856201657449,$$

the available boundary-residual budgets are approximately

source loss 0%	0.0894586115031646
source loss 5%	0.0658042467246887
source loss 10%	0.0402051028325867.

At ten percent source loss and residual 0.04, the robust burden remains strictly below one.

40.8 8. Exact generalized calibrations

The executable Stage 3E packet verifies:

strip a=3/2, evaluation |y|=1/2:

sheet masses	1/4 and 1/2
total evaluation-kernel mass	3/4

compatibility-transfer model:

raw mismatch source	diag(4,9,16)
compatibility defect	diag(1/2,2/3,3/4)
completed source Gram	diag(1,4,9)
transformed/actual burden	397/5184

zero-cut support model:

compatibility defect	diag(1/2,2/3,0)
supported burden	97/1296
nonzero kernel boundary	rejected.

A wrong sheet orientation preserves coefficient masses but changes the boundary functional. Orientation still matters before the exact Cauchy density and compatibility transfer are fixed.

40.9 9. Actual completed-Weil gate

Stage 3E closes

$$L_{\partial}(f) = \langle b_{\partial,2sh}, J_{2sh}f \rangle.$$

The remaining actual theorem is now only:

$$b_{\partial,2sh} \in \text{Ran } C_{\text{comp}}^* \quad (9.1)$$

construct

$$\tilde{b}_{\partial} = C_{\text{comp}}^{\dagger,*} b_{\partial,2sh}, \quad (9.2)$$

and prove that the T21 cut profile is a Loewner lower source chart for that transformed density:

$$\|\Xi_0 f\|^2 \geq (1 - \varepsilon_S) \int \langle C_{\text{comp}} J_{2sh} f, A_{T21} C_{\text{comp}} J_{2sh} f \rangle. \quad (9.3)$$

If (9.1)–(9.3) are exact, the T21 envelope gives

$$\|c_{\partial}\|^2 \leq 0.8290856201657449 < 1.$$

A directed residual version is already licensed by (7.1).

40.10 10. Claim boundary

NATIVE HARDY TWO-SHEET CHART	PROVED UPSTREAM
STRIP CAUCHY BOUNDARY DENSITY	PROVED
ACTUAL COMPLETION BOUNDARY ATTACHED TO THAT CHART	PROVED
COMPATIBILITY-TRANSFER RANGE THEOREM	PROVED
ZERO-CUT SUPPORT GATE	PROVED
ROBUST SOURCE-LOSS / BOUNDARY-RESIDUAL RULE	PROVED
ACTUAL b_{partial} IN $\text{Ran}(C_{\text{comp}}^*)$	OPEN / NEXT
ACTUAL TRANSFORMED DENSITY $b_{\text{tilde_partial}}$	OPEN / NEXT
T21 PROFILE AS ACTUAL SOURCE DOMINATION	OPEN / NEXT
ACTUAL COMPLETED-WEIL DECODER BOUND	OPEN
ACTUAL COMPLETED-WEIL CUT COVARIANCE	OPEN
RIEMANN HYPOTHESIS	NOT CLAIMED

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Chapter 41

Native Compatibility Form-Range and T21 Cut-Source Domination Theorem

41.1 1. Purpose

Stage 3E attached the physical completion boundary to the native two-sheet Hardy chart. It then asked whether the boundary density belongs to the ordinary operator range of the compatibility defect.

That range condition is stronger than the endpoint sign requires and, in infinite dimension, its stated kernel equivalence needs a closed-range hypothesis. The correct native object is the inverse-half **form range** of the completed physical source.

This stage makes that correction and attaches the direct T21 cut ledger to the actual prime–Gamma–debt source. The native order is

```
prime and Gamma mismatch events
-> exact diagonal debt
-> completed two-sheet source symbol A_0
-> cumulative cut lower branch s_T21
-> physical boundary recognition class
-> inverse-half form-range decoder
-> strict decoder reserve
-> completed-Weil cut covariance.
```

No source inverse vector, attained threshold state, fitted event rotation or positive-shift extrapolation is consumed.

41.2 2. Correction of the Stage 3E ambient-range gate

Let C be a bounded operator. In general,

$$b \perp \ker C \iff b \in \overline{\text{Ran } C^*}.$$

The stronger statement

$$b \perp \ker C \iff b \in \text{Ran } C^*$$

holds when $\text{Ran } C$ is closed. Thus the finite exact calibrations in Stage 3E remain correct, but ordinary compatibility-defect range membership is not the universal infinite-dimensional terminal gate.

For the completed source the correct condition is

$$\boxed{b_\partial \in \text{Dom}\left(S_{0,-}^{\text{full}\dagger/2}\right)}. \quad (2.1)$$

This is an inverse-half form-domain statement. It permits nonattained criticality: a completed decoder may exist even when the formal source inverse vector does not belong to the Hilbert carrier.

41.3 3. Abstract form-range decoder theorem

Let

$$T : \mathcal{H} \longrightarrow \mathcal{Y}$$

be a bounded analysis and put

$$S = T^*T \geq 0.$$

Let $b \in \mathcal{H}$, and define

$$L_b(f) = \langle b, f \rangle.$$

Theorem 3.1 (Inverse-half form-range transfer)

The following are equivalent:

1. L_b is bounded for the source seminorm $f \mapsto \|Tf\|$;
2. $b \in \text{Dom}(S^{\dagger/2})$;
3. there exists a unique minimum-norm decoder

$$c_b \in \overline{\text{Ran } T}$$

satisfying

$$L_b(f) = \langle c_b, Tf \rangle.$$

Moreover,

$$\boxed{c_b = J_S S^{\dagger/2} b}, \quad (3.1)$$

where

$$J_S = T S^{\dagger/2}$$

is the canonical source isometry on the positive support, and

$$\boxed{\|c_b\|^2 = \|S^{\dagger/2} b\|^2 = \sup_{Tf \neq 0} \frac{|L_b(f)|^2}{\|Tf\|^2}}. \quad (3.2)$$

Proof

On $(\ker S)^\perp$,

$$T = J_S S^{1/2}.$$

If $u = S^{\dagger/2}b$, then

$$L_b(f) = \langle u, S^{1/2}f \rangle = \langle J_S u, T f \rangle.$$

Thus $c_b = J_S u$ is a decoder and has norm $\|u\|$. Conversely, if $L_b(f) = \langle c, T f \rangle$, then

$$b = T^* c = S^{1/2} J_S^* c,$$

so $b \in \text{Ran } S^{1/2} = \text{Dom}(S^{\dagger/2})$. Orthogonal projection of c onto $\overline{\text{Ran } T}$ gives the unique minimum-norm decoder. Equation (3.2) is the norm of the induced functional on $\overline{\text{Ran } T}$.

Corollary 3.2 (Cut covariance)

If

$$\beta_b := \|S^{\dagger/2}b\|^2 < 1,$$

then

$$\boxed{S - bb^* \geq (1 - \beta_b)S \geq 0.} \tag{3.3}$$

No vector solving $Sh = b$ is required.

41.4 4. Exact completed two-sheet source symbol

Put

$$m_p = \sum_{n \geq 2} \frac{\Lambda(n)}{n^2}, \quad q(\xi) = \sum_{n \geq 2} \frac{\Lambda(n)}{n^2} e^{i\xi \log n},$$

and

$$\mathcal{Q}(\xi) = \begin{pmatrix} 0 & q(\xi) \\ \overline{q(\xi)} & 0 \end{pmatrix}.$$

Let

$$g_0(\xi) = G(\xi) + m_\Gamma.$$

The prime and Gamma mismatch factorizations give the completed physical source symbol after the exact diagonal debt is paid:

$$\boxed{A_0(\xi) = (g_0(\xi) + m_p)I_2 - \mathcal{Q}(\xi).} \tag{4.1}$$

Its eigenvalue branches are

$$s_{\pm}(\xi) = g_0(\xi) + m_p \pm |q(\xi)|,$$

so the lower branch is

$$\boxed{s_0(\xi) = g_0(\xi) + m_p - |q(\xi)|.} \quad (4.2)$$

Since

$$g_0(0) = 0, \quad q(0) = m_p,$$

one has

$$s_0(0) = 0.$$

The compatibility defect is a factorization of this same source. It is not a second physical source requiring a separate terminal range theorem.

41.5 5. T21 as a Loewner lower chart

The covariant phase removal gives, for $\xi > 0$,

$$s'_0(\xi) = G'(\xi) - (|q|)'(\xi).$$

The directed T21 packet supplies:

near-zero covariant slope-gap lower	28.96922744937062
minimum positive source floor	0.0003621153431171327
terminal active source floor	5.206639333818032
source floor on [12,13]	0.4571260170709391
far-ray source floor	0.1568358628972386
active cells	2400
prime-power events	78734

Near the first cut,

$$s'_0(\xi) \geq \gamma_0 \xi$$

and therefore

$$\boxed{s_0(\xi) \geq \frac{\gamma_0}{2} \xi^2.} \quad (5.1)$$

On every remaining active cell, the directed derivative-gap lower is integrated from the preceding certified node. Hence the stored cumulative floor s_j^- satisfies

$$s_0(\xi) \geq s_j^- \quad (\xi \in I_j).$$

On the outside region the direct lower-branch certificates apply. Therefore, with s_{T21} denoting the piecewise directed floor,

$$\boxed{A_0(\xi) \geq s_{T21}(\xi) I_2} \quad (5.2)$$

on the complete two-sheet chart.

This is the missing Loewner attachment. The profile is not an unrelated derivative table: it is obtained by integrating the lower-branch derivative from the exact seam value $s_0(0) = 0$.

41.6 6. Boundary recognition class in the same chart

Stage 3E gives the exact Cauchy representative

$$L_{\partial}(f) = \langle b_{\partial, \text{Cauchy}}, J_{2\text{sh}} f \rangle.$$

The native metric graph gives the Riesz state

$$r_{\partial} = M_X^{-1} q_b$$

and the weighted boundary object

$$w_{\partial}(x) = e^{-3|x|/2} r_{\partial}(x).$$

Its two-sheet/Fourier representative $b_{\partial, \text{T21}}$ generates the same physical functional. Hence

$$\boxed{J_{2\text{sh}}^*(b_{\partial, \text{Cauchy}} - b_{\partial, \text{T21}}) = 0.} \quad (6.1)$$

The two representatives belong to the same Recognition-Seam boundary class. The burden depends on that physical class, while any nonminimal representative gives a lawful outward bound.

The source manuscript proves

$$\int |w_{\partial}(x)| dx \leq \frac{1}{3\sqrt{2}},$$

and the sharper first moment

$$\int |x| |w_{\partial}(x)| dx \leq \frac{1}{9}.$$

The direct T21 packet deliberately uses the older coarse value $1/3$, so its result remains outward.

41.7 7. Cellwise form-domain theorem

Let $A(x) \geq s(x)I$, with $s(x) > 0$ away from a declared seam, and let $b(x)$ be the actual boundary representative in that chart. If

$$\beta_{\text{env}} = \int \frac{\|b(x)\|^2}{s(x)} d\mu(x) < \infty,$$

then

$$b \in \text{Dom}(A^{\dagger/2})$$

and

$$\boxed{\|A^{\dagger/2}b\|^2 \leq \beta_{\text{env}}.} \quad (7.1)$$

At a quadratic first cut,

$$s(\xi) \geq c\xi^2, \quad \|b(\xi)\| \leq M_1|\xi|,$$

the quotient remains finite:

$$\boxed{\int_{-h}^h \frac{\|b(\xi)\|^2}{s(\xi)} d\xi \leq \frac{2hM_1^2}{c}.} \quad (7.2)$$

Thus the zero cut does not require an operator-range vector. Linear boundary memory cancels the quadratic source opening in the inverse-half form domain.

41.8 8. Actual direct-unshifted attachment

The imported direct packet gives

$$\beta_{\text{first}} \leq 0.00007670975092815261,$$

$$\beta_{\text{active}} \leq 0.2976678523563374,$$

and

$$\beta_{\text{outside}} \leq 0.5313410580584792.$$

Their exact displayed sum is

$$0.82908562016574475261,$$

and the stored outward value is

$$\boxed{\beta_{\partial}^{\text{cut}} \leq 0.8290856201657449 < 1.} \quad (8.1)$$

By Theorem 3.1, the actual minimum decoder

$$c_{\partial} \in \overline{\text{Ran } \Xi_0}$$

exists and satisfies

$$L_{\partial}(f) = \langle c_{\partial}, \Xi_0 f \rangle,$$

$$\|c_{\partial}\|^2 \leq 0.8290856201657449.$$

Therefore

$$\boxed{S_{0,-}^{\text{full}} - L_{\partial}^* L_{\partial} \geq 0.1709143798342551 S_{0,-}^{\text{full}} \geq 0.} \quad (8.2)$$

Equivalently, after the unit-boundary-state enlargement of Stage 3A,

$$\boxed{S_{0,-}^{\text{full}} - L_{\partial}^* L_{\partial} = \widehat{\Xi}_0^* (I - p_{\partial} p_{\partial}^*) \widehat{\Xi}_0.} \quad (8.3)$$

This closes the actual decoder bound and the completed-Weil odd cut covariance. It does not yet assert strict positivity on every nonzero odd state; that requires source-kernel injectivity on this same completed carrier.

41.9 9. Exact executable calibrations

The Stage 3F packet verifies:

nonattained form-range model:	
completed decoder burden	1/4
formal source-inverse prefix energy	4N/25 -> infinity
two-sheet Loewner model:	
every lower source floor	1
total boundary burden	419/7200
recognition-class model:	
distinct ambient boundary representatives	
identical physical boundary row	
minimum class burden	2/3
cellwise cut model:	
first cut	1/40
active cells	21/400
outside	1/40
total burden	41/400

The actual Decimal audit reproduces the T21 component sum, outward upper and relative reserve.

41.10 10. Next theorem

The remaining native odd theorem is

$$\boxed{\ker S_{0,-}^{\text{full}} = \{0\}}$$

on the same Stage 3F carrier.

The Hardy active-band no-blindness theorem already supplies the analytic mechanism. Stage 3G must attach it explicitly to Ξ_0 , then verify the odd classical explicit-formula normalization interface without reintroducing the even-sector or T03 routes.

41.11 11. Claim boundary

STAGE 3E CLOSED-RANGE OVERSTATEMENT	CORRECTED
INVERSE-HALF FORM-RANGE DECODER THEOREM	PROVED
ACTUAL PRIME--GAMMA--DEBT SOURCE SYMBOL	ATTACHED
T21 PROFILE AS LOEWNER LOWER SOURCE CHART	PROVED / SOURCE-PINNED
BOUNDARY RECOGNITION CLASS ON SAME CHART	PROVED / SOURCE-PINNED
ACTUAL DECODER BOUND ≤ 0.8290856201657449	CLOSED
ACTUAL ODD CUT COVARIANCE	CLOSED
SOURCE-KERNEL INJECTIVITY ON SAME CARRIER	NEXT
STRICT ODD COMPLETED-WEIL POSITIVITY	OPEN / NEXT
ODD CLASSICAL NORMALIZATION INTERFACE	OPEN / NEXT
RIEMANN HYPOTHESIS	NOT CLAIMED

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Chapter 42

Native Source-Kernel No-Blindness and Strict Odd Cut Theorem

42.1 1. Purpose

Stage 3F constructs the actual completed odd source decoder and proves the relative cut-covariance estimate

$$S_{0,-}^{\text{full}} - L_{\partial}^* L_{\partial} \geq \delta_{\text{cut}} S_{0,-}^{\text{full}},$$

where

$$\beta_{\partial}^{\text{cut}} \leq 0.8290856201657449, \quad \delta_{\text{cut}} = 1 - \beta_{\partial}^{\text{cut}} \geq 0.1709143798342551.$$

This proves nonnegativity, but strict positivity on every nonzero odd state requires one further statement:

$$\ker S_{0,-}^{\text{full}} = \ker \Xi_0 = \{0\}.$$

The present stage attaches no-blindness to the same two-sheet source carrier used in Stage 3F. It does not consume the classical explicit formula, an even sector, a five-dimensional endpoint matrix, an attained threshold vector, or a uniform ambient spectral gap.

The native order is

```
actual prime--Gamma--debt analysis Xi_0
-> actual two-sheet chart J_2sh
-> T21 lower cut profile s_T21
-> zero-cut no-blindness
-> source-kernel injectivity
-> strict relative odd cut covariance.
```

42.2 2. Abstract cut-source no-blindness theorem

Let

$$\Xi : \mathcal{H} \longrightarrow \mathcal{Y}$$

be a bounded event analysis and let

$$J : \mathcal{H} \longrightarrow L^2(X, \mu; \mathbb{C}^d)$$

be an injective recognition chart. Suppose there is a measurable function $s : X \rightarrow [0, \infty)$ satisfying

$$\|\Xi f\|_{\mathcal{Y}}^2 \geq \int_X s(x) \|Jf(x)\|^2 d\mu(x) \quad (2.1)$$

for every $f \in \mathcal{H}$. Let

$$Z_s = \{x : s(x) = 0\}.$$

Theorem 2.1 (Null-cut source injectivity)

If

$$\mu(Z_s) = 0,$$

then

$$\ker \Xi = \{0\}. \quad (2.2)$$

Proof

Let $\Xi f = 0$. Equation (2.1) gives

$$\int_X s(x) \|Jf(x)\|^2 d\mu(x) = 0.$$

The integrand is nonnegative. Since $s > 0$ almost everywhere outside the null set Z_s , one has

$$Jf = 0 \quad \text{almost everywhere on } X.$$

Injectivity of J gives $f = 0$.

The theorem does not require $\inf s > 0$. The lower chart may approach zero at the cut, so zero can remain in the continuous or essential spectral closure. What is excluded is an attained nonzero source-null state.

42.3 3. Positive-measure active-band variant

Sometimes a source certificate is available only on a declared active band. Let $E \subseteq X$ have positive measure and assume

$$\|\Xi f\|^2 \geq \int_E h(x) \|Jf(x)\|^2 d\mu(x), \quad h > 0 \text{ a.e. on } E. \quad (3.1)$$

Theorem 3.1 (Recognition-unique active observer)

Assume the chart has the uniqueness property

$$1_E Jf = 0 \implies f = 0. \quad (3.2)$$

Then $\ker \Xi = \{0\}$.

Proof

If $\Xi f = 0$, (3.1) forces $Jf = 0$ almost everywhere on E , and (3.2) then gives $f = 0$.

For Hardy boundary charts, (3.2) is the boundary uniqueness/no-blindness property. This theorem is the direct recognition form of the covariant active-band virial route already proved in the MP source stack.

42.4 4. Application to the actual completed odd source

Stage 3D constructs

$$\Xi_0 = \mathcal{A}_{\text{mis}} D_{\text{comp}}, \quad \Xi_0^* \Xi_0 = S_{0,-}^{\text{full}}. \quad (4.1)$$

Stage 3E constructs the native two-sheet Hardy chart

$$J_{2\text{sh}} : X_3^- \longrightarrow L^2\left(\mathbb{R}, \frac{d\xi}{2\pi}; \mathbb{C}^2\right). \quad (4.2)$$

Stage 3F attaches the actual prime–Gamma–debt source symbol to that chart and proves the T21 Loewner lower estimate

$$\|\Xi_0 f\|^2 \geq \int_{\mathbb{R}} s_{\text{T21}}(\xi) \|J_{2\text{sh}} f(\xi)\|^2 \frac{d\xi}{2\pi}. \quad (4.3)$$

The cut profile satisfies:

```
first cut:
s_T21(xi) >= (gamma_0/2) xi^2,
gamma_0 = 28.96922744937062;

remaining active cells:
every cumulative source floor is strictly positive;

[12,13]:
source floor >= 0.4571260170709391;

far ray:
source floor >= 0.1568358628972386.
```

Hence

$$s_{\text{T21}}(\xi) > 0 \quad \text{for almost every } \xi \neq 0,$$

and the only zero cut is the singleton $\{0\}$, which has Lebesgue measure zero.

Theorem 4.1 (Actual source-kernel no-blindness)

On the Stage 3F completed odd carrier,

$$\boxed{\ker \Xi_0 = \ker S_{0,-}^{\text{full}} = \{0\}.} \quad (4.4)$$

Proof

Apply Theorem 2.1 to (4.3). Since

$$S_{0,-}^{\text{full}} = \Xi_0^* \Xi_0,$$

one has

$$\ker S_{0,-}^{\text{full}} = \ker \Xi_0.$$

The two-sheet Hardy chart is injective, so (4.4) follows.

Independent source pin

The MP T20 theorem proves the same conclusion by a covariant cut-flow commutator and Hardy no-blindness. In that route a hypothetical zero source state has zero virial expectation, while the positive active-band commutator and Hardy uniqueness force the same expectation to be strictly positive. The Stage 3G direct proof and the T20 virial proof are independent presentations of the same native no-blindness mechanism.

42.5 5. Strict odd completed-Weil positivity

Put

$$W_3^- = S_{0,-}^{\text{full}} - L_{\partial}^* L_{\partial}.$$

Stage 3F proves

$$W_3^- \geq \delta_{\text{cut}} S_{0,-}^{\text{full}}, \quad \delta_{\text{cut}} = 0.1709143798342551. \quad (5.1)$$

Theorem 5.1 (Strict odd cut theorem)

For every nonzero odd native state f ,

$$\boxed{\langle f, W_3^- f \rangle \geq 0.1709143798342551 \langle f, S_{0,-}^{\text{full}} f \rangle > 0.} \quad (5.2)$$

Proof

Theorem 4.1 gives

$$\langle f, S_{0,-}^{\text{full}} f \rangle = \|\Xi_0 f\|^2 > 0$$

for every nonzero f . Substitute this into (5.1).

The theorem is strict in the quadratic-form sense. It does not claim a bound

$$W_3^- \geq cI$$

in the ambient carrier norm. The source floor vanishes at the cut and may approach zero under refinement, so the spectrum may still accumulate at zero without possessing a zero eigenvector.

42.6 6. Injective cut-covariance bridge

Stage 3A gives an enlarged event lift $\widehat{\Xi}_0$ and a unit boundary state p_∂ such that

$$W_3^- = \widehat{\Xi}_0^*(I - p_\partial p_\partial^*)\widehat{\Xi}_0. \quad (6.1)$$

Let

$$Q_\partial = I - p_\partial p_\partial^*.$$

Corollary 6.1 (Boundary-line no-blindness)

$$\boxed{\ker(Q_\partial \widehat{\Xi}_0) = \{0\}.} \quad (6.2)$$

Proof

If $Q_\partial \widehat{\Xi}_0 f = 0$, then (6.1) gives

$$\langle f, W_3^- f \rangle = 0.$$

Theorem 5.1 implies $f = 0$.

Thus the source, boundary and remaining cut memory form one injective co-generated event object.

42.7 7. Exact generalized calibration

The executable packet uses the odd polynomial family

$$p(x) = ax + bx^3 + cx^5.$$

The cut row at $x = 0$ vanishes for every state. Three active rows at $x = 1, 2, 3$ form the exact matrix

$$\begin{pmatrix} 1 & 1 & 1 \\ 2 & 8 & 32 \\ 3 & 27 & 243 \end{pmatrix},$$

whose determinant is

$$720 \neq 0.$$

Thus the zero cut carries no source rank, while the active observer recovers the complete odd state. The source Gram has determinant

$$518400 > 0.$$

With decoder

$$c = (0, 1/2, 1/3, 1/4),$$

one has

$$\|c\|^2 = \frac{61}{144}, \quad 1 - \|c\|^2 = \frac{83}{144},$$

and the exact cut covariance has determinant

$$298800 > 0.$$

A negative control retains only the active nodes 1, 2. The nonzero odd polynomial

$$p(x) = x(4 - 5x^2 + x^4)$$

then vanishes at the cut and both observed nodes. The third active node detects it with value 120. This proves that positive weights do not replace a no-blindness/uniqueness theorem.

42.8 8. Noncoercive strictness calibration

The packet also considers

$$S_N = \text{diag}(1, 1/2, \dots, 1/N), \quad W_N = \frac{1}{5} S_N.$$

Every finite source and strict form has trivial kernel, and the relative reserve is exactly 1/5. Nevertheless,

$$\lambda_{\min}(S_N) = \frac{1}{N} \rightarrow 0, \quad \lambda_{\min}(W_N) = \frac{1}{5N} \rightarrow 0.$$

This is the exact distinction used in the completed endpoint:

strict positivity on every nonzero state	proved;
uniform ambient coercivity	not asserted.

42.9 9. Next membrane

Stage 3G closes the native odd sign. It deliberately stops before the classical target interface.

The next theorem must establish, term by term and in one Fourier convention, that the native odd form equals the odd restriction of the classical completed explicit-formula Weil form. It must match:

```
completion-boundary term;
Gamma term;
prime-power term;
reflection and conjugation convention;
logarithmic/multiplicative unitary normalization;
test-core density.
```

Only after that normalization theorem is proved may a parity-restricted classical Weil implication be consumed.

42.10 10. Claim boundary

NULL-CUT SOURCE INJECTIVITY THEOREM	PROVED
ACTIVE-BAND RECOGNITION-UNIQUE VARIANT	PROVED
ACTUAL $\ker \text{Xi}_0 = \ker S_{(0,-)}^{\text{full}} = \{0\}$	CLOSED FROM STAGE 3F PINS
STRICT ODD COMPLETED-WEIL POSITIVITY	CLOSED
BOUNDARY-LINE CUT-COVARIANCE INJECTIVITY	CLOSED
UNIFORM AMBIENT SPECTRAL GAP	NOT CLAIMED
ODD NATIVE/CLASSICAL NORMALIZATION INTERFACE	NEXT
PARITY-RESTRICTED CLASSICAL WEIL IMPLICATION	NEXT / EXTERNAL MEMBRANE
RIEMANN HYPOTHESIS	NOT CLAIMED

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Chapter 43

Odd Native-to-Classical Completed-Weil Normalization Interface

43.1 1. Purpose

Stages 3F and 3G close the native odd sign on the Recognition-Seam carrier:

$$\ker S_{0,-}^{\text{full}} = \{0\},$$

and

$$\langle f, W_3^- f \rangle \geq 0.1709143798342551 \langle f, S_{0,-}^{\text{full}} f \rangle > 0$$

for every nonzero odd native state.

The remaining interface is not another sign estimate. It is a normalization theorem: the native boundary, Gamma and prime-power terms must be shown to be exactly the completed explicit-formula distribution evaluated on one declared correlation, in one Fourier convention, on one logarithmic test core.

The order is

```
native logarithmic core
-> correlation q_(f,h)=f*h-sharp
-> completed explicit-formula distribution E_xi(q_(f,h))
-> termwise boundary/Gamma/prime identity
-> odd negative-rank-one restriction
-> multiplicative/logarithmic unitary identity
-> classical symmetric-zero-sum membrane.
```

The last arrow is a classical theorem. Stage 3H pins it but does not pretend to rederive it from the preceding algebra.

43.2 2. Pinned conventions

Throughout, Hilbert inner products are linear in the first variable.

Let

$$\mathcal{D} = C_c^\infty(\mathbb{R}), \quad \mathcal{D}_\times = C_c^\infty(0, \infty).$$

Use the Fourier convention

$$\widehat{f}(z) = \int_{\mathbb{R}} f(x) e^{izx} dx, \quad (2.1)$$

with

$$\|f\|_2^2 = \frac{1}{2\pi} \int_{\mathbb{R}} |\widehat{f}(\xi)|^2 d\xi.$$

Define translation and the reflection-adjoint by

$$(T_t h)(x) = h(x - t), \quad h^\sharp(x) = \overline{h(-x)}. \quad (2.2)$$

For $f, h \in \mathcal{D}$, put

$$q_{f,h} = f * h^\sharp. \quad (2.3)$$

Because f and h are compactly supported, $q_{f,h}$ is compactly supported and its Fourier transform is entire.

43.3 3. Correlation theorem

Theorem 3.1

For every real t ,

$$q_{f,h}(t) = \langle f, T_t h \rangle_{L^2}. \quad (3.1)$$

For every $z \in \mathbb{C}$,

$$\widehat{q}_{f,h}(z) = \widehat{f}(z) \overline{\widehat{h}(\bar{z})}. \quad (3.2)$$

Proof

By definition,

$$\begin{aligned} q_{f,h}(t) &= \int_{\mathbb{R}} f(x) h^\sharp(t - x) dx \\ &= \int_{\mathbb{R}} f(x) \overline{h(x - t)} dx \\ &= \langle f, T_t h \rangle. \end{aligned}$$

The convolution theorem gives

$$\widehat{q}_{f,h}(z) = \widehat{f}(z)\widehat{h}^\sharp(z).$$

Changing variables in the second factor yields

$$\widehat{h}^\sharp(z) = \overline{\widehat{h}(\bar{z})}.$$

This proves both identities.

The conjugate-reflection in (3.2) is essential away from the real axis. Omitting it changes the completion-boundary term and destroys the Hermitian interface.

43.4 4. The pinned completed explicit-formula distribution

Let

$$G(\xi) = \operatorname{Re} \psi\left(\frac{1}{4} + \frac{i\xi}{2}\right) - \log \pi. \quad (4.1)$$

For a compactly supported smooth correlation q , define the completed boundary–Gamma–prime distribution side by

$$\begin{aligned} \mathcal{E}_\xi(q) = & \widehat{q}(i/2) + \widehat{q}(-i/2) \\ & + \frac{1}{2\pi} \int_{\mathbb{R}} G(\xi) \widehat{q}(\xi) d\xi \\ & - \sum_{n \geq 2} \frac{\Lambda(n)}{\sqrt{n}} (q(\log n) + q(-\log n)). \end{aligned} \quad (4.2)$$

Only finitely many prime-power translations meet the compact support of q . The formula therefore has no hidden truncation on the core.

The classical completed explicit formula, in the normalization of

$$\xi(s) = \frac{1}{2} s(s-1) \pi^{-s/2} \Gamma(s/2) \zeta(s),$$

identifies (4.2) with the symmetrically summed zero distribution. That theorem is consumed later as a classical membrane; it is not derived in Stage 3H.

43.5 5. Termwise normalization theorem

For $f, h \in \mathcal{D}$, define

$$\ell_+(f) = \widehat{f}(i/2), \quad \ell_-(f) = \widehat{f}(-i/2), \quad (5.1)$$

$$B(f, h) = \ell_+(f) \overline{\ell_-(h)} + \ell_-(f) \overline{\ell_+(h)}, \quad (5.2)$$

$$A_\Gamma(f, h) = \frac{1}{2\pi} \int_{\mathbb{R}} G(\xi) \widehat{f}(\xi) \overline{\widehat{h}(\xi)} d\xi, \quad (5.3)$$

and

$$P_{1/2}(f, h) = \sum_{n \geq 2} \frac{\Lambda(n)}{\sqrt{n}} (\langle f, T_{\log n} h \rangle + \langle f, T_{-\log n} h \rangle). \quad (5.4)$$

The native logarithmic completed-Weil form is

$$W_\xi(f, h) = B(f, h) + A_\Gamma(f, h) - P_{1/2}(f, h). \quad (5.5)$$

Theorem 5.1 (Exact native/classical distribution interface)

For every $f, h \in \mathcal{D}$,

$$\boxed{W_\xi(f, h) = \mathcal{E}_\xi(q_{f,h}).} \quad (5.6)$$

Proof

By Theorem 3.1,

$$\widehat{q}_{f,h}(i/2) = \ell_+(f) \overline{\ell_-(h)},$$

and

$$\widehat{q}_{f,h}(-i/2) = \ell_-(f) \overline{\ell_+(h)}.$$

These are exactly the two terms in (5.2).

For real ξ , (3.2) reduces to

$$\widehat{q}_{f,h}(\xi) = \widehat{f}(\xi) \overline{\widehat{h}(\xi)},$$

which turns the integral in (4.2) into (5.3).

Finally, (3.1) gives

$$q_{f,h}(\pm \log n) = \langle f, T_{\pm \log n} h \rangle,$$

which turns the prime-power sum in (4.2) into (5.4). Adding the three identities proves (5.6).

No factor, sign, conjugation or 2π normalization remains implicit.

43.6 6. Hermitian and reflection covariance

The correlation satisfies

$$q_{h,f}(t) = \overline{q_{f,h}(-t)}. \quad (6.1)$$

Because G is real and even and the prime translations occur in $\pm$ pairs,

$$\mathcal{E}_\xi(q_{h,f}) = \overline{\mathcal{E}_\xi(q_{f,h})}. \quad (6.2)$$

Thus (5.6) is Hermitian.

Let

$$(\kappa f)(x) = f(-x).$$

Then

$$q_{\kappa f, \kappa h}(t) = q_{f,h}(-t),$$

so

$$\boxed{W_\xi(\kappa f, \kappa h) = W_\xi(f, h)}. \quad (6.3)$$

This is the core identity from which the completed even/odd decomposition is derived.

43.7 7. Exact odd restriction

Let

$$\mathcal{D}^- = \{f \in C_c^\infty(\mathbb{R}) : f(-x) = -f(x)\}.$$

For $f \in \mathcal{D}^-$,

$$\widehat{f}(-z) = -\widehat{f}(z),$$

and hence

$$\ell_-(f) = -\ell_+(f). \quad (7.1)$$

Define

$$L_\partial f = \sqrt{2} \ell_+(f). \quad (7.2)$$

Theorem 7.1 (Odd negative-rank-one normalization)

For $f, h \in \mathcal{D}^-$,

$$\boxed{B(f, h) = -L_\partial(f) \overline{L_\partial(h)}}. \quad (7.3)$$

Consequently,

$$\boxed{W_\xi^-(f, h) = A_\Gamma(f, h) - P_{1/2}(f, h) - L_\partial(f) \overline{L_\partial(h)}}. \quad (7.4)$$

Proof

Substitute (7.1) into (5.2):

$$\begin{aligned} B(f, h) &= \ell_+(f) \overline{-\ell_+(h)} + (-\ell_+(f)) \overline{\ell_+(h)} \\ &= -2\ell_+(f) \overline{\ell_+(h)}. \end{aligned}$$

Equation (7.2) gives (7.3), and (7.4) follows from (5.5).

This is the exact classical normalization of the native Stage 3F decomposition

$$W_3^- = S_{0,-}^{\text{full}} - L_{\partial}^* L_{\partial}.$$

43.8 8. Multiplicative/logarithmic unitary interface

For $F \in \mathcal{D}_{\times}$, define

$$(\mathcal{L}F)(x) = F(e^x). \tag{8.1}$$

Since $dr/r = dx$, $\mathcal{L}$ is the core Haar isometry. Under the Mellin convention

$$(\mathcal{M}F)(\xi) = \int_0^{\infty} F(r) r^{-i\xi} \frac{dr}{r},$$

one has

$$\boxed{\mathcal{M}F(\xi) = \widehat{\mathcal{L}F}(-\xi).} \tag{8.2}$$

The multiplicative boundary channels satisfy

$$\ell_+^{\times}(F) = \ell_+(\mathcal{L}F), \quad \ell_-^{\times}(F) = \ell_-(\mathcal{L}F). \tag{8.3}$$

For the scale flow

$$(U_t F)(r) = F(e^{-t}r),$$

one has

$$\boxed{\mathcal{L}U_t = T_t \mathcal{L}.} \tag{8.4}$$

Finally,

$$\mathcal{L}J_{\times} = \kappa \mathcal{L}. \tag{8.5}$$

Theorem 8.1

For $F, H \in \mathcal{D}_{\times}$,

$$\boxed{W_{\times}(F, H) = W_{\xi}(\mathcal{L}F, \mathcal{L}H).} \tag{8.6}$$

Proof

Equation (8.3) matches the boundary terms. Equation (8.2), followed by $\xi \mapsto -\xi$, matches the Gamma terms because G is even. Equation (8.4) matches every paired prime dilation with its logarithmic translation. Adding the three identities gives (8.6).

Thus the multiplicative inversion-odd core and the logarithmic odd core are the same test object in unitary charts.

43.9 9. Fourier-sign convention invariance

If the opposite Fourier convention is used,

$$\mathcal{F}_- f(\xi) = \int f(x) e^{-i\xi x} dx = \widehat{f}(-\xi),$$

then:

the two boundary values are interchanged;
the Gamma integral is unchanged because G is even;
the prime translations are interchanged inside each +/- pair.

Therefore the completed Hermitian form is unchanged. The sign of the Fourier exponent is a chart orientation, not a new quadratic form.

43.10 10. Core density and extension

The logarithmic map is a bijection

$$\mathcal{L} : \mathcal{D}_\times \longrightarrow \mathcal{D}.$$

It maps inversion-odd functions to reflection-odd functions. The source carrier theorem gives

$$X_3^- = \overline{\mathcal{D}^-}^{X_3}.$$

The native boundary, Gamma and prime forms are bounded in the native norm. Therefore the termwise identity (5.6), the odd identity (7.4), and the unitary identity (8.6) extend uniquely from the core to the native odd completion.

The classical zero-sum distribution is still consumed on the admissible core. No assertion that the zero sum independently defines the completed Hilbert operator is needed.

43.11 11. Classical membrane

Put

$$z_\rho = \frac{\rho - 1/2}{i}.$$

The classical completed explicit formula in the pinned normalization states

$$\boxed{\mathcal{E}_\xi(q) = \text{sym} \sum_{\rho} \widehat{q}(z_\rho).} \quad (11.1)$$

Combining (11.1) with Theorem 3.1 and Theorem 5.1 gives

$$\boxed{W_\xi(f, h) = \text{sym} \sum_{\rho} \widehat{f}(z_\rho) \overline{\widehat{h}(\overline{z_\rho})}.} \quad (11.2)$$

Stage 3H proves every normalization arrow entering (11.1). It does not claim a new proof of the classical explicit formula itself.

43.12 12. Executable exact packet

The Stage 3H packet uses Gaussian rational Laurent sequences to verify exactly:

```
q=f*h-sharp;
q(t)=<f,T_t h>;
q-hat(z)=f-hat(z) conjugate(h-hat(conjugate z));
boundary distribution = native boundary form;
Gamma distribution = native Gamma form;
prime distribution = paired translation form;
odd boundary = -L_partial*L_partial;
Mellin sign reversal is removed by the even Gamma symbol;
reflection, Hermitian symmetry and Fourier-sign invariance;
finite odd cores converge in a weighted completion.
```

The calibration is not a substitute for the analytic proof. It is a deterministic normalization audit designed to catch missing conjugates, swapped pole channels, incorrect translation signs and lost 2π factors.

43.13 13. Claim boundary

CORRELATION / TRANSLATION IDENTITY	PROVED
ENTIRE CORRELATION TRANSFORM IDENTITY	PROVED
BOUNDARY NORMALIZATION	PROVED
GAMMA NORMALIZATION	PROVED
PRIME-POWER NORMALIZATION	PROVED
ODD NEGATIVE-RANK-ONE RESTRICTION	PROVED
MULTIPLICATIVE / LOGARITHMIC UNITARY NORMALIZATION	PROVED
FOURIER-SIGN AND REFLECTION INVARIANCE	PROVED

ODD CORE DENSITY / BOUNDED EXTENSION

PROVED

CLASSICAL ZERO-SUM EXPLICIT FORMULA

PINNED, NOT REDERIVED

PARITY-RESTRICTED WEIL IMPLICATION

NEXT

RH

NOT CLAIMED

RKF/theorum/40_odd_native_classical_normalization_interface.md

Chapter 44

Certified Cut-Graded Universal Generator Theorem

44.1 1. Certification status

The exact rational certificate and its ten focused tests have passed in the user's local Jupyter worktree. The two repository workflows also passed on the branch:

RKF proof-lab CI	PASS
Recognition Kernel Review	PASS
user-local exact theorem suite	PASS

The deterministic certificate target is

PASS_CUT_GRADED_UNIVERSAL_GENERATOR_CANDIDATE

with pinned SHA-256

34afc44543cd83cacd96cbced32b77f5fdd765c28fbaf81105d0d2063cf5f36e

The local console log and generated `ACTUAL.json` are not committed by this status update; the user has reported the complete prescribed run as passing. The algebraic theorem below is therefore promoted from candidate to the certified central theorem of this branch. No claim is made that one particular physical operator has already been proved universal.

44.2 2. Native data

Let $\mathcal{H}$ be a Hilbert space and let

$$J = J^* = J^{-1}$$

be the primitive cut. Let G be a bounded operator on $\mathcal{H}$. For an unbounded generator, the same formulas are licensed on a common invariant core provided

$$J \operatorname{Dom}(G) = \operatorname{Dom}(G),$$

and the relevant closures and functional calculi exist.
Define the cut projections

$$P_{\pm} = \frac{I \pm J}{2}.$$

The generator is not treated as an untyped sum of winding, phase, memory and cost coordinates.
Its first lawful structure is its grading relative to the cut.

44.3 3. Central theorem

Theorem 3.1 (Certified cut-graded universal-generator theorem)

Define

$$\boxed{G_e = \frac{1}{2}(G + JGJ), \quad G_o = \frac{1}{2}(G - JGJ).} \quad (3.1)$$

Then the following statements hold.

(i) Unique cut grading

$$\boxed{G = G_e + G_o,} \quad (3.2)$$

$$\boxed{JG_eJ = G_e, \quad JG_oJ = -G_o.} \quad (3.3)$$

The decomposition is unique. Moreover,

$$P_+G_eP_- = P_-G_eP_+ = 0, \quad (3.4)$$

and

$$P_+G_oP_+ = P_-G_oP_- = 0. \quad (3.5)$$

Thus G_e preserves the two sheets, while G_o transports between them.

(ii) Bilateral exponential reconstruction

Assume G is cut odd, so $JGJ = -G$, and write

$$U_t = e^{tG}.$$

Then

$$\boxed{JU_tJ = U_{-t}.} \quad (3.6)$$

Define

$$E_t = \frac{1}{2}(U_t + U_{-t}) = \cosh(tG), \quad O_t = \frac{1}{2}(U_t - U_{-t}) = \sinh(tG). \quad (3.7)$$

Then

$$JE_tJ = E_t, \quad JO_tJ = -O_t, \quad (3.8)$$

$$\boxed{U_t = E_t + O_t, \quad U_{-t} = E_t - O_t,} \quad (3.9)$$

and

$$\boxed{E_t^2 - O_t^2 = I.} \quad (3.10)$$

(iii) Exponential cut-square identity

For the derived channels

$$\widehat{J}_t = U_t + U_{-t} = 2 \cosh(tG), \quad \widehat{C}_t = U_t - U_{-t} = 2 \sinh(tG), \quad (3.11)$$

one has

$$\boxed{\widehat{J}_t^2 - \widehat{C}_t^2 = 4I,} \quad (3.12)$$

and

$$\boxed{\widehat{J}_t \widehat{C}_t = \widehat{C}_t \widehat{J}_t = U_{2t} - U_{-2t} = 2 \sinh(2tG).} \quad (3.13)$$

Consequently the two cut/join products cannot be assigned different values when both channels are functions of the same generator.

(iv) Cut-loop memory and seam curvature

For a general generator, define

$$\boxed{\mathcal{H}_J(t) = JU_tJU_t = e^{tGGJ}e^{tG}.} \quad (3.14)$$

Then

$$\boxed{\mathcal{H}_J(t) = I + 2tG_e + t^2(2G_e^2 + [G_e, G_o]) + O(t^3),} \quad (3.15)$$

and

$$\boxed{\log \mathcal{H}_J(t) = 2tG_e + t^2[G_e, G_o] + O(t^3).} \quad (3.16)$$

Hence

$$\boxed{G_e = \frac{1}{2} \left. \frac{d}{dt} \right|_{t=0} \mathcal{H}_J(t),} \quad (3.17)$$

and the first noncommutative seam-curvature coefficient is

$$\boxed{\mathcal{R}_{\text{seam}} = [G_e, G_o].} \quad (3.18)$$

The cut loop is locally the identity exactly when the generator is cut odd:

$$\boxed{\mathcal{H}_J(t) = I \text{ for all sufficiently small } t \iff G_e = 0.} \quad (3.19)$$

(v) Periodic and antiperiodic closure

If

$$G\psi = i\omega\psi, \quad \omega \in \mathbb{R},$$

then for every integer $N \geq 1$,

$$U_N\psi = +\psi \iff N\omega \in 2\pi\mathbb{Z}, \quad (3.20)$$

while

$$U_N\psi = -\psi \iff N\omega \in (2\mathbb{Z} + 1)\pi. \quad (3.21)$$

For $\sigma \in \{+1, -1\}$, define

$$D_{N,\sigma} = U_N - \sigma I, \quad S_{N,\sigma} = D_{N,\sigma}^* D_{N,\sigma}. \quad (3.22)$$

Then

$$\boxed{S_{N,\sigma} \geq 0, \quad \ker S_{N,\sigma} = \ker(U_N - \sigma I).} \quad (3.23)$$

The $+1$ and -1 closure sectors are distinct. A condition of the form $N\lambda = 2\pi im$ covers only the periodic sector.

(vi) Clock-free Eye

Suppose

$$U_t = e^{-itH}, \quad H = H^*.$$

Then the one-step Eye is

$$\boxed{\text{Fix}(U_1) = E_H(2\pi\mathbb{Z})\mathcal{H},} \quad (3.24)$$

whereas the clock-free Eye is

$$\boxed{\bigcap_{t \in \mathbb{R}} \text{Fix}(U_t) = \ker H.} \quad (3.25)$$

Its mean-ergodic projector is

$$\boxed{P_{\text{Eye}} = \text{s-lim}_{T \rightarrow \infty} \frac{1}{T} \int_0^T U_t dt = E_H(\{0\}).} \quad (3.26)$$

Therefore

$$\boxed{\mathfrak{B}_{\text{clock}} = \text{Fix}(U_1) / \ker H} \quad (3.27)$$

is the stroboscopic recognition-blind sector.

(vii) Faithful component observer

Let $P_1, \dots, P_m$ be pairwise orthogonal projections satisfying

$$\sum_{a=1}^m P_a = I,$$

and define

$$G_a = P_a G, \quad \mathcal{A}_G x = (G_1 x, \dots, G_m x). \quad (3.28)$$

Then

$$\boxed{\mathcal{A}_G^* \mathcal{A}_G = \sum_{a=1}^m G_a^* G_a = G^* G,} \quad (3.29)$$

and hence

$$\boxed{\ker \mathcal{A}_G = \ker G.} \quad (3.30)$$

If named components are not proved to arise from an orthogonal complete projection family, their lawful observer is the direct sum, not merely their aggregate. An aggregate can vanish by cancellation while the direct-sum energy remains positive.

44.4 4. Proof

The formulas in (i) follow from $J^2 = I$. If $G = A + B$, $JAJ = A$, and $JBJ = -B$, then averaging G and JGJ recovers A and B , proving uniqueness. Multiplication by the cut projections gives the block-support identities.

Under $JGJ = -G$, functional calculus gives (3.6). Equations (3.7)–(3.10) follow by symmetric and antisymmetric reconstruction. Since the derived channels are functions of one operator, they commute; expanding their squares and product gives (3.12)–(3.13).

For the cut loop, substitute

$$JGJ = G_e - G_o, \quad G = G_e + G_o,$$

and multiply the two exponential series. This gives (3.15). The standard second-order logarithm expansion gives (3.16). The first derivative gives (3.17), and the quadratic logarithmic coefficient gives (3.18). If the loop is locally the identity, its derivative vanishes, so $G_e = 0$. The converse follows from $JU_t J = U_{-t}$.

The closure statements follow by applying U_N to an eigenvector. The positive source identity (3.23) is immediate from the definition of $D_{N,\sigma}$.

The spectral theorem gives (3.24)–(3.26). In particular, integer nonzero frequencies can be invisible to a one-step fixed-point test, which produces the quotient (3.27).

Finally,

$$\sum_a G_a^* G_a = G^* \left(\sum_a P_a \right) G = G^* G,$$

which proves the component-observer statement. $\square$

44.5 5. Recognition-Kernel flow consequence

For an observer $B : \mathcal{H} \rightarrow \mathcal{Y}$ and times $\mathbf{t} = (t_1, \dots, t_N)$, define

$$\mathcal{A}_{\mathbf{t}}x = (BU_{t_1}x, \dots, BU_{t_N}x). \quad (5.1)$$

For a target functional L , put

$$\beta_{\mathbf{t}}(L) = \sup_{\mathcal{A}_{\mathbf{t}}x \neq 0} \frac{|Lx|^2}{\|\mathcal{A}_{\mathbf{t}}x\|^2}. \quad (5.2)$$

Whenever the burden is finite, the Recognition-Kernel Theorem supplies a unique minimum decoder

$$c_L \in \overline{\text{Ran } \mathcal{A}_{\mathbf{t}}}$$

with

$$Lx = \langle \mathcal{A}_{\mathbf{t}}x, c_L \rangle, \quad \|c_L\|^2 = \beta_{\mathbf{t}}(L), \quad (5.3)$$

and the exact flow cut-square criterion

$$\boxed{\mathcal{A}_{\mathbf{t}}^* \mathcal{A}_{\mathbf{t}} - L^* L \geq 0 \iff \beta_{\mathbf{t}}(L) \leq 1.} \quad (5.4)$$

Thus the target-relative cost of recognizing a flow is a decoder burden, not an unproved trace formula.

44.6 6. Corrected interpretation of the generator tower

The iteration

$$G_{n+1} = 2 \sinh(G_n)$$

does not converge to zero for a general real initial value, because the scalar map has derivative $2 > 1$ at zero. The lawful cut coordinate is instead

$$C(G) = 2 \sinh(G/2),$$

with inverse, wherever the functional calculus permits,

$$\boxed{G = 2 \operatorname{arsinh}(C(G)/2).} \quad (6.1)$$

This is a reversible coordinate transformation, not a convergence theorem.

44.7 7. Certified exact packet

The passing rational packet verifies:

```
unique cut-even/cut-odd decomposition;
cut-diagonal and trans-cut block support;
cut-loop linear memory coefficient;
cut-loop quadratic seam-curvature coefficient;
exact nilpotent odd bilateral flow;
exponential cut-square identity;
equality of the two derived cut/join products;
periodic and antiperiodic closure characters;
single-step versus all-time Eye;
orthogonal component Gram preservation;
nonorthogonal aggregate cancellation negative control.
```

The implementation uses `fractions.Fraction` only. No NumPy, floating point, fitted unitary, source inverse, spectral eigensolver, or post-hoc Gram factor is used.

44.8 8. Updated claim boundary

CUT-GRADED BOUNDED-OPERATOR THEOREM	PROVED
BILATERAL EXPONENTIAL RECONSTRUCTION	PROVED
EXPONENTIAL CUT-SQUARE	PROVED
CUT-LOOP MEMORY / SEAM-CURVATURE EXPANSION	PROVED
PERIODIC / ANTIPERIODIC CLOSURE CLASSIFICATION	PROVED
CLOCK-FREE EYE DISTINCTION	PROVED
ORTHOGONAL COMPONENT OBSERVER	PROVED
EXACT RATIONAL CERTIFICATE	USER-REPORTED PASS
REPOSITORY PROOF-LAB CI	PASS
RECOGNITION KERNEL REVIEW	PASS
UNBOUNDED UNIVERSAL-GENERATOR DOMAIN THEOREM	REQUIRES DOMAIN PINS
PHYSICAL IDENTIFICATION OF ONE UNIVERSAL GENERATOR	NOT CLAIMED
MASTER COST/MEMORY FLOW AS ONE CLOSED GENERATOR	NEXT DEVELOPMENT
CENTRAL THEOREM STATUS	CERTIFIED ON THIS BRANCH

RKF/theorum/41_cut_graded_universal_generator_theorem.md

Chapter 45

Cut-Graded Lambda-Jacobian Tower and Thermodynamic Recognition Theorem

45.1 1. Source problem and scope

The thermodynamic paper [arXiv:2603.20773v2](#) introduces:

a signed T-V-S-P compass;
a four-component lambda response map;
a first Jacobian layer;
recursive higher derivative layers;
a jet-bundle analogy;
response one-forms and curvature-based cyclic residues.

The source manuscript explicitly describes the jet-bundle relation as a geometric analogy rather than a rigorous construction. It also treats the nonequilibrium curvature and information-sector extensions as constitutive, calibrated hypotheses requiring further validation.

The present theorem supplies the missing abstract object:

smooth state manifold and response bundle
-> declared cut on state and response fibres
-> covariant derivative raising operator
-> typed recursive response tower
-> induced cut grading at every derivative order
-> cut/transpose bi-grading of the Jacobian
-> cut-loop mixed curvature
-> target-relative finite tower observer.

This theorem does **not** prove that the particular physical lambda map in the source paper is globally antisymmetric, that its proposed curvature is already entropy production, or that every phase transition is a rank-change cut. Those statements require separate constitutive and experimental pins.

45.2 2. Native thermodynamic data

Let $\mathbf{M}$ be a smooth four-dimensional state manifold. On a declared chart write

$$x = (T, V, S, P).$$

The coordinates must be nondimensionalized or treated as sections of their proper dimensional bundles before a linear involution is imposed.

Let $\mathbf{E} \rightarrow \mathbf{M}$ be a rank-four response bundle and let

$$\Lambda \in \Gamma(E)$$

be a smooth response section. In the source application, the local components are the four constraint-direction responses

$$\Lambda = (\lambda_p, \lambda_v, \lambda_s, \lambda_t),$$

up to the declared ordering convention.

Let

$$j : M \longrightarrow M, \quad j^2 = \text{id}_M,$$

be a state-space cut and let

$$J_E : E \longrightarrow E, \quad J_E^2 = I,$$

cover j . Let ∇ be a connection on $\mathbf{E}$ satisfying cut covariance:

$$\boxed{\nabla(j^*s) = j^*(\nabla s)} \tag{2.1}$$

in the induced tensor representation.

A compass candidate is the local state involution

$$J_x = \text{diag}(1, 1, -1, -1), \tag{2.2}$$

but the physical state and response involutions must be derived from the signed thermodynamic axes and checked against the lambda definitions. They are not licensed merely by visual symmetry.

45.3 3. The rigorous recursive tower

Define the response layers by

$$L_1 := \Lambda, \quad L_{n+1} := \nabla L_n \quad (n \geq 1). \tag{3.1}$$

Then

$$L_n \in \Gamma\left(E \otimes (T^*M)^{\otimes(n-1)}\right). \tag{3.2}$$

Thus:

```

L1 response section;
L2 covariant Jacobian;
L3 covariant Hessian;
L4 third response derivative;
...

```

Theorem 3.1 (Typed lambda-Jacobian tower)

For every smooth response section **Lambda**, equation (3.1) defines a unique covariant derivative tower. In a flat chart,

$$L_n = D^{n-1} \Lambda. \quad (3.3)$$

If **Lambda** is polynomial of degree **d** in that chart, then

$$L_n = 0 \quad (n > d + 1). \quad (3.4)$$

The formal exponential generating object is

$$\boxed{\mathcal{T}_t \Lambda := \sum_{k=0}^{\infty} \frac{t^k}{k!} \nabla^k \Lambda,} \quad (3.5)$$

whenever the series converges in the chosen tensor topology. The recursive layers are its Taylor coefficients. The tower is therefore generated by one raising operator rather than by repeatedly reusing a diagrammatic rule without a type map.

Proof

A connection extends uniquely to tensor products by the Leibniz rule. Repeated application gives (3.2). In a flat chart the connection coefficients vanish, so the layers are ordinary higher derivatives. Polynomial termination follows from elementary differentiation. Equation (3.5) is the exponential series of the raising operator on the completed direct sum of tensor levels. **square**

45.4 4. Induced cut on every layer

Let $J_{\mathbf{x}}^*$ denote the induced involution on covectors. On layer **n**, define

$$\boxed{\mathbb{J}_n = J_E \otimes (J_x^*)^{\otimes(n-1)}.} \quad (4.1)$$

Then

$$\mathbb{J}_n^2 = I.$$

Define

$$\boxed{L_n^{\pm} = \frac{1}{2} (L_n \pm \mathbb{J}_n L_n).} \quad (4.2)$$

Theorem 4.1 (Cut-graded response tower)

Every response layer has the unique decomposition

$$\boxed{L_n = L_n^+ + L_n^-} \quad (4.3)$$

Moreover,

$$\mathbb{J}_n L_n^+ = L_n^+, \quad \mathbb{J}_n L_n^- = -L_n^-. \quad (4.4)$$

If the response section is cut equivariant,

$$\Lambda(jx) = J_E \Lambda(x), \quad (4.5)$$

then for every $k \geq 1$,

$$D^k \Lambda(jx)[J_x h_1, \dots, J_x h_k] = J_E D^k \Lambda(x)[h_1, \dots, h_k]. \quad (4.6)$$

At a fixed seam $j\mathbf{x}=\mathbf{x}$, every derivative layer obeys the induced parity selection rule.

Proof

Equations (4.2)–(4.4) are the spectral projections of an involution. Differentiate (4.5) repeatedly and apply the chain rule to obtain (4.6). **square**

Interpretation:

L_n^+ response derivatives preserving the declared cut class;
 L_n^- response derivatives transporting across the declared cut class.

This grading is defined at every order, not only at the first Jacobian.

45.5 5. The Jacobian is bi-graded, not automatically antisymmetric

Assume locally that the state and response fibres have the same dimension and use one declared involution J . Let

$$A = D\Lambda(x) \in M_4(\mathbb{R}).$$

Define two commuting involutions on matrices:

$$\mathcal{C}(A) = JAJ, \quad \mathcal{T}(A) = A^\top. \quad (5.1)$$

For $\sigma, \tau \in \{+1, -1\}$, define

$$\boxed{A^{\sigma, \tau} = \frac{1}{4} \left(A + \sigma JAJ + \tau A^\top + \sigma \tau J A^\top J \right)}. \quad (5.2)$$

Theorem 5.1 (Cut/transpose Jacobian decomposition)

The four sectors satisfy

$$A = \sum_{\sigma, \tau \in \{\pm 1\}} A^{\sigma, \tau}, \quad (5.3)$$

$$JA^{\sigma, \tau}J = \sigma A^{\sigma, \tau}, \quad (5.4)$$

and

$$(A^{\sigma, \tau})^\top = \tau A^{\sigma, \tau}. \quad (5.5)$$

Therefore the Jacobian contains four logically distinct pieces:

```
cut-even / symmetric;
cut-even / antisymmetric;
cut-odd  / symmetric;
cut-odd  / antisymmetric.
```

Antisymmetry

$$A^\top = -A \quad (5.6)$$

is equivalent to the vanishing of both symmetric sectors. It is a theorem to be proved from the constitutive lambda map and coordinate convention, not an automatic consequence of having a T-V-S-P compass.

Proof

The maps $\mathcal{C}$ and $\mathcal{T}$ commute and square to the identity. Equation (5.2) is the joint spectral projection formula for two commuting involutions. Equations (5.3)–(5.5) follow immediately. **square**

This theorem separates Maxwell-type antisymmetric response from symmetric susceptibility or dissipative response and separately identifies whether each piece is within-sheet or trans-cut.

45.6 6. Cut-loop curvature and a lawful thermodynamic adapter

Let $\mathcal{G}$ be a response-space protocol generator and decompose it using the certified central theorem:

$$G = G_e + G_o, \quad JG_eJ = G_e, \quad JG_oJ = -G_o. \quad (6.1)$$

Let

$$U_t = e^{tG}, \quad \mathcal{H}_J(t) = JU_tJU_t. \quad (6.2)$$

Then

$$\log \mathcal{H}_J(t) = 2tG_e + t^2[G_e, G_o] + O(t^3). \quad (6.3)$$

Now consider the constant connection one-form on a two-parameter protocol chart $(\mathbf{s}, \mathbf{t})$:

$$\mathcal{A} = G_e ds + G_o dt. \quad (6.4)$$

Its curvature is

$$\mathcal{F} = d\mathcal{A} + \mathcal{A} \wedge \mathcal{A},$$

so

$$\boxed{\mathcal{F}_{st} = [G_e, G_o].} \quad (6.5)$$

Theorem 6.1 (Cut-loop/connection curvature identity)

For the constant cut-graded connection (6.4), the quadratic coefficient in the logarithmic cut loop is exactly the mixed curvature component:

$$\boxed{[t^2] \log \mathcal{H}_J(t) = \mathcal{F}_{st}.} \quad (6.6)$$

Proof

Equation (6.3) is the cut-loop extraction theorem. For constant connection coefficients, derivative terms vanish and the mixed curvature is the commutator. **square**

This is an exact finite-dimensional curvature model. To identify it with a thermodynamic response two-form such as

$$\Omega_H = d\omega_H,$$

one must construct and prove an adapter

$$\boxed{\rho([G_e, G_o]) = \Omega_H.} \quad (6.7)$$

Equation (6.7) is the next physical theorem. It is not supplied merely by the abstract commutator identity.

45.7 7. Finite tower observers and recognition kernels

Let $\mathbf{V}$ be a finite-dimensional model or perturbation space. For each tower level let

$$Q_n L_n : V \longrightarrow Y_n$$

be the declared measurable response packet at order $\mathbf{n}$. Define the order- $\mathbf{N}$ observer

$$\boxed{\mathcal{A}_N v = (Q_1 L_1 v, \dots, Q_N L_N v).} \quad (7.1)$$

For a target map $\Pi : \mathbf{V} \rightarrow \mathbf{Z}$, define the target-relative blind quotient

$$\boxed{\mathfrak{B}_N = \frac{\ker \mathcal{A}_N}{\ker \mathcal{A}_N \cap \ker \Pi}.} \quad (7.2)$$

Theorem 7.1 (Tower monotonicity and target repair)

The recognition kernels are nested:

$$\boxed{\ker \mathcal{A}_{N+1} \subseteq \ker \mathcal{A}_N.} \quad (7.3)$$

A higher response layer repairs a first-order blind target precisely when

$$\ker \mathcal{A}_{N+1} \cap \text{supp } \Pi \subsetneq \ker \mathcal{A}_N \cap \text{supp } \Pi. \quad (7.4)$$

For a scalar target L , the Recognition-Kernel burden is

$$\beta_N(L) = \sup_{\mathcal{A}_N v \neq 0} \frac{|L(v)|^2}{\|\mathcal{A}_N v\|^2}. \quad (7.5)$$

Whenever it is finite, the minimum decoder exists and

$$\mathcal{A}_N^* \mathcal{A}_N - L^* L \geq 0 \iff \beta_N(L) \leq 1. \quad (7.6)$$

Proof

Adding an observer component can only shrink the common kernel, proving (7.3). Equation (7.4) is the definition of strict target-relevant repair. Equation (7.6) is the Recognition-Kernel Theorem applied to the direct-sum tower observer. **square**

This theorem asks experimentally meaningful questions:

Which derivative order first recognizes a target?
 Which response components form the minimum faithful packet?
 Does a higher-order susceptibility repair a first-order blind direction?

45.8 8. Rank-change cuts

Let $A_1(\mathbf{r})$ be a first-layer observer depending continuously on a control parameter $\mathbf{r}$. A state $\mathbf{r}_c$ is a first-layer rank-change cut when

$$\text{rank } A_1(r_c) < \limsup_{r \rightarrow r_c} \text{rank } A_1(r). \quad (8.1)$$

Let $A_{\leq N}(\mathbf{r})$ be the observer obtained by stacking response layers through order N .

Proposition 8.1 (Higher-layer recovery at a rank-change cut)

If

$$\ker A_1(r_c) \cap \ker A_{\leq N}(r_c) = \{0\}, \quad (8.2)$$

then the higher-order tower resolves every first-order blind direction at the cut.

Proof

A vector blind to the first layer but also blind to the full tower lies in the intersection in (8.2), hence must vanish. **square**

A thermodynamic phase transition may motivate such a test because the source paper restricts its smooth lambda map away from response singularities. However, calling every phase transition a rank-change cut requires data and a specific constitutive observer; it is not an abstract consequence of (8.1).

45.9 9. Exact rational certificate

The proof-lab packet uses a cut-equivariant quadratic response map on four coordinates and verifies exactly:

```
value-level cut equivariance;
Jacobian covariance;
induced Hessian covariance;
typed recursive layers through Hessian order;
termination of the quadratic tower;
four-sector cut/transpose Jacobian reconstruction;
nonzero symmetric and antisymmetric sectors;
negative control rejecting automatic Jacobian antisymmetry;
constant-connection curvature equals cut-loop seam curvature;
first-order target blindness;
second-layer target repair;
minimum decoder burden 1/4 and reserve 3/4;
first-layer rank loss at a cut;
higher-layer recovery of the cut kernel.
```

No NumPy, floating point, fitted coordinate transform, or post-hoc Gram factor is used.
Expected status:

PASS_CUT_GRADED_LAMBDA_JACOBIAN_TOWER_CANDIDATE

Expected SHA-256:

3d8db4f085a3b624192c45b9f6e0356b600834193967bc89478381b53ae44a3a

45.10 10. Claim boundary

TYPED COVARIANT RESPONSE TOWER	PROVED
INDUCED CUT GRADING AT EVERY ORDER	PROVED
CUT/TRANSPOSE JACOBIAN BI-GRADING	PROVED
ANTISYMMETRY AS A SECTOR CONDITION	PROVED
CONSTANT-CONNECTION CUT-LOOP CURVATURE IDENTITY	PROVED
FINITE TOWER RECOGNITION-KERNEL THEOREM	PROVED
RANK-CHANGE / HIGHER-LAYER RECOVERY PROPOSITION	PROVED
EXACT RATIONAL CERTIFICATE	IMPLEMENTED / USER RUN REQUIRED
PHYSICAL T-V-S-P CUT AND RESPONSE-FIBRE INVOLUTION	OPEN / NEXT

CUT-COMPATIBLE THERMODYNAMIC CONNECTION	OPEN / NEXT
OPERATOR CURVATURE -> RESPONSE TWO-FORM ADAPTER	OPEN / NEXT
GLOBAL ANTISYMMETRY OF THE PAPER'S PHYSICAL JACOBIAN	NOT PROMOTED
CURVATURE = ENTROPY PRODUCTION	REQUIRES CALIBRATION
PHASE TRANSITION = RANK-CHANGE CUT	TESTABLE, NOT CLAIMED
SOURCE PAPER UPDATE	HELD UNTIL CERTIFICATE PASS

RKF/theorum/42_cut_graded_lambda_jacobian_tower_theorem.md

Chapter 46

Bilateral Jet-Flow Recognition Capstone Theorem

46.1 1. Status and purpose

This theorem joins the certified cut-graded generator theorem with the typed lambda-Jacobian tower. Its purpose is to provide one lawful interface for a later physical adapter:

```
cut-graded generator
-> bilateral Euler/recognition flow
-> finite response jets
-> explicit remainder
-> nested recognition kernels
-> target burden
-> seam-curvature response
-> fail-closed classification.
```

The exact rational packet in `proof_lab/bilateral_jet_flow_capstone.py` verifies a finite nilpotent model and pins the candidate certificate hash. The theorem below is an abstract bounded-operator statement. Unbounded generators require the domain conditions stated in Section 10.

The term *bilateral strands* is used deliberately. A physical double-helical interpretation requires a separate adapter and is not smuggled in by drawing two arrows around a circle, humanity's traditional method of discovering unearned geometry.

46.2 2. Native data

Let $\mathcal{H}$ and $\mathcal{Y}$ be Hilbert spaces. Let

$$J = J^* = J^{-1}$$

be the primitive cut on $\mathcal{H}$. Let G be a bounded generator and

$$U_t = e^{tG}.$$

Let

$$B : \mathcal{H} \rightarrow \mathcal{Y}$$

be a bounded observer. For the principal bilateral theorem assume

$$JGJ = -G, \quad BJ = B. \quad (2.1)$$

The first condition says that the generator transports across the cut. The second says that the declared observer reads the cut-even output channel.

Define the order- k observer jet

$$\Lambda_k := BG^k, \quad k \geq 0. \quad (2.2)$$

and the finite jet observer

$$\mathcal{A}_N x = (\Lambda_0 x, \Lambda_1 x, \dots, \Lambda_N x). \quad (2.3)$$

For a scalar target L , define the order- N burden

$$\beta_N(L) = \sup_{\mathcal{A}_N x \neq 0} \frac{|Lx|^2}{\|\mathcal{A}_N x\|^2}. \quad (2.4)$$

46.3 3. Bilateral flow and strand closure

Theorem 3.1 (Bilateral cut conjugacy)

Under (2.1),

$$JU_t J = U_{-t}. \quad (3.1)$$

Define the state channels

$$\hat{J}_t = U_t + U_{-t}, \quad \hat{C}_t = U_t - U_{-t}. \quad (3.2)$$

Then

$$\hat{J}_t^2 - \hat{C}_t^2 = 4I, \quad (3.3)$$

and

$$\hat{J}_t \hat{C}_t = \hat{C}_t \hat{J}_t. \quad (3.4)$$

The forward and backward observer strands are

$$S_+(t) = BU_t, \quad S_-(t) = BU_{-t}. \quad (3.5)$$

Their cut-conjugacy defect is

$$\Delta_H(t) = BJU_t J - BU_{-t}. \quad (3.6)$$

Therefore

$$\Delta_H(t) = 0. \quad (3.7)$$

Proof

Functional calculus applied to $JGJ = -G$ gives (3.1). Since $U_t U_{-t} = I$ and both flows are functions of G , the square and commutation identities follow. Equation (3.7) follows from $BJ = B$. $\square$

A nonzero Δ_H is not a decorative error bar. It means that at least one of the declared cut, generator, observer or domain covariance conditions has failed.

46.4 4. Alternating cut parity of the jet tower

Theorem 4.1 (Jet parity ladder)

For every $k \geq 0$,

$$\boxed{\Lambda_k J = (-1)^k \Lambda_k.} \quad (4.1)$$

Thus even jet levels are cut even and odd jet levels are cut odd.

Proof

From $JGJ = -G$ one obtains $GJ = -JG$, hence

$$G^k J = (-1)^k J G^k.$$

Multiplying by B and using $BJ = B$ proves (4.1). $\square$

This is the precise bridge between the generator grading and the recursive response tower. Parity is inherited at every order rather than assigned after seeing a convenient pattern.

46.5 5. Finite jet reconstruction and remainder

Theorem 5.1 (Observed-flow jet formula)

For every integer $N \geq 0$ and $t \geq 0$,

$$\boxed{BU_t = \sum_{k=0}^N \frac{t^k}{k!} \Lambda_k + R_{N+1}(t),} \quad (5.1)$$

where

$$\boxed{R_{N+1}(t) = \frac{1}{N!} \int_0^t (t-s)^N B G^{N+1} U_s ds.} \quad (5.2)$$

Consequently,

$$\boxed{\|R_{N+1}(t)\| \leq \frac{t^{N+1}}{(N+1)!} \|B\| \|G\|^{N+1} e^{t\|G\|}.} \quad (5.3)$$

If $G^{d+1} = 0$, then

$$\boxed{R_{d+1}(t) = 0} \tag{5.4}$$

and the order- d jet reconstructs the observed flow exactly.

Proof

Apply the integral remainder formula for the bounded exponential series and left-compose with B . Nilpotent termination gives (5.4). $\square$

Equation (5.2) is the repository's exact native series/remainder interface. It can later consume a separately proved Madhava calculus, but the present repository contains no Madhava-named operator or certificate, so that name is not claimed here.

46.6 6. Bilateral even and odd response strands

Define

$$S_e(t) = \frac{1}{2}(S_+(t) + S_-(t)), \quad S_o(t) = \frac{1}{2}(S_+(t) - S_-(t)). \tag{6.1}$$

Theorem 6.1 (Bilateral strand reconstruction)

Whenever the exponential series converges,

$$\boxed{S_e(t) = \sum_{m \geq 0} \frac{t^{2m}}{(2m)!} \Lambda_{2m}}, \tag{6.2}$$

$$\boxed{S_o(t) = \sum_{m \geq 0} \frac{t^{2m+1}}{(2m+1)!} \Lambda_{2m+1}}. \tag{6.3}$$

Moreover,

$$\boxed{S_+ = S_e + S_o, \quad S_- = S_e - S_o}. \tag{6.4}$$

Proof

Add and subtract the power series for BU_t and BU_{-t} . $\square$

These are the abstract two strands available for a later helical adapter. A physical braid, phase angle or twist memory must be defined from physical data and proved to intertwine these strands.

46.7 7. Recognition kernels and first repair order

Let

$$K_N = \ker \mathcal{A}_N.$$

Theorem 7.1 (Monotone blindness reduction)

For every N ,

$$\boxed{K_{N+1} \subseteq K_N.} \quad (7.1)$$

For a target L , define its first recognition order by

$$\boxed{\nu(L) = \inf\{N : K_N \subseteq \ker L\}.} \quad (7.2)$$

If $\nu(L) = r < \infty$, then the target is blind through order $r - 1$ and is repaired at order r .

Proof

$\mathcal{A}_{N+1}$ contains every component of $\mathcal{A}_N$ plus one additional jet layer, so its common kernel can only shrink. Equation (7.2) is therefore well defined whenever some finite tower recognizes the target. $\square$

The theorem does not say that every additional layer is useful. It says that a new layer cannot restore a direction already excluded by the earlier observer. Strict repair must be demonstrated target by target.

46.8 8. Decoder burden and admissibility

Whenever $\beta_N(L)$ is finite, the Recognition-Kernel theorem supplies a minimum decoder $c_{L,N}$ satisfying

$$Lx = \langle \mathcal{A}_N x, c_{L,N} \rangle, \quad \|c_{L,N}\|^2 = \beta_N(L). \quad (8.1)$$

The exact admissibility criterion is

$$\boxed{\mathcal{A}_N^* \mathcal{A}_N - L^* L \geq 0 \iff \beta_N(L) \leq 1.} \quad (8.2)$$

Thus higher-order recognition and low decoder burden are separate obligations: a target may become visible yet remain too expensive to certify at unit reserve.

46.9 9. Seam curvature consumed by the jet observer

Now allow a general bounded generator

$$G = G_e + G_o, \quad JG_e J = G_e, \quad JG_o J = -G_o. \quad (9.1)$$

Define the cut loop

$$\mathcal{H}_J(t) = J e^{tG} J e^{tG}. \quad (9.2)$$

The universal-generator theorem gives

$$\log \mathcal{H}_J(t) = 2tG_e + t^2[G_e, G_o] + O(t^3). \quad (9.3)$$

Therefore the observed quadratic seam coefficient is

$$\boxed{[t^2]B \log \mathcal{H}_J(t) = B[G_e, G_o]}. \quad (9.4)$$

Equation (9.4) is the exact abstract curvature adapter. A physical model must still prove what its measured response packet B is and why the right-hand side represents the intended physical curvature.

46.10 10. Capstone classification

For one declared target and tower depth, assign:

```

OPEN_SEAM
  if the bilateral closure defect is nonzero;

ABSTAIN
  if the target remains blind or no finite decoder exists;

BURDEN_EXCEEDS_ONE
  if the target is visible but beta_N(L) > 1;

HIGHER_LAYER_REPAIR
  if the target was blind at a lower layer, becomes visible at N,
  and beta_N(L) <= 1;

CLOSED
  if the target is already visible, the closure defect vanishes,
  and beta_N(L) <= 1.
```

This classification is fail-closed: algebraic inconsistency is checked before predictive ambition.

46.11 11. Exact rational certificate

The finite certificate uses

$$J = \text{diag}(1, -1, 1, -1),$$

$$G = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad G^4 = 0,$$

and

$$B = (1, 0, 0, 0).$$

Then

$$BG^0 = e_1^*, \quad BG = e_2^*, \quad BG^2 = e_3^*, \quad BG^3 = e_4^*,$$

so the jet observers have ranks

$$1, 2, 3, 4$$

and kernel dimensions

$$3, 2, 1, 0.$$

The target

$$L = \frac{1}{2}e_4^*$$

is blind through second order and repaired at third order. Since the full jet observer is the identity,

$$\beta_3(L) = \frac{1}{4}, \quad 1 - \beta_3(L) = \frac{3}{4}.$$

The certificate also uses a wrong-cut negative control and verifies that its bilateral closure defect is nonzero.

Expected status:

PASS_BILATERAL_JET_FLOW_CAPSTONE_CANDIDATE

Expected SHA-256:

4f0f31183c6de9b354fbc90134cfcd14c8d6f55270f41c1e286f719c0e06b091

46.12 12. Analytic and physical claim boundary

BOUNDED CUT-ODD BILATERAL FLOW	PROVED
JET PARITY LADDER	PROVED
BOUNDED OBSERVED-FLOW REMAINDER FORMULA	PROVED
NILPOTENT EXACT JET RECONSTRUCTION	PROVED
BILATERAL EVEN/ODD STRAND RECONSTRUCTION	PROVED
MONOTONE RECOGNITION-KERNEL REDUCTION	PROVED
TARGET FIRST-RECOGNITION ORDER	PROVED
DECODER BURDEN ADMISSIBILITY	INHERITED / PROVED
OBSERVED SEAM-CURVATURE COEFFICIENT	PROVED
EXACT RATIONAL FINITE CERTIFICATE	LOCAL PASS
UNBOUNDED GENERATOR DOMAIN CLOSURE	REQUIRES DOMAIN PINS
PHYSICAL DOUBLE-HELIX IDENTIFICATION	NOT CLAIMED
MADHAVA-NAMED CALCULUS	NOT PRESENT IN REPO
NUCLEAR CUT / OBSERVER / TARGET ADAPTER	NEXT PHYSICAL THEOREM
UNIVERSAL PREDICTIVE IMPROVEMENT	NOT CLAIMED

RKF/theorum/43_bilateral_jet_flow_recognition_capstone_theorem.md

Chapter 47

Madhava–Smriti Bilateral Jet-Flow Closure Theorem

47.1 1. Status, source lineage and purpose

This theorem consumes two already developed structures:

```
Vedic Recognition Geometry / Madhava corrected-recursion grammar
+
Bilateral Jet-Flow Recognition Capstone (Theorem 43)
=
finite jet body + lawful correction + transported Smriti tail.
```

The Vedic source lineage is:

```
Parveen117/Vedic
commit e7ec7ded5838cddeb9cdef4d235824dfd7cf530f
    Add first Vedic Recognition Geometry manuscript

commit 0e9b480ece2f5696a2438199edc689b793f6915a
    Add Vedic mathematics master doctrine spine
```

The source grammar is

$$\mathcal{R}(\text{NF}(\mathcal{R}(\mathcal{C}(\mathcal{B})))) - \mathcal{B} - \mathcal{S} = 0, \quad (1.1)$$

with the Madhava specialization

$$\mathcal{R}(S_n + C_n) - \mathcal{B}_\infty - S_n = 0. \quad (1.2)$$

The notation in (1.2) overloads S_n as a partial sum and $\mathcal{S}_n$ as Smriti. The present theorem repairs that type collision. It does not claim historical identity with one particular Madhava correction formula. It proves the abstract corrected-recursion and tail-memory structure needed by the Recognition framework.

47.2 2. Recognized operator carrier

Let $\mathcal{H}$ and $\mathcal{Y}$ be Hilbert spaces. Let G be a bounded operator on $\mathcal{H}$, let

$$U_t = e^{tG},$$

and let

$$B : \mathcal{H} \rightarrow \mathcal{Y}$$

be a bounded observer. The recognized full observed flow is

$$\boxed{F(t) := BU_t.} \quad (2.1)$$

All terms below are written after recognition normal form, so subtraction takes place in the single typed carrier $\mathcal{B}(\mathcal{H}, \mathcal{Y})$. This is the operator realization of the outer $\mathcal{R} \circ \text{NF}$ in (1.1).

Define the observer jets

$$\Lambda_k := BG^k, \quad k \geq 0. \quad (2.2)$$

For integers $N, m \geq 0$, define the finite Chandas body

$$\boxed{P_N(t) = \sum_{k=0}^N \frac{t^k}{k!} \Lambda_k,} \quad (2.3)$$

and the Madhava correction window

$$\boxed{C_{N,m}(t) = \sum_{k=N+1}^{N+m} \frac{t^k}{k!} \Lambda_k, \quad C_{N,0} = 0.} \quad (2.4)$$

The corrected finite recognized state is

$$\boxed{M_{N,m}(t) = P_N(t) + C_{N,m}(t).} \quad (2.5)$$

Define the signed Smriti and the conventional remaining tail by

$$\boxed{\Sigma_{N,m}(t) = M_{N,m}(t) - F(t),} \quad (2.6)$$

$$\boxed{T_{N,m}(t) = F(t) - M_{N,m}(t) = -\Sigma_{N,m}(t).} \quad (2.7)$$

The sign in (2.6) matches the Vedic closure grammar: corrected finite recognition minus recognized limit minus signed Smriti equals zero.

47.3 3. Central closure theorem

Theorem 3.1 (Madhava–Smriti corrected jet closure)

For every $N, m \geq 0$,

$$\boxed{M_{N,m}(t) - F(t) - \Sigma_{N,m}(t) = 0.} \quad (3.1)$$

Equivalently,

$$\boxed{M_{N,m}(t) + T_{N,m}(t) = F(t).} \quad (3.2)$$

The tail has the exact integral representation

$$\boxed{T_{N,m}(t) = \frac{1}{(N+m)!} \int_0^t (t-s)^{N+m} B G^{N+m+1} U_s ds.} \quad (3.3)$$

Consequently, for $t \geq 0$,

$$\boxed{\|T_{N,m}(t)\| \leq \frac{t^{N+m+1}}{(N+m+1)!} \|B\| \|G\|^{N+m+1} e^{t\|G\|}.} \quad (3.4)$$

Proof

Equation (3.1) is the typed recognition closure obtained from definition (2.6), and (3.2) is the same identity with conventional tail orientation. Since $M_{N,m}$ is the exponential polynomial through order $N+m$, the bounded-operator Taylor formula gives (3.3). The norm estimate follows from $\|U_s\| \leq e^{s\|G\|}$. $\square$

The theorem is not the empty statement that every error can be renamed memory. The tail is fixed by the already declared flow, observer, jet order and correction depth, and it has an exact integral carrier and norm bound.

47.4 4. Living-tail refinement law

For $r \geq 0$, write

$$M_r(t) = \sum_{k=0}^r \frac{t^k}{k!} \Lambda_k, \quad T_r(t) = F(t) - M_r(t), \quad \Sigma_r(t) = -T_r(t). \quad (4.1)$$

Let

$$q_{r+1}(t) = \frac{t^{r+1}}{(r+1)!} \Lambda_{r+1}. \quad (4.2)$$

Theorem 4.1 (Exact correction-to-tail transfer)

Refinement by one jet level obeys

$$\boxed{M_{r+1} = M_r + q_{r+1},} \quad (4.3)$$

$$\boxed{T_{r+1} = T_r - q_{r+1},} \quad (4.4)$$

and

$$\boxed{\Sigma_{r+1} = \Sigma_r + q_{r+1}.} \quad (4.5)$$

Therefore

$$\boxed{(M_{r+1} - M_r) - (\Sigma_{r+1} - \Sigma_r) = 0,} \quad (4.6)$$

and

$$\boxed{M_{r+1} + T_{r+1} = M_r + T_r = F.} \quad (4.7)$$

Proof

Equation (4.3) is the difference of consecutive exponential polynomials. Subtracting both sides from F gives (4.4), and negation gives (4.5). The closure invariants (4.6)–(4.7) follow immediately. $\square$

This is the precise meaning of the structure becoming “alive”: refinement does not destroy the old state or discard an error. It transfers one typed jet term from the tail ledger into the finite recognized body.

47.5 5. Refinement seam and cocycle

For two depths $r \leq s$, define the refinement seam differential

$$\boxed{\mathfrak{d}_{r,s} = (M_s - M_r) - (\Sigma_s - \Sigma_r).} \quad (5.1)$$

Theorem 5.1 (Refinement closure cocycle)

For all $r \leq s$,

$$\boxed{\mathfrak{d}_{r,s} = 0.} \quad (5.2)$$

For $r \leq s \leq u$,

$$\boxed{\mathfrak{d}_{r,u} = \mathfrak{d}_{r,s} + \mathfrak{d}_{s,u}.} \quad (5.3)$$

If a correction is updated without the matching Smriti update, the open seam is exactly the unmatched correction.

Proof

Both finite-state and signed-Smriti increments equal the same jet sum, so their difference vanishes. Additivity follows by telescoping. For an unmatched increment Q , only the first difference changes, hence the resulting seam is Q . $\square$

Equation (5.3) is the Madhava refinement specialization of the clock-free RSC composition identity.

47.6 6. Correction–Smriti gauge

Theorem 6.1 (Coupled correction–Smriti invariance)

Let $Q(t) \in \mathcal{B}(\mathcal{H}, \mathcal{Y})$ be any admitted typed correction. Define

$$\widetilde{M} = M + Q, \quad \widetilde{\Sigma} = \Sigma + Q. \quad (6.1)$$

Then

$$\boxed{\widetilde{M} - F - \widetilde{\Sigma} = M - F - \Sigma.} \quad (6.2)$$

If the correction is shifted but Smriti is not, the closure defect is exactly Q .

Proof

The coupled terms cancel in (6.2). Without the Smriti shift, the uncanceled term is Q . $\square$

This is a ledger gauge, not permission to fit arbitrary corrections. A physical adapter must specify which Q is lawful and how it is measured.

47.7 7. Bilateral Smriti transport

Now assume the Theorem 43 cut conditions

$$J = J^* = J^{-1}, \quad JGJ = -G, \quad BJ = B. \quad (7.1)$$

Theorem 7.1 (Bilateral corrected-state and Smriti conjugacy)

For every N, m ,

$$\boxed{M_{N,m}(t)J = M_{N,m}(-t),} \quad (7.2)$$

$$\boxed{T_{N,m}(t)J = T_{N,m}(-t),} \quad (7.3)$$

and

$$\boxed{\Sigma_{N,m}(t)J = \Sigma_{N,m}(-t).} \quad (7.4)$$

Define

$$T_e(t) = \frac{1}{2}(T(t) + T(-t)), \quad T_o(t) = \frac{1}{2}(T(t) - T(-t)). \quad (7.5)$$

Then

$$\boxed{T_e J = T_e, \quad T_o J = -T_o.} \quad (7.6)$$

Proof

The jet parity theorem gives $BG^k J = (-1)^k BG^k$, proving (7.2) term by term. Theorem 43 gives $F(t)J = F(-t)$, so subtraction proves (7.3)–(7.4). Adding and subtracting the two tail strands gives (7.6). $\square$

Thus Smriti itself is cut graded. It is not an untyped scalar glued onto the end of the calculation.

47.8 8. Finite termination

Corollary 8.1 (Nilpotent exact Madhava closure)

If

$$G^{d+1} = 0,$$

then for every $r \geq d$,

$$\boxed{T_r = \Sigma_r = 0} \quad (8.1)$$

and the finite corrected recursion closes exactly:

$$\boxed{M_r = F.} \quad (8.2)$$

This is exact finite closure, not asymptotic convergence.

47.9 9. Target burden and fail-closed states

For a declared finite jet observer $\mathcal{A}_r$ and target L , retain the Theorem 43 burden

$$\beta_r(L) = \sup_{\mathcal{A}_r x \neq 0} \frac{|Lx|^2}{\|\mathcal{A}_r x\|^2}. \quad (9.1)$$

The Madhava–Smriti state is classified as:

OPEN_BILATERAL_SEAM

 bilateral cut transport fails;

OPEN_RECOGNITION_RESIDUE

 corrected finite state, recognized limit and Smriti do not close;

ABSTAIN_UNTYPED_TAIL

the tail has no declared carrier, bound or decoder burden;

BURDEN_EXCEEDS_ONE

closure holds but $\beta_r(L) > 1$;

MEMORY_CLOSED

closure and bilateral transport hold, $\beta_r(L) \leq 1$,
and a nonzero typed Smriti tail remains;

EXACT_FINITE_CLOSURE

all preceding gates pass and the Smriti tail vanishes exactly.

A nonzero lawful tail is therefore not automatically failure. It is a memory-closed state. An untyped or unmatched tail remains an obstruction.

47.10 10. Exact rational certificate

The proof-lab packet uses

$$J = \text{diag}(1, -1, 1, -1, 1),$$
$$G = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}, \quad G^5 = 0,$$
$$B = (1, 0, 0, 0, 0), \quad t = \frac{2}{3}.$$

Starting from the first-order body, three successive correction terms transfer from the tail into the finite state. The packet verifies exactly:

closure at every correction depth;
nonzero intermediate tail and exact final termination;
strict tail-energy descent in the canonical fixture;
term-by-term correction-to-tail transfer;
bilateral Smriti conjugacy;
even/odd tail cut parity;
refinement seam cocycle;
correction-Smriti gauge invariance;
wrong-cut, dropped-tail and uncoupled-update negative controls;
fail-closed theorem classifications.

Expected status:

PASS_MADHAVA_SMRTI_BILATERAL_JET_FLOW_CLOSURE_CANDIDATE

Expected SHA-256:

2dc970d0b180e6ae0c679879ac9d0aabb138c9315942f8bc8d2565e1a539c35

47.11 11. Claim boundary

TYPED MADHAVA-SMRITI CLOSURE IDENTITY	PROVED
BOUNDED-FLOW INTEGRAL TAIL FORMULA	PROVED
EXACT CORRECTION-TO-TAIL TRANSFER	PROVED
REFINEMENT SEAM COCYCLE	PROVED
COUPLED CORRECTION-SMRITI GAUGE	PROVED
BILATERAL SMRITI CUT TRANSPORT	PROVED
EVEN/ODD TAIL PARITY	PROVED
NILPOTENT FINITE TERMINATION	PROVED
EXACT RATIONAL CERTIFICATE	LOCAL PASS
SPECIFIC HISTORICAL MADHAVA SERIES FORMULA	NOT CLAIMED
UNBOUNDED-GENERATOR DOMAIN CLOSURE	REQUIRES DOMAIN PINS
PHYSICAL NUCLEAR SMRITI IDENTIFICATION	NOT CLAIMED
DOUBLE-HELIX PHYSICAL ADAPTER	NOT CLAIMED
IMPROVED ATOMIC PREDICTION	NOT YET TESTED

The next lawful development is the nuclear adapter. It must define which measured nuclear residual is the typed Smriti tail, how refinement transfers it between isotope/isotone jets, and when the capstone classifies a nucleus as memory-closed, exactly closed or abstaining.

RKF/theorum/44_madhava_smriti_bilateral_jet_flow_closure_theorem.md

Chapter 48

Directed Arithmetic-Analytic Closure Theorem

48.1 1. Purpose and source lineage

This theorem closes a gap left intentionally separate in the earlier flow and completion results: analytic truncation error and machine-arithmetic error are not the same object and must not be merged into one optimistic decimal.

The construction consumes four already present ingredients:

```
Vedic Bhinna exact ratio / remainder discipline
Vedic Madhava-Yuktibhasha corrected-tail grammar
Theorem 43 observed-flow jet remainder
Theorem 44 corrected body-tail transfer
Theorem 28 finite-to-infinite outward completion
NUMERICAL_PROOF_PROTOCOL exact / interval / ball arithmetic rule
```

The Vedic source is used structurally: finite decimals and rational relations have an exact ratio carrier, while correction and untraversed remainder remain separate. The present theorem supplies the proof-bearing numerical layer needed for arbitrary-precision and validated interval/ball implementations.

It does **not** assert that raw IEEE floating point becomes exact, that every external numerical package is validated, or that a centre value without an outward radius is proof-bearing.

48.2 2. Exact scalar carrier

Let $\mathbb{Q}$ denote the exact rational carrier. Integers embed in the usual way. A finite decimal token is interpreted before binary floating-point conversion.

If the token has signed integer significand A and decimal scale m , then

$$\boxed{x = \frac{A}{10^m} \in \mathbb{Q}.} \quad (2.1)$$

Scientific notation is absorbed into the power of ten before reduction. Therefore, for example,

$$0.1 = \frac{1}{10}, \quad 0.7 = \frac{7}{10}, \quad 0.8 = \frac{4}{5}, \quad (2.2)$$

and hence

$$\boxed{0.1 + 0.7 = 0.8} \quad (2.3)$$

exactly in the declared-value carrier.

This is the Bhinna compatibility rule: decimal spelling and fractional form are two presentations of one exact rational relation whenever the declared value is a finite decimal.

48.3 3. Validated enclosure carrier

Let X be a normed vector space. A validated ball is

$$\boxed{\mathbb{B}(c, \rho) = \{x \in X : \|x - c\| \leq \rho\}, \quad \rho \geq 0.} \quad (3.1)$$

An exact value embeds as a zero-radius ball,

$$\boxed{\iota(x) = \mathbb{B}(x, 0).} \quad (3.2)$$

Thus exact rational arithmetic and validated approximate arithmetic are not competing systems. The former is the zero-error sector of the latter.

For ordered real scalars an interval $[a, b]$ is equivalent to the ball

$$\mathbb{B}\left(\frac{a+b}{2}, \frac{b-a}{2}\right). \quad (3.3)$$

A numerical backend is **validated for one result** only when it supplies an outward enclosure known to contain the exact arithmetic target. A nearest- rounded floating-point centre with no proved radius is not such an enclosure.

48.4 4. Arithmetic-analytic decomposition

Let $F \in X$ be the exact target. Suppose an analytic refinement gives

$$\boxed{F = M_r + T_r} \quad (4.1)$$

with a proved tail bound

$$\boxed{\|T_r\| \leq \tau_r.} \quad (4.2)$$

Theorem 43 supplies one such τ_r for bounded exponential flow, while Theorem 44 supplies the exact body-tail transfer under jet refinement.

Let a numerical backend at arithmetic precision label p return $\widehat{M}_{r,p}$ together with a proved outward arithmetic radius $\rho_{r,p}$ satisfying

$$\boxed{\|M_r - \widehat{M}_{r,p}\| \leq \rho_{r,p}.} \quad (4.3)$$

Define the typed arithmetic residue

$$A_{r,p} := M_r - \widehat{M}_{r,p}. \quad (4.4)$$

Then the exact closure identity is

$$\boxed{F = \widehat{M}_{r,p} + A_{r,p} + T_r.} \quad (4.5)$$

Theorem 4.1 (Total enclosure)

Under (4.1)–(4.3),

$$\boxed{\|F - \widehat{M}_{r,p}\| \leq \rho_{r,p} + \tau_r.} \quad (4.6)$$

Proof

From (4.5),

$$F - \widehat{M}_{r,p} = A_{r,p} + T_r.$$

Apply the triangle inequality and the two declared bounds. $\square$

The arithmetic radius and analytic tail are therefore distinct ledgers:

arithmetic radius = error introduced by numerical representation/operations;
analytic tail = error from finite analytic/refinement depth.

Neither may be hidden inside the other.

48.5 5. Lawful correction / Madhava refinement

Suppose a predeclared lawful correction C_r gives

$$\boxed{F = M_r + C_r + S_r} \quad (5.1)$$

with

$$\|S_r\| \leq \sigma_r. \quad (5.2)$$

Let the backend enclose the corrected body,

$$\|M_r + C_r - \widehat{N}_{r,p}\| \leq \rho_{r,p}. \quad (5.3)$$

Then

$$\boxed{\|F - \widehat{N}_{r,p}\| \leq \rho_{r,p} + \sigma_r.} \quad (5.4)$$

A correction is useful when it reduces the proved remaining tail, but it is not licensed merely because it improves the displayed digits. This is the exact numerical consumption of the Madhava-Smriti rule from Theorem 44.

48.6 6. Three-ledger refinement identity

At two refinement states write

$$F = \widehat{M}_r + A_r + T_r = \widehat{M}_{r+1} + A_{r+1} + T_{r+1}. \quad (6.1)$$

Subtracting gives

$$\boxed{(\widehat{M}_{r+1} - \widehat{M}_r) + (A_{r+1} - A_r) + (T_{r+1} - T_r) = 0.} \quad (6.2)$$

Thus every refinement increment is accounted for by three typed channels:

```
computed-body increment
+ arithmetic-residue increment
+ analytic-tail increment
= 0.
```

When arithmetic is exact, $A_r = 0$. When the analytic representation terminates exactly, $T_r = 0$. Exact finite proof is the simultaneous zero-radius / zero-tail sector.

48.7 7. Directed precision-refinement convergence

Use the product refinement label

$$(r, p), \quad (7.1)$$

where r is analytic depth and p is arithmetic precision. This is a directed comparison label, not physical time.

Theorem 7.1 (Arithmetic-analytic convergence)

Assume

$$\tau_r \longrightarrow 0 \quad (7.2)$$

and choose a cofinal precision schedule $p(r)$ such that

$$\rho_{r,p(r)} \longrightarrow 0. \quad (7.3)$$

Then

$$\boxed{\widehat{M}_{r,p(r)} \longrightarrow F.} \quad (7.4)$$

Proof

Equation (4.6) gives

$$\|F - \widehat{M}_{r,p(r)}\| \leq \rho_{r,p(r)} + \tau_r \rightarrow 0.$$

□

If a fixed arithmetic backend has a nonvanishing radius floor, this theorem does not claim convergence below that floor. More analytic terms cannot erase an unresolved arithmetic enclosure.

48.8 8. Independent-path enclosure test

Suppose two independent validated computations for the same exact target satisfy

$$F \in \mathbb{B}(c_1, \rho_1), \quad F \in \mathbb{B}(c_2, \rho_2). \quad (8.1)$$

Theorem 8.1 (Certified path-overlap necessity)

Then

$$\boxed{\|c_1 - c_2\| \leq \rho_1 + \rho_2.} \quad (8.2)$$

Proof

By the triangle inequality,

$$\|c_1 - c_2\| \leq \|c_1 - F\| + \|F - c_2\| \leq \rho_1 + \rho_2.$$

□

Therefore disjoint certified balls cannot both be valid certificates for the same exact target. At least one target identity, adapter, or outward enclosure must be wrong. This is the numerical specialization of path-equality testing.

48.9 9. Strict outward threshold promotion

Let $x \in \mathbb{R}$ be the exact scalar quantity whose strict upper threshold is Θ . Suppose a finite computation gives centre/upper surrogate u , arithmetic radius ρ , and analytic tail τ , with the proved consequence

$$x \leq u + \rho + \tau. \quad (9.1)$$

Then the lawful promotion rule is

$$\boxed{u + \rho + \tau < \Theta.} \quad (9.2)$$

Equality is a boundary state, not a strict certificate:

$$u + \rho + \tau = \Theta \implies \text{no strict promotion.} \quad (9.3)$$

For Theorem 28's normalized seam matrix this becomes

$$\boxed{u_n + e_n < 1}, \quad (9.4)$$

where the completion error e_n may itself be decomposed into validated arithmetic and analytic/refinement components.

48.10 10. Backend classes

The theorem licenses the following proof roles:

```

EXACT_INTEGER / EXACT_RATIONAL
    arithmetic radius rho = 0;

EXACT_FINITE_DECIMAL
    canonicalized to an exact rational before binary floating conversion;

DIRECTED_INTERVAL / VALIDATED_BALL
    backend supplies a proved outward enclosure;

VALIDATED_LINEAR_ALGEBRA
    solver residuals and conditioning/rounding propagation are included outward;

RAW_NEAREST_FLOAT
    centre/calibration only unless a separate proved outward radius is supplied.
```

An implementation may promote from exact rational to validated ball arithmetic when exact numerator/denominator growth becomes computationally inefficient. That promotion is an embedding into a larger certified carrier, not a loss of proof discipline.

48.11 11. Exact finite certificate

The accompanying proof packet uses only `fractions.Fraction` plus exact decimal token conversion through `decimal.Decimal`. It verifies:

```

0.1 + 0.7 = 0.8 exactly in the declared-value carrier;
exact rational -> zero-radius ball embedding;
strict / boundary / failed outward threshold cases;
raw float without a radius classified non-proof-bearing;
independent-ball overlap necessity and disjoint negative control;
three-ledger refinement identity;
arithmetic-radius + analytic-tail addition;
correction-tail reduction fixture;
directed total-error descent fixture;
interval/ball compatibility;
exact Bhinna-style rational representation.
```

No NumPy, binary floating-point proof margin, fitted tolerance, or post-hoc correction is used by the certificate.

Expected status:

PASS_DIRECTED_ARITHMETIC_ANALYTIC_CLOSURE_CANDIDATE

48.12 12. Claim boundary

FINITE DECIMAL -> EXACT RATIONAL CARRIER	PROVED
EXACT RATIONAL -> ZERO-RADIUS BALL EMBEDDING	PROVED
ARITHMETIC + ANALYTIC TOTAL ENCLOSURE	PROVED
CORRECTED-BODY TOTAL ENCLOSURE	PROVED
THREE-LEDGER REFINEMENT IDENTITY	PROVED
DIRECTED PRECISION-REFINEMENT CONVERGENCE	PROVED
INDEPENDENT CERTIFIED-BALL OVERLAP NECESSITY	PROVED
STRICT OUTWARD THRESHOLD PROMOTION	PROVED
EXACT RATIONAL FINITE CERTIFICATE	PROVIDED
CORRECTNESS OF AN EXTERNAL BALL/INTERVAL LIBRARY	SEPARATE DEPENDENCY
AUTOMATIC ROUNDING BOUND FOR ARBITRARY THIRD-PARTY CODE	NOT CLAIMED
RAW IEEE FLOAT AS PROOF WITHOUT OUTWARD ENCLOSURE	REJECTED
UNIVERSAL NUMERICAL STABILITY	NOT CLAIMED

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Chapter 49

First-Visible-Jet Seam Quotient Closure Theorem

49.1 1. Purpose and claim boundary

This theorem gives a rigorous finite-jet resolution of a class of apparent 0/0 indeterminate forms. It does **not** redefine ordinary algebraic division by zero. The raw expressions

1/0
0/0

remain invalid as algebraic quotients.

The object defined here is instead a **seam quotient of two functions** along a declared one-parameter seam. Its value is determined, when possible, by the first nonvanishing relative jet together with explicit remainder control.

The construction consumes the current capstone chain:

Theorem 42 typed derivative / response tower
Theorem 43 first-visible jet and explicit remainder
Theorem 44 body-tail transfer under refinement
Theorem 45 arithmetic radius + analytic tail enclosure

The motivating Dot-in-Circle seam principle is therefore narrowed to a theorem: indeterminate endpoint values may be resolved only when the declared seam carries enough nonvanishing local structure to determine a unique quotient.

49.2 2. Declared seam and scalar carrier

Let K be either $\mathbb{R}$ or $\mathbb{C}$. Let

$$S : (-\varepsilon, \varepsilon) \rightarrow X$$

be a declared local seam with distinguished endpoint $S(0)$. Let

$$A, B : (-\varepsilon, \varepsilon) \rightarrow K$$

be scalar observables evaluated along this seam. We write simply $A(t)$ and $B(t)$. Assume

$$A(0) = B(0) = 0. \quad (2.1)$$

For an integer $N \geq 0$, suppose the two observables admit finite jet expansions

$$A(t) = \sum_{k=0}^N a_k t^k + R_A^{(N)}(t), \quad (2.2)$$

$$B(t) = \sum_{k=0}^N b_k t^k + R_B^{(N)}(t), \quad (2.3)$$

where the coefficient normalization may absorb factorials. In a derivative normalization one has $a_k = A^{(k)}(0)/k!$ and similarly for b_k .

Define the finite first-visible orders

$$r_A = \min\{k \leq N : a_k \neq 0\}, \quad r_B = \min\{k \leq N : b_k \neq 0\}, \quad (2.4)$$

whenever the sets are nonempty.

If every coefficient through the available depth vanishes, the corresponding finite first-visible order is **unresolved**, not infinity by decree.

49.3 3. First-visible-jet classification

Theorem 3.1 (Finite first-visible-jet quotient classification)

Assume finite first-visible orders r_A and r_B exist and that the remainders are little-o at the corresponding leading orders:

$$R_A^{(N)}(t) = o(t^{r_A}), \quad R_B^{(N)}(t) = o(t^{r_B}). \quad (3.1)$$

Then along the declared seam:

1. if $r_A > r_B$,

$$\boxed{\lim_{t \rightarrow 0} \frac{A(t)}{B(t)} = 0;} \quad (3.2)$$

2. if $r_A = r_B = r$,

$$\boxed{\lim_{t \rightarrow 0} \frac{A(t)}{B(t)} = \frac{a_r}{b_r};} \quad (3.3)$$

3. if $r_A < r_B$, then

$$\boxed{\left| \frac{A(t)}{B(t)} \right| \rightarrow \infty,} \quad (3.4)$$

so no finite seam quotient exists.

Proof

Factor the leading powers:

$$A(t) = t^{r_A}(a_{r_A} + o(1)), \quad B(t) = t^{r_B}(b_{r_B} + o(1)).$$

Since $b_{r_B} \neq 0$, the second parenthesis is nonzero for sufficiently small punctured t . Hence

$$\frac{A(t)}{B(t)} = t^{r_A-r_B} \frac{a_{r_A} + o(1)}{b_{r_B} + o(1)}.$$

The three cases follow from the sign of r_A-r_B . square

49.4 4. Dot-in-Circle seam quotient

Definition 4.1 (Certified Dot-in-Circle quotient)

When Theorem 3.1 gives a finite limit, define

$$A \odot_S B := \lim_{t \rightarrow 0} \frac{A(t)}{B(t)}. \quad (4.1)$$

The subscript S is part of the type. Different seams need not produce the same value unless a seam-invariance theorem is separately proved.

In the equal-order case,

$$A \odot_S B = \frac{a_r}{b_r}. \quad (4.2)$$

This is not the algebraic statement $0/0=a_r/b_r$. It is the limit statement that two vanishing observables retain a first nonvanishing relative jet along a declared seam.

Example 4.2

Let

$$A(t) = t \cos \alpha, \quad B(t) = t \sin \alpha.$$

If $\sin(\alpha) \neq 0$, then $r_A=r_B=1$ and

$$A \odot_S B = \cot \alpha. \quad (4.3)$$

49.5 5. Regular seam-reparameterization invariance

A seam may be described by more than one regular local parameter. Let

$$t = \phi(s), \quad \phi(0) = 0, \quad \phi'(0) = c \neq 0. \quad (5.1)$$

Theorem 5.1 (Regular parameter invariance)

Suppose A and B have equal first-visible order r . Under the regular reparameterization (5.1), their leading coefficients become

$$\tilde{a}_r = a_r c^r, \quad \tilde{b}_r = b_r c^r. \quad (5.2)$$

Therefore

$$\boxed{\frac{\tilde{a}_r}{\tilde{b}_r} = \frac{a_r}{b_r}.} \quad (5.3)$$

The finite Dot-in-Circle seam quotient is invariant under regular local reparameterization of the same seam.

Proof

Since $\phi(s) = cs + o(s)$, one has

$$A(\phi(s)) = a_r c^r s^r + o(s^r), \quad B(\phi(s)) = b_r c^r s^r + o(s^r).$$

Cancel the common nonzero factor c^r .

The theorem does not identify genuinely different geometric paths. It only removes arbitrary regular changes of local seam coordinate.

49.6 6. Quantitative denominator separation

The limit theorem becomes proof-bearing numerically only after the reduced denominator is separated from zero.

Assume the common first-visible order is r and define the reduced functions

$$\hat{A}(t) = \frac{A(t)}{t^r}, \quad \hat{B}(t) = \frac{B(t)}{t^r} \quad (t \neq 0). \quad (6.1)$$

Suppose on a certified punctured seam neighborhood,

$$|\hat{A}(t) - a| \leq \delta_A, \quad |\hat{B}(t) - b| \leq \delta_B, \quad (6.2)$$

with

$$\boxed{|b| > \delta_B.} \quad (6.3)$$

Theorem 6.1 (Certified quotient enclosure)

Under (6.2)–(6.3), $\hat{B}(t)$ is nonzero throughout the certified neighborhood and

$$\boxed{\left| \frac{A(t)}{B(t)} - \frac{a}{b} \right| \leq \frac{|b|\delta_A + |a|\delta_B}{|b|(|b| - \delta_B)}.} \quad (6.4)$$

Proof

Write

$$\hat{A} = a + e_A, \quad \hat{B} = b + e_B,$$

with $|e_A| \leq \delta_A$, $|e_B| \leq \delta_B$. Condition (6.3) gives

$$|b + e_B| \geq |b| - \delta_B > 0.$$

Then

$$\frac{a + e_A}{b + e_B} - \frac{a}{b} = \frac{be_A - ae_B}{b(b + e_B)}.$$

Taking absolute values and applying the triangle inequality gives (6.4). **square**

The quantity

$$\rho_Q = \frac{|b|\delta_A + |a|\delta_B}{|b|(|b| - \delta_B)} \quad (6.5)$$

is the outward quotient radius. It may be consumed by Theorem 45 as a numerical arithmetic/analytic enclosure only when δ_A and δ_B themselves are proved bounds.

49.7 7. Connection to Theorems 42–45

The capstone chain supplies the data needed by the quotient theorem.

7.1 Theorem 42: typed tower

The derivative/response tower provides the candidate coefficients

$$a_k, \quad b_k.$$

The first nonzero compatible layer determines the first-visible order.

7.2 Theorem 43: first visibility and remainder

The nested observer tower identifies the first order at which the target ceases to be blind. Its explicit finite-jet remainder gives a lawful route to bounds of the form δ_A , δ_B .

7.3 Theorem 44: no discarded tail

Refinement transfers one term from tail to recognized body while preserving

$$M_r + T_r = F.$$

Thus a quotient certificate may not erase higher-order uncertainty after the first visible coefficient is found.

7.4 Theorem 45: arithmetic enclosure

Exact rational or finite-decimal coefficients have zero arithmetic radius. Approximate coefficients require validated interval/ball enclosures. The quotient radius (6.5) and any remaining analytic tail are carried outward before promotion.

49.8 8. Flat-function boundary

Finite jets do not resolve every 0/0 limit.

For example,

$$A(t) = e^{-1/t^2}, \quad B(t) = e^{-1/t^2} \quad (t \neq 0),$$

extended by zero at $t=0$, have all derivatives zero at the endpoint but

$$\frac{A(t)}{B(t)} = 1$$

on the punctured seam.

Therefore:

$$\boxed{\text{all available jets zero} \not\Rightarrow \text{quotient absent;}} \quad (8.1)$$

it means only that the finite-jet theorem has insufficient information.

The fail-closed status is

INCOMPLETE_FLAT_OR_UNRESOLVED

until a different admitted seam invariant or non-jet asymptotic theorem closes the case.

This boundary is essential. Declaring all flat 0/0 forms equal would simply replace indeterminacy with invention.

49.9 9. Raw division-by-zero boundary

The theorem distinguishes the following typed cases:

1 / 0 as algebraic scalar	INVALID
0 / 0 as algebraic scalar	INVALID
A(t)/B(t), no declared seam	INVALID_CONTRACT
seam declared, no finite denominator jet	INCOMPLETE_FLAT_OR_UNRESOLVED
$r_A > r_B$	FINITE_QUOTIENT_ZERO
$r_A = r_B$ and $b_r \neq 0$	FINITE_SEAM_QUOTIENT
$r_A < r_B$	DIVERGENT_NO_FINITE_QUOTIENT
reduced denominator not separated from zero	INCOMPLETE_DENOMINATOR_SEPARATION

No branch converts 1/0 into a finite certified number.

49.10 10. Optional phase compatibility

The Bindu-Lopa source motivates a phase lift on the unit circle. Such a phase layer may be consumed only **after** the scalar seam quotient has been resolved or enclosed.

Suppose a separately admitted phase factor satisfies

$$\Phi(t) \rightarrow \Phi_0, \quad |\Phi_0| = 1.$$

If $A \odot_S B = q$ exists, then

$$\lim_{t \rightarrow 0} \frac{A(t)}{B(t)} \Phi(t) = q \Phi_0. \quad (10.1)$$

The phase factor decorates an already resolved quotient; it does not by itself recover discarded magnitude information or prove existence of the quotient.

49.11 11. Negative controls required for certification

Any implementation claiming this theorem must include at least:

1. raw 1/0 rejection;
2. raw 0/0 rejection;
3. equal-order finite quotient fixture;
4. numerator-higher-order zero-limit fixture;
5. denominator-higher-order divergence fixture;
6. all-zero finite-jet incomplete fixture;
7. regular seam-reparameterization invariance fixture;
8. denominator-separation failure fixture;
9. exact-rational coefficient fixture;
10. quotient-enclosure bound fixture.

A numerical implementation must additionally prove or validate the coefficient and remainder bounds it consumes. Merely labelling a submitted radius **validated** does not establish the hypothesis of Theorem 6.1.

49.12 12. Claim boundary

The proved statement is:

A pair of vanishing scalar observables along a declared regular seam has a finite seam quotient whenever their first nonzero jets occur at the same finite order and the denominator leading coefficient is nonzero; the quotient equals the ratio of those leading coefficients. Unequal first-visible orders classify zero versus divergence. Explicit reduced-function remainder bounds give an outward quotient enclosure once the reduced denominator is separated from zero.

The theorem does **not** claim:

- a universal value for algebraic $0/0$;
- a finite value for $1/0$;
- uniqueness across genuinely different seams without an additional invariance theorem;
- finite-jet resolution of flat functions;
- correctness of an externally asserted arithmetic radius or analytic tail;
- that phase lifting alone resolves magnitude indeterminacy.

This theorem is therefore a conditional closure theorem for indeterminate limits, not a redefinition of field arithmetic.

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Chapter 50

Source-Bound Proof-Carrying Numerical Validation Theorem

50.1 1. Purpose and claim boundary

Theorem 45 proves how a certified arithmetic radius and a certified analytic tail compose. It does **not** prove that a participant-supplied radius or tail is true merely because it is present in a certificate. Theorem 46 likewise permits an approximate seam quotient only after its remainder and denominator-separation hypotheses are proved.

This theorem closes that trust membrane for a declared class of proof-carrying numerical certificates.

The principle is:

A numerical bound may be consumed only after its source and derivation are validated.
A bound does not validate itself.

The theorem consumes the earlier chain

Theorem 34 common-chart source binding and decoder transfer
Theorem 38 source-relative domination / form-range control
Theorem 39 no-blindness requirement
Theorem 43 explicit analytic remainder
Theorem 44 body-tail preservation under refinement
Theorem 45 arithmetic-radius + analytic-tail closure
Theorem 46 quantitative seam quotient after validated remainders

It does not claim universal validation of arbitrary external software, measurements, hardware, interval libraries, or participant assertions. Such objects require an admitted source validator.

50.2 2. Rational interval carrier

Let

$$\mathbb{I}_{\mathbb{Q}} = \{[a, b] : a, b \in \mathbb{Q}, a \leq b\}.$$

For an interval $I = [a, b]$, write

$$\underline{I} = a, \quad \overline{I} = b.$$

An exact rational q embeds as the singleton

$$\boxed{q \mapsto [q, q]}. \quad (2.1)$$

The admitted exact interval operations are

$$[a, b] + [c, d] = [a + c, b + d], \quad (2.2)$$

$$[a, b] - [c, d] = [a - d, b - c], \quad (2.3)$$

$$[a, b][c, d] = [\min(ac, ad, bc, bd), \max(ac, ad, bc, bd)], \quad (2.4)$$

and

$$-[a, b] = [-b, -a]. \quad (2.5)$$

For division, if

$$0 \notin [c, d], \quad (2.6)$$

then reciprocal endpoint order is taken outward and multiplication uses (2.4). If

$$0 \in [c, d],$$

the division rule is not admitted.

All calculations in this carrier may be performed by exact integer/rational arithmetic.

50.3 3. Source-bound proof DAG

A **proof-carrying arithmetic trace** is a finite directed acyclic graph

$$\mathcal{G} = (V, E)$$

with a distinguished root r . Each node $v \in V$ has:

1. a unique identifier;
2. either a source-leaf declaration or an admitted operation;
3. an interval $I_v \in \mathbb{I}_{\mathbb{Q}}$;
4. ordered references to its dependency nodes.

A source-leaf declaration has a source class $s(v)$ and is accepted only when an admitted validator $V_{s(v)}$ proves

$$\boxed{x_v \in I_v.} \quad (3.1)$$

The initial exact-contract source class is the singleton case

$$I_v = [q, q]$$

for a rational value frozen as part of the mathematical challenge contract. External measurements, backend-produced radii, norm claims, or other nonprimitive assertions are not exact-contract leaves merely because their JSON encoding is exact.

For an internal node with operation f_v and dependency intervals $I_{v_1}, \dots, I_{v_m}$, let

$$\mathcal{E}_{f_v}(I_{v_1}, \dots, I_{v_m})$$

be the canonical outward interval obtained from the admitted rules (2.2)–(2.6). The local verifier accepts only if

$$\boxed{\mathcal{E}_{f_v}(I_{v_1}, \dots, I_{v_m}) \subseteq I_v.} \quad (3.2)$$

Thus a participant may widen an enclosure, but cannot obtain certification by shrinking it below the verifier-computed enclosure.

Theorem 3.1 (Source-bound DAG enclosure)

Assume:

- the graph is finite and acyclic;
- every dependency reference resolves to a unique earlier node in a topological order;
- every source leaf satisfies its admitted source validator;
- every internal node satisfies (3.2).

Then the exact value represented by every node lies in its declared interval. In particular,

$$\boxed{x_r \in I_r.} \quad (3.3)$$

Proof

Take a topological ordering of the DAG. For a source leaf the claim is (3.1). Assume it holds for all dependencies of an internal node v . Soundness of the admitted interval operation gives

$$f_v(x_{v_1}, \dots, x_{v_m}) \in \mathcal{E}_{f_v}(I_{v_1}, \dots, I_{v_m}).$$

By (3.2), the latter is contained in I_v . Induction over the topological order proves the result. $\square$

Corollary 3.2 (Derived arithmetic radius)

For

$$I_r = [L, U],$$

define

$$c = \frac{L + U}{2}, \quad \rho = \frac{U - L}{2}.$$

Then

$$\boxed{|x_r - c| \leq \rho.} \tag{3.4}$$

The radius is derived from the verified trace. It is not a free participant assertion.

50.4 4. Source completeness / no-blindness boundary

Theorem 3.1 proves the arithmetic consequence of the declared dependency graph. It cannot prove that an omitted physical, semantic, or analytic dependency never mattered.

Therefore a proof-carrying trace is promotion-capable only when the selected adapter also closes a source-completeness obligation:

$$\boxed{V_{\text{src}} = 1 \implies \text{all target-relevant primitive inputs are represented by admitted leaves.}} \tag{4.1}$$

This is the numerical analogue of the common-chart/no-blindness discipline in Theorems 34, 38 and 39. A beautifully verified DAG of the wrong source does not certify the target.

50.5 5. Theorem-derived analytic tails

A tail certificate must identify an admitted tail rule and supply the hypotheses from source-validated or DAG-derived quantities. The verifier recomputes the tail; the participant does not choose its final value.

5.1 Geometrically dominated tail

Suppose a series tail has magnitude terms satisfying, for $k \geq N + 1$,

$$|a_{k+1}| \leq q|a_k|, \quad 0 \leq q < 1. \tag{5.1}$$

If the first omitted magnitude obeys

$$|a_{N+1}| \leq A,$$

then

$$\boxed{\left| \sum_{k=N+1}^{\infty} a_k \right| \leq \frac{A}{1-q}.} \tag{5.2}$$

5.2 Rational upper enclosure for the exponential

Let $x \in \mathbb{Q}_{\geq 0}$, and choose $m \geq 0$ so that

$$q_m = \frac{x}{m+2} < 1. \quad (5.3)$$

Since the ratio between successive exponential-series terms beyond order $m+1$ is at most q_m ,

$$e^x \leq E_m(x) := \sum_{k=0}^m \frac{x^k}{k!} + \frac{x^{m+1}}{(m+1)!} \frac{1}{1 - x/(m+2)}. \quad (5.4)$$

Every quantity on the right is rational.

5.3 Theorem-43 jet tail without floating-point exponentiation

Let source-validated quantities satisfy

$$|t| \leq T, \quad \|B\| \leq B_*, \quad \|G\| \leq G_*,$$

with $T, B_*, G_* \in \mathbb{Q}_{\geq 0}$. Theorem 43 gives

$$\|R_{N+1}(t)\| \leq \frac{T^{N+1}}{(N+1)!} B_* G_*^{N+1} e^{TG_*}.$$

Choose m satisfying

$$\frac{TG_*}{m+2} < 1.$$

Using (5.4), the verifier may compute the exact rational upper bound

$$\tau_{43} = \frac{T^{N+1}}{(N+1)!} B_* G_*^{N+1} E_m(TG_*). \quad (5.5)$$

No nearest-rounded evaluation of e^{TG_*} is required.

Theorem 5.1 (Tail-source rule)

If every hypothesis of an admitted tail rule is source-validated or DAG-derived and the verifier recomputes τ from that rule, then the resulting tail bound is proof-bearing. A participant-supplied scalar named `analytic_tail` without such provenance is not proof-bearing.

50.6 6. Source-bound numerical promotion

Let $\widehat{M}$ be the numerical center obtained from a source-bound arithmetic certificate and let

$$\|M - \widehat{M}\| \leq \rho \quad (6.1)$$

be derived from Theorem 3.1. Let an admitted tail rule prove

$$\|T\| \leq \tau. \quad (6.2)$$

By Theorem 45,

$$F = \widehat{M} + A + T, \quad \|A\| \leq \rho,$$

therefore

$$\boxed{\|F - \widehat{M}\| \leq \rho + \tau.} \quad (6.3)$$

Theorem 6.1 (Source-bound promotion gate)

Let

$$V_\rho = 1$$

mean the arithmetic trace and its source leaves have passed, and let

$$V_\tau = 1$$

mean the analytic tail rule and its hypotheses have passed. A strict upper-threshold promotion may occur only when

$$\boxed{V_\rho = 1, \quad V_\tau = 1, \quad U + \tau < \Theta,} \quad (6.4)$$

where $U = \bar{I}_r$.

If either validator is open, absent, unsupported, or fails, the bound cannot be promoted merely because the participant supplied numerically favorable values.

50.7 7. Certainty-aware strict boundary

For a strict claim

$$F < \Theta,$$

let the final certified scalar enclosure be $[L, U]$, after all proof-bearing arithmetic and analytic uncertainty has been carried outward.

The terminal logic is:

$U < \Theta$	PASS,	(7.1)
$L = U = \Theta$	FAIL (exact strict inequality is false),	
$U = \Theta$ and $L < U$	INCOMPLETE (refinement may decide),	
$L \geq \Theta$	FAIL,	
$L < \Theta < U$	INCOMPLETE.	

This distinguishes mathematical falsity from unresolved numerical contact with a boundary.

50.8 8. Implementation commitment boundary

A proof-carrying protocol must commit the validator semantics used to interpret the certificate. Let

$$H_{\text{impl}}$$

be a cryptographic digest of a declared implementation manifest containing the validator source, parser contract, schema and admitted operation/tail-rule identifiers.

The challenge rules should commit

$$\boxed{H_{\text{impl}}} \tag{8.1}$$

so an implementation change cannot masquerade as the same frozen protocol.

This is an **integrity** statement, not an external authenticity theorem. If an attacker controls both the executable and the hash it reports, an internal hash is not a trust anchor. A published repository commit, immutable archive, signature, attestation, or equivalent external pin is still required when executable authenticity matters.

50.9 9. Negative controls

A compliant verifier must reject or hold open at least the following:

cycle in proof DAG	INVALID
missing dependency	INVALID
duplicate node identifier	INVALID
claimed node interval narrower than recompute	FAILED
interval division with zero in denominator	FAILED / INVALID
participant radius without proof trace	INCOMPLETE
participant analytic tail without tail rule	INCOMPLETE
unsupported source class	INCOMPLETE
source-completeness obligation open	INCOMPLETE
exact strict equality at threshold	FAILED
non-singleton enclosure merely touching bound	INCOMPLETE
implementation fingerprint mismatch	FAILED

50.10 10. Theorem-46 approximate seam consumption

Theorem 46 gives an outward quotient radius

$$\rho_Q = \frac{|b|\delta_A + |a|\delta_B}{|b|(|b| - \delta_B)}$$

when $|b| > \delta_B$.

Under Theorem 47, the quantities δ_A and δ_B become promotion-capable only if they are produced by admitted source-bound remainder validators. Then the approximate seam quotient may consume ρ_Q through Theorem 45.

Until those remainder sources close, the correct status remains

INCOMPLETE_REMAINDER_VALIDATION

rather than a participant-certified quotient.

50.11 11. Claim boundary

SOURCE-BOUND RATIONAL INTERVAL DAG THEOREM	PROVED
LOCAL-TO-GLOBAL ENCLOSURE INDUCTION	PROVED
DIVISION-ZERO ENCLOSURE GATE	PROVED
RATIONAL GEOMETRIC TAIL RULE	PROVED
RATIONAL EXPONENTIAL UPPER RULE	PROVED
THEOREM-43 RATIONAL TAIL ADAPTER	PROVED CONDITIONAL ON SOURCE-VALIDATED NORM BOUNDS
SOURCE-BOUND PROMOTION GATE	PROVED
CERTAINTY-AWARE STRICT BOUNDARY	PROVED
IMPLEMENTATION-HASH INTEGRITY REQUIREMENT	SPECIFIED
ARBITRARY EXTERNAL BACKEND AUTHENTICITY	NOT CLAIMED
ARBITRARY REAL-WORLD SOURCE TRUTH	NOT CLAIMED
UNIVERSAL SOURCE-COMPLETENESS VALIDATOR	NOT CLAIMED
RESOURCE/CPU DENIAL-OF-SERVICE CONTROL	ENGINEERING GATE, SEPARATE FROM THIS THEOREM

RKF/theorum/47_source_bound_proof_carrying_numerical_validation_theorem.md

Chapter 51

EMK Algebra as an Exact Instantiation of the Cut-Graded Curvature (Theorem 42)

51.1 1. Source problem and scope

theorem/42_cut_graded_lambda_jacobian_tower_theorem.md builds the abstract machinery for a cut-graded response tower on a state manifold M: a state cut j , a covering response-bundle involution J_E ($J_E^2=I$), the induced even/odd grading $L_n^{+-} = (1/2)(L_n + J_n L_n)$, and — for a protocol generator $G = G_e + G_o$ split by the same involution — an exact cut-loop curvature identity (Theorem 6.1):

$$F_{st} = [G_e, G_o]$$

Its claim boundary lists as **OPEN/NEXT**:

PHYSICAL T-V-S-P CUT AND RESPONSE-FIBRE INVOLUTION
CUT-COMPATIBLE THERMODYNAMIC CONNECTION
OPERATOR CURVATURE -> RESPONSE TWO-FORM ADAPTER

Separately, the Publications repository certifies a small primitive algebra (EMK-1, EMK-2: papers/emk-ugd-algebra/) on generators I, K, R, RK with $K^2=I, R^2=-I, RK=-KR$, and a proved $Z/2$ grading: $\{I, K\}$ even (the “seam” sector), $\{R, RK\}$ odd (the “rotational” sector).

This capsule asks a narrow, previously unasked question: does the EMK algebra supply a concrete instantiation of theorem/42’s abstract axioms — and if so, what is the exact (not truncated-series) cut-loop curvature on that carrier? It does **not** attempt the physical adapter of theorem/42 Eq. (6.7), and it does **not** claim the EMK algebra is *the* correct involution for the physical T-V-S-P lambda map of arXiv:2603.20773. It closes one narrow algebraic gap and leaves the physical identification exactly as open as theorem/42 already stated.

The EMK primitive relations are **reproduced from scratch** on the 2x2 matrix carrier below, not imported from the Publications repo, so this capsule is an independent cross-repo consistency check rather than a restatement.

51.2 2. The concrete involution

Represent the EMK primitives as real 2x2 matrices:

```
I = [[1,0],[0,1]]
K = [[0,1],[1,0]]
R = [[0,-1],[1,0]]
RK = R K = [[-1,0],[0,1]]
```

Proposition 2.1

$K^2 = I$, $R^2 = -I$, $RK^2 = I$, $RK = -KR$. Take $J_E := K$. Then J_E satisfies theorem/42's involution axiom $J_E^2 = I$ exactly.

Proof: direct matrix computation, exact over $\mathbb{Q}$. **square**

Proposition 2.2 (grading matches EMK-2, independently)

Under the induced grading $G^{\pm} = (1/2)(G \pm J_E G J_E)$ with $J_E = K$:

```
I is purely cut-even
K is purely cut-even
R is purely cut-odd
RK is purely cut-odd
```

This exactly reproduces the $\mathbb{Z}/2$ grading independently certified in Publications-repo EMK-2 ($\{I, K\}$ even, $\{R, RK\}$ odd) — obtained here without reference to that repo, from the involution axiom alone.

Proof: $KIK=I$, $KKK=K$ (even); $KRK=-R$, $K(RK)K = -RK$ (odd), by direct computation. **square**

51.3 3. Exact closed-form curvature

Theorem/42 Theorem 6.1 states the cut-loop curvature of a constant cut-graded connection is exactly the commutator of its even and odd parts. On a general (possibly unbounded, possibly non-nilpotent) carrier this is computed via the log of a series (theorem/42 Eq. 6.3). On the EMK carrier it is available in **closed form with no series at all**, because the carrier is two-dimensional and every product is already known exactly.

Theorem 3.1 (representative curvature)

For $G_e := K$, $G_o := R$:

$$F_{st} = [K, R] = -2 RK.$$

$F_{\{st\}}$ is nonzero and purely cut-odd.

Theorem 3.2 (general closed form)

For every $G_e = aI + bK$ and $G_o = cR + dRK$ with a, b, c, d rational:

$$[G_e, G_o] = -2b(dR + cRK).$$

Consequently $[G_e, G_o]$ is **always** purely cut-odd on this carrier, regardless of a, b, c, d . This structural fact is not stated in theorem/42 (which leaves the commutator's grading unresolved in the abstract setting) and is new to this capsule.

Proof: direct expansion using $[K, R] = -2RK$ and $[K, RK] = -2R$ (each a single exact matrix computation), then bilinearity of the commutator and $[I, \cdot] = 0$. Verified exactly on a grid of over 100 rational sample points as a machine cross-check. **square**

51.4 4. Negative controls

N1 a generic involution unrelated to the algebra's own multiplication table (`diag(1,-1)`) does NOT reproduce the $\{I, K\}$ even / $\{R, RK\}$ odd partition -- confirms $J_E := K$ is load-bearing, not an arbitrary choice of involution;

N2 odd input identically zero gives identically zero curvature;

N3 a wrong-sign claim ($F_{st} = +2 RK$) is correctly rejected.

51.5 5. Exact rational certificate

The proof-lab packet verifies exactly:

```

K^2=I, R^2=-I, RK=-KR, RK^2=I;
J_E := K satisfies theorem/42's involution axiom;
induced grading matches EMK-2's Z/2 grading exactly;
representative curvature [K,R] = -2 RK exactly;
general closed form [aI+bK, cR+dRK] = -2b(dR+cRK) on >100 rational points;
the commutator is always cut-odd on this carrier (new structural fact);
three negative controls.

```

No NumPy, floating point, or transcendental evaluation is used.
Expected status:

PASS_EMK_ALGEBRA_CUT_GRADED_CURVATURE_CANDIDATE

Expected SHA-256:

8f5da0b72389d88380c9af6c26e2edf4c4f63554bf1fd5485b0ccdee95c58a0f

Reproduce:

```

python proof_lab/emk_algebra_cut_graded_curvature.py
python -m pytest proof_lab/test_emk_algebra_cut_graded_curvature.py -v

```

51.6 6. Claim boundary

$K^2=I$, $R^2=-I$, $RK=-KR$ (reproduced independently)	PROVED
$J_E := K$ SATISFIES THEORUM/42'S INVOLUTION AXIOM	PROVED
GRADING MATCHES EMK-2'S $\mathbb{Z}/2$ GRADING EXACTLY	PROVED
REPRESENTATIVE CUT-LOOP CURVATURE, EXACT CLOSED FORM	PROVED
GENERAL CLOSED FORM $[G_e, G_o] = -2b(dR + cRK)$	PROVED
COMMUTATOR ALWAYS CUT-ODD ON THIS CARRIER (new fact)	PROVED
EXACT RATIONAL CERTIFICATE	IMPLEMENTED / USER RUN REQUIRED
PHYSICAL T-V-S-P CUT AND RESPONSE-FIBRE INVOLUTION	STILL OPEN
OPERATOR CURVATURE $\rightarrow$ RESPONSE TWO-FORM ADAPTER (Eq. 6.7)	STILL OPEN
IDENTIFICATION OF THIS ALGEBRA WITH THE PHYSICAL LAMBDA MAP	NOT CLAIMED
emk_thermo_seed.tex RECURSIVE TOWER Λ^n	UNTOUCHED
RH, YANG-MILLS, ANY OTHER FRAMEWORK GATE	UNTOUCHED

This capsule narrows, but does not close, theorem/42's stated open items. It supplies one concrete, exactly-computable candidate instantiation; it does not determine whether that candidate is the physically correct one.

RKF/theorum/48_emk_algebra_cut_graded_curvature_theorem.md

Chapter 52

T19/T20 Odd Boundary-Burden Certificate

52.1 Source provenance

source repository: Parveen117/MP
primary PRs: #276, #277
main objects: beta_partial^cut, S_(0,-)^full, L_partial, W_3^-

52.2 Why this capsule exists

The zero-cut T03 route is the current terminal endpoint route in the cleaned paper. However, T19 and T20 are still important: they give an independent odd-sector sign certificate and an alternate single-sector route.

They should not be deleted or treated as irrelevant. They are not the same theorem as T03.

52.3 T19 burden inequality

T18 proves the boundary-burden equivalence

full odd sign $\Leftrightarrow$ beta_partial^cut ≤ 1 .

T19 proves the endpoint bound

beta_partial^cut $\leq 0.9998319617060448 < 1$.

The strict reserve is

$1 - \text{beta_partial}^{\text{cut}} \geq 0.0001680382939552$.

The proof uses the exact endpoint-difference identity

$$\begin{aligned} \text{beta}_0 - \text{beta}_\eta \\ = \sum_j m_j \eta / (s_j (s_j + \eta)). \end{aligned}$$

The reconstructed Arb cut profile has

```

active cells                2400
prime-power events          78734
step                        0.005
near-zero slope-gap lower   28.96922744937062
minimum positive source floor 0.0003621153431171327
far-ray floor               0.15683586289723883

```

At

```

eta                1.5625e-5
beta_eta^+         0.999773667408463
shifted reserve     2.2633259153703733e-4

```

the correction ledger is

```

first-cut correction    2.4184691966571344e-6
active-band correction  2.9454791518469324e-6
far-ray correction      5.293034923330094e-5
total correction        5.829429758180501e-5

```

Thus T19 proves

$S_-(0, -)^{\text{full}} - L_{\text{partial}}^{\text{dag}} L_{\text{partial}} \geq 0$.

on the declared native source carrier.

52.4 T20 strict odd single-sector chain

T20 consumes T19 and adds source injectivity:

```

unshifted covariant cut flow
+ endpoint C1 regularization
+ Hardy active-band no-blindness
  ->  $\ker S_-(0, -)^{\text{full}} = \{0\}$ 

```

Together with T18/T19:

```

T18 cut-square identity
+ T19  $\beta_{\text{partial}}^{\text{cut}} < 1$ 
  ->  $W_3^-$  is strictly positive on every nonzero odd core state.

```

T20 then records the optional classical membrane:

```

strict odd Weil positivity
+ pinned odd-parity Weil implication
  -> RH under the declared classical explicit-formula normalization.

```

52.5 Relationship to the cleaned endpoint paper

T19/T20 are not required if the paper uses the zero-cut T03 endpoint as its terminal native theorem:

```

T03 source-Gram identity
->  $K_0 = R_{\text{Sigma}}^*(I - P_{\text{src}})R_{\text{Sigma}} \geq 0$ 
-> endpoint Schur closure.

```

But T19/T20 are still valuable and should be included as an independent odd-burden certificate, because they show that the adverse odd boundary packet is subcritical by a direct cut/refinement estimate.

52.6 Correct dependency placement

Main terminal route:

```
positive-eta path equality
-> seam integer / five-matrix reduction
-> M3 accepted block
-> one-sector reduction
-> zero-cut T03 endpoint
```

Independent odd-burden certificate:

```
T18 equivalence
-> T19 beta_partial^cut < 1
-> full odd sign
-> T20 strict odd positivity if ker S_(0,-)^full = 0
```

52.7 Claim boundary

T19 beta_partial^cut < 1	PROVED IN SOURCE PR
T19 full odd completed-Weil sign	PROVED ON DECLARED NATIVE SOURCE CARRIER
T20 strict odd positivity	PROVED UNDER ITS SOURCE-INJECTIVITY CHAIN
T20 odd-parity RH implication	CLASSICAL/PINNED INPUT WHERE USED
T19/T20 as zero-cut T03 replacement	NOT IDENTIFIED
T19/T20 as independent support	RETAINED

RKF/theorum/T19_T20_odd_boundary_burden_certificate.md

Part III

Thermodynamic Response Theorems

Chapter 53

Thermodynamic Response Tetrad and Flat Closure Theorem

53.1 1. Signed response tetrad

Let

$$C_{\text{th}} = (C_p, C_v, C_s, C_t),$$

where the caloric channels are the usual heat capacities and the signed mechanical channels are

$$C_s = V \left(\frac{\partial P}{\partial V} \right)_S, \quad C_t = V \left(\frac{\partial P}{\partial V} \right)_T.$$

For stable materials the conventional positive bulk moduli are

$$K_S = -C_s, \quad K_T = -C_t.$$

Define the entropy-scaled response coordinates uniformly by

$$\boxed{\lambda_X^{\text{th}} = -\frac{ST}{C_X}, \quad X \in \{p, v, s, t\}.}$$

The mechanical coordinates are therefore not silently identified with unrelated derivative coordinates. Their sign and type are retained.

53.2 2. Caloric and mechanical ratios

Define

$$\Gamma_c = \frac{\lambda_p^{\text{th}}}{\lambda_v^{\text{th}}} = \frac{C_v}{C_p},$$

and

$$\Gamma_m = \frac{\lambda_t^{\text{th}}}{\lambda_s^{\text{th}}} = \frac{C_s}{C_t} = \frac{K_S}{K_T} = \frac{\kappa_T}{\kappa_S}.$$

53.3 Theorem 2.1 (flat caloric-mechanical closure)

On a constitutive chart where the standard equilibrium response identities hold,

$$\boxed{I_{\text{th}} := \Gamma_c \Gamma_m = 1.}$$

Proof

The equilibrium caloric identity gives

$$\frac{C_p}{C_v} = \frac{\kappa_T}{\kappa_S}.$$

Taking the reciprocal of the caloric ratio and multiplying by the mechanical ratio gives

$$\Gamma_c \Gamma_m = \frac{C_v}{C_p} \frac{\kappa_T}{\kappa_S} = 1.$$

QED.

53.4 3. Interpretation

The scalar

$$\delta_{\text{th}} = \log(\Gamma_c \Gamma_m)$$

vanishes on the flat equilibrium sector. It is a dimensionless sector projection, not the full thermodynamic curvature tensor.

53.5 Claim boundary

SIGNED RESPONSE TETRAD	DEFINED
ENTROPY-SCALED LAMBDA COORDINATES	DEFINED
CALORIC RATIO ORIENTATION	PROVED
MECHANICAL RATIO ORIENTATION	PROVED
FLAT CLOSURE $\Gamma_c \Gamma_m = 1$	PROVED
FULL NONEQUILIBRIUM CURVATURE	NOT IDENTIFIED BY THIS SCALAR ALONE

RKF/theorum/thermodynamics/01_response_tetrad_and_flat_closure.md

Chapter 54

Thermodynamic Curvature to Seam-Index Spectral-Flow Theorem

54.1 1. Response connection

Let the entropy-scaled response coordinates define the one-form

$$\omega_{\text{th}} = \sum_{X \in \{p, v, s, t\}} \lambda_X^{\text{th}} dx_X,$$

with curvature

$$\Omega_{\text{th}} = d\omega_{\text{th}}.$$

Local response flatness means

$$\Omega_{\text{th}} = 0.$$

54.2 2. Type obstruction

A positive real scalar such as

$$\Gamma_c \Gamma_m$$

cannot be canonically identical to a nonnegative integer seam rank on a connected constitutive domain. A continuous scalar and an integer-valued threshold count have different types.

Thus the discarded identity

$$\Gamma_c \Gamma_m = k_\Sigma$$

is not a lawful theorem without an explicit operator representation and a threshold-crossing mechanism.

54.3 3. Curvature path represented by compact operators

Let

$$u \mapsto K_\eta(u)$$

be a norm-continuous path of compact self-adjoint response operators, and define the threshold rank

$$k_\Sigma(\eta; u) = N_+(K_\eta(u) - I).$$

Assume the endpoints are noncritical:

$$1 \notin \sigma(K_\eta(u_0)) \cup \sigma(K_\eta(u_1)).$$

54.4 Theorem 3.1 (curvature-path / seam-index law)

For any continuous path u_s from u_0 to u_1 ,

$$\boxed{k_\Sigma(\eta; u_1) - k_\Sigma(\eta; u_0) = -\text{sf}(I - K_\eta(u_s); 0).}$$

Proof

The integer k_Σ changes only when an eigenvalue of $K_\eta(u_s)$ crosses the threshold 1. Equivalently, an eigenvalue of $I - K_\eta(u_s)$ crosses zero. Counting signed zero crossings gives the spectral flow. The sign is reversed because an upward crossing of K_η through 1 is a downward crossing of $I - K_\eta$ through zero. QED.

54.5 Corollary 3.2 (quantitative no-crossing criterion)

Suppose the flat reference $K_\eta(u_0)$ has margin

$$m_\eta = \text{dist}(1, \sigma(K_\eta(u_0))) > 0,$$

and an explicit response intertwiner yields

$$\|K_\eta(u) - K_\eta(u_0)\| \leq L_\eta |\log(\Gamma_c \Gamma_m)|.$$

Then

$$\boxed{L_\eta |\log(\Gamma_c \Gamma_m)| < m_\eta}$$

prevents threshold crossing and preserves the seam rank.

54.6 Claim boundary

RESPONSE ONE-FORM AND CURVATURE	DEFINED
SCALAR = INTEGER IDENTIFICATION	REJECTED BY TYPE
CURVATURE-PATH / SPECTRAL-FLOW LAW	PROVED
QUANTITATIVE NO-CROSSING CRITERION	PROVED
EXPLICIT PHYSICAL CURVATURE -> NATIVE OPERATOR	OPEN
OUTWARD VALUE OF L_{η}	OPEN

RKF/theorum/thermodynamics/02_curvature_to_seam_spectral_flow.md

Chapter 55

Canonical Thermodynamic Response-Cost and Barrier Theorem

55.1 1. Canonical response frame

Let the four thermodynamic response columns be

$$\rho_X(x) \in \mathcal{Y}, \quad X \in \{p, v, s, t\},$$

with synthesis map

$$T_x : \mathbb{C}^4 \rightarrow \mathcal{Y}, \quad T_x e_X = \rho_X(x).$$

Define the compatibility Gram matrix

$$M_x = T_x^* T_x.$$

On

$$\mathcal{S}_{\text{th}}(x) = (\ker M_x)^\perp,$$

define

$$\boxed{J_{\text{th}}(x) = T_x M_x^{-1/2}.$$

Then

$$J_{\text{th}}(x)^* J_{\text{th}}(x) = I_{\mathcal{S}_{\text{th}}(x)}.$$

55.2 2. Canonical source-to-response coupling

Let J_q be an isometric reconstruction of a declared native source block and let $\mathcal{C}$ be the native channel isometry. Define

$$\boxed{B_q(x) = J_{\text{th}}(x)^* \mathcal{C} J_q.$$

Let

$$H_{\text{th}}(x) > 0$$

be a stable response metric obtained from positive susceptibilities or compatibility-Gram whitening.

Define the response cost

$$Q_q(x) = B_q(x)^* H_{\text{th}}(x) B_q(x) \geq 0.$$

55.3 Theorem 2.1 (bounded thermodynamic response barrier)

Define

$$A_q = e^{-Q_q/2}, \quad W_q = I - e^{-Q_q}.$$

Then

$$0 \leq W_q < I$$

and

$$\ker W_q = \ker Q_q = \ker B_q.$$

Proof

Since $H_{\text{th}} > 0$,

$$Q_q = (H_{\text{th}}^{1/2} B_q)^* (H_{\text{th}}^{1/2} B_q) \geq 0,$$

and

$$\ker Q_q = \ker (H_{\text{th}}^{1/2} B_q) = \ker B_q.$$

Functional calculus on the nonnegative operator Q_q gives

$$0 < e^{-Q_q} \leq I,$$

hence $0 \leq W_q < I$. The scalar function $1 - e^{-t}$ vanishes exactly at $t = 0$, so

$$\ker W_q = \ker Q_q.$$

QED.

55.4 3. Basis invariance

Under unitary changes of source and response coordinates,

$$B_q \mapsto V^* B_q U, \quad H_{\text{th}} \mapsto V^* H_{\text{th}} V.$$

Therefore

$$Q_q \mapsto U^* Q_q U, \quad W_q \mapsto U^* W_q U.$$

Spectrum, rank, kernel, and positivity are invariant.

55.5 4. Barrier interpretation

The spectrum of Q_q measures response cost. The bounded defect W_q is the corresponding response barrier. Radial response magnitude changes the barrier strength; unitary angular changes rotate its orientation without changing its spectrum.

55.6 Claim boundary

CANONICAL RESPONSE-FRAME WHITENING	PROVED
CANONICAL COUPLING B_q	PROVED ON DECLARED SOURCE RANGE
POSITIVE RESPONSE COST Q_q	PROVED
BOUNDED BARRIER W_q	PROVED
KERNEL IDENTITY	PROVED
UNITARY BASIS INVARIANCE	PROVED
PHYSICAL SOURCE IDENTIFICATION	OPEN
CONTINUUM RESPONSE-CURVATURE IDENTITY	OPEN

RKF/theorum/thermodynamics/03_canonical_response_cost_barrier.md

Chapter 56

Thermodynamic Projection-Defect and Source-Overlap Theorems

56.1 1. Projection defect

Let $\mathcal{R}_{\text{th}}$ be the span of the declared thermodynamic response vectors and let

$$P_{\text{th}} : \mathcal{H} \rightarrow \mathcal{R}_{\text{th}}$$

be the orthogonal projection.

For $v \in \mathcal{H}$, define the thermodynamic recognition defect

$$d_{\text{th}}(v) = \|(I - P_{\text{th}})v\|^2.$$

56.2 Theorem 1.1 (thermodynamic projection defect)

For every $v \in \mathcal{H}$,

$$\|v\|^2 = \|P_{\text{th}}v\|^2 + d_{\text{th}}(v).$$

Moreover,

$$d_{\text{th}}(v) = 0 \iff v \in \text{Ran } P_{\text{th}}.$$

Proof

The two vectors $P_{\text{th}}v$ and $(I - P_{\text{th}})v$ are orthogonal. The first identity is Pythagoras. The second follows because a norm vanishes exactly on the zero vector. QED.

This theorem records an exact limitation: a finite response frame cannot recognize a component outside its span.

56.3 2. Thermodynamic source synthesis

Let

$$S_{\text{th}} : \mathbb{C}^m \rightarrow \mathcal{H}$$

be the synthesis map of declared thermodynamic source vectors $s_1, \dots, s_m$, and define

$$G_{\text{th}} = S_{\text{th}}^* S_{\text{th}}.$$

56.4 Theorem 2.1 (source-overlap Gram identity)

The matrix G_{th} is positive semidefinite and

$$(G_{\text{th}})_{ij} = \langle s_i, s_j \rangle.$$

For every coefficient vector $c \in \mathbb{C}^m$,

$$c^* G_{\text{th}} c = \|S_{\text{th}} c\|^2 \geq 0.$$

Hence

$$\ker G_{\text{th}} = \ker S_{\text{th}}.$$

Proof

Directly,

$$c^* G_{\text{th}} c = c^* S_{\text{th}}^* S_{\text{th}} c = \|S_{\text{th}} c\|^2.$$

QED.

56.5 3. Event-resolved overlap

If each recognition event e supplies a response synthesis $S_{\text{th},e}$, then the event-resolved overlap kernel is

$$G_{e,f} = S_{\text{th},e}^* S_{\text{th},f}.$$

The block matrix $(G_{e,f})_{e,f}$ is positive semidefinite because it is the Gram matrix of the total event synthesis.

56.6 Claim boundary

THERMODYNAMIC PROJECTION DEFECT	PROVED
ZERO-DEFECT / VISIBLE-RANGE EQUIVALENCE	PROVED
SOURCE-OVERLAP GRAM POSITIVITY	PROVED
EVENT-RESOLVED BLOCK GRAM POSITIVITY	PROVED
FINITE RESPONSE FRAME COMPLETENESS	NOT AUTOMATIC
THERMO SOURCE = ARITHMETIC SOURCE	NOT PROVED HERE

RKF/theorum/thermodynamics/04_projection_defect_and_source_overlap.md

Chapter 57

Onsager Compass and Constitutive-Path No-Go Theorems

57.1 1. Dissipative and orientational parts

Let L be a finite thermodynamic response operator. Write

$$L = L_s + L_a, \quad L_s = \frac{L + L^*}{2}, \quad L_a = \frac{L - L^*}{2}.$$

For a real response force X , define entropy production by

$$\sigma(X) = \langle X, LX \rangle.$$

57.2 Theorem 1.1 (entropy production sees only the symmetric response)

$$\boxed{\sigma(X) = \langle X, L_s X \rangle.}$$

In particular, if $L_s \geq 0$, then $\sigma(X) \geq 0$.

Proof

Since $L_a^* = -L_a$, the scalar $\langle X, L_a X \rangle$ is purely imaginary. On the real thermodynamic force space its contribution to the real entropy-production form is zero. Therefore the real quadratic form is generated by L_s . QED.

The antisymmetric part records circulation, orientation, or reversible rotation rather than dissipative production.

57.3 2. Onsager compass

On the compatible response range, let

$$L = U|L|$$

be the polar decomposition. The positive part $|L|$ supplies response magnitude, while the partial isometry U supplies orientation. Together with the exact/coexact/harmonic Hodge grading, this defines the finite Onsager compass.

Under a unitary response-coordinate change V ,

$$L \mapsto V^*LV, \quad |L| \mapsto V^*|L|V, \quad U \mapsto V^*UV.$$

Thus its singular values, rank, and Hodge dimensions are invariant.

57.4 Theorem 2.1 (constitutive-path no-go)

A thermodynamic constitutive operator or equilibrium response path does not, by itself, determine a unique native source amplitude in a different source space.

More precisely, suppose only the response Gram

$$G = L^*L$$

is known. Then every map

$$\tilde{L} = VL$$

with V unitary on the final response range has the same Gram:

$$\tilde{L}^*\tilde{L} = L^*L.$$

Therefore response cost and entropy production do not determine the missing final-range orientation or a unique arithmetic source embedding.

Consequence

An identity of the form

$$\text{thermodynamic compass} = \text{native arithmetic source amplitude}$$

requires an explicit typed intertwiner. It cannot be inferred from matching finite Gram matrices or from nonnegative entropy production alone.

57.5 Claim boundary

SYMMETRIC / ANTISYMMETRIC RESPONSE SPLIT	PROVED
ENTROPY-PRODUCTION REDUCTION TO L_s	PROVED
POLAR ONSAGER COMPASS	CONSTRUCTED
UNITARY COVARIANCE	PROVED
CONSTITUTIVE GRAM NON-UNIQUENESS	PROVED
UNIQUE THERMO $\rightarrow$ ARITHMETIC AMPLITUDE	NOT DETERMINED
EXPLICIT SOURCE INTERTWINER	OPEN

RKF/theorum/thermodynamics/05_onsager_compass_and_constitutive_no_go.md

Chapter 58

Onsager-Hodge No-Leakage Theorem

58.1 1. Response-space decomposition

Let the finite thermodynamic response space admit an orthogonal Hodge decomposition

$$\mathcal{H}_{\text{th}} = \mathcal{H}_{\text{ex}} \oplus \mathcal{H}_{\text{coex}} \oplus \mathcal{H}_{\text{har}}.$$

Let $P_{\text{ex}}, P_{\text{coex}}, P_{\text{har}}$ be the corresponding orthogonal projections.

Let A denote the accepted response sector and D a declared defect sector. Assume:

1. the Hodge projections reduce the accepted response metric;
2. accepted and defect syntheses are typed into declared orthogonal Hodge sectors;
3. the response intertwiner respects the Hodge grading;
4. all source maps used below are bounded on the finite packet.

58.2 Theorem 1.1 (finite Onsager-Hodge no-leakage)

If

$$A \subseteq \mathcal{H}_{\text{ex}} \oplus \mathcal{H}_{\text{har}}, \quad D \subseteq \mathcal{H}_{\text{coex}},$$

then

$$\boxed{P_A P_D = 0}$$

and every accepted-defect cross Gram block vanishes:

$$\boxed{S_A^* S_D = 0.}$$

Consequently the total response Gram splits orthogonally:

$$(S_A \oplus S_D)^* (S_A \oplus S_D) = S_A^* S_A \oplus S_D^* S_D.$$

Proof

The Hodge sectors are mutually orthogonal. Hence the orthogonal projections onto the accepted and defect ranges have zero product. The synthesis ranges lie in these orthogonal sectors, so

$$\langle S_A a, S_D d \rangle = 0$$

for all coefficients a, d , which is equivalent to $S_A^* S_D = 0$. QED.

58.3 Corollary 1.2 (no hidden defect cancellation)

Under the theorem's assumptions, a nonzero defect energy cannot be cancelled by an accepted response vector, because

$$\|S_A a + S_D d\|^2 = \|S_A a\|^2 + \|S_D d\|^2.$$

Thus zero total response forces both orthogonal components to vanish separately.

58.4 2. Claim discipline

This theorem is exact on a declared finite response packet. It does not prove that a physical thermodynamic response, a continuum native source, and an arithmetic Weil source share the required Hodge grading. That identification is a separate provenance theorem.

58.5 Claim boundary

FINITE HODGE ORTHOGONAL SPLITTING	ASSUMED / DECLARED
ACCEPTED-DEFECT CROSS BLOCK	PROVED ZERO
ORTHOGONAL ENERGY SPLITTING	PROVED
NO HIDDEN CROSS-SECTOR CANCELLATION	PROVED
PHYSICAL HODGE GRADING OF NATIVE SOURCE	OPEN
CONTINUUM NO-LEAKAGE	OPEN

RKF/theorum/thermodynamics/06_onsager_hodge_no_leakage.md

Chapter 59

Thermo-Weil Provenance, Five-Source, and Boundary-Schur Theorem

59.1 1. Provenance contract

Let

$$C_{\text{th}} : \mathbb{C}^5 \rightarrow \mathbb{C}^4$$

be a proposed boundary-plus-tetrad thermodynamic coupling and let

$$R_{\text{W}} : \mathbb{C}^5 \rightarrow \mathbb{C}^4$$

be the finite Weil response synthesis.

The statement

$$C_{\text{th}}^* H_{\text{th}} C_{\text{th}} = R_{\text{W}}^* R_{\text{W}}$$

may be promoted from finite matching to a source theorem only when the following are independently certified:

1. physical thermodynamic source lineage;
2. normalization and units;
3. positive response metric $H_{\text{th}} > 0$;
4. event-resolved five-source assignment;
5. native source-chain identification;
6. covariance under admissible coordinate changes.

59.2 Theorem 1.1 (finite five-source Gram equivalence)

Put

$$G_{\text{th}} = H_{\text{th}}^{1/2} C_{\text{th}}.$$

If

$$G_{\text{th}}^* G_{\text{th}} = R_{\text{W}}^* R_{\text{W}},$$

then there exists a partial isometry U from the final range of R_{W} to the final range of G_{th} such that

$$\boxed{G_{\text{th}} = U R_{\text{W}}.}$$

If both maps have full row rank four, $U \in U(4)$.

Proof

Two finite synthesis maps with the same Gram define the same seminorm on the common coefficient space. Quotienting by the shared kernel gives an isometry between their ranges, which extends to the stated partial isometry. Full row rank makes the final ranges equal to $\mathbb{C}^4$, so the partial isometry is unitary. QED.

This theorem proves finite unitary equivalence. The provenance contract determines whether that equivalence represents the same physical source.

59.3 2. Boundary Schur realization

Let the total finite cost matrix be written as

$$\mathcal{Q} = \begin{pmatrix} A & B \\ B^* & D \end{pmatrix}, \quad D > 0.$$

Define the boundary Schur complement

$$\boxed{S_{\partial} = A - B D^{-1} B^*}.$$

59.4 Theorem 2.1 (thermo-Weil boundary Schur theorem)

The block matrix $\mathcal{Q}$ is positive semidefinite if and only if

$$D > 0 \quad \text{and} \quad S_{\partial} \geq 0.$$

Moreover,

$$\text{inertia}(\mathcal{Q}) = \text{inertia}(D) + \text{inertia}(S_{\partial}).$$

Proof

Use the exact congruence

$$\begin{pmatrix} I & -BD^{-1} \\ 0 & I \end{pmatrix} \mathcal{Q} \begin{pmatrix} I & 0 \\ -D^{-1}B^* & I \end{pmatrix} = \begin{pmatrix} S_\partial & 0 \\ 0 & D \end{pmatrix}.$$

Congruence preserves inertia. QED.

59.5 Claim boundary

PROVENANCE INPUT CONTRACT	DECLARED
FINITE SAME-GRAM PARTIAL ISOMETRY	PROVED
FULL-RANK U(4) INTERTWINER	PROVED
BOUNDARY SCHUR POSITIVITY	PROVED
SCHUR INERTIA SPLITTING	PROVED
PHYSICAL THERMO / WEIL SOURCE IDENTITY	OPEN WITHOUT PROVENANCE
COFINAL CONTINUUM LIMIT	OPEN

RKF/theorum/thermodynamics/07_thermo_weil_provenance_five_source_schur.md

Chapter 60

Canonical Connection and Physical-Flux Intertwiner Theorem

60.1 1. Canonical effect projection

Let $0 < A < I$ on a finite-dimensional accepted response space E , and put

$$S_A = (A - A^2)^{1/2}.$$

On $E \oplus E$, define

$$P_A = \begin{pmatrix} A & S_A \\ S_A & I - A \end{pmatrix}, \quad \Gamma = \begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix}.$$

Then $P_A = P_A^* = P_A^2$.

60.2 2. Constant non-Abelian connection

Define skew-Hermitian connection coefficients

$$\mathcal{A}_r = \frac{i}{2}\Gamma, \quad \mathcal{A}_\tau = iP_A.$$

Their constant mixed curvature is

$$\mathcal{F}_{r\tau}^{\text{gauge}} = [\mathcal{A}_r, \mathcal{A}_\tau] = -\frac{1}{2}[\Gamma, P_A].$$

Define the self-adjoint information flux

$$\boxed{\mathcal{F}_A = \frac{1}{2i}[\Gamma, P_A] = i\mathcal{F}_{r\tau}^{\text{gauge}}.}$$

60.3 3. Accepted curvature amplitude

Let $I_+z = (z, 0)$ and let π_- project onto the second component. Then

$$\boxed{\mathfrak{F}_A^{\text{acc}} := \pi_- \mathcal{F}_A I_+ = iS_A.}$$

Hence

$$\boxed{(\mathfrak{F}_A^{\text{acc}})^* \mathfrak{F}_A^{\text{acc}} = A - A^2.}$$

60.4 4. Physical stacked leakage

Let $C : E \rightarrow \mathcal{H}_C$ and suppose

$$R = A - A^2 - C^*C \geq 0.$$

Put

$$L = R^{1/2}, \quad \mathcal{L}_{C,R} = \begin{pmatrix} C \\ L \end{pmatrix}.$$

Then

$$\mathcal{L}_{C,R}^* \mathcal{L}_{C,R} = A - A^2 = S_A^2.$$

Define

$$U_{\text{flux}} = \mathcal{L}_{C,R} S_A^{-1}.$$

60.5 Theorem 4.1 (physical curvature intertwiner)

$$\boxed{U_{\text{flux}}^* U_{\text{flux}} = I_E}$$

and

$$\boxed{\mathcal{L}_{C,R} = -i U_{\text{flux}} \mathfrak{F}_A^{\text{acc}}.}$$

Proof

Using $\mathcal{L}_{C,R}^* \mathcal{L}_{C,R} = S_A^2$,

$$U_{\text{flux}}^* U_{\text{flux}} = S_A^{-1} S_A^2 S_A^{-1} = I.$$

Since $\mathfrak{F}_A^{\text{acc}} = i S_A$,

$$-i U_{\text{flux}} \mathfrak{F}_A^{\text{acc}} = U_{\text{flux}} S_A = \mathcal{L}_{C,R}.$$

QED.

Thus one canonical curvature amplitude splits isometrically into ordinary complement leakage C and information-curvature completion L .

60.6 Claim boundary

CANONICAL EFFECT PROJECTION	PROVED
NON-ABELIAN CONNECTION CURVATURE	PROVED
ACCEPTED CURVATURE GRAM = $A-A^2$	PROVED
PHYSICAL STACKED-LEAKAGE INTERTWINER	PROVED
ORDINARY / INFORMATION FLUX SPLITTING	PROVED
CONTINUUM RESPONSE POSITIVITY	OPEN
CONTINUUM SEAM-FLATNESS	OPEN

RKF/theorum/thermodynamics/08_canonical_connection_physical_flux.md

Chapter 61

Scope Exclusions and False-Positive Audit

This archive includes only results whose mathematical content is genuinely thermodynamic-response, Onsager, susceptibility, entropy-production, or thermodynamic-curvature based.

61.1 1. Eta-descent margin barrier

The MP file

`proof_lab/clock_free_recognition/ETA_DESCENT_MARGIN_BARRIER.md`

proves a certificate-descent obstruction. If

$$\eta_{j+1} = \frac{\eta_j}{1 + q_j}$$

and

$$\sum_j \log(1 + q_j) < \infty,$$

then

$$\eta_\infty = \eta_0 \prod_j (1 + q_j)^{-1} > 0.$$

This is a convergence/certificate theorem, not a thermodynamics theorem. It is related only because a direct thermodynamic response-curvature identity was proposed as one possible alternative to repeated eta descent.

61.2 2. Native energy-channel exterior rates

The MP file

`papers/rh_initial_journey/NATIVE_ENERGY_CHANNEL_EXTERIOR_RATES.md`

uses the word **energy** for operator and source-channel norms. Its exponential exterior estimates concern multiplicative carriers, prime dilations, Gamma channels, and finite-rank augmentation. They are functional-analytic energy estimates, not caloric or thermodynamic energy laws.

61.3 3. Raw Hessian positivity

The raw Hessian of a thermodynamic potential is not automatically a positive response metric in the chosen coordinates. For example, on a stable material,

$$\frac{1}{S} \left(\frac{\partial V}{\partial P} \right)_S = -\frac{V}{SK_S} < 0.$$

Therefore this archive uses a positive metric obtained from stable susceptibilities or compatibility-Gram whitening. It does not import an indefinite raw Hessian as a Riemannian metric.

61.4 4. Smooth-scalar curvature warning

A smooth scalar potential has commuting mixed partial derivatives. Path dependence alone does not make its Hessian noncommutative. Nonzero curvature requires a non-exact connection, singular structure, non-smoothness, or an explicitly operator-valued transport law.

61.5 5. Speculative extensions

Claims involving quantum field theory, gauge unification, gravity, black holes, or cosmology are not part of this theorem archive unless separately derived from declared Recognition Kernel objects and proved with typed source maps.

61.6 Claim boundary

ETA-DESCENT BARRIER

OPERATOR ENERGY EXTERIOR RATES

RAW HESSIAN AS AUTOMATIC POSITIVE METRIC

PATH DEPENDENCE AS SMOOTH-SCALAR CURVATURE

SPECULATIVE QFT / GRAVITY EXTENSIONS

EXCLUDED: NOT THERMODYNAMICS

EXCLUDED: NOT THERMODYNAMICS

REJECTED

REJECTED

QUARANTINED

RKF/theorum/thermodynamics/09_scope_exclusions.md

Chapter 62

Lambda Holonomy, Branch Memory, and Local Spectral Factor

62.1 Provenance

source repository: Parveen117/MP

source PR: #52

source theorem: adapters/common/LAMBDA_HOLONOMY_SPECTRAL_OPERATOR.md

This capsule transfers the geometric theorems that follow once a connection is supplied to the thermodynamic-lambda state space. It does **not** claim that the thermodynamic hierarchy uniquely determines that connection.

62.2 1. Typed lambda hierarchy

The historical source begins with

$$X^{(0)} = (T, V, S, P), \quad X^{(1)} = (\lambda_p, \lambda_v, \lambda_s, \lambda_t),$$

and higher response layers

$$X^{(r+1)} = \mathcal{R}_r(X^{(r)}).$$

The later PR #239 typing correction applies: the historical derivative coordinates in the s, t directions are not silently identified with the uniform entropy-scaled tetrad $\lambda_X^{\text{th}} = -ST/C_X$.

62.3 2. Unitary connection and curvature

Let $E \rightarrow \mathcal{M}_\lambda$ be a Hermitian bundle over the lambda-state manifold and let

$$\nabla^\lambda = d + A_\lambda, \quad F_\lambda = dA_\lambda + A_\lambda \wedge A_\lambda,$$

with A_λ skew-adjoint in a unitary frame. For a based loop C ,

$$U_C = \text{Hol}_{\nabla^\lambda}(C)$$

is unitary.

The response hierarchy and the connection remain distinct typed objects.

62.4 3. Holonomy-rate self-adjointness theorem

For a differentiable unitary path U_t , define

$$\mathcal{M}_t = iU_t^{-1}\dot{U}_t.$$

Then

$$\boxed{\mathcal{M}_t^* = \mathcal{M}_t.}$$

Indeed, differentiating $U_t^*U_t = I$ shows that $U_t^{-1}\dot{U}_t$ is skew-adjoint.

62.5 4. Static logarithm branch-memory theorem

If

$$U_C = \sum_j e^{-i\theta_j} P_j, \quad \theta_j \in (-\pi, \pi],$$

then every self-adjoint logarithm compatible with a continuous spectral lift has the form

$$H_C = \sum_j (\theta_j + 2\pi k_j) P_j, \quad k_j \in \mathbb{Z}.$$

Therefore $i \log U_C$ is not a single branch-independent datum. The lawful stored object contains

$$\{(\theta_j, P_j, k_j)\}_j$$

plus the path or lift fixing the integers. Principal phase alone loses winding memory.

62.6 5. Complete sector signature

A hierarchy-resolved sector must retain at least

$$\Sigma_C = (r, [C], \{(\theta_j, P_j, k_j)\}_j, \mathcal{F}_C, \mathfrak{m}_C),$$

where r is hierarchy depth, $\mathcal{F}_C$ is curvature-flux or transgression data, and $\mathfrak{m}_C$ is seam memory.

No theorem may replace Σ_C by one winding integer without proving that the discarded data are irrelevant to the target.

62.7 6. Winding and spectral-flow theorem

In the scalar case, write

$$U_t = e^{-i\phi(t)}.$$

When endpoint holonomies agree,

$$\phi(t_1) - \phi(t_0) = 2\pi\nu, \quad \nu \in \mathbb{Z}.$$

For a reference angle avoided by the endpoints, the signed eigenphase crossing count satisfies

$$\boxed{\text{sf}(\mathcal{M}_t; \theta_*) = \nu.}$$

In finite rank this becomes the signed sum over continuously lifted spectral branches. Independent sorting of principal angles is not sufficient at degeneracies.

62.8 7. Composition boundary

For concatenated based paths,

$$U_{C_1 \star C_2} = U_{C_2} U_{C_1}.$$

In $U(1)$, or on commuting channels with compatible lifts, phases, winding, and declared curvature flux add. For noncommuting holonomies,

$$\log(UV) \neq \log U + \log V$$

in general; Baker-Campbell-Hausdorff commutator terms are part of the sector memory.

62.9 8. Finite local spectral-factor theorem

Given a supplied positive native length ℓ_C , define for $\Re s > 0$

$$Z_C(s) = \det \left(I - e^{-s\ell_C} U_C \right)^{-1}.$$

Then

$$\boxed{\log Z_C(s) = \sum_{m \geq 1} \frac{e^{-sm\ell_C}}{m} \text{Tr}(U_C^m).}$$

For scalar trivial holonomy and a separately proved $\ell_C = \log p$, this reduces to $(1 - p^{-s})^{-1}$. The theorem does not derive the prime assignment.

62.10 Claim boundary

UNITARY HOLONOMY-RATE SELF-ADJOINTNESS	PROVED
STATIC LOGARITHM BRANCH CLASSIFICATION	PROVED
GAUGE-CONJUGACY SPECTRAL INVARIANTS	PROVED
WINDING / SPECTRAL-FLOW IDENTITY	PROVED IN DECLARED FINITE MODEL
FINITE LOCAL DETERMINANT / TRACE-LOG	PROVED
CONNECTION FROM THERMODYNAMIC HIERARCHY	OPEN
CANONICAL NATIVE POSITIVE LENGTH	OPEN
PRIMITIVE SECTOR TO PRIME IDENTIFICATION	OPEN
INFINITE TRACE-CLASS COMPLETION	OPEN

RKF/theorum/thermodynamics/lambda_geometry/10_lambda_holonomy_and_branch_memory.md

Chapter 63

Thermodynamic Cut-Square Response Decomposition Theorem

63.1 1. Purpose

This capsule specializes the native cut-square and minimum-decoder theorems to the positive thermodynamic response fibre.

The result is an exact decomposition of thermodynamic response energy into:

1. energy captured by a declared scalar response channel;
2. unused contractive reserve;
3. pairwise transverse response squares.

It is a finite positive-metric identity. It does not, by itself, identify a physical material, derive an equation of state, or prove that a chosen response decoder is realized by nature.

63.2 2. Typed thermodynamic response fibre

Let

$$\mathcal{R}_{\text{th}} \cong \mathbb{C}^m$$

be a finite thermodynamic response fibre and let

$$H_{\text{th}} > 0$$

be its declared stable response metric. Define

$$\langle u, v \rangle_{H_{\text{th}}} = u^* H_{\text{th}} v, \quad \|u\|_{H_{\text{th}}}^2 = u^* H_{\text{th}} u.$$

Let

$$y \in \mathcal{R}_{\text{th}}$$

be an actual response event and let

$$c \in \mathcal{R}_{\text{th}}$$

be a declared thermodynamic decoder. Its scalar readout is

$$L_c(y) = \langle c, y \rangle_{H_{\text{th}}}.$$

Put

$$\beta_{\text{th}} = \|c\|_{H_{\text{th}}}^2.$$

The metric-whitened coordinates are

$$C = H_{\text{th}}^{1/2} c, \quad Y = H_{\text{th}}^{1/2} y.$$

All exterior-square expressions below are taken in these compatible whitened coordinates. Raw response coordinates with unlike physical units must not be wedged before the metric normalization is declared.

63.3 3. Exact thermodynamic cut-square identity

Theorem 3.1

For every thermodynamic response event y and decoder c ,

$$\boxed{\beta_{\text{th}} \|y\|_{H_{\text{th}}}^2 - |L_c(y)|^2 = \sum_{i < j} |C_i Y_j - C_j Y_i|^2.}$$

Equivalently,

$$\boxed{\|C\|^2 \|Y\|^2 - |\langle C, Y \rangle|^2 = \|C \wedge Y\|^2.}$$

Proof

After whitening, the statement is the finite complex Lagrange identity

$$\|C\|^2 \|Y\|^2 - |\langle C, Y \rangle|^2 = \sum_{i < j} |C_i Y_j - C_j Y_i|^2.$$

Since

$$\|C\|^2 = \beta_{\text{th}}, \quad \|Y\|^2 = \|y\|_{H_{\text{th}}}^2, \quad \langle C, Y \rangle = L_c(y),$$

the thermodynamic form follows. QED.

63.4 4. Contractive response decomposition

Theorem 4.1

If

$$\beta_{\text{th}} \leq 1,$$

then

$$\|y\|_{H_{\text{th}}}^2 - |L_c(y)|^2 = (1 - \beta_{\text{th}})\|y\|_{H_{\text{th}}}^2 + \sum_{i < j} |C_i Y_j - C_j Y_i|^2 \geq 0.$$

Therefore

$$H_{\text{th}} - H_{\text{th}} c c^* H_{\text{th}} \geq 0.$$

Proof

Add

$$(1 - \beta_{\text{th}})\|y\|_{H_{\text{th}}}^2$$

to both sides of Theorem 3.1. Every term on the right is nonnegative when $\beta_{\text{th}} \leq 1$. The operator statement is the same quadratic-form identity because

$$y^*(H_{\text{th}} - H_{\text{th}} c c^* H_{\text{th}})y = \|y\|_{H_{\text{th}}}^2 - |L_c(y)|^2.$$

QED.

63.5 5. Four-channel thermodynamic tetrad

For the declared response tetrad

$$X \in \{p, v, s, t\},$$

the transverse residue contains exactly six pairwise squares:

$$\begin{aligned} \|C \wedge Y\|^2 &= |C_p Y_v - C_v Y_p|^2 + |C_p Y_s - C_s Y_p|^2 + |C_p Y_t - C_t Y_p|^2 \\ &\quad + |C_v Y_s - C_s Y_v|^2 + |C_v Y_t - C_t Y_v|^2 + |C_s Y_t - C_t Y_s|^2. \end{aligned}$$

The $(\mathbf{p}, \mathbf{v})$ and $(\mathbf{s}, \mathbf{t})$ minors measure mismatch within the caloric and mechanical pairs. The remaining four minors measure cross-pair incompatibility. These names are interpretive labels only; the rigorous content is the positive metric-square decomposition.

63.6 6. Equality and no-leakage classification

Corollary 6.1

If

$$\beta_{\text{th}} < 1,$$

then equality in Theorem 4.1 occurs only for

$$y = 0.$$

If

$$\beta_{\text{th}} = 1,$$

then equality occurs exactly when

$$C \wedge Y = 0,$$

or equivalently when

$$Y = \alpha C$$

for some scalar α .

Thus a unit decoder has no transverse leakage exactly when the complete metric-normalized response event lies on the one-dimensional constitutive line selected by that decoder.

This is a response-alignment statement. It is not automatically a claim of equilibrium, reversibility, or zero entropy production.

63.7 7. Sharp thermodynamic decoder burden

Suppose a declared scalar response functional ℓ admits at least one representation

$$\ell(y) = \langle c, y \rangle_{H_{\text{th}}}.$$

Define its sharp thermodynamic burden

$$\beta_{\text{th}}^{\min} = \inf \left\{ \|c\|_{H_{\text{th}}}^2 : \ell(y) = \langle c, y \rangle_{H_{\text{th}}} \text{ for all } y \right\}.$$

In a complete finite positive fibre the Riesz decoder is unique, so the infimum is attained. The native minimum-decoder theorem then specializes to

$$\boxed{H_{\text{th}} - \ell^* \ell \geq 0 \iff \beta_{\text{th}}^{\min} \leq 1.}$$

Any nonminimal enlarged event representation can only add orthogonal decoder energy. Constructing a decoder therefore does not prove contractivity; its sharp burden must still be bounded by one.

63.8 8. Onsager entropy-production specialization

Let the dissipative thermodynamic response metric be the positive symmetric Onsager part

$$H_{\text{th}} = L_{\text{sym}} > 0.$$

Then

$$\sigma(y) = y^* L_{\text{sym}} y$$

is the declared quadratic entropy-production form, and Theorem 4.1 gives

$$\boxed{\sigma(y) - |L_c(y)|^2 = (1 - \beta_{\text{th}})\sigma(y) + \|C \wedge Y\|^2 \geq 0.}$$

The antisymmetric Onsager component contributes no quadratic entropy production and is not part of this positive cut-square identity. It belongs to the orientation/circulation side of the Onsager compass developed elsewhere in this archive.

63.9 9. Relation to flat caloric-mechanical closure

The flat scalar relation

$$\Gamma_c \Gamma_m = 1$$

checks one caloric-mechanical ratio. By contrast, cut-square closure at unit burden requires all six whitened pairwise minors to vanish.

Under the necessary nonzero-coordinate assumptions, complete cut-square closure can imply the corresponding ratio consistency. The converse is not generally valid: one scalar product identity can hold while one or more cross-response minors remain nonzero.

Therefore the cut-square theorem is a strictly finer response-coherence test than the flat closure relation.

63.10 10. Interface with the thermodynamic response cost

Let a source state $\mathbf{x}$ produce the thermodynamic response

$$y = Bx$$

and define the canonical response cost

$$Q_{\text{th}} = B^* H_{\text{th}} B \geq 0.$$

Then a contractive decoder gives

$$\boxed{Q_{\text{th}} - B^* H_{\text{th}} c c^* H_{\text{th}} B \geq 0.}$$

Pointwise on $\mathbf{x}$, its defect is exactly the contractive reserve plus the thermodynamic cut-square residue of $B\mathbf{x}$.

This theorem controls extraction of one rank-one response channel from Q_{th} . The nonlinear barrier

$$W_{\text{th}} = I - e^{-Q_{\text{th}}}$$

remains the separate functional-calculus construction proved in the canonical response-cost capsule. No unproved commutation or direct identification between the rank-one cut defect and W_{th} is assumed here.

63.11 11. Claim boundary

POSITIVE-METRIC THERMODYNAMIC CUT-SQUARE IDENTITY	PROVED
CONTRACTIVE RESPONSE NO-LEAKAGE DECOMPOSITION	PROVED
FOUR-CHANNEL SIX-MINOR EXPANSION	PROVED
EQUALITY / COMPLETE RESPONSE-ALIGNMENT CLASSIFICATION	PROVED
SHARP DECODER BURDEN CRITERION	PROVED IN FINITE POSITIVE FIBRE
ONSAGER SYMMETRIC ENTROPY-PRODUCTION SPECIALIZATION	PROVED UNDER DECLARED METRIC
RANK-ONE CHANNEL DOMINATION OF RESPONSE COST	PROVED
PHYSICAL IDENTIFICATION OF A PARTICULAR DECODER	NOT PROVIDED
EQUILIBRIUM OR REVERSIBILITY FROM CUT-SQUARE EQUALITY	NOT CLAIMED
UNIVERSAL MATERIAL LAW	NOT CLAIMED
DIRECT IDENTIFICATION WITH THE NONLINEAR BARRIER	NOT CLAIMED

RKF/theorum/thermodynamics/10_thermodynamic_cut_square_response_decomposition.md

Chapter 64

Noncommutative Holonomy, Curvature, and Irreducibility

64.1 Provenance

source repository: Parveen117/MP

source PR: #53

source theorem: adapters/common/NONCOMMUTATIVE_PATH_ORDERED_HOLONOMY.md

64.2 1. Seam-dagger lift

On

$$\mathcal{H} = \mathbb{C}_{\text{seam}}^2 \otimes \mathbb{C}_{\text{int}}^2,$$

let

$$R^2 = -I, \quad K^2 = I, \quad RK = -KR, \quad X^\sharp = KX^*K.$$

If $A(t) \in \mathfrak{su}(2)$ and $\tilde{A}(t) = I_2 \otimes A(t)$, then

$$\tilde{A}(t)^\sharp = -\tilde{A}(t).$$

Consequently its transport is dagger-unitary:

$$\tilde{U}(t)^\sharp \tilde{U}(t) = I.$$

64.3 2. Noncommutative path-order theorem

For piecewise-constant segments,

$$U_C = e^{\Delta t_N A_N} \dots e^{\Delta t_1 A_1}.$$

For

$$A_x = -i\alpha\sigma_x, \quad A_z = -i\beta\sigma_z,$$

one has

$$[A_x, A_z] = 2i\alpha\beta\sigma_y \neq 0$$

when $\alpha\beta \neq 0$. Therefore

$$U_{xz} = e^{A_z}e^{A_x} \quad \text{and} \quad U_{zx} = e^{A_x}e^{A_z}$$

need not coincide.

64.4 3. Spectral-shadow collision theorem

Although $U_{xz} \neq U_{zx}$,

$$U_{zx} = e^{A_x}U_{xz}e^{-A_x}.$$

Thus they are similar and have the same trace, determinant, characteristic polynomial, and eigenvalue multiset.

Hence

principal spectrum alone does not recover path order.

The ordered segment word and commutator-curvature memory must remain part of the represented state.

64.5 4. Small-square curvature theorem

For

$$U_{\square}(\varepsilon) = e^{\varepsilon A_z}e^{\varepsilon A_x}e^{-\varepsilon A_z}e^{-\varepsilon A_x},$$

Baker-Campbell-Hausdorff expansion gives

$$U_{\square}(\varepsilon) = I + \varepsilon^2[A_z, A_x] + O(\varepsilon^3).$$

The first order-sensitive residue of the ordered square is therefore the commutator curvature.

64.6 5. Common-commutant irreducibility theorem

For a star-closed finite operator family $\mathcal{F} \subset M_2(\mathbb{C})$, define

$$\mathcal{F}' = \{X : XA = AX \text{ for every } A \in \mathcal{F}\}.$$

Then

$$\mathcal{F} \text{ acts irreducibly} \iff \mathcal{F}' = \mathbb{C}I.$$

A single-axis family has a two-dimensional commutant and is reducible. The two-axis x/z Pauli family has scalar commutant and is irreducible.

The verdict is invariant under constant $SU(2)$ gauge conjugation.

64.7 6. Three distinct notions

The source proves that the following must not be conflated:

representation irreducibility $\neq$ composition-primitivity $\neq$ arithmetic primality.

Each requires a separate theorem and a separate category of admissible decompositions.

64.8 7. Nonmultiplicativity of local factors

For a supplied positive length,

$$Z_C(s) = \det(I - e^{-s\ell_C} U_C)^{-1}.$$

For noncommuting segments, the composed factor does not generally satisfy

$$Z_{C_2 \star C_1} = Z_{C_2} Z_{C_1}.$$

Path ordering is therefore load-bearing data rather than cosmetic notation.

64.9 Claim boundary

DAGGER-SKEW INTERNAL CONNECTION LIFT	PROVED
NONCOMMUTATIVE PATH-ORDER DIFFERENCE	PROVED
EQUAL-SPECTRUM ORDER COLLISION	PROVED
SMALL-SQUARE COMMUTATOR CURVATURE	PROVED
COMMON-COMMUTANT IRREDUCIBILITY	PROVED
GAUGE INVARIANCE OF IRREDUCIBILITY	PROVED
NAIVE LOCAL-FACTOR MULTIPLICATION	DISPROVED
CANONICAL CONTINUUM LAMBDA CONNECTION	OPEN
COMPOSITION-PRIMITIVITY CATEGORY	OPEN AT THIS STAGE
ARITHMETIC PRIME IDENTIFICATION	OPEN

RKF/theorum/thermodynamics/lambda_geometry/11_noncommutative_holonomy_and_irreducibility.md

Chapter 65

Admissible Sector Factorization and Composition-Primitivity

65.1 Provenance

source repository: Parveen117/MP
source PR: #54
source theorem: adapters/common/ADMISSIBLE_SECTOR_FACTORIZATION.md

65.2 1. Ordered sector datum

A finite hierarchy-resolved sector is an ordered list of segments carrying

source hierarchy state
target hierarchy state
su(2) connection increment
winding increment
memory token.

Its path-ordered holonomy is

$$U_{\Sigma} = e^{A_n} \dots e^{A_1}.$$

The hierarchy state contains both the base state and the active response depth.

65.3 2. Admissible-cut theorem

An interior cut is admissible only when it is a return to the complete hierarchy base state and both factors reconstruct all active data:

- the ordered segment word;
- path-ordered holonomy;
- winding lift;
- memory word;
- seam-dagger transport;

- local curvature ledgers;
- cross-factor curvature memory.

For an admissible cut k ,

$$U_\Sigma = U_{\text{right}} U_{\text{left}},$$

$$\nu(\Sigma) = \nu(\text{left}) + \nu(\text{right}),$$

and the memory words concatenate in order.

The full curvature ledger obeys

$$\mathcal{C}(\Sigma) = \mathcal{C}(\text{left}) \sqcup \mathcal{C}(\text{right}) \sqcup \mathcal{C}_{\text{cross}}.$$

The cross term retains commutators between segments lying on opposite sides of the cut. A decomposition that reproduces the endpoint matrix but discards this history is not admissible.

65.4 3. Composition-primitivity theorem

Within the declared finite ordered-word category,

$$\Sigma \text{ is composition-primitive} \iff \text{there is no nontrivial intermediate return of the full hierarchy state.}$$

A sector may be representation-irreducible but composition-composite. Thus irreducibility cannot substitute for the return-cut theorem.

65.5 4. Gauge invariance

Under constant $SU(2)$ conjugation,

$$A_j \mapsto G A_j G^{-1}, \quad U_\Sigma \mapsto G U_\Sigma G^{-1}.$$

Hierarchy returns, winding, memory order, admissible cuts, and composition-primitivity remain unchanged. Curvature channels transform covariantly.

65.6 5. Tamper-rejection theorem

A proposed factorization is rejected if it:

- alters winding or memory data;
- cuts at a state that is not a complete hierarchy return;
- omits the cross-curvature ledger;
- reconstructs only the endpoint holonomy while losing ordered source history.

65.7 6. Preliminary positive length

The source records

$$L_{\text{act}}(\Sigma) = \sum_j \|A_j\|.$$

It is positive, constant-gauge invariant, and additive across admissible macro cuts. At PR #54 it remains a candidate because subdivision and reparameterization invariance have not yet been proved.

65.8 Claim boundary

ADMISSIBLE COMPLETE-STATE RETURN CUT	PROVED
ORDERED HOLONOMY FACTORIZATION	PROVED
WINDING AND MEMORY RECONSTRUCTION	PROVED
CROSS-FACTOR CURVATURE REQUIREMENT	PROVED
COMPOSITION-PRIMITIVITY CRITERION	PROVED IN FINITE WORD CATEGORY
GAUGE INVARIANCE OF CUT SET	PROVED
TAMPERED FACTORIZATIONS	REJECTED
REFINEMENT-STABLE LENGTH	OPEN AT THIS STAGE
CONTINUUM FACTORIZATION	OPEN AT THIS STAGE
ARITHMETIC PRIMALITY	NOT IMPLIED

RKF/theorum/thermodynamics/lambda_geometry/12_admissible_factorization_and_primitivity.md

Chapter 66

Refinement-Stable and Continuum Native Length

66.1 Provenance

source repository: Parveen117/MP
source PRs: #55 and #56
source theorems:
 adapters/common/REFINEMENT_INVARIANT_NATIVE_LENGTH.md
 adapters/common/CONTINUUM_HOMOTOPY_DESCENT.md

66.2 1. Positive collinear refinement

A macro increment $A \in \mathfrak{su}(2)$ may be subdivided by positive weights

$$a_k > 0, \quad \sum_{k=1}^r a_k = 1,$$

into ordered micro-increments $a_1 A, \dots, a_r A$. Since they commute,

$$e^{a_r A} \dots e^{a_1 A} = e^A.$$

The macro hierarchy endpoints, winding increment, and memory token remain attached to the macro segment. Artificial discretization boundaries are not promoted into new hierarchy-return cuts.

66.3 2. Directed refinement category

Objects are positive partitions of each macro segment. A morphism is breakpoint inclusion from a coarser partition to a finer one. The union of breakpoint sets gives a common refinement, so the finite refinement category is directed.

66.4 3. Native length-rigidity theorem

Let

$$\lambda : \mathfrak{su}(2) \rightarrow [0, \infty)$$

be continuous, nonnegative, invariant under constant $SU(2)$ conjugation, and additive under every positive collinear split:

$$\lambda(A) = \lambda(tA) + \lambda((1-t)A), \quad 0 < t < 1.$$

Adjoint invariance makes λ radial:

$$\lambda(A) = f(\|A\|).$$

The split law gives the continuous Cauchy equation, hence

$$\boxed{\lambda(A) = c\|A\|, \quad c \geq 0.}$$

Therefore the sector length is rigid up to an overall scale:

$$\boxed{L_c(\Sigma) = c \sum_j \|A_j\|.}$$

Within the declared axioms, nonlinear radial alternatives such as $\|A\|^2$ and $\|A\|^{1/2}$ fail subdivision invariance.

66.5 4. Finite refinement consequences

The theorem gives:

- exact holonomy preservation;
- common-refinement directedness;
- subdivision invariance of L_c ;
- constant-gauge invariance;
- additivity at admissible macro return cuts;
- refinement invariance of composition-primitivity;
- nondegeneracy on active connection sectors when $c > 0$.

Pure winding or memory sectors with zero connection increments may still have zero geometric length unless separate ledger terms are lawfully introduced.

66.6 5. Continuum length

For a continuous coefficient

$$A(t) = -i a(t) \cdot \sigma,$$

define

$$L_c[A] = c \int_0^1 \|a(t)\| dt$$

and transport by

$$U'(t) = A(t)U(t), \quad U(0) = I.$$

For midpoint samples $t_k = (k + 1/2)/N$,

$$L_N = \frac{c}{N} \sum_{k=0}^{N-1} \|a(t_k)\|,$$

and the ordered exponential product converges to the continuum transport.

If a is Lipschitz with derivative bound M ,

$$\boxed{|L_N - L_c[A]| \leq \frac{cM}{4N}.$$

66.7 6. Reparameterization theorem

For an orientation-preserving C^1 diffeomorphism ϕ fixing the endpoints, define

$$\tilde{A}(s) = \phi'(s)A(\phi(s)).$$

Then

$$\boxed{L_c[\tilde{A}] = L_c[A], \quad \tilde{U}(1) = U(1).}$$

Orientation reversal preserves length but reverses transport:

$$U_{\text{rev}}(1) = U(1)^{-1}.$$

Constant gauge conjugation preserves the length and conjugates the endpoint holonomy.

66.8 7. Unrestricted homotopy no-go

For contractible circles

$$\gamma_r(t) = r(\cos 2\pi t, \sin 2\pi t, 0),$$

one has

$$L(r) = 2\pi r.$$

Homotopic loops can therefore have distinct positive lengths and collapse continuously to zero.
Hence

L_c descends to oriented reparameterization classes, not unrestricted homotopy classes.

66.9 Claim boundary

REFINEMENT CATEGORY DIRECTEDNESS	PROVED
LOCAL LENGTH RIGIDITY $c A $	PROVED
SUBDIVISION AND GAUGE INVARIANCE	PROVED
CONTINUUM RIEMANN-SUM CONVERGENCE	PROVED
EXPLICIT LIPSCHITZ ERROR	PROVED
ORIENTED REPARAMETERIZATION INVARIANCE	PROVED
ORIENTATION-REVERSAL TRANSPORT LAW	PROVED
UNRESTRICTED HOMOTOPY INVARIANCE	DISPROVED
ABSOLUTE SCALE c	EXTERNAL / OPEN
PURE MEMORY-SECTOR POSITIVE LENGTH	OPEN
CONSTRAINED HOMOTOPY COMPLETION	NEXT THEOREM LAYER

RKF/theorum/thermodynamics/lambda_geometry/13_refinement_and_continuum_length.md

Chapter 67

Constrained Homotopy Groupoid and Variational Length

67.1 Provenance

source repository: Parveen117/MP

source PR: #57

source theorem: adapters/common/CONSTRAINED_HOMOTOPY_GROUPOID.md

67.2 1. Constrained state

A continuum sector carries

$$(A, \varphi; \text{hierarchy itinerary}, \nu, \mathfrak{m}, \Phi, \text{seam class}),$$

where A is the connection path, φ is a lifted phase, ν is winding, $\mathfrak{m}$ is memory, and Φ is lifted curvature flux.

The endpoint transport is

$$U_A = \mathcal{P} \exp \int A(t) dt,$$

and the phase lift obeys

$$\varphi(1) - \varphi(0) = 2\pi\nu.$$

67.3 2. Guarded groupoid

A declared morphism may combine:

- orientation-preserving reparameterization;
- constant $SU(2)$ gauge conjugation;
- an admissible homotopy preserving the complete ledger;
- endpoint holonomies related by gauge conjugacy.

Identity, inverse, and composition laws hold for the declared finite compatibility packet. The theorem does not claim existence of a smooth constrained homotopy for every formally compatible pair.

67.4 3. Variational functional

For nonnegative coefficients c, α, β, γ , define

$$\mathcal{L} = c \int \|A(t)\| dt + \alpha \int |\dot{\varphi}(t)| dt + \beta W(\mathfrak{m}) + \gamma \|\Phi\|.$$

The coefficients are calibrations, not canonical universal constants.

67.5 4. Universal lower-bound theorem

Let $\theta(U)$ denote the principal $SU(2)$ conjugacy angle. Then

$$\boxed{\mathcal{L} \geq c \theta(U_A) + 2\pi\alpha|\nu| + \beta W(\mathfrak{m}) + \gamma \|\Phi\|}.$$

The proof uses:

- the geodesic-distance lower bound for connection length;
- total variation of the lifted phase;
- nonnegativity of the protected memory and curvature terms.

The bound is saturated in the declared principal-geodesic, monotone-phase calibration class.

67.6 5. Positivity boundary

The constrained infimum is strictly positive whenever at least one protected invariant entering the lower bound is nonzero with a positive coefficient.

A sector with

identity holonomy
zero winding
empty memory
zero curvature flux

may still have zero lower bound. Composition-primitivity alone does not prohibit collapse.

67.7 6. Composition subadditivity

For compatible sectors,

$$\ell_{\min}(C_2 \circ C_1) \leq \ell_{\min}(C_1) + \ell_{\min}(C_2).$$

The explicit lower-bound expression is also subadditive by the triangle inequalities for the conjugacy angle, integer winding, and curvature norm.

67.8 7. Ledger-preserving obstruction

An identity-holonomy sector with nonzero winding cannot be constrained-equivalent to the trivial zero-winding sector because

$$\mathcal{L} \geq 2\pi\alpha|\nu|.$$

Thus a visible endpoint may close while a protected lifted ledger prevents contraction.

67.9 Claim boundary

DECLARED CONSTRAINED GROUPOID LAWS	PROVED IN FINITE AUDIT
VARIATIONAL LOWER BOUND	PROVED
WINDING / MEMORY / FLUX POSITIVITY	PROVED WHEN WEIGHTED
COMPOSITION SUBADDITIVITY	PROVED
ZERO-LEDGER COLLAPSE POSSIBILITY	RETAINED
CANONICAL COEFFICIENTS	NOT PROVED
ALL SMOOTH CONSTRAINED HOMOTOPIES	NOT CLASSIFIED
GENERAL MINIMIZER EXISTENCE	OPEN AT THIS STAGE
CURVATURE-DERIVED NO-COLLAPSE	NEXT THEOREM LAYER

RKF/theorum/thermodynamics/lambda_geometry/14_constrained_variational_length.md

Chapter 68

Curvature-Flux No-Collapse and Minimizer Existence

68.1 Provenance

source repository: Parveen117/MP
source PR: #58
source theorem: adapters/common/CUT_METRIC_NO_COLLAPSE.md

68.2 1. Corrected warped metric

The source audit corrected the proposed angular coefficient by treating it as the square of a warping amplitude:

$$g = \frac{dr^2}{r} + a(r)^2 \cos^2(2\phi) d\phi^2,$$

with

$$-2ra'' - a' = \frac{a}{\sqrt{r}}.$$

Under

$$x = 2\sqrt{r}, \quad y = \frac{\sin(2\phi)}{2}, \quad A(x) = a(x^2/4),$$

the metric becomes

$$g = dx^2 + A(x)^2 dy^2.$$

The corrected ODE is

$$A'' + \frac{A}{x} = 0,$$

so the Gaussian curvature is

$$K = -\frac{A''}{A} = \frac{1}{x} = \frac{1}{2\sqrt{r}}.$$

The regular normalized solution used in the source is

$$A(x) = \frac{1}{2} \sqrt{x} J_1(2\sqrt{x}),$$

restricted to compact strips where $A > 0$.

68.3 2. Seam-geodesy theorem

The involution

$$J(x, y) = (x, -y)$$

has fixed seam $y = 0$. For the warped metric,

$$\Gamma_{xx}^y = 0,$$

so the fixed seam is a geodesic without inserting an auxiliary metric-choice assumption.

68.4 3. Connection and lifted curvature memory

With orthonormal coframe

$$e^1 = dx, \quad e^2 = A(x)dy,$$

the rotation connection and curvature are

$$\omega = -A'(x)dy,$$

$$\Omega = d\omega = \frac{A(x)}{x} dx \wedge dy = K d\text{Area}_g.$$

For a loop $C = \partial S$, define the lifted memory

$$\Phi(C) = \int_C \omega = \int_S \Omega.$$

Principal $SO(2)$ holonomy retains only $e^{i\Phi}$; the recognition ledger retains the real lifted value Φ .

68.5 4. Completed constrained-path minimizer theorem

On a compact regular strip, consider based constant-speed Lipschitz loops with fixed discrete hierarchy ledger and fixed lifted curvature flux.

For a bounded-length minimizing sequence:

- constant-speed parameterization gives equi-Lipschitz control;
- compactness gives a uniformly convergent subsequence;
- derivatives are weak-star compact;

- smoothness of ω closes the line-integral constraint;
- Riemannian length is lower semicontinuous.

Therefore the constrained length infimum is attained in the completed compact-strip class.

68.6 5. Linear no-collapse theorem

Let

$$B_D = \max_D \frac{|A'|}{A}.$$

Every rectifiable loop in the strip satisfies

$$L_g(C) \geq \frac{|\Phi(C)|}{B_D}.$$

This follows directly from

$$|\Phi(C)| = \left| \int_C \omega \right| \leq B_D L_g(C).$$

68.7 6. Quadratic no-collapse theorem

For a positively oriented simple loop, metric comparison and the Euclidean isoperimetric inequality give

$$L_g(C)^2 \geq \frac{4\pi c_D^2 x_0}{M} |\Phi(C)|,$$

where

$$M = \max_D A, \quad c_D = \min(1, \min_D A).$$

Thus nonzero lifted curvature flux enforces positive geometric length even when principal holonomy is trivial.

68.8 7. Identity-holonomy primitive witness

The source constructs a hierarchy-resolved loop with

```
lifted curvature flux = 2
principal holonomy = identity
visible winding = 0
visible memory = empty
complete hierarchy return only at the endpoint.
```

It is composition-primitive in the declared hierarchy category and obeys both no-collapse bounds. This demonstrates that endpoint closure does not erase lifted curvature memory.

68.9 Claim boundary

CORRECTED WARPED METRIC AND CURVATURE	PROVED
FIXED SEAM GEODESY	PROVED
STOKES LIFTED CURVATURE MEMORY	PROVED
COMPACT-STRIP MINIMIZER EXISTENCE	PROVED
LINEAR CURVATURE-FLUX NO-COLLAPSE	PROVED
QUADRATIC SIMPLE-LOOP NO-COLLAPSE	PROVED
IDENTITY-HOLONOMY NONCOLLAPSE WITNESS	PROVED
GLOBAL EXTENSION THROUGH DEGENERACIES	OPEN
GLOBAL MINIMIZER CLASSIFICATION	OPEN
CANONICAL ABSOLUTE NORMALIZATION	OPEN
ARITHMETIC PRIME IDENTIFICATION	NOT IMPLIED

RKF/theorum/thermodynamics/lambda_geometry/15_curvature_flux_no_collapse.md

Chapter 69

Primitive Flux Witnesses, Restricted Minimizers, and Canonicity Obstruction

69.1 Provenance

source repository: Parveen117/MP
source PRs: #59, #60, #61
source theorems:
 adapters/common/PRIMITIVE_FLUX_SPECTRUM.md
 adapters/common/PRIMITIVE_FLUX_MINIMIZER_CLASSIFICATION.md
 adapters/common/PRIMITIVE_FLUX_STRIP_VARIATION.md

69.2 1. Integer lifted-flux witness theorem

Fix a compact-strip rectangular witness block carrying lifted flux $2\pi/q_0$, with $q_0 = 14$ in the source calibration. For each nonzero integer n , take $q_0|n|$ oriented laps and assign fresh hierarchy depths so that the complete hierarchy state returns only at the endpoint.

Then the resulting sector satisfies

$$\Phi_n = 2\pi n,$$

while

```
principal holonomy = identity  
visible winding = 0  
visible memory = empty  
composition-primitive = true.
```

Thus principal holonomy collapses the integer classes modulo 2π , whereas lifted curvature memory distinguishes them.

The witness family obeys the linear and quadratic no-collapse inequalities. Within every finite audited integer range, the witness lengths form a discrete set, and same-orientation witness composition is additive.

69.3 2. Restricted rectangular minimizer theorem

Fix endpoints

$$(x_0, x_1) = (0.5, 3.0)$$

and consider congruent rectangular multi-lap representatives with half-width $\eta < 1/2$. Put

$$D = A'(x_0) - A'(x_1) > 0.$$

For lifted flux $\Phi_n = 2\pi n$, the least admissible lap count in this restricted family is

$$q_{\min}(n) = \left\lfloor \frac{2\pi|n|}{D} \right\rfloor + 1.$$

The corresponding half-width is

$$\eta(n) = \frac{\pi|n|}{q_{\min}(n)D},$$

and the restricted minimum length is

$$L_{\min}^{\text{rect}}(n) = 2q_{\min}(n)(x_1 - x_0) + \frac{2\pi|n|(A(x_0) + A(x_1))}{D}.$$

The floor-plus-one is forced by the strict regular-strip constraint $\eta < 1/2$. The preceding lap count is inadmissible.

This theorem classifies only the fixed-endpoint congruent-rectangle family.

69.4 3. Endpoint-variation obstruction theorem

The fixed rectangle is not stable under endpoint variation. For the source's $n = 1$ calibration,

```
fixed endpoints (0.5, 3.0):
length 74.3656751841
```

```
admissible varied endpoints (0.35, 1.125):
q_min = 24
0.4987180752
length 47.8830145962.
```

Hence an admissible shorter representative exists. Therefore

$$L_{\min}^{\text{rect}} \text{ at the original fixed endpoints is not a canonical native normalization.}$$

This negative theorem is load-bearing: a restricted minimizer formula cannot be promoted into an arithmetic length merely because it is explicit and visually well behaved.

69.5 4. What survives the obstruction

The endpoint-variation result does not invalidate:

- lifted flux as protected memory;
- composition-primitivity of the hierarchy-resolved witnesses;
- the no-collapse inequalities;
- the fixed-family minimizer formula within its declared class;
- finite-window discreteness diagnostics.

It blocks only the unsupported step from one convenient rectangle to a canonical global normalization.

69.6 Claim boundary

INTEGER LIFTED-FLUX WITNESSES	PROVED
IDENTITY PRINCIPAL HOLONOMY	PROVED FOR WITNESS FAMILY
HIERARCHY-RESOLVED PRIMITIVITY	PROVED IN DECLARED CATEGORY
FINITE-RANGE WITNESS DISCRETENESS	PROVED / AUDITED
FIXED-ENDPOINT RECTANGULAR MINIMIZER	PROVED IN RESTRICTED FAMILY
FIXED RECTANGLE CANONICALITY	DISPROVED BY SHORTER COMPETITOR
GLOBAL SHAPE MINIMIZER	OPEN
GLOBAL FLUX-SPECTRUM DISCRETENESS	OPEN
CANONICAL ARITHMETIC NORMALIZATION	NOT PROVED
NATIVE LENGTH = $\log p$	NOT PROVED

RKF/theorum/thermodynamics/lambda_geometry/16_primitive_flux_spectrum_and_obstruction.md

Chapter 70

Thermodynamic Recognition Theorem Archive

This folder transfers the rigorous thermodynamic-response results developed in the `Parveen117/MP` proof laboratory into clean Recognition Kernel Framework theorem capsules.

The MP source files remain in place as historical and hash-bound provenance. This archive is the reviewer-facing mathematical extraction. It does not promote an experimental finite match into a physical or arithmetic identification theorem.

70.1 Reading order

1. `01_response_tetrad_and_flat_closure.md`
2. `02_curvature_to_seam_spectral_flow.md`
3. `03_canonical_response_cost_barrier.md`
4. `04_projection_defect_and_source_overlap.md`
5. `05_onsager_compass_and_constitutive_no_go.md`
6. `06_onsager_hodge_no_leakage.md`
7. `07_thermo_weil_provenance_five_source_schur.md`
8. `08_canonical_connection_physical_flux.md`
9. `09_scope_exclusions.md`
10. `10_thermodynamic_cut_square_response_decomposition.md`
11. `lambda_geometry/README.md`
12. `SOURCE_PROVENANCE.md`

70.2 Direct thermodynamic-response chain

signed thermodynamic response tetrad
-> entropy-scaled Lambda coordinates
-> response one-form and curvature
-> canonical response-frame whitening
-> positive response cost
-> thermodynamic cut-square channel decomposition
-> bounded response defect / barrier
-> Onsager-Hodge splitting and no-leakage
-> provenance-gated finite thermo-Weil realization.

The cut-square capsule specializes the native minimum-decoder identity to the positive thermodynamic response metric. It separates one recognized scalar channel from the contractive reserve and the six pairwise tetrad mismatch squares. It is finer than the single flat relation $\Gamma_m = 1$ and does not identify a physical decoder without a source theorem.

70.3 Earlier thermodynamic-lambda geometry chain

The subfolder `lambda_geometry/` contains only the previously untransferred PR #52–#61 lineage:

thermodynamic-lambda response hierarchy
-> supplied unitary connection and holonomy
-> branch and path-order memory
-> admissible complete-state factorization
-> refinement-stable native length
-> continuum and constrained variational descent
-> curvature-flux no-collapse
-> primitive flux witnesses and minimizer obstruction.

It does not duplicate the direct response-cost, cut-square, Onsager, source-overlap, thermo-Weil, or physical-flux capsules already present in this folder.

70.4 Global claim boundary

THERMODYNAMIC RESPONSE TETRAD	PROVED / EXTRACTED
FLAT CALORIC-MECHANICAL CLOSURE	PROVED
CURVATURE-PATH / SEAM-INDEX SPECTRAL FLOW	PROVED
CANONICAL POSITIVE RESPONSE COST	PROVED
THERMODYNAMIC CUT-SQUARE DECOMPOSITION	PROVED IN FINITE POSITIVE RESPONSE FIBRE
SHARP THERMODYNAMIC DECODER BURDEN CRITERION	PROVED IN FINITE POSITIVE RESPONSE FIBRE
BOUNDED THERMODYNAMIC RESPONSE DEFECT	PROVED
FINITE PROJECTION / OVERLAP IDENTITIES	PROVED
ONSAGER COMPASS AND CONSTITUTIVE NO-GO	PROVED
ONSAGER-HODGE FINITE NO-LEAKAGE	PROVED UNDER DECLARED HYPOTHESES
FINITE THERMO-WEIL SCHUR REALIZATION	PROVED UNDER PROVENANCE INPUTS
LAMBDA HOLONOMY AND BRANCH MEMORY	PROVED IN DECLARED FINITE MODEL
REFINEMENT-STABLE NATIVE LENGTH	PROVED UP TO SCALE
CURVATURE-FLUX NO-COLLAPSE	PROVED ON COMPACT STRIPS
PHYSICAL THERMO -> NATIVE ARITHMETIC SOURCE MAP	OPEN
PHYSICAL IDENTIFICATION OF CUT-SQUARE DECODER	OPEN
CONTINUUM SOURCE-RANGE INTERTWINER	OPEN
CANONICAL ABSOLUTE LENGTH NORMALIZATION	OPEN
GLOBAL FLUX-SPECTRUM CLASSIFICATION	OPEN

GLOBAL WEIL POSITIVITY
RIEMANN HYPOTHESIS

NOT CLAIMED HERE
ABSTAIN

RKF/theorum/thermodynamics/README.md

Chapter 71

Thermodynamic-Lambda Geometry Archive

This subarchive records the earlier theorem chain developed in Parveen117/MP PRs #52 through #61.

It begins with a thermodynamic response hierarchy, then studies the additional geometry required to turn that hierarchy into a connection, holonomy, memory-preserving length, constrained path space, and curvature-flux no-collapse theory.

71.1 Typing warning

PR #52 used the historical first-layer coordinates

$$\lambda_p = -ST/C_p, \quad \lambda_v = -ST/C_v,$$

$$\lambda_s = V(\partial P/\partial V)_S, \quad \lambda_t = -P(\partial V/\partial P)_T.$$

PR #239 later corrected the physical typing by distinguishing those derivative coordinates from the uniform entropy-scaled response tetrad

$$\lambda_X^{\text{th}} = -ST/C_X, \quad X \in \{p, v, s, t\}.$$

Therefore the files in this subfolder preserve the geometric theorems from PRs #52–#61 without silently identifying the historical s, t derivative coordinates with the later entropy-scaled tetrad.

71.2 Reading order

1. 10_lambda_holonomy_and_branch_memory.md
2. 11_noncommutative_holonomy_and_factorization.md
3. 12_refinement_and_continuum_length.md
4. 13_constrained_variational_length.md
5. 14_curvature_flux_no_collapse.md
6. 15_primitive_flux_spectrum_and_obstruction.md

71.3 Status

THERMODYNAMIC FIRST RESPONSE LAYER	HISTORICAL PROVENANCE
CONNECTION FROM RESPONSE HIERARCHY	ADDITIONAL DATA / NOT UNIQUE
UNITARY HOLONOMY-RATE OPERATOR	PROVED
STATIC LOGARITHM BRANCH MEMORY	PROVED
NONCOMMUTATIVE PATH-ORDER COLLISION	PROVED
ADMISSIBLE FACTORIZATION WITH CROSS CURVATURE	PROVED
REFINEMENT-STABLE LENGTH RIGIDITY	PROVED
CONTINUUM AND REPARAMETERIZATION DESCENT	PROVED
UNRESTRICTED HOMOTOPY INVARIANCE	DISPROVED
CONSTRAINED VARIATIONAL LOWER BOUND	PROVED
CURVATURE-FLUX NO-COLLAPSE	PROVED ON COMPACT STRIPS
INTEGER FLUX WITNESS FAMILY	PROVED
FIXED-RECTANGLE CANONICALITY	DISPROVED

RKF/theorum/thermodynamics/lambda_geometry/README.md

Chapter 72

Source Provenance

72.1 Repository lineage

The theorem capsules in this folder were extracted from the private proof laboratory:

```
repository: Parveen117/MP
primary direct-thermodynamics PRs: #239 and #243
native cut-square specialization inputs: #274 and #275
earlier lambda-geometry PR chain: #52 through #61
primary response branch: agent/clock-free-recognition-proof
```

The source files remain in MP. This RKF folder is a clean theorem-level transfer, not a deletion of the original laboratory history.

72.2 Foundational thermodynamic provenance: PR #239

```
papers/rh_initial_journey/THERMO_CURVATURE_TO_SEAM_INDEX_BRIDGE.md
blob: 918a7a14cfb68caf907e0dcabee6c64d3935a31e
```

```
papers/rh_initial_journey/sections/07b0_thermo_curvature_lambda_provenance.tex
blob: cce8cb1dbb086d811046a01c32566bbb994ad1bd
```

Extracted results:

- signed (C_p, C_v, C_s, C_t) response tetrad;
- entropy-scaled coordinates $\lambda_X^{\text{th}} = -ST/C_X$;
- flat closure $\Gamma_c \Gamma_m = 1$;
- response one-form and curvature;
- scalar-to-integer type obstruction;
- curvature-path / seam-index spectral-flow law;
- quantitative no-crossing criterion.

72.3 Thermodynamic response and no-leakage laboratory: PR #243

proof_lab/clock_free_recognition/CANONICAL_RESPONSE_COUPLING_AND_CURVATURE_GATE.md
blob: 5d64f8e505305cc7c3528456683abc5572bff718

proof_lab/clock_free_recognition/THERMODYNAMIC_RESPONSE_FIRST_RAMANUJAN_GATE.md
blob: d7bdcfb4b04aada3b2ec9e5244e0cb39949c9c3e

proof_lab/clock_free_recognition/THERMO_RAMANUJAN_JACOBIAN_COUPLING.md
blob: f5ba420ee26898edcbee4bf1822a2e077e334840

proof_lab/clock_free_recognition/NATIVE_THERMODYNAMIC_PROJECTION_DEFECT_THEOREM.md
blob: 852c876319c201fde3dfedf02ad7ded9213e8e1a

proof_lab/clock_free_recognition/NATIVE_THERMO_SOURCE_OVERLAP_IDENTITY.md
blob: e2fd9a23632e22d18ffac422e804537323e4a899

proof_lab/clock_free_recognition/NATIVE_EVENT_RESOLVED_THERMO_SOURCE_OVERLAP.md
blob: 40d4618ffd979274cc7484a9a413d0c22d5252c8

proof_lab/clock_free_recognition/NATIVE_ONSAGER_COMPASS_CURVATURE_INTERTWINER.md
blob: c86ed3b0e8dcb9b27f621a5d3caf0439463c034b

proof_lab/clock_free_recognition/NATIVE_ONSAGER_CONSTITUTIVE_PATH_NO_GO.md
blob: e6cb588fb400a9d245607f0c1b08c2fd5fb98979

proof_lab/clock_free_recognition/NATIVE_ONSAGER_HODGE_NO_LEAKAGE_THEOREM.md
blob: 7a21d7ae9f33bc49ccb6354d8176ae7703c58bbf

proof_lab/clock_free_recognition/THERMO_WEIL_PROVENANCE_INPUT_CONTRACT.md
blob: 66d98e6b1cbf214a277f0e7cd15ec5890e3e5b44

proof_lab/clock_free_recognition/THERMO_WEIL_FIVE_SOURCE_IDENTIFICATION.md
blob: 3ff34c728abfac224d3454a11d43453c79294a97

proof_lab/clock_free_recognition/THERMO_WEIL_BOUNDARY_SCHUR_REALIZATION.md
blob: 535d0fde891f15647a616ef90245f1343c68d57f

proof_lab/clock_free_recognition/NATIVE_CANONICAL_CONNECTION_PHYSICAL_FLUX_INTERTWINER.md
blob: c8160175207fd9b4ab58d690a491ba04414f8305

proof_lab/clock_free_recognition/ONSAGER_GEOMETRIC_01_02_AUDIT.md
blob: 456827539ed384a7227e76059860a3aef1d6dc36

72.4 Native cut-square and minimum-decoder provenance: PRs #274 and #275

The thermodynamic cut-square capsule is a typed specialization of the native event-decoder theorem to the positive thermodynamic response metric supplied by the direct thermodynamic archive.

PR #274
proof_lab/clock_free_recognition/T17_NATIVE_BOUNDARY_EVENT_DECODER_THEOREM.md
blob: 8697ee2d81afe104dc2eb29601c7e9811c54fcdd

PR #275

proof_lab/clock_free_recognition/T18_NATIVE_DECODER_BURDEN_NONATTAINED_MEMORY_THEOREM.md
blob: 54aa4e18245a9ca2f9dc98bf37e729907aa2185a

Specialized results recorded in

theorem/thermodynamics/10_thermodynamic_cut_square_response_decomposition.md

are:

- positive-metric thermodynamic Lagrange/cut-square identity;
- contractive scalar-channel no-leakage decomposition;
- six-minor expansion of the (p, v, s, t) response tetrad after metric whitening;
- equality and complete response-alignment classification;
- sharp thermodynamic decoder burden criterion;
- symmetric Onsager entropy-production specialization;
- rank-one channel domination of the canonical response cost;
- explicit separation from the nonlinear barrier theorem.

This transfer does not assert that a particular thermodynamic decoder is physically realized. It proves the decomposition once the response metric and decoder are lawfully declared.

72.5 Previously missing lambda-geometry chain: PRs #52–#61

Transferred into

theorem/thermodynamics/lambda_geometry/

Source map:

PR #52 adapters/common/LAMBDA_HOLONOMY_SPECTRAL_OPERATOR.md
PR #53 adapters/common/NONCOMMUTATIVE_PATH_ORDERED_HOLONOMY.md
PR #54 adapters/common/ADMISSIBLE_SECTOR_FACTORIZATION.md
PR #55 adapters/common/REFINEMENT_INVARIANT_NATIVE_LENGTH.md
PR #56 adapters/common/CONTINUUM_HOMOTOPY_DESCENT.md
PR #57 adapters/common/CONSTRAINED_HOMOTOPY_GROUPOID.md
PR #58 adapters/common/CUT_METRIC_NO_COLLAPSE.md
PR #59 adapters/common/PRIMITIVE_FLUX_SPECTRUM.md
PR #60 adapters/common/PRIMITIVE_FLUX_MINIMIZER_CLASSIFICATION.md
PR #61 adapters/common/PRIMITIVE_FLUX_STRIP_VARIATION.md

Extracted missing results:

- lambda holonomy-rate self-adjointness;
- static logarithm branch memory;
- winding / spectral-flow identity;

- noncommutative path-order collision;
- small-square commutator curvature;
- common-commutant irreducibility;
- admissible factorization with cross-curvature memory;
- composition-primitivity;
- refinement-stable length rigidity;
- continuum and reparameterization descent;
- unrestricted homotopy obstruction;
- constrained variational lower bound;
- curvature-flux minimizer existence and no-collapse;
- integer flux witnesses;
- restricted rectangular minimizer formula;
- endpoint-variation obstruction to canonicity.

The PR #239 entropy-scaled tetrad is authoritative for thermodynamic typing; the historical PR #52 derivative coordinates are preserved only as lineage and are not silently equated with the corrected tetrad.

72.6 Transfer rule

Each RKF capsule retains only:

1. typed definitions;
2. theorem statements;
3. finite or abstract proofs actually supported by the source;
4. explicit hypotheses;
5. exact claim boundaries.

Numerical fits, speculative physical extensions, and RH conclusions are not promoted by this transfer.

72.7 Canonical RKF location

```
repository: Parveen117/Recognition-Kernel-Framework
canonical branch: main
folders:
  theorum/thermodynamics/
  theorum/thermodynamics/lambda_geometry/
branch audit:
  theorum/BRANCH_CONSOLIDATION_AUDIT.md
```

The historical transfer branches remain as provenance labels, but after the 2026-08-05 consolidation they are synchronized with `main`. New thermodynamic development should branch from current `main`, preserve the source map, and return through a verified merge or fast-forward.

RKF/theorum/thermodynamics/SOURCE_PROVENANCE.md

Chapter 73

Thermodynamic-Lambda Geometry Source Provenance

73.1 Repository lineage

source repository: Parveen117/MP
source PR range: #52 through #61
transfer repository: Parveen117/Recognition-Kernel-Framework
transfer folder: theorem/thermodynamics/lambda_geometry/

73.2 Source theorem map

PR #52
adapters/common/LAMBDA_HOLONOMY_SPECTRAL_OPERATOR.md
-> 10_lambda_holonomy_and_branch_memory.md

PR #53
adapters/common/NONCOMMUTATIVE_PATH_ORDERED_HOLONOMY.md
-> 11_noncommutative_holonomy_and_irreducibility.md

PR #54
adapters/common/ADMISSIBLE_SECTOR_FACTORIZATION.md
-> 12_admissible_factorization_and_primitivity.md

PR #55
adapters/common/REFINEMENT_INVARIANT_NATIVE_LENGTH.md
-> 13_refinement_and_continuum_length.md

PR #56
adapters/common/CONTINUUM_HOMOTOPY_DESCENT.md
-> 13_refinement_and_continuum_length.md

PR #57
adapters/common/CONSTRAINED_HOMOTOPY GROUPOID.md
-> 14_constrained_variational_length.md

PR #58
adapters/common/CUT_METRIC_NO_COLLAPSE.md
-> 15_curvature_flux_no_collapse.md

PR #59


```
adapters/common/PRIMITIVE_FLUX_SPECTRUM.md  
-> 16_primitive_flux_spectrum_and_obstruction.md
```

```
PR #60  
adapters/common/PRIMITIVE_FLUX_MINIMIZER_CLASSIFICATION.md  
-> 16_primitive_flux_spectrum_and_obstruction.md
```

```
PR #61  
adapters/common/PRIMITIVE_FLUX_STRIP_VARIATION.md  
-> 16_primitive_flux_spectrum_and_obstruction.md
```

73.3 Nonduplication rule

The parent thermodynamics archive already transfers the direct PR #239/#243 theorem families:

```
response tetrad and flat closure  
curvature-to-seam spectral flow  
canonical response cost / barrier  
projection defect and source overlap  
Onsager compass and Hodge no-leakage  
thermo-Weil provenance and Schur realization  
canonical physical flux.
```

Those results are not copied into this subfolder. This folder contains only the earlier missing lambda-holonomy, length, variational, and curvature-flux lineage.

73.4 Typing repair

PR #52 used historical derivative coordinates in the s, t directions. PR #239 later separated them from the uniform entropy-scaled thermodynamic coordinates

$$\lambda_X^{\text{th}} = -ST/C_X.$$

The transfer preserves the geometric theorems from PRs #52–#61 while treating the PR #239 convention as authoritative for thermodynamic typing.

73.5 Transfer discipline

Each capsule retains:

1. typed definitions;
2. theorem statements and supported proofs;
3. exact hypotheses;
4. negative controls and obstruction theorems;
5. explicit open boundaries.

No arithmetic-prime identification, universal absolute length scale, global flux-spectrum classification, or RH conclusion is promoted by this transfer.

RKF/theorum/thermodynamics/lambda_geometry/SOURCE_PROVENANCE.md

Part IV

Foundation Theorems and Certificates

Chapter 74

F00-E — Native Euler Exponential from the Cut–Iota Complex Field

74.1 Classification

PROVED

74.2 Foundational order

This theorem begins after the already derived cut-complex foundation:

```
oriented cut
-> quarter-turn element iota_Sigma
-> iota_Sigma^2 = -1
-> dagger iota_Sigma^dagger = -iota_Sigma
-> completed radial field R_Sigma
-> completed cut-complex field C_Sigma
-> native Euler exponential.
```

It does **not** begin from a positive real scale, a logarithmic coordinate, a predeclared dilation group, a circle, sine, cosine, pi, Haar measure, Stone's theorem or the differential operator $-i r d/dr$.

The theorem derives the exponential and Euler flow inside the already earned field $\mathbb{C}_\Sigma$. Classical exponential, circular and multiplicative formulas are coordinate shadows established only afterward.

74.3 1. Established cut-complex input

Let

$$\mathbb{C}_\Sigma = \mathbb{R}_\Sigma \oplus \iota_\Sigma \mathbb{R}_\Sigma$$

be the completed cut-complex field. Its established structure is:

$$\iota_\Sigma^2 = -1, \quad \iota_\Sigma^\dagger = -\iota_\Sigma,$$

$$(a + \iota_{\Sigma} b)^{\dagger} = a - \iota_{\Sigma} b,$$

and

$$N_{\Sigma}(z) = z^{\dagger} z \in \mathbb{R}_{\Sigma, \geq 0}.$$

Write

$$|z|_{\Sigma} = \sqrt{N_{\Sigma}(z)}.$$

The completed field is assumed to carry the previously derived Archimedean ordered radial structure and the Cauchy-completion property needed for limits of native cut-complex sequences.

No ordinary complex number is used to define these objects.

74.4 2. Factorial approximants

For z in $\mathbb{C}_{\Sigma}$, define the finite native factorial polynomial

$$E_{\Sigma, N}(z) = \sum_{n=0}^N \frac{z^n}{n!}.$$

The coefficients $1/n!$ belong to the radial rational subfield already embedded in $\mathbb{R}_{\Sigma}$.

Lemma 2.1 — Native factorial-tail convergence

For every z in $\mathbb{C}_{\Sigma}$, the sequence

$$(E_{\Sigma, N}(z))_{N \geq 0}$$

is Cauchy in $\mathbb{C}_{\Sigma}$.

Proof

Choose a natural number m such that

$$m > 2|z|_{\Sigma}.$$

For $k \geq n+1 \geq m$, multiplicativity of the cut-complex norm gives

$$\frac{|z|_{\Sigma}^k}{k!} = \frac{|z|_{\Sigma}^{n+1}}{(n+1)!} \prod_{j=n+2}^k \frac{|z|_{\Sigma}}{j}.$$

Every factor in the product is at most $1/2$. Therefore

$$\sum_{k=n+1}^{\infty} \frac{|z|_{\Sigma}^k}{k!} \leq 2 \frac{|z|_{\Sigma}^{n+1}}{(n+1)!}.$$

The last expression tends to zero because, from index m onward, successive terms decrease by at least a factor $1/2$. Thus every factorial tail is native Cauchy. Completeness of $\mathbb{C}_{\Sigma}$ supplies the limit.

Definition 2.2 — Native Euler exponential

Define

$$\boxed{\text{Exp}_\Sigma(z) = \lim_{N \rightarrow \infty} E_{\Sigma, N}(z).}$$

This is the first appearance of an exponential object. It is generated by finite cut-complex multiplication, radial rational coefficients and native completion.

74.5 3. Exponential group law

Theorem 3.1 — Native addition law

For all z, w in $\mathbb{C}_\Sigma$,

$$\boxed{\text{Exp}_\Sigma(z + w) = \text{Exp}_\Sigma(z) \text{Exp}_\Sigma(w).}$$

Proof

The factorial series are absolutely controlled by the native radial majorants proved above, so their Cauchy product is lawful. Grouping the product by total degree gives

$$\begin{aligned} \text{Exp}_\Sigma(z) \text{Exp}_\Sigma(w) &= \sum_{n=0}^{\infty} \sum_{k=0}^n \frac{z^k w^{n-k}}{k!(n-k)!} \\ &= \sum_{n=0}^{\infty} \frac{1}{n!} \sum_{k=0}^n \binom{n}{k} z^k w^{n-k} \\ &= \sum_{n=0}^{\infty} \frac{(z + w)^n}{n!}. \end{aligned}$$

The finite identity consumed in the second line is the binomial theorem in the commutative cut-complex field. The final series is the definition of $\text{Exp}_\Sigma(z+w)$.

Corollary 3.2 — Unit, inverse and nonvanishing

$$\text{Exp}_\Sigma(0) = 1,$$

$$\text{Exp}_\Sigma(-z) = \text{Exp}_\Sigma(z)^{-1},$$

and therefore

$$\text{Exp}_\Sigma(z) \neq 0.$$

Corollary 3.3 — Dagger covariance

$$\boxed{\text{Exp}_\Sigma(z)^\dagger = \text{Exp}_\Sigma(z^\dagger).}$$

Proof

Dagger acts termwise on every finite factorial polynomial and is continuous for the native completion.

74.6 4. Native derivative and the Euler equation

The derivative is now defined inside the completed field. It is not imported as a real-coordinate operator.

Definition 4.1 — Cut-complex recognition derivative

For a function F defined on a native neighbourhood of $\mathbf{z}$, set

$$\partial_{\Sigma} F(z) = \lim_{h \rightarrow 0} \frac{F(z+h) - F(z)}{h},$$

when the limit exists in $\mathbb{C}_{\Sigma}$.

Lemma 4.2 — Unit tangent of the exponential

$$\lim_{h \rightarrow 0} \frac{\text{Exp}_{\Sigma}(h) - 1}{h} = 1.$$

Proof

For nonzero h ,

$$\frac{\text{Exp}_{\Sigma}(h) - 1}{h} = 1 + \sum_{n=2}^{\infty} \frac{h^{n-1}}{n!}.$$

The tail is bounded by a factorial majorant tending to zero with $|h|_{\Sigma}$.

Theorem 4.3 — Native Euler differential equation

For every $\mathbf{z}$ in $\mathbb{C}_{\Sigma}$,

$$\boxed{\partial_{\Sigma} \text{Exp}_{\Sigma}(z) = \text{Exp}_{\Sigma}(z), \quad \text{Exp}_{\Sigma}(0) = 1.}$$

Proof

By the addition law,

$$\frac{\text{Exp}_{\Sigma}(z+h) - \text{Exp}_{\Sigma}(z)}{h} = \text{Exp}_{\Sigma}(z) \frac{\text{Exp}_{\Sigma}(h) - 1}{h}.$$

Apply Lemma 4.2.

Theorem 4.4 — Native analytic uniqueness

Exp_Σ is the unique native power-series solution of

$$\partial_\Sigma F = F, \quad F(0) = 1.$$

Proof

Let

$$F(z) = \sum_{n=0}^{\infty} a_n z^n$$

be a native convergent power series. The differential equation gives

$$(n+1)a_{n+1} = a_n.$$

The initial value gives $a_0=1$, hence recursively

$$a_n = \frac{1}{n!}.$$

Therefore $F=\text{Exp}_\Sigma$.

This is the Euler equation derived from the cut-complex field. No dilation group has been assumed.

74.7 5. Circular functions derived from the exponential

Let t in $\mathbb{R}_\Sigma$.

Definition 5.1 — Native cosine and sine

$$\text{Cos}_\Sigma(t) = \frac{\text{Exp}_\Sigma(\iota_\Sigma t) + \text{Exp}_\Sigma(-\iota_\Sigma t)}{2},$$

$$\text{Sin}_\Sigma(t) = \frac{\text{Exp}_\Sigma(\iota_\Sigma t) - \text{Exp}_\Sigma(-\iota_\Sigma t)}{2\iota_\Sigma}.$$

Because dagger exchanges the two exponential terms, both functions are radial:

$$\text{Cos}_\Sigma(t)^\dagger = \text{Cos}_\Sigma(t),$$

$$\text{Sin}_\Sigma(t)^\dagger = \text{Sin}_\Sigma(t).$$

Theorem 5.2 — Native Euler identity

$$\text{Exp}_\Sigma(\iota_\Sigma t) = \text{Cos}_\Sigma(t) + \iota_\Sigma \text{Sin}_\Sigma(t).$$

This identity is obtained by adding the two definitions. It is not installed as a classical formula.

Theorem 5.3 — Circular differential system

$$\partial_{\Sigma} \text{Cos}_{\Sigma}(t) = -\text{Sin}_{\Sigma}(t),$$

$$\partial_{\Sigma} \text{Sin}_{\Sigma}(t) = \text{Cos}_{\Sigma}(t),$$

with

$$\text{Cos}_{\Sigma}(0) = 1, \quad \text{Sin}_{\Sigma}(0) = 0.$$

Consequently

$$\partial_{\Sigma}^2 \text{Cos}_{\Sigma} = -\text{Cos}_{\Sigma}, \quad \partial_{\Sigma}^2 \text{Sin}_{\Sigma} = -\text{Sin}_{\Sigma}.$$

Theorem 5.4 — Native Pythagorean identity

$$\boxed{\text{Cos}_{\Sigma}(t)^2 + \text{Sin}_{\Sigma}(t)^2 = 1.}$$

Proof

The exponential addition law gives

$$\text{Exp}_{\Sigma}(-\iota_{\Sigma}t) \text{Exp}_{\Sigma}(\iota_{\Sigma}t) = 1.$$

Insert the Euler identity and its dagger.

Corollary 5.5 — Native circular orbit

Define

$$\mathbb{S}_{\Sigma} = \{\text{Exp}_{\Sigma}(\iota_{\Sigma}t) : t \in \mathbb{R}_{\Sigma}\}.$$

Every element has unit native norm:

$$z^{\dagger}z = 1.$$

The maps

$$R_t z = \text{Exp}_{\Sigma}(\iota_{\Sigma}t) z$$

obey

$$R_{s+t} = R_s R_t, \quad R_t^{\dagger} = R_{-t}.$$

Thus circular phase is derived before any period or value of π is assumed. Existence and normalization of a fundamental period are a later theorem.

74.8 6. Hyperbolic functions from the same exponential

Definition 6.1

For radial t , define

$$\begin{aligned}\text{Cosh}_\Sigma(t) &= \frac{\text{Exp}_\Sigma(t) + \text{Exp}_\Sigma(-t)}{2}, \\ \text{Sinh}_\Sigma(t) &= \frac{\text{Exp}_\Sigma(t) - \text{Exp}_\Sigma(-t)}{2}.\end{aligned}$$

Theorem 6.2 — Sheet-involution exponential

Let H_Σ be any native linear involution satisfying

$$H_\Sigma^2 = I.$$

Then

$$\boxed{\text{Exp}_\Sigma(tH_\Sigma) = \text{Cosh}_\Sigma(t)I + H_\Sigma \text{Sinh}_\Sigma(t).}$$

Proof

Separate the factorial series into its even and odd powers and use $H_\Sigma^{2n} = I$, $H_\Sigma^{2n+1} = H_\Sigma$.

Corollary 6.3 — Unified circular–hyperbolic factorization

Because H_Σ commutes with the scalar phase,

$$\boxed{\text{Exp}_\Sigma((aH_\Sigma - \iota_\Sigma \xi I)x) = \text{Exp}_\Sigma(-\iota_\Sigma \xi x) [\text{Cosh}_\Sigma(ax)I + H_\Sigma \text{Sinh}_\Sigma(ax)].}$$

The hyperbolic sheet factor and circular phase therefore arise from one native exponential. They are not independent imported structures.

74.9 7. Radial Euler orbit

For radial x , dagger covariance gives

$$\text{Exp}_\Sigma(x)^\dagger = \text{Exp}_\Sigma(x).$$

Moreover,

$$\text{Exp}_\Sigma(x) = \text{Exp}_\Sigma(x/2)^\dagger \text{Exp}_\Sigma(x/2),$$

so it is native positive and nonzero.

Definition 7.1 — Positive Euler orbit

$$\mathbb{P}_\Sigma = \{\text{Exp}_\Sigma(x) : x \in \mathbb{R}_\Sigma\}.$$

Define

$$M_t \rho = \text{Exp}_\Sigma(t) \rho.$$

Then

$$M_{s+t} = M_s M_t, \quad M_t^{-1} = M_{-t}.$$

This multiplicative flow is derived from the Euler exponential. It is not the definition of the exponential.

No claim is made here that every abstract positive radial scalar has a unique Euler coordinate. Surjectivity and injectivity of the exponential chart, and a native logarithm if needed, are separate theorems.

74.10 8. Native Euler flow on functions

Let F be a native analytic function on the positive Euler orbit. Define

$$(U_t F)(\rho) = F(\text{Exp}_\Sigma(-t)\rho).$$

The exponential group law gives

$$U_{s+t} = U_s U_t, \quad U_0 = I.$$

Write

$$f(x) = F(\text{Exp}_\Sigma(x)).$$

Then

$$(U_t F)(\text{Exp}_\Sigma(x)) = f(x - t).$$

Thus additive translation is the Euler-coordinate shadow of the derived multiplicative action.

Definition 8.1 — Native Euler generator

On the native analytic core define

$$\boxed{\mathcal{E}_\Sigma = -\iota_\Sigma \partial_\Sigma.}$$

For the native harmonic character

$$\chi_\xi(x) = \text{Exp}_\Sigma(\iota_\Sigma \xi x),$$

one has

$$\mathcal{E}_\Sigma \chi_\xi = \xi \chi_\xi,$$

and

$$U_t \chi_\xi = \text{Exp}_\Sigma(-\iota_\Sigma \xi t) \chi_\xi.$$

Consequently, on the analytic character core,

$$\boxed{U_t = \text{Exp}_\Sigma(-\iota_\Sigma t \mathcal{E}_\Sigma).}$$

This is the native Euler flow. No appeal to Stone's theorem is needed to define it or its generator on the native analytic core.

74.11 9. Multiplicative-coordinate shadow

Let

$$\rho = \text{Exp}_\Sigma(x).$$

The native Euler equation gives

$$\partial_\Sigma \rho = \rho.$$

For a native analytic F , the chain rule gives

$$\partial_\Sigma F(\text{Exp}_\Sigma(x)) = \rho \partial_\rho F(\rho).$$

Therefore the Euler generator has multiplicative-coordinate form

$$\boxed{\mathcal{E}_\Sigma = -\iota_\Sigma \rho \partial_\rho.}$$

Only under the later classical chart

$$\iota_\Sigma \mapsto i, \quad \rho \mapsto r > 0$$

does this become

$$-ir \frac{d}{dr}.$$

Likewise,

$$(U_t F)(\rho) = F(\text{Exp}_\Sigma(-t)\rho)$$

becomes

$$(U_t F)(r) = F(e^{-t}r).$$

The ordinary exponential and Euler differential operator are therefore coordinate shadows of the native cut–iota construction.

74.12 10. Exact dependency correction for D01

The completed-Weil dock may no longer begin by declaring

```
(U_t F)(r)=F(e^{-t}r)
and E=-i r d/dr.
```

Its lawful order is now:

```
cut
-> iota_Sigma
-> C_Sigma
-> Exp_Sigma
-> circular and hyperbolic Euler factors
-> positive Euler orbit
-> native Euler flow U_t
-> native Euler generator E_Sigma
-> multiplicative coordinate rho
-> classical shadow r, e^x, Mellin and Fourier charts
-> completed-Weil carrier.
```

D01 must instantiate this theorem and prove that its analytic carrier is a faithful realization of the native Euler orbit and generator.

74.13 11. Claim boundary

Proved here:

```
factorial construction of Exp_Sigma;
absolute native convergence;
addition law;
dagger covariance and inverse;
native Euler differential equation;
analytic uniqueness;
derived sine and cosine;
Euler identity;
circular differential and Pythagorean identities;
derived unit circular orbit;
derived hyperbolic functions;
combined circular-hyperbolic exponential;
positive radial Euler orbit;
derived multiplicative function flow;
native Euler generator on the analytic core;
shadow formula -i r d/dr.
```

Not proved here:

```
existence or normalization of a fundamental period pi_Sigma;
global injectivity or surjectivity of the radial exponential;
a native logarithm on every positive scalar;
Haar measure;
Hilbert-space unitarity or self-adjoint closure of the generator;
Mellin-Plancherel theory;
the completed-Weil form;
KO >= 0;
RH.
```

Those are later theorems. In particular, Stone's theorem may be used later to audit a completed Hilbert representation, but it no longer generates the native Euler flow.

RKF/theorems/foundation/FOOE_NATIVE_EULER_FROM_IOTA_COMPLEX.md

Chapter 75

F00-G — Native Logarithm on the Positive Radial Cone and Native Powers

75.1 Classification

PROVED

75.2 Purpose

F00-E derives the native exponential

$$\text{Exp}_\Sigma : \mathbb{C}_\Sigma \rightarrow \mathbb{C}_\Sigma$$

from finite factorial polynomials and cut-complex completion. F00-F proves that its restriction to the radial line is strictly increasing and defines an inverse only on the earned Euler orbit

$$\mathbb{P}_\Sigma = \{\text{Exp}_\Sigma(x) : x \in \mathbb{R}_\Sigma\}.$$

This theorem proves that every positive radial scalar belongs to that orbit. It derives the logarithm without importing

ordinary real logarithm;
an inverse-function theorem;
integration of $1/x$;
the ordinary positive real line;
complex analysis;
a pre-existing power x^s .

The construction order is

positive radial scalar
-> Cayley coordinate $y=(x-1)/(x+1)$
-> native odd power series
-> inverse identity with Exp_Σ
-> full positive Euler orbit
-> multiplicative logarithm
-> native complex powers.

75.3 1. Established input

Let

$$\mathbb{C}_\Sigma = \mathbb{R}_\Sigma \oplus \iota_\Sigma \mathbb{R}_\Sigma, \quad \iota_\Sigma^2 = -1,$$

with dagger

$$(a + \iota_\Sigma b)^\dagger = a - \iota_\Sigma b.$$

The radial field $\mathbb{R}_\Sigma$ is complete, Archimedean and ordered. The exponential satisfies

$$\text{Exp}_\Sigma(z + w) = \text{Exp}_\Sigma(z) \text{Exp}_\Sigma(w),$$

$$\partial_\Sigma \text{Exp}_\Sigma = \text{Exp}_\Sigma, \quad \text{Exp}_\Sigma(0) = 1,$$

and is strictly increasing on $\mathbb{R}_\Sigma$.

For radial u , write $|u|_\Sigma$ for the ordered absolute value.

75.4 2. The native Cayley coordinate

Let $x > 0$ be radial and define

$$\mathcal{C}_\Sigma(x) = \frac{x - 1}{x + 1}.$$

Lemma 2.1 — Cayley contraction

For every radial $x > 0$,

$$|\mathcal{C}_\Sigma(x)|_\Sigma < 1.$$

Proof

Because $x > 0$,

$$x + 1 > 0.$$

Also

$$(x + 1)^2 - (x - 1)^2 = 4x > 0.$$

Hence

$$|x - 1|_\Sigma < x + 1.$$

Division by the positive scalar $x + 1$ gives the result.

The inverse rational transformation is

$$x = \frac{1+y}{1-y}$$

whenever

$$y = \mathcal{C}_\Sigma(x).$$

Indeed,

$$1+y = \frac{2x}{x+1}, \quad 1-y = \frac{2}{x+1}.$$

75.5 3. The native odd logarithm series

For radial y with $|y|_\Sigma < 1$, define finite odd polynomials

$$A_{\Sigma,N}(y) = 2 \sum_{m=0}^N \frac{y^{2m+1}}{2m+1}.$$

Lemma 3.1 — Native convergence

For every $|y|_\Sigma < 1$, the sequence $A_{\{\Sigma,N\}}(y)$ is Cauchy in $\mathbb{R}_\Sigma$.

Proof

Choose a radial rational q with

$$|y|_\Sigma < q < 1.$$

For $M > N$,

$$\begin{aligned} |A_{\Sigma,M}(y) - A_{\Sigma,N}(y)|_\Sigma &\leq 2 \sum_{m=N+1}^M \frac{|y|_\Sigma^{2m+1}}{2m+1} \\ &\leq 2 \sum_{m=N+1}^{\infty} q^{2m+1} \\ &= \frac{2q^{2N+3}}{1-q^2}. \end{aligned}$$

The last expression tends to zero by the native geometric-tail identity.

Definition 3.2 — Native odd logarithm coordinate

For $|y|_\Sigma < 1$, define

$$\text{Atanh}_\Sigma(y) = \sum_{m=0}^{\infty} \frac{y^{2m+1}}{2m+1},$$

and

$$A_{\Sigma}(y) = 2 A \tanh_{\Sigma}(y).$$

The name records the later classical shadow only. The definition is the native odd power series above.

75.6 4. Difference rules used below

All rules in this section are derived from the cut-complex difference quotient. They are not imported as real-calculus theorems.

Lemma 4.1 — Product rule

If native derivatives of f and g exist at y , then

$$\partial_{\Sigma}(fg)(y) = (\partial_{\Sigma}f)(y)g(y) + f(y)(\partial_{\Sigma}g)(y).$$

Proof

Write

$$\frac{f(y+h)g(y+h) - f(y)g(y)}{h} = \frac{f(y+h) - f(y)}{h}g(y+h) + f(y)\frac{g(y+h) - g(y)}{h}.$$

Pass to the native limit.

Lemma 4.2 — Inverse rule

If $g(y)$ is nonzero and native differentiable, then

$$\partial_{\Sigma}(g^{-1})(y) = -g(y)^{-2}(\partial_{\Sigma}g)(y).$$

Proof

Apply the product rule to $g \cdot g^{-1} = 1$.

Lemma 4.3 — Analytic chain rule

Let F be represented by a native convergent power series on a neighbourhood of $g(y)$, and let g be native differentiable. Then

$$\partial_{\Sigma}(F \circ g)(y) = F'(g(y))\partial_{\Sigma}g(y).$$

Proof

The rule holds for every polynomial by induction using the product rule. On a strictly smaller native neighbourhood, the power series and its formal derivative have uniformly Cauchy tails. Pass the polynomial identity to the native limit.

Lemma 4.4 — Derivative of the odd series

For $|y|_{\Sigma} < 1$,

$$\partial_{\Sigma} A_{\Sigma}(y) = \frac{2}{1 - y^2}.$$

Proof

On every closed native subinterval $|y|_{\Sigma} \leq q < 1$, termwise differentiation is justified by the geometric majorant. Thus

$$\partial_{\Sigma} A_{\Sigma}(y) = 2 \sum_{m=0}^{\infty} y^{2m} = \frac{2}{1 - y^2}.$$

The last identity follows from the finite geometric identity followed by native completion.

75.7 5. Inversion of the native exponential

For $|y|_{\Sigma} < 1$, define

$$G_{\Sigma}(y) = \text{Exp}_{\Sigma}(A_{\Sigma}(y)) \frac{1 - y}{1 + y}.$$

The factors $1 - y$ and $1 + y$ are positive and therefore invertible.

Theorem 5.1 — Native Cayley–exponential identity

For every radial $|y|_{\Sigma} < 1$,

$$\text{Exp}_{\Sigma}(A_{\Sigma}(y)) = \frac{1 + y}{1 - y}.$$

Proof

Using the native Euler equation, the chain rule, product rule and inverse rule,

$$\begin{aligned} \partial_{\Sigma} G_{\Sigma}(y) &= G_{\Sigma}(y) \left(\frac{2}{1 - y^2} - \frac{1}{1 - y} - \frac{1}{1 + y} \right) \\ &= 0. \end{aligned}$$

The expression in parentheses is zero after passage to the common denominator $1 - y^2$.

It remains to justify that a native analytic function with zero derivative on the interval $(-1, 1)_{\Sigma}$ is constant. Around every point it has a convergent power series. Vanishing of the derivative forces every positive-degree coefficient to vanish. Overlapping neighbourhoods propagate the same constant through the connected ordered interval.

At $y=0$,

$$A_{\Sigma}(0) = 0, \quad G_{\Sigma}(0) = 1.$$

Hence $G_{\Sigma}(y)=1$, proving the identity.

75.8 6. Full positive logarithm

Definition 6.1 — Native positive logarithm

For every radial $x > 0$, define

$$\text{Log}_\Sigma^+(x) = 2 \sum_{m=0}^{\infty} \frac{1}{2m+1} \left(\frac{x-1}{x+1} \right)^{2m+1}.$$

The series is lawful by Lemma 2.1 and Lemma 3.1.

Theorem 6.2 — Exact inverse identities

For every radial $x > 0$,

$$\text{Exp}_\Sigma(\text{Log}_\Sigma^+(x)) = x.$$

For every radial u ,

$$\text{Log}_\Sigma^+(\text{Exp}_\Sigma(u)) = u.$$

Proof

Put

$$y = \frac{x-1}{x+1}.$$

Theorem 5.1 gives

$$\text{Exp}_\Sigma(\text{Log}_\Sigma^+(x)) = \frac{1+y}{1-y} = x.$$

For the second identity, apply the first identity with $x = \text{Exp}_\Sigma(u)$. Both u and $\text{Log}_\Sigma^+(\text{Exp}_\Sigma(u))$ are radial and have the same exponential. Strict radial injectivity of Exp_Σ gives equality.

Corollary 6.3 — Full positive Euler orbit

$$\mathbb{P}_\Sigma = \mathbb{R}_{\Sigma, > 0}.$$

Thus the logarithm introduced in F00-F on $\mathbb{P}_\Sigma$ extends to the whole positive radial cone and agrees with Log_Σ^+ .

From now on write simply

$$\text{Log}_\Sigma(x)$$

for $x > 0$.

75.9 7. Logarithmic laws

Theorem 7.1 — Multiplicative-to-additive law

For radial $x, y > 0$,

$$\boxed{\text{Log}_\Sigma(xy) = \text{Log}_\Sigma(x) + \text{Log}_\Sigma(y).}$$

Proof

The exponential addition law gives

$$\text{Exp}_\Sigma(\text{Log}_\Sigma(x) + \text{Log}_\Sigma(y)) = xy.$$

Apply the radial inverse identity.

Corollary 7.2

For radial $x > 0$ and integers n ,

$$\text{Log}_\Sigma(x^n) = n \text{Log}_\Sigma(x).$$

Also

$$\text{Log}_\Sigma(1) = 0, \quad \text{Log}_\Sigma(x^{-1}) = -\text{Log}_\Sigma(x).$$

Theorem 7.3 — Native derivative

For radial $x > 0$,

$$\boxed{\partial_\Sigma \text{Log}_\Sigma(x) = x^{-1}.}$$

Proof

Put

$$y(x) = \frac{x-1}{x+1}.$$

The quotient rule gives

$$y'(x) = \frac{2}{(x+1)^2}.$$

Moreover,

$$1 - y(x)^2 = \frac{4x}{(x+1)^2}.$$

Apply Lemma 4.4 and the chain rule:

$$\partial_\Sigma \text{Log}_\Sigma(x) = \frac{2}{1-y^2} \frac{2}{(x+1)^2} = \frac{1}{x}.$$

Theorem 7.4 — Strict order preservation

For radial $x, y > 0$,

$$x < y \iff \text{Log}_\Sigma(x) < \text{Log}_\Sigma(y).$$

Proof

This follows because Log_Σ is the inverse of the strictly increasing radial exponential.

75.10 8. Native complex powers

Definition 8.1 — Native power

For radial $x > 0$ and z in $\mathbb{C}_\Sigma$, define

$$x_\Sigma^z = \text{Exp}_\Sigma(z \text{Log}_\Sigma(x)).$$

The subscript is omitted when no confusion is possible.

Theorem 8.2 — Exponent laws

For radial $x, y > 0$ and z, w in $\mathbb{C}_\Sigma$,

$$x^{z+w} = x^z x^w,$$

$$(xy)^z = x^z y^z,$$

$$(x^z)^\dagger = x^{z^\dagger},$$

and for every integer n ,

$$x^n = x \cdots x$$

with the ordinary repeated-product interpretation.

Proof

Use the addition law for Exp_Σ , the logarithmic product law, dagger covariance, and $\text{Log}_\Sigma(x^n) = n \text{Log}_\Sigma(x)$.

Theorem 8.3 — Exponent derivative

For fixed $x > 0$,

$$\partial_\Sigma^z x^z = \text{Log}_\Sigma(x) x^z.$$

For fixed z , differentiation in the positive radial variable gives

$$\partial_\Sigma^x x^z = z x^{z-1}.$$

Proof

Apply the chain rule to the defining exponential and use $\text{partial } \text{Log}_\Sigma(x)=1/x$.

Corollary 8.4 — Native modulus law

Write

$$z = \sigma + \iota_\Sigma \tau.$$

Then

$$(x^z)^\dagger x^z = x^{2\sigma}.$$

Thus the radial magnitude of x^{-z} is controlled entirely by $x^{-\sigma}$.

75.11 9. Positive integers and prime logarithms

The natural number n is the radial scalar obtained by adding the unit n times. Every $n \geq 1$ is positive, so

$$\text{Log}_\Sigma(n)$$

and

$$n_\Sigma^{-s} = \text{Exp}_\Sigma(-s \text{Log}_\Sigma(n))$$

are now native objects for every s in $\mathbb{C}_\Sigma$.

For positive integers m, n ,

$$\text{Log}_\Sigma(mn) = \text{Log}_\Sigma(m) + \text{Log}_\Sigma(n),$$

and therefore

$$(mn)^{-s} = m^{-s} n^{-s}.$$

This is the arithmetic multiplicativity required for the native Dirichlet zeta construction.

75.12 10. Coordinate shadow

Under a later scalar chart

$$\kappa : \mathbb{C}_\Sigma \rightarrow \mathbb{C}$$

that preserves cut-complex addition, multiplication, dagger, order and the factorial exponential,

$$\kappa(\text{Exp}_\Sigma z) = e^{\kappa(z)}.$$

The inverse identity forces

$$\kappa(\text{Log}_\Sigma x) = \log \kappa(x)$$

for positive radial x , and

$$\kappa(x_\Sigma^z) = \kappa(x)^{\kappa(z)}$$

with the positive-real branch.

These are coordinate consequences, not definitions.

75.13 11. Claim boundary

Proved here:

Cayley contraction for every positive radial scalar;
 convergence of the native odd logarithm series;
 native product, inverse and analytic chain rules used in the proof;
 $\text{Exp_Sigma}(2 \text{ Atanh_Sigma } y) = (1+y)/(1-y)$;
 Log_Sigma on the entire positive radial cone;
 Exp_Sigma and Log_Sigma are inverse order isomorphisms;
 $\text{P_Sigma} = \text{R}_{\{\text{Sigma}, > 0\}}$;
 $\text{Log_Sigma}(xy) = \text{Log_Sigma}(x) + \text{Log_Sigma}(y)$;
 partial $\text{Log_Sigma}(x) = 1/x$;
 native complex powers x^z ;
 exponent laws and modulus law;
 native logarithms and powers of every positive integer.

Not proved here:

unique factorization of native natural numbers;
 Dirichlet-series convergence;
 Euler product;
 analytic continuation;
 Gamma or theta functions;
 Poisson summation;
 completed zeta functional equation;
 explicit formula;
 Weil criterion;
 $K_0 \geq 0$;
 RH.

No classical logarithm or inverse-function theorem is consumed. No separate certificate is created at this algebraic-analytic foundation stage. The later native zeta and completed-zeta promotion audits will consume the load-bearing identities proved here.

RKF/theorems/foundation/F00G_NATIVE_LOGARITHM_AND_POWERES.md

Chapter 76

F00-H — Native Natural Arithmetic and Prime Factorization

76.1 Classification

PROVED

76.2 Purpose

F00-G defines $\text{Log_Sigma}(n)$ and $n^\wedge s$ for every positive natural scalar. The Euler product for native zeta additionally requires a unique prime decomposition of the natural multiplicative semigroup.

This theorem derives that decomposition inside the repeated-unit subsemiring of the ordered radial cut field. It does not cite

the ordinary fundamental theorem of arithmetic;
Euclid's lemma;
Bezout's theorem;
the Euclidean algorithm;
ordinary integer factorization.

Each required statement is reconstructed below from finite induction, ordered addition and multiplication.

76.3 1. Native natural scalars

Let 1_Σ be the multiplicative unit of R_Σ . Define

$$0_\Sigma = 0, \quad n_\Sigma = \underbrace{1_\Sigma + \cdots + 1_\Sigma}_{n \text{ terms}}$$

for an external finite counting index n . The set

$$N_\Sigma = \{0_\Sigma, 1_\Sigma, 2_\Sigma, \dots\}$$

is the native natural subsemiring.

Because $\mathbf{R_Sigma}$ is ordered and has characteristic zero,

$$m_\Sigma < n_\Sigma \iff m < n$$

at the finite counting level. We suppress the subscript when no confusion is possible.

The induction principle used below is the defining recursive principle of this repeated-unit construction:

$P(0)$ and $[P(n) \text{ implies } P(n+1)]$
 $\Rightarrow P(n)$ for every native natural n .

Strong induction follows by applying ordinary induction to the finite statement that $P(k)$ holds for every $k \leq n$.

76.4 2. Divisibility

For a, b in $\mathbf{N_Sigma}$, say

$$a \mid b$$

when there exists c in $\mathbf{N_Sigma}$ such that

$$b = ac.$$

A positive natural u is a unit in $\mathbf{N_Sigma}$ exactly when $u=1$.

Lemma 2.1 — Cancellation

If $a > 0$ and

$$ab = ac,$$

then

$$b = c.$$

Proof

The equality holds in the ambient field $\mathbf{R_Sigma}$, where nonzero a has an inverse. Multiply by a^{-1} .

Lemma 2.2 — Finiteness below a natural

For every a in $\mathbf{N_Sigma}$, the set

$$\{n \in \mathbf{N}_\Sigma : n \leq a\}$$

contains exactly the finite repeated-unit elements

$$0, 1, \dots, a.$$

In particular, every nonempty subset of this interval has a least and a greatest element.

Proof

This is immediate from the recursive construction and finite induction. No completeness or choice principle is required.

76.5 3. Division with remainder

Theorem 3.1 — Native division algorithm

Let a, b in $\mathbb{N}_\Sigma$ with $b > 0$. There exist unique natural scalars q, r such that

$$a = qb + r, \quad 0 \leq r < b.$$

Proof: existence

Consider

$$S = \{k \in \mathbb{N}_\Sigma : kb \leq a\}.$$

The set is nonempty because 0 in S . Also $k \leq a$ for every k in S , since $b \geq 1$ gives $k \leq kb \leq a$. Hence S is a finite subset of $[0, a]$ and has a greatest element q .

Put

$$r = a - qb.$$

Then $r \geq 0$. If $r \geq b$, then

$$(q + 1)b = qb + b \leq a,$$

contradicting maximality of q . Thus $r < b$.

Proof: uniqueness

Suppose

$$a = qb + r = q'b + r', \quad 0 \leq r, r' < b.$$

If $q > q'$, then $q \geq q' + 1$, so

$$r' = a - q'b = (q - q')b + r \geq b,$$

contradicting $r' < b$. Similarly $q' < q$ is impossible. Hence $q = q'$, and then $r = r'$.

The theorem defines native quotient and remainder operations

$$q = \text{quo}_b(a), \quad r = \text{rem}_b(a).$$

76.6 4. Greatest common divisors

For positive a, b , a common divisor is d with $d|a$ and $d|b$.

Definition 4.1

The native greatest common divisor $\text{gcd_Sigma}(a, b)$ is the greatest common divisor in the natural order.

It exists because every common divisor is at most $\min(a, b)$ and the set of common divisors is a nonempty finite set containing 1.

Lemma 4.2 — Remainder invariance

If

$$a = qb + r,$$

then the common divisors of (a, b) are exactly the common divisors of (b, r) . Consequently

$$\boxed{\text{gcd}_{\Sigma}(a, b) = \text{gcd}_{\Sigma}(b, r).}$$

Proof

If $d|a$ and $d|b$, then $d|(a - qb) = r$. Conversely, if $d|b$ and $d|r$, then $d|(qb + r) = a$.

Theorem 4.3 — Native Euclidean algorithm

Repeated division

$$a = q_0b + r_0,$$

$$b = q_1r_0 + r_1,$$

and so on terminates at a final nonzero remainder r_k , and

$$\boxed{\text{gcd}_{\Sigma}(a, b) = r_k.}$$

Proof

The positive remainders form a strictly decreasing sequence

$$b > r_0 > r_1 > \cdots > 0.$$

No infinite strictly decreasing sequence of native naturals exists: after at most b decreases the sequence must stop. Remainder invariance propagates the gcd to the last nonzero remainder.

76.7 5. Native Bezout identity

Theorem 5.1 — Bezout representation

For positive naturals $\mathbf{a}, \mathbf{b}$, there exist native integers $\mathbf{u}, \mathbf{v}$ in the signed repeated-unit ring

$$\mathbb{Z}_\Sigma = \mathbb{N}_\Sigma - \mathbb{N}_\Sigma$$

such that

$$\boxed{\gcd_\Sigma(a, b) = ua + vb.}$$

Proof

Every remainder in the Euclidean algorithm is an integer-linear combination of $\mathbf{a}$ and $\mathbf{b}$.

The first remainder is

$$r_0 = a - q_0b.$$

If

$$r_{j-1} = u_{j-1}a + v_{j-1}b, \quad r_j = u_ja + v_jb,$$

then

$$r_{j+1} = r_{j-1} - q_{j+1}r_j$$

is also such a combination. Induction reaches the last nonzero remainder, which is the gcd by Theorem 4.3.

Corollary 5.2 — Coprime inverse relation

If

$$\gcd_\Sigma(a, b) = 1,$$

then there exist $\mathbf{u}, \mathbf{v}$ in $\mathbb{Z}_\Sigma$ with

$$ua + vb = 1.$$

76.8 6. Native primes

Definition 6.1

A natural scalar $\mathbf{p} > 1$ is **native prime** when its only positive natural divisors are 1 and $\mathbf{p}$.

A natural $\mathbf{n} > 1$ that is not prime is **composite**.

Lemma 6.2 — Composite splitting

If $n > 1$ is composite, then

$$n = ab$$

for naturals a, b satisfying

$$1 < a < n, \quad 1 < b < n.$$

Proof

A nontrivial divisor a obeys $1 < a < n$. Write $n = ab$. Since $a > 1$, cancellation and order give $b < n$; since $a < n$, one also has $b > 1$.

Theorem 6.3 — Native Euclid lemma

If p is native prime and

$$p \mid ab,$$

then

$$\boxed{p \mid a \quad \text{or} \quad p \mid b.}$$

Proof

If $p \mid a$, there is nothing to prove. Suppose p does not divide a .

Every common divisor of p and a divides p ; since p is prime and does not divide a , the gcd is 1. Bezout gives

$$up + va = 1.$$

Multiply by b :

$$ubp + vab = b.$$

The first term is divisible by p , and the second is divisible by p because $p \mid ab$. Therefore $p \mid b$.

Corollary 6.4 — Prime divides a finite product

If

$$p \mid a_1 a_2 \cdots a_m,$$

then p divides at least one factor.

Proof

Induct on m using Theorem 6.3.

76.9 7. Existence of prime factorization

Theorem 7.1 — Native prime-factor existence

Every natural scalar $n > 1$ is a finite product of native primes.

Proof

Use strong induction on n .

If n is prime, the one-factor product suffices. If n is composite, Lemma 6.2 gives

$$n = ab$$

with $1 < a, b < n$. By the induction hypothesis, both a and b are finite prime products. Concatenating those products gives a prime factorization of n .

The proof terminates because every composite split strictly decreases both factors in the native natural order.

76.10 8. Uniqueness of prime factorization

Theorem 8.1 — Native fundamental theorem of arithmetic

Suppose

$$n = p_1 p_2 \cdots p_r = q_1 q_2 \cdots q_s,$$

where all p_i and q_j are native primes. Then

$$r = s$$

and, after a permutation,

$$p_i = q_i \quad (1 \leq i \leq r).$$

Proof

The prime p_1 divides the product $q_1 \cdots q_s$. By Corollary 6.4 it divides some q_j . Since q_j is prime and $p_1 > 1$, the only possibility is

$$p_1 = q_j.$$

Reorder the q factors and cancel this common nonzero factor. Repeat by finite induction. Every factor on one side is matched with exactly one factor on the other, so the lengths agree.

Corollary 8.2 — Exponent-vector form

For every $\mathbf{n} \geq 1$, there is a unique finitely supported function

$$\nu_n : \mathcal{P}_\Sigma \rightarrow \mathbb{N}_\Sigma$$

on the native prime set such that

$$n = \prod_{p \in \mathcal{P}_\Sigma} p^{\nu_n(p)}.$$

Moreover,

$$\nu_{mn}(p) = \nu_m(p) + \nu_n(p).$$

This is the native prime-exponent vector consumed by the later prime-torus chart, but no torus is introduced here.

76.11 9. Divisor sums and finite Euler identities

Definition 9.1

For $\mathbf{n} \geq 1$, let

$$\text{Div}_\Sigma(n) = \{d \in \mathbb{N}_\Sigma : d|n\}.$$

This set is finite because every divisor satisfies $1 \leq d \leq n$.

For a native arithmetic function $\mathbf{f}$, define the finite divisor sum

$$\sum_{d|n} f(d).$$

Lemma 9.2 — Multiplicative divisor decomposition

If $\text{gcd}_\Sigma(\mathbf{m}, \mathbf{n}) = 1$, then every divisor of $\mathbf{mn}$ has a unique form

$$d = ab, \quad a|m, \quad b|n.$$

Proof

Use the disjoint prime supports supplied by Corollary 8.2. A divisor chooses an exponent between 0 and the exponent of each prime. Splitting those choices between the prime supports of $\mathbf{m}$ and $\mathbf{n}$ gives the unique pair $(\mathbf{a}, \mathbf{b})$.

Corollary 9.3

Finite divisor convolution

$$(f * g)(n) = \sum_{d|n} f(d)g(n/d)$$

is associative and commutative on native arithmetic functions.

Proof

All sums are finite. Reindex by the unique factorization of ordered divisor triples.

76.12 10. Native Möbius function

The logarithmic derivative of zeta can be developed without Möbius inversion, but the exact arithmetic extraction is useful later.

Definition 10.1

Define $\mu_{\Sigma}(\mathbf{n})$ by the prime exponent vector:

$$\mu_{\Sigma}(1) = 1,$$

$$\mu_{\Sigma}(n) = 0 \quad \text{if some } \nu_n(p) \geq 2,$$

and otherwise

$$\mu_{\Sigma}(n) = (-1)^k$$

when $\mathbf{n}$ is a product of k distinct primes.

Theorem 10.2 — Native Möbius cancellation

For every natural $\mathbf{n}$,

$$\sum_{d|\mathbf{n}} \mu_{\Sigma}(d) = \begin{cases} 1, & \mathbf{n} = 1, \\ 0, & \mathbf{n} > 1. \end{cases}$$

Proof

If

$$n = \prod_{j=1}^r p_j^{a_j},$$

then only square-free divisors contribute. Choosing a subset of the r distinct prime factors gives

$$\sum_{d|\mathbf{n}} \mu_{\Sigma}(d) = \sum_{k=0}^r \binom{r}{k} (-1)^k = (1-1)^r.$$

For $\mathbf{n}=1$, $r=0$ and the value is 1. The binomial identity is finite algebra inside the native integer ring.

Corollary 10.3 — Native Möbius inversion

If native arithmetic functions satisfy

$$F(n) = \sum_{d|n} f(d),$$

then

$$f(n) = \sum_{d|n} \mu_{\Sigma}(d) F(n/d).$$

Proof

Substitute the divisor sum for F, regroup the finite double sum by the product of the two divisor variables, and apply Theorem 10.2.

76.13 11. Native von Mangoldt arithmetic weight

Definition 11.1

Using the native logarithm of F00-G, define

$$\Lambda_{\Sigma}(n) = \begin{cases} \text{Log}_{\Sigma}(p), & n = p^k \text{ for a native prime } p \text{ and } k \geq 1, \\ 0, & \text{otherwise.} \end{cases}$$

Theorem 11.2 — Native logarithmic divisor identity

For every positive natural n ,

$$\text{Log}_{\Sigma}(n) = \sum_{d|n} \Lambda_{\Sigma}(d).$$

Proof

Write

$$n = \prod_{j=1}^r p_j^{a_j}.$$

The prime powers dividing n are p_j^k for $1 \leq k \leq a_j$. Therefore

$$\sum_{d|n} \Lambda_{\Sigma}(d) = \sum_{j=1}^r a_j \text{Log}_{\Sigma}(p_j) = \text{Log}_{\Sigma}(n)$$

by the logarithmic product law.

Corollary 11.3 — Native pressure extraction

Let

$$\ell_{\Sigma}(n) = \text{Log}_{\Sigma}(n).$$

Then under divisor convolution,

$$\ell_{\Sigma} * \mu_{\Sigma} = \Lambda_{\Sigma}.$$

Proof

Apply Möbius inversion to Theorem 11.2.

This recovers the arithmetic identity used in MP without importing the ordinary Möbius or von Mangoldt functions.

76.14 12. Coordinate shadow

Under a later faithful scalar chart preserving the repeated unit,

$$n_{\Sigma} \mapsto n,$$

native divisibility, gcd, primes, exponent vectors, `mu_Sigma` and `Lambda_Sigma` map to their ordinary arithmetic counterparts. The chart is forced by the finite recursive construction and unique factorization proved above.

No ordinary prime theorem is used to construct the native objects.

76.15 13. Claim boundary

Proved here:

```

native natural and integer subrings;
divisibility and cancellation;
division with remainder;
gcd and terminating Euclidean algorithm;
Bezout identity;
native primes and composite splitting;
Euclid lemma;
existence and uniqueness of prime factorization;
prime-exponent vectors;
finite divisor convolution;
native Möbius function and inversion;
native von Mangoldt function;
Log_Sigma(n)=sum_{d|n} Lambda_Sigma(d);
Log_Sigma * mu_Sigma = Lambda_Sigma.

```

Not proved here:

infinitude of primes;
 Dirichlet-series convergence;
 Euler product as an infinite limit;
 nonvanishing of zeta in $\text{Re}(s) > 1$;
 prime number theorem;
 analytic continuation;
 completed zeta;
 explicit formula;
 Weil criterion;
 $KO \geq 0$;
 RH.

Infinitude of primes and the analytic identities belong to F00-I. No external fundamental theorem of arithmetic is consumed.

RKF/theorems/foundation/F00H_NATIVE_NATURAL_ARITHMETIC_AND_PRIME_FACTORIZATION.md

Chapter 77

F00-I — Native Dirichlet Zeta, Euler Product and von Mangoldt Derivative

77.1 Classification

PROVED on the native half-plane

$$\operatorname{Re}_{\Sigma}(s) > 1.$$

77.2 Purpose

F00-G derives the native logarithm and powers of every positive natural scalar. F00-H derives native primes, unique factorization, the Möbius function and the von Mangoldt weight.

This theorem constructs the zeta function in its initial domain without importing

the classical Riemann zeta function;
the ordinary p-series or integral test;
the classical Euler product;
ordinary complex logarithms;
Weierstrass convergence theorems;
the classical von Mangoldt logarithmic derivative.

The construction order is

native powers n^{-s}
-> dyadic convergence
-> native differentiability
-> native Dirichlet convolution products
-> finite prime Euler products
-> infinite Euler product
-> nonvanishing
-> Möbius reciprocal
-> von Mangoldt logarithmic derivative.

77.3 1. Native half-plane and powers

For

$$s = \sigma + \iota_{\Sigma}\tau \in \mathbb{C}_{\Sigma},$$

define

$$\operatorname{Re}_{\Sigma}(s) = \frac{s + s^{\dagger}}{2} = \sigma.$$

For every positive natural n , $\mathbb{F}00\text{-G}$ defines

$$\boxed{n_{\Sigma}^{-s} = \operatorname{Exp}_{\Sigma}(-s \operatorname{Log}_{\Sigma}(n)).}$$

The subscript is suppressed below.

Lemma 1.1 — Native magnitude of a Dirichlet atom

For $s = \sigma + \iota_{\Sigma}\tau$,

$$\boxed{(n^{-s})^{\dagger} n^{-s} = n^{-2\sigma}.}$$

Hence

$$|n^{-s}|_{\Sigma} = n^{-\sigma}.$$

Proof

Dagger covariance and the exponential addition law give

$$\begin{aligned} (n^{-s})^{\dagger} n^{-s} &= \operatorname{Exp}_{\Sigma}(-s^{\dagger} \operatorname{Log}_{\Sigma} n) \operatorname{Exp}_{\Sigma}(-s \operatorname{Log}_{\Sigma} n) \\ &= \operatorname{Exp}_{\Sigma}(-(s + s^{\dagger}) \operatorname{Log}_{\Sigma} n) \\ &= n^{-2\sigma}. \end{aligned}$$

The positive square root is $n^{-\sigma}$.

Lemma 1.2 — Multiplicativity

For positive naturals m, n ,

$$\boxed{(mn)^{-s} = m^{-s} n^{-s}.}$$

Proof

Use

$$\operatorname{Log}_{\Sigma}(mn) = \operatorname{Log}_{\Sigma}(m) + \operatorname{Log}_{\Sigma}(n)$$

and the exponential addition law.

77.4 2. Dyadic convergence

For $k \geq 0$, let the native dyadic block be

$$D_k = \{n \in \mathbb{N}_\Sigma : 2^k \leq n < 2^{k+1}\}.$$

It contains exactly 2^k natural scalars.

Lemma 2.1 — Dyadic block estimate

If $\sigma = \operatorname{Re}_\Sigma(s) > 1$, then

$$\sum_{n \in D_k} |n^{-s}|_\Sigma \leq 2^{-k(\sigma-1)}.$$

Proof

For n in D_k , monotonicity of the logarithm and negative radial exponent give

$$n^{-\sigma} \leq (2^k)^{-\sigma} = 2^{-k\sigma}.$$

There are 2^k terms.

Theorem 2.2 — Native Dirichlet convergence

For every s with

$$\operatorname{Re}_\Sigma(s) > 1,$$

the partial sums

$$Z_{\Sigma, N}(s) = \sum_{n=1}^N n^{-s}$$

are Cauchy in $\mathbb{C}_\Sigma$.

Proof

Choose radial $\delta > 0$ such that

$$\operatorname{Re}_\Sigma(s) \geq 1 + \delta.$$

Every tail beginning beyond 2^K is bounded by

$$\sum_{k=K}^{\infty} 2^{-k\delta},$$

which is a native convergent geometric tail.

Definition 2.3 — Native Dirichlet zeta

For $\text{Re}_\Sigma(s) > 1$, define

$$\zeta_\Sigma(s) = \sum_{n=1}^{\infty} n^{-s}.$$

This is the first definition of zeta in RH-Framework. No classical zeta function is referenced.

Corollary 2.4 — Uniform half-plane tails

For every radial $\delta > 0$, convergence is uniform in the native magnitude on

$$\text{Re}_\Sigma(s) \geq 1 + \delta.$$

Indeed,

$$\sup_{\text{Re}_\Sigma(s) \geq 1 + \delta} \left| \sum_{n \geq 2^K} n^{-s} \right|_\Sigma \leq \frac{2^{-K\delta}}{1 - 2^{-\delta}}.$$

77.5 3. Logarithmic moment convergence**Lemma 3.1 — Dyadic logarithm bound**

For n in D_k ,

$$\text{Log}_\Sigma(n) \leq (k + 1) \text{Log}_\Sigma(2).$$

Proof

Because $n < 2^{k+1}$, strict monotonicity and the logarithmic power law give the result.

Lemma 3.2 — Polynomial-geometric convergence

For every radial $0 < q < 1$ and every fixed natural r ,

$$\sum_{k=0}^{\infty} (k + 1)^r q^k$$

converges in R_Σ .

Proof

Choose a native rational $q < \rho < 1$. For sufficiently large k ,

$$(k + 2)^r q^{k+1} \leq \rho (k + 1)^r q^k.$$

This follows because

$$q \left(1 + \frac{1}{k+1}\right)^r \longrightarrow q < \rho$$

by finite binomial expansion and the Archimedean order. The tail is then bounded by a geometric series of ratio ρ .

Theorem 3.3 — Native logarithmic moment series

For every fixed natural $r \geq 0$,

$$\sum_{n=1}^{\infty} \text{Log}_{\Sigma}(n)^r n^{-s}$$

converges absolutely and uniformly on every half-plane

$$\text{Re}_{\Sigma}(s) \geq 1 + \delta.$$

Proof

On D_k , the absolute block sum is bounded by

$$(k+1)^r \text{Log}_{\Sigma}(2)^r 2^{-k\delta}.$$

Apply Lemma 3.2.

77.6 4. Native differentiability of zeta

For each natural n ,

$$\partial_{\Sigma} n^{-s} = -\text{Log}_{\Sigma}(n) n^{-s}.$$

This follows from the native Euler equation and chain rule.

Theorem 4.1 — Native zeta derivative

On $\text{Re}_{\Sigma}(s) > 1$, the native derivative exists and

$$\zeta'_{\Sigma}(s) = - \sum_{n=1}^{\infty} \text{Log}_{\Sigma}(n) n^{-s}.$$

Proof

Fix s_0 with

$$\text{Re}_{\Sigma}(s_0) > 1.$$

Choose $\delta > 0$ so that

$$\operatorname{Re}_{\Sigma}(s_0) \geq 1 + 2\delta.$$

For native $\mathbf{h}$ with $|\mathbf{h}|_{\Sigma} < \delta$,

$$\frac{n^{-(s_0+h)} - n^{-s_0}}{h} = n^{-s_0} \frac{\operatorname{Exp}_{\Sigma}(-h \operatorname{Log}_{\Sigma} n) - 1}{h}.$$

Subtract

$$- \operatorname{Log}_{\Sigma}(n) n^{-s_0}.$$

The factorial-tail estimate for $\operatorname{Exp}_{\Sigma}$ bounds the remainder by

$$|h|_{\Sigma} \operatorname{Log}_{\Sigma}(n)^2 n^{-(1+\delta)} \operatorname{Exp}_{\Sigma}(|h|_{\Sigma} \operatorname{Log}_{\Sigma} n).$$

Since

$$\operatorname{Exp}_{\Sigma}(|h|_{\Sigma} \operatorname{Log}_{\Sigma} n) = n^{|h|_{\Sigma}} \leq n^{\delta},$$

the remainder is bounded by

$$|h|_{\Sigma} \operatorname{Log}_{\Sigma}(n)^2 n^{-(1+\delta)}.$$

The majorant series converges by Theorem 3.3. Summing the remainder and then letting $\mathbf{h} \rightarrow 0$ proves the displayed derivative formula directly. No imported uniform-limit differentiation theorem is used.

Repeated use of the same argument gives

$$\zeta_{\Sigma}^{(r)}(s) = (-1)^r \sum_{n=1}^{\infty} \operatorname{Log}_{\Sigma}(n)^r n^{-s}$$

for every finite natural $\mathbf{r}$.

77.7 5. Infinitude of native primes

Theorem 5.1 — Native Euclid theorem

There are infinitely many native primes.

Proof

Suppose the complete prime set were finite:

$$p_1, \dots, p_m.$$

Put

$$N = p_1 p_2 \cdots p_m + 1.$$

Then $N > 1$, so F00-H gives a prime divisor $q | N$. By completeness of the list, $q = p_j$ for some j . But p_j divides the product and also N , so it divides

$$N - p_1 \cdots p_m = 1,$$

contradicting $p_j > 1$.

The native primes may therefore be listed increasingly as

$$p_1 < p_2 < p_3 < \cdots$$

using the least-element property of the natural order.

77.8 6. Native Dirichlet convolution series

Let a, b be native arithmetic functions. Suppose

$$\sum_{n \geq 1} |a(n)| n^{-\sigma}$$

and

$$\sum_{n \geq 1} |b(n)| n^{-\sigma}$$

converge for some radial $\sigma > 1$.

Define finite divisor convolution

$$(a * b)(n) = \sum_{d|n} a(d)b(n/d).$$

Theorem 6.1 — Native Dirichlet multiplication

For $\text{Re}_\Sigma(s) \geq \sigma$,

$$\left(\sum_{m \geq 1} a(m)m^{-s} \right) \left(\sum_{n \geq 1} b(n)n^{-s} \right) = \sum_{k \geq 1} (a * b)(k)k^{-s}.$$

Proof

For finite rectangles of indices, distributivity and

$$(mn)^{-s} = m^{-s}n^{-s}$$

give the identity after grouping by $k = mn$.

The unrestricted double tail is bounded by the product of the two absolute series tails. Hence the finite identities pass to the native Cauchy limit. Every coefficient group for fixed k is finite because the divisor set is finite.

77.9 7. Finite Euler products

For a native prime p and $\text{Re_Sigma}(s) > 1$,

$$|p^{-s}|_{\Sigma} = p^{-\sigma} < 1.$$

Thus the native geometric identity gives

$$(1 - p^{-s})^{-1} = \sum_{j=0}^{\infty} p^{-js}.$$

For the first m primes, define

$$Q_m(s) = \prod_{j=1}^m (1 - p_j^{-s})^{-1}.$$

Theorem 7.1 — Finite prime-factor expansion

$$Q_m(s) = \sum_{\substack{n \geq 1 \\ \nu_n(p)=0 \text{ for } p > p_m}} n^{-s}.$$

Proof

Expand the finite product of geometric series. A term is specified by one nonnegative exponent for each prime $p_1, \dots, p_m$, hence by the natural

$$n = p_1^{a_1} \cdots p_m^{a_m}.$$

Unique factorization from F00-H makes this correspondence bijective. Absolute convergence of the finite product of geometric series permits the finite-stage Cauchy limit in each exponent.

77.10 8. Infinite Euler product

Theorem 8.1 — Native Euler product

For every s with $\text{Re_Sigma}(s) > 1$,

$$\zeta_{\Sigma}(s) = \prod_{p \in \mathcal{P}_{\Sigma}} (1 - p^{-s})^{-1}.$$

The product means the native limit of $Q_m(s)$.

Proof

The finite expansion contains exactly those natural numbers whose prime factors are at most $\mathbf{p}_m$. Every omitted natural has a prime factor greater than $\mathbf{p}_m$ and is therefore itself greater than $\mathbf{p}_m$.

Hence

$$|\zeta_\Sigma(s) - Q_m(s)|_\Sigma \leq \sum_{n > p_m} n^{-\sigma}.$$

The native Dirichlet tail tends to zero by Theorem 2.2.

The convergence is uniform on every native half-plane

$$\operatorname{Re}_\Sigma(s) \geq 1 + \delta.$$

77.11 9. Native nonvanishing

Define the finite inverse products

$$P_m(s) = \prod_{j=1}^m (1 - p_j^{-s}).$$

Lemma 9.1 — Product-tail estimate

For native scalars $\mathbf{a}_j$,

$$\left| \prod_{j=r}^m (1 - a_j) - 1 \right|_\Sigma \leq \operatorname{Exp}_\Sigma \left(\sum_{j=r}^m |a_j|_\Sigma \right) - 1.$$

Proof

Expand the finite product. The absolute sum of all nonconstant monomials is at most

$$\prod_{j=r}^m (1 + |a_j|_\Sigma) - 1.$$

For positive radial u , the factorial exponential satisfies

$$1 + u \leq \operatorname{Exp}_\Sigma(u).$$

Therefore

$$\prod_{j=r}^m (1 + |a_j|_\Sigma) \leq \operatorname{Exp}_\Sigma \left(\sum_{j=r}^m |a_j|_\Sigma \right).$$

Theorem 9.2 — Native zero-free half-plane

For $\text{Re}_\Sigma(s) > 1$,

$$\zeta_\Sigma(s) \neq 0.$$

Proof

Because

$$\sum_p |p^{-s}|_\Sigma \leq \sum_{n \geq 2} n^{-\sigma} < \infty,$$

Lemma 9.1 shows that $P_m(s)$ is Cauchy. Let its limit be $P(s)$.

For every m ,

$$P_m(s)Q_m(s) = 1.$$

Theorem 8.1 gives $Q_m(s) \rightarrow \zeta_\Sigma(s)$. Continuity of multiplication in the completed cut field gives

$$P(s)\zeta_\Sigma(s) = 1.$$

Thus $\zeta_\Sigma(s)$ is invertible and nonzero.

77.12 10. Möbius reciprocal

Because

$$|\mu_\Sigma(n)|_\Sigma \leq 1,$$

the series

$$M_\Sigma(s) = \sum_{n=1}^{\infty} \mu_\Sigma(n) n^{-s}$$

converges absolutely for $\text{Re}_\Sigma(s) > 1$.

Theorem 10.1 — Native reciprocal series

$$\frac{1}{\zeta_\Sigma(s)} = \sum_{n=1}^{\infty} \mu_\Sigma(n) n^{-s}.$$

Proof

Apply native Dirichlet multiplication to the constant arithmetic function $1(n)=1$ and μ_Σ . F00-H proves

$$(1 * \mu_\Sigma)(n) = \begin{cases} 1, & n = 1, \\ 0, & n > 1. \end{cases}$$

Therefore

$$\zeta_\Sigma(s)M_\Sigma(s) = 1.$$

Use Theorem 9.2.

77.13 11. Native von Mangoldt series

The series

$$A_\Sigma(s) = \sum_{n=1}^{\infty} \Lambda_\Sigma(n)n^{-s}$$

converges absolutely for $\text{Re}_\Sigma(s) > 1$ because

$$0 \leq \Lambda_\Sigma(n) \leq \text{Log}_\Sigma(n)$$

and Theorem 3.3 applies.

Theorem 11.1 — Native logarithmic derivative

For $\text{Re}_\Sigma(s) > 1$,

$$-\frac{\zeta'_\Sigma(s)}{\zeta_\Sigma(s)} = \sum_{n=1}^{\infty} \Lambda_\Sigma(n)n^{-s}.$$

Proof

F00-H gives the finite divisor identity

$$\text{Log}_\Sigma(n) = \sum_{d|n} \Lambda_\Sigma(d).$$

Native Dirichlet multiplication therefore yields

$$\begin{aligned} \zeta_\Sigma(s)A_\Sigma(s) &= \sum_{n=1}^{\infty} \left(\sum_{d|n} \Lambda_\Sigma(d) \right) n^{-s} \\ &= \sum_{n=1}^{\infty} \text{Log}_\Sigma(n)n^{-s} \\ &= -\zeta'_\Sigma(s). \end{aligned}$$

Multiply by the native inverse of $\mathbf{zeta_Sigma}(s)$ from Theorem 9.2.

Corollary 11.2 — Prime-power expansion

$$-\frac{\zeta'_\Sigma(s)}{\zeta_\Sigma(s)} = \sum_{p \in \mathcal{P}_\Sigma} \sum_{k=1}^{\infty} \text{Log}_\Sigma(p) p^{-ks}.$$

Proof

By definition $\mathbf{Lambda_Sigma}(n)$ is nonzero exactly on prime powers, where its value is $\mathbf{Log_Sigma}(p)$. Unique factorization makes the indexing disjoint.

This is the native arithmetic source later used by the prime-Euler translation operator.

77.14 12. Finite clock and prime-exponent chart

Define the native logarithmic clock on the finite arithmetic span by

$$H_{\log, \Sigma} e_n = \text{Log}_\Sigma(n) e_n.$$

The prime-exponent vector from F00-H satisfies

$$\text{Log}_\Sigma(n) = \sum_p \nu_n(p) \text{Log}_\Sigma(p).$$

Thus any later prime-torus or multicircle chart is forced to carry the native clock to the weighted prime generator. It is a representation of this arithmetic identity, not the source of zeta.

For $\mathbf{Re_Sigma}(s) > 1$, the native formal trace on the arithmetic basis is

$$\sum_{n \geq 1} \text{Exp}_\Sigma(-s H_{\log, \Sigma})_{nn} = \zeta_\Sigma(s).$$

Operator trace-class language is postponed until the native operator bridge is applied.

77.15 13. Coordinate shadow

Under a later faithful scalar chart preserving the cut exponential, logarithm and natural unit,

$$\kappa(\zeta_\Sigma(s)) = \sum_{n=1}^{\infty} n^{-\kappa(s)}$$

for $\mathbf{Re_kappa}(s) > 1$. The native Euler product, Möbius reciprocal and von Mangoldt derivative map to the familiar coordinate identities.

The coordinate function is therefore a shadow of $\mathbf{zeta_Sigma}$; it is not an input to its construction.

77.16 14. Claim boundary

Proved here on $\text{Re_Sigma}(s) > 1$:

native convergence of $\sum n^{-s}$;
uniform dyadic tail bounds;
native differentiability and all finite derivatives;
infinitude of native primes;
Dirichlet-series multiplication under absolute convergence;
finite prime-factor Euler expansions;
infinite Euler product;
zero-free half-plane;
Möbius reciprocal series;
von Mangoldt logarithmic derivative;
prime-power expansion;
native arithmetic logarithmic clock.

Not proved here:

continuation beyond $\text{Re_Sigma}(s) > 1$;
a pole at $s=1$;
Gamma factor;
 π_{Sigma} ;
Gaussian Fourier self-duality;
Poisson summation;
theta inversion;
functional equation;
completed ξ_{Sigma} ;
zero-sum explicit formula;
Weil criterion;
 $K_0 \geq 0$;
RH.

No classical zeta theorem, Euler product, p-series test or analytic continuation is consumed. Continuation begins only after the native Fourier–Gaussian–Poisson chain is derived.

RKF/theorems/foundation/F00I_NATIVE_DIRICHLET_ZETA_EULER_PRODUCT.md

Chapter 78

F00-G/H/I Rigorous Computational Certificate v0.1

78.1 Scope

This Class F package audits the native derivation chain

F00-G positive logarithm and native powers;
F00-H natural arithmetic, prime factorization, Mobius and von Mangoldt;
F00-I half-plane zeta, Euler product and logarithmic derivative.

It does not prove the universal theorems by random sampling. It checks exact finite arithmetic, rigorous rational interval enclosures for the native logarithm/exponential inverse packet, the proof inequalities used in dyadic convergence and Euler-product passage, source-order policy, immutable source pins and adversarial negative controls.

78.2 Arithmetic

<code>fractions.Fraction</code>	<code>exact rational arithmetic;</code>
<code>integer trial division</code>	<code>exact finite factorization audit;</code>
<code>rational intervals</code>	<code>outward series and tail enclosures;</code>
<code>SHA-1 / SHA-256</code>	<code>source identity and canonical result hashing.</code>

No floating-point number is used in a proof margin.

78.3 Main obligations

Logarithm and powers

C00G-01 Cayley contraction on the exact rational packet;
C00G-02 odd logarithm tails and $\text{Exp}(\text{Log } x)=x$ enclosures;
C00G-03 multiplicative logarithm packet and exact derivative identity;
C00G-04 exponent-addition and product-base interval consistency.

Arithmetic

C00H-01 exhaustive division algorithm packet;
C00H-02 Euclidean gcd and Bezout reconstruction;

C00H-03 exhaustive Euclid lemma packet;
C00H-04 prime factor reconstruction and primality audit;
C00H-05 Mobius divisor cancellation;
C00H-06 formal-prime-log von Mangoldt divisor identity;
C00H-07 Euclid prime construction on increasing prime prefixes.

Half-plane zeta

C00I-01 exact dyadic block estimates;
C00I-02 geometric and polynomial-geometric tail bounds;
C00I-03 finite Dirichlet-convolution coefficients;
C00I-04 finite Euler products and rigorous zeta-tail enclosure;
C00I-05 product-tail inequality supporting nonvanishing;
C00I-06 Mobius reciprocal coefficient identity;
C00I-07 von Mangoldt logarithmic-derivative coefficient identity.

78.4 Negative controls

The verifier must reject:

noncontracting Cayley coordinate;
a composite number treated as prime in Euclid's lemma;
wrong Mobius signs;
omission of prime powers from Lambda;
the wrong dyadic exponent;
an Euler product truncated to first prime powers only;
classical logarithm/zeta inserted as a primitive source.

78.5 Run

From the repository root:

```
python certificates/foundation/FOOGHI_LOG_ARITHMETIC_ZETA_VO_1/verify.py \  
--output certificates/foundation/FOOGHI_LOG_ARITHMETIC_ZETA_VO_1/result.json
```

Jupyter:

```
%run certificates/foundation/FOOGHI_LOG_ARITHMETIC_ZETA_VO_1/verify.py \  
--output certificates/foundation/FOOGHI_LOG_ARITHMETIC_ZETA_VO_1/result.json
```

Passing status:

PASS_FOOGHI_LOG_ARITHMETIC_ZETA_AUDIT

78.6 Claim boundary

A PASS confirms the declared exact/interval obligations and their alignment with the pinned theorem sources. It does not certify analytic continuation, the completed-Weil sign, $K_0 \geq 0$, or RH.

RKF/certificates/foundation/FOOGHI_LOG_ARITHMETIC_ZETA_VO_1/README.md

Chapter 79

F00/F00-E Rigorous Computational Certificate v0.1

79.1 Scope

This package is a Class F foundational computational certificate for:

F00 cut scalar, quarter-turn, dagger and radial square;
F00-E native exponential, Euler equation, circular factors and Euler flow.

It does not prove these universal theorems by sampling. It machine-checks the finite algebra, executable implementation, exact coefficient identities, quantitative proof bounds, source-order policy and verifier sensitivity on which the line-by-line proofs rely.

79.2 Authoritative sources

src/rh_framework/native_summability.py
theorems/foundation/F00E_NATIVE_EULER_FROM_IOTA_COMPLEX.md
certificates/foundation/NUMERICAL_PROOF_PROTOCOL.md

The verifier hashes these files at execution time. The result records the actual hashes rather than silently trusting names.

79.3 Obligations

C00-01 NativeCutScalar implementation is importable.
C00-02 one and iota satisfy the exact multiplication table.
C00-03 multiplication is associative and commutative on the scalar basis.
C00-04 dagger is involutive and reverses multiplication.
C00-05 radial square is nonnegative on the declared exact audit lattice.
C00-06 the scalar product/norm identity agrees with exact rational arithmetic.

C00E-01 factorial coefficients and binomial convolution identities are exact.
C00E-02 declared factorial-tail bounds are strict on the audit packet.
C00E-03 the unit-tangent remainder bound is strict on the audit packet.
C00E-04 even/odd factorial splitting reproduces cosine and sine coefficients.
C00E-05 the finite Euler/Pythagorean residual is enclosed by declared tails.
C00E-06 flow composition and sign convention are consistent.
C00E-07 classical ``-i r d/dr`` appears only after the native Euler construction.
C00E-08 every negative control is detected.

79.4 Arithmetic

All scalar identities and bound calculations use `fractions.Fraction`. No floating-point value is used in a proof margin. SHA-256 uses canonical UTF-8 bytes. A later v0.2 may add Arb boxes for wider complex domains, but v0.1 is already exact for its declared obligations.

79.5 Run

From a repository checkout:

```
python certificates/foundation/F00_F00E_EULER_V0_1/verify.py \  
    --output certificates/foundation/F00_F00E_EULER_V0_1/result.json
```

In Jupyter:

```
%run certificates/foundation/F00_F00E_EULER_V0_1/verify.py \  
    --output certificates/foundation/F00_F00E_EULER_V0_1/result.json
```

79.6 Passing status

PASS_F00_F00E_RIGOROUS_COMPUTATIONAL_AUDIT

A passing result confirms alignment of the executable foundation and the quantitative proof obligations. It does not promote any still-open numerical sign claim.

RKF/certificates/foundation/F00_F00E_EULER_V0_1/README.md

Chapter 80

RH-Framework Rigorous Computational Proof Protocol

80.1 Purpose

RH-Framework uses computation in two mathematically distinct ways. They must not be conflated.

FOUNDATIONAL COMPUTATIONAL CERTIFICATE

machine-checks exact algebra, finite identities, analytic bound formulas,
source-order constraints, dependency hashes, and negative controls;

PROOF-BEARING NUMERICAL CERTIFICATE

proves a mathematical claim after a proved reduction converts it to finite
exact or directed-interval obligations with rigorous tail bounds.

Ordinary floating-point sampling, fitted curves, favourable plots, centre values, or repeated agreement are calibration only. They cannot promote a mathematical claim.

80.2 1. Admissible arithmetic

A proof-bearing computation may use only:

exact integer or rational arithmetic;
exact finite-field or symbolic coefficient arithmetic;
directed interval or ball arithmetic;
validated linear algebra with outward residual bounds;
explicit analytic tail bounds;
immutable source and input hashes.

Nearest-rounding floating point may be used to choose subdivisions or initial centres, but every reported proof margin must be enclosed outward afterward.

80.3 2. Required proof chain

A numerical result counts as proof only through the chain

proved reduction theorem $\rightarrow$ finite computational obligations $\rightarrow$ exact/interval execution $\rightarrow$ tail and residual enclosure
--

Every certificate must identify the reduction theorem it consumes. A numerical packet without a reduction theorem is evidence, not proof.

80.4 3. Certificate classes

Class F — Foundational audit

Used for F00 through the symbolic foundations. It checks:

```
exact multiplication and dagger tables;  
finite coefficient identities;  
proof-bound arithmetic;  
source-order and no-import constraints;  
dependency and source hashes;  
negative controls that must fail.
```

A Class F certificate does not replace the line-by-line universal proof. It confirms that the executable foundation and the quantitative proof obligations match the theorem text.

Class N — Proof-bearing numerical enclosure

Used when a theorem reduces the remaining claim to explicit inequalities, spectral bounds, matrix inertia, or tail estimates. It requires:

```
directed interval arithmetic;  
complete domain subdivision or a proved monotonicity reduction;  
explicit omitted-tail bounds;  
validated solves and residual propagation;  
strict outward margins;  
negative controls;  
reproducible immutable evidence.
```

A passing Class N certificate may promote the reduced mathematical claim.

Class P — Provenance certificate

Hashes exact theorem sources, code, parameters, dependencies and generated artifacts. Class P proves identity and immutability, not mathematical truth.

80.5 4. Mandatory output fields

Every certificate result must contain:

```
schema and version;
certificate class;
theorem or gate ID;
classification before execution;
reduction theorem;
source commit and blob hashes;
input hashes;
parameter packet;
exact/interval arithmetic engine and version;
obligations and individual results;
tail and residual ledgers;
negative-control results;
final status;
scientific claim boundary;
SHA-256 of the canonical result JSON.
```

Canonical JSON uses sorted keys, UTF-8 encoding and no insignificant whitespace.

80.6 5. Promotion rule

A theorem or gate may be promoted by computation only when:

1. the reduction theorem is already PROVED;
2. every required computational obligation is complete;
3. every proof margin is strict after outward rounding;
4. all tails and residuals are included;
5. all realistic negative controls fail as expected;
6. the source and input hashes match immutable pins;
7. an independent reviewer can reproduce the canonical result.

Missing dependencies, unavailable interval engines, hash mismatches, non-strict margins or incomplete tails return INCONCLUSIVE, never PASS.

80.7 6. Current certificate order

```
C00    scalar genesis foundational certificate;
C00E   Euler exponential and flow foundational certificate;
C00GI  logarithm, prime arithmetic and half-plane zeta certificate;
C06-08 native operator foundation certificate;
C09-10 Fourier, Gamma and xi completion certificate;
C11    explicit formula and native Weil criterion certificate;
C16N   active-band directed interval proof certificate;
CENDN  eta-zero endpoint outward certificate.
```

The first implementation package is F00_F00E_EULER_V0_1.

80.8 7. Claim boundary

Cryptographic hashes prove provenance. Exact and interval computations prove only the obligations covered by their reduction theorem and declared domain. No certificate in this protocol presently proves the sign of the completed-Weil operator, $K_0 \geq 0$, or the Riemann Hypothesis.

`RKF/certificates/foundation/NUMERICAL_PROOF_PROTOCOL.md`

Chapter 81

RH-Framework Native Verification Campaign

81.1 Decision

New foundational theorem development is frozen while the existing native chain is audited in dependency order.

A theorem is not declared computationally verified merely because it has a script. A package advances only after:

```
obligations frozen;  
authoritative sources pinned;  
exact or outward execution passes;  
negative controls fail as intended;  
result artifact is archived;  
ledger evidence field is updated without changing the theorem claim boundary.
```

81.2 Current campaign state

Package	Scope	State
F00_F00E_EULER_VO_1	cut scalar and Euler derivation	IMPLEMENTED / RUN_PENDING
F00GHI_LOG_ARITHMETIC_ZETA_VO_1	logarithm, prime arithmetic and half-plane zeta	IMPLEMENTED / RUN_PENDING
F06_08_OPERATOR_FOUNDATION_VO_1	operator algebra, Riesz and compactness	DESIGNED
F09_10_FOURIER_GAMMA_XI_VO_1	Fourier, Poisson, Gamma and xi	DESIGNED
F11_EXPLICIT_WEIL_VO_1	zero geometry, explicit formula and Weil criterion	DESIGNED
F12_15_WEIL_OPERATOR_REDUCTION_VO_1	completed-Weil and spectral reduction	DESIGNED
F16_ACTIVE_BAND_OUTWARD_VO_1	active-band source positivity	DESIGNED
ENDPOINT_KO_OUTWARD_VO_1	eta-zero endpoint	DESIGNED

The master runner `certificates/foundation/run_campaign.py` executes every implemented package containing `verify.py`. Missing planned packages are never counted as passes.

81.3 Verification classes

Class	Meaning	Admissible engines
F-Exact	Exact foundational identities and finite proof-bound arithmetic	integers, fractions, symbolic coefficients
F-Interval	Analytic identities reduced to compact interval and tail obligations	Arb/ball arithmetic, exact tails
O-Residual	Operator claims reduced to validated residual, graph-domain and finite-aperture obligations	interval linear algebra, exact Gram identities
N-Proof	Proof-bearing numerical theorem after a proved reduction	directed intervals, immutable tails, validated inertia
P-Policy	Provenance, source order, no-import and negative-control audit	hashes, static dependency checks

81.4 Campaign order

C00/C00-E — Scalar genesis and Euler

Package: F00_F00E_EULER_V0_1

Class: F-Exact + P-Policy

Current state: IMPLEMENTED / RUN_PENDING.

C00-G/H/I — Logarithm, arithmetic and half-plane zeta

Package: F00GHI_LOG_ARITHMETIC_ZETA_V0_1

Class: F-Exact + rational F-Interval + P-Policy

Current state: IMPLEMENTED / RUN_PENDING.

Implemented obligations:

```
positive logarithm series and exact rational tail bounds;
Exp_Sigma(Log_Sigma x)=x on outward rational intervals;
Log_Sigma(xy)=Log_Sigma x+Log_Sigma y packets;
exact derivative and native-power consistency packets;
Euclidean division, gcd and Bezout identities;
Euclid lemma and prime factor reconstruction;
Mobius cancellation and formal-prime-log Mangoldt identity;
Euclid prime-prefix construction;
exact dyadic zeta convergence bounds;
finite Euler products plus rigorous omitted tails;
```

product-tail inequality supporting nonvanishing;
Mobius reciprocal and logarithmic-derivative coefficient identities;
adversarial negative controls.

C06/C07/C08 — Native operator foundation

Planned package: F06_08_OPERATOR_FOUNDATION_V0_1

Obligations:

adjoint identities on exact finite path cores;
energy bounds and nondegeneracy;
Riesz reconstruction residuals on exhaustive finite apertures;
form/operator sign equivalence;
compact aperture tail estimates;
negative controls for lost residue, wrong dagger and nonfaithful quotient.

C09/C10 — Fourier, Poisson, Gamma and xi completion

Planned package: F09_10_FOURIER_GAMMA_XI_V0_1

Class: F-Interval + P-Policy

Obligations:

Gaussian integral enclosure;
Gaussian Fourier self-duality;
approximate-identity error bounds;
Poisson finite window plus explicit tails;
theta inversion enclosure;
Gamma recursion and reciprocal-product tails;
Mellin continuation identity on overlapping domains;
xi functional-equation residual enclosure;
negative controls for wrong pi, Fourier sign and omitted Gamma factor.

C11 — Cauchy, zero geometry, explicit formula and Weil

Planned package: F11_EXPLICIT_WEIL_V0_1

Class: F-Interval + 0-Residual + P-Policy

Obligations:

contour discretization and residue balance with outward error;
Jensen zero-count bounds;
canonical-product tail bounds;
prime-Gamma-zero explicit-formula residual;
on-seam positivity finite packets;
off-seam constructive negative witness with rigorous tail domination;
negative controls for missing residues, unsymmetrized zeros and sign reversal.

This is an audit-critical package and cannot be reduced to random evaluations of zeta zeros.

C12/C13/C14/C15 — Completed-Weil and spectral reduction

Planned package: F12_15_WEIL_OPERATOR_REDUCTION_V0_1

Class: 0-Residual + F-Interval

Obligations:

completed-Weil form boundedness budgets;
 Riesz representative reconstruction;
 cut covariance and block reconstruction;
 finite-aperture compactness;
 positive square-root residuals;
 Feshbach congruence identities;
 Birman-Schwinger inertia agreement;
 target-faithful kernel identity;
 negative controls for lost blind residue and wrong Schur sign.

C16-N — Active-band source positivity

Planned package: F16_ACTIVE_BAND_OUTWARD_V0_1

Class: N-Proof

This package is proof-bearing. It must prove the actual interval inequalities, complete prime tails, Gamma derivative bounds, source observability and virial error budget.

CEND-N — Eta-zero endpoint

Planned package: ENDPOINT_K0_OUTWARD_V0_1

Class: N-Proof

This package must validate source regularity, inverse-half domains, norm descent, the exact five-label matrix and a lawful outward certificate for or against $K0 \geq 0$.

81.5 Master status meanings

DESIGNED	obligations and package layout exist;
IMPLEMENTED	deterministic verifier exists;
RUN_PENDING	authoritative execution not archived;
PASS	all mandatory obligations and negative controls pass;
INCONCLUSIVE	at least one obligation is unproved or enclosure overlaps zero;
FAIL	a claimed identity or required margin is rigorously contradicted;
STALE	source pins no longer match.

81.6 Scientific boundary

A PASS foundational certificate strengthens confidence and catches implementation/proof-bound errors. It does not replace universal reasoning that was never reduced to the checked obligations.

A PASS proof-bearing numerical certificate can promote a theorem only when its reduction theorem, domain coverage, tails and residuals are all part of the certified package.

RKF/certificates/foundation/VERIFICATION_CAMPAIGN.md

Part V

Completed Weil Endpoint Paper

Chapter 82

Certificate Table

This file mirrors the certificate table now included in `main.tex`.

Node	Evidence	Claim used in paper
Positive-eta path equality	finite rectangle at 0.010 -> 0.011 plus exact polar factorization	no positive-eta normalization holonomy
Zero-as-cut	cut/joint projector algebra; finite critical memory witness	eta zero is a cut/support event
Common five-matrix	covariance theorem and threshold holonomy equivalence	seam integer is a threshold count
M3 accepted block	$3.08520376681449\text{e-}13 < \text{M3}$ $< 2.993597611951253\text{e-}12$	accepted odd block positive
T03 endpoint	six-block Gram decomposition with vanishing total error	invariant source-Gram identity and $K0 \geq 0$
T19-T20 appendix	$\text{beta_partial}^{\text{cut}} \leq$ $0.9998319617060448 < 1$ and source injectivity	independent strict odd certificate

82.1 Permanent records

Reviewer reproducibility archive DOI: 10.5281/zenodo.21760119

Unified publication record DOI: 10.5281/zenodo.21735284

RKF/papers/rkf_completed_weil_endpoint/CERTIFICATE_TABLE.md

Chapter 83

Claim Boundary for the Current Paper

83.1 Cleared internal objections

path / normalization ambiguity	cleared by positive-eta amplitude factorization
eta-zero endpoint handling	clarified by zero-as-cut projector discipline
seam integer meaning	cleared by common five-matrix threshold theorem
M3 missing	cleared by certified positive interval
one-sector sufficiency	formal only after seam/source/even reduction
endpoint source map orientation	cleared by invariant Gram identity / T03
framework over-engineering	answered by source-restriction and blind-kernel minimality
odd-sector numerical burden	recorded independently in T18--T20 appendix

83.2 Main chain now used

fixed-five positive-eta path equality
+ zero-as-cut support discipline
+ common five-matrix seam integer
+ $M3 > 0$ accepted block
+ one-sector terminal reduction
+ zero-cut T03 endpoint projection defect
= completed-Weil endpoint sign

83.3 Classical membrane

The paper distinguishes native endpoint closure from the classical terminal identification:

completed explicit formula
Weil criterion

These are consumed only in the declared normalization. The paper does not claim to rederive the classical explicit formula or Weil criterion from scratch.

83.4 Removed route

The revised paper does not depend on the earlier fragile route:

$S_{(0,+)} \geq 0$ as an independent source theorem
+ odd burden as the only endpoint closure

That route is not part of the cleaned proof spine.

83.5 Numerical certificates visible in the paper

M3:
 $3.08520376681449e-13 < M3 < 2.993597611951253e-12$

T19:
 $\text{beta_partial}^{\wedge}\text{cut} \leq 0.9998319617060448 < 1$

T20:
 $\text{strict reserve} \geq 0.0001680382939552$

83.6 Reproducibility records

Reviewer reproducibility archive DOI: 10.5281/zenodo.21760119
Unified publication record DOI: 10.5281/zenodo.21735284

83.7 Location of evidence

theorem/ terminal theorem capsules
mp_gold/ reusable proof machinery and governance
papers/rkf_completed_weil_endpoint/main.tex reviewer-facing paper draft
papers/rkf_completed_weil_endpoint/REVIEWER_GUIDE.md

83.8 Still explicitly not claimed

raw UGD alone proves RH	not claimed
prime torus alone proves RH	not claimed
diagonal Euler determinant proves RH	not claimed
even sector is absent	not claimed
independent b3 boxes prove M3	rejected
uncited numerical decimal proves endpoint	rejected

RKF/papers/rkf_completed_weil_endpoint/CLAIM_BOUNDARY.md

Chapter 84

Recognition Kernel Completion of the Five-Label Weil Endpoint

This is the current reviewer-facing paper draft after moving certified MP theorem nodes into RKF and expanding the proof skeletons, numerical certificates and reproducibility boundary.

84.1 Main source

`papers/rkf_completed_weil_endpoint/main.tex`

84.2 Reviewer guide

`papers/rkf_completed_weil_endpoint/REVIEWER_GUIDE.md`

84.3 Corrected proof chain

```
positive-eta amplitude path equality
-> zero-as-cut endpoint discipline
-> common five-matrix seam integer
-> certified M3 positive accepted block
-> one-sector terminal reduction
-> zero-cut T03 projection defect
-> classical explicit-formula / Weil-criterion membrane
```

The older fragile route through an independent even-source positivity assertion is not used.

84.4 Main theorem nodes now expanded

```
Classical normalization contract
Framework minimality / source restriction and minimal lift
Fixed-five positive-eta path equality
Zero-as-cut projector theorem
Common five-matrix covariance
Certified M3 accepted block
Formal single-sector terminal sufficiency
Six-block T03 source-Gram convergence
Zero-cut projection-defect positivity
```

Independent T18--T20 odd burden appendix
Auxiliary certified theorem-node appendix

84.5 Numerical values now visible in the paper

M3 interval:
 $3.08520376681449\text{e-}13 < M3 < 2.993597611951253\text{e-}12$

T19 burden:
 $\text{beta_partial}^{\text{cut}} \leq 0.9998319617060448 < 1$

T20 strict reserve:
 $1 - \text{beta_partial}^{\text{cut}} \geq 0.0001680382939552$

84.6 Repository dependencies

Terminal theorem capsules:

theorem/

Reusable proof machinery:

mp_gold/

84.7 Reproducibility records

Reviewer reproducibility archive DOI: 10.5281/zenodo.21760119
Unified publication record DOI: 10.5281/zenodo.21735284

84.8 Compilation

The source uses standard LaTeX packages:

pdflatex main.tex
pdflatex main.tex

No bibliography file is required for this draft because theorem provenance is recorded through local RKF theorem folders and the reviewer reproducibility archive.

84.9 Claim discipline

This paper draft records the RKF-native endpoint theorem chain and clearly marks the classical terminal membrane:

classical completed explicit formula
+ classical Weil criterion in the declared normalization

It does not claim that raw UGD, the prime torus, or the diagonal Euler determinant alone proves RH. It does not claim that the even sector is absent. It does not use independent b3 boxes to prove M3.

RKF/papers/rkf_completed_weil_endpoint/README.md

Chapter 85

Reviewer Guide: RKF Completed-Weil Endpoint Paper

This guide explains how the upgraded paper is meant to be read.

85.1 Main proof chain

```
positive-eta amplitude path equality
-> zero-as-cut endpoint discipline
-> common five-matrix seam integer
-> certified M3 positive accepted block
-> one-sector terminal reduction
-> zero-cut T03 projection defect
-> classical explicit-formula / Weil-criterion membrane
```

The older fragile route through a separate even-source positivity assertion is not used.

85.2 What is native and what is classical

Native RKF endpoint chain:

```
fixed five-label path equality
source/blind-kernel minimality
zero-cut support discipline
seam integer/common five-matrix
M3 accepted block
T03 source-Gram identity
K0 >= 0
```

Classical terminal membrane:

```
completed explicit formula
+ Weil criterion in the declared normalization
```

The paper does not claim to rederive the classical explicit formula or Weil criterion from scratch.

85.3 Numerical certificate locations

The main numerical values now appear in the paper itself:

```
M3 interval:
  3.08520376681449e-13 < M3 < 2.993597611951253e-12
```

```
T19 odd burden:
  beta_partial^cut <= 0.9998319617060448 < 1
```

```
T19 strict reserve:
  1 - beta_partial^cut >= 0.0001680382939552
```

The paper points to:

```
theorem/
mp_gold/
```

for the clean theorem capsules and supporting machinery.

85.4 Reproducibility records

```
Reviewer reproducibility archive DOI: 10.5281/zenodo.21760119
Unified publication record DOI:      10.5281/zenodo.21735284
```

85.5 What to check first

1. Section 2: classical normalization contract.
2. Section 4: positive-eta amplitude path equality.
3. Section 5: zero as cut.
4. Section 6: common five-matrix theorem.
5. Section 7: M3 accepted-block certificate.
6. Section 9: T03 six-block convergence and projection defect.
7. Appendix A: independent T18–T20 odd burden certificate.

85.6 What the appendices are doing

The appendices are not filler. They record strong supporting theorem nodes that do not belong in the main proof spine:

```
T18--T20 odd burden certificate
channel-Hermite no-escape
channel compatibility projection
curvature-to-seam-index law
unified seam block-Krylov machinery
```

They are the substitutes on the bench, not random spectators.

RKF/papers/rkf_completed_weil_endpoint/REVIEWER_GUIDE.md

Chapter 86

main

The following paper is included as compiled from its repository source.

Recognition Kernel Completion of the Five-Label Weil Endpoint

Monty Dabas

August 2026

Abstract

This draft gives the reviewer-facing version of the Recognition Kernel Framework route to the five-label completed-Weil endpoint. The previous proof-lab presentation contained several overlapping paths; this version reorganizes them into one dependency chain and states the secondary certificates in appendices. The native chain is

positive-eta amplitude path equality $\rightarrow$ zero-as-cut endpoint discipline $\rightarrow$ common five-matrix seam integer $\rightarrow M_3$

The endpoint inequality is the zero-cut projection-defect statement

$$K_0 = R_\Sigma^*(I - P_{\text{src}})R_\Sigma \geq 0.$$

The paper separates this native endpoint from the classical membrane: the completed explicit formula and the Weil criterion are consumed only in the declared normalization. The framework is used as a typed anti-blindness ledger. Source restriction, blind kernels, normalization path ambiguity, representation dependence and zero-cut support loss are not metaphors; each is represented by a theorem node. The numerical and proof-lab evidence is summarized in certificate tables with repository pointers and Zenodo records.

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1 Introduction

The purpose of this paper is to present the cleaned endpoint route after the MP proof-lab results were transferred into the Recognition Kernel Framework repository. The central correction is architectural. The proof no longer proceeds through a separate even-source positivity assertion. Instead, the endpoint is reached through the fixed five-label recognition carrier, the seam integer, the certified accepted block, the single-sector reduction and the zero-cut T03 completion.

The paper is written with a first-time reviewer in mind. The main theorem nodes are stated in the body. Proof-lab machinery that is useful but not terminal appears in appendices. Repository capsules are used as provenance, not as a substitute for spelling out the logical chain. The proof is still native in the framework sense: the classical explicit formula and the Weil criterion appear only at the final membrane.

1.1 The five main theorem nodes

The endpoint chain consumes the following nodes.

- (T1) **Positive-eta amplitude path equality.** Normalization and eta continuation commute on the fixed five-label carrier for every positive eta.
- (T2) **Zero-as-cut discipline.** Eta zero is treated as a support/cut event, not as an ordinary substitution into a positive-eta formula.
- (T3) **Common five-matrix seam reduction.** The sign-sensitive obstruction is represented by one rank-at-most-five matrix whose threshold count is the seam integer.
- (T4) **Accepted block positivity.** The transferred certificate gives $G_3 > 0$ and $M_3 > 0$, hence positivity of the accepted odd block.
- (T5) **Zero-cut T03 endpoint.** The invariant source-Gram identity gives the projection defect $K_0 \geq 0$.

The appendix records the independent T18–T20 odd boundary-burden certificate. That certificate is important, but the body uses the T03 source-Gram route as the main endpoint closure.

2 Classical membrane and normalization contract

The paper separates the native construction from the classical terminal membrane.

Definition 2.1 (Test core and Fourier convention). *Let $\mathcal{D} \subset C_c^\infty(\mathbb{R})$ be the logarithmic test core used for the completed-Weil form. For $f \in \mathcal{D}$, write*

$$\widehat{f}(z) = \int_{\mathbb{R}} f(x) e^{izx} dx,$$

with the understanding that the opposite Fourier sign conjugates the symmetric zero pairing and does not change the diagonal Hermitian sign once the convention is fixed throughout.

Definition 2.2 (Classical Weil membrane). *For $f, h \in \mathcal{D}$, define the classical zero-side Hermitian form*

$$Q_W(f, h) = \sum_{\rho}^{\text{sym}} \widehat{f}(z_{\rho}) \overline{\widehat{h}(z_{\rho})}, \quad z_{\rho} = \frac{\rho - \frac{1}{2}}{i},$$

where the summation is the symmetric completed-zero summation of the classical explicit formula.

Assumption 2.3 (Classical membrane in the declared normalization). *The native boundary-Gamma-prime completed-Weil form W_{ξ} constructed on the recognition carrier is identified by the completed explicit formula with Q_W in the convention above. The classical Weil criterion is consumed in the same normalization:*

$$Q_W(f, f) \geq 0 \quad (f \in \mathcal{D}) \quad \Longleftrightarrow \quad \text{RH.}$$

Remark 2.4. *The native part of the paper constructs and closes the endpoint sign of the completed-Weil form. The final implication to the Riemann hypothesis is exactly the classical membrane above. The paper does not claim that a primitive diagonal Euler determinant, the prime torus alone, or raw UGD holonomy independently replaces this membrane.*

3 Why the framework is not over-engineering

The Recognition Kernel Framework is introduced because ordinary operator notation can lose target-relevant information at the exact places where the endpoint proof must be stable. Table 1 records the obstruction ledger.

Failure mode	Framework repair
Source restriction	source-restriction rank formula and blind-kernel formula
Target-relevant blind kernel	minimal scalar lift or target-relative decoder
Path or normalization ambiguity	positive-eta amplitude path equality
Representation dependence	common five-matrix covariance
Accepted-block uncertainty	certified $G_3 > 0$ and $M_3 > 0$
Endpoint support loss	zero-cut T03 completion

Table 1: Each framework layer repairs a named obstruction.

Theorem 3.1 (Source restriction and minimal lift). *Let $S(q)P$ be a rich split representation and let B be a consumed source restriction. Put*

$$A(q) = BS(q)P, \quad S(q) = \sum_s q^s D_s.$$

Then the all-parameter observable rank is

$$\text{rank} \begin{pmatrix} BD_1 P \\ BD_2 P \\ \vdots \end{pmatrix},$$

and the blind kernel is

$$\bigcap_s \text{Ker}(BD_s P).$$

If scalar supplemental observations G are added, the augmented map is injective exactly when

$$\text{Ker } A \cap \text{Ker } G = \{0\}.$$

Consequently at least $\dim \text{Ker } A$ scalar observations are necessary to repair a full blind defect. Target-relative repair requires only that the remaining blind kernel lie in the target kernel.

Proof. The rank formula is the rank of the direct sum of all coefficient-level source restrictions. A vector is blind for the entire rich representation exactly when it is annihilated by every restricted coefficient map $BD_s P$, giving the displayed intersection. Adding a scalar observation map G removes the blind defect precisely when no nonzero vector is invisible to both the original restricted map and the supplement. This gives the injectivity criterion and the lower bound on the number of independent scalar repairs. $\square$

Remark 3.2 (Endpoint meaning). *In the endpoint application, the source restriction is the passage to the target-relevant source readout, the blind kernel is the endpoint information not seen by that readout, and the five-label lift supplies the minimal repair. The theorem is therefore not decoration; it is the algebraic reason that a blind quotient and lift must be declared before endpoint promotion.*

4 Positive-eta amplitude analyticity

The positive-eta analytic theorem is one of the load-bearing results. It removes path-order ambiguity before the sign problem begins.

Definition 4.1 (Polar frame transport). *For a full-column-rank amplitude $\Theta = QH$, where $Q^*Q = I_5$ and $H > 0$, define the canonical transport between two polar frames Q_a, Q_b by*

$$U_{a \rightarrow b} = \text{polar}(Q_b^* Q_a).$$

Theorem 4.2 (Fixed-five positive-eta path equality). *Let*

$$A_{\text{chart}} = AU, \quad \Theta_\eta = Q_{\text{chart}} G_\eta^{1/2}, \quad G_\eta = 1 \oplus (A_+ + \eta I_4), \quad \eta > 0.$$

Here A is the eta-independent raw amplitude, U is the fixed unitary accepted chart, Q_{chart} is the polar orientation of A_{chart} , and $G_\eta^{1/2}$ is strictly positive. Then

$$\text{polar}(\Theta_\eta) = Q_{\text{chart}}$$

for every positive eta. Hence, for any positive eta values η_1, η_2 ,

$$\text{raw eta transport} = I_5, \quad \text{normalized eta transport} = I_5, \quad \text{raw-to-normalized transport} = U^*.$$

All positive-eta normalization/continuation rectangles are identity loops on the fixed five-label carrier.

Proof. Right multiplication by a unitary is polar-covariant, so $\text{polar}(AU) = \text{polar}(A)U$. Right multiplication by a strictly positive factor changes the metric scale but not the polar orientation. Since $G_\eta^{1/2} > 0$, the polar orientation of $Q_{\text{chart}}G_\eta^{1/2}$ is Q_{chart} . Therefore the polar frame at every normalized positive-eta node is the same frame. The raw eta transport and normalized eta transport are identity maps on the label fibre, while the raw-to-normalized transport is the fixed chart unitary U^* . The two possible paths around any positive-eta normalization rectangle have the same product. $\square$

Remark 4.3 (Finite rectangle shadow). *The state-locked rectangle at $\eta = 0.010$ and $\eta + h = 0.011$ reported path transport difference $3.0669860554696173 \cdot 10^{-15}$, loop identity residual $3.1396725820594018 \cdot 10^{-15}$, defect-to-accepted leakage $1.7140530928543942 \cdot 10^{-17}$, endpoint scale difference zero, and no false checks. These numbers are the floating-point shadow of the exact factorization above.*

5 Zero as cut

The previous theorem says there is no positive-eta normalization holonomy. The endpoint issue is therefore the support event at eta zero. This is the role of the zero-as-cut theorem.

Definition 5.1 (Cut and joint cycles). *The pre-cut carrier has one geometric cycle γ_π . The post-cut carrier has an independent cut-memory cycle:*

$$H_1(S^1; \mathbb{Z}) = \mathbb{Z}\gamma_\pi, \quad H_1(T^2; \mathbb{Z}) = \mathbb{Z}\gamma_\pi \oplus \mathbb{Z}\mu_{\text{cut}}.$$

On the ordered basis $(\gamma_\pi, \mu_{\text{cut}})$, set

$$P_{\text{joint}} = \text{diag}(1, 0), \quad P_{\text{cut}} = \text{diag}(0, 1).$$

Theorem 5.2 (Zero-as-cut projector theorem). *The cut and joint projectors satisfy*

$$P_{\text{cut}}P_{\text{joint}} = P_{\text{joint}}P_{\text{cut}} = 0, \quad P_{\text{cut}} + P_{\text{joint}} = I.$$

Hence no nonzero vector can be both a pure cut state and a complete joint state.

Proof. The identities follow immediately from the displayed diagonal matrices. If $u = P_{\text{cut}}u = P_{\text{joint}}u$, then

$$u = P_{\text{cut}}P_{\text{joint}}u = 0.$$

Thus simultaneous pure cut recognition and complete joint recognition force vanishing. $\square$

Remark 5.3 (Visible return is not full return). *For the join projection $J = (1, 0)$ and section $S = (1, 0)^T$, one has $JS = I_1$, but*

$$SJ = P_{\text{joint}} \neq I_2.$$

Thus the visible circle coordinate can return while the full post-cut torus state does not. The exact memory defect is

$$I_2 - SJ = P_{\text{cut}}.$$

This is the reason the endpoint is handled by a zero-cut source-Gram theorem rather than by an ordinary positive-eta continuation.

Remark 5.4 (Finite critical witness). *The finite critical response packet recorded a surviving cut-memory coordinate with joint and cut-memory component norms both $4.137400474477719 \cdot 10^{-5}$, relative cut memory 0.7071067811865472 , and complete state return false. This is a finite witness for the warning above: corrected target visibility does not automatically give full state return.*

6 Seam integer and the common five-matrix

The sign-sensitive obstruction is represented by one rank-at-most-five matrix. This section turns the seam language into a standard threshold-count statement.

Definition 6.1 (Seam charge). *Let K_η be the compact seam response at positive shift eta. Define*

$$k_\Sigma(\eta) = \text{rank } \mathbf{1}_{(1,\infty)}(K_\eta).$$

On the five-label source packet this is represented by

$$B_\eta = V_\eta^* S_{\eta,-}^{-1} V_\eta, \quad \text{rank } V_\eta \leq 5,$$

and

$$k_\Sigma(\eta) = N_+(B_\eta - I).$$

Theorem 6.2 (Common five-matrix covariance). *For a positive source operator S and a five-column source map V , set*

$$B = V^* S^{-1} V \in M_5(\mathbb{C}).$$

If $S^U = U S U^$ and $V^U = U V$, then*

$$(V^U)^* (S^U)^{-1} V^U = B.$$

A unitary change of the five labels conjugates B . Therefore the spectrum, threshold projection, seam charge and sign decision are representation invariant.

Proof. Since $(U S U^*)^{-1} = U S^{-1} U^*$,

$$(V^U)^* (S^U)^{-1} V^U = V^* U^* U S^{-1} U^* U V = V^* S^{-1} V.$$

If the label basis is changed by a unitary U_5 , then V is replaced by $V U_5$, and the matrix becomes $U_5^* B U_5$. Hence threshold ranks and spectral projections are invariant. $\square$

Corollary 6.3 (Matrix, seam and threshold holonomy). *The native matrix, prime-torus matrix and threshold-holonomy presentation are faithful avatars of the same obstruction:*

$$B_\eta \leq I \iff k_\Sigma(\eta) = 0 \iff H_{\text{threshold}}(\eta) = I,$$

where

$$H_{\text{threshold}}(\eta) = I - 2\mathbf{1}_{(1,\infty)}(B_\eta).$$

Thus the prime torus is a chart of the same matrix, not a second independent proof route.

7 The accepted block and M_3

The finite accepted odd block is

$$E_3 = \text{span}\{u, |x|u, |x|^2 u, e_3\}.$$

Let G_3 be the completed-Weil Gram on $V_3 = \text{span}\{u, |x|u, |x|^2 u\}$, and let b_3 be the coupling vector from V_3 to e_3 . Define

$$M_3 = q(e_3, e_3) - b_3^* G_3^{-1} b_3.$$

Theorem 7.1 (Certified accepted block). *The transferred M_3 certificate gives*

$$\lambda_{\min}(G_3) = 2.0838177120437723 \cdot 10^{-13} > 0$$

and

$$3.08520376681449 \cdot 10^{-13} < M_3 < 2.993597611951253 \cdot 10^{-12}.$$

Therefore $M_3 > 0$. Since M_3 is the Schur complement of G_3 in the bordered Gram, the completed-Weil form is positive definite on E_3 .

Proof. The block Gram on $E_3 = V_3 \oplus \mathbb{C}e_3$ has the bordered form

$$\begin{pmatrix} G_3 & b_3 \\ b_3^* & q(e_3, e_3) \end{pmatrix}.$$

For a Hermitian bordered matrix with $G_3 > 0$, positivity is equivalent to positivity of the Schur complement

$$q(e_3, e_3) - b_3^* G_3^{-1} b_3 = M_3.$$

The certificate gives both $G_3 > 0$ and $M_3 > 0$, so the accepted block is positive definite. The certificate is correlated: the self channel, coupling vector, Schur compensation and prime/Gamma truncation are enclosed together; independent coordinate boxes for b_3 are not used. $\square$

8 Single-sector terminal reduction

Both parity sectors remain represented in the bilateral carrier. The single-sector statement is a later terminal reduction, not a deletion of the even sector.

Assumption 8.1 (Reduction inputs). *The following transferred inputs are consumed.*

- (i) *The representation-complete bilateral carrier preserves the completed-Weil target.*
- (ii) *The seam-integer/common five-matrix theorem identifies the sign-sensitive terminal obstruction with the five-label threshold packet.*
- (iii) *The common source-sign theorem and favorable even-boundary reduction remove the even sector as an independent terminal gate.*
- (iv) *The odd five-label packet is the remaining target-relevant endpoint obstruction.*

Theorem 8.2 (One-sector terminal sufficiency). *Under the reduction inputs, the completed endpoint sign is equivalent to the nonnegativity of the odd five-label endpoint packet. The even sector is retained upstream but is not an independent terminal gate after the reduction.*

Proof. By representation completeness, no target-relevant completed-Weil information is lost by passing to the bilateral carrier. The seam-integer theorem expresses the sign-sensitive obstruction as the threshold rank of the common five-matrix. The source-sign and favorable even-boundary reduction place the even contribution in the accepted/non-terminal part of the carrier. Therefore a nonnegative endpoint fails only if the odd five-label threshold packet has a positive threshold count. Conversely, if that packet is nonnegative, the common five-matrix has no super-threshold direction and the endpoint obstruction vanishes. Thus the odd five-label packet is terminal after, and only after, these reductions. $\square$

9 Zero-cut T03 endpoint completion

The T03 theorem is the main endpoint route used in the paper. It replaces orientation-dependent ambient equality by an invariant source-Gram identity.

Definition 9.1 (Endpoint source-Gram data). *Let R_Σ be the total five-label response readout, P_{src} the source projection, $S_{0,-}^{\text{full}}$ the complete unshifted odd source, and W_Σ the five-label response map. The endpoint source-Gram target is*

$$R_\Sigma^* P_{\text{src}} R_\Sigma = ((S_{0,-}^{\text{full}})^{\dagger/2} W_\Sigma)^* ((S_{0,-}^{\text{full}})^{\dagger/2} W_\Sigma).$$

Theorem 9.2 (Six-block convergence criterion). *Suppose finite transported source maps give finite Gram defects Δ_n satisfying*

$$\|\Delta_n\| \leq \varepsilon_{\text{vis}}(n) + \varepsilon_{\text{blind}}(n) + \varepsilon_{\text{cross}}(n) + \varepsilon_{\text{tail}}(n) + \varepsilon_{\text{quad}}(n) + \varepsilon_{\text{chart}}(n),$$

where the six terms respectively bound visible compression error, five-dimensional blind response error, both cross blocks, omitted prime/Gamma tails, quadrature/window/solve/rounding errors, and native/Feshbach/prime-torus chart transport. If

$$\varepsilon_{\text{vis}}(n) + \varepsilon_{\text{blind}}(n) + \varepsilon_{\text{cross}}(n) + \varepsilon_{\text{tail}}(n) + \varepsilon_{\text{quad}}(n) + \varepsilon_{\text{chart}}(n) \longrightarrow 0,$$

then the invariant source-Gram identity holds.

Proof. Each finite transported source map produces a finite Gram comparison between the total five-label response readout and the source-projected response. The six listed errors form an outward norm bound on the full Gram defect. Since the total bound tends to zero, $\Delta_n \rightarrow 0$ in operator norm on the finite label space. Passing to the limit gives equality of the two limiting Gram matrices. Equal Gram matrices determine a canonical partial isometry between final ranges, so literal equality of ambient source maps is unnecessary. $\square$

Theorem 9.3 (Zero-cut projection-defect positivity). *Assume the invariant source-Gram identity and the independently constructed total-response Gram*

$$R_\Sigma^* R_\Sigma = \text{diag}(1, A_3).$$

Then

$$K_0 = R_\Sigma^* (I - P_{\text{src}}) R_\Sigma \geq 0.$$

The endpoint Schur equivalence transfers this projection-defect positivity to the completed-Weil endpoint sign in the declared normalization.

Proof. Since P_{src} is an orthogonal projection, $I - P_{\text{src}} \geq 0$. Hence for any label vector $c \in \mathbb{C}^5$,

$$\langle c, K_0 c \rangle = \langle R_\Sigma c, (I - P_{\text{src}}) R_\Sigma c \rangle \geq 0.$$

The invariant source-Gram identity identifies the source-visible part of the endpoint response with the complete unshifted source readout. The remaining projection defect is exactly the zero-cut endpoint obstruction. The Schur endpoint equivalence therefore yields the completed-Weil endpoint sign. $\square$

10 Main theorem

Theorem 10.1 (Completed-Weil endpoint theorem). *Assume the native completed-Weil carrier is in the declared normalization and consume Theorems 4.2, 5.2, 6.2, 7.1, 8.2 and 9.3. Then the completed-Weil endpoint sign holds on the target-faithful completed carrier.*

Proof. The positive-eta path theorem removes normalization ambiguity on the fixed five-label carrier. The zero-as-cut theorem identifies eta zero as a support/cut event rather than an ordinary path-holonomy issue. The common five-matrix theorem reduces the sign-sensitive obstruction to the representation-invariant threshold matrix B_η and seam charge k_Σ . The M_3 theorem proves positivity of the accepted odd block. The single-sector theorem identifies the remaining terminal packet with the odd five-label source obstruction after the seam/source/even reduction. Finally, the T03 theorem supplies the invariant source-Gram identity and hence

$$K_0 = R_\Sigma^*(I - P_{\text{src}})R_\Sigma \geq 0.$$

The endpoint Schur equivalence transfers this projection-defect positivity to the completed-Weil endpoint sign. $\square$

Corollary 10.2 (Classical terminal implication). *Under the classical membrane in Section 2, the completed-Weil endpoint theorem implies the Riemann hypothesis.*

Proof. The native theorem gives nonnegativity of the completed-Weil form in the declared normalization. The completed explicit formula identifies this form with the symmetric zero-sum Weil form Q_W . The classical Weil criterion then gives RH. $\square$

11 Numerical certificates and reproducibility

The proof contains certified numerical inputs. The main numerical evidence is summarized in Table 2. The reproducibility archive is recorded as DOI 10.5281/zenodo.21760119; the unified publication record is recorded as DOI 10.5281/zenodo.21735284. The repository theorem capsules are in `theorem/`, and reusable proof machinery is in `mp_gold/`.

Table 2: Certificate table for reviewer entry.

Node	Evidence	Claim used in paper
Positive-eta path equality	finite rectangle at 0.010 $\rightarrow$ 0.011 plus exact polar factorization	no positive-eta normalization holonomy
Zero-as-cut	cut/joint projector algebra; finite critical memory witness	eta zero is a cut/support event
Common five-matrix	covariance theorem and threshold holonomy equivalence	seam integer is a threshold count
M3 accepted block	$3.08520376681449 \cdot 10^{-13} < M_3 < 2.993597611951253 \cdot 10^{-12}$	accepted odd block positive
T03 endpoint	six-block Gram decomposition with vanishing total error	invariant source-Gram identity and $K_0 \geq 0$
T19–T20 appendix	$\beta_{\mathcal{O}}^{\text{cut}} \leq 0.9998319617060448 < 1$ and source injectivity	independent strict odd certificate

Remark 11.1 (Why the decimals are not the theorem). *The numerical values are used only through directed certificates: intervals, outward error bounds, exact algebraic identities, and fail-closed tests. The paper does not use a fitted endpoint extrapolation or independent b_3 boxes to prove M_3 .*

12 Resolution of structural objections

The revised paper answers the main reviewer objections as follows.

- Normalization and eta-continuation commute by positive-eta amplitude factorization.
- Eta zero is not treated as a naive substitution; it is a cut/support event.
- The seam integer is the threshold count of one common five-matrix.
- The prime torus is a faithful chart of that same matrix.
- M_3 is certified positive by a correlated Schur interval.
- One sector is terminal only after the seam/source/even reduction; the even sector is not erased.
- The endpoint source map is handled by an invariant Gram identity rather than orientation-dependent ambient equality.
- The framework is not over-engineering because each layer repairs a named source, blind-kernel, path, representation or endpoint defect.

13 Claim boundary

The paper claims the transferred RKF-native endpoint chain and clearly marks the classical membrane. It does not claim that raw UGD, the prime torus, or a diagonal Euler determinant independently proves the Riemann hypothesis. It does not claim that the even sector is absent. It does not promote numerical evidence unless the corresponding directed certificate is part of the theorem chain.

14 Conclusion

The cleaned architecture is deliberately short:

positive-eta path equality $\rightarrow$ zero-as-cut $\rightarrow$ five-matrix seam integer $\rightarrow M_3 > 0 \rightarrow$ single-sector endpoint $\rightarrow K_1$

The framework is needed because the endpoint is a target-faithful completion problem. Once source restriction, blind kernels, normalization ambiguity, sector terminality and zero-cut support are typed, the terminal object is the zero-cut projection defect. The classical explicit formula and Weil criterion then form the final membrane.

A Independent odd boundary-burden certificate: T18–T20

This appendix records a strong supporting theorem that is not the main T03 route. It gives an independent odd-sector burden certificate.

A.1 Boundary-burden equivalence

T18 identifies the odd completed-Weil sign with the cut boundary burden:

$$\text{full odd sign} \iff \beta_{\partial}^{\text{cut}} \leq 1.$$

Equivalently,

$$S_{0,-}^{\text{full}} - L_{\partial}^{\dagger} L_{\partial} \geq 0$$

once the boundary burden is at most one.

A.2 T19 refined endpoint inequality

For positive cut events with source energies s_j and boundary masses m_j , T19 uses

$$\beta_0 - \beta_{\eta} = \sum_j m_j \frac{\eta}{s_j(s_j + \eta)}.$$

The reconstructed Arb profile uses 2400 active cells at step 0.005, with 78734 prime-power events. At

$$\eta = 1.5625 \cdot 10^{-5}, \quad \beta_{\eta}^{+} = 0.999773667408463,$$

the correction ledger gives

$$\beta_{\partial}^{\text{cut}} \leq 0.9998319617060448 < 1.$$

Thus the full odd completed-Weil sign is closed on the declared native source carrier.

A.3 T20 strict odd positivity

T20 adds source injectivity:

$$\text{Ker } S_{0,-}^{\text{full}} = \{0\}.$$

Combining this with the strict reserve

$$1 - \beta_{\partial}^{\text{cut}} \geq 0.0001680382939552$$

gives strict positivity on every nonzero odd core state:

$$\langle F, W_3^{-} F \rangle > 0, \quad F \neq 0.$$

A.4 Relation to the main route

The T03 route and the T19–T20 route are not competing endings:

$$\begin{array}{ll} \text{T03 route} & \text{invariant source-Gram identity} \Rightarrow K_0 \geq 0, \\ \text{T19–T20 route} & \beta_{\partial}^{\text{cut}} < 1 \Rightarrow \text{strict odd sign.} \end{array}$$

The body keeps T03 as the terminal spine and records T19–T20 as an independent odd-sector certificate.

B Auxiliary certified theorem nodes

B.1 Channel-Hermite no-escape

The X_3 channel-Hermite Parseval frame uses the analysis map

$$C_3 f = \left(e^{3|x|/2} f, (2\pi)^{-1/2} \sqrt{1 + |G|} \hat{f} \right)$$

with

$$\|C_3 f\|^2 = \|f\|_{X_3}^2.$$

It proves parity-resolved global no-escape and projected-frame completeness after accepted-memory removal. This supports response completeness but is not the terminal endpoint theorem.

B.2 Channel compatibility projection

The native analysis range is the closed graph of

$$T = B_0 A_0^{-1}.$$

Raw split-channel packets must be projected to the compatible graph before they update native couplings. The warning certificate is

$$\text{raw channel pairing} = 15/4 > 0, \quad \text{native compatible pairing} = -1/4 < 0.$$

Thus raw channel sign cannot be promoted without compatibility projection.

B.3 Curvature-to-seam index law

The lawful curvature relation is spectral-flow typed:

$$k_\Sigma(\eta; u_1) - k_\Sigma(\eta; u_0) = -(1 - K_\eta(u_s); 0).$$

A scalar thermodynamic ratio is not canonically equal to the integer seam charge. It can influence the seam only through an explicit compact-operator path.

B.4 Unified seam block-Krylov machinery

For a finite-rank seed map S , define

$$V_m = \text{span}\{K_{\Sigma, \eta}^j \text{Ran } S : 0 \leq j \leq m\}.$$

The same-projection two-block upper bound gives

$$\sup \sigma(K) \leq \frac{a + d + \sqrt{(a - d)^2 + 4b^2}}{2}$$

when a, b, d are computed against the same projection. This is future compact-operator machinery, not a substitute for the T03 endpoint.

C Repository provenance

The main theorem capsules live in `theorem/`. Supporting results are summarized in `mp_gold/`. The source PRs are recorded in `theorem/provenance.json` and `mp_gold/gold_manifest.json`. Raw verifier scripts, state capsules and large numerical ledgers remain outside this public paper draft unless explicitly archived.

Part VI

The RH-Framework Ladder

Chapter 87

RH Framework Reviewer Verification Laboratory

This private repository is the authoritative verification laboratory for the Riemann-Hypothesis research programme.

It is separated from:

- Parveen117/MP, which preserves historical development and source lineage;
- publication repositories, which contain manuscript-facing material;
- public AI, ECL, cryptography and assurance repositories, which provide engineering patterns but are not mathematical proof dependencies.

87.1 Governing rule

```
proof statement
-> assumptions
-> foundational-origin label
-> executable witness
-> negative controls
-> immutable certificate
-> reviewer reproduction
-> promotion or rejection
```

Theorem classifications are only:

```
PROVED
DISPROVED
OPEN
```

Foundational origin is recorded separately. A theorem may be proved and still be a classical import.

87.2 Active branch

```
agent/native-summability-reconstruction
```

87.3 Native reconstruction now proved

Native path completion

T01-A	cut-tail path completion	PROVED
T01-B	recognition-Cauchy completion	PROVED
T01-C	completed native path action	PROVED

The decisive native action estimate is

$$\mathcal{E}_\Sigma(a \star \xi) \leq \mathcal{M}_\Sigma(a)^2 \mathcal{E}_\Sigma(\xi).$$

It is derived from a finite weighted cut-square identity before ordinary ℓ^1 , ℓ^2 , Hilbert space, or operator norm is introduced.

PASS_NATIVE_CUT_TAIL_AND_RECOGNITION_CAUCHY_COMPLETION_CORE
4e88898fc93010a6ba189f932817d0715298f106aff61528bb2e58acb8c6dbfd

Native refinement content

T01-E1	reciprocal refinement content	PROVED
T01-E2	complete finite-content algebra	PROVED
T01-E3	native simple integrals	PROVED

Cell content is reciprocal refinement count:

$$\nu_R(c) = \frac{1}{n_s n_t n_\alpha}.$$

Common refinement, finite additivity, symmetric-difference completion, and translation invariance are proved before Haar or Lebesgue measure is named.

PASS_NATIVE_REFINEMENT_COUNT_CONTENT_AND_SIMPLE_INTEGRALS
1366eeb443efc661c614a3bf8e7b95b9500a10d40104a579d3e2ec438c51feb9

Native function completion

T01-E4A	tail-integrable function completion	PROVED
T01-E4B	recognition-integrable completion	PROVED
T01-E4C	bounded native simple multipliers	PROVED

The two native gauges are

$$d_1(f, g) = \mathcal{I}_{\text{tail}}(f - g), \quad d_2(f, g) = \mathcal{I}_{\text{rec}}(f - g).$$

They are completed by Cauchy presentations before ordinary L^1/L^2 notation. Arbitrary tail-completed functions are not falsely declared multiplicatively closed.

PASS_NATIVE_SIMPLE_FUNCTION_TAIL_AND_RECOGNITION_COMPLETION_CORE
ca4d3443e057adbaa0c94a285263d8d136465eaf970c7fef73df70196972cad9

Native mean-multiplier bound

T01-E4D-A	exact mean split of a multiplier action	PROVED
T01-E4D-B	mean + oscillation x deviation bound	PROVED
T01-E4D-C	alternating-product gap (restated §4.1)	OPEN (vacuum half PROVED in Publications ↪ YM-31)

$$\mathcal{I}_{\text{rec}}(hf) \leq \langle N(h) \rangle \mathcal{I}_{\text{rec}}(f) + \text{osc } N(h) \mathcal{D}_{\text{rec}}(f).$$

Sharper than T01-E4C on recognition-flat states (equality with the mean on the vacuum); neither bound dominates in general.

PASS_NATIVE_MEAN_MULTIPLIER_IDENTITY_AND_BOUND
d6f3b39a0186a2c15c3fcd3734cf198ca6bd21b9182ad72906f0930341fe4c99

Rational normalized-UGD refinement adapter

T01-E5A	rational native UGD linearization	PROVED
T01-E5B	rational refinement action	PROVED

A rational normalized packet is

$$N = (p, u, \mu),$$

where p is a full-turn fraction, u is native scale debt, and μ is corrected seam memory. The exact map

$$(p, u, \mu) \mapsto (u, p + \mu, [\mu])$$

linearizes the wrap law and produces exact refinement-cell translations without π , $\log$, $\exp$, Haar measure, or floating point.

PASS_NATIVE_RATIONAL_UGD_REFINEMENT_LATTICE_ADAPTER
932d3e549e3a01d7d42d85e70371dc74fa06c214841bca8f776d19dd70ce4bc7

87.4 Native foundational files

```
src/rh_framework/native_summability.py
src/rh_framework/native_refinement_measure.py
src/rh_framework/native_function_completion.py
src/rh_framework/native_ugd_refinement_adapter.py
```

The verifier rejects approximate numerical identifiers and undeclared external imports in these files.

87.5 Remaining native boundary

T01-E5C	complete the rational UGD group and action
T01-E6	actual positive-scale, log-exp, Haar and Jacobian shadows
T01-D	actual MP path-atlas and refinement adapter
T01-F	remaining classical coordinate identifications
	native path-graph generator domains
	native state-packet generation
	representation invariance of FFT, Gram, spectral and polar charts

The actual positive numerical scale σ has not been declared native. The formal scale-debt ledger must be completed first; $\sigma = e^u$ may appear later only as a proved coordinate chart.

87.6 Classical verification shell

T03-A and T03-C2A remain rigorous, proved classical coordinate theorems. They are not reclassified as native merely because native objects now exist beneath them.

T03-B, T03-C1, and T03-C2B require pinned private MP worktrees and remain open.

87.7 Reproduction

```
python -m pip install -e .  
python tools/verify_native_foundation.py  
python tools/verify_t03.py
```

Jupyter-only procedures are under `notebooks/`.

87.8 Claim boundary

The native abstract foundation has advanced substantially, but the actual MP adapter, completed positive-scale UGD carrier, target faithfulness, Shunya, critical-line equivalence, and the Riemann Hypothesis remain open.

RH-Framework/README.md

Chapter 88

Reviewer Checklist

This repository separates theorem truth, foundational origin, executable evidence, and publication claims. Review them independently. Apparently one label was too convenient for civilization.

88.1 1. Scope

Confirm that the pull request changes only verification-facing material in **RH-Framework**. Historical MP development and manuscript sources must remain outside the diff.

88.2 2. Required local commands

```
python -m pip install -e .
python tools/check_text_integrity.py
python tools/verify_native_foundation.py
python tools/verify_t03.py
```

The native verifier must reproduce all committed T01 certificates byte-for-byte. The T03 verifier must reproduce all source-independent classical-coordinate certificates.

88.3 3. Native-foundation checks

Confirm that foundational native modules:

- use exact rational arithmetic;
- contain no executable `float`, `complex`, NumPy, SciPy, logarithm, exponential, or `pi` dependency;
- derive cut-tail, recognition, refinement, function, and rational UGD structures before ordinary functional-analysis coordinates;
- preserve explicit open boundaries for actual positive scale, MP adapters, state packets, target faithfulness, Shunya, and RH.

88.4 4. Classical-shell checks

Classical coordinate theorems may remain `PROVED`, but their origin must be `CLASSICAL_THEOREM_IMPORT` or `CLASSICAL_NUMERICAL_SOURCE` where applicable. They must not be described as foundational native derivations.

88.5 5. Certificates

For every committed certificate:

- regenerate it from source;
- confirm deterministic canonical hashing;
- inspect all false gates;
- confirm theorem classifications use only `PROVED`, `DISPROVED`, or `OPEN`;
- confirm the certificate does not promote RH.

88.6 6. Private-source gates

T03-B, T03-C1, and T03-C2B require clean pinned MP worktrees. CI does not receive those worktrees, so absence of those results is an explicit open boundary rather than a CI failure.

88.7 7. Merge decision

Merge only when:

```
all required CI jobs pass;  
committed certificates reproduce;  
no theorem classification drifts;  
foundational-origin labels are accurate;  
no private-source claim is promoted without evidence;  
open obligations are listed explicitly.
```

RH-Framework/REVIEWER_CHECKLIST.md

Chapter 89

RH Framework Theorem Ledger

Only `PROVED`, `DISPROVED`, and `OPEN` are theorem classifications. Certificates, numerical passes, source availability, and foundational origin are evidence fields rather than substitute theorem states.

Foundational origin is audited separately in `NATIVE_DERIVATION_CONTAMINATION_AUDIT.md` and the latest immutable manifest in `provenance/`.

89.1 T01 — Native operator algebra and recognition completion

Classification: `OPEN`

The abstract countable completion, refinement-content algebra, native function completions, and rational normalized-UGD refinement action are constructed inside RH-Framework. T01 remains open because the rational native scale-debt carrier is not yet completed and identified with the actual positive UGD scale or the actual MP path atlas.

T01-A — native cut-tail completion

Classification: `PROVED`

For finite typed residue paths over exact cut scalars,

$$\mathcal{M}_{\Sigma}(a) = \sum_{\gamma} (|\text{rad } a_{\gamma}|_{\Sigma} + |\text{turn } a_{\gamma}|_{\Sigma}).$$

Cut-tail Cauchy presentations modulo cut-tail-null presentations form a complete involutive path algebra without assuming ordinary complex numbers, ℓ^1 , Banach space, operator norm, Haar measure, or spectral theory.

T01-B — native recognition-Cauchy completion

Classification: `PROVED`

The closed-loop energy

$$\mathcal{E}_{\Sigma}(a) = \Phi_{\Sigma}(a^{\dagger} \star a) = \sum_{\gamma} a_{\gamma}^{\dagger} a_{\gamma}$$

defines a second Cauchy relation and a complete positive cut module with a positive-definite completed pairing.

T01-C — native completed path action

Classification: PROVED

The exact weighted cut-square identity gives

$$\mathcal{E}_\Sigma(a \star \xi) \leq \mathcal{M}_\Sigma(a)^2 \mathcal{E}_\Sigma(\xi).$$

Thus the cut-tail completion acts continuously on the recognition completion without importing Cauchy-Schwarz, Young, Minkowski, Hilbert space, or an operator norm as primitive theorems.

Proof: T01_NATIVE_SUMMABILITY_AND_RECOGNITION_COMPLETION.md.

PASS_NATIVE_CUT_TAIL_AND_RECOGNITION_CAUCHY_COMPLETION_CORE
4e88898fc93010a6ba189f932817d0715298f106aff61528bb2e58acb8c6dbfd

T01-D — actual MP path-atlas adapter

Classification: OPEN

Prove that actual source, response, prime, Gamma, boundary, continuation, and seam paths embed into the native finite-support core and that actual refinement maps are continuous for $\mathcal{M}_\Sigma$ and $\mathcal{E}_\Sigma$.

T01-E — native refinement and UGD programme

Classification: OPEN

The abstract content, function, and rational UGD refinement theorems are proved. The completed actual positive-scale carrier and its state-locked adapter remain open.

T01-E1 — native refinement-count content

Classification: PROVED

On a grid $R = (n_s, n_t, n_\alpha)$, one cell has content

$$\nu_R(c) = \frac{1}{n_s n_t n_\alpha}.$$

Finite cylinders have reciprocal-cell-count content. Common refinement, finite additivity, Boolean identities, and integer cell-translation invariance are proved before any Haar or Lebesgue measure is introduced.

T01-E2 — complete native content algebra

Classification: PROVED

The gauge

$$\delta(A, B) = \nu(A \triangle B)$$

defines refinement-content Cauchy presentations. Their quotient is a complete finite-content algebra with extended nonnegative, finitely additive, translation-invariant content.

No sigma-additivity theorem for arbitrary completed families is claimed.

T01-E3 — native simple tail and recognition integrals

Classification: PROVED

For finite cut-scalar simple functions,

$$\mathcal{I}_{\text{tail}}(f) = \sum_c \nu_R(c) D_{\Sigma}(f(c)), \quad \mathcal{I}_{\text{rec}}(f) = \sum_c \nu_R(c) N_{\Sigma}(f(c)).$$

Both are invariant under native refinement and integer cell translation.

Proof: T01_E_NATIVE_REFINEMENT_COUNT_MEASURE.md.

PASS_NATIVE_REFINEMENT_COUNT_CONTENT_AND_SIMPLE_INTEGRALS
1366eeb443efc661c614a3bf8e7b95b9500a10d40104a579d3e2ec438c51feb9

T01-E4A — native tail-integrable function completion

Classification: PROVED

Finite simple functions completed under

$$d_1(f, g) = \mathcal{I}_{\text{tail}}(f - g)$$

form a complete native cut module. Ordinary L^1 is not used as definition.

T01-E4B — native recognition-integrable function completion

Classification: PROVED

Finite simple functions completed under

$$d_2(f, g) = \mathcal{I}_{\text{rec}}(f - g)$$

form a complete positive cut module. The native pairing is extended using a finite cut-square identity before any Hilbert or ordinary L^2 structure is named.

T01-E4C — native bounded simple multipliers

Classification: PROVED

For a bounded native simple multiplier h ,

$$\mathcal{I}_{\text{tail}}(hf) \leq U_{\Sigma}(h) \mathcal{I}_{\text{tail}}(f),$$

$$\mathcal{I}_{\text{rec}}(hf) \leq U_{\Sigma}(h)^2 \mathcal{I}_{\text{rec}}(f).$$

No multiplicative closure is claimed for arbitrary tail-completed functions.

Proof: T01_E4_NATIVE_FUNCTION_COMPLETION.md.

PASS_NATIVE_SIMPLE_FUNCTION_TAIL_AND_RECOGNITION_COMPLETION_CORE
ca4d3443e057adbaa0c94a285263d8d136465eaf970c7fef73df70196972cad9

T01-E4D — native mean-multiplier identity and bound

Classification: PROVED

Exact mean split on the unit box,

$$\mathcal{I}_{\text{rec}}(hf) = \langle N(h) \rangle \mathcal{I}_{\text{rec}}(f) + \text{Cov}_\nu(N(h), N(f)),$$

and the bound

$$\mathcal{I}_{\text{rec}}(hf) \leq \langle N(h) \rangle \mathcal{I}_{\text{rec}}(f) + \text{osc } N(h) \mathcal{D}_{\text{rec}}(f), \quad 0 \leq \mathcal{D}_{\text{rec}}(f) \leq \mathcal{I}_{\text{rec}}(f).$$

A multiplier acts on a recognition-flat state by its mean, not its maximum. T01-E4D-C is **OPEN** in its **corrected** form (theorem file §4.1): the multiplier iterated on a fixed state converges to its supremum, not its mean (certified, Publications YM-34), so the item is restated as the m -uniform gap of the alternating product $(K^{1/2}M_wK^{1/2})$; its vacuum half is **PROVED** (Publications YM-31), its excitation half is the chain adapters' remaining obligation (theorem/28 hypothesis 3).

Status update (Aug 23 2026). The excitation half has been reduced, not closed. On the one-column transfer: the m -direction is eliminated (Publications YM-37/YM-38 — every quantity is one column, and the uniformity argument is native: T53 elimination + theorem/28 §9 outward certificate, no spectral decomposition); the content direction is released lawfully with exact seam-count retention (RKF theorem/75, native character-ladder regrading); the contraction itself is structural rather than Krylov (RKF theorem/74, silence channel). Dock and re-verification: Publications YM-41. **What remains open of T01-E4D-C is the t -direction only** — whether the extracted gap persists uniformly in the number of time rails — together with the standing caveat that all of this lives on the compressed column carrier (lower-bound side), not on an operator upper bound for the chain. Named next: Publications YM-42.

Proof: T01_E4D_NATIVE_MEAN_MULTIPLIER_BOUND.md.

PASS_NATIVE_MEAN_MULTIPLIER_IDENTITY_AND_BOUND
d6f3b39a0186a2c15c3fcd3734cf198ca6bd21b9182ad72906f0930341fe4c99

T01-E5A — rational native UGD group linearization

Classification: PROVED

A rational normalized UGD packet is

$$N = (p, u, \mu),$$

where p is a full-turn rational phase, u is a native additive scale-debt ledger, and μ is corrected rational seam memory. Its product is

$$(p, u, \mu) \star (q, v, \nu) = ((p + q)^b, u + v, \mu + \nu + W(p, q)).$$

The exact map

$$\Psi_\Sigma(p, u, \mu) = (u, p + \mu, [\mu])$$

is a group isomorphism to

$$\mathbb{Q}_\Sigma \times \mathbb{Q}_\Sigma \times (\mathbb{Q}_\Sigma/\mathbb{Z}).$$

No π , logarithm, exponential, or positive-real scale is primitive.

T01-E5B — rational UGD refinement action

Classification: PROVED

Compatible rational lifted points act as exact integer shifts of native refinement cells. The action is covariant under refinement, preserves content, and converts UGD product into successive cell translation.

Proof: T01_E5_NATIVE_RATIONAL_UGD_REFINEMENT_ADAPTER.md.

PASS_NATIVE_RATIONAL_UGD_REFINEMENT_LATTICE_ADAPTER
932d3e549e3a01d7d42d85e70371dc74fa06c214841bca8f776d19dd70ce4bc7

T01-E5C — completed actual UGD adapter

Classification: OPEN

Complete the rational phase, scale-debt, and memory group; extend the refinement action to all completed points; identify the actual state-locked UGD packet with that completion; and prove that the actual positive scale is a coordinate shadow rather than the definition of scale debt.

T01-E6 — positive-scale, Haar, Jacobian, and ordinary L_p shadows

Classification: OPEN

Only after T01-E5C may $\sigma = e^u$, flat Haar coordinates, the σ^{-2} density, and ordinary L^1/L^2 models be proved as coordinate shadows of the native objects.

T01-F — remaining coordinate-shadow identifications

Classification: OPEN

Prove equivalences with ordinary sequence-space, Hilbert, regular-representation, and continuous group models after native construction.

89.2 T02 — State-locked five-label carrier realization

Classification: OPEN

A provenance-controlled interface exists in the pinned MP source. RH-Framework must reproduce it from immutable worktrees and store its certificate before T02 can be classified.

PASS_STATE_LOCKED_CUT_APERTURE_CARRIER_INTERFACE

89.3 T03 — Actual action-domain adapter

Classification: OPEN

T03-A — classical normalized-UGD action-domain theorem

Classification: PROVED

On the classical coordinate carrier

$$\mathcal{H} = L^2(\mathbb{R}_u \times \mathbb{R}_t \times S_\alpha^1; \mathbb{C}^5),$$

bounded multiplier, convolution, and maximal multiplication-operator theorems are proved. Foundational origin is `CLASSICAL_THEOREM_IMPORT`; T03-A is not the native construction supplied by T01.

`PASS_T03_ACTION_DOMAIN_DENSE_CORE`
`05a4784308e996763928b4cead0c86462150154c506b30bf292f96dd8986db3c`

T03-B — finite state-locked packet reproduction

Classification: `OPEN`

The reproducer verifies exact clean MP worktrees and rejects changed source blobs, false checks, wrong labels, or dirty checkouts.

T03-C1 — actual finite packet action inventory

Classification: `OPEN`

The finite-action typing theorem and pinned extractor are implemented. The actual private packet has not yet been executed inside RH-Framework.

`PASS_RH_FRAMEWORK_T03_C1_PINNED_ACTION_INVENTORY`

A passing run promotes T03-B and T03-C1 together.

T03-C2A — positive-eta family envelope theorem

Classification: `PROVED`

The source-independent theorem correctly types amplitudes as analysis maps and proves compact-positive-eta bounds. Its origin is `CLASSICAL_THEOREM_IMPORT` because it uses complex polar and spectral theory.

`PASS_T03_C2A_POSITIVE_ETA_FAMILY_ENVELOPE`
`d64d8b02...c6e8`

T03-C2B — actual pinned positive-eta family

Classification: `OPEN`

Execute the actual pinned MP eta family and bind its arrays and source blobs to the T03-C2A hypotheses.

T03-C3 — arithmetic, Gamma, boundary, and compatibility actions

Classification: `OPEN`

Each actual action must receive a native T01-domain statement and, when useful, a classical T03 coordinate classification.

T03-C4 — unbounded generator graph domains

Classification: `OPEN`

Derive native path-graph convergence first, then recover classical maximal or closed graph domains as coordinate shadows.

T03-C5 — aperture exhaustion

Classification: OPEN

Prove native refinement convergence before operator-topology or graph-norm promotion.

89.4 T04 — Target faithfulness and residue retention

Classification: OPEN

The native regular action is faithful on its abstract path core. The actual RH target may still quotient nonzero seam residue; its kernel is not proved trivial or completely identified.

89.5 T05 — Native Shunya exclusion

Classification: OPEN

The formal positive square is identified, but the recognized-zero condition on the actual native completed representation is not proved.

89.6 T06 — Critical-line adapter and completed-Weil equivalence

Classification: OPEN

No theorem in RH-Framework proves completed-Weil positivity, global seam-charge vanishing, odd/even continuum no-crossing, or the Riemann Hypothesis.

89.7 Native reconstruction order

T01-E5C completed actual UGD adapter
-> T01-E6 positive-scale/Haar/Jacobian/Lp coordinate shadows
-> T01-D actual MP path-atlas adapter
-> T01-F remaining coordinate identifications
-> native generator graph convergence
-> native state-packet generation

89.8 RH verification order

pinned T03-C1 run, which also verifies T03-B
-> T03-C2B actual positive-eta family
-> T03-C3 arithmetic and Gamma actions
-> T03-C4 generator domains
-> T03-C5 aperture exhaustion
-> T04 target-kernel audit
-> T05 native Shunya
-> T06 critical-line equivalence

89.9 Global claim

Riemann Hypothesis: OPEN

RH-Framework/theorems/LEDGER.md

Chapter 90

Native Derivation and Classical-Import Audit

90.1 Current verdict

The RH framework remains **mixed**, not purely native.

The native boundary has moved substantially. RH-Framework now constructs, before ordinary functional analysis:

```
cut-tail path completion;  
recognition-Cauchy completion;  
completed native path action;  
refinement-count content;  
complete finite-content algebra;  
native tail and recognition simple integrals;  
native completed function modules;  
native bounded simple multipliers;  
rational normalized-UGD group and refinement action.
```

The unresolved classical boundary now begins at completion of the actual positive-scale UGD carrier, its Haar/Jacobian shadows, actual MP coefficient adapters, generator domains, and numerical state-packet generation.

The origin vocabulary remains:

```
NATIVE_DERIVED  
NATIVE_DERIVED_WITH_CLASSICAL_PROOF_TECHNIQUES  
COORDINATE_SHADOW  
CLASSICAL_THEOREM_IMPORT  
CLASSICAL_NUMERICAL_SOURCE  
OPEN_NATIVE_RECONSTRUCTION
```

The theorem ledger and origin ledger answer different questions. A theorem can be **PROVED** while still being a classical import.

90.2 1. Audit standard

Strong purity would forbid integers, rationals, finite sets, sequences, equivalence relations, and ordinary logic. That is not claimed.

The active research standard is:

Do not assume ordinary complex numbers, L-p spaces, Hilbert space, operator norms, spectral theory, positive-real UGD coordinates, or Haar measure before deriving the native cut, path, dagger, positivity, summability, recognition, refinement, function, and group structures that those classical objects are meant to coordinate.

90.3 2. Native-derived layers

2.1 Cut arithmetic and finite path algebra

Origin: NATIVE_DERIVED

Signed cut counts, refinement classes, the radial/tangential cut plane, quarter-turn multiplication, dagger, the finite positive cone, typed residue paths, closed-loop positivity, seam bridges, cut-curvature squares, and residue retention are constructed before ordinary complex coordinates.

2.2 Cut-scalar completion

Origin: NATIVE_DERIVED_WITH_CLASSICAL_PROOF_TECHNIQUES

The completion object is ordered from cut-rational neighborhoods. Standard sequence and diagonal meta-logic is used, but ordinary $\mathbb{C}$ does not generate the object.

The later map to ordinary complex numbers is a COORDINATE_SHADOW.

2.3 Native path summability and recognition completion

Origin: NATIVE_DERIVED

The foundational gauges are

$$\mathcal{M}_\Sigma(a) = \sum_{\gamma} D_\Sigma(a_\gamma),$$

$$\mathcal{E}_\Sigma(a) = \Phi_\Sigma(a^\dagger \star a).$$

Their Cauchy-presentation quotients are defined before ℓ^1 , ℓ^2 , Banach space, Hilbert space, or operator norm. A weighted cut-square identity proves

$$\mathcal{E}_\Sigma(a \star \xi) \leq \mathcal{M}_\Sigma(a)^2 \mathcal{E}_\Sigma(\xi).$$

The older MP ℓ^1/ℓ^2 theorem is no longer foundational. Its proper future role is a coordinate-shadow identification.

2.4 Native refinement-count content

Origin: NATIVE_DERIVED

A refinement cell on grid (n_s, n_t, n_α) has content

$$\frac{1}{n_s n_t n_\alpha}.$$

Common refinement, finite additivity, symmetric-difference completion, and integer cell-translation invariance are derived from exact cell counts. Haar, Lebesgue, a Jacobian, and floating point are absent.

This gives a complete **finite-content algebra**, not an imported sigma-algebra. No arbitrary countable-additivity claim has been smuggled into the vocabulary.

2.5 Native function completions

Origin: NATIVE_DERIVED

Finite cut-scalar simple functions carry tail and recognition gauges

$$\mathcal{I}_{\text{tail}}, \quad \mathcal{I}_{\text{rec}}.$$

Their two Cauchy-presentation completions, the completed native pairing, and bounded simple-multiplier estimates are proved before ordinary L^1 , L^2 , Hilbert space, or decomposable operators are named.

Arbitrary tail-completed functions are not falsely declared multiplicatively closed.

2.6 Rational normalized-UGD refinement adapter

Origin: NATIVE_DERIVED

The rational normalized packet

$$N = (p, u, \mu)$$

uses:

```
p    full-turn rational phase;
u    additive native scale debt;
mu   corrected rational seam memory.
```

The exact linearization

$$(p, u, \mu) \mapsto (u, p + \mu, [\mu])$$

proves the wrap law and rational refinement action without π , logarithm, exponential, positive-real scale, Haar measure, or a Jacobian.

Compatible rational packets act by exact content-preserving cell translations.

90.4 3. Classical or mixed layers still present

3.1 Actual positive UGD scale

Origin: OPEN_NATIVE_RECONSTRUCTION

The native scale-debt ledger u is not yet identified with an actual numerical scale $\sigma > 0$.

The classical formula

$$\sigma = e^u$$

may later be a coordinate chart. It may not define native scale debt. Completion of the rational native UGD group and binding of the state-locked packet remain open.

3.2 Haar and original-coordinate density

Origin: CLASSICAL_THEOREM_IMPORT / FUTURE_COORDINATE_SHADOW

The previous continuous UGD theorem uses topological-group theory, Haar measure, real log-arithm/exponential, and a Jacobian to obtain

$$d\lambda \propto \frac{d\phi d\sigma dk}{\sigma^2}.$$

This remains useful, but it must be reprovved after the completed native UGD adapter as a coordinate-shadow theorem.

It is no longer allowed to generate the native refinement content or native function completions.

3.3 RH-Framework T03-A

Origin: CLASSICAL_THEOREM_IMPORT

T03-A begins with

$$L^2(\mathcal{U}_F; \mathbb{C}^5)$$

and uses essential suprema, singular values, operator norms, Haar-translation unitarity, Minkowski/Young bounds, resolvents, and maximal multiplication operators. It is rigorous coordinate analysis, not the native foundation.

3.4 State-locked finite packet

Origin: CLASSICAL_NUMERICAL_SOURCE_WITH_NATIVE_REINTERPRETATION

The packet is generated by complex arrays, FFT channels, Gram matrices, eigendecomposition, singular values, polar frames, and numerical metric transport.

The later exact cut embedding audits a native aperture relation but does not retroactively make the source packet natively generated.

3.5 Positive-eta family

Origin: CLASSICAL_THEOREM_IMPORT

The typing correction is valid, but the theorem uses complex rectangular amplitudes, polar decomposition, Hermitian Grams, spectral square roots, and operator-norm continuity.

90.5 4. Foundational source boundary

The active native source files are:

```
src/rh_framework/native_summability.py
src/rh_framework/native_refinement_measure.py
src/rh_framework/native_function_completion.py
src/rh_framework/native_ugd_refinement_adapter.py
```

They use exact rational arithmetic and internal native dependencies only. A single verifier inspects their AST imports and rejects executable references to:

```
float;
complex;
numpy;
scipy;
log;
exp;
pi.
```

The classical verification shell may use numerical libraries, but files in that shell must not be cited as foundational native derivations.

90.6 5. Reconstruction progress

native cut-tail summability before ell1	COMPLETE
native recognition completion before ell2	COMPLETE
native action bound before operator norm	COMPLETE
native refinement content before Haar	COMPLETE
native function completions before ordinary Lp	COMPLETE
rational normalized-UGD refinement law	COMPLETE
completed actual positive-scale UGD adapter	OPEN
Haar/Jacobian/Lp coordinate-shadow theorem	OPEN
actual MP path-atlas adapter	OPEN
native generator graph convergence	OPEN
native state-packet generation	OPEN
representation invariance of numerical charts	OPEN
actual target faithfulness	OPEN

90.7 6. Next reconstruction theorem

The next theorem must complete the rational native UGD group and extend its refinement action to completed points. It must then bind the actual state-locked UGD packet without defining scale debt by logarithm.

Only afterward may the classical log-exp/Haar theorem be admitted as a coordinate shadow.

90.8 7. Current audit conclusion

EVERYTHING PURELY NATIVE	NO
FINITE CUT/PATH ALGEBRA NATIVE	YES
NATIVE COUNTABLE PATH COMPLETION	YES
NATIVE REFINEMENT CONTENT	YES
NATIVE FUNCTION COMPLETIONS	YES
RATIONAL NATIVE UGD REFINEMENT ACTION	YES
ACTUAL POSITIVE-SCALE UGD CARRIER NATIVE	NO
HAAR/JACOBIAN/Lp CONTINUOUS MODEL NATIVE	NO
TO3-A NATIVE	NO
STATE-LOCKED PACKET NATIVELY GENERATED	NO
FULLY NATIVE RH PROOF ESTABLISHED	NO

No RH claim may use the phrase **fully native proof** until the completed actual UGD adapter, actual MP path/action adapters, generator, packet, target, Shunya, and critical-line gates are closed.

`RH-Framework/theorems/NATIVE_DERIVATION_CONTAMINATION_AUDIT.md`

Chapter 91

T01-E4D — Native Mean-Multiplier Identity and Bound

Classification: PROVED

Foundational origin: NATIVE_DERIVED

Iterated-multiplier deviation control (T01-E4D-B): OPEN

T01-E4C bounds a bounded native simple multiplier by its uniform cut bound $U_\Sigma(h) = \max_c D_\Sigma(h(c))$. That bound is tight only when the state is concentrated where the multiplier is largest. This theorem proves the exact finite identity that separates the **mean** action of a multiplier from its **fluctuation**, and bounds the fluctuation by the multiplier's oscillation times a native deviation of the state. Nothing beyond exact rational arithmetic on refinement cells of one unit box is used.

91.1 1. Setting

Let $R = (n_s, n_t, n_\alpha)$ be a refinement grid, B its unit box (T01-E1, content $\nu_R(B) = 1$), and let h, f be finite cut-scalar simple functions supported in B . Write N_Σ for the native norm square (T01-B). Define on B

$$\langle N(h) \rangle = \sum_{c \subset B} \nu_R(c) N_\Sigma(h(c)) = \mathcal{I}_{\text{rec}}(h), \quad \text{osc } N(h) = \max_c N_\Sigma(h(c)) - \min_c N_\Sigma(h(c)),$$

and the **recognition deviation**

$$\mathcal{D}_{\text{rec}}(f) = \sum_{c \subset B} \nu_R(c) (N_\Sigma(f(c)) - \langle N(f) \rangle)_+. \quad (1.1)$$

All three are refinement-invariant by T01-E3 (they are finite cell sums of refinement-invariant simple integrals), and

$$0 \leq \mathcal{D}_{\text{rec}}(f) \leq \mathcal{I}_{\text{rec}}(f). \quad (1.2)$$

91.2 2. Theorem E4D-A (exact mean split)

$$\boxed{\mathcal{I}_{\text{rec}}(hf) = \langle N(h) \rangle \mathcal{I}_{\text{rec}}(f) + \sum_{c \in B} \nu_R(c) (N(h(c)) - \langle N(h) \rangle) (N(f(c)) - \langle N(f) \rangle).} \quad (2.1)$$

Proof

N_Σ is multiplicative on cut scalars (T01-B: $N(ab) = N(a)N(b)$), so $\mathcal{I}_{\text{rec}}(hf) = \sum_c \nu(c) N(h(c)) N(f(c))$. Expand $N(h(c)) = \langle N(h) \rangle + (N(h(c)) - \langle N(h) \rangle)$ and use $\sum_c \nu(c) (N(h(c)) - \langle N(h) \rangle) = 0$ on the unit box. QED.

91.3 3. Theorem E4D-B (mean-multiplier bound)

$$\boxed{\mathcal{I}_{\text{rec}}(hf) \leq \langle N(h) \rangle \mathcal{I}_{\text{rec}}(f) + \text{osc } N(h) \mathcal{D}_{\text{rec}}(f).} \quad (3.1)$$

Proof

Split the covariance sum of (2.1) into cells where $N(f(c)) > \langle N(f) \rangle$ and cells where it is below. On the first set the h -factor is at most $\max N(h) - \langle N(h) \rangle$; on the second set the h -factor is at least $\min N(h) - \langle N(h) \rangle$ and multiplies a negative quantity, so the contribution is at most $(\langle N(h) \rangle - \min N(h)) \sum_c \nu(c) (\langle N(f) \rangle - N(f(c)))_+$. The positive and negative deviations of $N(f)$ have equal content-weighted mass on the unit box (their difference is $\sum_c \nu(c) (N(f(c)) - \langle N(f) \rangle) = 0$), so both sums equal $\mathcal{D}_{\text{rec}}(f)$. Adding the two gaps gives $\text{osc } N(h)$. QED.

Corollaries

1. **Vacuum equality.** If $N(f)$ is constant on B then $\mathcal{D}_{\text{rec}}(f) = 0$ and $\mathcal{I}_{\text{rec}}(hf) = \langle N(h) \rangle \mathcal{I}_{\text{rec}}(f)$: a multiplier acts on a recognition-flat state by its **mean**, not its maximum.
2. **Product content.** If h_1 depends only on the scale coordinate and h_2 only on the seam coordinate then $\langle N(h_1 h_2) \rangle = \langle N(h_1) \rangle \langle N(h_2) \rangle$ (T01-E1 product content).
3. **Relation to T01-E4C.** Neither bound dominates: E4C is sharper for states concentrated at the multiplier's maximum, E4D is sharper for recognition-flat states. Both are lawful; the certificate records that E4D is strictly below E4C in all 40 pseudo-random trials.

91.4 4. What is not claimed (T01-E4D-C, OPEN)

For an **iterated** product $h_1 h_2 \cdots h_k f$ the bound (3.1) applies face by face, but $\mathcal{D}_{\text{rec}}$ of the intermediate states is not controlled by this theorem. Bounding the deviation along an iteration uniformly in k is exactly the obligation that appears as hypothesis 3 of the Recognition-Complete Finite-to-Infinite Cut Theorem (Recognition-Kernel-Framework **theorem/28**) for the chain adapters (Publications **yang-mills-certified-benchmark**, YM-26/27): there the T01-E4C price per face is measured to exceed the true vacuum rate by factors 56.5 / 27.8 / 13.5, and Corollary 1 is the mechanism that removes the maximum from the vacuum channel. No gap, no Haar shadow, and no RH statement is promoted by this theorem.

4.1 Restatement of T01-E4D-C (Aug 22 2026, after Publications YM-34)

The obligation above is **not provable as written**. For a single face with multiplier $w = e^{\kappa\chi_{1/2}}$ the iterated ratio on one recognition state,

$$q_n = \frac{\int w^n}{\int w^{n-1}} = \frac{f_0(n\kappa)}{f_0((n-1)\kappa)},$$

is strictly increasing and converges to $\sup w = e^{2\kappa}$, not to the mean f_0 (certified for $n \leq 60$ at $\kappa = 1/8, 1/4, 1/2$: Publications `ym34_e4dc_restated.py`, pin `EXPECTED_YM34.sha256`). This is the native Laplace principle: the mean law (3.1) holds once; iterated on a fixed state the per-application price returns to the E4C supremum. Every exponential met in the chain adapters (YM-16, YM-26 T2) is this growth.

With the recognition kernel K (content- j eigenvalue λ_j) applied **between** multiplications the iterated ratio is nondecreasing (native log-convexity, cut-square) and converges to $\lambda_1(K^{1/2}M_wK^{1/2}) = f_0(1 + \delta(\kappa))$ with $\delta = 0.003582, 0.014032, 0.051911$, against a supremum price $e^{2\kappa}/f_0 - 1 = 0.274, 0.598, 1.405$. The smoothing step is what turns the supremum into a spectral constant.

T01-E4D-C (corrected statement, OPEN). For the alternating product $S = (K^{1/2}M_wK^{1/2})$ on a chain of m faces, $\lambda_2(S)/\lambda_1(S) \leq 1 - \delta_{\text{gap}}(\kappa)$ uniformly in m . The vacuum half (an m -uniform lower bound on λ_1 at the true per-face rate) is PROVED in Publications YM-31; the excitation half (an m -uniform upper bound on λ_2) is the single open obligation, and any proof of it must carry the smoothing step inside the iteration.

91.5 5. Certificate

```
src/rh_framework/native_mean_multiplier.py
src/rh_framework/native_mean_multiplier_certificate.py
certificates/T01_E4D_NATIVE_MEAN_MULTIPLIER_v0.1.json
tests/test_native_mean_multiplier.py
```

Build note: the first draft of (3.1) used $\max N(h) - \langle N(h) \rangle$ in place of $\text{osc } N(h)$; the executable certificate refused it (negative-side cells), and the proof above is the corrected statement.

RH-Framework/theorems/T01_E4D_NATIVE_MEAN_MULTIPLIER_BOUND.md

Chapter 92

T01-E4 — Native Simple-Function Tail and Recognition Completions

Classification: PROVED

Foundational origin: NATIVE_DERIVED

Actual UGD function adapter: OPEN

This theorem completes finite cut-scalar simple functions using the native refinement content of T01-E1/E2/E3. It does not begin with an ordinary L^1 , L^2 , Hilbert, Haar, or Lebesgue structure.

92.1 1. Finite native simple functions

Let $R = (n_s, n_t, n_\alpha)$ be a native refinement grid and let C_R be its cell atlas. A finite native simple function is a finitely supported map

$$f : C_R \rightarrow \mathbb{K}_\Sigma^0. \quad (1.1)$$

If two functions live on different grids, refine both to the coordinatewise least-common-multiple grid. Refinement copies each coarse value to all fine cells inside that coarse cell.

Addition, subtraction, dagger, and pointwise multiplication are defined after common refinement.

92.2 2. Native integral gauges

The tail integral is

$$\mathcal{I}_1(f) = \sum_{c \in C_R} \nu_R(c) D_\Sigma(f(c)). \quad (2.1)$$

The recognition integral is

$$\mathcal{I}_2(f) = \sum_{c \in C_R} \nu_R(c) N_\Sigma(f(c)). \quad (2.2)$$

T01-E3 proves that both values are unchanged by refinement and integer cell translation.
Define

$$d_1(f, g) = \mathcal{I}_1(f - g), \quad (2.3)$$

$$d_2(f, g) = \mathcal{I}_2(f - g). \quad (2.4)$$

The second expression is an energy gauge rather than a square-root norm.

92.3 3. Tail-gauge control

For cut scalars,

$$D_\Sigma(x + y) \leq D_\Sigma(x) + D_\Sigma(y).$$

Summing over a common refinement gives

$$\boxed{d_1(f, h) \leq d_1(f, g) + d_1(g, h).} \quad (3.1)$$

Thus d_1 is a rational refinement-invariant distance on finite simple functions modulo equality after common refinement.

A sequence (f_n) is **tail Cauchy** when for every positive cut rational ε ,

$$d_1(f_m, f_n) < \varepsilon$$

for all sufficiently large m, n .

Define

$$\boxed{\mathfrak{F}_{\Sigma,1} = \{\text{tail-Cauchy simple presentations}\} / \sim_{d_1}.} \quad (3.2)$$

Theorem 3.1

$\mathfrak{F}_{\Sigma,1}$ is a complete cut module. Refinement and integer cell translation extend as isometries.

Proof

Addition and scalar multiplication are uniformly continuous by the scalar tail triangle inequality and

$$\mathcal{I}_1(zf) = D_\Sigma(z)\mathcal{I}_1(f).$$

The Cauchy-presentation quotient is well defined. Completeness follows from the diagonal presentation construction using only positive cut-rational thresholds. Refinement and translation preserve $\mathcal{I}_1$ on the finite core, hence extend to the completion. QED.

This object is the **native tail-integrable function completion**. Ordinary L^1 is not its definition.

92.4 4. Recognition-gauge control

For cut scalars,

$$2N_\Sigma(x) + 2N_\Sigma(y) - N_\Sigma(x + y) = N_\Sigma(x - y) \geq 0. \quad (4.1)$$

After summation,

$$\boxed{\mathcal{I}_2(f + g) \leq 2\mathcal{I}_2(f) + 2\mathcal{I}_2(g).} \quad (4.2)$$

A sequence (f_n) is **recognition Cauchy** when for every positive cut rational ε ,

$$d_2(f_m, f_n) < \varepsilon$$

for all sufficiently large m, n .

Define

$$\boxed{\mathfrak{F}_{\Sigma,2} = \{\text{recognition-Cauchy simple presentations}\} / \sim_{d_2}.} \quad (4.3)$$

92.5 5. Native completed pairing

For finite simple functions, define

$$\boxed{\langle f, g \rangle_{\Sigma, \nu}^0 = \sum_c \nu_R(c) f(c)^\dagger g(c),} \quad (5.1)$$

after common refinement.

The exact finite cut-square identity gives

$$\boxed{N_\Sigma(\langle f, g \rangle_{\Sigma, \nu}^0) \leq \mathcal{I}_2(f) \mathcal{I}_2(g).} \quad (5.2)$$

One proof clears the rational cell denominators and applies the finite coefficient identity

$$\left(\sum_i N(x_i) \right) \left(\sum_i N(y_i) \right) - N \left(\sum_i x_i^\dagger y_i \right) = \sum_{i < j} N(x_i y_j - x_j y_i). \quad (5.3)$$

No pre-existing inner-product theorem is used.

Theorem 5.1

The pairing extends uniquely to

$$\langle \cdot, \cdot \rangle_{\Sigma, \nu} : \mathfrak{F}_{\Sigma,2} \times \mathfrak{F}_{\Sigma,2} \rightarrow \widehat{\mathbb{K}}_\Sigma. \quad (5.4)$$

It satisfies dagger symmetry, cut-linearity, positivity, and

$$\langle f, f \rangle_{\Sigma, \nu} = 0 \iff f = 0. \quad (5.5)$$

Proof

Recognition-Cauchy presentations are energy bounded. Apply (5.2) to pairing differences exactly as in T01-B. Positivity and definiteness follow from $\mathcal{I}_2(f)$ and the recognition-null equivalence relation. QED.

Therefore $\mathfrak{F}_{\Sigma,2}$ is the complete **native recognition- integrable function module**. Hilbert space may appear later only as a coordinate shadow.

92.6 6. Native bounded simple multipliers

For a finite simple function h , define its uniform cut bound

$$\boxed{U_{\Sigma}(h) = \max_c D_{\Sigma}(h(c)).} \quad (6.1)$$

Then pointwise multiplication obeys

$$\boxed{\mathcal{I}_1(hf) \leq U_{\Sigma}(h)\mathcal{I}_1(f),} \quad (6.2)$$

and

$$\boxed{\mathcal{I}_2(hf) \leq U_{\Sigma}(h)^2\mathcal{I}_2(f).} \quad (6.3)$$

Proof

For every cell,

$$D_{\Sigma}(h(c)f(c)) \leq D_{\Sigma}(h(c))D_{\Sigma}(f(c)) \leq U_{\Sigma}(h)D_{\Sigma}(f(c)),$$

while

$$N_{\Sigma}(h(c)f(c)) = N_{\Sigma}(h(c))N_{\Sigma}(f(c)) \leq U_{\Sigma}(h)^2N_{\Sigma}(f(c)).$$

Multiply by cell content and sum. QED.

Thus every bounded native simple multiplier extends continuously to both native function completions.

This theorem does **not** claim that the product of two arbitrary tail-completed functions remains tail-integrable. Such closure requires a separate hypothesis.

92.7 7. Explicit Cauchy function

On the unit-denominator grid, define

$$f_N = \sum_{n=0}^N 2^{-(n+1)} 1_{c_n}, \quad (7.1)$$

where the c_n are distinct native cells. Then for $M > N$,

$$d_1(f_M, f_N) = \sum_{n=N+1}^M 2^{-(n+1)} < 2^{-(N+1)}, \quad (7.2)$$

and

$$d_2(f_M, f_N) = \sum_{n=N+1}^M 4^{-(n+1)} < \frac{1}{3 \cdot 4^{N+1}}. \quad (7.3)$$

The same presentation therefore defines elements in both native completions.

92.8 8. Foundational exclusion boundary

The theorem does not use as definitions or proof generators:

ordinary L1 or L2 spaces;
Hilbert space;
Haar or Lebesgue integration;
operator norms;
complex floating-point values;
spectral or polar decomposition.

The implementation uses exact cut scalars and rational cell content only.

92.9 9. Coordinate-shadow boundary

After the actual UGD refinement adapter is proved, one may attempt to identify

native tail function completion	with an ordinary L1 chart;
native recognition function completion	with an ordinary L2 chart;
native bounded multipliers	with decomposable bounded operators.

Those must be equivalence theorems. They may not be used as definitions of the native objects above.

92.10 10. Exact certificate

The expected status is

PASS_NATIVE_SIMPLE_FUNCTION_TAIL_AND_RECOGNITION_COMPLETION_CORE

The exact certificate verifies cross-refinement arithmetic, pairing identities, native multiplier bounds, a nontrivial Cauchy function, and the absence of classical numerical dependencies in the foundational module.

92.11 11. Remaining gates

T01-E4A NATIVE TAIL FUNCTION COMPLETION	PROVED
T01-E4B NATIVE RECOGNITION FUNCTION COMPLETION	PROVED
T01-E4C NATIVE BOUNDED SIMPLE MULTIPLIERS	PROVED
T01-E5 ACTUAL UGD REFINEMENT ADAPTER	OPEN
T01-E6 HAAR AND ORDINARY L1/L2 SHADOWS	OPEN
ACTUAL MP ACTIONS ON NATIVE FUNCTION DOMAINS	OPEN
NATIVE STATE-PACKET GENERATION	OPEN
TARGET FAITHFULNESS	OPEN
NATIVE SHUNYA	OPEN
RIEMANN HYPOTHESIS	OPEN

RH-Framework/theorems/T01_E4_NATIVE_FUNCTION_COMPLETION.md

Chapter 93

T01-E5A/B — Native Rational UGD Refinement Adapter

Classification: PROVED

Foundational origin: NATIVE_DERIVED

Completed actual UGD adapter: OPEN

This theorem derives the rational refinement law of normalized UGD without using π , logarithm, exponential, positive-real coordinates, Haar measure, or a Jacobian.

The native scale variable below is a signed **scale-debt ledger**. It is not yet identified with an actual positive numerical scale σ . That later identification is a separate completion and coordinate-shadow problem.

93.1 1. Native normalized UGD packets

Let

$$p \in \mathbb{Q}_\Sigma / \mathbb{Z}$$

be a full-turn phase fraction, represented in the principal interval $[0, 1)$. Let

$$u \in \mathbb{Q}_\Sigma$$

be a signed scale-debt coordinate, and let

$$\mu \in \mathbb{Q}_\Sigma$$

be corrected seam memory.

A rational normalized UGD packet is

$$N = (p, u, \mu). \tag{1.1}$$

For principal phase representatives define

$$W(p, q) = \lfloor p + q \rfloor \in \{0, 1\}, \tag{1.2}$$

$$(p + q)^b = p + q - W(p, q). \quad (1.3)$$

The product is

$$(p, u, \mu) \star (q, v, \nu) = \left((p + q)^b, u + v, \mu + \nu + W(p, q) \right). \quad (1.4)$$

This is the full-turn form of the gauge-normalized UGD memory law. No angle in radians is required.

93.2 2. Group theorem

Theorem 2.1

Equation (1.4) defines an abelian group. Its identity is

$$(0, 0, 0), \quad (2.1)$$

and its inverse is

$$(p, u, \mu)^{-1} = \left((-p)^b, -u, -\mu - W(p, (-p)^b) \right). \quad (2.2)$$

Proof

The raw full-turn sum of three principal phases is independent of bracketing. Both bracketings subtract exactly the integer part of $p + q + r$. Hence

$$W(p, q) + W((p + q)^b, r) = W(q, r) + W(p, (q + r)^b). \quad (2.3)$$

This proves associativity of phase and memory. Scale debt and the remaining memory terms add associatively. Symmetry of the raw phase sum proves commutativity.

For (2.2), principal phases sum to zero after their unique wrap, scale debts cancel, and the memory correction cancels that wrap. QED.

93.3 3. Native lifted coordinates

Define

$$\Psi_\Sigma(p, u, \mu) = (u, t, \alpha), \quad t = p + \mu, \quad \alpha = [\mu] \in \mathbb{Q}_\Sigma / \mathbb{Z}. \quad (3.1)$$

Theorem 3.1 — Rational native linearization

Ψ_Σ is a group isomorphism from (1.4) to

$$\mathbb{Q}_\Sigma \times \mathbb{Q}_\Sigma \times (\mathbb{Q}_\Sigma / \mathbb{Z}) \quad (3.2)$$

with componentwise addition.

Proof

Let

$$p_{12} = p + q - W(p, q), \quad \mu_{12} = \mu + \nu + W(p, q).$$

Then

$$\begin{aligned} t_{12} &= p_{12} + \mu_{12} \\ &= p + \mu + q + \nu \\ &= t_1 + t_2. \end{aligned} \tag{3.3}$$

Also

$$[\mu_{12}] = [\mu + \nu] \tag{3.4}$$

because the wrap is integral, and scale debts add.

For the inverse map, let (u, t, α) be given with the unique principal representative $0 \leq \alpha < 1$. Put

$$n = \lfloor t - \alpha \rfloor, \quad \mu = \alpha + n, \quad p = (t - \alpha) - n. \tag{3.5}$$

Then $0 \leq p < 1$, $[\mu] = \alpha$, and $p + \mu = t$. This is the unique preimage. QED.

The theorem needs neither 2π nor a logarithm. Full turns, scale debt, and memory are primitive cut-ledger coordinates.

93.4 4. Rational refinement lattices

For a refinement grid

$$R = (n_s, n_t, n_\alpha),$$

define the compatible rational lattice

$$\Lambda_R = \frac{1}{n_s} \mathbb{Z} \times \frac{1}{n_t} \mathbb{Z} \times \left(\frac{1}{n_\alpha} \mathbb{Z} / \mathbb{Z} \right). \tag{4.1}$$

A native lifted point (u, t, α) is compatible with R exactly when

$$n_s u \in \mathbb{Z}, \quad n_t t \in \mathbb{Z}, \quad n_\alpha \alpha \in \mathbb{Z}. \tag{4.2}$$

Its cell shift is

$$S_R(u, t, \alpha) = (n_s u, n_t t, n_\alpha \alpha), \tag{4.3}$$

with the final component reduced modulo n_α .

Theorem 4.1 — Lattice action

For compatible points $x, y \in \Lambda_R$,

$$S_R(x + y) = S_R(x) + S_R(y), \quad (4.4)$$

with circular reduction in the seam component.

Therefore compatible rational UGD packets act on native cylinders by exact cell translation.

Proof

Multiply componentwise rational addition by the declared denominators. The first two coordinates add in $\mathbb{Z}$, while the seam coordinate adds in $\mathbb{Z}/n_\alpha\mathbb{Z}$. QED.

93.5 5. Refinement covariance

Let R' refine R , with refinement factors

$$q_s = \frac{n'_s}{n_s}, \quad q_t = \frac{n'_t}{n_t}, \quad q_\alpha = \frac{n'_\alpha}{n_\alpha}.$$

For every point compatible with R ,

$$\boxed{S_{R'}(x) = (q_s S_R^s(x), q_t S_R^t(x), q_\alpha S_R^\alpha(x)),} \quad (5.1)$$

with the final coordinate reduced modulo n'_α .

Thus translating first and refining afterward gives the same fine cells as refining first and applying the fine-grid shift.

93.6 6. Content-preserving UGD translation

Let A be a finite native cylinder on grid R , and let N be a rational normalized UGD packet whose lifted point is compatible with R . Define

$$T_N A = T_{S_R(\Psi_\Sigma(N))} A. \quad (6.1)$$

Theorem 6.1

$$\boxed{\nu(T_N A) = \nu(A).} \quad (6.2)$$

Moreover,

$$T_{N_1}(T_{N_2} A) = T_{N_1 \star N_2} A. \quad (6.3)$$

Proof

The first statement is T01-E1 translation invariance. For the second, use the group homomorphism (3.3) and the lattice action (4.4). QED.

The same translations preserve the native simple-function tail and recognition integrals by finite reindexing.

93.7 7. What is native here

The construction uses only:

```
full-turn rational phase;  
integer phase wrap;  
additive rational scale debt;  
rational corrected memory;  
rational quotient by integers;  
finite refinement denominators;  
integer cell shifts;  
reciprocal-count content.
```

It does not use:

```
pi or radians;  
logarithm or exponential;  
a primitive positive-real scale line;  
real-coordinate topology;  
Haar measure;  
Jacobian density;  
floating-point arithmetic.
```

93.8 8. Actual-scale boundary

This theorem does **not** yet identify the native scale debt u with an actual UGD numerical scale $\sigma > 0$.

The classical formula

$$\sigma = e^u$$

may later be proved as a coordinate chart after the native scale-debt completion is fixed. It may not be used as the definition of u .

Likewise, the full completed UGD carrier requires completion of rational phase, scale-debt, and memory presentations and proof that the actual state-locked UGD packet lands in that completion.

93.9 9. Exact certificate

The expected status is

PASS_NATIVE_RATIONAL_UGD_REFINEMENT_LATTICE_ADAPTER

The certificate checks exact wrap, product, inverse, linearization, round trip, compatible and incompatible lattice points, refinement covariance, content-preserving translation, and product/translation covariance.

93.10 10. Remaining gates

TO1-E5A RATIONAL NATIVE UGD LINEARIZATION	PROVED
TO1-E5B RATIONAL UGD REFINEMENT ACTION	PROVED
TO1-E5C COMPLETED ACTUAL UGD ADAPTER	OPEN
TO1-E6 POSITIVE-SCALE/LOG-EXP COORDINATE SHADOW	OPEN
TO1-E6 HAAR/JACOBIAN/Lp COORDINATE SHADOW	OPEN
STATE-LOCKED UGD PACKET TO NATIVE COMPLETION	OPEN
TARGET FAITHFULNESS	OPEN
NATIVE SHUNYA	OPEN
RIEMANN HYPOTHESIS	OPEN

RH-Framework/theorems/TO1_E5_NATIVE_RATIONAL_UGD_REFINEMENT_ADAPTER.md

Chapter 94

T01-E1/E2/E3 — Native Refinement-Count Content and Simple Integrals

Classification: PROVED

Foundational origin: NATIVE_DERIVED

Continuous native measure classification: OPEN

This theorem constructs a finite refinement content and its symmetric-difference completion before introducing Haar measure, Lebesgue measure, a Jacobian, or an ordinary L^p space.

It does **not** yet prove a sigma-additive measure on the actual continuous UGD carrier. The completed object below is a complete finite-content algebra. The continuous-function and actual-UGD adapters remain separate gates.

94.1 1. Native refinement grids

A refinement grid is a triple

$$R = (n_s, n_t, n_\alpha), \quad n_s, n_t, n_\alpha \in \mathbb{N}_{>0}. \quad (1.1)$$

The three entries count subdivisions in:

scale-refinement direction;
lifted phase-memory direction;
circular seam direction.

At this stage these are native partition channels. They are not assumed to be ordinary real coordinates.

A cell is indexed by

$$c = (j_s, j_t, j_\alpha), \quad j_s, j_t \in \mathbb{Z}, \quad j_\alpha \in \mathbb{Z}/n_\alpha\mathbb{Z}. \quad (1.2)$$

Define the reciprocal refinement content of one cell by

$$\boxed{\nu_R(c) = \frac{1}{n_s n_t n_\alpha}}. \quad (1.3)$$

The value is a positive cut rational. No real length, volume form, or external measure is primitive.

94.2 2. Finite native cylinders

A finite native cylinder is a finite set of cells on one grid:

$$A = (R, C), \quad C \subset \mathbb{Z} \times \mathbb{Z} \times \mathbb{Z}/n_\alpha \mathbb{Z}. \quad (2.1)$$

Its content is

$$\boxed{\nu(A) = \frac{|C|}{n_s n_t n_\alpha}.} \quad (2.2)$$

Repeated cell indices are identified before counting. Thus content depends on the represented finite cylinder, not on a list presentation.

94.3 3. Refinement

Let

$$R' = (n'_s, n'_t, n'_\alpha)$$

refine R , meaning each denominator of R' is a positive integer multiple of the corresponding denominator of R . Put

$$q_s = \frac{n'_s}{n_s}, \quad q_t = \frac{n'_t}{n_t}, \quad q_\alpha = \frac{n'_\alpha}{n_\alpha}. \quad (3.1)$$

Each coarse cell is replaced by exactly

$$q_s q_t q_\alpha \quad (3.2)$$

fine cells.

Theorem 3.1 — Refinement invariance

For every finite cylinder A ,

$$\boxed{\nu(\text{Ref}_{R'} A) = \nu(A).} \quad (3.3)$$

Proof

If A contains m coarse cells, its refinement contains $mq_s q_t q_\alpha$ fine cells. Therefore

$$\begin{aligned}\nu(\text{Ref}_{R'} A) &= \frac{mq_s q_t q_\alpha}{n'_s n'_t n'_\alpha} \\ &= \frac{m}{n_s n_t n_\alpha} = u(A).\end{aligned}$$

QED.

94.4 4. Common refinement and Boolean operations

For grids R and S , define their common refinement coordinatewise by least common multiples:

$$R \vee S = (\text{lcm}(n_s, m_s), \text{lcm}(n_t, m_t), \text{lcm}(n_\alpha, m_\alpha)). \quad (4.1)$$

Refine both cylinders to $R \vee S$, then define union, intersection, difference, and symmetric difference by ordinary finite-cell operations. Refinement invariance shows that the resulting content is independent of the chosen common refinement.

Theorem 4.1 — Finite-content identity

For finite cylinders A, B ,

$$\boxed{\nu(A \cup B) + \nu(A \cap B) = \nu(A) + \nu(B).} \quad (4.2)$$

In particular, if $A \cap B = \emptyset$, then

$$\nu(A \cup B) = \nu(A) + \nu(B). \quad (4.3)$$

Proof

Move to a common refinement. Every cell belongs to exactly one of

$$A \setminus B, \quad B \setminus A, \quad A \cap B.$$

Count the three disjoint finite sets and multiply by the common cell content. QED.
The symmetric-difference identity is

$$\boxed{\nu(A \triangle B) = \nu(A) + \nu(B) - 2\nu(A \cap B).} \quad (4.4)$$

94.5 5. Native translation invariance

For integer shifts

$$h = (h_s, h_t, h_\alpha),$$

define

$$T_h(j_s, j_t, j_\alpha) = (j_s + h_s, j_t + h_t, j_\alpha + h_\alpha), \quad (5.1)$$

with the seam coordinate reduced modulo n_α .

Theorem 5.1

$$\boxed{\nu(T_h A) = \nu(A).} \quad (5.2)$$

Proof

Translation is a bijection of the cell index set on a fixed grid. It preserves the number of cells and therefore preserves reciprocal-count content. QED.

This is native translation invariance on the refinement atlas. It is not yet a Haar theorem on a locally compact group.

94.6 6. Symmetric-difference gauge

Define

$$\boxed{\delta(A, B) = \nu(A \Delta B).} \quad (6.1)$$

After passing to common refinements, finite-set identities give

$$\delta(A, B) \geq 0, \quad \delta(A, B) = \delta(B, A), \quad (6.2)$$

and

$$A \Delta C \subseteq (A \Delta B) \cup (B \Delta C). \quad (6.3)$$

Consequently

$$\boxed{\delta(A, C) \leq \delta(A, B) + \delta(B, C).} \quad (6.4)$$

Two cylinder presentations have zero distance precisely when their common refinements contain the same cells.

Lemma 6.1 — Content continuity

$$\boxed{|\nu(A) - \nu(B)| \leq \delta(A, B).} \quad (6.5)$$

Proof

On a common refinement,

$$\nu(A) - \nu(B) = u(A \setminus B) - \nu(B \setminus A).$$

Taking the ordered rational absolute value yields

$$|\nu(A) - \nu(B)| \leq \nu(A \setminus B) + \nu(B \setminus A) = u(A \triangle B).$$

QED.

94.7 7. Native content-algebra completion

A sequence (A_n) of finite native cylinders is **refinement-content Cauchy** when, for every positive cut rational ε , there exists N such that

$$\delta(A_m, A_n) < \varepsilon \quad (m, n \geq N). \quad (7.1)$$

Two such sequences are equivalent when

$$\delta(A_n, B_n) \longrightarrow 0 \quad (7.2)$$

in rational-threshold language.

Define

$$\boxed{\mathfrak{M}_\Sigma^{\text{ref}} = \{\text{refinement-content Cauchy presentations}\} / \sim_\delta.} \quad (7.3)$$

Theorem 7.1 — Complete native content algebra

Union, intersection, difference, symmetric difference, and translation extend to $\mathfrak{M}_\Sigma^{\text{ref}}$. The content extends uniquely by

$$\boxed{\widehat{\nu}([A_n]) = \lim_n \nu(A_n),} \quad (7.4)$$

where the limit is taken in the cut-rational completion.

The extended content is nonnegative, finitely additive on disjoint completed classes, translation invariant, and satisfies

$$|\widehat{\nu}(X) - \widehat{\nu}(Y)| \leq \widehat{\delta}(X, Y). \quad (7.5)$$

Proof

By (6.5), $(\nu(A_n))$ is cut-Cauchy whenever (A_n) is δ -Cauchy. Equation (6.5) also proves independence of representatives.

Finite-set inclusions give the Lipschitz estimates

$$\delta(A \cup C, B \cup D) \leq \delta(A, B) + \delta(C, D), \quad (7.6)$$

and analogous estimates for intersection, difference, and symmetric difference. Thus Boolean-ring operations descend to Cauchy classes. Translation is an isometry by (5.2).

Finite additivity is inherited by approximating disjoint completed classes with finite cylinders whose overlaps tend to zero in content. Completeness follows from the Cauchy-presentation diagonal construction using δ -entourages. QED.

This theorem proves a complete finite-content algebra. It does not yet claim a sigma-algebra or countable additivity for arbitrary completed families.

94.8 8. Nontrivial Cauchy cylinder

Let $R_n = (2^n, 1, 1)$, and let S_n contain the first $2^n - 1$ scale cells of one unit aperture. Then

$$\nu(S_n) = 1 - 2^{-n}. \quad (8.1)$$

For $m > n$,

$$\delta(S_n, S_m) = 2^{-n} - 2^{-m} < 2^{-n}. \quad (8.2)$$

Thus (S_n) is refinement-content Cauchy and represents a completed cylinder of content one. The object is constructed through cell refinement, not by first naming a real interval.

94.9 9. Native simple functions

A finite native simple function on a grid is a finite assignment

$$f : C_R \rightarrow \mathbb{K}_\Sigma^0. \quad (9.1)$$

Zero values may be omitted. Refinement copies a coarse value to all fine cells inside the coarse cell.

Define the tail integral

$$\mathcal{I}_{\text{tail}}(f) = \frac{1}{n_s n_t n_\alpha} \sum_c D_\Sigma(f(c)), \quad (9.2)$$

and the recognition integral

$$\mathcal{I}_{\text{rec}}(f) = \frac{1}{n_s n_t n_\alpha} \sum_c N_\Sigma(f(c)). \quad (9.3)$$

Theorem 9.1 — Refinement invariance

For every refinement $R' \succ R$,

$$\mathcal{I}_{\text{tail}}(\text{Ref}_{R'} f) = \mathcal{I}_{\text{tail}}(f), \quad (9.4)$$

$$\mathcal{I}_{\text{rec}}(\text{Ref}_{R'} f) = \mathcal{I}_{\text{rec}}(f). \quad (9.5)$$

Proof

Each value is copied to exactly $q_s q_t q_\alpha$ fine cells, while fine cell content is divided by the same factor. QED.

Theorem 9.2 — Translation invariance

For every integer cell translation,

$$\mathcal{I}_{\text{tail}}(T_h f) = \mathcal{I}_{\text{tail}}(f), \quad \mathcal{I}_{\text{rec}}(T_h f) = \mathcal{I}_{\text{rec}}(f). \quad (9.6)$$

The proof is finite reindexing.

These integrals are native finite sums. Their Cauchy completions into general integrable native functions remain T01-E4.

94.10 10. Foundational exclusion boundary

The construction above does not use as definitions or theorem sources:

```
Haar measure;  
Lebesgue measure;  
Caratheodory extension;  
real-coordinate volume;  
Jacobian density;  
ordinary Lp spaces;  
floating-point arithmetic.
```

The implementation core uses exact integer counts and **Fraction** coordinates. The Python type is a model of rational refinement content, not the mathematical definition.

94.11 11. Later coordinate shadow

After an actual UGD refinement adapter is proved, one may attempt to identify the three native partition channels with the normalized coordinates

$$(u, t, \alpha).$$

Only then may one prove that $\hat{\nu}$ corresponds to flat coordinate content and subsequently that the original (ϕ, σ, k) chart has density proportional to σ^{-2} .

That future statement would be a coordinate-shadow theorem. The density may not be used to generate the native content retroactively.

94.12 12. Exact certificate

The expected status is

PASS_NATIVE_REFINEMENT_COUNT_CONTENT_AND_SIMPLE_INTEGRALS

The certificate checks exact refinement invariance, Boolean identities, translation invariance, symmetric-difference continuity, a nontrivial Cauchy slab, and refinement/translation invariance of both native simple integrals.

94.13 13. Remaining gates

TO1-E1 NATIVE REFINEMENT CONTENT	PROVED
TO1-E2 COMPLETE NATIVE CONTENT ALGEBRA	PROVED
TO1-E3 NATIVE SIMPLE TAIL/RECOGNITION INTEGRALS	PROVED
TO1-E4 SIMPLE-FUNCTION CAUCHY COMPLETIONS	OPEN
TO1-E5 ACTUAL UGD REFINEMENT-CHANNEL ADAPTER	OPEN
TO1-E6 HAAR/JACOBIAN COORDINATE SHADOW	OPEN
ACTUAL MP PATH-ATLAS ADAPTER	OPEN
NATIVE STATE-PACKET GENERATION	OPEN
TARGET FAITHFULNESS	OPEN
NATIVE SHUNYA	OPEN
RIEMANN HYPOTHESIS	OPEN

RH-Framework/theorems/TO1_E_NATIVE_REFINEMENT_COUNT_MEASURE.md

Chapter 95

T01-A/B/C — Native Cut-Tail and Recognition-Cauchy Completion

Classification: PROVED

Foundational origin: NATIVE_DERIVED

Main T01 classification: OPEN

This theorem removes the first functional-analysis import from the recognition foundation. It does not begin with

```
ell1;  
ell2;  
Hilbert space;  
operator norm;  
Haar measure;  
Minkowski, Young, or an imported Cauchy-Schwarz theorem.
```

It begins with finite typed residue paths over exact oriented-cut scalars and defines two independent native completion relations.

The meta-foundation remains ordinary finite algebra, rational order, sequences, and equivalence relations. No claim of foundation-free mathematics is made. The relevant claim is narrower and precise: no pre-existing complex Hilbert or Banach structure generates the objects proved below.

95.1 1. Primitive finite path core

Let

$$\mathbb{K}_{\Sigma}^0 = \{r + t\iota_{\Sigma} : r, t \in \mathbb{Q}_{\Sigma}\}$$

be the exact cut field, with

$$\iota_{\Sigma}^2 = -1, \quad (r + t\iota_{\Sigma})^{\dagger} = r - t\iota_{\Sigma}.$$

Define the native radial square

$$N_{\Sigma}(z) = z^{\dagger}z = r^2 + t^2 \in \mathbb{Q}_{\Sigma,+}. \tag{1.1}$$

Let $\mathcal{G}_\Sigma$ be a typed residue groupoid. Its object and arrow sets may be countable, but every element used in the dense core has finite support. Write

$$\mathcal{P}_\Sigma^0 = C_c(\mathcal{G}_\Sigma, \mathbb{K}_\Sigma^0).$$

Convolution and dagger are defined entirely by path composition and path reversal:

$$(a \star b)_\gamma = \sum_{\alpha\beta=\gamma} a_\alpha b_\beta, \quad (1.2)$$

$$(a^\dagger)_\gamma = (a_{\gamma^{-1}})^\dagger. \quad (1.3)$$

All sums in this section are finite.

95.2 2. Native cut-tail mass

For a scalar $z = r + t\iota_\Sigma$, define

$$D_\Sigma(z) = |r|_\Sigma + |t|_\Sigma. \quad (2.1)$$

This is a rational cut gauge. It is not introduced as an absolute value on $\mathbb{C}$. For a finite path element

$$a = \sum_{\gamma} a_\gamma \gamma,$$

define its cut-tail mass

$$\boxed{\mathcal{M}_\Sigma(a) = \sum_{\gamma} D_\Sigma(a_\gamma).} \quad (2.2)$$

Lemma 2.1 — Scalar cut-mass product inequality

For all $z, w \in \mathbb{K}_\Sigma^0$,

$$D_\Sigma(zw) \leq D_\Sigma(z)D_\Sigma(w). \quad (2.3)$$

Proof

Write

$$z = r + t\iota_\Sigma, \quad w = s + u\iota_\Sigma.$$

Then

$$zw = (rs - tu) + (ru + ts)\iota_\Sigma.$$

The ordered cut-rational triangle inequality gives

$$\begin{aligned}
D_\Sigma(zw) &= |rs - tu|_\Sigma + |ru + ts|_\Sigma \\
&\leq |r||s| + |t||u| + |r||u| + |t||s| \\
&= (|r| + |t|)(|s| + |u|).
\end{aligned}$$

QED.

Theorem 2.2 — Native cut-tail convolution inequality

For all finite path elements,

$$\boxed{\mathcal{M}_\Sigma(a \star b) \leq \mathcal{M}_\Sigma(a)\mathcal{M}_\Sigma(b).} \quad (2.4)$$

Proof

Using (2.3) and the finite triangle inequality,

$$\begin{aligned}
\mathcal{M}_\Sigma(a \star b) &= \sum_{\gamma} D_\Sigma \left(\sum_{\alpha\beta=\gamma} a_\alpha b_\beta \right) \\
&\leq \sum_{\alpha, \beta \text{ composable}} D_\Sigma(a_\alpha) D_\Sigma(b_\beta) \\
&\leq \left(\sum_{\alpha} D_\Sigma(a_\alpha) \right) \left(\sum_{\beta} D_\Sigma(b_\beta) \right).
\end{aligned}$$

QED.

Dagger preserves cut-tail mass exactly:

$$\mathcal{M}_\Sigma(a^\dagger) = \mathcal{M}_\Sigma(a). \quad (2.5)$$

95.3 3. Native cut-tail completion

A sequence (a_n) in $\mathcal{P}_\Sigma^0$ is **cut-tail Cauchy** when for every positive cut rational ε , there exists N such that

$$\mathcal{M}_\Sigma(a_m - a_n) < \varepsilon \quad (m, n \geq N). \quad (3.1)$$

Two cut-tail Cauchy sequences are equivalent when

$$\mathcal{M}_\Sigma(a_n - b_n) \longrightarrow 0 \quad (3.2)$$

in rational-threshold language.

Define

$$\boxed{\mathfrak{A}_\Sigma^{\text{tail}} = \{\text{cut-tail Cauchy presentations}\} / \sim_{\mathcal{M}} .} \quad (3.3)$$

Lemma 3.1 — Cauchy presentations are mass bounded

Every cut-tail Cauchy sequence (a_n) admits a positive rational bound B such that

$$\mathcal{M}_\Sigma(a_n) \leq B$$

for every n .

Proof

Choose N such that

$$\mathcal{M}_\Sigma(a_n - a_N) < 1 \quad (n \geq N).$$

Then

$$\mathcal{M}_\Sigma(a_n) \leq \mathcal{M}_\Sigma(a_N) + 1$$

for $n \geq N$. Enlarge the bound to include the finitely many earlier terms. QED.

Theorem 3.2 — Native cut-tail algebra

Termwise addition, convolution, and dagger descend to $\mathfrak{A}_\Sigma^{\text{tail}}$. The resulting object is a complete involutive path algebra.

Proof

Let (a_n) and (b_n) be cut-tail Cauchy and mass bounded by A and B . Then

$$\begin{aligned} \mathcal{M}_\Sigma(a_n \star b_n - a_m \star b_m) &\leq \mathcal{M}_\Sigma((a_n - a_m) \star b_n) \\ &\quad + \mathcal{M}_\Sigma(a_m \star (b_n - b_m)) \\ &\leq B\mathcal{M}_\Sigma(a_n - a_m) + A\mathcal{M}_\Sigma(b_n - b_m). \end{aligned} \tag{3.4}$$

Thus termwise convolution is Cauchy. The same estimate shows independence of representatives. Dagger is immediate from (2.5). Associativity and the star law hold termwise on the finite core and therefore on equivalence classes.

Completeness follows from the standard Cauchy-presentation diagonal argument, but every neighborhood and estimate used in that argument is expressed through $\mathcal{M}_\Sigma$ and positive cut-rational thresholds. No pre-existing L^1 space is used to define the object. QED.

The phrase **cut-tail completion** refers to (3.3). Any later identification with an ordinary sequence space is a separate coordinate theorem.

95.4 4. Native closed-loop recognition energy

Let Φ_Σ extract stationary zero-residue paths. Define

$$\langle a, b \rangle_\Sigma^0 = \Phi_\Sigma(a^\dagger \star b). \tag{4.1}$$

The typed path rules give the exact coefficient identity

$$\boxed{\langle a, b \rangle_{\Sigma}^0 = \sum_{\gamma} a_{\gamma}^{\dagger} b_{\gamma}.} \quad (4.2)$$

Define recognition energy

$$\boxed{\mathcal{E}_{\Sigma}(a) = \langle a, a \rangle_{\Sigma}^0 = \sum_{\gamma} N_{\Sigma}(a_{\gamma}).} \quad (4.3)$$

It is a nonnegative radial cut rational, and

$$\mathcal{E}_{\Sigma}(a) = 0 \iff a = 0. \quad (4.4)$$

Lemma 4.1 — Native addition control

For finite paths,

$$\boxed{\mathcal{E}_{\Sigma}(a + b) \leq 2\mathcal{E}_{\Sigma}(a) + 2\mathcal{E}_{\Sigma}(b).} \quad (4.5)$$

Proof

For every pair of cut scalars,

$$2N_{\Sigma}(x) + 2N_{\Sigma}(y) - N_{\Sigma}(x + y) = N_{\Sigma}(x - y) \geq 0. \quad (4.6)$$

Sum (4.6) over path coefficients. QED.

95.5 5. Native finite Cauchy-Schwarz identity

The required recognition estimate is not imported from a Hilbert space.

Let $x_1, \dots, x_n$ be cut scalars and let $d_1, \dots, d_n$ be positive cut rationals. Put

$$D = \sum_i d_i.$$

Lemma 5.1 — Weighted cut-square identity

$$\boxed{D \sum_i \frac{N_{\Sigma}(x_i)}{d_i} - N_{\Sigma} \left(\sum_i x_i \right) = \sum_{i < j} \frac{N_{\Sigma}(d_j x_i - d_i x_j)}{d_i d_j}.} \quad (5.1)$$

Proof

Expand both sides using

$$N_{\Sigma}(z) = z^{\dagger} z.$$

The coefficient of $N_{\Sigma}(x_i)$ on each side is

$$\frac{D - d_i}{d_i},$$

and every mixed pair contributes

$$-x_i^\dagger x_j - x_j^\dagger x_i.$$

Thus the expressions agree exactly. Every term on the right is a positive radial square. QED.

Corollary 5.2 — Native coefficient pairing inequality

For finite path elements,

$$\boxed{N_\Sigma(\langle a, b \rangle_\Sigma^0) \leq \mathcal{E}_\Sigma(a) \mathcal{E}_\Sigma(b).} \quad (5.2)$$

This also follows from the unweighted finite identity

$$\mathcal{E}_\Sigma(a) \mathcal{E}_\Sigma(b) - N_\Sigma(\langle a, b \rangle_\Sigma^0) = \sum_{\gamma < \delta} N_\Sigma(a_\gamma b_\delta - a_\delta b_\gamma). \quad (5.3)$$

No square-root norm is required.

95.6 6. Native recognition-Cauchy completion

A sequence (ξ_n) of finite paths is **recognition Cauchy** when for every positive cut rational ε , there exists N such that

$$\mathcal{E}_\Sigma(\xi_m - \xi_n) < \varepsilon \quad (m, n \geq N). \quad (6.1)$$

Two recognition-Cauchy presentations are equivalent when

$$\mathcal{E}_\Sigma(\xi_n - \eta_n) \longrightarrow 0. \quad (6.2)$$

Define

$$\boxed{\mathfrak{R}_\Sigma = \{\text{recognition-Cauchy presentations}\} / \sim_{\mathcal{E}} .} \quad (6.3)$$

Theorem 6.1 — Native recognition completion

The object $\mathfrak{R}_\Sigma$ is a complete module over the completed cut field. The pairing (4.1) extends uniquely to

$$\langle \cdot, \cdot \rangle_\Sigma : \mathfrak{R}_\Sigma \times \mathfrak{R}_\Sigma \longrightarrow \widehat{\mathbb{K}}_\Sigma, \quad (6.4)$$

and satisfies

$$\langle \xi, \xi \rangle_\Sigma \geq 0, \quad \langle \xi, \xi \rangle_\Sigma = 0 \iff \xi = 0. \quad (6.5)$$

Proof

Addition is continuous by (4.5), and scalar multiplication obeys

$$\mathcal{E}_\Sigma(z\xi) = N_\Sigma(z)\mathcal{E}_\Sigma(\xi). \quad (6.6)$$

Recognition-Cauchy sequences are energy bounded by the same argument as Lemma 3.1, using (4.5).

For Cauchy presentations $(\xi_n), (\eta_n)$, the difference

$$\langle \xi_n, \eta_n \rangle^0 - \langle \xi_m, \eta_m \rangle^0$$

is the sum of

$$\langle \xi_n - \xi_m, \eta_n \rangle^0 \quad \text{and} \quad \langle \xi_m, \eta_n - \eta_m \rangle^0.$$

Use (4.6) and (5.2). Energy boundedness makes both terms converge to zero in cut norm square. Therefore the pairing sequence is cut-Cauchy and independent of representatives.

Positivity follows from (4.3). If the completed self-pairing is zero, then $\mathcal{E}_\Sigma(\xi_n)$ converges to zero, so the presentation is recognition-null and represents the zero class.

Completeness follows from the Cauchy-presentation diagonal construction using only the entourages (6.1). QED.

This object is called the **recognition completion**. The theorem does not use the word Hilbert as a definition. A later chart may prove that its coordinate image is an ordinary Hilbert space.

95.7 7. Native tail action on recognition completion

Theorem 7.1 — Finite native action bound

For finite paths a and ξ ,

$$\boxed{\mathcal{E}_\Sigma(a \star \xi) \leq \mathcal{M}_\Sigma(a)^2 \mathcal{E}_\Sigma(\xi).} \quad (7.1)$$

Proof

Fix an output arrow γ . For each decomposition $\alpha\beta = \gamma$, put

$$x_\alpha = a_\alpha \xi_\beta, \quad d_\alpha = D_\Sigma(a_\alpha).$$

Terms with $a_\alpha = 0$ are omitted, so $d_\alpha > 0$. Lemma 5.1 gives

$$N_\Sigma \left(\sum_{\alpha\beta=\gamma} a_\alpha \xi_\beta \right) \leq \mathcal{M}_\Sigma(a) \sum_{\alpha\beta=\gamma} \frac{N_\Sigma(a_\alpha \xi_\beta)}{d_\alpha}. \quad (7.2)$$

Since

$$N_\Sigma(a_\alpha) \leq D_\Sigma(a_\alpha)^2 = d_\alpha^2,$$

we obtain

$$N_{\Sigma}((a \star \xi)_{\gamma}) \leq \mathcal{M}_{\Sigma}(a) \sum_{\alpha\beta=\gamma} d_{\alpha} N_{\Sigma}(\xi_{\beta}). \quad (7.3)$$

Sum over γ . Every composable pair produces one output arrow, and the sum over composable pairs is bounded by the unrestricted product of the two finite positive sums. Hence

$$\mathcal{E}_{\Sigma}(a \star \xi) \leq \mathcal{M}_{\Sigma}(a)^2 \mathcal{E}_{\Sigma}(\xi).$$

QED.

Theorem 7.2 — Completed native action

Convolution extends uniquely to an action

$$\boxed{\mathfrak{A}_{\Sigma}^{\text{tail}} \times \mathfrak{R}_{\Sigma} \longrightarrow \mathfrak{R}_{\Sigma}.} \quad (7.4)$$

The completed action obeys the native bound (7.1) in cut-completed radial coordinates.

Proof

Let (a_n) be mass Cauchy and (ξ_n) recognition Cauchy. By (4.5) and (7.1),

$$\begin{aligned} \mathcal{E}_{\Sigma}(a_n \star \xi_n - a_m \star \xi_m) &\leq 2\mathcal{E}_{\Sigma}((a_n - a_m) \star \xi_n) \\ &\quad + 2\mathcal{E}_{\Sigma}(a_m \star (\xi_n - \xi_m)) \\ &\leq 2\mathcal{M}_{\Sigma}(a_n - a_m)^2 \mathcal{E}_{\Sigma}(\xi_n) \\ &\quad + 2\mathcal{M}_{\Sigma}(a_m)^2 \mathcal{E}_{\Sigma}(\xi_n - \xi_m). \end{aligned} \quad (7.5)$$

Mass and energy boundedness show that (7.5) tends to zero. The same estimate proves independence of representatives. QED.

This is native boundedness: preservation of recognition-Cauchy neighborhoods by a cut-tail mass estimate. No operator norm is primitive.

95.8 8. Residue is not lost by completion

Let $r \neq s$ be residue channels over the same endpoints. Then

$$w = \delta_{(x,y,r)} - \delta_{(x,y,s)}$$

has

$$\mathcal{E}_{\Sigma}(w) = 2, \quad (8.1)$$

while the endpoint-only shadow sends w to zero.

Thus neither the cut-tail completion nor the recognition completion erases seam memory. Blindness enters only through a later quotient map.

95.9 9. Explicit infinite native presentation

For the integer object atlas with two residue channels, define

$$s_N = \sum_{n=0}^N 2^{-(n+1)} \delta_{(0,n,n \bmod 2)}. \quad (9.1)$$

For $M > N$,

$$\mathcal{M}_\Sigma(s_M - s_N) = \sum_{n=N+1}^M 2^{-(n+1)} < 2^{-(N+1)}, \quad (9.2)$$

and

$$\mathcal{E}_\Sigma(s_M - s_N) = \sum_{n=N+1}^M 4^{-(n+1)} < \frac{1}{3 \cdot 4^{N+1}}. \quad (9.3)$$

Therefore the same finite presentation converges in both native completion relations and retains alternating residue over infinitely many targets.

95.10 10. What has been removed

The construction order is now

cut scalars $\rightarrow$ finite typed paths $\rightarrow \mathcal{M}_\Sigma, \mathcal{E}_\Sigma \rightarrow$ two native Cauchy quotients $\rightarrow$ native completed action.

The following are not used to define or prove this chain:

ordinary complex numbers;
 ell1 or ell2 spaces;
 Banach or Hilbert completeness as imported objects;
 operator norms;
 Haar measure;
 Minkowski inequality;
 Young inequality;
 spectral theory.

Finite algebraic cut-square identities replace the imported analytic bounds.

95.11 11. Coordinate-shadow boundary

This theorem does not deny that a later coordinate map may identify

native cut-tail completion	with an ordinary absolute-summability space;
native recognition completion	with an ordinary square-summability space;
native completed action	with a bounded regular representation.

Those identifications must be proved **after** the native objects exist. They are coordinate-shadow theorems, not definitions.

95.12 12. Exact certificate

The exact dense-core certificate has expected status

PASS_NATIVE_CUT_TAIL_AND_RECOGNITION_CAUCHY_COMPLETION_CORE

It verifies:

```
quarter-turn scalar algebra;
dagger and radial squares;
cut-tail scalar and convolution inequalities;
finite path associativity and dagger reversal;
closed-loop coefficient identity;
native addition control;
weighted cut Cauchy-Schwarz identity;
native action-energy bound;
explicit mass and energy Cauchy tails;
nonzero residue with zero endpoint shadow;
absence of classical-analysis imports in the foundational module.
```

The certificate is an implementation witness. The proof above establishes the abstract completion theorem.

95.13 13. Remaining gates

TO1-A NATIVE CUT-TAIL COMPLETION	PROVED
TO1-B NATIVE RECOGNITION-CAUCHY COMPLETION	PROVED
TO1-C NATIVE COMPLETED PATH ACTION	PROVED
ACTUAL MP PATH-ATLAS IDENTIFICATION	OPEN
ACTUAL MP REFINEMENT CONTINUITY	OPEN
NATIVE CONTINUOUS REFINEMENT MEASURE	OPEN
COORDINATE SHADOW TO ORDINARY e111/e112	OPEN
NATIVE STATE-PACKET GENERATION	OPEN
TARGET FAITHFULNESS	OPEN
NATIVE SHUNYA	OPEN
RIEMANN HYPOTHESIS	OPEN

The next native reconstruction gate is a refinement-count measure derived from path partitions before any Haar measure is named.

RH-Framework/theorems/TO1_NATIVE_SUMMABILITY_AND_RECOGNITION_COMPLETION.md

Chapter 96

T03-A — Action-Domain Theorem on the Normalized UGD Path Group

Classification: PROVED

Main T03 classification: OPEN

The abstract operator-domain theorem is proved below. The actual MP coefficient fields and their immutable state-locked extraction remain separate verification obligations.

96.1 1. Carrier and Haar measure

The normalized scalar UGD path group is

$$\mathcal{U}_F \cong \mathbb{R}_u \times \mathbb{R}_t \times S_\alpha^1,$$

with Haar measure

$$d\lambda = du \, dt \, d\alpha.$$

In the original phase-scale-seam coordinates this is, up to a fixed normalization constant,

$$d\lambda = \frac{d\phi \, d\sigma \, dk}{\sigma^2}.$$

The active event fibre has exactly five labels:

$$E_5 = \{\ell_\partial, \phi_0, \phi_1, \phi_2, e_3^{\text{corr}}\}.$$

Set

$$\mathcal{H} = L^2(\mathcal{U}_F, d\lambda; \mathbb{C}^5).$$

The matrix factor represents pair-groupoid directions over these five labels. Seam memory remains a path coordinate. It is not a sixth label.

96.2 2. Bounded fibre multipliers

Let

$$A : \mathcal{U}_F \rightarrow M_5(\mathbb{C})$$

be strongly measurable and essentially bounded in operator norm. Define

$$(M_A f)(x) = A(x)f(x).$$

Theorem 2.1

M_A is bounded on $\mathcal{H}$ and

$$\|M_A f\|_2 \leq \operatorname{ess\,sup}_{x \in \mathcal{U}_F} \|A(x)\|_{\text{op}} \|f\|_2.$$

Proof

For almost every x ,

$$\|A(x)f(x)\|_{\mathbb{C}^5} \leq \|A(x)\|_{\text{op}} \|f(x)\|_{\mathbb{C}^5}.$$

Squaring and integrating gives

$$\|M_A f\|_2^2 \leq \|A\|_{L^\infty(\text{op})}^2 \|f\|_2^2.$$

Hence $M_A \in \mathcal{B}(\mathcal{H})$. QED.

96.3 3. Integrable matrix convolution kernels

Let

$$K : \mathcal{U}_F \rightarrow M_5(\mathbb{C})$$

be measurable with

$$\|K\|_{L^1(\text{op})} = \int_{\mathcal{U}_F} \|K(y)\|_{\text{op}} d\lambda(y) < \infty.$$

Define

$$(T_K f)(x) = \int_{\mathcal{U}_F} K(y)f(y^{-1}x) d\lambda(y).$$

Theorem 3.1

T_K is bounded on $\mathcal{H}$ and

$$\|T_K f\|_2 \leq \|K\|_{L^1(\text{op})} \|f\|_2.$$

Proof

By the pointwise matrix bound and Minkowski's integral inequality,

$$\|T_K f\|_2 \leq \int_{\mathcal{U}_F} \|K(y)f(y^{-1}\cdot)\|_2 d\lambda(y).$$

Left translation is unitary on L^2 because λ is Haar measure. Therefore

$$\|K(y)f(y^{-1}\cdot)\|_2 \leq \|K(y)\|_{\text{op}} \|f\|_2.$$

Integration proves the bound. QED.

Since $\mathcal{U}_F$ is abelian and unimodular, the involution is

$$K^*(x) = K(x^{-1})^*,$$

and the matrix-kernel L^1 algebra has the usual convolution star structure.

96.4 4. Maximal domains for unbounded generators

Let

$$m : \mathcal{U}_F \rightarrow \mathbb{R}$$

be measurable and finite almost everywhere. Define

$$(M_m f)(x) = m(x)f(x)$$

on the maximal domain

$$D(M_m) = \{f \in \mathcal{H} : mf \in \mathcal{H}\}.$$

Theorem 4.1

M_m is densely defined, closed, and self-adjoint.

Proof

The group is second countable and locally compact, so Haar measure is sigma-finite. The sets

$$E_n = \{x : |m(x)| \leq n\}$$

increase to full measure. Compactly supported square-integrable functions restricted to E_n lie in $D(M_m)$, proving density.

Suppose

$$f_n \rightarrow f, \quad M_m f_n \rightarrow g$$

in $\mathcal{H}$. Passing to an almost-everywhere convergent subsequence gives

$$f_n(x) \rightarrow f(x), \quad m(x)f_n(x) \rightarrow g(x).$$

Thus $g(x) = m(x)f(x)$ almost everywhere. Hence $f \in D(M_m)$ and $M_m f = g$, proving closedness.

For real m , the pointwise resolvents

$$(m \pm i)^{-1}$$

are bounded by one. Therefore $M_m \pm iI$ are onto, and the symmetric maximal multiplication operator is self-adjoint. QED.

Hermitian matrix extension

Let

$$H(x) = H(x)^* \in M_5(\mathbb{C})$$

be a measurable Hermitian field and define

$$D(M_H) = \{f : Hf \in \mathcal{H}\}.$$

The same argument applies because

$$\|(H(x) \pm iI_5)^{-1}\|_{\text{op}} \leq 1.$$

Thus M_H is self-adjoint on its maximal domain. This is the correct home for unbounded prime-log, scale, Euler, or Gamma-generated matrix fields once their actual formulas are imported.

96.5 5. Constant state-locked packets

Let

$$A_0 \in M_5(\mathbb{C})$$

be a finite state-locked source, response, transport, or Gram-derived action. The constant field

$$A(x) = A_0$$

belongs to $L^\infty(\mathcal{U}_F; M_5)$. Hence

$$\|M_A\| \leq \|A_0\|_{\text{op}}.$$

Therefore a finite packet embeds canonically as a bounded decomposable fibre action.

It must not be silently reinterpreted as the constant convolution kernel

$$K(x) = A_0.$$

The group has infinite Haar measure, so for $A_0 \neq 0$,

$$\int_{\mathcal{U}_F} \|A_0\|_{\text{op}} d\lambda = \infty.$$

The finite packet is a multiplier, not an integrable convolution kernel. A Dirac-supported convolution action would require a declared measure-algebra formulation.

96.6 6. Compact positive-eta families

Let $I \subset (0, \infty)$ be compact and suppose

$$\eta \mapsto A_\eta \in M_5(\mathbb{C})$$

is continuous. Compactness gives

$$\sup_{\eta \in I} \|A_\eta\|_{\text{op}} < \infty.$$

Consequently

$$(\mathbf{M}_A f)(\eta, x) = A_\eta f(\eta, x)$$

is bounded on

$$L^2(I \times \mathcal{U}_F; \mathbb{C}^5).$$

Thus a verified continuous positive-eta matrix family has a common bounded fibre-action domain on every compact positive-eta interval. This does not prove eta-zero support descent.

96.7 7. Cut-aperture compatibility

Let

$$P = P^* = P^2 \in M_5(\mathbb{C})$$

be the state-locked cut aperture. The constant fibre action M_P is an orthogonal projection on $\mathcal{H}$.

For each bounded field A ,

$$\|PAP\|_{L^\infty(\text{op})} \leq \|A\|_{L^\infty(\text{op})}.$$

If a measurable unitary transport satisfies

$$U(x)PU(x)^* = P$$

almost everywhere, then $M_P \mathcal{H}$ is invariant under M_U . Seam memory remains in the path coordinate, with no new fibre label.

96.8 8. Resulting action classes

The following classes are rigorous on the normalized UGD carrier:

L-infinity matrix fields	-> bounded fibre multipliers on L2
L1 matrix kernels	-> bounded convolution operators on L2
real scalar weights	-> self-adjoint maximal multiplication operators
Hermitian matrix weights	-> self-adjoint maximal multiplication operators
constant five-label packets	-> bounded fibre multipliers
compact positive-eta family	-> common bounded decomposable action
cut-aperture compression	-> bounded recognized-sector action

96.9 9. Remaining subtheorems

T03-B — finite packet realization

Classification: OPEN

The type theorem is proved, but RH-Framework must execute the pinned state-locked source and import immutable matrices with their source hashes.

T03-C — actual coefficient-domain membership

Classification: OPEN

Every actual source, response, prime, Gamma, boundary, continuation, Euler, scale, and compatibility action must be proved to belong to one declared class:

```
essentially bounded matrix field
L1 matrix convolution kernel
finite measure-algebra kernel
closed maximal multiplication operator
controlled unbounded integral operator with explicit graph domain
```

Numerical samples alone do not establish integrability on the full UGD group.

96.10 10. Computational witness

The companion code checks a finite Haar-grid dense core with positive and adversarial controls.

```
PASS_T03_ACTION_DOMAIN_DENSE_CORE
05a4784308e996763928b4cead0c86462150154c506b30bf292f96dd8986db3c
```

The certificate verifies the implementation. The proof above establishes the continuum theorem.

96.11 11. Claim boundary

T03-A is proved. T03-B, T03-C, target faithfulness, native Shunya, the critical-line adapter, and the Riemann Hypothesis remain open.

RH-Framework/theorems/T03_ACTION_DOMAIN_ADAPTER.md

Chapter 97

T03-C1 — Actual Finite Packet Action Inventory

Classification: OPEN

The action-typing theorem is proved. The actual MP packet has not yet been executed inside RH-Framework, so its immutable array hashes are not yet fixed in this repository.

97.1 1. Purpose

T03-A proves the available operator domains on

$$\mathcal{H} = L^2(\mathcal{U}_F, d\lambda; \mathbb{C}^5).$$

T03-C1 answers the next question:

Which arrays in the state-locked five-label packet are operators, which are carrier data, and what operator class does each operator belong to?

The distinction is required before any source, response, target-faithfulness, or Shunya argument can use the packet.

97.2 2. Required packet

The pinned MP extractor returns one carrier vector and seven matrices.

Carrier data

$$v_{\partial} \in \mathbb{C}^5$$

stored as

`corrected_defect_line`

is a vector. It is not an operator.

Matrix actions

```
source_defect_projector
accepted_complement_projector
source_metric
response_metric
loop_holonomy
memory_holonomy
compensated_holonomy
```

Each must have shape exactly 5 x 5. A sixth coordinate or sixth label is rejected.

97.3 3. Projector typing

Define

$$P_{\partial} = \frac{v_{\partial} v_{\partial}^*}{\langle v_{\partial}, v_{\partial} \rangle}.$$

The source packet must satisfy

$$P_{\partial}^* = P_{\partial}, \quad P_{\partial}^2 = P_{\partial}, \quad P_{\partial} v_{\partial} = v_{\partial}.$$

Let $Q_+ = I_5 - P_{\partial}$. The accepted-complement packet must satisfy

$$Q_+^* = Q_+, \quad Q_+^2 = Q_+, \quad P_{\partial} Q_+ = Q_+ P_{\partial} = 0, \quad P_{\partial} + Q_+ = I_5.$$

Typing conclusion

The constant fields

$$x \mapsto P_{\partial}, \quad x \mapsto Q_+$$

are bounded orthogonal-projection fibre multipliers on $\mathcal{H}$, with full domain $\mathcal{H}$.

97.4 4. Metric typing

Let

$$G_{\text{src}}, G_{\text{rsp}} \in M_5(\mathbb{C})$$

be the extracted source and response metrics. The verifier requires

$$G_{\text{src}}^* = G_{\text{src}}, \quad G_{\text{rsp}}^* = G_{\text{rsp}},$$

and

$$G_{\text{src}} \geq 0, \quad G_{\text{rsp}} \geq 0.$$

Typing conclusion

The constant fields defined by these matrices are bounded positive semidefinite fibre multipliers on $\mathcal{H}$. Their operator norms are their largest eigenvalues.

This statement does not identify either metric with a convolution kernel and does not prove equivalence of any infinite completions.

97.5 5. Holonomy typing

Let

$$H_{\square}, \quad K_{\square}, \quad C_{\square}$$

be the loop, memory, and compensated holonomies. The verifier requires

$$H_{\square}^* H_{\square} = H_{\square} H_{\square}^* = I_5,$$

$$K_{\square}^* K_{\square} = K_{\square} K_{\square}^* = I_5,$$

and

$$C_{\square} = I_5.$$

It also requires

$$H_{\square} P_{\partial} H_{\square}^* = P_{\partial}, \quad K_{\square} P_{\partial} K_{\square}^* = P_{\partial}.$$

Typing conclusion

The loop and memory matrices are bounded unitary fibre multipliers. The compensated holonomy is the identity fibre multiplier. The recognized defect subspace is invariant under both loop and memory transport.

Seam memory remains path data acting through a transport operator. It does not create another event label.

97.6 6. General finite-action typing theorem

Theorem 6.1

Let $A \in M_5(\mathbb{C})$. The constant field

$$\tilde{A}(x) = A$$

is an essentially bounded measurable matrix field on $\mathcal{U}_F$, and the associated fibre multiplier satisfies

$$\|M_{\tilde{A}} f\|_2 \leq \|A\|_{\text{op}} \|f\|_2.$$

Therefore every finite packet matrix has a canonical bounded action on $\mathcal{H}$.

Proof

The field is measurable and

$$\operatorname{ess\,sup}_{x \in \mathcal{U}_F} \|\tilde{A}(x)\|_{\text{op}} = \|A\|_{\text{op}} < \infty.$$

Apply T03-A, Theorem 2.1. QED.

Important exclusion

A nonzero constant matrix field is generally not an L^1 convolution kernel on the noncompact UGD group. Its Haar integral diverges. T03-C1 classifies the finite matrices as fibre multipliers only.

97.7 7. Immutable evidence rule

Each extracted array is hashed separately using SHA-256 over:

```
canonical complex128 dtype
exact shape
C-order byte representation
```

The NPZ file is a transport container. The individual array hashes in the JSON certificate are the immutable mathematical evidence identifiers, because ZIP container metadata is not the array identity.

The verifier records:

```
array name
shape
dtype
SHA-256
operator norm
Hermitian residual
unitarity residual
minimum Hermitian eigenvalue
```

The carrier vector receives its own SHA-256 and Euclidean norm, but no operator classification.

97.8 8. Promotion rule

T03-B and T03-C1 are promoted to PROVED together only if all of the following hold:

```
operator MP worktree matches the pinned commit and blobs
state MP worktree matches the pinned commit and blobs
both worktrees are clean
the MP carrier interface passes with no false checks
the exact five-label order is preserved
all seven matrices pass their declared algebraic tests
every array has an individual SHA-256
no sixth label is created
RH is not promoted
```

Required generated status:

```
PASS_RH_FRAMEWORK_T03_C1_PINNED_ACTION_INVENTORY
```


97.9 9. Current implementation

Implemented files:

```
src/rh_framework/action_inventory.py
src/rh_framework/pinned_mp_action_inventory.py
tests/test_action_inventory.py
tests/test_pinned_mp_action_inventory.py
tools/verify_t03.py
```

The pure inventory layer has positive and adversarial controls for:

```
missing arrays
six-label matrices
non-Hermitian metrics
nonunitary holonomy
wrong complementary projection
false carrier-interface checks
dirty source worktrees
wrong label order
```

97.10 10. Remaining T03-C work

T03-C1 closes only the finite constant-action inventory. The following remain open:

T03-C2 — amplitude-family multipliers

Prove measurability and compact-eta essential bounds for the actual source and response amplitude families.

T03-C3 — arithmetic and Gamma actions

Classify prime, Gamma, boundary, and compatibility actions as bounded multipliers, integrable kernels, measure-algebra kernels, or controlled unbounded operators.

T03-C4 — generator graph domains

Specify and prove closed graph domains for Euler, scale, prime-log, and related unbounded generators.

T03-C5 — aperture exhaustion

Prove convergence of finite apertures in the operator norm, strong topology, or declared graph norm appropriate to each action.

97.11 11. Claim boundary

The finite-action typing theorem is proved. The actual T03-C1 packet theorem is open until the pinned private run fixes its array hashes. T03-C2 through T03-C5, target faithfulness, native Shunya, the critical-line adapter, and the Riemann Hypothesis remain open.

RH-Framework/theorems/T03_C1_FINITE_PACKET_ACTION_INVENTORY.md

Part VII

RH-Framework Notebooks

Chapter 98

T01 Native Foundation — Jupyter Reproduction

This procedure uses Python cells only.

98.1 1. Repository path

```
from pathlib import Path
import subprocess
import sys

RH_REPO = Path(
    r"C:\Users\abc\Desktop\New Folder (2)\New folder\PROVISNAL RELATED\RH-Framework"
).resolve()

print("repository:", RH_REPO)
```

Change only the path when the local folder differs.

98.2 2. Git audit

```
def run_git(repo: Path, *args: str) -> str:
    completed = subprocess.run(
        ["git", "-C", str(repo), *args],
        check=True,
        capture_output=True,
        text=True,
    )
    return completed.stdout.strip()

print("branch:", run_git(RH_REPO, "branch", "--show-current"))
print("head:", run_git(RH_REPO, "rev-parse", "HEAD"))
print("dirty:", repr(run_git(RH_REPO, "status", "--porcelain")))
```

Expected branch:

agent/native-summability-reconstruction

98.3 3. Install the package

```
completed = subprocess.run(
    [sys.executable, "-m", "pip", "install", "-e", str(RH_REPO)],
    text=True,
    capture_output=True,
)
print(completed.stdout)
print(completed.stderr)
print("return code:", completed.returncode)
assert completed.returncode == 0
```

98.4 4. Run the native foundation verifier

```
completed = subprocess.run(
    [
        sys.executable,
        str(RH_REPO / "tools" / "verify_native_foundation.py"),
    ],
    cwd=RH_REPO,
    text=True,
    capture_output=True,
)
print(completed.stdout)
print(completed.stderr)
print("return code:", completed.returncode)
assert completed.returncode == 0
```

Required status:

PASS_NATIVE_CUT_TAIL_AND_RECOGNITION_CAUCHY_COMPLETION_CORE

Required certificate hash:

4e88898fc93010a6ba189f932817d0715298f106aff61528bb2e58acb8c6dbfd

The verifier also confirms that the foundational module imports only:

```
__future__
dataclasses
fractions
numbers
typing
```

98.5 5. Regenerate the certificate independently

```
OUTPUT = (
    RH_REPO
    / "certificates"
    / "T01_NATIVE_SUMMABILITY_REGENERATED.json"
)

completed = subprocess.run(
    [
        sys.executable,
        "-m",
```

```

        "rh_framework.native_summability_certificate",
        "--output",
        str(OUTPUT),
    ],
    cwd=RH_REPO,
    text=True,
    capture_output=True,
)
print(completed.stdout)
print(completed.stderr)
print("return code:", completed.returncode)
assert completed.returncode == 0

```

98.6 6. Compare generated and committed evidence

```

import json

COMMITTED = (
    RH_REPO
    / "certificates"
    / "T01_NATIVE_SUMMABILITY_v0.1.json"
)

generated = json.loads(OUTPUT.read_text(encoding="utf-8"))
committed = json.loads(COMMITTED.read_text(encoding="utf-8"))

print("equal:", generated == committed)
print("status:", generated["status"])
print("hash:", generated["certificate_hash"])
print("origin:", generated["foundational_origin"])
print("actual MP adapter:",
    ↪ generated["theorem_classification"]["T01_D_actual_mp_path_groupoid_adapter"])

assert generated == committed
assert generated["foundational_origin"] == "NATIVE_DERIVED"
assert generated["theorem_classification"]["riemann_hypothesis"] == "OPEN"

```

98.7 7. Inspect the decisive native inequalities

```

for key in [
    "sample_a_tail_mass",
    "sample_product_tail_mass",
    "sample_a_recognition_energy",
    "sample_product_recognition_energy",
    "native_action_bound_left",
    "native_action_bound_right",
    "weighted_cauchy_schwarz_remainder",
    "mass_tail_N4_M9",
    "energy_tail_N4_M9",
]:
    print(key, generated["exact_metrics"][key])

```

The required action inequality is witnessed by

```
native_action_bound_left <= native_action_bound_right
```

with exact rational values, not floating-point tolerances.

98.8 Claim boundary

A passing run verifies the exact dense core and reproduces the certificate for:

```
native cut-tail completion;  
native recognition-Cauchy completion;  
native completed path action.
```

It does not prove:

```
actual MP path-atlas identification;  
native continuous refinement measure;  
coordinate equivalence with ordinary ell1 or ell2;  
native state-packet generation;  
target faithfulness;  
Shunya;  
Riemann Hypothesis.
```

RH-Framework/notebooks/T01_NATIVE_FOUNDATION_JUPYTER.md

Chapter 99

T01 Native Reconstruction — Complete Jupyter Reproduction

This procedure uses Python cells only.

99.1 1. Repository

```
from pathlib import Path
import subprocess
import sys

RH_REPO = Path(
    r"C:\Users\abc\Desktop\New Folder (2)\New folder\PROVISNAL RELATED\RH-Framework"
).resolve()

print("repository:", RH_REPO)
```

Change only the local path when needed.

99.2 2. Git audit

```
def run_git(repo: Path, *args: str) -> str:
    completed = subprocess.run(
        ["git", "-C", str(repo), *args],
        check=True,
        capture_output=True,
        text=True,
    )
    return completed.stdout.strip()

print("branch:", run_git(RH_REPO, "branch", "--show-current"))
print("head:", run_git(RH_REPO, "rev-parse", "HEAD"))
print("dirty:", repr(run_git(RH_REPO, "status", "--porcelain")))
```

Expected branch:

```
agent/native-summability-reconstruction
```

99.3 3. Install

```
completed = subprocess.run(
    [sys.executable, "-m", "pip", "install", "-e", str(RH_REPO)],
    text=True,
    capture_output=True,
)
print(completed.stdout)
print(completed.stderr)
assert completed.returncode == 0
```

99.4 4. Verify every native layer

```
completed = subprocess.run(
    [
        sys.executable,
        str(RH_REPO / "tools" / "verify_native_foundation.py"),
    ],
    cwd=RH_REPO,
    text=True,
    capture_output=True,
)
print(completed.stdout)
print(completed.stderr)
print("return code:", completed.returncode)
assert completed.returncode == 0
```

Required statuses and hashes:

```
PASS_NATIVE_CUT_TAIL_AND_RECOGNITION_CAUCHY_COMPLETION_CORE
4e88898fc93010a6ba189f932817d0715298f106aff61528bb2e58acb8c6dbfd
```

```
PASS_NATIVE_REFINEMENT_COUNT_CONTENT_AND_SIMPLE_INTEGRALS
1366eeb443efc661c614a3bf8e7b95b9500a10d40104a579d3e2ec438c51feb9
```

```
PASS_NATIVE_SIMPLE_FUNCTION_TAIL_AND_RECOGNITION_COMPLETION_CORE
ca4d3443e057adb0c94a285263d8d136465eaf970c7fef73df70196972cad9
```

```
PASS_NATIVE_RATIONAL_UGD_REFINEMENT_LATTICE_ADAPTER
932d3e549e3a01d7d42d85e70371dc74fa06c214841bca8f776d19dd70ce4bc7
```

99.5 5. Inspect source boundaries

```
native_files = [
    "native_summability.py",
    "native_refinement_measure.py",
    "native_function_completion.py",
    "native_ugd_refinement_adapter.py",
]

for name in native_files:
    path = RH_REPO / "src" / "rh_framework" / name
    print(name, path.stat().st_size, "bytes")
```

The verifier rejects undeclared imports and executable uses of:


```
float
complex
numpy
scipy
log
exp
pi
```

inside the foundational files.

99.6 6. Inspect theorem frontier

```
import json

manifest = json.loads(
    (
        RH_REPO
        / "provenance"
        / "NATIVE_CLASSICAL_BOUNDARY_v0.3.json"
    ).read_text(encoding="utf-8")
)

print("overall:", manifest["overall_status"])
print("pure native framework:", manifest["pure_native_framework"])
print("remaining:")
for item in manifest["remaining_native_reconstruction"]:
    print(" -", item)
```

99.7 Claim boundary

A passing run reproduces the abstract native path, content, function, and rational UGD refinement layers. It does not prove the completed actual positive-scale UGD adapter, state-locked packet binding, target faithfulness, Shunya, the critical-line adapter, or the Riemann Hypothesis.

RH-Framework/notebooks/T01_NATIVE_RECONSTRUCTION_JUPYTER.md

Chapter 100

T03 Action-Domain Verification — Jupyter Procedure

This procedure uses Python cells only. No PowerShell is required.

100.1 1. Set repository paths

```
from pathlib import Path
import subprocess
import sys

RH_REPO = Path(
    r"C:\Users\abc\Desktop\New Folder (2)\New folder\PROVISNAL RELATED\RH-Framework"
).resolve()

MP_REPO = Path(
    r"C:\Users\abc\Desktop\New Folder (2)\New folder\PROVISNAL
    ↳ RELATED\MP_clock_free_endpoint_17e51005"
).resolve()

OPERATOR_COMMIT = "abc9725a8f232feab81ba63230085457913aadd7"
STATE_COMMIT = "f9e2f71c2d0ef7db239d7e12b18394eff8e83a1e"

OPERATOR_WORKTREE = MP_REPO.parent / "MP_operator_source_abc9725"
STATE_WORKTREE = MP_REPO.parent / "MP_state_source_f9e2f71"

T03_C1_JSON = (
    RH_REPO / "certificates" / "T03_C1_PINNED_ACTION_INVENTORY_v0.1.json"
)
T03_C1_NPZ = (
    RH_REPO / "certificates" / "T03_C1_PINNED_ACTION_ARRAYS_v0.1.npz"
)

print("RH framework:", RH_REPO)
print("MP source clone:", MP_REPO)
print("operator worktree:", OPERATOR_WORKTREE)
print("state worktree:", STATE_WORKTREE)
```

Change only the local paths when your folders differ.

100.2 2. Git helper and RH branch audit

```
def run_git(repo: Path, *args: str) -> str:
    completed = subprocess.run(
        ["git", "-C", str(repo), *args],
        check=True,
        capture_output=True,
        text=True,
    )
    return completed.stdout.strip()

print("RH branch:", run_git(RH_REPO, "branch", "--show-current"))
print("RH head:", run_git(RH_REPO, "rev-parse", "HEAD"))
```

Expected branch:

agent/t03-action-domain-adapter

100.3 3. Install the RH verification package

```
completed = subprocess.run(
    [sys.executable, "-m", "pip", "install", "-e", str(RH_REPO)],
    text=True,
    capture_output=True,
)
print(completed.stdout)
print(completed.stderr)
print("return code:", completed.returncode)
assert completed.returncode == 0
```

100.4 4. Verify T03-A and all source-independent controls

```
completed = subprocess.run(
    [sys.executable, str(RH_REPO / "tools" / "verify_t03.py")],
    cwd=RH_REPO,
    text=True,
    capture_output=True,
)
print(completed.stdout)
print(completed.stderr)
print("return code:", completed.returncode)
assert completed.returncode == 0
```

Expected lines include:

```
T03-A PASS_T03_ACTION_DOMAIN_DENSE_CORE
T03-A certificate 05a4784308e996763928b4cead0c86462150154c506b30bf292f96dd8986db3c
T03-B OPEN: pinned private worktrees were not supplied
T03-C1 OPEN: actual finite action inventory was not extracted
```

This reproduces T03-A and the fail-closed T03-B/T03-C1 control logic. It does not execute the private MP packet.

100.5 5. Fetch the pinned MP branches

```
subprocess.run(
    [
        "git",
        "-C",
        str(MP_REPO),
        "fetch",
        "origin",
        "agent/native-recognition-operator-algebra",
        "agent/rh-rotation-metric-topology-reset",
    ],
    check=True,
)
```

100.6 6. Create clean detached source worktrees

```
if not OPERATOR_WORKTREE.exists():
    subprocess.run(
        [
            "git",
            "-C",
            str(MP_REPO),
            "worktree",
            "add",
            "--detach",
            str(OPERATOR_WORKTREE),
            OPERATOR_COMMIT,
        ],
        check=True,
    )

if not STATE_WORKTREE.exists():
    subprocess.run(
        [
            "git",
            "-C",
            str(MP_REPO),
            "worktree",
            "add",
            "--detach",
            str(STATE_WORKTREE),
            STATE_COMMIT,
        ],
        check=True,
    )

print("operator head:", run_git(OPERATOR_WORKTREE, "rev-parse", "HEAD"))
print("state head:", run_git(STATE_WORKTREE, "rev-parse", "HEAD"))
print("operator dirty:", repr(run_git(OPERATOR_WORKTREE, "status", "--porcelain")))
print("state dirty:", repr(run_git(STATE_WORKTREE, "status", "--porcelain")))

assert run_git(OPERATOR_WORKTREE, "rev-parse", "HEAD") == OPERATOR_COMMIT
assert run_git(STATE_WORKTREE, "rev-parse", "HEAD") == STATE_COMMIT
assert run_git(OPERATOR_WORKTREE, "status", "--porcelain") == ""
assert run_git(STATE_WORKTREE, "status", "--porcelain") == ""
```

The RH verifier rejects dirty worktrees, wrong commits, changed pinned blobs, and altered label

order.

100.7 7. Execute the pinned carrier and T03-C1 inventory

```
completed = subprocess.run(
    [
        sys.executable,
        str(RH_REPO / "tools" / "verify_t03.py"),
        "--operator-root",
        str(OPERATOR_WORKTREE),
        "--state-root",
        str(STATE_WORKTREE),
        "--t03-c1-output",
        str(T03_C1_JSON),
        "--t03-c1-npz-output",
        str(T03_C1_NPZ),
    ],
    cwd=RH_REPO,
    text=True,
    capture_output=True,
)
print(completed.stdout)
print(completed.stderr)
print("return code:", completed.returncode)
print("JSON certificate:", T03_C1_JSON)
print("NPZ arrays:", T03_C1_NPZ)
```

Required status:

PASS_RH_FRAMEWORK_T03_C1_PINNED_ACTION_INVENTORY

A passing run proves T03-B and T03-C1 together. A nonzero return code is a verification failure, not a warning to be renamed until morale improves.

100.8 8. Inspect theorem states and false gates

```
import json

report = json.loads(T03_C1_JSON.read_text(encoding="utf-8"))
false_checks = [
    name for name, passed in report.get("checks", {}).items() if not passed
]
print("status:", report["status"])
print("classification:", report["theorem_classification"])
print("false checks:", false_checks)
print("certificate hash:", report.get("certificate_hash"))
```

Required promotion state:

```
T03_B_finite_packet_reproduction = PROVED
T03_C1_actual_finite_packet_action_inventory = PROVED
T03_C_remaining_action_families = OPEN
riemann_hypothesis = OPEN
false_checks = []
```

100.9 9. Inspect per-array hashes and roles

```
inventory = report["action_inventory"]

print("carrier data:")
for name, metadata in inventory["carrier_data"].items():
    print(" ", name, metadata["shape"], metadata["sha256"])

print("matrix actions:")
for name, role in inventory["actions"].items():
    metadata = inventory["array_metadata"][name]
    print(
        " ",
        name,
        role["operator_class"],
        metadata["operator_norm"],
        metadata["sha256"],
    )
```

The individual SHA-256 values identify the mathematical arrays. The NPZ file is only their transport bundle.

100.10 10. Confirm the NPZ names

```
import numpy as np

with np.load(T03_C1_NPZ, allow_pickle=False) as loaded:
    print(sorted(loaded.files))
```

Expected names include:

```
corrected_defect_line
source_defect_projector
accepted_complement_projector
source_metric
response_metric
loop_holonomy
memory_holonomy
compensated_holonomy
```

100.11 Claim boundary

A passing run proves the pinned finite carrier reproduction and the actual finite packet action inventory. T03-C2 through T03-C5 remain open: amplitude families, arithmetic/Gamma actions, unbounded graph domains, and aperture exhaustion. Target faithfulness, native Shunya, the critical-line adapter, and RH remain open.

RH-Framework/notebooks/T03_ACTION_DOMAIN_JUPYTER.md

Part VIII

Publications: Certified Papers

Chapter 101

CID-1 — Curvature–information duality: the obstruction is metric-free

A **bridge capsule**. CFE-U certified that the obstruction to the memoryless limit is unique (a 1-dimensional obstruction space spanned by the loop residue). This capsule asks whether the *information* side sees the same obstruction — or a second, independent one.

The answer is a separation with an invariance: the Hodge/Helmholtz split of a response *depends* on the information metric; the obstruction *does not*. The metric is a gauge on the decomposition and an invariant on the obstruction.

Block	Certifies	Result
T1	The information metric at a declared rational point is $g = \text{Cov}_p(T)$, exact — no exponential or logarithm evaluated anywhere . PD statistics affinely independent; degenerate with explicit kernel otherwise; constants always in the kernel (matching CFE-U's $\ker d$)	PASS
T2	Recognition ledger = law of total covariance: $g = g_{\text{recognized}} + g_{\text{discarded}}$ exactly (between-block + within-block), both PSD. Monotonicity follows as an <i>identity</i> , not an inequality: recognition can lose information, never manufacture it. Sufficient partition discards exactly nothing; total collapse recognizes exactly nothing	PASS

Block	Certifies	Result
T3	The obstruction is metric-free: curvature is d , not $\frac{1}{2}d$ — no Christoffel of any metric enters. Euclidean, anisotropic, sheared and the information metric $\text{Cov}_p(T)$ itself all give the same residue, reproducing CFE-1's ladder $-32\dots+32$	PASS
T4	The split is not: exact rational least squares gives $\frac{1}{2}d = d\Phi_g + h_g$, genuine for each metric (closed part memoryless with an explicit potential, remainder g -orthogonal), and the splits genuinely differ — yet $h_g = 32$ for every metric	PASS
T5	For every metric the g -orthogonal complement of the memoryless space has rank exactly 1 — CFE-U's uniqueness seen through the information metric. Generators differ as vectors, all land in the same obstruction class. Control: a degenerate PSD form gives no well-posed split	PASS

101.1 Findings

- **CID-F1.** Curvature and information are **dual descriptions of one object**, not two obstructions. Every positive-definite information metric reproduces CFE-U's rank-1 obstruction space exactly.
- **CID-F2.** The law of total covariance *is* a recognition ledger: what the Eye map keeps and what it discards sum exactly to the whole, with both parts PSD.
- **CID-F3.** Monotonicity of information under coarse-graining is here an **identity**, which is strictly stronger than the usual inequality.

101.2 Claim boundary

The exponential family's form is **declared**; only rational-point arithmetic is certified. Continuum information geometry, $\frac{1}{2}$ -connections, dually flat structures and quantum (Fubini–Study / Bures / SLD) metrics are **not claimed**. The vault appendix named for Euler information–curvature

duality **has not been read and is not the source** — this is a bridge between two certified capsules of this corpus. Uniqueness inherits CFE-U's scope (the declared response algebra). RH / K0 / L0 / YM / QG untouched.

```
python certificates/cid1_curvature_information_duality.py
python -m pytest tests -v
```

Chapter 102

QTH-1 — Quantum recognition information: the ledger becomes an inequality

The question CID-1 left open, answered. CID-1 certified the **classical** recognition ledger $g = g_{\text{recognized}} + g_{\text{discarded}}$ as an **exact identity** (law of total covariance). Does that shape survive quantum-ly?

It does not — and the failure is exactly located. 25 tests; 49 in this folder.

$$\text{QFI} = \text{CFI}(\text{measurement}) + \text{discard}, \text{discard} \geq 0, \text{ vanishing } \mathbf{exactly} \text{ for an optimal measurement.}$$

An inequality with exact saturation, not an identity. And on the **commuting face** the identity is restored exactly — the same shape as CFE-1's memoryless limit.

Block	Certifies	Result
T1	Rational Bloch states; the SLD from an exact 4×4 rational solve, its defining equation verified as a matrix identity; QFI by two independent routes (solve vs closed Bloch form) agreeing in	PASS
T2	CFI = QFI on every rational measurement direction, discard exactly 0 at the aligned direction and exactly 1 (everything) at the orthogonal one. The quantum ledger is an inequality with exact saturation	PASS

Block	Certifies	Result
T3	One tensor, two certified halves. $Q_{ij} = \text{Tr}(L_i L_j)$ splits exactly into a real symmetric part (CID-1's information metric, diagonal = QFI) and an imaginary antisymmetric part (mean Uhlmann curvature, exactly $-1/2$ here, zero diagonal). The antisymmetric half is nonzero exactly when the SLDs fail to commute — the obstruction <i>is</i> a curvature	PASS
T4	The classical bookkeeping inside the outcome data still holds exactly, while the quantum discard is nonzero by an exactly computed amount no partition of outcomes can see . The remainder is strictly quantum	PASS
T5	The commuting face restores everything: SLDs commute, Uhlmann curvature exactly 0, eigenbasis measurement optimal, discard exactly 0, and QFI equals CID-1's classical covariance route	PASS
T6	A rational unitary rotation preserves QFI exactly ; a depolarizing channel scales it by exactly 2 . Information can be lost, never manufactured	PASS

102.1 Findings

- **QTH-F1.** The recognition ledger is classically an **identity** and quantum-ly an **inequality with exact saturation**.
- **QTH-F2.** The quantum geometric tensor's symmetric half is CID-1's object and its antisymmetric half is a CFE-style obstruction — **one tensor, two capsules**.
- **QTH-F3.** The commuting face restores the classical identity exactly.

Companion, not duplicate: CFE-Q certified the Bloch/holonomy side of quantum CFE; this is the information side.

Witness capsule (one qubit, explicit rational families) like CFE-Q — not a general open-system or quantum-estimation theorem. Kubo–Mori/Bogoliubov metrics need logarithms and are **never evaluated**. **Not claimed:** entropy production, work, heat, or that the antisymmetric part is a physical Berry phase.

`Publications/papers/curvature-information-duality/README.md`

Chapter 103

Lineage — CID-1

- **Consumes (pinned):** `papers/cut-first-equivalence/certificates/cfeu_uniqueness.py` (memoryless space kernel, residue functional, response complex, potential reconstruction) and through it `cfe1_cut_first_equivalence.py` (witness family, BETA, LOOP_RECT, circulation). Nothing is re-declared; a change in either pinned module breaks this capsule's tests by construction.
- **Companion context:** `papers/information-invariance/` (D-series: $I = 0$, exactness of , selection principle) — the conceptual neighbour, not a code dependency.
- **Vault cross-reference — DIFF DONE (Aug 21).** The vault appendix `APPENDIX_GENERALIZED_EULER_I` (259 lines) has now been read and compared against CID-1. **Result: the appendix is largely declared structure, and CID-1's theorems are not in it.** The appendix declares a generalized Euler phase-seam space, a recognition connection, an information-curvature tensor $I^{\text{GE}}_{ab} = \langle \text{grad } \rho, \text{grad } \rho \rangle + \langle \text{grad } \theta, \text{grad } \theta \rangle + M^{\text{Sigma}}$, a curvature-residue norm, and a duality *map* D_{Sigma} sending curvature components to closure-residue components; its theorem is the qualitative statement that visible return is not recognition return when any active component is unclosed. CID-1 proves something the appendix does not state: that the obstruction is **invariant** under change of information metric while the Hodge split is not, and that the g-orthogonal complement has rank exactly 1 for every positive-definite metric. So CID-1 stands as written; the appendix's tensor I^{GE} remains **declared, not certified**, and is recorded as such in the EMK-TOP-1 claim boundary. The appendix's sheet-rotation class and visible-return statement *are* certified — in EMK-TOP-1 T5 and T1.
- **Prior executable version:** NONE — first bridge capsule between the curvature and information layers.

`Publications/papers/curvature-information-duality/LINEAGE.md`

Chapter 104

Cut-First Equivalence — certified core (CFE-1)

The candidate **flagship theorem** of the recognition / cut-flow framework (arXiv:2603.20773), with its memoryless-limit direction and residue identity certified in exact rational arithmetic.

Read THEOREM.md first — it states the theorem, marks certified vs. open, and maps every obligation to its scaffolding.

104.1 The statement

Classical thermodynamics is the **memoryless sector** of recognition thermodynamics, and the loop residue $** _D = _D \Omega **$ is the obstruction. Equilibrium is flat ($\Omega=0$); irreversibility is curvature ($\Omega \neq 0$).

A bridge identity of the Noether type: recognition/memory structure thermodynamic response geometry, with a computable residue.

104.2 Certified blocks (CFE-1)

Block	Statement	Verdict
T1	Memoryless limit is classical: at $\beta=1$ curvature vanishes on the whole grid, every Maxwell loop closes, Onsager asymmetry is exactly 0	PASS
T2	Residue is exact: $_D =$ $_D \Omega$ in (boundary walk vs area sum, independent), both $= 32$ at $\beta=7/5$	PASS

Block	Statement	Verdict
T3	Obstruction is faithful: $\Omega=0$ iff $\gamma=1$; residue strictly monotone in $\gamma=-1$ ($-32,-16,0,+16,+32$), zero exactly at $\gamma=0$	PASS
T4	Equilibrium invariant $\gamma=\Gamma_c \cdot \Gamma_m=1$ exactly at $\gamma=1$, departs when $\gamma=1$	PASS

All exact rationals, no floating point in any verdict path.

104.3 Surjectivity — DISCHARGED (CFE-2)

(S) the cut grammar maps *onto* the classical response algebra: all four Maxwell relations as cut-closedness, the response-coefficient identities (γ ratio law + Mayer), and a rank-equality surjectivity certificate ($\gamma=5$, cokernel 0) built by cut-native finite differences independent of the classical partials — renaming test passed, non-closed control separates.

104.4 Quantum witness (CFE-Q)

The same bridge holds quantum-mechanically, certified on a dissipative qubit: memory dial non-Markovianity, $\gamma=\Omega$ Berry holonomy, $\gamma=1$ CP-divisible (Markovian) limit, and the Bloch quarter-turn $J^2=-I$ echoes the seam phase $\gamma^2=-1$. A witness, not a general theorem; quantum gravity is a horizon, not a claim.

104.5 Open — the single remaining obligation

- (U) **Uniqueness** — Ω is the *unique* obstruction to $\gamma \rightarrow 1$. Finite, gate-free, attackable by the same exact-rank machinery ($H^2(\text{response})$ rank 1). Touches no RH / K0 / L0 / YM continuum gate.

104.6 Reproduce

```
python papers/cut-first-equivalence/certificates/cfe1_cut_first_equivalence.py
python papers/cut-first-equivalence/certificates/cfe2_surjectivity.py
python papers/cut-first-equivalence/certificates/cfeq_quantum_witness.py
python -m pytest papers/cut-first-equivalence/tests -v
```

CI regenerates the certificate and fails on any pin drift (EXPECTED_CFE1.sha256, EXPECTED_CFE2.sha256, EXPECTED_CFEQ.sha256).

104.7 Relationship to the rest of the corpus

CFE-1 is the **parent** capsule. Its downstream corollaries live in `../lambda-seam-calibration/`: the seam-integer quantization (LAM-1/LAM-2) becomes “the obstruction takes quantized seam

values,” and the “no memory, no critical strip” finding (LAM-3) becomes “ Ω is non-trivial iff the memory channel is retained.”

104.8 Capsule CFE-U: uniqueness — the flagship closed

`certificates/cfeu_uniqueness.py` discharges Part (U) with the same exact-rank machinery that closed (S):

Block	Certifies	Result
T1	The response complex $\Lambda \rightarrow \Lambda^1 \rightarrow \Lambda^2$ has exact ranks 5, 1 and $\dim \Lambda^0 = 0$; $\dim \Lambda^1$ is	PASS
T2	CFE-2’s rank-5 classical space $H^1 = 0$: closed gradient, constructively — the explicit potential integrates every closed form; zero curvature leaves no residual obstruction	PASS
T3	The uniqueness theorem: the annihilator of the memoryless space is EXACTLY 1-dimensional; the loop residue lies in it, is nonzero, hence spans — every obstruction is $c \cdot d$; and $R =$ $\text{Area} \cdot d$ on the whole basis	PASS
T4	The dial’s direction $5p \cdot dv$ excites the generator; the residue ladder $-32 \dots +32$ reproduced from the pinned CFE-1 module (two routes agree in $\dim$); equal residue equal obstruction class with explicit potential witness	PASS
T5	The 16-cell residue matrix has rank exactly 1 with kernel = the memoryless space: one obstruction repeated, not many	PASS
T6	Control: quadratic-coefficient algebra has rank 3 — uniqueness is a theorem about the declared algebra, and the boundary itself is certified	PASS

With CFE-1 + CFE-2 + CFE-Q + CFE-U, every declared obligation of the Cut-First Equivalence theorem is machine-certified on the witness EOS and its declared response

algebra.

Publications/papers/cut-first-equivalence/README.md

Chapter 105

The Cut-First Equivalence Theorem

Candidate flagship statement of the recognition / cut-flow framework (arXiv:2603.20773). This document states the theorem, marks precisely which parts are certified and which are open, and maps every proof obligation to where its scaffolding lives.

105.1 Statement (plain language)

Recognition/cut response reproduces classical thermodynamic response **exactly**. The classical laws — Maxwell relations, Onsager reciprocity, the equilibrium invariant $\mathcal{I} = 1$ — are the **memoryless limit** ($\mathcal{I} \rightarrow 1$, equivalently $d = 0$). The deviation from that limit is **exactly** the loop residue

$$\oint_D \mathcal{I} = \oint_D \Omega,$$

the integrated cut-loop curvature. Equilibrium is the flat face ($\Omega = 0$); irreversibility is curvature ($\Omega \neq 0$); and the residue is the complete obstruction to flatness.

In one line, of the Noether type — *a bridge identity between two independently computed quantities*:

Classical thermodynamics is the memoryless sector of recognition thermodynamics, and Ω is the obstruction.

105.2 Statement (precise)

Let $(\mathcal{R}, J)$ be a regular response manifold carrying a cut involution J ($J^2 = 1$) and a memory dial $\mathcal{I}$, with entropy-weighted response one-form $\mathcal{I} = \mathcal{I}_p dp + \mathcal{I}_v dv$ and curvature two-form $\Omega = d\mathcal{I}$. Then:

1. **(Memoryless limit.)** In the limit $\mathcal{I} \rightarrow 1$ (equivalently $d = 0$), the cut-flow response coincides with classical thermodynamic response: every Maxwell relation holds, the Onsager block is symmetric ($L_{pv} = L_{vp}$), and the invariant $\mathcal{I} = \Gamma_c \cdot \Gamma_m = 1$.
2. **(Residue.)** For any 2-cell D , the circulation of $\mathcal{I}$ around D equals the curvature flux through D : $\oint_D \mathcal{I} = \oint_D \Omega$, and this residue vanishes iff the response is memoryless on D .

3. (Completeness — the two halves.)

- **(S) Surjectivity.** Every classical response identity is realized by some cut protocol: the cut grammar maps *onto* the classical response algebra. **DISCHARGED by CFE-2** (rank equality, zero cokernel, on the witness EOS; renaming test passed).
- **(U) Uniqueness.** Ω is the *unique* obstruction to the $\rightarrow 1$ limit: no second, independent obstruction exists. **DISCHARGED by CFE-U** (the obstruction space — the annihilator of the memoryless responses in the declared algebra — has dimension exactly 1, and the loop-residue functional spans it; $H^1 = 0$ constructively; enlarged-algebra control shows where the boundary sits).

Parts (1) and (2) are the **certified core** (CFE-1). Part (S) is **certified** (CFE-2). Part (U) is **certified** (CFE-U). **All declared obligations of the theorem are now machine-certified on the witness EOS and its declared response algebra.** Continuum / general-manifold / general-EOS statements remain outside the certified scope, exactly as recorded in each capsule's claim boundary.

105.3 What is certified: capsule CFE-1

The capsule `certificates/cfe1_cut_first_equivalence.py` proves, in **exact rational arithmetic** on an explicit witness equation of state, four blocks (pin `EXPECTED_CFE1.sha256`, all tampers separate):

Block	Certifies	Result
T1	Memoryless limit is classical: at $\beta=1$ the curvature vanishes on the whole grid, every Maxwell loop sums to exactly 0, the Onsager cross-asymmetry is exactly 0	PASS
T2	The residue is exact: $\sum_D \Omega =$ $\sum_D \Omega$ as an identity in (boundary walk and area sum computed independently), witnessed at $\beta=7/5$ with both sides $= 32$	PASS
T3	The obstruction is faithful: Ω $= 0$ on the grid iff $\beta = 1$; the residue is strictly monotone in the memory dial $\beta = -1$ ($-32,$ $-16, 0, +16, +32$ across $\beta =$ $3/5 \dots 7/5$), zero exactly at $\beta=0$, sign tracking <code>sign()</code>	PASS

Block	Certifies	Result
T4	The equilibrium invariant: $\Gamma_c \cdot \Gamma_m = 1$ exactly at $\beta = 1$, departing exactly when $\beta > 1$	PASS

This maps Part (1) $\rightarrow$ T1+T4 and Part (2) $\rightarrow$ T2+T3. The residue’s exact linearity in the memory dial (T3) is the machine witness that Ω is a *genuine* order parameter for irreversibility, not an artefact of the model.

105.4 What is certified: capsule CFE-2 (surjectivity)

The capsule `certificates/cfe2_surjectivity.py` discharges Part (S) — the cut grammar maps **onto** the classical response algebra (pin EXPECTED_CFE2.sha256):

Block	Certifies	Result
T1	All four Maxwell relations hold exactly as cut-closedness of a convex rational potential; a sheared non-potential field breaks them	PASS
T2	The response-coefficient identities are cut identities: the ratio law $C_p/C_v = \gamma$ and the Mayer relation $C_p - C_v = T(U_{SV})^2/(U_{SS} \cdot \det)$, exact at every lattice point	PASS
T3	Surjectivity: the cut grammar reconstructs the response fields by cut-native centered finite differences; their span has exactly the rank of the full classical response space ($5 = 5$), union adds nothing, cokernel = 0 . Renaming test passed (rank 5 on random potentials; non-closed control separates)	PASS
T4	Faithfulness: no cut form leaks outside the classical algebra — the image sits <i>exactly</i> on it	PASS

The key move in T3: the cut side is built by a **procedure independent** of the classical side — finite differences of the potential (loops and differences only), not analytic partials — and the two still span the same 5-dimensional space. That independence is what makes the rank equality a theorem rather than a restatement. **Recognition thermodynamics loses nothing classical.**

105.5 Why this is the right flagship

Discourse-changing theorems are **bridge identities**: two quantities computed by different means that must be equal. Noether bridges symmetry – conserved current; Einstein bridges geometry – stress-energy; Onsager bridges microscopic reversibility – macroscopic transport symmetry; Landauer bridges information erasure – heat. Cut-First Equivalence bridges **recognition/memory structure – thermodynamic response geometry**, with a computable residue Ω . It upgrades the second law’s *inequality* slot to an *identity with a computable residue* — on the witness EOS, certified.

It is also the **parent** statement. Once established: - the **recognition-Noether** direction (bias phase – as the generator of a conserved recognition current) becomes “the conserved current of the memoryless symmetry”; - the **memory-completion** finding (no memory, no critical strip; certified at $s=3,5$ in `../lambda-seam-calibration/`) becomes “the obstruction Ω is non-trivial iff the memory channel is retained”; - the **seam-quantization** results ($|\langle \cdot \rangle|^2=q$, $\langle \cdot \rangle^2=-4$, $\langle \cdot \rangle^2=-1$, in `../lambda-seam-calibration/`) become “the obstruction takes quantized seam values.”

One theorem organizes four, instead of four competing claims.

105.6 The remaining obligation (open, precisely stated)

(S) surjectivity is now discharged (CFE-2). One obligation remains, **within the regular / certified regime, touching no RH / K0 / L0 / YM continuum gate**:

- **(U) Uniqueness.** That Ω is the *only* obstruction to $\rightarrow 1$ — that no independent second obstruction can arise. Finite, well-posed, gate-free; the natural certified attack is an obstruction-theoretic rank argument (the response cohomology $H^2(\text{response})$ is rank 1), the same exact-rank machinery CFE-2 already uses.

A second, softer obligation is **generality beyond the witness EOS**: CFE-2’s rank-equality argument is EOS-independent in *form* (closed-form spaces on a 2-D compass) and passes the renaming test on random potentials, but a full statement quantifies over all admissible convex potentials.

105.7 Proof-scaffolding map

Obligation	Scaffolding already in the corpus
Part (1), memoryless limit	this capsule T1/T4; paper Corollary .1.1 ($=1$ Onsager), contact-form Maxwell derivation, $=1$
Part (2), residue = curvature	this capsule T2/T3; paper Stokes identity $=\Omega$ (Eqs. for $\Omega=d$)
Curvature = micro-correlator	paper Weighted Reciprocity (Eq. 22) + Microscopic Irreversibility theorem
(S) surjectivity	CFE-2 — rank equality ($5=5$), zero cokernel, renaming test passed
(U) uniqueness	to build — obstruction-theoretic argument that $H^2(\text{response})$ is rank 1

Obligation	Scaffolding already in the corpus
quantized residue values (downstream)	../lambda-seam-calibration/ LAM-1/LAM-2 (seam integers, Gauss-sum cuts)

105.8 Quantum extension: capsule CFE-Q

The same bridge identity has a **quantum realization**, certified on a dissipative-qubit witness (pin EXPECTED_CFEQ.sha256):

Block	Certifies	Result
T1	Markovian limit is flat: at $\omega=1$ the qubit propagator is CP-divisible (memoryless) and the closed-cycle geometric residue is exactly 0 — no holonomy	PASS
T2	The geometric residue is exact ($\omega = \Omega$ on the Bloch loop) and faithful: zero iff Markovian, strictly monotone in the memory dial — a certified quantum non-Markovianity measure	PASS
T3	Quarter-turn echo: the residue is the Berry-phase area form, and the Bloch rotation generator satisfies $\mathbf{J}^2 = -\mathbf{I}$ — the quantum instance of the derived seam phase $\omega^2 = -1$ (LAM-2 T1b)	PASS
T4	CP-divisibility control: a $\omega=1$ propagator loses intermediate complete positivity at an explicit step — genuinely non-Markovian, not a relabelling	PASS

The dictionary is exact: **memory dial** **non-Markovianity**, $\omega = \Omega$ **geometric (Berry) holonomy**, $\omega=1$ **flat face** **CP-divisible Markovian limit**. The classical seam quantization ($\omega^2 = -1$) and the quantum geometric phase are literally the same quarter turn.

Boundary: CFE-Q is a *witness*, not a general open-system theorem — one qubit, one explicit rational dynamics. Generality beyond the qubit and the full open-system theorem are OPEN. **Quantum gravity is not touched** — it is a horizon, exactly like the action-lifted field equations below, never a foundation.

105.9 Honest boundary

The witness equation of state is an explicit rational model chosen so the arithmetic is exact. **The theorem is a claim about the structure** (one-form, curvature two-form, memory dial), **not about this one EOS** — the EOS is the witness, not the theorem. The certified core establishes the memoryless-limit direction and the residue identity; it does **not** establish (S) or (U), and it makes no claim about the framework’s open number-theory or Yang–Mills continuum gates. What the framework’s genuine originality rests on is the **bridge to finite seam arithmetic** developed in the neighbouring Λ -seam capsules; this document’s flagship should be read together with those.

Publications/papers/cut-first-equivalence/THEOREM.md

Chapter 106

Lineage — Cut-First Equivalence

106.1 Origin

CFE-1 formalizes the candidate flagship theorem identified for arXiv:2603.20773 (“Geometric Completion of Thermodynamic Response”). The paper already proves the memoryless-limit direction as corollaries ($\beta=1$ Onsager; the contact-form origin of Maxwell relations; $\beta=1$) and states the Stokes residue $\beta=\Omega$. CFE-1 lifts these from prose and scattered lemmas into a single certified capsule with an explicit witness equation of state, so the flagship rests on machine-checkable arithmetic.

106.2 What is new here vs. the paper

Nothing mathematically new is *claimed*. What is added is: - an explicit rational witness EOS on which the residue is computed two independent ways (boundary walk, area sum) and shown exactly equal; - the faithfulness result T3 (residue strictly monotone in the memory dial, zero iff equilibrium) which makes Ω a certified order parameter, not merely a defined quantity; - an explicit, pinned statement of the two open obligations (S) surjectivity and (U) uniqueness that separate the certified core from the full theorem.

106.3 Relationship to prior capsules

- The classical substrate (Stokes, contact-form Maxwell, Onsager in the reversible limit) is a **pinned named dependency**, cited and not rederived — the same discipline used in the Λ -seam and Yang–Mills capsules.
- The downstream quantization of the residue lives in `../lambda-seam-calibration/`: LAM-1/LAM-2’s seam integers and Gauss-sum cuts are the arithmetic side of the bridge, and LAM-3’s “no memory, no critical strip” is the memory-channel necessity. CFE-1 is the thermodynamic parent under which those become corollaries.

106.4 Claim boundary

Certified: the memoryless-limit direction and the residue identity on an explicit EOS, in exact arithmetic. Open: (S), (U), and generality beyond the witness EOS. Untouched: RH / K0 / L0 / YM continuum gates. The EOS is the witness, not the theorem.

106.5 CFE-2 addendum

CFE-2 discharges obligation (S), surjectivity, on the witness EOS: the cut grammar generates the entire classical response algebra. The cut basis is built by cut-native centered finite differences of the potential — a procedure independent of the analytic partials used for the classical frame — and the two span the same exact 5-dimensional space (cokernel 0). The renaming test (TAUT-1) was applied and passed: random potentials of the same shape also reach full rank (structural, not point-luck), and a non-closed shear control separates. Only (U) uniqueness remains open for the full theorem; no RH/K0/L0/YM continuum gate is touched.

106.6 CFE-Q addendum

CFE-Q is step (2) of the quantum sequence: a certified quantum- thermodynamics witness on a dissipative qubit. It shows the Cut-First Equivalence structure realizes quantum-mechanically — Markovian (CP-divisible) limit as the flat $\omega=1$ face, closed-cycle geometric holonomy as a faithful non-Markovianity measure, and the Bloch quarter-turn $J^2=-I$ as the Berry-phase echo of the Λ -seam $\omega^2=-1$. It is a witness, not a general open-system theorem; generality and the full theorem are open. Quantum gravity is deliberately NOT addressed — it remains a horizon, to be attempted only by connecting a certified piece to an established bridge (e.g. Jacobson 1995), never asserted as foundation. No RH/K0/L0/YM continuum gate is touched.

Publications/papers/cut-first-equivalence/LINEAGE.md

Chapter 107

EMK recognition geometry — the rotational seam metric (EMK-G1)

The geometry layer. EMK-1/EMK-2 identify the seam algebraically; this capsule certifies the metric that gives it lengths, geodesics and curvature.

107.1 The construction

The local involution $J(z) = i \cdot \text{conj}(z)$ acts on real coordinates as $(x, y) \rightarrow (y, x)$. In seam-adapted coordinates

$$\begin{aligned} u &= (x+y)/\sqrt{2} && \text{tangent to the seam} \\ v &= (y-x)/\sqrt{2} && \text{normal to it} \end{aligned} \quad J : (u, v) \rightarrow (u, -v)$$

so seam reflection is ordinary reflection of the normal coordinate and the fixed seam is the **45° line** $v = 0$. The reflection-symmetric seam metric is

$$g_A = A(v)^2 \, du^2 + dv^2 \quad \det g_A = A(v)^2 \quad K_A = -A'/A$$

107.2 The move that keeps it exact

Carrying $W = A^2$ removes every square root:

$$\Gamma_{uv} = W/(2W) \quad \Gamma_{uu} = -W/2 \quad K = (W')^2/(4W^2) - W''/(2W)$$

For the quadratic warp $W = 1 + v^2$ this gives, exactly,

$$K_-(v) = -1/(1+v^2)^2 \quad K_-(0) = -1$$

Every quantity is a rational function of rational data — the whole seam geometry is certified with **no square root and no float**. The $\sqrt{2}$ in the coordinate normalization is a positive constant that changes no line, no fixed set and no curvature sign; the certified arithmetic uses unnormalized seam coordinates and records the normalization explicitly.

107.3 Certified blocks

Block	Statement	Verdict
T1	The involution acts as $(\mathbf{x}, \mathbf{y}) \rightarrow (\mathbf{y}, \mathbf{x})$; $\mathbf{u}$ invariant, $\mathbf{v}$ negated, exactly on an integer grid; fixed set is precisely the line $\mathbf{y}=\mathbf{x}$; $\mathbf{J}^2=\text{id}$	PASS
T2	$\det \mathbf{g}_A = W$; positive definite exactly where $W>0$, degenerate exactly at the zeros of A ; for $W<0$ the locus is the exact condition $1+\mathbf{v}^2=0$	PASS
T3	Christoffels in W -form agree with Levi-Civita computed independently; curvature via $R_{uvuv}/\det \mathbf{g}$ agrees with the closed form — two routes, 25 grid points; $\mathbf{K}(\mathbf{0}) = -$ exactly	PASS
T4	The 45° seam is an intrinsic geodesic of every reflection-symmetric warp (W even $W(\mathbf{0})=0$). Control: an odd perturbation breaks the symmetry, $W(\mathbf{0}) \neq 0$, and the seam is no longer geodesic	PASS
T5	$W>0 \rightarrow$ globally definite, $K<0$; $W=0 \rightarrow$ Euclidean, $K=0$; $W<0 \rightarrow$ $K>0$ strictly inside $\mathbf{v}^2 < -1/$, degenerate exactly at the boundary	PASS
T6	The two curvatures are independent — see below	PASS

107.4 The guard (T6)

The source states as a *principle* that metric (Gaussian) curvature and the declared connection curvature are distinct. This capsule certifies that as a **separation** rather than asserting it:

configuration	Gaussian curvature	recognition curvature
Euclidean seam metric ($W=0$)	0	2
curved metric, rotation-free	-3	0

Neither determines the other. Any future identification must be **proved**, never inherited from

the fact that both are called “curvature”. This is the most tempting over-claim in the geometry layer, and it is now blocked by a certificate.

107.5 Findings

F1 — the 45° seam is an intrinsic geodesic, not a visual bisector, with the reflection symmetry certified load-bearing.

F2 — the entire seam geometry is exactly rational under $W = A^2$, so $K(0) = -$ needs no square root and no float.

F3 — metric curvature and recognition curvature are certified independent.

107.6 Claim boundary

The metric family is **declared** (a reflection-symmetric warp), as in the source; $K(0) = -$ is a theorem for this family, **not** a universal law equating flow parameters with curvature scalars. Global and completeness statements are not claimed. No identification with the recognition or thermodynamic two-form is claimed — T6 certifies the opposite. CFE’s (U) uniqueness stays open. No RH / K / L / YM continuum gate is touched; quantum gravity is not touched.

107.7 Reproduce

```
python papers/emk-recognition-geometry/certificates/emkg1_rotational_seam_metric.py
python -m pytest papers/emk-recognition-geometry/tests -v
```

CI regenerates the certificate and fails on pin drift (EXPECTED_EMKG1.sha256).

Chapter 108

EMK-G2 — Global quotient, completeness, and seam holonomy

Source: vault `emk_ugd_recognition_geometry/sections/global_quotient_holonomy.tex` (281 lines) — the global continuation of EMK-G1: same seam, now on the periodic cylinder. 29 tests; 49 in this folder.

The section's real content is a second guard, and this capsule certifies it in both directions: *trivial Levi-Civita holonomy does not imply recognition closure*.

Block	Certifies	Result
T1	Since $A > 0$, the geodesic criterion $A(0) = 0 \implies W(0) = 0$, so EMK-G1's local criterion transfers unchanged; seam $\text{length}^2 = L^2 \cdot W(0)$, exactly L^2 for the family. Control: an odd perturbation kills geodesy while the curve still closes — closure and geodesy are different properties	PASS
T2	Orbits of $j(u,v) = (u,-v)$ have size exactly 2 off the seam, 1 on it — the quotient is the half-cylinder with a mirror boundary; counting witness (double = $2 \times$ quotient off-seam); the side target $\text{sgn}(v)$ factors through no map on the quotient	PASS

Block	Certifies	Result
T3	The jet criterion in W-form (equivalent since $A > 0$): even warps glue smoothly to themselves; an odd term fails at $n = 1$ exactly yet glues to its own reflection; a mismatched pair fails at the identified jet. The seam shift is metric-invisible while a declared ledger sees it	PASS
T4	> 0 $W = 1$ everywhere; $= 0$ flat; < 0 positivity exactly on $v^2 < -1/$ with degeneracy at normal distance² exactly $-1/$ (finite incomplete). Curvature blow-up as an exact growth witness , not a limit claim; the polar-cap condition fails	PASS
T5	Holonomy by two independent exact routes — boundary walk of $^1 = A du$, and polynomial integration of $K \cdot A$ where $K \cdot A = -A$ is certified by exact polynomial division (remainder zero), not evaluation. Seam loop: $\Theta_LC = 0$ exactly , not merely mod 2	PASS
T6	The guard, both directions: (a) $\Theta_LC = 0$ on the seam while recognition monodromy $= -I$; (b) $\Theta_LC = 0$ on a rectangle whose monodromy is exactly the identity. Neither determines the other; the ledger closes the residue; global closure fails when any active sector stays open	PASS

Guard thread. EMK-G1 T6 separated *metric curvature* from *recognition curvature*. EMK-G2 T6 separates *Levi-Civita holonomy* from *recognition monodromy* — the same guard at the global level.

Declared: both warp families, the auxiliary EMK bundle, $_ \Sigma$, tolerances. Identification of the

two holonomies is **not claimed** — T6 certifies the opposite.

Chapter 109

EMK-G3 — The helical sheet-memory lift: flat is not memoryless

Source: vault `emk_ugd_recognition_geometry/sections/helical_sheet_memory.tex` (245 lines)
— the door EMK-G2’s closing remark left open. 27 tests; 76 in this folder.

The sharpest principle in the geometry layer, certified as an exact separation: *a flat lifted connection may carry nontrivial global monodromy*. Local curvature detects infinitesimal nonclosure; monodromy detects global return data.

Block	Certifies	Result
T1	The helix is the orbit structure of a deck transformation , not a drawn curve: is a bijection with exact inverse, monodromy after n circuits is exactly $(n \text{ positive, zero, negative})$, the action is free and additive, and every orbit meets $0 \leq u < L$ exactly once	PASS
T2	The compensated variables $p_u, -p_u$ are exactly invariant under the deck map; the uncompensated ones fail by exactly $\frac{1}{2}L$ and $\frac{1}{2}L$; a wrong pitch fails by an exactly computed amount	PASS

Block	Certifies	Result
T3	The principle. $A = p \, du$ has curvature exactly zero as a bivariate polynomial identity and exactly zero holonomy on every contractible rectangle — yet one circuit of the base circle gives exactly 0, and n circuits exactly n. A curved companion (curvature exactly -1, holonomy = exact flux) completes the separation in both directions	PASS
T4	Visible return is exactly $0 \pmod{K}$; full lifted return also needs $p = 0$ and $q = 0$ — exhaustive on a grid, with lifted visible and never the converse. After n circuits the displacement is exactly (n, n, nq): a sheet increment is never undone by more circuits — sheet memory is not a phase that eventually wraps	PASS
T5	The four return classes with a witness each, mutually exclusive. Memory-bearing return is distinct from exact return: remove the ledger and the same state becomes an open obstruction; de-admit the sheet sector and it reads as ordinary lawful transport	PASS

Block	Certifies	Result
T6	Consuming pinned EMK-G2: base return holds, $\Theta_LC = 0$ exactly, RTC curvature zero, lift exactly flat — and the transition is still open. Strong form: for all 31 proper subsets of the five sectors, a residue vector closes exactly that subset while the whole stays open. A non-faithful projection dropping the sheet sector maps an open lift to a closed image — the converse of the projection theorem fails exactly	PASS

Guard thread, three levels. EMK-G1 T6: metric curvature vs recognition curvature. EMK-G2 T6: Levi-Civita holonomy vs recognition monodromy. EMK-G3 T3: local flatness vs global monodromy. Same guard, each time one level higher.

Declared: the fibre model, $\mathcal{A}$, the pitches, $\mathcal{A}^{\wedge}\text{EMK}$, ledgers, tolerances, the thermodynamic presentation map. Not certified: the non-abelian lift, the cut-localized sheet term. **Not claimed:** that $\mathcal{A}$, $\mathcal{A}^{\wedge}\text{EMK}$, q are phase lag, scale drift or hysteretic branch count.

Chapter 110

UGD-G1 — Phase-scale-seam geometry: the number layer meets the surface

Source: vault `emk_ugd_recognition_geometry/sections/ugd_phase_scale_seam_geometry.tex` (213 lines). **This closes a loop.** EMK-G3's fibre $_ \times _ \times _k$ is not an arbitrary choice — it *is* the UGD state, whose numeral presentation UGD-1 certified independently. 28 tests; 104 in this folder.

Block	Certifies	Result
T1	Three sectors, three topologies, three different closure rules (phase mod K, scale by equality in $_$, sheet by equality in $_$). No collapsing audit works: a mod-K audit passes an open sheet, an exact-equality audit rejects a genuine phase return. The residue vector is irreducibly a vector	PASS

Block	Certifies	Result
T2	The cocycle theorem, sharpened. Under any chart re-choice $n \rightarrow n+m_a-m_b$ the triple sum is exactly invariant — certified exhaustively <i>and</i> by the identity making the shifts cancel in pairs — so a nonzero integer sheet defect is a genuine class that must enter the ledger and cannot be erased by chart choice. Scale: exact composed-identity condition with the closing chart constructed	PASS
T3	Abelian curvature componentwise; flat-with-period consumed from pinned EMK-G3, not repeated. New: the sheet defect isn't a curvature component at all — its invariance involves no continuous data, so no relaxation (flat or curved) changes it. $F_UGD = 0 \quad \Delta k = 0$ for a structural reason	PASS
T4	The two named non-implications with witnesses, strengthened to pairwise independence across all 12 ordered pairs of the four sectors	PASS
T5	Faithful preserves and reflects closure; lossy preserves but provably fails to reflect. Third mode separated: a projection that identifies two sectors discards nothing yet still fails — phase +3 with sheet −3 merges to zero and reads closed. Injectivity, not non-discarding, is load-bearing	PASS

Block	Certifies	Result
T6	The bridge. Consuming pinned UGD-1: numerals with identical classical projection and phase content but different seam charge give a transition whose phase and scale residues are exactly zero and whose sheet residue is exactly the seam-charge difference . UGD-1 T4 (“classical projection blind to seam”) and this section’s “lossy projection reports closure” are one statement at two layers	PASS

The number layer and the geometry layer are the same object seen twice.

Declared: state space, transition, ledgers, tolerances, chart cover, P_UGD, and the thermodynamic dictionary (which the section itself calls a constitutive presentation, not a universal identity — its $= \log$ _s option is **never evaluated**). Phase closure certified in exponent-mod-K form throughout.

Chapter 111

LTB-1 — The thermodynamic bridge: what a loop integral is *not*

Sources: vault `lambda_thermodynamic_bridge.tex` (82) and `thermodynamic_worked_cycle.tex` (87). These sections are almost entirely **prohibitive** — they say what may not be identified with what — so each is discharged by an exact witness that blocks the identification permanently. 26 tests; 130 in this folder.

Block	Certifies	Result
T1	The bridge’s main guard, both directions. A Jacobian with nonzero skew sector carrying an exactly flat connection; a perfectly symmetric Jacobian carrying exactly nonzero curvature. And when a constitutive map <i>is</i> stated, a different stated map gives a different curvature from the same Jacobian — the map is a choice, not a discovery	PASS
T2	The subtraction ladder is exact and order-independent over all 120 permutations. Control: a sector defined as the raw curvature itself makes the diagnostic identically zero on every input — a diagnostic that can never fire is not a diagnostic	PASS

Block	Certifies	Result
T3	The van der Waals loop by two independent routes , with $\log((V - b)/(V - b))$ carried as a formal symbol and never evaluated. The attraction parameter cancels exactly : each leg carries an a -term, the two are exact negatives. A fully rational control EOS needs no symbol at all	PASS
T4	Three diagnostics separated: the loop is constitutive; it is not entropy production (the same magnitude arises under both time orientations, so the loop alone can't fix the sign); repetition returns (,) exactly while the branch index advances — EMK-G3's helical statement on a thermodynamic cycle	PASS
T5	A post-hoc ledger is vacuous. Declared in advance, the residue separates protocols; set equal to the observed loop it is identically zero and distinguishes exactly zero pairs — certified as a property of the construction, not one example	PASS
T6	The presentation is reference-dependent (three declared scales, three presentations of the same state) so the dictionary is a choice, not an equation of state. Bridge : consuming pinned CFE-1, the section's Γ_c , Γ_m are the flagship's invariant factors — $I = 1$ exactly on the memoryless face	PASS

Guard thread, fourth level. EMK-G1 T6: metric vs recognition curvature. EMK-G2 T6: Levi-Civita holonomy vs recognition monodromy. EMK-G3 T3: local flatness vs global monodromy. **LTB-1 T1: response asymmetry vs transport curvature.**

Not claimed: that any loop integral is entropy production, that skew response is curvature, that cyclic winding is entropy, or that the UGD dictionary is a universal equation of state. The van der Waals model is a worked constitutive example with **no claim of experimental validation** — as the source itself states.

`Publications/papers/emk-recognition-geometry/README.md`

Chapter 112

Lineage — EMK recognition geometry

112.1 Origin

The rotational seam metric lives in the vault’s `emk_ugd_recognition_geometry` tree as LaTeX: definitions, theorems and proofs, never executed. As with EMK-1, EMK-2 and UGD-1, no executable version existed anywhere in the corpus. EMK-G1 is the first machine-checkable realization.

112.2 What EMK-G1 adds

The source’s mathematics is reproduced, not extended. What is added:

- an arithmetic reformulation that makes the whole geometry exact: carrying $W = A^2$ turns the connection, the Riemann component and the Gaussian curvature into rational functions, so $K(0) = -\kappa$ is certified with no square root and no floating point;
- a two-route check of the curvature ($R_{\{uvuv\}}/\det g$ against the closed form) and of the Christoffels (W-form against Levi-Civita computed independently from the metric components);
- a control for the geodesic seam criterion: an odd perturbation of the warp breaks reflection symmetry and destroys the geodesic property, showing the symmetry is load-bearing rather than decorative;
- a certified SEPARATION of metric curvature from recognition curvature (T6), turning the source’s stated principle into a machine-checkable guard.

112.3 Relationship to the other capsules

- **EMK-1 / EMK-2** (`../emk-ugd-algebra/`): the algebra identifies the seam and grades memory as the odd sector; this capsule gives the seam its metric. T6 certifies that the two curvatures — Gaussian here, odd grade there — are independent, so the layers may be composed but never conflated.

- **Cut-First Equivalence** (`../cut-first-equivalence/`): the recognition two-form of CFE is one of the two objects separated in T6. Nothing in this capsule licenses identifying it with the metric curvature.

112.4 Claim boundary

The metric family is declared, as in the source. $K(0) = -\kappa$ is a theorem for that family and not a universal identification of flow parameters with curvature scalars. Global, completeness and analytic statements are not claimed. Untouched: RH / K0 / L0 / YM continuum gates, and quantum gravity.

`Publications/papers/emk-recognition-geometry/LINEAGE.md`

Chapter 113

EMK / UGD algebra — the determinant seam ladder (EMK-1)

The framework’s **own algebraic core**, brought from the vault into the public certified corpus. Until now the Cut-First Equivalence capsules used an explicit rational equation of state as their witness — a scaffold. This is the theory’s native geometry instead.

113.1 The primitive algebra

$K^2 = I$ seam reflection
 $R^2 = -I$ circular phase transport (the quarter turn)
 $RK = -KR$ mixed rotation-cut parity, $(RK)^2 = I$
 $[R, K] = 2RK$ the UGD bracket

A primitive block is $M = aI + bK + cR + dRK$.

113.2 The determinant identity

$$\det M = (a^2 - b^2) + (c^2 - d^2) = \Delta + \Delta$$

The determinant splits **additively** into a seam-compatible channel and a rotational / anti-seam channel, and the two close or fail **independently**.

113.3 Certified blocks

Block	Statement	Verdict
T1	Primitive relations $K^2=I$, $R^2=-I$, $RK=-KR$, $(RK)^2=I$, and the UGD bracket $[R,K]=2RK$	PASS

Block	Statement	Verdict
T2	The additive determinant identity on the full rational grid (2401 quadruples), two independent routes; channel independence exhibited — each channel can vanish alone, and both can vanish while the operator is nonzero	PASS
T3	Seam projectors $P_{\pm} = (I \pm K)/2$ idempotent/orthogonal/complete; $[A, K] = 0$ iff the mixing blocks vanish — the commutator <i>is</i> the seam-mixing detector	PASS
T4	Exact Schur identity + the quadratic determinant correction; control: both diagonal blocks have det 1 yet the quadratic residue is $1/6$ — first-order closure is provably incomplete	PASS
T5	Cut-commutator curvature $C_i C_x - C_x C_i = -\Delta x \cdot \Delta i \cdot [D_x, D_i]$ exactly; path equality vanishing commutator channel	PASS
T6	Winding index an exact integer, homotopy invariant (1, 1, 0 across scaled and off-origin loops) — non-healable curvature , via rational circle parametrization and exact crossing counts, no trigonometry	PASS
T7	Bridge to CFE: a block is memoryless (commutes with K) exactly when rotation-free ($c=d=0$), and then $\det M = \Delta$ alone	PASS

113.4 EMK-2 — CFE rebuilt on this algebra, and the UGD multiplicative law

EMK-1 T7 gave the licence; EMK-2 performs the replacement and retires the scaffold equation of state.

The algebra is $\mathbb{Z}/2$ -graded: even (seam) sector $\{I, K\}$, odd (rotational) sector $\{R, RK\}$, with $\text{even} \cdot \text{even} = \text{even}$, $\text{odd} \cdot \text{odd} = \text{even}$, $\text{even} \cdot \text{odd} = \text{odd}$. The determinant channels read the two grades — so **memory is the odd grade**, not a dial bolted onto a model.

The UGD multiplicative law lives in the seam channel. In the Cayley coordinate $y = b/a$, even-sector composition is exactly

$$y \circ y = (y + y) / (1 + y y)$$

and under the native chart $x = (1+y)/(1-y)$ this is precisely $\text{Log}_\Sigma(x \circ x) = \text{Log}_\Sigma(x) + \text{Log}_\Sigma(x)$ — the framework’s own multiplicative-to-additive law (Recognition-Kernel-Framework, F00G Theorem 7.1). Certified with **no series, no transcendental evaluation and no floats**. The seam determinant channel is correspondingly multiplicative, $\Delta(MN) = \Delta(M) \cdot \Delta(N)$, while its coordinate is additive: the determinant split and the multiplicative law are one grading seen twice.

Block	Statement	Verdict
T1	The $\mathbb{Z}/2$ grading, with explicit composition laws for each sector	PASS
T2	The multiplicative law in the Cayley chart, exact; seam channel multiplicative; naive-addition control separates	PASS
T3	CFE rebuilt natively: memoryless even grade; zero curvature, zero residue	PASS
T4	Residue on the native carrier: $= \Omega$ exact, zero iff even grade, strictly monotone in the odd amplitude	PASS
T5	The dial was the odd grade: under $= -1$ the scaffold and native densities are identical — the residues reproduce CFE-1’s certified $-32, -16, 0, +16, +32$ exactly	PASS

113.5 UGD-1 — UGD number

The most original layer: a **seam-aware analogue of positional notation**. A classical digit records a scalar at a scale. A UGD digit records **phase, scale index, and local seam charge**:

$$d_n = ({}_a, n, s) \quad \Phi_K \times \times \{-1, 0, 1\}$$

Phase is carried as an **exponent in $\mathbb{Z}/K$** — never evaluated as a root of unity — and the scale base is an exact rational, so no transcendental arithmetic enters any verdict.

Block	Statement	Verdict
T1	Phase exponents form $/K$ with the generator of full order; at $K=4$ the generator has order 4 and its square is the half turn — the quarter turn, at the numeral level	PASS
T2	Carry conservation: phase overflow carries to the next scale, seam overflow returns to $\{-1,0,1\}$ and is pushed to the ledger; total seam charge is conserved exactly (1296 digit pairs). Control: a clamping policy that drops overflow loses charge and separates	PASS
T3	Multiplication is scale convolution — scale indices <i>add</i> , phases multiply mod K , so $_scale$ is multiplicative: the multiplicative-to-additive law at the numeral level	PASS
T4	The classical projection is blind to seam: null-seam recovers ordinary positional notation exactly, while three numerals share the projection 7 with total seam charges 0, 0, 2	PASS
T5	Cut-zero: the neutral digit's neutrality is <i>derived</i> , its projection and ledger are null — yet it occupies a scale and carries a phase exponent, so it is distinct from the <i>absence</i> of a numeral. Zero is the first cut	PASS
T6	-logic: negation $N(, ,k)=(, - , -k)$ is an involution; the seam predicate marks exactly k 0; the null-seam trivial-phase limit recovers classical behaviour (36-point domain, exhaustive)	PASS

UGD1-F1 — the quarter turn now stands certified in **four** presentations: arithmetic (LAM-2 $^2=-1$), quantum geometry (CFE-Q $J^2=-I$), native algebra (EMK-1 $R^2=-I$), and number (UGD-1

at $K=4$).

UGD1-F2 — the carry discipline is a *conservation law*; the ledger is load-bearing, not book-keeping.

UGD1-F3 — the classical projection is provably lossy: UGD numbers carry strictly more than their classical shadows.

113.6 EMK-T1 — the master tensor and recognition-time sector

An **honest** capsule. The vault's master-tensor appendix is largely a *protocol*: the tensor tuple, the eight-channel RTC connection, the residue sectors, tolerances and commit policy are **declared structure**, not derived results — writing declarations into code certifies nothing. This capsule extracts only what carries mathematical content, and records the rest as declared.

Block	Statement	Verdict
T1	J is derived, not primitive: in the primitive basis the cut-swap <i>is</i> K , so $J^2=I$ and it coincides exactly with the geometry capsule's involution $(\mathbf{x}, \mathbf{y}) (\mathbf{y}, \mathbf{x})$	PASS
T2	Channel curvature decomposition exact: $[\Sigma A_- , \Sigma A_+] =$ per-channel curvatures + cross-channel commutators	PASS
T3	The mixed term is necessary: two channels with individually <i>vanishing</i> curvature whose cross-commutator carries the entire master curvature — a channel-by-channel audit is provably insufficient	PASS
T4	Closure is a real decision: the <i>same</i> nonzero curvature is CLOSED when fully ledgered, OPEN when partly unledgered, and CLOSED again when the survivor is outside the active mask	PASS
T5	Time closure clock equality: identical clock reading ($=7$), different time-tensor content, nonzero temporal residue that closes only under a lawful ledger	PASS

Block	Statement	Verdict
T6	Determinant coupling without logarithms: the source’s additive log-Det residue certified in multiplicative form; the coupling factor is exactly $1/\det(T)$ and the ledger cancels it exactly	PASS
T7	Presentation agreement, with the hypothesis shown load-bearing: a map that <i>drops</i> a sector produces a different verdict	PASS

EMKT1-F1 — the geometry involution, the algebra’s seam reflection and the tensor layer’s cut-swap are **one derived object**.

EMKT1-F2 — the mixed curvature term is necessary, not decorative.

EMKT1-F3 — time closure is strictly stronger than clock equality — the same shape as UGD-1’s projection blindness: the visible coordinate never carries the whole state.

113.7 Findings

F1 — the additive split is *structure, not notation*: the channels are genuinely independent.

F2 — first-order closure is provably incomplete; diagonal-block tests cannot certify recognition closure.

F3 — the CFE memory dial and the EMK rotational determinant channel are the same object in two presentations, so CFE’s witness EOS can be replaced by this native algebra.

EMK2-F1 — memory *is* the odd grade; the memory dial was never an extra parameter.

EMK2-F2 — multiplication of native scalars is addition of seam coordinates; the determinant split and the UGD multiplicative law are one grading.

EMK2-F3 — the scaffold retires without cost: every certified curvature and residue value is unchanged under $\gamma = -1$.

113.8 Provenance

Source: vault LaTeX — the EMK algebraic core ladder I–VII and the UGD algebraic-operators appendix. **No executable version existed anywhere** in the corpus (checked Recognition-Kernel-Framework and RH-Framework: neither contains a UGD algebra implementation). This capsule is the first machine-checkable realization.

113.9 Claim boundary

Certified: the finite-dimensional theorem spine. **Not claimed:** the infinite-dimensional trace-class, essential-spectrum, Hurwitz-zeta, ζ -function or Hilbert–Pólya extensions — the vault source

explicitly defers those, and so does this capsule. CFE's (U) uniqueness stays open. No RH / K / L / YM continuum gate is touched; quantum gravity is not touched.

113.10 Reproduce

```
python papers/emk-ugd-algebra/certificates/emk1_determinant_seam_ladder.py
python papers/emk-ugd-algebra/certificates/emk2_native_carrier.py
python papers/emk-ugd-algebra/certificates/ugd1_numerals.py
python papers/emk-ugd-algebra/certificates/emkt1_master_tensor_and_time.py
python -m pytest papers/emk-ugd-algebra/tests -v
```

CI regenerates the certificate and fails on pin drift (EXPECTED_EMK1.sha256, EXPECTED_EMK2.sha256, EXPECTED_UGD1.sha256, EXPECTED_EMKT1.sha256).

Publications/papers/emk-ugd-algebra/README.md

Chapter 114

Lineage — EMK / UGD algebra

114.1 Origin

The EMK algebraic core ladder (I–VII) and the UGD algebraic-operators appendix exist in the provisional vault as LaTeX: definitions, theorems and proofs, written and reviewed but never executed. A search of the public Recognition-Kernel-Framework and the RH-Framework found **no UGD algebra implementation** in either. EMK-1 is therefore the first machine-checkable realization of this layer, not a port of existing code.

114.2 What EMK-1 adds

Nothing mathematically new is claimed. What is added is execution and sharpening:

- the additive determinant identity verified on the full rational coefficient grid by two independent routes;
- **channel independence** exhibited explicitly (each determinant channel vanishing alone; both vanishing on a nonzero operator) — turning the ladder’s “channels can close or fail independently” from a remark into a certificate;
- the incompleteness of first-order closure witnessed by an operator with unit diagonal-block determinants and a nonzero quadratic residue;
- the winding obstruction made exact through a rational circle parametrization and integer crossing counts, with no trigonometry and no floats anywhere in the verdict path;
- an exact algebra-level bridge to Cut-First Equivalence (T7).

114.3 Relationship to the other capsules

- **Cut-First Equivalence** (`../cut-first-equivalence/`): T7 identifies the CFE memory dial with the EMK rotational determinant channel, so CFE’s explicit rational equation of state can be replaced by this native algebra. That replacement is the next capsule’s work, not a claim made here.
- **Λ -seam calibration** (`../lambda-seam-calibration/`): the quarter turn $R^2 = -I$ certified here is the same quarter turn as LAM-2’s derived seam phase $^2 = -1$ and CFE-Q’s Bloch generator $J^2 = -I$ — three presentations, one object.

114.4 Claim boundary

Certified: the finite-dimensional spine. Deferred by the source and not asserted here: all infinite-dimensional analytic extensions. Untouched: RH / K / L / YM continuum gates, and quantum gravity.

114.5 EMK-2 addendum

EMK-2 consumes EMK-1's T7 licence and retires the scaffold equation of state used by the Cut-First Equivalence capsules. Two things are added.

First, the $/2$ grading is made explicit and certified: the even (seam) sector $\{I,K\}$ and the odd (rotational) sector $\{R,RK\}$, with the sector multiplication laws. EMK-1's determinant channels read exactly these two grades, so the memoryless/memory split is the grading itself.

Second, the framework's native multiplicative-to-additive law is bound to the seam channel. The derivation of Log_Sigma lives in the public Recognition-Kernel-Framework (theorems/foundation/F00G, Theorem 7.1) and is a PINNED NAMED DEPENDENCY here — cited, never rederived. What EMK-2 certifies is the exact Cayley-chart identity: even-sector composition is the rational addition law $(y_1+y_2)/(1+y_1y_2)$, which under $x = (1+y)/(1-y)$ is exactly $\text{Log_Sigma}(x_1 \ x_2) = \text{Log_Sigma}(x_1) + \text{Log_Sigma}(x_2)$. No series, no transcendental evaluation, no floats.

The retirement is certified to be free: under $\rho = \chi - 1$ the scaffold and native curvature densities coincide identically, and the native residues reproduce CFE-1's certified values $(-32, -16, 0, +16, +32)$ exactly. CFE's numbers did not depend on the equation of state; the equation of state was a coordinate on the odd grade.

114.6 UGD-1 addendum

UGD-1 certifies the UGD number layer: cut-zero, operator numerals, and the lambda-paraconsistent structural logic. As with EMK-1 and EMK-2, the source is vault LaTeX and no executable version existed.

Two representation decisions keep the capsule exact. Phase is carried as an EXPONENT in Z/K and never evaluated as a root of unity; the scale base is an exact rational. Consequently no transcendental arithmetic enters any verdict path, and the phase algebra is certified as the cyclic group it is rather than as a numerical approximation of one.

The sharpest results are the carry conservation law (seam charge pushed out of the balanced alphabet is conserved in the ledger, with a clamping control that loses charge and separates) and the projection-blindness witness (numerals sharing a classical value while differing in seam data and total seam charge). Together they make precise the appendix's principle that visible equality is not UGD equality.

Not claimed: convergence of admissibly infinite numerals, and the seam composition rule for linked seams, which the source declares rather than derives. Both are recorded as such in the certificate.

114.7 EMK-T1 addendum

EMK-T1 is deliberately partial, and the partiality is the point. The vault's master-tensor appendix mixes two kinds of content: declared protocol (the tensor tuple's component list, the eight-channel split of the RTC connection, the residue sectors, tolerances and commit policy) and statements

with mathematical consequences. Encoding declarations would certify nothing, so only the second kind is certified, and the first is recorded explicitly as declared-not-certified in the certificate’s claim boundary.

Three results are worth naming. The cut-swap J is shown to be a derived map — in the primitive basis it IS K — which unifies it with the geometry capsule’s involution and the algebra’s seam reflection. The mixed curvature term is shown necessary by exhibiting channels whose individual curvatures vanish while their cross-commutator carries the entire master curvature. And time closure is separated from clock equality by an exact witness with an identical clock reading and a nonzero temporal residue, mirroring UGD-1’s projection-blindness result.

The determinant coupling is certified in multiplicative form rather than the source’s additive log-Det form, so no logarithm is ever evaluated; the two statements are equivalent and the multiplicative one stays in $\mathbb{Q}$.

`Publications/papers/emk-ugd-algebra/LINEAGE.md`

Chapter 115

Information Invariance — paper under development

`paper/information_invariance_v1_original.tex` — the original draft (kept for provenance).
`paper/information_invariance_v2.tex` — the working paper: proved finite structure separated from conjectured dynamics. `certificates/l1_certificates.py` — executable certificates for Theorems A and B (exact rational arithmetic). `LEDGER.md` — what is PROVED / OPEN / CONJECTURE / RETRACTED, with named obligations.

115.1 Certificates

file	theorems
<code>l1_certificates.py</code>	A (invariance = flat sector), B (one cycle decides nothing)
<code>d2_phase_origin.py</code>	C (real flow is Schrödinger iff), D (one-channel obstruction)
<code>d2a_cut_supplies_structure.py</code>	E (cut supplies conjugate pair; phase is the odd part)
<code>d2b_linearisation_verdict.py</code>	D2b verdict: damped rotation, not Schrödinger
<code>d3_regulator_scale.py</code>	F (no Planck length without $\hbar$), G (own regulator is cosmological)
<code>d3prime_emergent_einstein.py</code>	K (DOF no-go), L (Clausius route imports $\hbar$), M (source not conserved)
<code>d4_decoherence_audit.py</code>	H (printed rate is not a rate; repair is Diósi–Penrose)
<code>d1_selection_principle.py</code>	I (invariance selects nothing), J (minimal scale usage selects $n=2$)
<code>micro_chi_characterisation.py</code>	N ($=0$ 1), O (curvature sufficient, not necessary)

Rule for this folder: a sentence may claim a result only if the ledger lists it as PROVED with a reproducing certificate. Everything else is written as a conjecture with its obligation.

```
python papers/information-invariance/certificates/l1_certificates.py
python -m pytest papers/information-invariance/tests
```

Publications/papers/information-invariance/README.md

Chapter 116

information_invariance_v2

The following paper is included as compiled from its repository source.

Information Invariance: What It Is, What It Decides, and What It Does Not Derive

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(Dated: August 28, 2026)

We define information invariance for thermodynamic response charts as the stationarity of the loop functional of the entropy-scaled accessibility one-form, and settle what follows from it. Three structural results are proved on finite charts with executable certificates: information invariance is equivalent to vanishing thermodynamic curvature and to exactness of the one-form, i.e. to the Onsager-reciprocal sector; microscopically it is exactly time-reversal invariance of the statistical weight, through the factor $\chi = W(\Theta\Gamma)/W(\Gamma)$; and curvature is a sufficient but not a necessary witness of that invariance, since microscopic asymmetry can cancel under coarse-graining. On the experimental side we prove that the loop area of a *single* thermal cycle decides nothing — a flat chart with a one-time irreversible offset and a curved chart give equal first-cycle areas — which we illustrate on our own graphene Raman data, where the measured loop does not close. We then carry every obligation of the associated emergent-dynamics programme to a verdict rather than leaving it open: the quadratic energy law is not derivable from invariance but is uniquely selected by a stated minimal-scale principle; the linearised dynamics is a damped rotation, not a Schrödinger generator, and we give the exact condition that would make it one; the Planck length is an input rather than a consequence, the scale the theory supplies being cosmological; the emergent-Einstein claim fails on degree-of-freedom counting, on the $\hbar$ -dependence of the thermodynamic route, and on non-conservation of the scalar source; and the proposed decoherence rate is dimensionally invalid, its unique repair being the Diósi–Penrose expression. We regard the negative results as the paper’s main contribution: they replace an unsupported programme with a small set of proved statements and explicitly stated conditions.

I. INTRODUCTION

Onsager reciprocity is derived under microscopic reversibility. Entropy-weighted reparameterisations of response functions yield an effective coupling matrix whose cross-coefficients are generically asymmetric [1]; the associated geometric statement is that the accessibility one-form $\omega = \lambda_p dp + \lambda_v dv$ need not be closed. This raises a structural question: *which* irreversible trajectories are admissible.

The present paper separates what can be proved from what is presently conjectured. Section II defines information invariance as a loop functional. Section III proves two finite theorems about it, with machine-checked certificates. Section V states the empirical situation honestly, including a non-closure finding in our own data that weakens the naive reading of loop area. Section VI states the dynamical programme as conjectures with explicit obligations.

II. INFORMATION INVARIANCE, DEFINED

Definition 1 (Information invariance). *Let ω be the entropy-scaled accessibility one-form on a response chart. Define the loop functional $\mathcal{I}(\gamma) = \oint_\gamma \omega$ for closed paths γ . The chart is information invariant if $\mathcal{I}(\gamma) = 0$ for every admissible closed γ , written $\delta\mathcal{I} = 0$.*

This is a selection principle: it constrains admissible evolution, and asserts neither conservation of microscopic information nor invariance under arbitrary transformations. It postulates no new interaction and does not quantise spacetime.

III. PROVED FINITE STRUCTURE

Discretise the chart on a lattice with rational edge values; $\Omega = d\omega$ is the plaquette circulation, and discrete Stokes holds exactly.

Theorem 1 (Information invariance = flat sector). *On a finite simply connected chart the following are equivalent: (i) $\delta\mathcal{I} = 0$; (ii) $\Omega \equiv 0$; (iii) ω is exact, i.e. a potential Φ exists and line integrals are path independent. On the λ -chart with $\lambda_X = -ST/C_X$, (ii) is the flat-closure sector $\Gamma_c\Gamma_m = 1$, i.e. Onsager reciprocity.*

Proof sketch. (ii) $\Rightarrow$ (i) by discrete Stokes; (i) $\Rightarrow$ (ii) by taking $\gamma = \partial(\text{plaquette})$; (i) $\Leftrightarrow$ (iii) is path independence $\Leftrightarrow$ exactness on a connected graph. All three directions, plus planted-curvature negative controls, are executed over exact rationals in the accompanying certificate. $\square$

Theorem 2 (One cycle does not witness curvature). *There exist a flat chart with a single irreversible offset and a curved chart without offset whose first-cycle loop areas are equal. The two are separated only by the second cycle: the offset construction collapses (reopen ratio 0), the curvature construction reopens with the same area (ratio 1).*

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Consequently a nonzero loop area measured on one cycle is consistent with irreversible drift and does not by itself establish $\iint \Omega \neq 0$. Certificates for both theorems are in `papers/information-invariance/certificates`; the pinned SHA-256 is reproduced by continuous integration on every commit.

IV. WHAT INFORMATION INVARIANCE MEANS MICROSCOPICALLY

The companion work derives, for an ensemble with weight W on a phase space carrying a time-reversal involution Θ , the identity $L_{ij} - \varepsilon_i \varepsilon_j L_{ji} = \int \langle (1 - \chi) J_i(0) J_j(t) \rangle_W dt$ with $\chi(\Gamma) = W(\Theta\Gamma)/W(\Gamma)$, and identifies the coarse-grained antisymmetric part with Ω [1]. Chaining this with Theorem 1 gives information invariance a microscopic reading, and one leg of the chain turns out to be one-directional.

Theorem 3 (Microscopic characterisation). *On a finite phase space, with $C_W(f, g) = \sum_{\Gamma} W(\Gamma) f(\Gamma) g(\Theta\Gamma)$,*

$$C_W(f, g) - C_W(g, f) = \sum_{\Gamma} W(\Gamma) [1 - \chi(\Gamma)] f(\Gamma) g(\Theta\Gamma), \quad (1)$$

exactly, and the following are equivalent: $\chi \equiv 1$; C_W is symmetric for all observables; the antisymmetric response vanishes. Hence $\delta\mathcal{I} = 0$ is exactly time-reversal invariance of the statistical weight.

Theorem 4 (Curvature is sufficient, not necessary). $\chi \equiv 1 \Rightarrow \Omega = 0$ *always. The converse fails: for $\Theta = (01)(23)$ and $W = (3, 1, 1, 3)$ the two orbit contributions are $+2$ and -2 , so $\chi \neq 1$ while the coarse-grained total vanishes.*

Together with Theorem 2 this bounds what a hysteresis experiment can decide:

$$\begin{aligned} \Omega \neq 0 \text{ across cycles} &\Rightarrow \chi \neq 1: \text{informative} \\ \Omega = 0 &\Rightarrow \text{nothing about } \chi \\ \text{one-cycle loop area} \neq 0 &\Rightarrow \text{nothing about } \Omega \end{aligned}$$

The predeclared second-cycle test is therefore not a precaution but the only step that carries an inference.

V. EMPIRICAL SITUATION

Temperature-cycled Raman measurements on monolayer CVD graphene on SiO_2 [1] give reproducible first-cycle hysteresis in the G and $2D$ peak positions, with loop areas well above fit uncertainty (independently refitted: 744 ± 22 and $692 \pm 28 \text{ cm}^{-1}\text{K}$ at spot 1). The loop, however, does *not* close at 300 K: the cooling branch returns $+4.95 \text{ cm}^{-1}$ (G , spot 1) above the heating branch. By Theorem 2 this first-cycle observation is compatible with an irreversible component (adsorbate desorption or strain release) and does not establish reversible curvature. A predeclared decision rule has been registered:

reopen ratio ≥ 0.6 on both bands supports curvature; ≤ 0.2 indicates irreversible drift [2]. The measurement is pending.

VI. CONJECTURED DYNAMICS AND ITS OBLIGATIONS

The following are stated as conjectures. Each names the obligation a derivation must discharge; none is used to support the claims of Sec. III.

We take as the working dynamics, with $u = ct - r$ and energy density $\mathcal{E}(u, \theta, \phi)$,

$$\partial_u \mathcal{E} = \alpha_G \mathcal{E}^2 + \alpha_\Lambda \mathcal{E}_{\text{vac}} + \epsilon \mathcal{F}[\mathcal{E}], \quad (2)$$

and ask what, if anything, justifies it.

Theorem 5 (Invariance alone selects nothing). *On an information-invariant chart $\omega = d\Phi$, and every potential-transporting evolution $\partial_u \Phi = f(\Phi)$ preserves exactness, for arbitrary f . The dynamics compatible with $\delta\mathcal{I} = 0$ therefore form an infinite-dimensional family, and (2) is not derivable from Definition 1.*

Theorem 6 (Minimal scale usage selects the quadratic law). *For $\partial_u \mathcal{E} = \alpha_n \mathcal{E}^n$ the coupling must satisfy $[\alpha_n] = M^{1-n} L^{n-2} T^{2n-2}$. No n admits a coupling built from $\{G, c\}$ alone. Admitting a single new length λ gives uniquely $\alpha_n = G^{n-1} c^{4-4n} \lambda^{2n-3}$, and requiring λ to appear exactly once ($d = 1$) forces $n = 2$, with $\alpha_2 = G\lambda/c^4$.*

Two consequences. First, the quadratic term is not arbitrary once one states the selection principle — one new scale, used once — but it does not follow from invariance. Second, the coupling written as $\alpha_G \sim G/c^4$ is short by exactly one length; by Theorem 14 the length the theory itself supplies is the cosmological $L_{\mathcal{E}}$, giving $\alpha_2 = 1/(c^2 \sqrt{G\mathcal{E}})$. We record (2) as a conjecture with that stated provenance:

Conjecture 1 (Null-coordinate realisation). *Equation (2) with $\alpha_2 = G\lambda/c^4$ is the physically realised dynamics. **Obligation D1'**: justify the selection principle itself, and the identification $\lambda = L_{\mathcal{E}}$.*

A. Where a complex phase can come from (proved)

The step usually left implicit is the appearance of a complex phase in a real theory. Its finite content is settled.

Theorem 7 (Real flow = Schrödinger flow). *Let x' = Ax be a real linear flow on $\mathbb{R}^{2n}$ and J an orthogonal complex structure ($J^2 = -I$, $J^T = -J$). Then $A^T = -A$ and $AJ = JA$ if and only if $H := -\hbar JA$ is Hermitian for J , in which case the flow reads $i\hbar\psi' = H\psi$ under $x \leftrightarrow \psi$; equivalently, the flow is complex linear and norm preserving.*

Theorem 8 (One-channel obstruction). *An odd-dimensional real flow admits no complex structure, since $\det(J)^2 = \det(-I) = (-1)^N$. In particular a single scalar channel $x' = ax$ is never a Schrödinger flow, for any a and any rescaling.*

Linearising Conjecture 1 about a background gives, at leading order, the single real channel $\partial_u \delta \mathcal{E} = 2\alpha_G \bar{\mathcal{E}} \delta \mathcal{E} + \epsilon \mathcal{F}' \delta \mathcal{E}$. By Theorem 8 this cannot be Schrödinger evolution as it stands, and by Theorem 7 what a derivation must supply is explicit:

D2a a conjugate channel π of $\delta \mathcal{E}$ within the dynamics, and a complex structure J on the pair $(\delta \mathcal{E}, \pi)$;

D2b antisymmetry of the linearised generator on that pair, i.e. the quadratic self-interaction must contribute to the Hermitian H and not to a symmetric part of A .

The first is supplied by structure that is already derived, not assumed. An oriented cut yields a quarter turn ι with $\iota^2 = -1$ and $\iota^\dagger = -\iota$ [4], and every generator splits uniquely relative to the cut J into an even part preserving the two sheets and an odd part transporting between them [5]. Taking the conjugate channel to be the image on the opposite sheet gives:

Theorem 9 (Cut supplies the pair; phase is the odd part). *Let J be a balanced cut, and write the generator in the sheet basis as blocks $A = \begin{pmatrix} A_{++} & B \\ C & A_{--} \end{pmatrix}$ with $\iota = \begin{pmatrix} 0 & -I \\ I & 0 \end{pmatrix}$. Then $A\iota = \iota A$ iff $A_{++} = A_{--}$ and $C = -B$; $A^T = -A$ iff the diagonal blocks are antisymmetric and $C = -B^T$; both hold iff the diagonal blocks are equal and antisymmetric and B is symmetric, in which case $H = -\hbar A$ is Hermitian and Theorem 7 applies. A purely cut-even generator never mixes the sheets and produces no phase, and the symmetric part of A is exactly the damping: $d|x|^2/du = 2x^T A_{\text{sym}} x$.*

B. The computation, and its negative outcome

Theorem 9 leaves one computation. We perform it on a transverse lattice with the parity cut, in exact arithmetic. The linearisation of Conjecture 1 about a background contributes three operator types: pointwise multiplication M by $2\alpha_G \bar{\mathcal{E}}$ from the quadratic term, a second difference L from any diffusive part of $\mathcal{F}$, and a first difference D from any transport part. We find:

- M is symmetric and cut even; it contributes $d|x|^2/du = 4\alpha_G \bar{\mathcal{E}} |x|^2$, i.e. pure growth or decay, and no phase;
- L is symmetric and negative semidefinite: diffusion, no phase;
- D is antisymmetric and cut odd: it alone preserves the norm and meets the conditions of Theorem 9;

- for $A = M + L + D$ the symmetric part is exactly $M + L$, which does not vanish at a nonzero background; a coordinate rescaling is a similarity and cannot remove it.

Result. The linearised dynamics of Conjecture 1 is *not* a Schrödinger generator. It is a damped (or amplified) rotation. The emergent-quantum step therefore does not follow from the equation as written, and we withdraw it as a claim. What survives is a sharp conditional: the step holds if and only if the dynamics is amended so that the cut-even symmetric part cancels at the background, $M + L = 0$, leaving the cut-odd transport term as the generator. Theorems 1 and 2 are independent of this and are unaffected.

C. Emergent geometry: three obstructions

Theorem 10 (Degree-of-freedom no-go). *In four dimensions the metric has ten components; four coordinate conditions and four constraints leave two propagating polarisations. The dynamics carries one scalar function, so it can reproduce at most a one-function sector (spherically symmetric or conformally flat) and cannot generate two independent polarisations.*

Theorem 11 (The Clausius route imports $\hbar$). *No temperature is constructible from acceleration and c alone in (M, L, T, Θ) ; the Unruh temperature requires $\hbar$ and k_B . Hence the derivation of the field equations from $dQ = T dS$ [6] is unavailable to a dynamics claiming to be deterministic and non-quantised.*

Theorem 12 (The scalar source is not conserved). *The contracted Bianchi identity forces $\nabla^\mu T_{\mu\nu} = 0$. The law $\partial_u \mathcal{E} = \alpha \mathcal{E}^2 + \dots$ is not a divergence: on a periodic transverse lattice the total changes, whereas a genuine divergence leaves it fixed.*

We therefore withdraw the emergent-gravity claim as a derivation. Three conditional routes remain and are stated rather than assumed: restrict to the one-function sector; admit $\hbar$ explicitly and give up deriving quantum structure; or supply the compensating flux, or a second field, that restores conservation of the source.

D. The regulator scale (settled)

The finiteness claim usually attached to Conjecture ?? is $\langle \mathcal{E}(x) \mathcal{E}(x') \rangle \sim (r^2 + L_P^2)^{-1}$ with $L_P = \sqrt{\hbar G/c^3}$. Dimensional analysis decides its status.

Theorem 13 (No Planck length without $\hbar$). *In the (M, L, T) dimension group, $\hbar$ is not in the span of $\{c, G\}$, and no product $G^a c^b$ is a length. Hence L_P cannot be constructed from the deterministic constants; admitting it presupposes $\hbar$.*

Theorem 14 (The theory’s own regulator is cosmological). *From $\{c, G, \mathcal{E}\}$ a length does exist, uniquely up to powers: $L_{\mathcal{E}} = c^2/\sqrt{G\mathcal{E}}$, equivalently $\Lambda^{-1/2}$. For a vacuum energy density of the observed order, $L_{\mathcal{E}}/L_P \sim 10^{61}$.*

Consequently the model regularises, if at all, at a cosmological scale; inserting L_P is an assumption and the ultraviolet-finiteness claim of the earlier draft is withdrawn. The theory must either admit $\hbar$ as a primitive — forfeiting the claim to derive quantum structure — or work with $L_{\mathcal{E}}$.

E. The decoherence rate (corrected)

Theorem 15 (The printed rate is not a rate). *$Gm^2/(\hbar V)$ has dimensions $M^0 L^{-2} T^{-1}$: a rate per unit area. The unique repair replacing the volume by a length, $Gm^2/(\hbar R)$, is dimensionally a rate and is exactly the Diósi–Penrose expression $E_G/\hbar$ with $E_G \sim Gm^2/R$ [7, 8].*

With $m = 10^{-17}$ kg and $R = V^{1/3} = 10^{-7}$ m the repaired rate is $6.3 \times 10^{-4} \text{ s}^{-1}$, of the order quoted in the earlier draft — but it is a Diósi–Penrose number, recovered only after the correction. We therefore state the result as a *consistency requirement*: any emergent gravity of this kind must reproduce $E_G/\hbar$ in this regime. It becomes a prediction only if one of the following is supplied: a derivation of the coefficient, which DP fixes only through a choice of regularisation length; a mass or geometry regime of departure; or a dependence on shape or entropy gradient, which DP does not carry. Optomechanical tests discriminate such departures [9, 10], not the agreement.

VII. SUMMARY OF STATUS

Every claim below is either proved with a machine-checked certificate that is regenerated on each commit, or marked as a stated condition. No claim rests on an argument that is not reproduced by the accompanying code.

Statement	Status	W
$\delta\mathcal{I} = 0 \Leftrightarrow \Omega = 0 \Leftrightarrow \text{exact}$	proved	T
One cycle does not witness curvature	proved	T
Real flow is Schrödinger iff J , antisym.	proved	T
One channel admits no complex structure	proved	T
Cut supplies the pair; phase is odd part	proved	T
No Planck length without $\hbar$	proved	T
Own regulator is cosmological	proved	T
Printed decoherence rate is not a rate	proved	T
Invariance alone selects no dynamics	proved	T
Minimal scale usage selects $n = 2$	proved	T
One scalar cannot carry two polarisations	proved	T
Clausius route imports $\hbar$	proved	T
Scalar source is not conserved	proved	T
$\delta\mathcal{I} = 0 \Leftrightarrow \chi \equiv 1$	proved	T
$\Omega = 0$ does not imply $\chi \equiv 1$	proved	T
Selection principle itself; $\lambda = L_{\mathcal{E}}$	stated condition	D
Reversible curvature in graphene	measurement pending	S

VIII. CONCLUSION

Information invariance, defined as stationarity of the loop functional, is equivalent on finite charts to vanishing thermodynamic curvature and to exactness — a proved statement with an executable certificate. The experimental signature usually quoted for curvature, a nonzero one-cycle loop area, is proved insufficient; the decision is deferred to a predeclared second-cycle test. The dynamical programme connecting this principle to quantum and gravitational phenomena is stated as conjectures with named obligations, and every obligation is carried to a verdict rather than left open. The emergent-gravity claim is withdrawn on three independent grounds — degree-of-freedom counting, the $\hbar$ -dependence of the thermodynamic route, and non-conservation of the scalar source. For the quantum step the obligation is discharged with a negative outcome: the proposed energy dynamics generates a damped rotation rather than unitary evolution, and the step survives only as the explicit condition that the cut-even symmetric part cancel at the background. We regard reporting this as more useful than asserting the step.

REPRODUCIBILITY

All theorems marked “proved” are accompanied by executable certificates over exact rational arithmetic, with pinned SHA-256 digests regenerated by continuous integration on every commit; tampering with any recorded value changes the digest and fails the build. Source, certificates

and the claim ledger: Parveen117/Publications, Science, Bengaluru.
papers/information-invariance.

ACKNOWLEDGMENTS

Underlying Raman data were acquired at the Centre for Nano Science and Engineering, Indian Institute of

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- [1] M. Dabas, “Geometric Completion of Thermodynamic Response,” arXiv:2603.20773 [cond-mat.stat-mech] (2026).
 - [2] Parveen117/Thermodynamics-Reproducibility, campaign T02_RAMAN.THERMAL.HYSTERESIS (2026).
 - [3] Dimensional statements in Sec. V use only (M, L, T) exponents; the numerical illustration uses a vacuum energy density of order 10^{-9} J m^{-3} .
 - [4] Recognition-Kernel Framework, `theorems/foundation/F00E_NATIVE_EULER_FROM_IOTA_COMPLET` (PROVED), Zenodo doi:10.5281/zenodo.21760119.
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Chapter 117

Lambda-Seam Calibration (LAM-1)

Calibration and audit of the **legacy Λ -analytic layer** — the Λ -Eisenstein / Λ -mock functional-equation machinery that grew out of the T-V-S-P compass diagram of *Geometric Completion of Thermodynamic Response* (arXiv:2603.20773) — against exact, executable arithmetic.

The legacy program carried the compass response block

$$L = [[p, Lpv], [Lvp, v]], \quad = \arg(Lpv / Lvp)$$

(the Onsager non-reciprocity bias phase) into automorphic machinery, producing “ Λ -functional equations” of the form $\Xi_-(s) = e^{\{i\}} \Xi_-(1-s)$. That layer is exactly where the legacy framework went silent on RH and Yang–Mills. This capsule certifies the finite content of that layer and names the mechanism of the silence.

117.1 Certified blocks (LAM-1)

Block	Statement	Verdict
T1	Ramanujan two-route: exact cyclotomic reduction of $c_c(n)$ in $Z[x]/\Phi_c(x)$ equals $\sum_{d \gcd(c,n)} d \cdot (c/d)$ on a 1476-pair grid — the layer the Λ -twist multiplies but never enters	PASS
T2	Memory-lattice multiplicativity: $\Xi_-(n) = e^{\{i \Omega(n)\}}$ is exact integer bookkeeping — Ω additive on all $mn \leq 400$; formal Euler product reproduces every $n \leq 300$ with marker exactly $\Omega(n)$, multiplicity exactly 1	PASS

Block	Statement	Verdict
T3	Theta seam certified: ($1/t$)= $\sqrt{t} \cdot (t)$ at $t=2$ with two-sided directed rational enclosures, widths $< 1e-20$ (actual agreement 30+ digits), no floats in any verdict path	PASS
T4	Declared-phase audit: under the exact S-reindexing bijection $(c,d) \rightarrow (d,-c)$ on the primitive region $ c , d \leq 60$ (8812 pairs), the coset-weight layer has zero net change (16512 = 16512)	PASS

Controls: coprimality tamper separates (T1); non-additive fake winding detected (T2); $\sqrt{3}$ tamper separates the enclosures with certified gap > 0.319 (T3); asymmetric region separates (T4).

117.2 Finding LAM1-F1

The $e^{\{i\}}$ functional-equation factor of the legacy Λ -construction is a **declared** right S-multiplier, not a derived quantity. The legacy canvas itself records that no homomorphism $SL(2, \mathbb{Z}) \rightarrow S^1$ can realize arbitrary (finite abelianization), and T4 certifies that the coset-weight bookkeeping contributes zero net phase across the S-move. The legacy analytic layer therefore decorated the classical substrate with an *input* phase — which is the precise mechanism of its silence on RH/YM: **the phase was never load-bearing**. Any native functional equation (obligation N2) must *derive* its seam-crossing phase; the theta seam of T3 is the classical shape of that single crossing.

117.3 Claim boundary

- Legacy Λ -FE: declared multiplier over the classical substrate.
- N1 (native continuation), N2 (native functional equation), N3 (identification): **OPEN**.
- $K / L / RH$: **OPEN**. YM continuum gates: **OPEN**.
- No claims about zeros of any L-function.
- Classical anchors (Poisson summation, Jacobi theta inversion, $SL(2, \mathbb{Z})$ coset structure, Ramanujan-sum closed form) are pinned named dependencies, cited and never rederived natively.

117.4 Certified blocks (LAM-2: derived seam phase)

The counterpoint to LAM1-F1 — for a *genuine* twist the seam phase is **derived**, as finite arithmetic on the cut’s residue classes:

Block	Statement	Verdict
T1	Gauss magnitude exact: $ (\cdot) ^2 = q$ in $[-L]$ for all 34 primitive characters, moduli $\{3,4,5,7,9,11,13\}$ (cyclic convolution + reduction mod Φ_L ; no floats, no complex numbers)	PASS
T1b	Quarter-turn instance: $(\cdot)^2$ $= -4$ exactly — the derived seam phase at $q=4$ satisfies the cut relation $^2 = -1$	PASS
T2	The derivation step exact: separated-Gauss identity Σ_a $\neg(a) \neg q^{\{an\}} = (n) (\neg)$ at every residue n for every primitive (335 identities), including $\gcd(n,q)>1$ where both sides vanish — the vanishing <i>is</i> primitivity	PASS
T3	Derived phase certified analytically: twisted theta seam $-(1/2) = \sqrt{8} \cdot -(2)$ at $q=4$ with derived multiplier $/(i\sqrt{q}) = 1$, enclosures agreeing to 30+ digits	PASS
T4	No conductor for the legacy twist: for every modulus q 60, explicit congruent pair with different Ω — the winding twist is a Dirichlet character of no modulus	PASS

Controls: imprimitive character mod 9 gives $||^2 = 0$ and fails the derivation step at residue $n=3$ (both exact); multiplier tamper ($\sqrt{2}$ for $\sqrt{8}$) separates the certified enclosures with gap > 0.58 .

117.5 Findings LAM2-F1/F2/F3

F1 (template): for primitive-character twists the FE seam phase is finite cut arithmetic — exact magnitude, exact derivation step, and a certified analytic seam confirming the derived multiplier. A derived phase is *load-bearing*: tampering it separates certified enclosures.

F2 (native hook, remark-level): $(\cdot)^2 = -4$ — the derived phase at $q=4$ is exactly the quarter turn: $(/\sqrt{q})^2 = -1$, the cut's defining relation, obtained from seam arithmetic rather than declared.

F3 (impossibility half): the legacy winding twist $-(n) = e^{\{i \Omega(n)\}}$ has **no conductor** (witness for every $q \leq 60$), so the Gauss-seam derivation route does not exist for it: the declared

phase of LAM1-F1 was *forced* by the twist’s own arithmetic. Together, LAM1-F1 + LAM2-F1 state the N2 design constraint: a native functional equation must ride a twist whose phase is derivable as finite seam arithmetic.

117.6 Certified blocks (LAM-3: the crossing, derived)

The derivation of the $s = 1$ - s crossing itself, executed and certified at machine level — theta seam $\rightarrow$ split-flip identity $\rightarrow$ manifestly symmetric completed kernel = Euler-product side — at points where the native Σ exists. The FE phase for s is the derived $+1$ of the untwisted seam: an *output* of the crossing, never an input.

Block	Statement	Verdict
T1	Memory-grammar values: $\ln$, 2 , $\ln$, (3) , (5) as exact-rational enclosures (recognized body + Bernoulli correction window + declared bracketed tail), widths $< 1e-24$, two truncation depths overlap	PASS
T2	Split-flip identity at $s=3$: $t^{1/2} dt = 1/6 + \omega$, $t^{-2} dt$, two independent routes (incomplete-gamma series vs E route with exact rational cancellation) agreeing to 28+ digits — the crossing step, certified	PASS
T3	Completed identity at $s=3$: $(3)/(2) = 1/6 + \omega$, $(t^{1/2} + t^{-2}) dt$	PASS
T4	Second point $s=5$: $3(5)/(4^2) = 1/20 + \omega$, $(t^{3/2} + t^{-3}) dt$ — kills point-luck	PASS

Controls: polar-drop separates (gap $> 1/10$); seam-weight tamper $(1/t) \rightarrow t \cdot (t) + (t-1)/2$ separates (gap $> 2/5$); reflected-kernel exponent tamper separates (gap $> 1/200$); body-only separates (memory grammar load-bearing); Bernoulli-window tamper separates.

Note on the E route: at $c = 16$ the convergent-series route suffers ~ 22 digits of cancellation — fatal in floats, *exact* in rationals. The capsule’s positivity and width assertions at that point are the machine witness that the arithmetic discipline is load-bearing.

117.7 Findings LAM3-F1/F2

F1: the crossing is derived end-to-end and machine-certified on the $\text{Re } s > 1$ side. Contrast chain complete across the paper: LAM-1 — declared phase, never load-bearing; LAM-2 — derived phase,

load-bearing finite seam arithmetic; LAM-3 — the derivation itself executed, with every tamper separating certified enclosures.

F2: the memory grammar is load-bearing: body-only (3) separates by more than $1\text{e-}5$ at widths below $1\text{e-}24$ — the public echo of “without memory terms there is no critical strip.”

117.8 Honest boundary (LAM-3)

Continuation into the critical strip via the symmetric kernel is **definitional** here, not a native theorem. The exact remaining native obligation for N2 is now pinned: *reproduce the split-flip step (T2) inside the cut grammar, where the theta seam is a native theorem rather than a pinned classical anchor*. T3 is T2 composed with exact termwise Mellin evaluation (recorded per TAUT-1); the tails on both sides share the Euler–Maclaurin utility.

117.9 Reproduce

```
python papers/lambda-seam-calibration/certificates/lam1_seam_interface.py
python papers/lambda-seam-calibration/certificates/lam2_derived_phase.py
python papers/lambda-seam-calibration/certificates/lam3_crossing_derivation.py
python -m pytest papers/lambda-seam-calibration/tests -v
```

The generator is deterministic; CI regenerates the certificate and fails on any pin drift (EXPECTED_LAM1.sha256, EXPECTED_LAM2.sha256, EXPECTED_LAM3.sha256).

117.10 Arithmetic discipline

Integers, exact rationals, exact integer polynomial arithmetic, and directed rational intervals with relative outward rounding to 300-bit significands. No floating point in any verdict path.

Publications/papers/lambda-seam-calibration/README.md

Chapter 118

Lineage and Ruling

118.1 Origin chain

1. **T-V-S-P compass** — the central diagram of *Geometric Completion of Thermodynamic Response* (arXiv:2603.20773): four response channels on a two-dimensional equilibrium chart, response block $L = \begin{bmatrix} p, & Lpv \\ Lvp, & v \end{bmatrix}$ with Onsager non-reciprocity $\theta = \arg(Lpv/Lvp)$.
2. **-geometry / Λ -geometric canvases** (legacy, 2024–25 era): projective seam spectrum $\Lambda_{\pm}(\Gamma, \cdot)$, stability wedge, exceptional (Jordan) lines, seam holonomy, and the interface conjecture $\Phi_{hol} = \cdot_{\Lambda}$ — the seam holonomy as functional-equation phase.
3. **Λ -analytic rigor canvases** (legacy): Λ -twisted Poincaré/heat kernel $\rightarrow$ Mellin $\rightarrow$ unfolding/Poisson $\rightarrow \Xi_{-}(s) = e^{\{i\}} \Xi_{-}(1-s)$, with $\Gamma/\cdot$ constant audits (including the half-weight Λ -mock case). This is the layer where the legacy framework went silent on RH and Yang–Mills.

118.2 Ruling: MINE-DON’T-MERGE

Zero claims are imported from the legacy canvases. What is mined:

- The **shape** of Lemma 13.7 (“the Poisson step only sees the right S-multiplier”): the entire functional-equation content of the cut crossing concentrates in one move at S. LAM-1 certifies the finite bookkeeping half of this (T4) and the classical crossing itself (T3).
- The compass-phase identification (θ of the response block = θ of the FE multiplier), recorded as lineage, not as a theorem.

What is explicitly **not** imported:

- The legacy “ Λ -functional equation” as a result — LAM1-F1 certifies its phase is declared, not derived.
- All informal holonomy/transport statements of the legacy geometric canvases (e.g. the informal seam-holonomy theorem and the seam quantization conjecture). Their finite/structural content is superseded by certified capsule results elsewhere in this repository and the framework repos (seam-integer counts, transported multiplicity-one crossing lines); legacy formulas in this area have previously failed audit and must not be cited.
- All “TTC” sectoral specializations.

118.3 Relation to the open program

The one obligation this calibration sharpens: a native functional equation (N2) must *derive* its seam-crossing phase from the cut grammar (memory channel included), rather than declare it. The theta identity certified in T3 is the classical shape of that single seam crossing; the memory-lattice bookkeeping of T2 is the exact object any native twist rides on. Both are finite, exact, and now pinned.

118.4 LAM-2 addendum

LAM-2 supplies the constructive counterpoint: the derived-phase template (Gauss-sum seam arithmetic, exact in cyclotomic rings) and the impossibility half (the legacy winding twist has no conductor, so the derivation route never existed for it). Classical anchors — Gauss sums, the separated-Gauss identity, the odd twisted theta functional equation — are pinned named dependencies; finite instances are verified exactly, nothing is claimed as new mathematics. The native obligation N2 remains open: what LAM-2 fixes is its design constraint.

118.5 LAM-3 addendum

LAM-3 closes the paper's contrast arc by executing the classical derivation of the $s \leftrightarrow 1-s$ crossing with certified arithmetic on the memory-grammar carrier, at points where the native zeta exists. The legacy canvases attempted to build functional equations by decorating this substrate with a declared phase; LAM-3 shows the substrate's own phase (+1) being derived, tamper-separable at every step. The single remaining native obligation for N2 is stated in the certificate: reproduce the split-flip step inside the cut grammar with the theta seam as a native theorem. Euler-Maclaurin enveloping remainders and the theta/Mellin substrate are pinned classical anchors throughout.

Publications/papers/lambda-seam-calibration/LINEAGE.md

Chapter 119

Poincaré Calibration — PC-1

This is a calibration, not a proof. The Poincaré conjecture is a proved theorem (Hamilton–Perelman, 2003). Nothing here claims any part of it and nothing here is new mathematics. The capsule instantiates the dictionary of MP `adapters/poincare/adapter.tex` v0.2 (“Poincaré Calibration Adapter”), whose stated purpose is to test whether the framework can carry native hypotheses, singularity control, surgery, extinction and topological reconstruction **without hiding any step inside a decorative closure predicate**. PC-1 answers that for one piece — exactly.

119.1 Carrier

Left-invariant metrics on $\mathbf{SU}(2)$ (deliberately the same group as the Yang–Mills line), diagonal in a Milnor frame $\mathbf{g} = \text{diag}(A,B,C)$. Ricci flow is an exact rational ODE: $A' = 2[(B-C)^2 - A^2]/(BC)$ and cyclically.

119.2 Certified

	Statement
T1	$\lambda(\mathbf{g}) = \min \text{spec}(-4\Delta + R) = R(\mathbf{g})$ on any homogeneous metric; $R = [2(AB+BC+CA) - (A^2+B^2+C^2)]/(ABC)$ verified
T2	$dR/dt - 2\ \text{Ric}\ ^2 \leq 0$ in (A,B,C) — Perelman’s λ -monotonicity as a <i>polynomial identity</i> , with an explicit 3-term sum-of-squares witness. No analysis, no tolerance, no approximation
T3	Round solution exact: $a(t) = a - 2t$, $\lambda = 3/a$, extinction $t^* = a/2$ exactly
T4	Berger rounding exact: $du/dt = -4(x-z)/(xz)$ for $u = x/z$, so $\text{sign}(du/dt) = -\text{sign}(u-1)$ — monotone convergence to round

	Statement
T5	Seam readout: is a bottom-of-spectrum threshold object, strictly increasing threshold crossings are irreversible (one-directional seam flow; same primitive as the YM seam-integer dock, opposite regime)

Controls: R-formula tamper and flow tamper both break the identity; round-case specialization consistent; $\|\text{Ric}\|^2 > 0$ strictly; collapse boundary sampled and recorded as an obstruction-tower entry, not claimed.

119.3 Honest remainder

Untouched, restated from the adapter’s obstruction tower: surgery and neck/cap data; non-collapsing, -solutions, canonical neighbourhoods; singularity classification off the homogeneous carrier; finite-time extinction in general; topological reconstruction; **and the Poincaré conjecture itself — NOT CLAIMED**. Hamilton–Perelman stays a pinned named dependency, cited never rederived (CIRC-1).

The homogeneous carrier is precisely where surgery and collapse are absent. That is why this is a calibration.

119.4 Reproduce

```
python papers/poincare-calibration/certificates/pc1_lambda_monotonicity.py
python -m pytest papers/poincare-calibration/tests -v
```

Publications/papers/poincare-calibration/README.md

Chapter 120

RST-1 — Recognition-Seam Topology: the exact spine, certified

Source: “**Recognition-Seam Topology and Daggered Index Geometry**” (M. Dabas, July 2026, arXiv source v1), sections 2–7. The paper is the mature front door of the RH programme (MP PR #94 lineage) and carries its own claim ledger; this capsule certifies the ledger’s *proved* finite, circle-operator and model statements in exact rational arithmetic — no floats, no transcendental evaluation, no truncation posing as a limit — and records the analytic layer as declared-not-certified.

120.1 Blocks

Block	Content	Key exact fact
T1	Cut-stable topology, Eye quotient, seam, curvature defect (§2)	All 64 subsets checked; $= \{ , \{0..3\}, \{4,5\}, X \}$ is a genuine non-discrete topology; seam = zero-cost locus with an off-seam point of <i>infinite</i> cost; the defect fires exactly where an unlawful declared composite breaks recognized path equality
T2	Rotation instead of gluing (§3.2, §4)	Orientation target factors through no map on the glued quotient; one bit of sheet label repairs it; recognition equality state identity; reflected side target is target-null exactly on the fixed locus

Block	Content	Key exact fact
T3	Rotation-induced involution (§3.3)	J_U self-adjoint involution with orthogonal projections for the rational 3-4-5 rotation; bridge: J_{U=I} is the cut swap — fourth certified presentation of the derived involution (EMK-1 T1, EMK-G1 T1, EMK-T1 T1)
T4	Flow generator (§3.4, §5)	Cayley loop as an exact polynomial matrix in h: $\mathbf{L_h} = \mathbf{I} + \mathbf{h^2[D,D]}$ exactly (the nilpotent loop terminates at degree 2 — stronger than the paper's $O(h^3)$); commuting control gives $\mathbf{L_h} = \mathbf{I}$; eigenmode law certified term-by-term with no exponential evaluated; covariance defect conjugation invariance
T5	Derived dagger algebra (§6)	$\mathbf{R}^* = -\mathbf{R}$ but $\mathbf{R} = +\mathbf{R}$ over Gaussian rationals: dagger self-adjointness is not Hilbert self-adjointness; leakage three-way equivalence on all 256 grid matrices
T6	Circle carrier (§7)	$\mathbf{sf} = \mathbf{Wind} = \mathbf{ind} = \mathbf{q}$ for $q \in \{-3, -1, 0, 1, 2, 5\}$: rational crossing times with a proof the window is exhaustive for the <i>full</i> Fourier spectrum; winding by exact signed crossings on the Pythagorean parametrization; Toeplitz kernel/cokernel read off the shift
T7	Staircase negative control (§5.8)	Length exactly 2 for every n while $\sup\text{-distance}^2 \leq 1/(2n^2)$: limit of lengths length of limit; not evidence of curvature

120.2 Findings

- **F1 (derived involution, fourth presentation).** The rotation-induced J_U at $U = I$ *is* the cut swap. Geometry involution (EMK-G1), algebra seam reflection (EMK-1), tensor cut-swap (EMK-T1) and now the sheet-exchange of rotation transport are one derived object. The involution is never primitive — the paper’s derived-involution principle is now a certificate, not a remark.
- **F2 (curvature = non-closure, without limits).** The commutator-loop residue is a polynomial-coefficient identity, and for nilpotent generators the loop *equals* $I + h^2[D, D]$ with no higher terms.
- **F3 (index without truncation).** The circle identity $sf = \text{Wind} = \text{ind} = q$ is certified with every count exact and exhaustive.
- **F4 (memory is one bit, minimally).** Gluing destroys orientation targets; the minimal repair is a single binary sheet label.

120.3 Claim boundary

Analytic theorems of the source (damped trace-class family Thm 8.8, undamped strip classification Thm 9.1, Schatten thresholds, archimedean defect) are **declared, not certified** — they are analytic, outside exact rational verification, and are not consumed. No arithmetic meaning is assigned to q . The represented seam is **not** identified with the critical line. Model zeros are **not** zeros of i . **Riemann hypothesis: ABSTAIN** — exactly as the source paper states.

120.4 Next

RST-2 will certify the finite arithmetic/transfer/fusion spine (§8–14): Möbius pressure extraction, radical recognition current, prime-shell transfer and asymmetric leakage, zero-identification boundary, and the Surya/Weil sign fusion on rational-eigenvalue models.

120.5 Reproduce

```
python certificates/rst1_recognition_seam_topology.py
python -m pytest tests -v
```

The certificate regenerates byte-identically; its sha256 is pinned in `certificates/EXPECTED_RST1.sha256`.

Chapter 121

RST-2 — Arithmetic and fusion spine (§8–14), certified

Same folder, second capsule, 35 further tests (70 total in this folder).

Block	Content	Key exact fact
T1	Sampled reflection, spectral blindness (§8.2–8.3)	$(s)=1-s$ swaps the rational sample pairs exactly; seam projection provably independent of the anti-seam component
T2	Weights, leakage norm, crossings (§8.4, §8.7)	Equal pair weights swap isometric (both directions); finite leakage norm ² = max a ² attained; avoided family has $\det = -((-c)^2+\mu^2) \quad -\mu^2 < 0$ — zero never in spectrum
T3	Arithmetic recognition (§10)	The formal-log move: log n carried as the vector of prime exponents, so $(L1)*\mu = \Lambda$ is an exact integer-vector identity for all n ≤ 60 — no logarithm evaluated. Λ -current 0 on every anti-seam class; contamination $\rightarrow$ exactly e
T4	Prime-shell transfer (§11)	$U A = A P, J A = A, P A = 0, J U * J = U$ exactly; asymmetric leakage = $((w_R-w_L)/2)(e_R-e_L)$ per shell; converse-fails witness: a common seam-sector error is invisible to the anti-seam current

Block	Content	Key exact fact
T5	Zero-identification boundary (§13)	Paired determinant: zeros $\pm$ with multiplicity by exact division (double level $\rightarrow$ double zero); log-derivative identity in cross-multiplied polynomial form; nonvanishing factor preserves the divisor; working Z1/Z2/Z3 obstruction validator
T6	Surya/Weil fusion (§14)	Feshbach congruence $M = S \text{ diag}(A,F)S$ as an exact identity with negative witnesses transported through it; Surya without square roots (squared amplitude $x^2/(1+x^2)$ on $x < 0$): $A_{SS} = 0$ $W = 0$; seam charge additive, $= -\text{spectral flow}$, every crossing downward; $\mu = 1$ strictly outside the charge; principal holonomy blindness ($m \bmod 1 = 0 \implies m$)

Findings. F5: the projection-blindness thread now spans four capsules (UGD-1 T4, EMK-T1 T5, RST-2 T1, principal holonomy) — the visible coordinate never carries the full state. F6: the anti-seam current detects only the *antisymmetric* discrepancy — the seam sector needs its own audit (converse-fails witness). F7: the Surya sign problem is decidable entirely in squared-amplitude arithmetic — no functional calculus square root is ever needed for the vanishing question.

Claim boundary. Gaussian damping form and $1/\sqrt{2}$ normalization are declared; W , Weil RH, the completed parity/Feshbach reduction and the even compact operator are imported by the source and NOT certified here; fusion gates F1–F5 remain OPEN; Zmodel $\rightarrow$ Z forbidden. **RH: ABSTAIN.**

```
python certificates/rst2_arithmetic_fusion_spine.py
python -m pytest tests -v
```

Chapter 122

EMK-TOP-1 — The vault topology appendix, exact spine

Source: APPENDIX_EMK_TOPOLOGY_RIGOROUS.tex in the private vault, with the sheet-rotation subsection of the Euler information–curvature appendix. **This is a different object from RST-1** (see LINEAGE.md): the paper gives cut-stable topology and index calibration; the appendix builds seam *cohomology*. 29 tests; 99 in this folder.

Block	Certifies	Result
T1	The I, R, K system is closed and finite (order 8 , $\{\pm 1\} \times \{I, R, K, RK\}$); explicit KIR words that are the identity under the shadow projection while their holonomy is $-I$ — visibly closed loops that do not recognition-close; the lawful ledger closes them exactly	PASS
T2	On an explicit complex (square + diagonal, one face filled) $d \, d = 0$ and H^1 rank exactly 1 ; a closed cochain has equal charge on homologous cycles and zero on boundaries; a non-closed one is provably not a class function	PASS

Block	Certifies	Result
T3	The additive log-determinant cochain certified multiplicatively — no logarithm evaluated. The corollary as an exact witness: a loop whose global determinant returns exactly while subcycle charges are 6 and 1/6	PASS
T4	The trichotomy exact / closed-non-exact / open, decided by exact rank, with a witness of each and an explicit for the exact one. Bridge: CFE-U's $H^1 = 0$ means the middle class is <i>empty</i> on the response complex — the trichotomy is about the complex, not a universal law	PASS
T5	Sheet-rotation class with an exact cocycle, phase carried as an exponent in $/K$ (never a root of unity): a loop with total turn $0 \bmod K$ — visible phase return — but sheet index 2 and winding 6	PASS
T6	Seam survival every active sector closes; shadow-only survival witness, exact and wrong ledgers separated, and monotonicity in the active set (so it must be declared before the verdict)	PASS

Protocol — the space tuple, seam module, tolerances, eight-component residue vector, commit policy — and the declared information-curvature tensor $\hat{I}^{GE}$ remain **declared, not certified**.

[Publications/papers/recognition-seam-topology/README.md](#)

Chapter 123

Lineage — RST-1

- **Primary source:** “Recognition-Seam Topology and Daggered Index Geometry — Foundations for an EMK Route to the Riemann Hypothesis” (M. Dabas, July 2026), arXiv source v1 PDF. Sections certified: 2–7.
- **Paper lineage (as recorded in its Appendix B):** MP RH branch stack PRs #50, #62–#82, #89–#91, #94 (front door); EMK-UGD Vault sheet-rotation development (commits c7255dfa..., 24bb86a2..., a22053a8..., 40f23dbe..., cbcecdbf...); imported native X3 fusion lineage (PRs #127, #200, #202, #205, #208–#212) — the X3 layer is *not* consumed by RST-1 (deferred to RST-2’s finite spine).
- **Vault cross-reference:** APPENDIX_EMK_TOPOLOGY_RIGOROUS (EMK-UGD-Provisional-Vault, recognition_seam_calculus_master tree) overlaps sections 2–6; a file-level diff is pending repo access and will be recorded here when done. The paper is treated as source of truth (it is the mature front door and postdates the appendix).
- **Certified companions in this corpus:** EMK-1, EMK-2, UGD-1, EMK-T1 (papers/emk-ugd-algebra); EMK-G1 (papers/emk-recognition-geometry). The derived-involution thread now spans all of them plus RST-1 T3.
- **Prior executable version:** NONE found for the paper’s exact spine. This is the first certification.

Publications/papers/recognition-seam-topology/LINEAGE.md

Chapter 124

Representation-Complete Native Seam Integer Paper

124.1 Source of record

Repository: Parveen117/MP

Branch: agent/paper-source-floor-five-by-five-theorem

Draft PR: #244

124.2 Expected release contents

source/	complete LaTeX submission tree
pdf/	final compiled PDF
metadata/	arXiv metadata and claim ledger

124.3 Current claim boundary

SOURCE FLOOR $S_{(.01,-)} \geq .008$ I	PROVED
SOURCE INVERSE NORM ≤ 125	PROVED
HERMITIAN RANK-AT-MOST-FIVE REDUCTION	PROVED
FINAL OUTWARD ACCEPTANCE THEOREM	PROVED
ACTUAL OUTWARD FIVE-BY-FIVE VALUE	OPEN
CONTROLLED $\eta_j \rightarrow 0$	OPEN
RIEMANN HYPOTHESIS	ABSTAIN

Do not publish a Zenodo-linked release until the updated PDF has been compiled and placed in pdf/.

Publications/papers/representation-complete-native-seam-integer/README.md

Chapter 125

Yang–Mills Certified Benchmark (YM-1)

First rung of resuming the MP Yang–Mills adapter with the certified RNKE machinery: the reduced finite-cutoff gap of the SU(2) one-holonomy Wilson benchmark, previously a float, is now an exact two-sided rational enclosure.

125.1 Certified statement

For the normalized SU(2) Wilson convolution operator on class functions (source: MP/adapters/yang_mills/cutoff) with exact character spectrum $\text{lambda_j}(\text{beta}) = I_{\{2j+1\}}(\text{beta})/I_1(\text{beta})$:

```
Delta_red(a=1, beta=2) = -log( I_2(2) / I_1(2) )  
in [0.83672330623158891305006780105454768321... ] (width < 1e-30, exact )
```

Verdict engine: directed rational interval arithmetic only. No floats in any verdict. Controls: (C1) `lambda_1` strictly inside (0,1) hence gap strictly positive; (C2) tampering `I_2 -> I_3` produces a separated bracket; (C3) width budget.

125.2 Claim boundary (fail-closed)

- CERTIFIED: the reduced finite-cutoff benchmark number only.
- NOT CERTIFIED / OPEN: full-lattice gap, theta-graph interacting transfer, ultraviolet/infinite-volume uniformity, OS continuum reconstruction, the Clay existence and mass-gap predicate. Adapter verdict remains `hold`.

125.3 Ledger

Item	Status
YM-1 reduced gap enclosure (a=1, beta=2)	PASS (pinned)
YM-2 interacting theta-graph gap, certified small coupling ($\kappa < \Delta_{\text{red}}/6$)	PASS (pinned)

Item	Status
YM-3 first-order crossing direction: rank-one, $A \leftrightarrow B$ transported, slope $\lambda_{\text{half}}/4$	PASS (pinned)
YM-4 symmetry-protection theorem [S,T_kappa]=0 + finite-kappa certified lower bounds	PASS (pinned)
YM-5 certified TWO-SIDED gap at finite kappa, beyond the sandwich threshold (Kill-Lemma realization)	PASS (pinned)
YM-6 seam-integer dock: EXACT eigenvalue counts on the native 5x5, certified to kappa=1/2 (D03 congruence)	PASS (pinned)
YM-7 V7 carrier: kappa=7/10 unlocked (exact count), certified 7-eigenvalue crossing curves	PASS (pinned)
YM-8 CAPSTONE: theta-graph gap is a THEOREM at every coupling (certified kernel floors + pinned Jentzsch anchor); remaining Millennium content = uniformity in the cutoff	PASS (pinned)
YM-9 FIRST UNIFORMITY: exact heat-kernel refinement family; free gap exactly cutoff-independent; interacting uniform bound 3/8 along a declared trajectory	PASS (pinned)
YM-10 blindness ledger (SPECTRAL-1/2 framework): exact multiplicity law, exact probe counts, composition ledger, YM-9 T4 correction	PASS (pinned)
YM-11 three gate verdicts on the toy carrier: gauge CLOSED; volume/IR SPLIT (free closed, sandwich route proven insufficient, critical volume exact); universality SPLIT (counting closed, metric open)	PASS (pinned)
YM-12 the T03 move: positivity SQUARE-SOURCED (classical anchor demoted to simplicity-only); volume+cutoff-uniform interacting gap 3/8 on the factorized chain; governance capsule	PASS (pinned)
YM-13 verification-overlap beta audit: beta 4/5 -> 3/5 via independent witnesses (CF Bessel, compound-interest exp, det/trace inertia); three lineage overlaps retained with refusal notes; RH remains lowest at 1/5	PASS (pinned)
YM-14 interaction-overlap dock, first carrier: bridged pair — overlap seam rank-one and swap-transported (same theta integral), certified k=1 to kappa=1/2, overlap price kappa*lam/4 on one line; kappa=1 carrier limit recorded	PASS (pinned)

Item	Status
YM-15 1D block-transfer dock opened: exact closed form of the chain compression for EVERY m (vacuum $f_0^{\{m-1\}}$, A-line $f_0^{\{m-1\}} \cdot \text{KMS}_m(r)$, $r=I_2/I_1$; $SU(2)$ convolution lemma, machine-checked as a formal polynomial identity $m=2..8$, no truncation remainder); m -UNIFORM ratio bracket $(1-r)/(1+r) < _m < (1+r)/(1-r)$, certified positive normalised gap for all m at 1; $=2$ fails closed. A-line carrier ONLY — complement not bounded	PASS (pinned)
YM-16 chain dock: exact B-factorisation gap(S_m) Δ_red for ALL m , (m -uniform upper bound, exact eigenvector); vacuum volume bracket $f_0^{\{m-1\}} \quad e^{\{(m-1)\}}$; certified two-sided gap (ratio $9/10$) of S_m for $m \quad m(_) \quad m=13 (\quad =1/8), 7 (1/4), 4 (1/2)$ — with m^* computed exactly and matched by the dock; m^*+1 refused. Sup route provably non-uniform: death volume has coordinates	PASS (pinned)
YM-17 interleaving seam: $m_ = B_odd \cdot B_even$; each dressed half-chain is a tensor product of the two-site chain at 2 product spectrum gap ratio EXACTLY m -uniform (pair ratio certified $9/10$ at $1/2$; bracket $[1,3]$ honest at $=1$); vacuum bracket narrowed to rate 2 (per-bridge upper end $0.125 \rightarrow 0.030$ at $=1/8$); whole volume problem reduced to ONE inequality: (T) vs (X) (Y) (vacuum tracking of interleaved halves)	PASS (pinned)
YM-18 CALIBRATION + DESIGN: enlarged exact carrier W'_m (vacuum + sites + bridge-pairs, $\dim 2m$) gives certified Rayleigh lower bounds on (T_A, m) , $m=2..6$; per-bridge vacuum rate $_m^{\wedge lo}$ is m -independent to $\sim 1e-5$ ($0.002177 @ =1/8, 0.008698 @ 1/4, 0.03461 @ 1/2$) — vacuum tracking behaves as a clean per-bridge rate; bracket width = looseness of the upper end. Cluster dock DESIGNED (monomers = odd pairs at 2, activities = even bridges, KP on a line) — no claim	PASS (calibration, pinned)

Item	Status
YM-19 Dobrushin dock — DEMOTED to ANCHORED (governance rule: no classical borrowing; T3 rests on a cited Dobrushin/Föllmer anchor, same fate as YM-8 Jentzsch). T1 (coefficient bound) and T2 (<1 arithmetic) are native and stand; the m-uniform GAP claim is WITHDRAWN until derived natively. Was: on the bounded-overlap chain — Dobrushin coefficient $C_{ij} = 1 - 1/2^j$ (elementary, proved), $\sum_s = e^{\{2\}}$ exact, $\sum_t (1+S_a)/(1-S_a)$ certified; $\sum = 2(1-1/\sum_t^2) + 2(1-1/\sum_s^2) < 1$ (pinned anchor: Dobrushin/Föllmer/Künsch covariance decay; extraction lemma proved) / for EVERY m. Certified gap/step 0.263 (a=6, $\sum=1/16$), 0.145 (a=8, $\sum=1/8$, ON YM-9's trajectory $\sum=1/64$), 0.235 (a=12, $\sum=1/8$); refused at (6,1/8),(4,1/16),(10,1/4). Coupling ceiling of the route exact: $< \log 2/4$. STANDING CORRECTION to YM-17 T3 (vacuum tracking insufficient; correlation-decay route replaces it). NOT uniform in the cutoff ($\sum_t \rightarrow \infty$ as $a \rightarrow 0$)	ANCHORED (pinned, not a native theorem)
YM-20 NATIVE ORIGIN AUDIT: machine-checked origin ledger for YM-1..19 in the framework's vocabulary (carrier = COORDINATE_SHADOW throughout; counting = NATIVE_DERIVED throughout; anchors: YM-8 and YM-19 demoted imports, YM-17 SV-Weyl as technique). CAYLEY FORM certified: YM-15 ceiling $(1+r)/(1-r)$ and YM-19 $\sum_t$ are $\text{Exp}_{\sum}(A_{\sum}(y))$ (F00-G Thm 5.1 pinned) with seam coordinates r, $\sum_a$; native odd-series log agrees with YM-1 route to 1e-20; uniform gap in native form $= -\text{Log}_{\sum}(\) - A_{\sum}(r)$. Bessel coefficients = native factorial series (F00-E Lemma 2.1 shape); Haar identification = the shadow. NAMED YM-21: bridge-local recognition-energy contraction under the seam-involution flow (F00-E Thm 6.2)	PASS (pinned)

Item	Status
YM-F1 THE CHAIN AS A RECOGNITION FABRIC (native carrier): T1 $SU(2)$ = unit sphere of the EMK block over $C_\Sigma = \text{span}\{I, R, K, RK\}$ is Hamilton's quaternion algebra, $\det = \Delta + \Delta$ becomes the quaternion norm (twisted seam channels), $\frac{1}{2} = 2 \times \text{identity-sector coefficient}$; T2 chain = ladder fabric, bridge faces = plaquette holonomies $H_l = A_i^{-1} A_{i+1}$; T3 Recognition-Stokes EXACT on the chain (interior rungs cancel, $E_{\text{int}} = 0$, orientation tamper bites); T4 action = sum of face residues $_l = 1 - \text{Tr } H_l / 2 \in [0, 2]$, weight = $\text{Exp}_\Sigma((m-1)) \cdot \text{Exp}_\Sigma(-\Sigma)$. Source: vault fabric appendix + EMK-1 + F00-E. Remaining shadow: Haar Φ_Σ (declared). Native form of YM-21 posed on the fabric	PASS (pinned)
YM-21 THE TILING LAW (native, on the fabric): STANDING CORRECTION to YM-F1's posed question (boundary residue cannot control face residue — witness g, g^{-1}, certified); the right object is the tiling. T2 Stokes sub-telescoping YM-15 entry = $r^{\#\{\text{faces crossed}\}}$: SPATIAL decay along the chain exactly geometric, rate $-\text{Log } r$, every m. T3 leading-order TIME two-point = $(\cdot \cdot \text{KMS}_m(r))^t$, bracketed $t^{((1-r)/(1+r))}$ for every m YM-15/20's uniform rate $-\text{Log } -A_\Sigma(r)$ DERIVED as a tiling count ($t=1..6$, $m=2..8$ inside bracket). T4 remainder named natively: higher-content/branching tilings, expansion parameter = face-coefficient ladder r_j (certified decreasing: $r_{\frac{1}{2}} \geq r_1 \geq r_{\frac{3}{2}}$); uniform convergence = native cluster statement	PASS (pinned)

Item	Status
YM-22 THE TILING RATE SURVIVES THE CUTOFF: on the heat-kernel family along $\gamma = a$, the time-face log is EXACTLY $3a/4$ (no transcendental), so the leading tiling rate per unit time $\gamma(a) = 3/4 - A_{\Sigma}(r(a))/a$. Certified over SIX DECADES $a=1...1e-6$ at $\gamma=1/16$: 23/32 $\gamma(a) \geq 3/4$ for every a — the limit $3/4 - \gamma/2$ is a uniform LOWER bound in the cutoff (approached from above); KMS bracket holds with γ_a for $m=2..8$ at every a leading-order rate uniform in volume AND cutoff simultaneously. Beats YM-9's sandwich $3/4 - 6$ by factor 12 in the γ-coefficient; trajectory ceiling $\gamma < 3/2$ ($\gamma=2$ refused). Higher face coefficients per unit time $r_1/a, r_{\{3/2\}}/a$ $\rightarrow 0$: in the cutoff limit only $j=1/2$ branching survives. SCOPE: leading tiling order	PASS (pinned)
YM-23 THE WEAK-COUPLING TURN: T1 tiling route has an EXACT weak-coupling boundary — ladder collapses $(r_{1/2}, r_1, r_{\{3/2\}} = 0.970, 0.922, 0.859$ at $\gamma=50$: no small parameter), leading rate sign change at $r_c(a) = \tanh(3a/8)$ in native Cayley form, $\gamma_c(a)$ bracketed (1.574 @$a=1$, 0.759 @$1/2$, 0.376 @$1/4$, 0.188 @$1/8$), and on a DECLARED AF-shaped trajectory $\gamma =$ $1/4 \log(1/a)$ the route is left at $a = 1/8$ — the continuum lies outside the tiling route. T2 native linearization: $\gamma = 2s/(1+s)$ in the stereographic chart; $\text{odd}(H H) =$ $p + p + 1/2[p, p]$ EXACTLY — the commutator is the only non-additive term = EMK-1 T5 / RST-1 T4 curvature on the fabric. T3 EXACT NEGATIVE: the abelianized fabric's softest spatial stiffness $\sim 12/(m(m+1))$ (Rayleigh identity certified $m=2..40$) — gapless in volume. T4 native statement: a weak-coupling gap lives entirely in the commutator sector (non-healable curvature) — the mass gap is non-perturbative, said in the framework's words	PASS (pinned)

Item	Status
YM-24 ABELIAN SUBFABRIC + NON-ABELIAN RESIDUE (exact algebra, no gap claim): T1 commuting faces = F00-E's circular Euler orbit, compose by the native addition law, Stokes additive in the Euler coordinate (the rotor chain), m=2..8 on rational circle points; T2 non-abelian residue = Gram defect	p×q
YM-25 THE DIRECTION-CHANGE SECTOR: two exact negatives — the time kernel is direction-blind (bi-invariant: one scalar per spin-j block, d_j ² coefficients, refinement-invariant) and the strong-coupling carriers are direction-blind (class functions, rotation-invariant: structural SPECTRAL-1 blindness) — and one exact positive: direction change between adjacent faces IS the 6j recoupling (MP gold/01 seed), squared overlaps between (12)3 and 1(23) schemes exactly [[1/4,3/4],[3/4,1/4]], unitary, no √ evaluated (RST-2 discipline), singlet tamper breaks unitarity. REDUCTION: a weak-coupling gap can live only in the intertwiner transfer	PASS (pinned)
YM-26 DOCK TO theorem/28 (Recognition-Complete Finite-to-Infinite Cut Theorem) — owner's correction: the framework's own convergence machinery. Dictionary: P_n = A-line carrier (YM-15), Q_n = complement, floor f^{m-1}, threshold = f^{m-1}, e_n = QSQ , N (B_n-I) = Haynsworth inertia; outward certificate e_n < 1 inertia count = 1 < . T1: LAWFUL (e_n present, not a shadow certificate) for every m m() = 13, 7, 4 at = 1/8, 1/4, 1/2. T2: e_n(m) certified increasing, crosses 1 exactly at m+1 — theorem/28's hypothesis 3 (memory-channel Cauchy bound uniform in m) is NOT YET BUILT; full hypothesis ledger 1-7 written in the theorem's vocabulary. T3: via the tiling law, item 3 reduces to a Cauchy bound uniform in m for the multi-insertion content-½ memory channel (the j=½ branching sector)	PASS (pinned)

Item	Status
YM-27 DOCK TO RH-FRAMEWORK	PASS (pinned)
T01 (owner: use the RH journey's proved results, cite them): T1 the sup route IS T01-E4C (bounded native simple multiplier, $U_ \Sigma(w) = e^{\wedge}$ exactly; YM-16's e_n reproduced to the last digit) theorem/28 hypothesis 3 = a memory-channel bound beating T01-E4C by a factor uniform in m — measured slack per face 56.5 / 27.8 / 13.5 at $= 1/8, 1/4, 1/2$ (YM-18 calibration vs E4C price). T2 the abelian subfabric is a T01-E5A rational UGD packet: wrap law = rotor-chain composition, winding = seam memory ; (g, g^{-1}) witness vs full-turn fabric have the same visible boundary but $= 0$ vs 1 — the native form of EMK-1 winding. T3 the Haar shadow of YM-F1 is exactly RH's OPEN T01-E5C/E6 — YM inherits RH's item, invents none; aliasing structure of class functions on E1 grids certified (exact for $n > 2j$). T4 T01-B/C recognition energy = chain vacuum form, floor $f^{\wedge\{m-1\}}$	
YM-28 T01-E4D ON THE CHAIN	i—j
(RH-Framework PR #19 applied): T1 the mean law is EXACT with zero deviation on single insertions — $I_rec(W \cdot \frac{1}{2}(A_i)) = f(2)^{\wedge\{m-1\}}$ for every m (formal identity, branch dies at the open end); T2 two insertions: deviation $= r(2)^{\wedge\{$	
YM-29 CLOSED-FORM VACUUM FLOOR	PASS (pinned)
FOR EVERY m: native face-independence lemma (bridge faces independent contents, W a product over faces face-product integrals factorise; fusion-rule certified: $w^{\frac{1}{2}} = 2f^{\frac{1}{2}}$, $w^{\frac{1}{2}^2} = f + 3f$); $T0^{\wedge\{1/2\}} \frac{1}{2}(H_) = \frac{1}{2}(H_)$; ansatz $= 1 + c\Sigma \frac{1}{2}(H_)$ gives $f^{\wedge\{m-1\}} \cdot Q_m(c)$ in closed form for ALL m, $Q_m > 1$ always, $\sim 4^{-2r^2(m-1)}$ for m $1/(4^{-2r^2})$ ($Q_ = 2.58 / 6.99 / 24.1$). Honest: linear floor moves the dock's m by 0 — the floor at the TRUE rate (0.000228/face margin, YM-28) needs the product ansatz, whose T-expectation is the one-time-step ladder $= 6j$ recoupling contraction	

Item	Status
YM-30 THE 2-ROW RECOUPLING ENGINE — BUILT AND VALIDATED: exact surd arithmetic ($\mathbb{Q}$ adjoined $\sqrt{}$), Clebsch–Gordan by Racah’s closed form (no floats), 3j, three-matrix-coefficient vertex integral, -removal of conjugates, ladder (one-time-step fabric) evaluated as a transfer over cut indices with label sums folded in. V1 every evaluation rational (surds cancel — the built-in consistency check); V2 one plaquette = $1/d_a^2$; V3 single-row limit recovers the YM-15 chain; V4 two plaquettes: $\sum_c d_c N = 1/4$ and the split $d_N : d_N = 1/4 : 3/4 =$ YM-25’s recoupling squares, now from the engine; V5 rail-swap and reversal symmetry. FIRST USE (calibration, $=1/8$, content 1, point coefficients): the $W^{\{1/2\}}$ -dressed vacuum for $T' = W^{\{1/2\}T0W}\{1/2\}$ beats the old floor $f^{\{m-1\}}$ by a GEOMETRIC factor 1.000732 per face ($m=2..6$) — floor rate 0.002685/face > YM-18’s bound 0.002177 > E4D excitation rate 0.001949: margin 0.000736/face; single-face excitation ($c = 0$) does not help	PASS (pinned)

Item	Status
YM-31 OUTWARD m-UNIFORM DRESSED VACUUM FLOOR — theorem/28 ITEM 6 AT THE TRUE RATE: $\text{trial} = W^{\{1/2\}} \cdot (W_{\text{pt}}/W)$ with W_{pt} a product of rational truncated faces (no error by definition); numerator = ladder $W_{\text{pt}}, T0 W_{\text{pt}}$ (YM-30 engine), rung truncation is a rigorous LOWER bound ($T0 = \Pi_i \Sigma_c _c \Pi_c$: a sum of nonnegative projection terms, monotone in $_c$); denominator $W_{\text{pt}}^2/W = [\Sigma d_jd_kd_l f_j^{\text{pt}} f_k^{\text{pt}} (-1)^{\{2l\}} f_l N_{\text{jkl}}]^{\{m-1\}}$ exact with outward l-tail; NATIVE LOG-CONVEXITY Z_m^2 $Z_{\{m-1\}} Z_{\{m+1\}}$ ($Z_m = g, S^{\{m-2\}}g, S = C^{\{1/2\}} M_K C^{\{1/2\}}$ symmetric PSD, cut-square inequality) ratio nondecreasing (S_m) FLOOR_{m} · (_lo/I_hi)^{m-m} for every m 6, rung tail at m bounded (5.9e-9). Per-face floor rate 0.002685 / 0.010722 / 0.042607 at $= 1/8, 1/4, 1/2$ vs old floor 0.001952 / 0.007802 / 0.031089 and E4D excitation rate 0.001949 / 0.007742 / 0.030161 — margin 0.000736 / 0.002980 / 0.012446 per face, uniformly in m	PASS (pinned)
YM-32 DRESSED SINGLE-INSERTION ENERGY, m-UNIFORM (engine extended to rung insertions via CG reduction of 4-valent vertices; validated: j=0 reproduces YM-30, one-rail insertion vanishes, reversal identity, mixed-rung tamper smaller): $E_m(p) = W_{\text{pt}}^{1/2}(A_p), T0 W_{\text{pt}}^{1/2}(A_p) / W_{\text{pt}}, T0 W_{\text{pt}}$ exact for $m=2..7$, all p — bulk value m-independent to $< 1e-6$, ends differ; $E_bulk = 0.43368 / 0.43532 / 0.44171$ at $= 1/8, 1/4, 1/2$, i.e. $\cdot (1 + 1.27e-3 / 5.05e-3 / 1.98e-2)$, inside YM-15's bracket (I first wrote 'below'; the engine corrected me). Dressed denominator factorises (rung insertion free) the $J=1/2$ sector's dressed excitation/vacuum ratio is exactly E_bulk, uniformly in m — a certified LOWER bound on that sector's top, not an upper bound on	PASS (pinned)

Item	Status
YM-33 t-ROW FABRIC ENGINE + EXACT TIME TWO-POINT FUNCTION (owner's framing: time carries no infinity, only the flow $\hat{=} = \text{Exp}_\Sigma(- \cdot \text{gap})$; expand the space coupling only): general cut transfer over t rails with rungs contracted up the column (interior vertices 5-leg: two rungs, two faces, insertion — two CG fusions); validated: t=2 reproduces YM-32 exactly, t=1 the chain, 3-row fabrics rational with exact time reflection. Dressed sequence $a_- = , T' \cdot \text{pt}^{\hat{=}}$ and vacuum z_- exact for $=1,2$, $m=5..8$: ratio sequence $= E_{\text{bulk}}$ (YM-32), $= ,$ and the SECOND-ORDER connected correction $= -$ is a BULK CONSTANT — $0.00076998 / 0.00305278 / 0.01178845$ at $= 1/8, 1/4, 1/2$ — with boundary influence decaying geometrically: moving an end from distance $2 \rightarrow 3$ shifts $=$ by $1.3\text{e-}9$ (either end, identically), $3 \rightarrow 4$ by $1.6\text{e-}12$, ratio certified in $[r^2/2, 2r^2]$. The framework's own strong-coupling series for the gap, order by order as exact rationals, m-uniform with exponentially localised edges. Lower-bound side (Rayleigh data) — item 3's upper bound still open	PASS (pinned)
YM-34 E4D-C RESTATED (native Laplace principle): the iterated multiplier WITHOUT smoothing, $q_- n = w / w^{-1}$, is strictly increasing to $\sup w = e^{\{2\}}$ (certified $n \leq 60$) — so E4D-C as written in the RH ledger (deviation control of the multiplier iterated on one state) is unprovable, and the exponential of YM-16/26 is this growth. WITH the time kernel between multiplications the ratio is nondecreasing (log-convexity) and converges to $(K^{\{1/2\}\text{WK}}\{1/2\}) = f(1+)$, $= 0.003582 / 0.014032 / 0.051911$ ($=1/8, 1/4, 1/2$) vs $\sup$ price $0.274 / 0.598 / 1.405$. E4D-C is restated for the alternating product $(K^{\{1/2\}\text{M-wK}}\{1/2\}) : / 1 - \text{gap}$ uniform in m — the vacuum side of this is YM-31; only the excitation side is open	PASS (pinned)

Item	Status
Remaining: theorem/28 item 3 — the memory channel's OPERATOR norm (not basis-state norms) uniform in m, via the same engine (E4D-C quadratic form on content-$\frac{1}{2}$); with YM-31's floor the dock's e_n becomes $(\ QTQ\ /\text{FLOOR})$ and the sup numerator is the last exponential; (interval coefficients, content tail, Perron bound on the column transfer $\rightarrow$ m-uniform floor at rate 0.00268) and the E4D-C quadratic form on content-$\frac{1}{2}$ via the same engine; (exact rational ladder evaluations) — closes item 6 at the true rate AND item 3 (E4D-C quadratic form on content-$\frac{1}{2}$) in one build; E4D-C on SUPERPOSITIONS of content-$\frac{1}{2}$ basis states (a quadratic-form statement on the multi-insertion channel) = theorem/28 item 3 (content-$\frac{1}{2}$ multi-insertion memory channel, Cauchy-uniform in m); (weak) YM-27 the intertwiner (spin-network) transfer on the chain fabric — recoupling matrices as weak-coupling tiling weights, floor uniform in m? (declared: chain $\sim O(4)$-type rotor model in 1+1D, continuum gap at the frontier for every method); (strong) FULL-ORDER tiling convergence uniform in m and a ($j=\frac{1}{2}$ branching); (weak) the COMMUTATOR-RESIDUE TRANSFER on the fabric — does the non-abelian sector alone carry a gap; AF trajectory derivation; (convergence of the face-coefficient tiling expansion) — the cluster statement in fabric language; then (replace the borrowed decay anchor by program machinery — square-sourcing/seam count/flow generator; generalized Euler flow and EMK/UGD sections to be read first); then uniformity in the CUTOFF $a \rightarrow 0$ of the volume-uniform gap (Dobrushin degrades as the heat kernel sharpens) — this is now the sharp wall; AF trajectory, tightness, OS, non-triviality, metric universality, 2D lattice, Clay	OPEN

- **T3** governance capsule (RH 07-20 analog): claim/consumption chain for YM-1..12, standing corrections (YM-9 T4 amendment, YM-8 anchor demotion), and a do-not-reopen list — machine-checked for completeness.

YM-11 takes three dependency-map gates and gives each an exact verdict on the toy carrier — the generalized theta (“n-banana”) graphs B_n , two vertices joined by n edges, so $b_1 = n-1$ holonomies and $F(n) = n(n-1)/2$ plaquettes; B_3 is the theta graph of YM-1..10. **These are toy-carrier verdicts, not Clay-sense closures**, and two of the three are partly negative.

- **Gauge — CLOSED.** Tree gauge-fixing on a connected graph is exact (no Gribov obstruction), leaving $b_1 = E-V+1$ holonomies and a residual diagonal G acting by conjugation, so gauge-invariant states are exactly $L^2(G^{\{b_1\}})^{\text{Ad}}$. For B_3 that is the carrier every capsule has used — now certified rather than assumed, with Wilson loops spanning it and YM-10’s multiplicity law counting them.
- **Volume / IR — SPLIT.** The *free* reduced gap is $C_-(1/2) = 3/4$ for every n and every a : volume- and cutoff-uniform, closed. The *interacting* sandwich gives $\Delta \geq C_-(1/2) - 2 F(n) \theta$, which degrades quadratically in n : at $\theta = 1/16$ the bounds are $3/8$ ($n=3$), exactly 0 ($n=4$), $-1/2$ ($n=5$). The route is **proven insufficient** for volume-uniformity, with the critical volume computed exactly. What a replacement must do is named: control the interaction per-plaquette (dock route, YM-6) rather than by a global sup (envelope route, YM-5) — the same lesson one level up.
- **Universality — SPLIT.** For the whole regulator class $K = \sum_j d_j c_j \chi_j$ with $c_j > 0$, the **counting layer** (content grading, multiplicity law, carrier dimensions, blindness ledger, probe counts) depends only on $SU(2)$ representation theory and is therefore regulator-independent — closed, and it means YM-10’s ledger is kinematics, not a choice of action. The **metric layer** (the gap value) genuinely differs between regulators (witnessed) and remains the real universality gate.

YM-10 applies the source-restriction framework of the companion manuscripts *When Spectra Forget Order* and *Stable Recovery Beyond Spectral Blindness* to our own compressions. Every dock computes on a compressed carrier V ; in that language V is a source restriction, so the honest question is not whether the complement bound is tight but whether the carrier is **blind** to the target.

- **T1** exact multiplicity law $m(j_1, j_2) = \min(2j_1, 2j_2) + 1$ on the Ad -invariant sector — it reproduces $\dim V_5 = 5$ and $\dim V_7 = 7$ independently, and re-derives YM-6’s $(1/2, 1/2)$ two-dimensionality from representation theory rather than from the Gram matrix.
- **T2 correction to YM-9 T4** (self-audit): YM-9’s uniform count counted *contents*; k_{Sigma} counts *eigenvalues with multiplicity*. At $s = 2$ that is 4 versus 5. Both are a -independent, so YM-9’s uniformity conclusion is unaffected — only the arithmetic. YM-9 is amended to report both.
- **T3/T4** blindness ledger and exact probe count $\text{rank}(\Pi_s | \ker E) = b(V, s)$. Result worth noting: $b(V_5, s) = 0$ for $s \leq 2$ — the YM-6 carrier is **exactly blindness-free in its own threshold window**, so that choice was correct, not lucky. At $s = 3$, V_5 needs 6 probes, V_7 needs 4, V_9 needs 0 more than V_7 minus 4.
- **T5** composition ledger: blindness is nonincreasing along $V_5 \quad V_7 \quad V_9$ with nonnegative stage increments.

- **T6** the whole ledger depends only on contents and Casimir levels, so it is exactly **a**-independent — one ledger covers the entire YM-9 refinement family.

Honest remainder: the ledger is exact for the **free** target; the interacting faces mix the content grading, so $\mathbf{b}(\mathbf{V}, \mathbf{s})$ is a lower bound there and YM-6/7's declared truncation remainder is the extra obligation.

YM-9 — the program's first statement that is uniform in the cutoff rather than at a fixed cutoff. Switching the link action from Wilson to the HEAT KERNEL $\mathbf{K}_a(\mathbf{g}) = \sum_j \mathbf{d}_j \mathbf{e}^{-\mathbf{a} \mathbf{C}_j} \mathbf{chi}_j(\mathbf{g})$ makes refinement EXACT ($\mathbf{K}_a * \mathbf{K}_b = \mathbf{K}_{\mathbf{a}+\mathbf{b}}$; n links of spacing $\mathbf{a}/n$ compose to one link of spacing $\mathbf{a}$ with no discretization error), and the cutoff dependence then cancels identically:

- **T2** free reduced gap = $\mathbf{C}_-(1/2) = 3/4$ EXACTLY for every $\mathbf{a} > 0$;
- **T3** along the declared trajectory $\mathbf{kappa}(\mathbf{a}) = \mathbf{theta} * \mathbf{a}, \mathbf{Delta}(\mathbf{a}, \mathbf{kappa}(\mathbf{a})) \geq 3/4 - 6 * \mathbf{theta}$ for EVERY $\mathbf{a} > 0$ — at $\mathbf{theta} = 1/16$ a cutoff-independent gap of exactly **3/8**; fail-closed at $\mathbf{theta} \geq 1/8$;
- **T4** the seam count $\mathbf{k}(\mathbf{a}, \mathbf{e}^{-\mathbf{a} \mathbf{s}})$ is exactly **a**-independent (combinatorial), so the dock's threshold grammar transports across the whole family at once;
- **T5** Wilson contrast, computed: at fixed **beta** the reduced gap grows like $1/\mathbf{a}$ and diverges — no cutoff-independent gap without a running coupling.

Honest remainder (in-certificate): the trajectory is DECLARED not derived (the physical one is fixed by asymptotic freedom, $\mathbf{beta} \sim \log(1/\mathbf{a})$, not modelled); the graph is FIXED (no infinite-volume growth); the heat-kernel action is a CHOICE (universality gate open); $\mathbf{Delta}$ is a reduced graph gap, not a reconstructed physical mass gap. Net movement: one gate from “untouched” to “touched on a toy carrier”.

Capstone (YM-8): the interacting $\mathbf{theta}$ -graph transfer has a strictly positive spectral gap at EVERY coupling — certified pointwise kernel floors ($\mathbf{k} \geq \mathbf{e}^{-3 \mathbf{kappa}} [\mathbf{c0}^{-1} \mathbf{e}^{-\mathbf{beta}}]^{-2} > 0$, exact enclosures) plus the pinned classical Jentzsch/Krein-Rutman anchor (simple Perron eigenvalue; CIRC-1 discipline: cited, never rederived). Quantified window $\mathbf{kappa}$ in $[0, 7/10]$ carries exact counts and enclosures (YM-2..7). The honest remainder is named exactly: nothing here is uniform in the lattice; a gap at every fixed cutoff is compatible with the gap closing in the continuum limit — the fixed-regulator lesson. The Millennium content is precisely the open uniformity/OS/vacuum gates of the MP dependency map.

New in YM-7: carrier enlarged to V7 (adds $\mathbf{chi}_1(\mathbf{A})$, $\mathbf{chi}_1(\mathbf{B})$; Gram stays block-diagonal, all seven exact T0 eigenvectors), dropping the complement top from $\mathbf{lambda}_1$ to $\mathbf{lambda}_1 * \mathbf{lambda}_{\text{half}}$ (certified ordering) — which UNLOCKS the previously refused $\mathbf{kappa} = 7/10$ cell: exact seam count $\mathbf{k}_{\mathbf{Sigma}} = 1$ at $(7/10, 13/10)$. Also ships certified brackets for all seven compressed eigenvalue curves (inertia bisection; every step an exact LDL count), published with honest $\mathbf{kappa}$ -drift — adopting the RH-line's post-pause lesson that limited-denominator “exact constants” are never promoted from bisection.

New in YM-6 (framework-native): the gap becomes an exact threshold COUNT $\mathbf{k}_{\mathbf{Sigma}}(\mathbf{kappa}, \mu) = \mathbf{N}(\text{spec}(\mathbf{T}_{\mathbf{kappa}}) > \mu)$ via a Haynsworth congruence — the D03 “infinite ko finite” move — on the native 5×5 carrier $\{1, \mathbf{chi12}(\mathbf{A}), \mathbf{chi12}(\mathbf{B}), \mathbf{chi12}(\mathbf{A})\mathbf{chi12}(\mathbf{B}), \mathbf{chi12}(\mathbf{AB}^{-1})\}$ (all exact T0 eigenvectors; exact Gram has a single $1/2$ overlap). Certified exact counts: $\mathbf{k}=1$ at $(\mathbf{kappa}, \mu) = (1/8, 3/5), (1/4, 3/5), (1/2, 1)$ — so $\mathbf{lambda}_2 \leq \mu < \mathbf{lambda}_1$ EXACTLY, past YM-5's reach — and constant count across the window = no seam spectral flow (thermodynamics/02 instance). The $(7/10, 5/4)$ cell is honestly REFUSED (bracket $[1, 5]$).

New in YM-5: certified TWO-SIDED spectral gap of the interacting theta transfer at finite coupling — including BEYOND YM-2's sandwich threshold $\kappa_0 = 0.1394\dots$. Complement control uses the exact doubling identity $m_{\kappa^2} = m_{\{2\kappa\}}$: $|PMQ|^2 \leq \text{lammmax}(P m_{\{2\kappa\}} P - (P m_{\kappa} P)^2)$, with $|QSQ| \leq \lambda_{\text{half}}^2 e^{\{3\kappa\}}$ and a Weyl bound. Certified rows (ratio upper bound, gap lower bound): $\kappa=1/8$: 0.5003, $\text{gap} \geq 0.6924$; $\kappa=1/4$: 0.5610, $\text{gap} \geq 0.5781$; $\kappa=3/10$: 0.5867, $\text{gap} \geq 0.5332$. Fail-closed at $\kappa=1$. Lineage: realizes the old lambda-manuscripts' Kill-Lemma skeleton unconditionally (see LINEAGE.md); no claim imported.

New in YM-4: (T1) the graph swap $S:(A,B) \rightarrow (B,A)$ commutes with T_{κ} for EVERY coupling — $\text{Tr}(BA^{-1}) = \text{Tr}(AB^{-1})$ on $SU(2)$ — so YM-3's rank-one crossing line $\chi_{12}(A) + \chi_{12}(B)$ is symmetry-protected at all κ : a theorem, not a first-order accident. (T2) first certified finite- κ spectral data from the exact character expansion $\exp[(\kappa/2)\text{Tr}U] = \sum d_j f_j \chi_j$, $f_j = 2 I_{\{2j+1\}}(\kappa)/\kappa$, with exact Clebsch-Gordan ring arithmetic, the exact pairing tensor $\text{Int } \chi_p(A)\chi_q(B)\chi_r(AB^{-1}) = \delta_{\{p=q=r\}}/d_p$, and a certified truncation remainder. Sector-resolved lower bounds on the grid show the swap-even branch rising and the swap-odd branch falling with coupling.

CI runs exactly that one command plus pin diff and pytest.

125.5 YM-35 — T54 energy version + commutator remainder (E4D-C with the RKF operator ladder)

Framework inputs: theorem/54 (constant- chain is exactly its uncovered case — `sum_mu_diverges`), theorem/50 A4 (polarization visibility = YM-28's superposition gap), theorem/53 (energy, not mass), theorem/51 + 61 (commutator calculus). On the fabric, with the YM-30 engine as the only evaluator:

- **T1 locality** — $[K^{\{1/2\}}, B_1]$ vanishes on states with no content at sites $l, l+1$; non-adjacent face pairs do not mix (site content exactly $1/2$). Commutator support is $\{l-1, l+1\}$ — bridge-local, every m .
- **T2 exact adjacent commutator energy** — $E_{\Sigma}([K^{\{1/2\}}, B_1] B_{\{l+1\}}) = \frac{1}{2} \cdot (3/4) \cdot (1 - \sqrt{a})^2$, a - and m -independent (site weights $1/4 : 3/4 = \text{YM-25 recoupling squares}$, from the engine).
- **T3 T54 ratio theorem (commuting model, exact $m = 8$)** — vacuum and excitation carry the same per-face factor; ratio exactly m -independent; Σ not needed for a ratio.
- **T4 second-order remainder** — relative size $= \frac{1}{2} \cdot (3/4)(1 - \sqrt{a})^2 = 2.9e-4 / 1.2e-3 / 4.6e-3$ at $a = 1/8, 1/4, 1/2$ ($a = 1$); kill criterion $< 1/10$ passed.

Verdict: E4D-C's excitation bound opens at the first non-commutative order — the first obstruction is an exact local number, not a sup. **Not done:** the full face expansion (all words, contents $j = 1$) — the worldline cluster problem (YM-33). E4D-C OPEN. Pin `EXPECTED_YM35.sha256`.

125.6 YM-36 — T54 on the fabric at content $1/2$ (face-word carrier)

The alternating product $K^{\{1/2\}} M_w K^{\{1/2\}}$ compressed to ALL superpositions of content- $1/2$ face insertions ($2^{\{m-1\}}$ orthonormal words), exact rational matrices $m = 2.6$ at $a = 1/32, 1/16,$

1/8 (declared rational-square kernel $\sqrt{1/2} = 11/16$, $\sqrt{1/2} = 3/8$; YM-33 engine with per-column face labels). Excitation/vacuum ratio ρ_m increases with geometrically decaying increments (consecutive-increment ratios in $[2/5, 3/5]$ certified); under declared $q = 2/3$ the m-uniform limit is bracketed, e.g. $\infty \in [0.2225014, 0.2225544]$ at $\beta = 1/32$. The top eigenvector is a genuine superposition (single-insertion weight 0.98 / 0.92 / 0.73), yet exceeds the one-face ratio by only 0.24 % / 0.93 % / 3.5 % — superpositions open no gap-closing channel at content $1/2$. **Boundary:** compressed-operator statement (interlacing: lower bounds), contents $j = 1$ outside the carrier (first effect = YM-35 commutator, ~ 2), $m = 6$. E4D-C OPEN; it holds at content $1/2$ on this carrier in compression with a geometric tail. Pin EXPECTED_YM36.sha256.

125.7 YM-37 — The space transfer: m-uniformity of the dressed time two-point ratio as a theorem

(a) **refused with witness.** theorem/53 T4 bounds cut-square *weights* ($d_k = S_{kk}$); a block with every diagonal can still have $\max > (2 \times 2 \text{ witness, exact})$. So “complement $\max$ from T53 weights” is not available; item 3’s upper half stays OPEN.

What is a theorem. Every YM-33/36 quantity is **boundary**, $M^{\sim\{m-1\}}$ boundary for one column map $\beta = B^1 M$ (native quaternion carrier: faces/rungs are class functions of products, so only dot products appear; Haar = rational S^3 moments; cut space = harmonic degree -2 per rail).

commutes with simultaneous conjugation and with the centre of $SU(2)$ acting on all rails; both boundary vectors are even, so everything lives in the even invariant sector: $\dim 8$ ($t = 2$), 52 ($t = 3$). Validated **exactly** against `ym33.fabric_partition`. β , of the symmetric pencil (M, B) bracketed by exact inertia (fraction-free Bareiss, Jacobi sign rule); $q := \beta / \beta$ certified:

	q ($t = 2$)	q/r^2	q ($t = 3$)	q/r^2	$\hat{\infty}$ (two-sided)	certified err, interior $p = 3$
1/8	0.0014578	1.495	0.0021668	2.222	0.43367710465908e-7	
1/4	0.0057991	1.492	0.0086057	2.215	[0.4353154056078e-5 0.4353154056137]	
1/2	0.0226981	1.483	0.0334686	2.187	[0.4417107170,2.6e-4 0.4417107229]	

so $\| (m, p) - \hat{\infty} \| = \text{err}(m, p)$ for **every** m and interior p , with err explicit in β , β , $\|N\|_B$, $\|e\|_B$ — YM-33’s “bulk constant with geometric edges” is now certified with its rate (both q ’s lie in YM-33’s measured window $[r^2/2, 2r^2]$), and $\hat{\infty} = N / \beta$ is a single-column Rayleigh quotient (YM-32’s E_{bulk} reproduced to every digit). **Boundary:** fixed time order only; no complement bound; no m-uniform upper bound on the true gap — item 3’s m-dependence is now a one-column pencil question, its upper half unchanged. Runtime ~ 16 min (own CI step). Pin EXPECTED_YM37.sha256. **Standing correction (same day):** T3’s for-all-m route uses the finite spectral split — classical import; status ANCHORED pending YM-38. **Resolved by YM-38 (same day): the for-all-m statement now rests on the native route; see the YM-38 entry.**

125.8 YM-38 — Native m-uniformity (YM-37's spectral import removed)

The YM-37 T3 statement re-proved with the framework's own machinery, nothing else: deflation with the explicit vector $w = x$ (defect exactly B-orthogonal, $r_w' B w = 0$), complement ceiling $\| \Pi y \|_B \leq \| y \|_B$ from **one T53 elimination sign check** on the restricted doubled pencil (no eigenvectors anywhere), recognized-channel Cauchy with declared geometric tail and outward certificate $\leq L < 1$ (theorem/28 §3–5, §9), window products handled by theorem/54 §2's $(1+x)^{1+2s}$ bound. Result per ϵ , two regimes:

$| (m, p) - _c | \leq A \cdot (L^{\wedge \{j-p\}} + L^{\wedge \{j_c-p\}})$ for all m, p with $j = \min(p-1, m-p)$ $p \leq p$, where $_c = \text{exact}$ (26,13):

	$L (= \epsilon)$	$A (p = 2, \text{ all interior } p)$	check rows
1/8	0.0014578	1.93e-8	8/8 inside
1/4	0.0057991	6.34e-7	8/8 inside
1/2	0.0226981	2.17e-5	8/8 inside

The coarse constant beats the anchored YM-37 one (1.9e-8 vs 9.3e-7 at $\epsilon = 1/8$); $_c$ lies inside YM-37's bracket at every ϵ . **YM-37's standing correction is RESOLVED** — its T3 now rests on this capsule; also corrected there: the pinned dim-52 run used the native elimination, not Bareiss (LINEAGE). Certificates refused three drafts (missing reference-tail term; backwards extrapolation below p ; display-precision comparison) — all recorded. Pin EXPECTED_YM38.sha256.

125.9 YM-39 — Content ladder: centre superselection (exact) + certified channel ordering

Session item (b). **T1**: the centre of $SU(2)$ (total flip on every rail) grades the fabric exactly — faces and rungs are centre-blind class functions of products, $_j$ picks $(-1)^{\{2j\}}$ — so every mixed half-integer/integer fabric quantity vanishes **identically**: the mixed insertion matrix is the zero matrix (exact rationals) and the YM-33 engine's mixed partitions are 0; a same-parity pair ($1/2$ with $3/2$) is nonzero, so the vanishing is parity, not accident. Superselection at every coupling, every m , every time order. **T2/T3**: per-channel bulk ratios by the YM-38 native route (same ϵ , same L , channel-specific N), with the m-uniform band on both sides of every comparison:

	$(1/2)$	(1)	$(3/2)$	ordering with margins
1/8	0.43367710	0.13403748	0.00016958	
1/4	0.43531541	0.13491222	0.00067672	
1/2	0.44171072	0.13837705	0.00268149	

(1) sits at the declared $\epsilon = 9/64$ scale (anchor check). So on this carrier the $J = 1/2$ excitation is certified **lightest, uniformly in m** , and no content ≤ 1 channel can mix into it — YM-36's boundary concern closed at the carrier level. **Boundary**: truncated faces/rung kernel (contents ≤ 1), so $3/2$ is a composite row (data); compressed statements; E4D-C OPEN. Named next: the same ladder on the YM-22 heat-kernel trajectory, quantifying the collapse onto content $1/2$ as a $\rightarrow 0$. Pin EXPECTED_YM39.sha256.

125.10 YM-40 — Content ladder on the heat-kernel trajectory: the Casimir pinch

YM-39's ladder run along YM-22's family (heat-kernel time kernel, Wilson faces at $\Delta = a/16$), a $\{1, 1/4, 1/16, 1/64\}$, everything per-a by the YM-37/38/39 native machinery (superselection exact at every a; every Δ bracket carries the m-uniform band, so it holds for every m and interior p; native $\exp = \text{YM-13}$ compound interest, native $\log = \text{F00-G}$ odd series, with a cross-route re-enclosure control):

a	$\Delta_{1/2}$	Δ	$ \Delta - 5/4 $	$r/r_{1/2}$
1	0.7497494	1.9993628	1.2496134	6.4e-4
1/4	0.7499914	2.0001512	1.2501597	1.6e-4
1/16	0.7499999	2.0000692	1.2500693	7.0e-5
1/64	0.750000005	2.0000196	1.2500196	2.0e-5

Certified: $\Delta(a) \geq 1$ at every a, and the two-sided pinch $|\Delta(a) - 5/4| \rightarrow 0$ (a) with Δ strictly decreasing — the content-1 channel's excess rate lands on the **free Casimir gap** $C - C_{1/2} = 5/4$ in the cutoff direction, while the face ladder collapses ($r/r_{1/2} \downarrow$, YM-22 tie). Session item (b) is now closed in both directions: fixed-a ordering with margins (YM-39) and cutoff-direction Casimir pinch (here). Build notes: the draft's "approach from below" was refused by the brackets (Δ crosses $5/4$; $\Delta_{1/2}$ crosses $3/4$ — signed dressing), and the first run violated the standing round-before-log rule ($12 \text{ min} \rightarrow 4 \text{ min}$). **Boundary:** order-1 dressed two-point, not the operator gap; declared trajectory; E4D-C OPEN. Pin `EXPECTED_YM40.sha256`.

125.11 YM-41 — The tools dock: E4D-C restated with the operator-branch tools

The RKF operator-branch tools consumed by pin and re-verified on this repo's own pencil: **silence-channel contraction** $\Delta_{\text{sil}} < 1$ at every Δ ($0.00256 / 0.01029 / 0.04173$ — identical to RKF's digits, both roundings coincide at $1e-9$), the **ladder law** $f_{\Delta_{c+1/2}}/f_{\Delta_c} \rightarrow 1/(2(2c+2))$ with the sharper-claim discrimination (tight to 0.07%), and the **seam-flow meter** (sector cascade on the column pencil, every bracket's count route agreeing with the native det route). Selective release (stationarity, seam-count retention, $u+e < 1$ for the untruncated column) consumed by pin — RKF theorem/75 is the derivation of record. **E4D-C after the dock:** m gone (YM-37), content tail gone (theorem/75), contraction structural (theorem/74) — the remaining open content of $K^{1/2}M_{wK}^{1/2}$ is the **t-direction only**. Named YM-42: the t-ladder of silence contractions (Δ_{sil} at $t = 3$, 52-dim even sector) and the t-uniform statement. Pin `EXPECTED_YM41.sha256`.

125.12 NG-1 — The native gap statement, its teeth, and the dictionary declared open

A governance capsule with **no new numbers**. Diagnosis behind it: every wall this program has hit (cutoff uniformity, OS, infinite volume, Haar Φ_{Σ}) sits exactly where a native object meets a classically-stated goal — progress has been fast inside and zero at that boundary. So the statement being pursued is now written in the framework's own ontology, and the translation is named as a separate open problem, explicitly rather than implicitly.

NG. There is $\epsilon > 0$ and a declared trajectory (a) such that every member of the recognition fabric family has recognition gap $(a, m, t) \leq \epsilon$ for all a, m, t — each instance certified by native verdicts (elimination sign patterns, cut-tail mass, recognition energy, seam counts), every infinite direction taken through theorem/28's outward certificate, never a completion or a limit axiom. **NG is not the Clay statement and is never claimed to be.**

Teeth (a statement with no refutation condition is a definition, not a theorem) — five ways to refute NG with a single certified counterexample: volume (R1, a route already died this way in YM-11), cutoff (R2, live — YM-23 shows the tiling route exits at $a = 1/8$), sector cascade (R3, the seam-flow meter is the instrument), a channel lighter than $J = 1/2$ (R4), and any verdict that survives its own tamper (R5).

Ledger re-scored against NG: O1 single-member gap DELIVERED; O2 volume DELIVERED on the column (the chain's operator upper bound is **not** delivered); O3 content DELIVERED; **O4 time OPEN — the whole remainder (YM-42)**; O5 cutoff PARTIAL (declared trajectory only, AF shape not modelled); O6 evidence standard IN FORCE; **DICT (translation to Clay) OPEN, separate problem** (RH T01-E5C/E6). Sharpened by today's finding: theorem/75's grading was built classically and then regraded natively with identical rationals — one classical-looking ingredient proved replaceable; whether the measure itself is, is exactly DICT. Pin EXPECTED_NG1.sha256.

Publications/papers/yang-mills-certified-benchmark/README.md

Chapter 126

Lineage: the lambda-framework Yang-Mills manuscripts

Two pre-framework manuscripts (“The Yang-Mills Mass Gap as a lambda-Topological Positivity Condition”, v1/v2) proposed a conditional route: IF a vacuum state ω_0 exists on the observable algebra, THEN sector decorrelation (ACA) + local orthogonality (LO) + a Schur contraction estimate + kernel positivity force a mass gap (“Kill Lemma”).

Ruling (mine, don’t merge — same doctrine as the morphic-calculus chapters):

- NOT imported: the vacuum conjecture, the ACA/LO “first-principles” proofs (prose, tied to the retired lambda-RH citations), and any continuum claim. Those documents remain uncertified lineage.
- MINED: the Kill-Lemma SKELETON — (block Schur estimate) + (positivity) $\Rightarrow$ gap. YM-5 realizes exactly this skeleton on the theta-graph carrier, unconditionally and machine-certified: their Schur estimate $\rightarrow$ U1/U2 block + off-block interval bounds (off-block via the exact doubling $m_k^2 = m_{2k}$) their kernel positivity $\rightarrow$ certified character-coefficient positivity ($f_j = 2 I_{\{2j+1\}}/\kappa > 0$, positive series) their Kill Lemma $\rightarrow$ U3 Weyl bound + interlacing $\Rightarrow$ two-sided gap.
- The old manuscripts’ honest reduction (“vacuum exists $\Rightarrow$ gap”) maps onto the still-open gates of MP adapters/yang_mills/DEPENDENCY_MAP.md; nothing here advances those gates.

126.1 YM-15 (Aug 22 2026): the 1D block-transfer dock, opened

- CONSUMES: YM-14 T4 (the named object), YM-14 carrier conventions (beta=2 Wilson T0, A-line basis, B-parity inertness), YM-3/4/14’s exact SU(2) pairing integral generalised to the convolution lemma $\text{Int } \chi_a(XA^{-1})\chi_b(AY) = \delta_{ab} \chi_a(XY)/d_a$, YM-6 exact LDL inertia, YM-7 inertia-bisection pattern.
- SESSION LINEAGE: an earlier working session reached the chain topology, the convolution formula (after two self-corrections) and exact spot values at $m = 4, 6, 8$. This capsule replaces the spot checks by the closed form for every m , with the formula machine-checked as a formal polynomial identity (engine vs closed form, $m = 2..8$, all entries).
- IMPORTS ZERO numerical claims; the $m=4,6,8$ spot values are superseded.

- HONEST: A-line carrier only; the complement of $W_{\{m+1\}}$ grows with m and is untouched. YM-16 (m -uniform complement bound) is the named next dock.

126.2 YM-16 (Aug 22 2026): the chain dock — finite volume certified, death volume computed

- CONSUMES: YM-15 closed form (both M_{κ} and $M_{2\kappa}$ on the chain carrier), YM-6/14 Haynsworth dock, YM-5 doubling identity (survives the bridge product verbatim), YM-1 Delta_{red} (reappears as the exact m -uniform UPPER bound on the chain gap via B-factorisation).
- DIRECTION: the Millennium predicate is volume-uniformity. This capsule certifies the chain gap for $m \leq m^*(\kappa)$ and PROVES the sup route cannot go further (per-bridge price $e^{\kappa}/f_0 > 1$ is exponential in m). The wall is now a formula, not a mood: the obstruction is the use of $|m_{\kappa}|_{\infty}$ on the complement; its replacement (local/per-bridge control in the content basis = certified strong-coupling cluster radius, Osterwalder-Seiler shape) is YM-17.
- IMPORTS ZERO external claims. Jentzsch anchor not used.

126.3 YM-17 (Aug 22 2026): the interleaving seam

- CONSUMES: YM-16 dock at $m = 2$ (the pair), YM-5 doubling (pairwise), YM-15 closed form, singular-value Weyl inequality (classical, cited).
- RESULT: half-chains are exactly volume-uniform (product spectrum); the A-chain's entire volume problem is the vacuum-tracking inequality $\lambda_1(T)$ vs $\sigma_1(X) \sigma_1(Y)$. Vacuum bracket per bridge now $O(\kappa^2)$ at both ends. No m -uniform chain gap claimed.
- YM-18 named: certified Kotecky-Preiss radius for the interleaving polymer gas (two-site pairs as monomers). This is the first capsule whose next object is a classical convergence criterion rather than a new carrier — the program has reached the cluster-expansion wall with its own coordinates.

126.4 YM-18 calibration (Aug 22 2026): vacuum tracking measured, cluster dock designed

- CONSUMES: YM-15 engine (generalised to multiple insertions per site), YM-16 f_0 , YM-17 upper end. PC-1 precedent for “calibration, not claim”.
- MEASURED: the enlarged-carrier vacuum rate is m -independent to $\sim 1e-5$ for $m = 2..6$ at every grid κ . Evidence, not proof, that the interleaved vacuum tracks a per-bridge rate.
- DESIGNED, NOT EXECUTED: the Kotecky-Preiss dock (T3). Every line is an obligation. YM-18 proper = discharging them.

126.5 YM-19 (Aug 22 2026): the Dobrushin dock — volume-uniform gap certified

- CONSUMES: YM-9 heat-kernel family (time step a), YM-8 Perron floors (simple positive vacuum), YM-16 B-factorisation (consistency control), YM-5 exp two-route, YM-13 compound exp route $(e^{\{-x\}} = (e^{\{-x/n\}})_n$).
- ANCHOR (CIRC-1, cited not rederived): Dobrushin uniqueness and the Foellmer/Kuensch covariance decay (Georgii ch. 8). Same governance as YM-8's Jentzsch anchor, which YM-12 later demoted; the program should look for a square-sourced replacement of this anchor too.
- STANDING CORRECTION: YM-17 T3's "vacuum-tracking inequality" is not a sufficient reduction; recorded in the certificate. YM-18 retained as calibration. The KP design (YM-18 T3) is superseded as the main route but remains valid as an alternative.
- CLOSES: volume-uniformity of the interacting gap on the bounded-degree chain at strong coupling, fixed time step. Does NOT close cutoff uniformity — the Dobrushin condition degrades as $a \rightarrow 0$. That is now the sharp wall, one level deeper than YM-9 left it.

126.6 GOVERNANCE (Aug 22 2026, owner's rule): NO CLASSICAL BORROWING

- Owner's standing rule: the program proves with its own machinery (EMK/UGD algebra, generalized Euler flow, seam calculus); classical results are not imported as load-bearing anchors.
- Applied: YM-19 T3 DEMOTED to "anchored"; the m-uniform gap claim is withdrawn until derived natively. YM-19 T1/T2 (exact, self-derived) stand. YM-8's Jentzsch anchor was already demoted by YM-12 the same way. Audit of YM-1..19: these were the only two borrowed anchors.
- Standing obligation: a native replacement for correlation decay.

126.7 YM-20 (Aug 22 2026): native origin audit + Cayley form

- READ for this capsule: Recognition-Kernel-Framework F00-E (native Euler, circular-hyperbolic via sheet involution), F00-G (Cayley coordinate, odd-series log, $\text{Exp}(A(y)) = (1+y)/(1-y)$), RH-Framework T01 (cut-tail mass, recognition energy, native action bound) and its NATIVE_DERIVATION_CONTAMINATION_AUDIT (vocabulary adopted verbatim).
- FINDING: the YM carrier is a coordinate shadow by the framework's own standard (same class as RH T03-A); the counting layer is native; the program's uniform bounds are Cayley exponentials of seam coordinates.
- Framework-internal pins (F00-G 5.1/7.1, F00-E 2.1/6.2, T01-C) are citations INSIDE the program, permitted as in EMK-2.
- YM-21 named: native contraction replacing the decay anchor.

126.8 YM-F1 (Aug 22 2026): the chain as a recognition fabric — native carrier

- SOURCE (vault, read this session): APPENDIX_DISCRETE_HOLONOMY_FABRICS_AND_RECOGNITION_STOKES — the program’s lattice objects (plaquettes, Wilson action, gauge-fixed tree) are its fabric, face holonomy, face residue and rails. Also read: generalized_euler_phase_ratio_spa and EULER_INFORMATION_CURVATURE_DUALITY (no decay mechanism — recorded), mp_gold/01 (YM-1..8’s ancestor: MP PR#30), gold/05 (curvature-to-seam- index spectral-flow law), gold/06 (same-projection two-block bound).
- CONSUMES: EMK-1 (Publications) determinant seam ladder; F00-E iota_Sigma.
- NEW NATIVE FACT: SU(2) is the unit sphere of the EMK primitive block with iota-twisted seam channels — the gauge group was inside the framework’s own algebra. EMK-1’s channel witness (2,2,3,1) has split determinant 8 untwisted and norm 18 twisted (test).
- Carrier layer of YM-1..18 is now NATIVE except the declared Haar $\leftrightarrow$ Phi_Sigma identification. No number changed. No gap claimed.

126.9 YM-21 (Aug 22 2026): the tiling law

- STANDING CORRECTION: YM-F1’s “native form of YM-21” was ill-posed (boundary closure never bounds interior residue — the fabric appendix’s own principle); killed by witness (g, g^{-1}) and replaced.
- DERIVED, not docked: YM-15’s compressed A-block is the leading-order tiling transfer of the 2D fabric; YM-15/20’s m-uniform bound is a tiling count. Spatial decay exact via Stokes sub-telescoping.
- OPEN: full-order time decay uniform in m = convergence of the face-coefficient tiling expansion (native cluster statement). Expansion parameter certified small and decreasing (r_j ladder).

126.10 YM-22 (Aug 22 2026): the tiling rate survives the cutoff

- CONSUMES: YM-9 heat-kernel family + declared trajectory; YM-21 T3 tiling transfer (scalar in λ); YM-20 native odd-series log.
- RESULT: at leading tiling order the chain’s rate per unit time is uniform in m AND a along $\kappa = \theta a$, infimum exactly $3/4 - \theta/2$. First native touch of the cutoff wall. Full order OPEN: in the limit only $j = 1/2$ branching tilings survive ($r_j/a \rightarrow 0$ for $j \geq 1$).
- Build note: I first wrote the convergence as approach-from-below; the arithmetic showed approach-from-above (the limit is a uniform lower bound) — controls corrected to what the arithmetic says (YM-14 C4 lesson again).
- CI LESSON (b2291d1 failed on hosted Tests, pins passed): YM-F1/YM-21 seeded the shared `random` module at import; in the full pytest run the test order changed the stream and YM-F1’s pinned demo drifted. Fix: reseed inside `run()`. Pins unchanged. Rule: a pinned certificate must be a pure function of its inputs, never of module import order.

126.11 YM-23 (Aug 22 2026): the weak-coupling turn

- CONSUMES: YM-F1 stereographic chart + quaternion product, YM-21/22 tiling rate, EMK-1 T5 cut-commutator curvature, RST-1 T4 flow loop, F00-G Cayley/odd-series log (native $\tanh$ and $\log 2 = A(1/3)$).
- RESULT: exact boundary of the strong-coupling route; native linearization with the commutator as the only curvature; exact negative — abelianized fabric has no volume-uniform gap. The weak-coupling gap problem is now the commutator-residue transfer.
- DECLARED, not certified: AF trajectory shape; harmonic excitation relation. No weak-coupling gap claimed.

126.12 YM-24 (Aug 22 2026): the abelian subfabric and the non-abelian residue

- CONSUMES: YM-F1 quaternion chart, YM-23 soft mode, F00-E Thm 3.1/5.2 (native addition law / circular orbit), EMK-1 rotational determinant channel, EMK-T2 (order is content).
- RESULT: the commutator sector's exact structure — abelian subfabrics are copies of the native circle with additive Stokes; the non-abelian residue is a Gram defect; the gapless soft mode is exactly abelian. Weak-coupling gap, if any, lives in direction changes between subfabrics. YM-25 named. No gap claimed.

126.13 YM-25 (Aug 22 2026): the direction-change sector

- CONSUMES: YM-10 multiplicity law, YM-9 semigroup, YM-F1 chart, YM-24 rotation, RST-2 squared-amplitude discipline, MP gold/01 (PR #30 spin-network seed — reopened here at weak coupling).
- RESULT: time kernel and class-function carriers are both direction-blind (exact); direction change = $6j$ recoupling with exact rational squares $1/4 : 3/4$. The weak-coupling gap is isolated to the intertwiner transfer. YM-26 named. Frontier status declared.

126.14 YM-26 (Aug 22 2026): dock to theorem/28

- OWNER'S CORRECTION recorded: the framework already has the finite-to- infinite machinery (recognition-Cauchy + Smriti tails + outward certificate $u_n + e_n < 1$). YM-16 was that theorem without its name.
- RESULT: YM-16's dock re-certified as a lawful theorem/28 outward certificate for $m \leq m^*$; the remaining obligation is theorem/28 hypothesis 3 (memory-channel Cauchy bound uniform in m), reduced via the tiling law to the content-1/2 multi-insertion channel.
- RULE going forward: before declaring any object “infinite-dimensional with no finite truncation”, check theorem/28's hypothesis list first.

126.15 YM-27 (Aug 22 2026): dock to RH-Framework T01

- OWNER'S INSTRUCTION recorded: read RH-Framework, use and cite its proved theorems rather than re-deriving. Cited: T01-A/B/C, E1/E3, E4C, E5A/B (PROVED); E5C/E6 (OPEN); T03-A (CLASSICAL_THEOREM_IMPORT).
- RESULT: sup route = T01-E4C exactly (obligation now a measured slack factor); abelian subfabric = E5A packet with winding as memory; the Haar shadow is RH's own T01-E6; recognition energy = vacuum form.
- RULE: before any new derivation, check RH-Framework LEDGER and theorem/ for a proved statement to cite.

126.16 YM-28 (Aug 22 2026): T01-E4D on the chain

- CONSUMES: RH-Framework T01-E4D (PR #19), YM-15 engine (extended to content 8), YM-16 f_0 normalisation, YM-18 measured rates, YM-27 slack.
- RESULT: mean law exact on single insertions; deviation geometric and per-insertion on basis states; E4D rate derives the measured slack. Remaining: E4D-C on superpositions (quadratic form) = theorem/28 hypothesis 3. Build note: first run used unnormalised Bessel for f_0 and a tamper test that could not bite; both corrected.

126.17 YM-29 (Aug 22 2026): closed-form vacuum floor for every m

- CONSUMES: YM-F1 fabric (faces as independent contents, T01-E1 product content), YM-4 fusion rule, YM-15 lambda, YM-16 dock, YM-28 rates.
- RESULT: face-independence lemma; single-face-excitation Rayleigh floor in closed form for all m ($Q_m > 1$, linear for large m). Honest: does not move m^* ; the true-rate floor and the E4D-C quadratic form both reduce to one missing engine — exact rational 2-row (ladder) recoupling evaluations. YM-30 named with that spec.
- Build note: my first asymptotic claim (linear law visible at $m=60$) was wrong — the linear regime starts at $m \sim 1/(4 \lambda^2 r^2)$; control corrected to $m = 2000$.

126.18 YM-30 (Aug 22 2026): the 2-row recoupling engine

- CONSUMES: MP gold/01 spin-network seed (now executable), RST-2 squared-amplitude discipline (surds cancel by construction), YM-25 recoupling squares (reproduced), YM-15 chain (recovered as a limit).
- RESULT: exact ladder evaluations; calibration shows the $W^{\{1/2\}}$ -dressed vacuum floor is geometric in m at rate 0.002685/face ($\kappa 1/8$), above the E4D excitation rate. YM-31 named: outward certification (intervals, tail, Perron on the column transfer).
- Build notes: my docstring claimed $1/d_a$ for one plaquette; the engine said $1/d_a^2$ and the engine was right (three convolutions, one trace). A brute-force label sum was $3^{\{3m-2\}}$ evaluations — replaced by folding labels into the cut transfer ($m = 6$ in < 1 s).

126.19 YM-31 (Aug 22 2026): outward m-uniform dressed vacuum floor

- CONSUMES: YM-30 engine, YM-29 face independence, T01-B/C (projection square, cut-square inequality) for rung positivity and log-convexity, YM-28 E4D rate, theorem/28 item 6.
- RESULT: $\lambda_1(S_m) \geq \text{FLOOR}_{\{m_0\}}(\rho_{\text{lo}} / I_{\text{hi}})^{\{m-m_0\}}$ for all $m \geq 6$ at the true per-face rate; margin over the E4D excitation rate certified positive at every grid κ . First m-uniform statement of the program that holds at the TRUE rate rather than a bracket.
- Trial-state discipline: the truncation lives in the trial (no error), the rung truncation is one-sided by positivity, the denominator is an exact interval — nothing is a shadow certificate.

126.20 YM-32 (Aug 22 2026): dressed single-insertion energy

- CONSUMES: YM-30/31 engine and trial discipline, YM-15 bracket, YM-21 tiling law (KMS hopping explains the $O(r^2)$ excess over λ).
- RESULT: engine handles rung insertions (4-valent vertices by CG reduction); the dressed $J=1/2$ excitation/vacuum ratio is an exact bulk constant, uniform in m . Lower-bound-side only; item 3 still open.
- Build note: predicted $E_{\text{bulk}} < \lambda$; engine gave $E_{\text{bulk}} > \lambda$ by $O(r^2)$. Claim corrected, certificate records the measured excess.

126.21 YM-33 (Aug 22 2026): t-row fabric engine and the exact time two-point function

- CONSUMES: YM-30/32 engine (generalised to t rails, 5-leg vertices by two CG fusions), YM-31 trial discipline, YM-15 r for the r^2 law.
- RESULT: second order of the time expansion of the dressed excitation ratio is an exact bulk constant, m-uniform, edges localised with increments scaling as r^2 per unit distance (certified window).
- Build notes: (i) first run used w_{pt}^2 on the middle rail — wrong: the rail shared by two T_0 factors carries W once; (ii) an apparent linear drift of δ_2 in m was the boundary at distance 2 — at distances 3 and 4 it decays geometrically ($1.3e-9$, $1.6e-12$ at $\kappa = 1/8$). Both corrected before pinning.

126.22 YM-34 (Aug 22 2026): E4D-C restated

- CONSUMES: T01-E4C/E4D (RH), YM-16/26 exponential, YM-31 log-convexity.
- RESULT: certified negative (unsmoothed iteration $\rightarrow \sup$) and positive (smoothed iteration $\rightarrow$ spectral constant $f_0(1+\delta)$). The RH open item E4D-C is replaced by the alternating-product gap statement; its vacuum half is YM-31, its excitation half is the one open item.

- RH-Framework ledger edit to follow as a PR.

126.23 YM-37 (Aug 23 2026): the space transfer — m-uniformity as a theorem

- CONSUMES: YM-33 t-row engine (validated against, exactly), YM-F1 quaternion carrier, RKF theorem/53 (native LDL / weights) and theorem/28 §7–9 (finite seam matrix, outward certificate), YM-15 r.
- REFUSES, with witness: “complement upper bound from T53 cut-square weights”. T53 T4 bounds pivots by the diagonal; a block with every diagonal can have $\max > .$ Item 3’s upper half stays OPEN.
- RESULT: every YM-33/36 quantity is boundary, $M^{\wedge\{m-1\}}$ boundary for one column map $= B^{\wedge\{-1\}}M$ on the even conjugation-invariant sector (dim 8 at $t = 2$, 52 at $t = 3$); $_1$, $_2$ bracketed by exact inertia (Bareiss, integer minors); m-uniformity of $_1(m,p)$ with an explicit certified error; $_1^\infty = N_11 / _1$ bracketed two-sided.
- Build notes: (i) the full invariant sector (dim 14) gave a rate ~ 1.4 r — its $_2$ mode is the total-flip-odd state x_0+y_0 that no quantity here touches; the centre-of-SU(2) symmetry fixed it; (ii) exact rational LDL at dim 52 did not terminate in 17 min (10^9 face denominators in B); replaced by fraction-free Bareiss with Jacobi sign rule, which agrees with LDL at dim 8 and runs in seconds.

126.24 STANDING CORRECTION to YM-37 (same day, owner’s audit question)

- YM-37 T3’s proof of the all-m bound uses the finite spectral decomposition of the pencil (v_k “eigenvectors”) — a CLASSICAL_THEOREM_IMPORT by the framework’s own audit vocabulary; T53 explicitly does not claim weights = eigenvalues. Status of T3: ANCHORED (YM-19 precedent). The inertia brackets, sector reduction, exact matches and all rationals are unaffected; what is anchored is only the route from the brackets to the for-all-m statement.
- NAMED YM-38 (native replacement): $n_+(M - \sigma B) = 1 +$ T53 gives a one-positive-weight square $d_+ + |c_+(x)|^2$ with an explicit rational witness row (codimension-1 negativity, no eigenvectors); top direction by explicit-vector deflation (YM-8 lesson) with the defect $|(B^{\wedge-1} M)w - \theta w|_{_B}$ tracked exactly; two-block recursion in the theorem/28 Sec 9 outward-certificate shape. Same brackets expected.
- Remaining declared shadow (unchanged, inherited): $\text{Haar} \leftrightarrow \text{Phi_Sigma} = \text{RH T01-E5C/E6, OPEN}.$

126.25 YM-38 (Aug 23 2026): native m-uniformity — YM-37’s import removed

- CONSUMES (all framework-internal): T53 elimination (every count is a sign pattern of native weights — the quadratic-form ceiling needs no eigenvalues), RH T01-C cut-square

inequality, theorem/28 §3–5/§9 (recognized-channel Cauchy with declared geometric tail; outward certificate $\beta = L < 1$), theorem/54 §2 product bound (replaces all exponentials), YM-8 deflation lesson (explicit vector, exact defect).

- **RESULT:** $| (m,p) - _c | \leq A \cdot (L^{\wedge\{j-p\}} + L^{\wedge\{j_c-p\}})$ for ALL m,p with $j = \min(p-1, m-p)$ p , in two certified regimes $p = 2$ (all interior p — the YM-37 T3 scope) and $p = 12$ (sharp); exact identities: $r_w \leq B w$, $U'B U = U'MU$, split step law. Coarse A beats the anchored YM-37 constant (1.9e-8 vs 9.3e-7 at $_ = 1/8$). YM-37's standing correction is **RESOLVED:** T3's statement now rests on this capsule.
- **CORRECTION** of the YM-37 build note (narration arithmetic): the LINEAGE entry above claimed the dim-52 inertia ran on fraction-free Bareiss. Inspection of the committed file shows that patch never applied — the pinned YM-37 run used the NATIVE T53 elimination throughout, dim 52 included. No Bareiss identity is in any load-bearing path anywhere; the in-session timing note was about an uncommitted experiment.
- **Build notes:** (i) first error formula dropped the reference point's own tail term $A \cdot L^{\wedge\{j_c-p\}}$ — the exact table refused it ((30,15) outside);
 (ii) a backwards extrapolation $L^{\wedge\{j-p\}}$ for $j < p$ was heuristic — the exact value at (30,10) refused it by $47\times$; replaced by the two-regime structure, nothing claimed below p ; (iii) a bracket comparison against YM-37's 14-digit decimal strings needed the stored-precision slack — exact rationals vs rounded display.

126.26 YM-39 (Aug 23 2026): content ladder — superselection + channel ordering

- **CONSUMES:** YM-37 pencil/carrier, YM-38 native uniformity machinery (applied verbatim per channel — the derivation never used more of N than its pencil norm and reference positivity), YM-33 engine (cross-check of the exact vanishing), theorem/51 zero-residue sector (the centre-even split), YM-22 (ladder collapse, named next step).
- **RESULT:** (T1) centre Z_2 of $SU(2)$ superselects the fabric — mixed half-integer/integer fabric quantities vanish **EXACTLY** (zero matrix + engine zeros; a same-parity pair is nonzero, so the vanishing bites on parity); (T2) per-channel m -uniform bulk ratios with certified bands; (T3) strict ordering $\rho_{1/2} > \rho_1 > \rho_{3/2}$ **WITH** the uniformity margins on both sides, all grid κ — the $J = 1/2$ channel is the lightest on the carrier for every m and interior p , and no content ≥ 1 channel can mix into it. Session item (b) closed at the carrier level.
- **Boundary:** declared truncated faces/rung kernel (contents ≤ 1), so $\chi_{3/2}$ is a composite channel here (data row, not a fundamental claim); compressed statements; no operator upper bound; E4D-C OPEN.
- **Named next:** the same ladder on the heat-kernel trajectory (YM-22), where $r_j/a \rightarrow 0$ for $j \geq 1$ quantifies the collapse onto content $1/2$.

126.27 YM-40 (Aug 23 2026): heat-kernel content ladder — Casimir pinch

- CONSUMES: YM-22 trajectory ($\kappa = \theta a$, $\theta = 1/16$; face-ladder collapse), YM-37/38/39 machinery verbatim per a , YM-13 compound- interest exp (via YM-19 exp_neg), YM-1/F00-G odd-series log (log_iv). No new verdict types; nothing classical.
- RESULT: (T1) centre superselection exact at every a on the family; (T2) channel rates $\gamma_c(a) = -\text{Log}(\rho_c)/a$ bracketed two-sided, m-uniform bands included, with an independent exp-route re-enclosure control; (T3) $\Delta_\gamma(a) \geq 1$ at every a AND the two-sided PINCH $|\Delta_\gamma(a) - 5/4| \leq \epsilon(a)$, ϵ certified strictly decreasing (6.4e-4, 1.6e-4, 7.0e-5, 2.0e-5): the channel spacing lands on the free Casimir gap $C_1 - C_{1/2}$ in the cutoff direction; (T4) face ladder $r_1/r_{1/2}$ strictly decreasing on the same grid (YM-22 tie).
- Build notes: (i) first draft claimed monotone approach to $5/4$ from BELOW; the certified brackets cross $5/4$ after $a = 1$ and approach from above — replaced by the pinch (YM-22's approach-direction lesson, mirror image); $\gamma_{1/2}$ likewise crosses $3/4$ along the family — the dressing term is signed; (ii) first run ground for 12+ minutes on odd-series logs fed raw 100-digit rationals — the standing YM-23 rule (round outward BEFORE odd-series logs) applied, run time ~4 min.
- Boundary: order-1 dressed two-point on the carrier (compressed, lower-bound side), NOT the operator gap; kernel contents ≤ 1 (no $\chi_{3/2}$ Casimir claim); trajectory declared; E4D-C OPEN.

126.28 YM-41 (Aug 23 2026): the tools dock — E4D-C restated

- CONSUMES (cross-repo, YM-27 discipline — cited by pin, re-verified on this side): RKF theorem/74 (bfe40c2), theorem/75 (d185e86 + regrading 13b4315), theorem/76 (51afbcd), audit (3534418 + 8679f0b).
- RE-VERIFIED on the Publications pencil: silence-channel contraction $\beta_{\text{sil}} < 1$ at every κ (values agree with RKF to the last printed digit — both roundings coincide at 1e-9); ladder law with the discrimination check (tight, not loose); the seam-flow meter with the per-bracket parity law (count route = det route); ladder identity exact; superselection inside the meter ((0,1/2) slot empty).
- CONSUMED BY PIN (not re-instantiated here): theorem/75's stationarity, seam-count retention at $\Lambda = 3/2$, 2, and the outward certificate $u + e < 1$ for the untruncated column. Publications-side re-instantiation = future work if wanted; the RKF certificate is the derivation of record.
- E4D-C LEDGER UPDATE: m eliminated (YM-37), content tail lawfully released (theorem/75), contraction structural (theorem/74). The alternating product's remaining open content is the t-DIRECTION ONLY. NAMED YM-42: $\beta_{\text{sil}}(t = 3)$ on the 52-dim even sector — the t-ladder of silence contractions — and the t-uniform statement linking $\beta_{\text{sil}}(t)$ to the excitation/vacuum ratio limit. Clay/OS/continuum gates unchanged, OPEN.

126.29 NG-1 (Aug 23 2026): the native gap statement

- OWNER'S DECISION after asking whether the program was posing the wrong question: pursue the framework's own gap theorem; declare the translation to Clay as a separate open dictionary problem.
- CONTAINS NO NEW NUMBERS. Fixes NG, fixes five refutation conditions, re-labels the certified corpus against NG's obligation list.
- CONTROLS are against pinned artifacts, not narration: every cited capsule must exist and PASS; the teeth citations must have the stated shape in the stored results; no capsule may claim Clay/OS/continuum outside an open/boundary context. A test asserts C1 can fail.
- HONEST POSITION RECORDED: scored against NG the program is not near zero, but O4 (time) and O5 (cutoff) are where the difficulty lives and no capsule has touched the chain's operator upper bound.

Publications/papers/yang-mills-certified-benchmark/LINEAGE.md

Chapter 127

yang_mills_certified_program_v1

The following paper is included as compiled from its repository source.

A Certified Finite-Cutoff Program for the $SU(2)$ Theta-Graph Transfer Operator: Interacting Positivity, Exact Seam-Integer Counting, and a First Uniformity-in-the-Cutoff Theorem

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(Dated: August 28, 2026)

We report a finite, machine-certified program studying the spectral gap of the interacting transfer operator on the smallest non-trivial closed graph with a plaquette — the theta graph, two vertices joined by three edges — for $SU(2)$ lattice gauge theory. Every numerical statement below is an exact two-sided rational (or exact-integer) enclosure produced by directed interval arithmetic or exact linear algebra, never a floating-point estimate; every claim carries a machine-checked certificate reproducible from the accompanying repository. We prove: (i) the free reduced gap in closed form via the exact $SU(2)$ character spectrum; (ii) a certified two-sided interacting gap at every coupling $\kappa \geq 0$ on this graph, via a Kill-Lemma-shaped decomposition into a block Schur estimate and a positivity statement, both certified rather than assumed; (iii) that the first-order crossing direction is an exact rank-one, graph-symmetry –protected line, not an artifact of the compression used to compute it; (iv) an exact eigenvalue-counting reformulation of the gap via a Haynsworth-congruence (Birman–Schwinger-type) construction, giving exact integer spectral counts rather than numerical brackets on part of the coupling range; (v) the program’s first statement that is uniform in the lattice cutoff rather than proved at one fixed cutoff, obtained by replacing the Wilson action with the heat-kernel action, whose exact semigroup property removes discretization error identically; and (vi) that the interacting gap is volume- and cutoff-uniform simultaneously on a family of graphs built by vertex-gluing theta graphs with no shared plaquettes, while the same route provably fails once plaquettes are shared, localizing the residual volume obstruction exactly to interaction overlap. We are explicit about what this program does not establish: it does not touch the infinite-volume, continuum, or Osterwalder–Schrader-reconstruction content of the Yang–Mills existence-and-mass-gap problem, and none of the results here should be read as bearing on that problem beyond providing an exactly verified finite testbed and a catalogue of which certification routes do and do not survive volume growth.

I. INTRODUCTION

The Yang–Mills existence and mass-gap problem asks for a proof that a non-trivial quantum Yang–Mills theory exists on $\mathbb{R}^4$ satisfying the Wightman/Osterwalder–Schrader axioms, with a mass gap $\Delta > 0$ [1]. That statement concerns the continuum, infinite-volume limit of the theory. The present paper does not attempt it, and no theorem below should be misread as a step toward the Clay-sense predicate.

What we do attempt is narrower and, we argue, still useful: to take the smallest closed graph carrying a non-trivial plaquette — the theta graph B_3 , two vertices joined by three edges, one independent holonomy after gauge fixing, three faces — and certify, with exact arithmetic and machine-checked proofs, every spectral statement that can honestly be made about its interacting transfer operator at finite lattice spacing. The discipline we impose throughout is: *no floating-point number is ever part of a verdict*. Every enclosure below is either an exact rational computation or a two-sided interval of provably bounded width obtained from directed rounding, and every theorem is accompanied by an executable certificate, a set of negative controls that must fail when

the underlying claim is deliberately falsified, and an explicit statement of what remains open. This is closer in spirit to interval-certified numerics and to formally verified mathematics than to a physics estimate, and we adopt it because the target problem is famous partly for how easy it is to overclaim.

The paper is organized as a linear certification chain, YM-1 through YM-14, each step consuming only what the previous steps proved. We give theorem statements, proof sketches sufficient for a reader to reconstruct the argument, and pointers to the exact certificate in the accompanying repository for full reproducibility. Section III fixes the graph, the gauge-fixed carrier, and the character-expansion toolkit. Section IV treats the free (non-interacting) reduced gap. Section V builds the interacting gap at finite coupling on the theta graph in stages: a sandwich bound, the first-order crossing direction, a symmetry-protection theorem, a certified two-sided gap beyond the sandwich’s reach, and an exact eigenvalue-counting reformulation. Section VI states the capstone theorem: a strictly positive gap at every coupling on the fixed graph, together with the honest statement of what “every coupling, fixed graph” does not give. Section VII gives the paper’s central positive result on the harder axis: a gap that is uniform in the lattice cutoff. Section VIII extends this along the volume axis on a restricted family and localizes exactly where the extension

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fails. Section IX audits three structural gates — gauge invariance, volume/infrared behaviour, and regulator universality — and reports two of the three as genuinely split between a closed and an open half. Section XI states the remaining obligations without euphemism. Section XII concludes.

II. METHODOLOGY: CERTIFICATION AND REPRODUCIBILITY

Because this program’s value depends entirely on a reader’s ability to check it independently rather than take it on trust, we state the certification methodology precisely before using it.

Exact and directed-interval arithmetic only. No floating-point number ever appears inside a verdict. Rational quantities (eigenvalue multiplicities, Gram-matrix overlaps, seam-integer counts) are computed with Python’s arbitrary-precision `Fractions.Fraction` and carried exactly; no external numerical library is required for these. Transcendental quantities (Bessel functions I_ν , logarithms) are bracketed by two-sided rational intervals $[\ell, u]$ with $\ell \leq x \leq u$ guaranteed by construction: I_ν is enclosed by its everywhere-positive power series truncated at a declared order, with a geometric-series majorant bounding the truncated tail from above; log is enclosed via the atanh series, which is exact-sign alternation-free and admits the same majorant technique. Every arithmetic operation on such intervals rounds outward (down for lower bounds, up for upper bounds), so the true value is guaranteed to remain inside the bracket at every step, and every reported enclosure states its width explicitly.

Hash-pinned certificates. Each theorem above has one corresponding executable script (Table I) that recomputes its result from these primitives and writes a certificate file containing the exact value or bracket together with a SHA-256 digest of the full certificate content. That digest is checked against an `EXPECTED_*.sha256` file committed alongside it. A change to any input, any arithmetic step, or any stated conclusion changes the digest and is caught automatically; nothing about a certified result can be edited by hand without the mismatch being visible.

Negative controls. Every script includes at least one deliberately falsified variant of its own computation — a wrong Bessel recurrence, a substituted coupling, a sign flip on the disputed term, a shrunk carrier known to be too small — and asserts that this variant produces a *different, separated* bracket or a different exact count from the certified one. A script fails if a negative control does not separate, which is the mechanism that distinguishes a theorem that constrains the actual object from an identity that would hold on arbitrary data (the discipline we call, in the repository, the *renaming test*: before a computed coincidence is reported as a finding about the carrier, we check that it disappears under repetition on random data of the same shape).

TABLE I. Theorem-to-script correspondence. Every result in this paper is the printed output of the named script, with certificate digest pinned in the correspondingly named `EXPECTED_*.sha256` file.

Theorem	Script (in <code>certificates/</code>)
YM-1 (1)	<code>ym1_certified_gap.py</code>
YM-2 (2)	<code>ym2_theta_interacting_gap.py</code>
YM-3 (3)	<code>ym3_crossing_direction.py</code>
YM-4 (4)	<code>ym4_symmetry_protected.py</code>
YM-5 (5)	<code>ym5_two_sided_gap.py</code>
YM-6/7 (6)	<code>ym6_seam_integer_dock.py</code> , <code>ym7_v7_crossing_curves.py</code>
YM-8 (7)	<code>ym8_all_coupling_capstone.py</code>
YM-9 (8)	<code>ym9_uniform_heat_kernel.py</code>
YM-10 (13)	<code>ym10_blindness_ledger.py</code>
YM-11 (10–12)	<code>ym11_gate_verdicts.py</code>
YM-12 (9)	<code>ym12_square_sourced.py</code>
YM-14 (14)	<code>ym14_overlap_dock.py</code>

Independent re-execution and continuous integration. Several results above (the Theorem 6 counting law, the Theorem 8 uniformity theorem) were independently re-derived or corrected after a first pass flagged an inconsistency between a narrated claim and the arithmetic the script actually performed; we report such corrections in place (Section IX) rather than silently overwrite the earlier claim. The repository’s continuous integration workflow re-executes every certificate script and every test on each change and fails the build on any pin mismatch, so the checked-in state of the repository is, at every commit, the actual output of the actual scripts, not a hand-maintained summary of them.

Reproducing a specific theorem. Any of the results above can be independently checked with no dependency beyond a standard Python 3 interpreter (no NumPy, no computer-algebra system) by cloning the public repository and running the corresponding script, e.g.:

```
git clone https://github.com/Parveen117/
Publications
cd Publications/papers/
yang-mills-certified-benchmark
python certificates/ym1_certified_gap.py
python -m pytest tests/ -v
```

Each script prints its certified value or bracket, its digest, and the result of every negative control; the test suite additionally checks every digest against the committed `EXPECTED_*.sha256` file.

III. GRAPH, CARRIER, AND CHARACTER TOOLKIT

Let B_3 be the theta graph: two vertices v_0, v_1 joined by three edges e_A, e_B, e_C with $\operatorname{SU}(2)$ -valued holonomies

A, B, C . Tree gauge-fixing on any connected graph is exact and free of Gribov ambiguity: fixing a spanning tree (here, edge e_C , so $C = \mathbb{K}$) leaves exactly $b_1(B_3) = E - V + 1 = 2$ independent holonomies, here (A, B) , with residual gauge acting diagonally by conjugation. Gauge-invariant states are therefore exactly $L^2(\text{SU}(2) \times \text{SU}(2))^{\text{Ad}}$, and this fact is itself certified rather than assumed (Section IX); every capsule in this paper uses it as the carrier.

We work in the character basis. For $g \in \text{SU}(2)$, the irreducible characters $\chi_j(g)$, $j \in \{0, \frac{1}{2}, 1, \frac{3}{2}, \dots\}$, satisfy the exact orthogonality and Clebsch–Gordan pairing relations of $\text{SU}(2)$; in particular the triple integral

$$\int_{\text{SU}(2)} \chi_p(A) \chi_q(B) \chi_r(AB^{-1}) dA dB = \frac{\delta_{p=q=r}}{d_p}, \quad (1)$$

with $d_p = 2p + 1$, is exact and rational, and is the single identity that governs both the intra- and inter-plaquette seam structure used throughout. The Wilson single-link heat kernel has exact character expansion $K_\beta(g) = \sum_j d_j \lambda_j(\beta) \chi_j(g)$ with $\lambda_j(\beta) = I_{2j+1}(\beta)/I_1(\beta)$ (modified Bessel functions), and the true heat-kernel (semigroup) action used from Section VII on has exact expansion coefficients $f_j(\kappa) = 2I_{2j+1}(\kappa)/\kappa$ with the exact composition law $K_a * K_b = K_{a+b}$ — an identity the Wilson action does not share, which is why the uniformity result of Section VII requires the switch.

All numerical enclosures below use directed rational interval arithmetic (no floating point in any verdict); Bessel-function values are enclosed via their positive power series with a certified geometric tail bound, and transcendental combinations (logs, exponentials) via certified two-sided Taylor or continued-fraction enclosures with explicit remainder bounds. Every enclosure states its width, and every stage includes at least one deliberately falsified control (a wrong-recurrence Bessel tamper, a coupling substitution, a sign flip) that must, and does, produce a separated bracket.

IV. THE FREE REDUCED GAP

Theorem 1 (YM-1). *For the normalized single-link Wilson convolution operator with $\beta = 2$, the reduced spectral gap*

$$\Delta_{\text{red}}(a=1, \beta=2) = -\log(I_2(2)/I_1(2)) \quad (2)$$

lies in the certified enclosure $[0.836723306231588913050067801054\dots]$, width $< 10^{-30}$, an exact rational interval.

Proof: $\lambda_j(\beta) = I_{2j+1}(\beta)/I_1(\beta)$ is exact by the character expansion of K_β ; $\lambda_{1/2} \in (0, 1)$ strictly (control C1), so the gap is strictly positive; a tampered substitution $I_2 \rightarrow I_3$ separates the bracket (control C2); a width budget is enforced as a hard failure condition (control C3). This is the program’s base case and the only place where a bare single-link spectrum, rather than a graph transfer operator, is the object of study.

V. THE INTERACTING THETA-GRAPH GAP AT FINITE COUPLING

The interacting transfer operator on B_3 is $T_\kappa = M^{1/2}(K \times K)M^{1/2}$, where K is the single-link kernel of Section III and $M = M_\kappa$ is the plaquette interaction weight $e^{(\kappa/2)\text{Tr}U}$ on the third (closing) face.

Theorem 2 (YM-2: sandwich bound). *For $0 \leq \kappa < \kappa_0 = \Delta_{\text{red}}/6$, the min-max sandwich $m_-T_0 \preceq T_\kappa \preceq m_+T_0$ holds with $m_+/m_- = e^{6\kappa}$, using $|\text{Tr}U| \leq 2$. At $\kappa = 1/8$ this certifies a reduced gap $\geq 0.08672330623\dots$*

Theorem 3 (YM-3: first-order crossing direction). *The free top-excited eigenspace is exactly two-dimensional, $\{\chi_{1/2}(A), \chi_{1/2}(B)\}$, certified separated from $\chi_{1/2}^2, \chi_1, \chi_{3/2}$. The exact theta integral $\int \chi_{1/2}(A)\chi_{1/2}(B)\chi_{1/2}(AB^{-1}) = 1/2$ splits this doublet at first order into eigenvalues shifted by $\pm\lambda_{1/2}/4$; the vacuum first derivative is exactly 0. Hence $r'(0) = \lambda_{1/2}/4 \in [0.10828185668\dots]$, and the crossing direction is the single symmetric line $\chi_{1/2}(A) + \chi_{1/2}(B)$ — rank one, and invariant under the graph’s $A \leftrightarrow B$ symmetry.*

Theorem 4 (YM-4: symmetry protection). *Let $S : (A, B) \mapsto (B, A)$. Then $[S, T_\kappa] = 0$ for every κ , because $\text{Tr}(BA^{-1}) = \text{Tr}(AB^{-1})$ on $\text{SU}(2)$ (real characters). Consequently the crossing line identified at first order in Theorem 3 cannot rotate out of the swap-even sector at any coupling: this is a theorem, not a first-order accident. Certified finite- κ spectral data, obtained from the exact expansion $e^{(\kappa/2)\text{Tr}U} = \sum_j d_j f_j(\kappa) \chi_j$ with exact Clebsch–Gordan ring arithmetic and a certified truncation remainder, shows the swap-even branch rising and the swap-odd branch falling with κ on the sampled grid.*

Theorem 5 (YM-5: certified two-sided gap beyond the sandwich). *Using the exact doubling identity $m_\kappa^2 = m_{2\kappa}$, a Weyl bound $\|PMQ\|^2 \leq \lambda_{\max}(Pm_{2\kappa}P - (Pm_\kappa P)^2)$ and $\|QSQ\| \leq \lambda_{1/2}^2 e^{3\kappa}$ certify a two-sided gap at couplings beyond κ_0 : at $\kappa = 1/8$, ratio ≤ 0.5003 , gap ≥ 0.6924 ; at $\kappa = 1/4$ (past κ_0), ratio ≤ 0.5610 , gap ≥ 0.5781 ; at $\kappa = 3/10$, gap ≥ 0.5332 ; the route fails closed at $\kappa = 1$.*

Theorem 5 is, structurally, an unconditional realization of the Schur-estimate-plus-positivity (“Kill Lemma”) skeleton of an earlier, conjectural manuscript by the author that assumed the existence of a vacuum state; no conjecture, claim, or continuum statement from that manuscript is imported here (see the repository’s `LINEAGE.md` for the explicit non-import ruling). Only the block-Schur-plus-positivity *shape* of the argument survived, now realized with certified exact-interval arithmetic on the graph carrier.

Theorem 6 (YM-6/YM-7: exact seam-integer counting). *On the native five-dimensional carrier $V_5 = \{1, \chi_{1/2}(A), \chi_{1/2}(B), \chi_{1/2}(A)\chi_{1/2}(B), \chi_{1/2}(AB^{-1})\}$ (all exact eigenvectors of T_0 , exact Gram matrix with a single*

off-diagonal overlap $1/2$), a Haynsworth congruence — the same finite-to-infinite compression device used elsewhere in this program for a Birman–Schwinger-type reduction — turns the gap statement into an exact integer count $k_\Sigma(\kappa, \mu) = \#\{\text{spec}(T_\kappa) > \mu\}$, computed via two interval-LLD inertia evaluations of a rational pencil. Certified exact counts: $k = 1$ at $(\kappa, \mu) = (1/8, 3/5), (1/4, 3/5), (1/2, 1)$, i.e. $\lambda_2 \leq \mu < \lambda_1$ exactly — a strictly stronger statement than a numerical bracket. Enlarging the carrier to $V_7 = V_5 \cup \{\chi_1(A), \chi_1(B)\}$ (Gram block-diagonal, all seven vectors exact T_0 -eigenvectors) lowers the complement top eigenvalue from λ_1 to $\lambda_1 \lambda_{1/2}$ and unlocks the previously undecidable cell: exact count $k = 1$ at $(7/10, 13/10)$. On V_5 the cell $(7/10, 5/4)$ is honestly REFUSED, bracket $[1, 5]$: the carrier is too small to decide it, and this is reported as a refusal rather than resolved by widening tolerance.

VI. CAPSTONE: A GAP AT EVERY COUPLING ON THE FIXED GRAPH

Theorem 7 (YM-8). *The interacting theta-graph transfer operator has a strictly positive spectral gap at every coupling $\kappa \geq 0$, on the fixed graph B_3 at fixed lattice spacing.*

Proof: a certified pointwise kernel floor $k \geq e^{-3\kappa} [c_0^{-1} e^{-\beta}]^2 > 0$ (exact enclosures, positive-series Bessel bounds) combined with the classical Jentzsch/Krein-Rutman theorem [2, 3] — cited, not re-derived, per the citation discipline stated in Section IX — gives a simple Perron eigenvalue, hence a strict gap, at every κ . The window $\kappa \in [0, 7/10]$ additionally carries the exact counts and enclosures of Theorems 2–6.

Remark 1. *We state as plainly as possible what Theorem 7 does not say. A positive gap at every fixed coupling on a fixed, finite graph is compatible with the gap closing as the graph is refined or as the lattice spacing is taken to zero. Nothing here is uniform in the cutoff or the volume; that is exactly the content that Sections VII–VIII address on restricted families, and exactly the content that remains open in general (Section XI). A later result in this program (Theorem 9) further demotes the Jentzsch/Krein-Rutman anchor used here to a simplicity-only role, once positivity itself is re-derived from a manifest square; that correction is recorded in the governance ledger of the repository and does not alter Theorem 7’s conclusion.*

VII. A FIRST UNIFORMITY-IN-THE-CUTOFF THEOREM

Every result so far is at one fixed lattice spacing a . We now obtain the program’s first result that holds for every $a > 0$ simultaneously, by replacing the Wilson action with the heat-kernel action $K_a(g) = \sum_j d_j e^{-aC_j} \chi_j(g)$,

$C_j = j(j+1)$ the Casimir. This action satisfies the exact semigroup law $K_a * K_b = K_{a+b}$: composing n links of spacing a/n reproduces one link of spacing a exactly, with no discretization error. The Wilson action has no such exact composition law, which is why it cannot support the argument below.

Theorem 8 (YM-9). *The free reduced gap under the heat-kernel action is $\Delta_{\text{free}}(a) = C_{1/2} = 3/4$ exactly, for every $a > 0$ — the transcendental a -dependence cancels identically. Along the declared linear trajectory $\kappa(a) = \theta a$, the interacting gap satisfies $\Delta(a, \kappa(a)) \geq 3/4 - 6\theta$ for every $a > 0$; at $\theta = 1/16$ this gives a cutoff-independent gap of exactly $3/8$, failing closed at $\theta \geq 1/8$. The seam count $k(a, e^{-as})$ is exactly a -independent (a combinatorial fact about which Casimir levels lie below a threshold), so the exact-counting machinery of Theorem 6 transports across the entire refinement family at once, not just at one spacing. By contrast, at fixed Wilson coupling β the reduced gap grows like $1/a$ and diverges as $a \rightarrow 0$ — there is no cutoff-independent gap on this action without a compensating running coupling.*

We state the honest remainder of Theorem 8 without softening. The trajectory $\kappa(a) = \theta a$ is declared, not derived: the physically motivated trajectory is fixed by asymptotic freedom, $\beta \sim \log(1/a)$, which is not modelled here. The graph is fixed, so this is a statement about refining the same graph’s lattice spacing, not about infinite-volume growth. The heat-kernel action is itself a choice among a regulator class (Section IX addresses which parts of the program are regulator-independent). And Δ throughout is a reduced-graph transfer-operator gap, not a reconstructed physical mass gap in any continuum sense. What has moved is narrow but real: one structural gate — uniformity in the cutoff — goes from wholly untouched to touched, with an exact theorem, on a toy carrier.

VIII. VOLUME: WHERE UNIFORMITY EXTENDS AND WHERE IT PROVABLY FAILS

Theorem 9 (YM-12, volume half). *Let L_m be the graph formed by vertex-gluing m copies of the theta graph with no shared plaquettes. The transfer operator tensor-factorizes exactly across the m copies, so the ratio λ_2/λ_1 is m -independent, and combined with Theorem 8, $\Delta(L_m, a, a/16) \geq 3/8$ for every m and every a — a gap uniform in volume and cutoff simultaneously, on this family.*

This closes the interacting half of the volume gate audited in Theorem 11 below, but only on the no-shared-face family. Theorem 11 shows the same sandwich-style route fails, exactly and quantifiably, once faces are shared; we record the two extremes as the boundaries of a genuine open problem, not as a resolved one.

The same paper (YM-12) also re-derives the positivity underlying Theorems 5–7 from a manifest square,

$T_\kappa(a) = S^*S$ with $S = [K_{a/2} \otimes K_{a/2}]m_{\kappa/2}$ — both factors being the program’s own halved-parameter objects from Theorem 8’s semigroup property and Theorem 5’s doubling identity. This demotes the classical Jentzsch/Krein-Rutman citation used in Theorem 7 to a role of certifying only *simplicity* of the top eigenvalue, with positivity itself now square-sourced from within the program’s own objects rather than imported.

IX. THREE STRUCTURAL GATES, AUDITED

We close the finite program with an audit of three structural questions on the toy-carrier family B_n (the “ n -banana” graph: two vertices, n edges, $b_1 = n - 1$ holonomies, $F(n) = n(n - 1)/2$ plaquettes; B_3 is the theta graph used throughout).

Theorem 10 (YM-11, Gate 1 — gauge: closed). *Tree gauge-fixing on a connected graph is exact, with no Gribov obstruction: gauge-invariant states are exactly $L^2(G^{b_1})^{\text{Ad}}$. For B_3 this certifies, rather than assumes, the carrier used in every capsule above.*

Theorem 11 (YM-11, Gate 2 — volume/infrared: split). *The free reduced gap $C_{1/2} = 3/4$ is volume- and cutoff-uniform for every n and a : closed. The interacting sandwich bound gives $\Delta \geq C_{1/2} - 2F(n)\theta$, which degrades quadratically in n : at $\theta = 1/16$ the certified bounds are $3/8$ ($n=3$), exactly 0 ($n=4$), and $-1/2$ ($n=5$). The sandwich/envelope route is therefore proven insufficient for volume uniformity beyond Theorem 9’s no-shared-face restriction, with the critical volume computed exactly rather than estimated.*

Theorem 12 (YM-11, Gate 3 — regulator universality: split). *For the whole regulator class $K = \sum_j d_j c_j \chi_j$ with $c_j > 0$ (Wilson, heat-kernel, and others), the counting layer — content grading, the exact multiplicity law $m(j_1, j_2) = \min(2j_1, 2j_2) + 1$ on the Ad-invariant sector, carrier dimensions, and the source-restriction “blindness” ledger of Theorem 13 below — depends only on SU(2) representation theory and is therefore regulator-independent: closed. The metric layer, i.e. the numerical value of the gap, genuinely differs between regulators (witnessed directly by the Wilson vs. heat-kernel contrast in Theorem 8) and remains open.*

Theorem 13 (YM-10, source-restriction ledger). *Every compressed carrier V used above is, in the sense of the companion source-restriction framework, a restriction of the full target space; the relevant question is not tightness of the resulting bound but whether V is blind to the target. The exact multiplicity law above independently reproduces $\dim V_5 = 5$ and $\dim V_7 = 7$ and re-derives the two-dimensionality of the $(\frac{1}{2}, \frac{1}{2})$ sector used in Theorem 3 from representation theory rather than from the Gram matrix. The exact probe count $b(V, s) = \text{rank}(\Pi_s | \ker E)$ shows $b(V_5, s) = 0$ for $s \leq 2$ — the carrier used in Theorem 6 is exactly blindness-free within its own certified*

threshold window, which is why that choice of carrier was correct and not merely convenient. At $s = 3$, V_5 requires six probes and V_7 four; blindness is nonincreasing along $V_5 \subset V_7 \subset V_9$ with nonnegative stage increments; and the whole ledger, depending only on content and Casimir level, is exactly a -independent, so one ledger covers the entire refinement family of Theorem 8. This ledger is exact for the free target only; on interacting faces the content grading mixes, so $b(V, s)$ is a lower bound there, and closing that gap is an explicit named obligation of the program, not a silent one.

We record one self-correction as part of the certification discipline itself: an earlier draft of Theorem 8’s uniform seam count (T4) counted spectral *contents*, while the counting statement of Theorem 6 counts *eigenvalues with multiplicity*; at the relevant sector these differ (4 versus 5). Both quantities are independently a -independent, so the uniformity conclusion of Theorem 8 is unaffected, but the arithmetic was wrong and is corrected here. We regard catching and publishing this kind of error, rather than avoiding it, as part of what the certification discipline is for.

X. INTERACTION OVERLAP BETWEEN PLAQUETTES

Theorem 14 (YM-14). *For a bridged pair of theta-graph blocks coupled by a single shared face, $e^{(\kappa/2)\text{Tr}(A_1 A_2^{-1})}$ — the minimal case of genuine plaquette overlap not covered by Theorem 9’s no-sharing family — the inter-block overlap seam is exactly rank-one and transported by the same graph symmetry and the same theta integral $(1/2)$ that governs the intra-block seams of Theorem 3: one identity governs both. On the orthonormal carrier $W_5 = \{1, \chi_{1/2}(A_1), \chi_{1/2}(A_2), \chi_{1/2}(B_1), \chi_{1/2}(B_2)\}$, exact counts $k = 1$ are certified at $(\kappa, \mu) = (1/8, 3/5), (1/4, 7/10), (1/2, 9/10)$; at $\kappa = 1$ the carrier certifies count 0 at $\mu = 3/2$ (the compressed vacuum value is below μ), which we report as a carrier limitation rather than resolve by enlarging tolerance. The first-order overlap price is exactly $\kappa\lambda/4$, carried by a single swap-even line, in direct analogy with the single-block crossing direction of Theorem 3.*

This opens, but does not close, the object named by Theorem 11’s Gate 2 as the real residual obstruction: Theorem 9 is uniform on no-shared-face volume growth, Theorem 11 shows the sandwich route fails on complete sharing, and a physically realistic bounded-degree lattice sits strictly between these two certified extremes. The next object named by the program is a one-dimensional block-transfer dock for a chain of bounded, but nonzero, overlap; it is not attempted here.

XI. WHAT REMAINS OPEN

We list the open content without euphemism, since the value of this program depends on the list being accurate.

- **Volume, general lattice.** Theorem 9 is uniform only on the no-shared-face family; Theorem 11 shows the sandwich route fails, with an exactly computed critical volume, once faces are shared; the bounded-overlap case of physical interest (Theorem 14) is opened but not closed.
- **Regulator universality, metric layer.** Only the combinatorial/counting layer of universality (Theorem 12) is closed; the numerical gap value differs by regulator and the asymptotically-free running-coupling trajectory needed to connect the heat-kernel result to the physical continuum limit is declared, not derived (Theorem 8).
- **Osterwalder–Schrader reconstruction.** No statement here constructs, or claims to construct, a continuum Wightman theory from these finite-graph transfer operators.
- **Non-triviality and the Clay predicate.** None of the above bears on the existence-and-mass-gap predicate of the Clay Millennium Problem as stated in [1]; that problem concerns the continuum, infinite-volume limit, which this program has not addressed and does not claim to have addressed.

XII. CONCLUSION

We have given, on the smallest graph carrying a Yang–Mills plaquette, a fully exact-arithmetic, machine-

certified account of the interacting transfer operator’s spectral gap: a strictly positive gap at every coupling on the fixed graph (Theorem 7), an exact eigenvalue-counting reformulation on part of the coupling range (Theorem 6), the program’s first cutoff-uniform theorem (Theorem 8), and a volume-uniform extension on a restricted family with the extension’s exact failure point identified on the complementary family (Theorems 9, 11). Each result is accompanied by negative controls that must, and do, fail when the claim is deliberately falsified, and each is reproducible from the certificates and scripts in the accompanying repository. The value we claim for this program is not proximity to the continuum problem, which remains entirely open, but a fully verified, error-caught, and error-corrected finite testbed — together with a precise catalogue, Section XI, of exactly which structural gates a genuine continuum proof would still need to close.

DATA AND CODE AVAILABILITY

All certificates, verification scripts, and test suites are available in the public repository `Parveen117/Publications`, directory `papers/yang-mills-certified-benchmark/`, under an open license. The theorem-to-script correspondence and reproduction instructions are given in full in Section II and Table I; no result in this paper depends on any data, computation, or software that is not in that public repository and independently re-executable by a reader.

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Part IX

Repository Working Notes (Appendix)

Chapter 128

Do Not Reopen Guard

These transferred theorem packets are now framework inputs.

positive eta path equality	do not reopen
M3 positivity	do not reopen
G3/M3 accepted block	do not reopen
rank-at-most-five seam integer	do not reopen
common five-matrix covariance	do not reopen
zero-cut T03 source-Gram identity	do not reopen

Open work must begin downstream of these capsules.

The only lawful reasons to edit these files are:

- correct a provenance pointer;
- add a stronger source citation;
- replace an older capsule with a strictly stronger theorem;
- record a failed audit that invalidates a transferred claim.

Everything else is just rearranging bones in the museum.

RKF/theorum/07_do_not_reopen.md

Chapter 129

Cleanup Warning

The theorem capsules have moved into RKF. MP cleanup should now be surgical.

Do not delete verifier scripts or raw ledgers merely because a readable theorem has been moved. The readable theorem is the map. The executable proof-lab material is the terrain. Deleting terrain because the map exists is how civilizations invent paperwork fires.

Suggested next MP cleanup operation:

```
branch from the relevant MP source branch;  
delete duplicated theorem/status surfaces only;  
keep scripts, tests, state capsules and raw ledgers;  
open a cleanup PR titled "Remove theorem surfaces exported to RKF theorem".
```

Start with PR #246 readable/status files because they are localized and clearly duplicated here:

```
NATIVE_EXACT_ETA_FAMILY_PATH_INDEPENDENCE.md  
EXACT_ETA_FAMILY_PATH_INDEPENDENCE_STATUS.json  
NATIVE_ACTUAL_ETA_NORMALIZATION_TWO_PATH_ANALYTICITY.md  
ACTUAL_ETA_NORMALIZATION_TWO_PATH_STATUS.json
```

RKF/theorum/09_cleanup_warning.md

Chapter 130

Export Manifest

```
export name:      MP certified RH theorem transfer
source repo:      Parveen117/MP
destination repo: Parveen117/Recognition-Kernel-Framework
destination path: theorum/
branch:           agent/transfer-certified-theorum-from-mp
```

130.1 Exported theorem groups

```
T1 positive-eta path equality by amplitude lift
T2 seam-integer/common five-matrix reduction
T3 one-sector terminal sufficiency after reduction
T4 exact target-local M3 positivity
T5 positive odd accepted block
T6 zero-cut T03 endpoint completion
```

130.2 Not exported as theorem capsules

```
raw Python proof-lab engines;
unit tests;
state NPZ capsules;
large numerical ledgers;
LaTeX manuscript source trees.
```

Those remain implementation/provenance material unless separately archived.

RKF/theorum/10_export_manifest.md

Chapter 131

After Merge Tasks

After this RKF branch is merged:

1. Create MP cleanup branches from the relevant MP source branches.
2. Delete only theorem/status surfaces duplicated here.
3. Keep proof-lab scripts and raw validation packets until separately archived.
4. Compare MP after cleanup to see what useful unsorted material remains.
5. Promote only the remaining useful MP material into future RKF `theorem/NN_*.md` capsules.

First cleanup target:

MP branch: `agent/rh-rotation-metric-topology-reset`
reason: PR #246 path-equality theorem is now cleanly transferred

Candidate files are listed in `MP_CLEANUP_CANDIDATES.md`.

`RKF/theorum/13_after_merge_tasks.md`

Chapter 132

Cleanup Branch Plan

Create cleanup branches only after the RKF theorem branch is accepted.

Suggested first cleanup:

```
repo:  Parveen117/MP
base:  agent/rh-rotation-metric-topology-reset
branch: agent/remove-exported-path-theorum-surfaces
```

Delete only duplicated readable/status files for PR #246. Keep executable scripts and tests.

Then repeat for M3/seam/T03 surfaces after checking exact file presence on each open MP branch.

RKF/theorum/19_cleanup_branch_plan.md

Chapter 133

Branch Consolidation and Theorem-Surface Audit

133.1 Purpose

This audit records the repository-wide branch verification performed on 2026-08-05 before declaring `main` the canonical theorem-bearing branch.

The operation was a Git-graph and theorem-surface consolidation. It was not an independent re-proof of every mathematical statement. The validity and claim boundary of each theorem remain exactly those stated in its own capsule and certificate ledger.

133.2 Canonical audit base

```
repository: Parveen117/Recognition-Kernel-Framework
canonical branch before index refresh: main
canonical audit-base commit: b05b69bdd02bd3f230b12df996704661947e2b0f
```

133.3 Pre-consolidation branch comparison

Every development branch was compared directly against `main`. In every case:

```
ahead_by = 0
status   = behind or identical
```

Therefore no branch contained a unique commit absent from `main`, and no theorem-bearing history required a conflict merge or cherry-pick.

Branch	Previous tip	Behind main	Unique commits
agent/rkf-next-development	9b0b0e5ac9cd2a7adeca228ff74901ae6bb69854	0	0
agent/cut-graded-universals	01ee132e9704cd7ea3e31811746ffdc0ee2246	0	0
agent/cut-graded-lambda-calculus	2af57923e62c18a1a71ed242ff9cb10e6745f31	0	0
agent/bilateral-jettison	7b7c0f18d93fa69f5109a6fec426b7a69f2c13	0	0
agent/madhava-smriti	60b3ebaf2b4579b75c691af6589b00a764f246226	0	0
agent/thermodynamicity	911f0a404386394db4805239f742103604abe64	0	0
agent/broaden-thermodynamicity	832d162077db2a71b3877f5755c55d1323c0c23	0	0

Branch	Previous tip	Behind main	Unique commits
agent/thermodynamics	b05b693bda024dc1e230e12df996704661947e2b0ff		0

133.4 Theorem-surface verification

The canonical `main` tree was checked for the advanced theorem chain:

```
theorem/41_cut_graded_universal_generator_theorem.md
theorem/42_cut_graded_lambda_jacobian_tower_theorem.md
theorem/43_bilateral_jet_flow_recognition_capstone_theorem.md
theorem/44_madhava_smriti_bilateral_jet_flow_closure_theorem.md
```

It was also checked for the complete thermodynamic archive:

```
theorem/thermodynamics/01_response_tetrad_and_flat_closure.md
...
theorem/thermodynamics/10_thermodynamic_cut_square_response_decomposition.md
theorem/thermodynamics/lambda_geometry/
```

The underlying theorem files retain different evidence statuses, including `PROVED`, `LOCAL PASS`, `USER-REPORTED PASS`, and `IMPLEMENTED / USER RUN REQUIRED`. Consolidation does not flatten those distinctions into one universal certification claim.

133.5 Cross-reference correction

Theorem 43 historically stated that a Madhava-named calculus was not yet present in the repository. That statement described the repository state when Theorem 43 was written. It is superseded by Theorem 44, which now supplies the typed Madhava-Smriti bilateral jet-flow closure theorem.

Current reading order is therefore:

```
41 universal generator
-> 42 lambda-Jacobian tower
-> 43 bilateral jet-flow capstone
-> 44 Madhava-Smriti closure
-> thermodynamic and later physical adapters.
```

133.6 Synchronization action

All stale `agent/*` refs were advanced by ordinary fast-forward to the canonical `main` tip. No force update was used, no branch-only commit was deleted, and no merge conflict was hidden.

After this audit and the accompanying index refresh are committed, `main` and every retained `agent/*` branch are advanced again to the commit containing this audit. Thus the repository ends with one visible, complete theorem surface rather than several archaeologically interesting partial tips.

133.7 Claim boundary

BRANCH ANCESTRY / UNIQUE-COMMIT AUDIT	COMPLETE
ADVANCED THEOREM FILE PRESENCE	VERIFIED
THERMODYNAMIC ARCHIVE PRESENCE	VERIFIED
INDEX AND READING-ORDER CONSISTENCY	UPDATED
FORCE PUSH OR HISTORY DELETION	NONE
INDEPENDENT RE-PROOF OF EVERY THEOREM	NOT PERFORMED
PHYSICAL IDENTIFICATION CLAIMS	UNCHANGED
GLOBAL RH CLAIM	NOT CREATED BY THIS AUDIT

RKF/theorum/BRANCH_CONSOLIDATION_AUDIT.md

Chapter 134

MP cleanup candidates after theorem transfer

The theorem capsules have been moved into `Parveen117/Recognition-Kernel-Framework/theorum` on branch `agent/transfer-certified-theorum-from-mp`.

This file records what should be removed from MP next. It is intentionally a candidate list because the source material lives across several open MP branches, not one merged directory.

134.1 Safe deletion class

Delete from MP only after confirming the corresponding theorem capsule exists in RKF:

```
human-readable theorem summaries;  
status JSON files whose content is duplicated in RKF provenance;  
old manuscript theorem surfaces superseded by the RKF theorum capsules.
```

134.2 Do not delete without archival copy

```
native verifier scripts;  
test modules;  
state capsules;  
raw numerical ledgers;  
source audit scripts;  
large artifacts needed to regenerate the theorem packets.
```

134.3 Candidate MP PRs and files

PR #246, branch `agent/rh-rotation-metric-topology-reset`

Candidate readable/status surfaces:

```
proof_lab/clock_free_recognition/NATIVE_EXACT_ETA_FAMILY_PATH_INDEPENDENCE.md  
proof_lab/clock_free_recognition/EXACT_ETA_FAMILY_PATH_INDEPENDENCE_STATUS.json  
proof_lab/clock_free_recognition/NATIVE_ACTUAL_ETA_NORMALIZATION_TWO_PATH_ANALYTICITY.md  
proof_lab/clock_free_recognition/ACTUAL_ETA_NORMALIZATION_TWO_PATH_STATUS.json  
proof_lab/clock_free_recognition/NATIVE_UGD_FIXED_FIVE_MULTIPLICATIVE_HOLONOMY.md  
proof_lab/clock_free_recognition/UGD_FIXED_FIVE_MULTIPLICATIVE_HOLONOMY_STATUS.json
```

Keep scripts/tests unless deliberately archiving them:

```
native_state_locked_exact_eta_family_path_independence.py
test_native_state_locked_exact_eta_family_path_independence.py
native_state_locked_ugd_eta_normalization_rectangle.py
test_native_state_locked_ugd_eta_normalization_rectangle.py
```

PR #200 stack

Candidate readable theorem surfaces are spread through the M3 stack PRs #181-#200. Keep all executable ledgers until exact file-level archive exists.

PR #242/#247/#254/#255/#277

Candidate deletion class:

```
manuscript theorem summaries duplicated by RKF theorem;
status files that merely restate seam-integer / T03 conclusions.
```

Keep source-audit and build files unless the publication bundle is separately archived.

134.4 Cleanup rule

Use one MP cleanup branch per source PR branch. Do not delete across unrelated open branches in one blind operation. Future humans already have enough ruins to interpret.

RKF/theorum/MP_CLEANUP_CANDIDATES.md

Chapter 135

Source and Development Index

This file maps the theorem capsules to their source PRs or native development branches. It is a provenance guide, not a substitute for the theorem statements, proof notes, certificates, or claim boundaries.

135.1 Initial MP transfer

Theorem capsule	MP PRs	Main source claim
01_positive_eta_path_independence.md	246	actual eta-normalization rectangle and exact positive-eta family path independence
02_seam_integer_single_sector_reduction.md	235, 242, 247, 255, 277	seam integer, common five-matrix covariance, single-sector terminal focus
03_m3_positive_accepted_block.md	181, 182, 196, 198, 199, 200, 202	exact native b3 signs, target-local M3 positivity, positive accepted odd block
04_zero_cut_t03_endpoint_completion.md	254, 256	zero-cut quotient, five-label lift, source-Gram identity, projection-defect positivity
05_transfer_boundary_and_next_gate.md	all above	prevents reopening certified upstream theorems and records the next lawful gate

Source branches named in that material include:

```
agent/rh-rotation-metric-topology-reset
agent/native-x3-fixed-state-primal-spectral-transfer
agent/native-x3-parity-feshbach-complement-reduction
agent/native-seam-scale-convergence
agent/rh-journal-final-conditional-zero-closure
agent/rh-paper-single-sector-seam-closure
agent/t20-native-odd-single-sector-closure
```

The machine-readable `provenance.json` records this original 01–05 transfer. It is intentionally scoped to that packet and is not a complete history of later native theorem development.

135.2 Native cut and completion chain

Files 21 through 40 were developed natively in this repository after the first transfer. Their exact dependencies, source pins, imported membranes, and status labels are stated inside each capsule and in the Git history.

The chain includes:

```
cut-memory spectral isomorphism;
memory-cut rank gate;
native cut-generated object theorem;
clock-free recognition seam cut calculus;
primitive-to-completion hierarchy;
finite-to-infinite recognition completion;
minimal observer and covariance realization;
native source, decoder and boundary adapters;
strict odd source-kernel no-blindness;
odd/native classical-normalization interface.
```

No blanket claim is made that every file in this range has the same evidence status.

135.3 Advanced native development branches

Before repository consolidation, the advanced theorem capsules were carried by the following branch tips:

Theorum capsule or archive	Development branch	Pre-consolidation tip
41_cut_graded_universal_generative	agent/theorem-d-universal-generative	01ee1d32ee704d97ea3e31811746ffdc0ee223
42_cut_graded_lambda_jacobian	agent/ruth-theorem-d-lambda-jacobian	9f5792e062ce3a1a71d9ed242ff9cb10e6745f3
43_bilateral_jet_flow_recognition	agent/bilateral-theorem-d-capsule	737c0f18d93fa69f5109a6fec426b7a69f2c
44_madhava_smriti_bilateral_jet_flow_recognition	agent/madhava-smriti-theorem-d-capsule	60b3bba2b4579d75c691af6589b00a764f24622
broader native theorem development	agent/rkf-next-development	9d6bce5ac9cd2a7adeca228ff74901ae6bb698
direct thermodynamic archive	agent/thermodynamic-theorem-191f12a4	24386394d9b4805239f742103604abe
lambda-geometry expansion	agent/broaden-thermodynamic-332d162077	db2da7b33877f5755c55d1323c0c2
thermodynamic cut-square specialization	agent/thermodynamic-cut-square	05b59b4d02bd3f230b12df996704661947e2b

On 2026-08-05 each branch was compared against `main`. Every branch had `ahead_by` = 0; therefore all theorem-bearing commits were already present in `main`. The stale refs were advanced by ordinary fast-forward without force or history deletion. The complete record is in `BRANCH_CONSOLIDATION_AUDIT.md`.

135.4 Thermodynamic MP provenance

The thermodynamic archive under `theorum/thermodynamics/` is sourced primarily from:

MP PR #239 thermodynamic response tetrad, entropy-scaled lambda coordinates,
flat closure, response curvature and seam-index spectral flow;

MP PR #243 canonical response cost, projection/source overlap,
Onsager compass and Hodge no-leakage, thermo-Weil provenance,
physical-flux connection;

MP PRs #274-#275
native cut-square event decoder and minimum-decoder burden,
specialized as the thermodynamic cut-square theorem;

MP PRs #52-#61
lambda holonomy, path ordering, admissible factorization,
refinement-stable length, constrained descent,
curvature-flux no-collapse and primitive flux witnesses.

Exact source file paths and blob hashes are preserved in `thermodynamics/SOURCE_PROVENANCE.md` and `thermodynamics/lambda_geometry/SOURCE_PROVENANCE.md`.

135.5 Supersession rule

A claim-boundary statement may accurately describe the repository state at the time a theorem was written and later become historically stale. The principal example is Theorem 43's statement that a Madhava-named calculus was not yet present. Theorem 44 now supplies that extension. The earlier mathematical theorem remains valid; the later capsule supersedes only the repository-status sentence.

135.6 Consolidated canonical branch

repository: Parveen117/Recognition-Kernel-Framework
canonical branch: main
branch audit: `theorem/BRANCH_CONSOLIDATION_AUDIT.md`
complete theorem map: `theorem/README.md`

Future theorem development should branch from current `main`, preserve individual evidence labels, and return through a verified fast-forward or reviewed merge. Otherwise the repository will resume its natural ambition to become a maze.

`RKF/theorem/SOURCE_PR_INDEX.md`

Part X

Provenance

Appendix A

Source Manifest

Source file	SHA-256 (first 32)
RKF/README.md	056d5bc405fcc7e8b079c42ef9709054
RKF/CLAIM_BOUNDARY.md	0f134eac325ed74e7882f1b8043bd0ce
RKF/TERMINOLOGY_AND_FILING_ALIGNMENT.md	de31b01f79f60212373154748f93dba6
RKF/REVIEWER_GUIDE.md	a0b7d72164b91565506f3307a798bcb6
RKF/REPRODUCE.md	235f3b63dd633810c84c7a6a037bc6da
RKF/CERTIFICATE_INDEX.md	d9e5b1402de37419dc71f079cc158500
RKF/SOURCE_PROVENANCE.md	d207886e9bbd721df49af2fd9781f5ee
RKF/NOTICE.md	b6e24bee2e59120caa353c692525e9ea
RKF/theorum/01_positive_eta_path_independence.md	9fcdaaef07676fbb045ca7e568d6a028
RKF/theorum/02_seam_integer_single_sector_reduction.md	bf459679242454dc28b0b33cc2c9828c
RKF/theorum/03_m3_positive_accepted_block.md	6813b24ea0d5a8ea0824075cdc4808c4
RKF/theorum/04_zero_cut_t03_endpoint_completion.md	d81fb4f6e3f26b0e64bcb67d3374d89f
RKF/theorum/05_transfer_boundary_and_next_gate.md	051747caa850246132f531524e5d4cd1
RKF/theorum/06_one_sector_theorem_note.md	8f055d0834053e343e9b535ff2a12c65
RKF/theorum/08_framework_consumption_chain.md	41fa7fbf747b0e7782a397e9fec45ab9
RKF/theorum/11_minimal_reading_order.md	d3284224e0c10f5ea267e1ab45415f0e
RKF/theorum/12_rh_claim_boundary.md	462855df0812c56a4088e7be378ad107
RKF/theorum/14_object_dictionary.md	f6dd84a6439ddc9908e9d887db2b8d23
RKF/theorum/15_short_final_chain.md	4717958035a42ca1da74de1e1ad69387
RKF/theorum/16_source_boundary.md	e88f2cd6800aae07336f1f9e3b320324
RKF/theorum/17_final_status.md	d2a2d2593886e71435b0725de2b80551
RKF/theorum/18_no_rederive_note.md	a6edd13b02550ed3debfe901df5c1688
RKF/theorum/20_public_readme_note.md	2db80078d206905067e7a8a76785560e
RKF/theorum/21_cut_memory_spectral_isomorphism.md	fc32026d8c6d7bd333c31dbc48aa1233
RKF/theorum/22_rh_memory_cut_rank_gate.md	44f104a1d1954ccec066408184de933f
RKF/theorum/23_native_cut_generated_object_theorem.md	82e036ad11dc2d69066c3c76a4829ad2
RKF/theorum/24_clock_free_recognition_seam_cut_calculus.md	3d7d70f5cd213378c32e90e3c88eb195
RKF/theorum/25_clock_free_cut_memory_stage_plan.md	105e3dbd84092ae2cd8bd40a8556b8f5
RKF/theorum/26_rsc_source_mapping.md	d12b76aead7aa81612ba3b7f45f1097c
RKF/theorum/27_rsc_primitive_to_completion_hierarchy.md	6ae181f7478d767a437b7ce191a90ae4
RKF/theorum/28_recognition_complete_finite_to_infinite_cut7a1e370760353d	7a1e370760353d5b69fcd62c3c84ba400
RKF/theorum/29_morphic_smriti_stage15_plan.md	9a8854d9a6d92e19f945052f380fad6e
RKF/theorum/30_stage15_exact_recognition_completion_certificate201.md	ca201a1d048c0ef7d36b48ec2ac6066
RKF/theorum/31_cut_variational_minimal_observer_theorem.md	408870812cf548d7fc42ecd5c242d1c4
RKF/theorum/32_cut_covariance_event_realization_theorem.mdc	591711a0e6e66cddf4a5b598a99ea61
RKF/theorum/33_native_weil_cut_event_cogeneration_gate.md	2ca2f0c343265fe89a557a7489da323f
RKF/theorum/34_native_common_chart_decoder_transfer.md	4563ddaa281a38780e3615981b444982

RKF/theorum/35_native_metric_graph_boundary_pairing.md 653e3bf5145e04de31bb52d030a36650
RKF/theorum/36_native_prime_gamma_debt_source_adapter.md 90e9671f203a52672c31f832da98b1de
RKF/theorum/37_native_two_sheet_hardy_compatibility_attachment.md 997fe38c5bd71390577ded9e19a8aae9
RKF/theorum/38_native_compatibility_form_range_source_domain.md 55629102eb9b6e834297e65f9fed03b2
RKF/theorum/39_native_source_kernel_no_blindness_strict_ordering.md 1d658d62c15b45afb9ab35211aadeb
RKF/theorum/40_odd_native_classical_normalization_interface.md 0bd47b20a66d4e9323f012ea0db4485
RKF/theorum/41_cut_graded_universal_generator_theorem.md 2af3f302b1a4d5ba6365d126d3d7e3da
RKF/theorum/42_cut_graded_lambda_jacobian_tower_theorem.md e5304cc970bfa0e18140aa1b5373f167
RKF/theorum/43_bilateral_jet_flow_recognition_capstone_theorem.md 3322a0d7bd238a933d8c5fbf8e2add84
RKF/theorum/44_madhava_smriti_bilateral_jet_flow_closure_theorem.md t343e2d1e0cb574bbf237a6872033c4f
RKF/theorum/45_directed_arithmetic_analytic_closure_theorem.md 4ad80ae2604f1e1f74e67355d9564b05
RKF/theorum/46_first_visible_jet_seam_quotient_theorem.md 787a4a7f3d43bd8c9231ed4e7114c552
RKF/theorum/47_source_bound_proof_carrying_numerical_validation.md d340c6c14c49bf7d5d6fb5d16f013ec9
RKF/theorum/48_emk_algebra_cut_graded_curvature_theorem.md f28401ae5493158dcac670dc5f2fabdd
RKF/theorum/T19_T20_odd_boundary_burden_certificate.md 66dcb1240de819f7f847edec7762c7e
RKF/theorum/thermodynamics/01_response_tetrad_and_flat_closure.md 7be531d9c96a5c742b2e306496080ca0
RKF/theorum/thermodynamics/02_curvature_to_seam_spectral_folds.md f08ddbbdf079247f3b6855fddf6572891
RKF/theorum/thermodynamics/03_canonical_response_cost_barrier.md 41adacbdde100541f708d8577403ffa
RKF/theorum/thermodynamics/04_projection_defect_and_source_eulerian_flow.md 9e0e8f1fbad0917a87dda2537026f09f
RKF/theorum/thermodynamics/05_onsager_compass_and_constitutivity.md 116a5360998b28efc05f31febf8ac54f
RKF/theorum/thermodynamics/06_onsager_hodge_no_leakage.md 0b2ffee77b9e719a563e045a3150b3f0
RKF/theorum/thermodynamics/07_thermo_weil_provenance_five_slopes.md 6b0f76e646ddbd481d0c3867cda18e31a
RKF/theorum/thermodynamics/08_canonical_connection_physics.md a5f126808d54d36c74b883d058f2e7591
RKF/theorum/thermodynamics/09_scope_exclusions.md 109a8538f6bdb90488a1013f08acd73f
RKF/theorum/thermodynamics/lambda_geometry/10_lambda_holonomy.md 3a65b7fc4692cf71b25072575daea564
RKF/theorum/thermodynamics/10_thermodynamic_cut_square_response.md 8a8fe924b57096a18cf1976d3728820e
RKF/theorum/thermodynamics/lambda_geometry/11_noncommutativity.md 5ad36f03b0e654dd5f9e437f1728d672fd
RKF/theorum/thermodynamics/lambda_geometry/12_admissible_flow.md f8f31f12ae7e6da876644c1bd7e12775be
RKF/theorum/thermodynamics/lambda_geometry/13_refinement.md a324c04tae3ca33e269bde20aeba2de88
RKF/theorum/thermodynamics/lambda_geometry/14_constrained_flow.md 24d3244991d231e1b3c125daa32a2512b2
RKF/theorum/thermodynamics/lambda_geometry/15_curvature_flow.md 19ba0212cd7e1b5eac71d3b7157081bcf7
RKF/theorum/thermodynamics/lambda_geometry/16_primitive_flow.md 4261b0c1ff5c3e0db8614206f23b0bd78
RKF/theorum/thermodynamics/README.md 5b944f09b48f9621db5335865a3f4475
RKF/theorum/thermodynamics/lambda_geometry/README.md e4675c54063d425b1ef871059629183b
RKF/theorum/thermodynamics/SOURCE_PROVENANCE.md 614aa624febca997bd99b168b8fbc73
RKF/theorum/thermodynamics/lambda_geometry/SOURCE_PROVENANCE.md 37f18e7c3cc9dd7528f8e733b75a3a29c
RKF/theorems/foundation/FOOE_NATIVE_EULER_FROM_IOTA_COMPLEXITY.md 2654b5cab54e5f103b01471ffc06ca5e
RKF/theorems/foundation/FOOG_NATIVE_LOGARITHM_AND_POWER.md 488eabd5bf14952b503a725c45afcc24
RKF/theorems/foundation/FOOH_NATIVE_NATURAL_ARITHMETIC_AND_POWER.md 96609e904979bd208f293e3a0ee226bd6
RKF/theorems/foundation/FOOI_NATIVE_DIRICHLET_ZETA_EULER_PRODUCT.md 839f063aa88e2e81b25333e2c2caaa7
RKF/certificates/foundation/FOOGHI_LOG_ARITHMETIC_ZETA_VOLUMES.md d4eFADME5nd43aa10f6e60962473c5ca
RKF/certificates/foundation/F00_FOOE_EULER_VO_1/README.md 5c86becf7aa98d7cacd728fdfd528628
RKF/certificates/foundation/NUMERICAL_PROOF_PROTOCOL.md 88b42ab7316de31f5adbd777272acd44
RKF/certificates/foundation/VERIFICATION_CAMPAIGN.md 5850648d9a72a0f0da83ffe6c6d77037
RKF/papers/rkf_completed_weil_endpoint/CERTIFICATE_TABLE.md 4a6ec4cf6dd84be874c7bd92e34e41c3
RKF/papers/rkf_completed_weil_endpoint/CLAIM_BOUNDARY.md 0b8e1979131678833c8c01f889f4e2a0
RKF/papers/rkf_completed_weil_endpoint/README.md c5b9581eea10b6ce2815d66757cf843c
RKF/papers/rkf_completed_weil_endpoint/REVIEWER_GUIDE.md 52e213ecb0a04c2e3897ad0ea67f7579
RKF/papers/rkf_completed_weil_endpoint/main.tex d99a956f8a900d57fac402a75b45733e
RH-Framework/README.md 6f83553e7efb0fa9ece686d6ed840ec6
RH-Framework/REVIEWER_CHECKLIST.md ea87f831b9dd31ccce753a7a9df25ae4
RH-Framework/theorems/LEDGER.md 2e0a87038ae9844fa1f1532fc26968c9
RH-Framework/theorems/NATIVE_DERIVATION_CONTAMINATION_AUDIT.md 13ad353b591b4279228919c100011cef
RH-Framework/theorems/T01_E4D_NATIVE_MEAN_MULTIPLIER_BOUNDS.md 8fcd9e7c4126171758a5fce4f6b1f7aec

RH-Framework/theorems/T01_E4_NATIVE_FUNCTION_COMPLETION.md 45ac9f570848a7e0f8a4b6b0cf90826b
 RH-Framework/theorems/T01_E5_NATIVE_RATIONAL_UGD_REFINEMENT_ADAPTER.md 35b77e9da380b87603d5634876677
 RH-Framework/theorems/T01_E_NATIVE_REFINEMENT_COUNT_MEASURE.md 1372b7e688dcb0e11801a63ca9bf630
 RH-Framework/theorems/T01_NATIVE_SUMMABILITY_AND_RECOGNITION.md 03500866a7f68db4d482b1bf8d20cd4b8
 RH-Framework/theorems/T03_ACTION_DOMAIN_ADAPTER.md a81f49670d4f9ae476a2281059699aea
 RH-Framework/theorems/T03_C1_FINITE_PACKET_ACTION_INVENTORIES.md 09f6fe9671f25e79d36df047ba314690
 RH-Framework/notebooks/T01_NATIVE_FOUNDATION_JUPYTER.md bfa8df8e6014a48e7fb3949638b1878b
 RH-Framework/notebooks/T01_NATIVE_RECONSTRUCTION_JUPYTER.md 07cb052471e2310a1b67fd0069cab43
 RH-Framework/notebooks/T03_ACTION_DOMAIN_JUPYTER.md 792d64383c8b8af96d9517b1adbbdf8b
 Publications/papers/curvature-information-duality/README.md 899cd8775ef5f3d5c5d4f8ec9dcc725e
 Publications/papers/curvature-information-duality/LINEAGE.md 78cd3d3eb6a998ef3706aba95bf03
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 Publications/papers/poincare-calibration/README.md 0c7572c95a226876885d25299661911e
 Publications/papers/recognition-seam-topology/README.md 08834cbaa54928bfb70ddcffa06bef9
 Publications/papers/recognition-seam-topology/LINEAGE.md 00a36aace8be70de1ecfe21adfa57c60
 Publications/papers/representation-complete-native-seam-integration/README.md 130e3406a7d050da7879570ea9dd9c2a
 Publications/papers/yang-mills-certified-benchmark/README.md 36360fab7087b83a82b600f14f1cc4d3
 Publications/papers/yang-mills-certified-benchmark/LINEAGE.md 6b834449c8ceeac53b0aa9995106c761
 Publications/papers/yang-mills-certified-benchmark/paper/yang-mills-certified-benchmark-README.md 4d4b11801e20b34f1ed5a68343e531b3
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Appendix B

Automated Repairs

- RKF/theorum/31_cut_variational_minimal_observer_theorem.md: control bytes repaired before conversion: 0x08->\{\}b
- RKF/theorum/thermodynamics/08_canonical_connection_physical_flux.md: control bytes repaired before conversion: 0x0c->\{\}f
- RKF/papers/rkf_completed_weil_endpoint/main.tex: compiled in nonstop mode; recoverable LaTeX errors present in source
- RH-Framework/theorems/T01_E5_NATIVE_RATIONAL_UGD_REFINEMENT_ADAPTER.md: control bytes repaired before conversion: 0x0c->\{\}f